Clinical Trials in Practice

From Foundations to AI Agents

Alexandra Sokolova Vadim Sokolov

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Preface

A compound that looked decisive on the bench can still turn out inert, unsafe, or no better than placebo in real people. Laboratory results do not settle what helps patients. The clinical studies that do are large, slow, expensive, and tightly regulated. There is no substitute for them.

A well-run trial can show that a treatment extends life or eases suffering. A poorly designed one can hide a real benefit, miss a serious harm, or burn years and produce nothing usable. When you account for failed programs and the decade of capital tied up along the way, bringing a drug to market costs billions of dollars (DiMasi, Grabowski, and Hansen 2016).

The work is also getting harder. Trials enroll fewer patients, with tighter entry criteria, because the diseases are rarer or the treatments more selective, while the protocols grow denser: biomarkers, multiple arms, sites across many countries. Adaptive designs, Bayesian methods, and master protocols exist to meet that pressure. They work only when the planning and oversight are as serious as the mathematics.

This book treats clinical trials as science and as regulated operations, and it gives Phase III the most room, because that is where approval is won or lost. Methodology, operations, and technology are treated together, not as separate subjects. Where it turns to automation and AI, one test applies: the tool must stand up in an inspection. Without validation, governance, and a named person accountable for the decision, it does not belong.

Who This Book Is For

Medical directors, study managers, and others in clinical development who need design, execution, and technology in one place. Investigators and site staff who want to understand what sponsors mean by quality. Students and managers deciding whether AI has a place in regulated research.

Organization

Part I: Foundations covers how clinical research took its present shape, where a trial sits in drug development, the regulatory frame, and the ethics of research with human subjects.

Part II: Clinical Trial Phases takes up development economics and the aims, designs, and decision points from Phase I through Phase IV.

Part III: Phase III Deep Dive is the core of the book: confirmatory methods and the daily work of a study, from design and endpoints through sites, data, monitoring, safety, and documents, then the systems and automation underneath that work.

Part IV: Trial Completion and Beyond ends with closeout, the clinical study report, and submission: the last steps that decide whether years of clinical work become an approved medicine.

Part I.

Part I: Foundations of Clinical Trials

1. Historical Context and Evolution

The rules that govern clinical trials were not written in the abstract. They were written after people were harmed. A liquid antibacterial mixed with a toxic solvent killed more than a hundred patients in 1937, many of them children. Nazi physicians ran experiments on concentration camp prisoners without consent. In Tuskegee, researchers watched syphilis progress in Black men for forty years and, once penicillin was the cure, kept it from them. Thalidomide left thousands of infants with ruined limbs after their mothers took it for morning sickness. Each failure forced a new protection into law or ethics: safety testing before marketing, voluntary consent, independent review, proof of efficacy. The protocols, monitors, and oversight committees of the modern trial are scar tissue from wounds inflicted on subjects who had no voice.

1.1. The Elixir Sulfanilamide Disaster (1937)

Sulfanilamide worked. To reach patients who had difficulty swallowing tablets, the S.E. Massengill Company dissolved the antibacterial in diethylene glycol, a sweet-tasting but toxic industrial solvent related to the ethylene glycol used in antifreeze. What followed was the first disaster to show that commercial drug manufacturing without safety testing could kill on a large scale (U.S. Department of Agriculture 1937).

The Secretary of Agriculture's report to Congress traced the deaths to a gap in the law (U.S. Department of Agriculture 1937):

> "Before the 'elixir' was put on the market, it was tested for flavor but not for its effect on human life. The existing Federal Food and Drugs Act does not require that new drugs be tested before they are placed on sale."

The chief chemist testified that *no tests were made to determine the toxicity* of either the separate ingredients or of the finished product (U.S. Department of Agriculture 1937). The "control laboratory" checked the elixir for *appearance, flavor, and fragrance*, not for safety. The formula used (see Table 1.1) was:

Table 1.1.: Elixir Sulfanilamide Formula (1937)

Ingredient	Amount
Sulfanilamide	58.5 pounds
Diethylene Glycol	60 gallons
Elixir Flavor	1 gallon
Raspberry Extract	1 pint
Saccharin Soluble	1 pound
Amaranth Solution 1-16	1.5 pints
Caramel	2 fluidounces
Water	to make 80 gallons total

The product was labeled simply as "Elixir Sulfanilamide," with no disclosure of diethylene glycol and no warning of danger, and most of it was administered on physicians' prescriptions (U.S. Department of Agriculture 1937). Of the 240 gallons manufactured and distributed, all but about 6 gallons were recovered before they could be consumed; that small quantity was enough to kill, and the report to Congress counted 93 deaths in fifteen states as it was written in late 1937 (U.S. Department of Agriculture 1937). The final toll is generally put at more than 100 people, many of them children (Ballentine 1981).

Seizure had to rest on a technicality: the word "elixir" implies an alcoholic solution, but this product was a diethylene glycol solution. Had it been called a "solution," no charge could have been brought.

Congress responded with the *Federal Food, Drug, and Cosmetic Act of 1938*, which for the first time required manufacturers to demonstrate the safety of new drugs before marketing them. The 1938 Act left the boundary between physician-prescribed and self-medication drugs unsettled; the *Durham-Humphrey Amendment of 1951* fixed that line in statute, defining which drugs require a practitioner's supervision and may be dispensed only on prescription (United States Congress 1951).

1.2. The Nuremberg Code (1947)

After the war, a U.S. military tribunal put 23 German physicians and administrators in the dock for war crimes and crimes against humanity, among them experiments on concentration camp prisoners, without consent, that caused suffering and death under the banner of medical science. Before naming who was guilty, the judges set down what lawful human research requires. That statement became the *Nuremberg Code*: ten principles forged in a criminal judgment (Nuremberg Military Tribunal 1949; University of North Carolina at Chapel Hill 2024). The first principle is still the one that matters most: no one may be enrolled without free, informed consent.

The rest follow from it. The study must be worth doing, the risks must not outrun the possible gain, the investigators must be competent, and the subject must be free to walk away. If the experiment turns harmful, the physician in charge must stop it.

1.3. The Thalidomide Tragedy (1961-1962)

FDA medical officer Dr. Frances Kelsey refused to approve thalidomide for marketing, citing insufficient safety data. The drug was already selling in Europe as a sedative for sleep and nausea, including morning sickness, and it caused severe birth defects, primarily phocomelia (limb malformations), in an estimated 10,000 infants worldwide. Kelsey's refusal kept it off the U.S. market; about 17 birth-defect cases were still reported from investigational use.

The episode showed that proof of safety alone was not enough: drugs needed demonstrated *efficacy*, and there were still no systematic rules for studying drugs in pregnancy or for reporting adverse events. The *Kefauver-Harris Drug Amendments of 1962* followed directly (see Figure 1.1). Manufacturers now had to prove efficacy as well as safety before marketing, and proof of efficacy had to rest on *substantial evidence* from *adequate and well-controlled* investigations: the statutory phrase that still anchors modern Phase III practice (Chapter 9). Clinical pharmacologist Louis Lasagna was among those who pressed Congress for that standard, testifying that adequately controlled drug comparisons were almost impossible to find under the old regime (Junod 2008). The Amendments also established the formal Investigational New Drug (IND) application. Clinical work had to be grounded in prior animal studies, adverse reactions had to be reported, and research subjects had to give informed consent.

1.4. The Declaration of Helsinki (1964)

The Nuremberg Code came out of a criminal judgment. The Declaration of Helsinki came from the medical profession itself: the *World Medical Association (WMA)* adopted it in 1964 as detailed ethical guidance for physicians conducting medical research (World Medical Association 2013). The 1964 text distinguished therapeutic from non-therapeutic research, required that studies rest on adequately performed laboratory and animal work and be conducted by scientifically qualified persons, and sharpened Nuremberg's vague proportionality principle into a more operationalized standard: research risk must be outweighed by the scientific importance of the objective.

Several protections now regarded as central to the Declaration came later. Independent committee review of protocols before enrollment entered with the 1975 Tokyo revision, the conceptual ancestor of today's ethics committees and IRBs; explicit obligations around trial registration and the publication of results, including negative findings, were strengthened in the revisions of the 2000s. Revision has continued, most recently in 2024 (World Medical Association 2025).

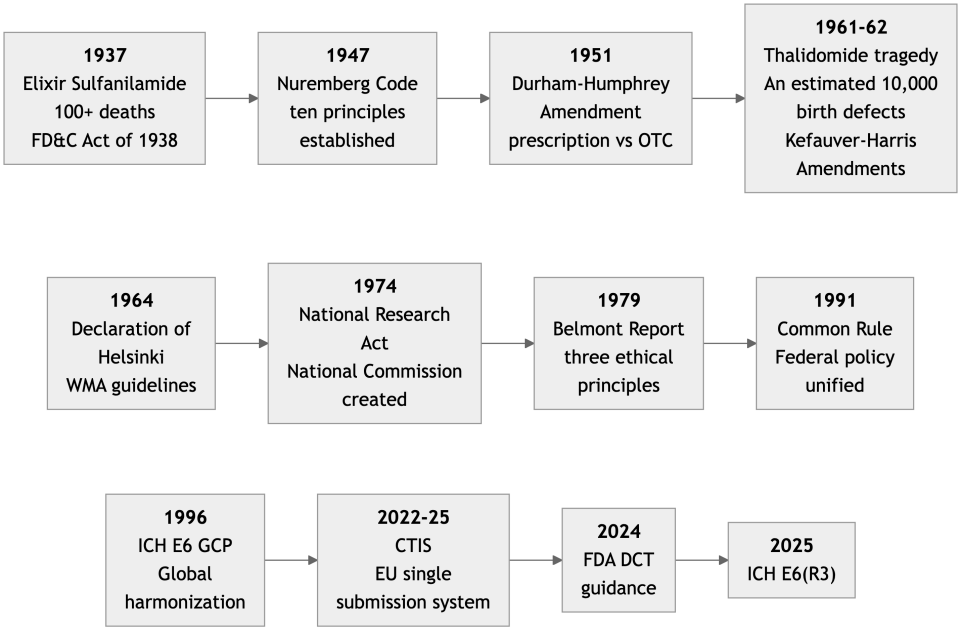


Figure 1.1.: Timeline of Major Regulatory Milestones in Clinical Research

1.5. The Tuskegee Syphilis Study (1932-1972)

Between 1932 and 1972, the U.S. Public Health Service conducted the Tuskegee Syphilis Study on approximately 600 low-income African-American men in rural Alabama (399 with syphilis and 201 controls without) to observe the natural progression of untreated syphilis. Participants were misled into believing they were being treated for "bad blood," a local catch-all that obscured syphilis. After penicillin became standard therapy in the mid-1940s, researchers both withheld it and blocked men from getting treatment elsewhere. The study ran until it was exposed in 1972.

1.6. The Belmont Report (1979)

In response to Tuskegee and other ethical violations, the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research produced the *Belmont Report* in 1979 (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979). Its three principles (Table 1.2) still guide clinical research:

Table 1.2.: The Three Principles of the Belmont Report

Principle	Core Requirement	Application to Research
Respect for Persons	Individuals should be treated as autonomous agents; persons with diminished autonomy are entitled to protection	*Informed consent*: Subjects must voluntarily agree to participate with full understanding of risks, benefits, and alternatives. Vulnerable populations require additional protections.
Beneficence	Do not harm; maximize possible benefits and minimize possible harms	*Risk-benefit assessment*: Research risks must be reasonable in relation to anticipated benefits. Study design must maximize benefits and minimize risks.
Justice	Fair distribution of research burdens and benefits	*Equitable subject selection*: The selection of research subjects must be fair. Those who bear the burdens should potentially benefit. Vulnerable groups should not be exploited.

In the United States they became binding federal rules through the *Federal Policy for the Protection of Human Subjects*, the Common Rule, adopted in 1991 and codified for HHS (U.S. Department of Health and Human Services 2024). It requires prior IRB review and approval, documented informed consent, and additional protections for pregnant women, prisoners, and children; at adoption it aligned those requirements across fifteen federal departments and agencies. For clinical investigations of drugs and devices, FDA imposes parallel obligations for informed consent and IRB review (U.S. Food and Drug Administration 2023b, 2023c). The 2018 revision, among other changes, generally requires a single IRB for U.S. multi-site research funded by Common Rule agencies and a concise "key information" section at the start of consent forms. How IRBs exercise that authority is taken up in Chapter 4. Internationally, the Declaration of Helsinki and the CIOMS guidelines fill the corresponding ethical role.

1.7. ICH and Global Harmonization (1990s-Present)

Pharmaceutical development had gone global; its standards had not. The *International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH)* was established in 1990 to bring together regulatory authorities and pharmaceutical industries from Europe, Japan, and the United States.

1.7.1. ICH E6 GCP

The ICH *Guideline for Good Clinical Practice (E6)* was first adopted in 1996 as an international ethical and scientific quality standard for designing, conducting, recording, and reporting clinical trials (International Council for Harmonisation 1996). It rests on the Declaration of Helsinki, and it starts where this chapter

started: subject welfare takes priority over scientific objectives. Risks must be justified by expected benefits; trials need clear protocols, qualified investigators, and documented consent. E6(R2) in 2016 added risk-based monitoring and quality management (International Council for Harmonisation 2016a). E6(R3) restructured GCP into Principles plus Annexes (*Quality by Design*, risk-proportionate oversight, decentralized trials, technology-neutral data standards); Principles and Annex 1 became effective in the EU on July 23, 2025, while Annex 2 (decentralized, pragmatic, and other non-traditional interventional trials) reached Step 4 in June 2026 and takes effect in the EU on January 15, 2027 (International Council for Harmonisation 2025b; European Medicines Agency 2024a).

1.8. Recent Developments (2020s)

COVID-19 forced sponsors and sites to improvise continuity (remote visits, local labs and care, shipment logistics, documented deviations), and FDA captured much of that practice in its public-health-emergency trial-conduct guidance (U.S. Food and Drug Administration 2021b). What began as contingency stuck. FDA finalized *digital health technology* guidance for remote data acquisition in 2023 (U.S. Food and Drug Administration 2023g), then in 2024 a framework for *decentralized elements* (telehealth, in-home visits, local healthcare providers, direct-to-participant investigational product, electronic consent) that treats decentralization as design choices, not a separate trial species (U.S. Food and Drug Administration 2024n).

Method and evidence rules moved with the operations. Adaptive designs, already framed by FDA's 2019 guidance on pre-specified adaptations, integrity protections, and simulation of operating characteristics (U.S. Food and Drug Administration 2019a), gained a draft international counterpart in ICH E20 (Step 2b, June 2025) (International Council for Harmonisation 2025a). Building on its 2010 device guidance (U.S. Food and Drug Administration 2010b), FDA's January 2026 draft for drugs and biologics treats Bayesian primary inference, including borrowing from external information, as acceptable when sponsors pre-specify success criteria, justify priors, evaluate operating characteristics by simulation, plan sensitivity analyses for prior-data conflict, and document the work for review (U.S. Food and Drug Administration 2026g).

Real-world data followed the same arc from permission to proof: 2023 guidance on using RWD/RWE for regulatory decisions (U.S. Food and Drug Administration 2023f), then 2024 expectations for EHR and claims sources that demand provenance, completeness, linkage quality, and outcome ascertainment (U.S. Food and Drug Administration 2024k). Master protocols and platforms, accelerated in the pandemic, drew their own FDA guidance on governance and multiplicity, issued as a draft in December 2023 and revised in June 2026 (U.S. Food and Drug Administration 2026e).

Who gets into trials became a statutory fight. FDORA (2022) required *Diversity Action Plans* for trials that start enrolling more than 180 days after FDA finalizes

its guidance on them; FDA's 2024 draft said when and how (U.S. Food and Drug Administration 2024e). After the January 2025 executive order on diversity programs, FDA pulled that draft from its website and restored it weeks later under court order (Grossi 2025). The statutory deadline for the final guidance, June 26, 2025, passed with the draft still in place, so the requirement has yet to take effect (Grossi 2025; U.S. Food and Drug Administration 2024e). A December 2025 final guidance on trial participation, revising older eligibility-criteria advice under a different mandate, kept the demand for representative enrollment while dropping the "diversity" label (U.S. Food and Drug Administration 2025c).

Oversight got more analytical. Risk-based quality management and centralized monitoring now lean on statistical signals and machine learning to prioritize sites and source review (Association of Clinical Research Organizations 2025). The same stack is used for protocol drafting, feasibility, safety triage, and TMF work, and whatever it produces still has to be validated, attributable, and inspectable (National Institute of Standards and Technology 2023).

In Europe, the 2020s turned the *Clinical Trials Regulation* (2014) into working infrastructure. The *Clinical Trials Information System (CTIS)* became the single submission and coordinated-assessment channel for multinational trials; it went live on January 31, 2022, became mandatory for new applications on January 31, 2023, and finished transitioning legacy trials by January 31, 2025 (European Medicines Agency 2022).

Remote visits, CTIS portals, and model-assisted monitoring change how trials run. They do not change what the history in this chapter was for: keeping subjects from being harmed so that someone else can finish a study.

2. Lifecycle of a Clinical Trial

The molecule that enters a Phase I trial has already survived years of discovery research and preclinical testing, and the Clinical Study Report that trial produces is just one element of a regulatory submission that synthesizes evidence across multiple studies. In between sits a sequence of interconnected stages, each with its own deliverables and regulatory checkpoints. This chapter maps that sequence from initial concept through regulatory approval, with pointers to the chapters that take each stage in depth.

2.1. The Big Picture

Clinical development, the stretch of the arc in Figure 2.1 where human trials occur, typically spans many years and represents the largest single investment in bringing a drug to market (U.S. Food and Drug Administration 2018b; DiMasi, Grabowski, and Hansen 2016).

Most candidates fail during clinical development, which makes the design and execution of each trial consequential for the entire program.

2.1.1. The Pipeline Misnomer

The linear "chevron" model (Figure 2.3) is a useful high-level roadmap and a "deeply engrained misnomer" that connotes a consistent, unidirectional flow (Wagner et al. 2018a). In reality, therapeutic development is a "diverse web of failure-prone iterative learning loops" traversed by scientists, physicians, regulators, payers, and patients.

The Forum on Drug Discovery, Development, and Translation of the US National Academies of Sciences, Engineering, and Medicine proposed the **Drug Discovery, Development and Deployment Map (4DM)** to capture that complexity (see Figure 2.2). The 4DM replaces the pipeline with a set of interacting "neighborhoods" (the full map has eight; the simplified view here highlights five: basic science, biomarker development, clinical research, regulatory review, and the medical landscape) in which discoveries in one area can render earlier results irrelevant or open new shortcuts to later stages. The map captures this reality with "reiterative loops," indicating that many "shots on goal" are often required before success is achieved at any step (Wagner et al. 2018a, 2018b).

Failure rates are not uniform across therapeutic areas. In the Biotechnology Innovation Organization (BIO)'s analysis of 2011-2020 development programs, the

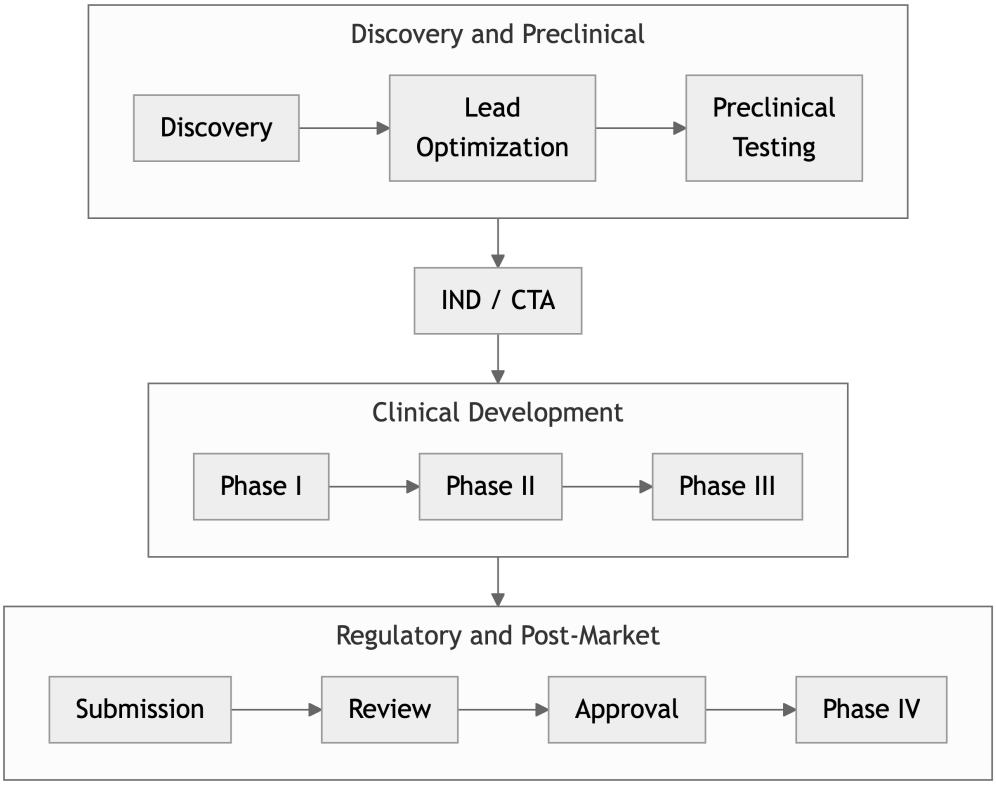


Figure 2.1.: Clinical trials within the drug development lifecycle

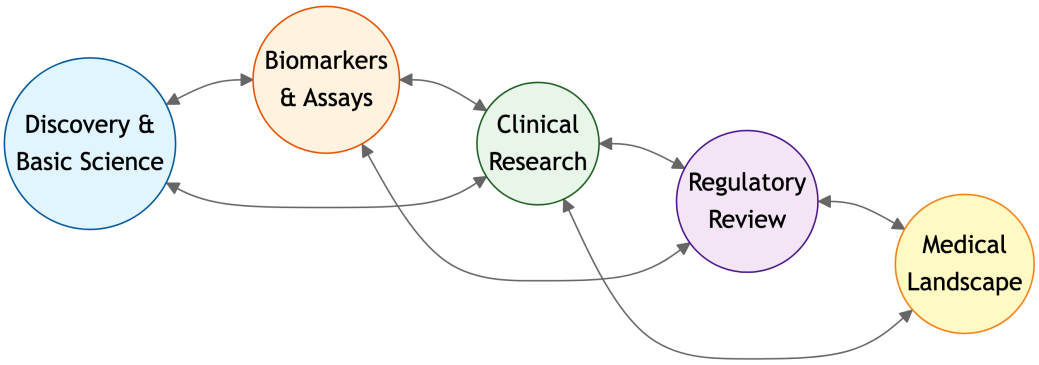


Figure 2.2.: Simplified 'Neighborhood' view of the 4DM Map (adapted from Wagner et al., 2018)

overall *likelihood of approval (LOA) from Phase I* was **5.3%** for oncology versus **9.3%** for non-oncology programs, with the largest attrition occurring at the Phase II transition (oncology Phase II-to-III success **24.6%**) (Biotechnology Innovation Organization, QLS Advisors, and Informa Pharma Intelligence 2021). Oncology carries high scientific uncertainty, heterogeneous patient populations, and intense competition all at once. Academic analyses of trial outcomes reach a similar qualitative conclusion: oncology has among the lowest probabilities of advancing from early clinical testing to approval, even as biomarker-enriched programs can outperform historical averages (Wong, Siah, and Lo 2019).

2.2. Trial Phases at a Glance

The phases are not as sharply bounded as Figure 2.3 and Table 2.1 make them look. Phase I/II combination studies are common, and some programs run Phase II and Phase III concurrently. The conceptual framework remains useful because each phase still has a primary purpose.

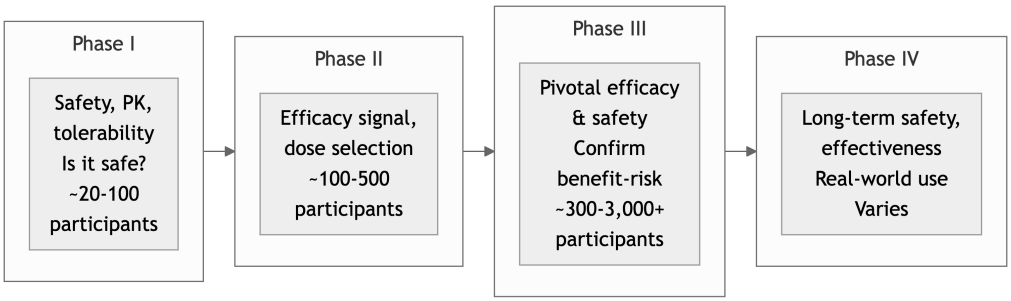


Figure 2.3.: Clinical trial phases and their primary objectives

Table 2.1.: Clinical Trial Phases Overview

Phase	Primary Question	Typical Size	Duration	Key Decisions
Phase I	Is it safe in humans?	20-100	6-12 months	Dose escalation, stopping rules
Phase II	Does it show efficacy?	100-500	1-2 years	Go/no-go, dose selection
Phase III	Does benefit outweigh risk?	300-3,000+	2-4 years	Primary endpoint, label claims
Phase IV	How does it perform post-market?	Varies	Ongoing	Safety signals, new indications

The sequence is cumulative. Phase I establishes that the drug can be given to humans at tolerable doses; Phase II produces the first evidence of therapeutic effect

and identifies the dose(s) to carry forward; Phase III confirms that the drug works in a larger, more representative population and characterizes the safety profile sufficiently to support labeling. Phase IV continues surveillance after approval. Each phase has a chapter of its own (Chapter 7, Chapter 8, Chapter 9, Chapter 10).

2.3. Anatomy of a Phase III Trial

Phase III trials represent the largest investment and the highest operational complexity, and the lifecycle of a single one (Figure 2.4) is a template that applies, with variations, to earlier phases.

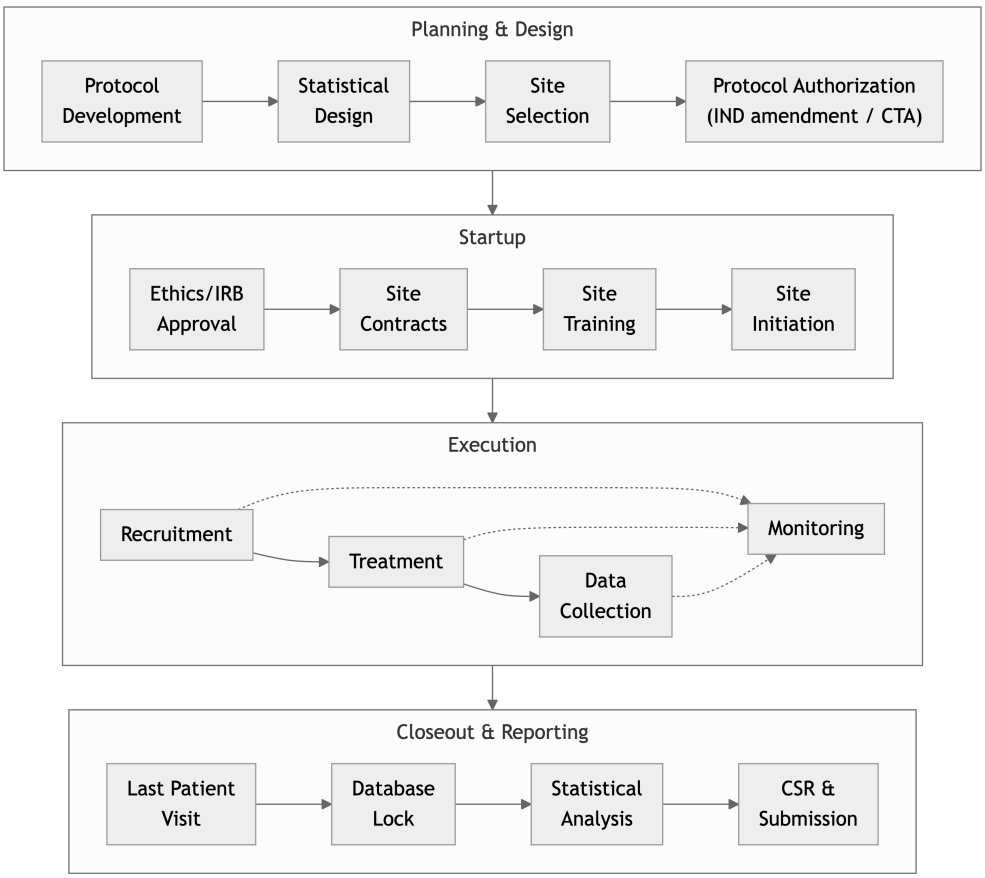


Figure 2.4.: Lifecycle of a Phase III clinical trial

2.3.1. Planning and Design

A poorly designed protocol creates problems that cannot be fixed during execution, which is why months of work precede the first enrollment. The *protocol* (the scientific and operational blueprint for the trial) specifies the objectives, design, patient

population, treatments, assessments, endpoints, and analysis plan, and drafting it takes clinical, statistical, regulatory, and operational input.

Statistical design (Chapter 11) determines sample size, randomization scheme, and analysis approach, while *adaptive and Bayesian methods* (Chapter 12) extend it to platform trials and designs that update as data accumulate. *Study design* (Chapter 13) addresses the broader scientific architecture: parallel vs. crossover, superiority vs. non-inferiority, adaptive vs. fixed. When a concurrent randomized control is impractical, *real-world evidence and external control arms* (Chapter 14) offer an alternative source of comparison. *Protocol benchmarks* (Chapter 15) examines what makes a protocol operationally feasible.

Site selection (Chapter 18) identifies the investigators and institutions that will conduct the trial. Regulatory authorization for a Phase III protocol usually rides on an existing IND (as a protocol amendment) in the US, plus Clinical Trial Applications (CTAs) in countries that require them; a new IND is the exception, not the rule, by this stage.

2.3.2. Startup

Startup runs from regulatory approval to first patient enrollment, and it is one of the more variable and time-consuming phases to plan. The work includes obtaining *ethics committee/IRB approval* (Chapter 4), negotiating *site contracts* and budgets, *training site staff* on the protocol and systems, and conducting *site initiation visits* to verify readiness; the *logistics and project management* chapter (Chapter 20) covers the operational mechanics of site activation and vendor coordination. Recent Tufts CSDD benchmarking of site activation and patient enrollment gives sponsors reference metrics for setting realistic startup timelines (Lamberti et al. 2024).

2.3.3. Execution

Recruitment (Chapter 19) is frequently the most challenging operational element. Enrollment performance varies widely across studies and sites, and recruitment delays remain a leading cause of timeline slippage (Lamberti et al. 2024). Once patients are enrolled, *randomization and blinding* (Chapter 16) assign them to treatment arms while protecting the integrity of the comparison, and *endpoints* (Chapter 17) define what is measured and how success is determined.

Data management (Chapter 21) ensures that the information collected is accurate, complete, and ready for analysis, while *monitoring* (Chapter 22) provides oversight: verifying that sites are following the protocol, that data matches source documents, and that patient safety is protected. *Safety* (Chapter 23) is a continuous concern throughout execution, since adverse events must be identified, documented, and reported according to regulatory requirements.

The *Clinical Trial IT Infrastructure* chapter (Chapter 25) examines the technology systems that carry all of this: Electronic Data Capture (EDC), Clinical Trial

Management Systems (CTMS), electronic Trial Master File (eTMF), and Randomization and Trial Supply Management (RTSM). The *AI Agents in Clinical Development* chapter (Chapter 26) explores how AI is beginning to change operational workflows.

2.3.4. Closeout and Reporting

After the last patient completes the last visit, closeout begins. *Database lock* freezes the data for analysis. Biostatisticians execute the pre-specified analysis plan. Medical writers prepare the *Clinical Study Report (CSR)* (Chapter 28), the document that presents the trial's findings.

The CSR becomes part of the *regulatory submission* (Chapter 29): the New Drug Application (NDA), Biologics License Application (BLA), or Marketing Authorization Application (MAA) that requests marketing authorization. The *essential documents* chapter (Chapter 24) covers the documentation requirements that must be maintained throughout and archived after the trial, and the *closeout* chapter (Chapter 27) addresses the operational side: site closeout visits, document archiving, and the administrative closure of the trial.

2.4. Cross-Cutting Concerns

Four concerns run the entire length of a trial (Figure 2.5).

Regulatory compliance shapes everything from initial protocol design to long-term data retention. *Ethics and patient protection* requires informed consent and IRB oversight maintained from first participant to last. *Quality and data integrity* rests on adherence to the Attributable, Legible, Contemporaneous, Original, and Accurate (ALCOA) principles. *Economic constraints* are the multi-billion-dollar investments that define the feasibility of modern clinical development.

2.5. How to Use This Book

Figure 2.6 lays out the structure, and readers with different backgrounds will want different entry points into it. For a conceptual foundation, start with Part I: history (Chapter 1), regulatory (Chapter 3), and ethics (Chapter 4). Part II covers drug development, economics, and the progression through Phases I-IV. Part III treats Phase III design and execution in operational detail, from statistical methods through monitoring and safety, and Part IV covers closeout, the CSR, and regulatory submission. Technology has two chapters of its own: IT infrastructure (Chapter 25) and AI agents (Chapter 26).

Clinical trials are both a scientific method and an operational system. The science determines what questions can be answered; the operations determine whether the trial can actually answer them.

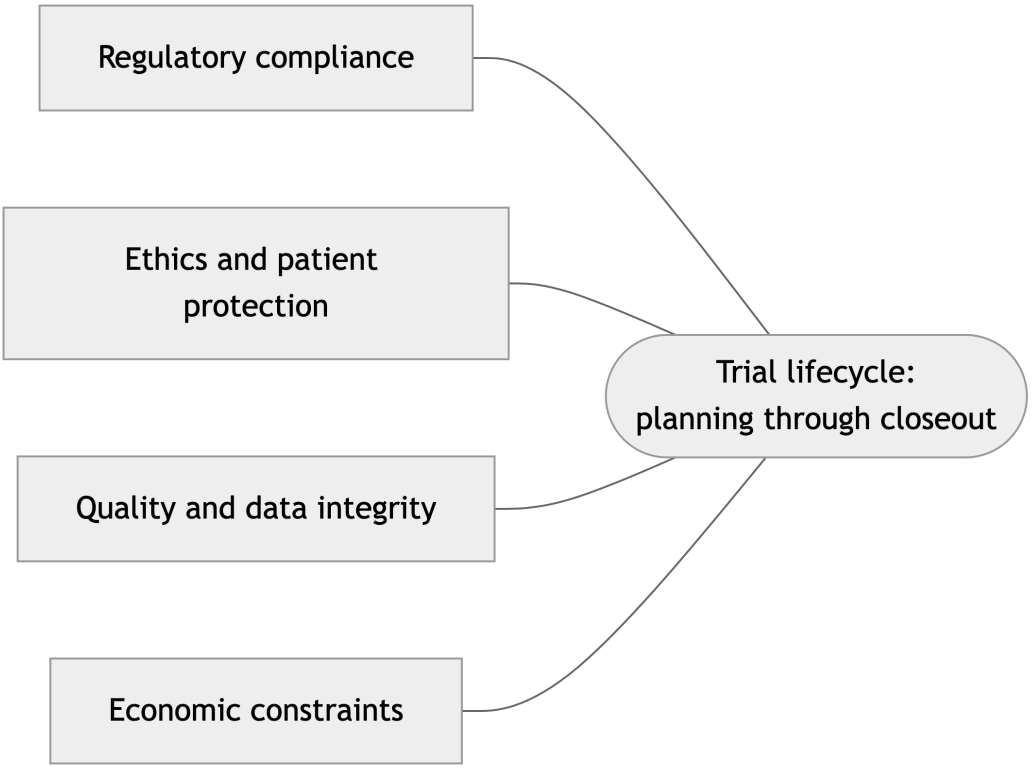


Figure 2.5.: Cross-cutting concerns that span the trial lifecycle

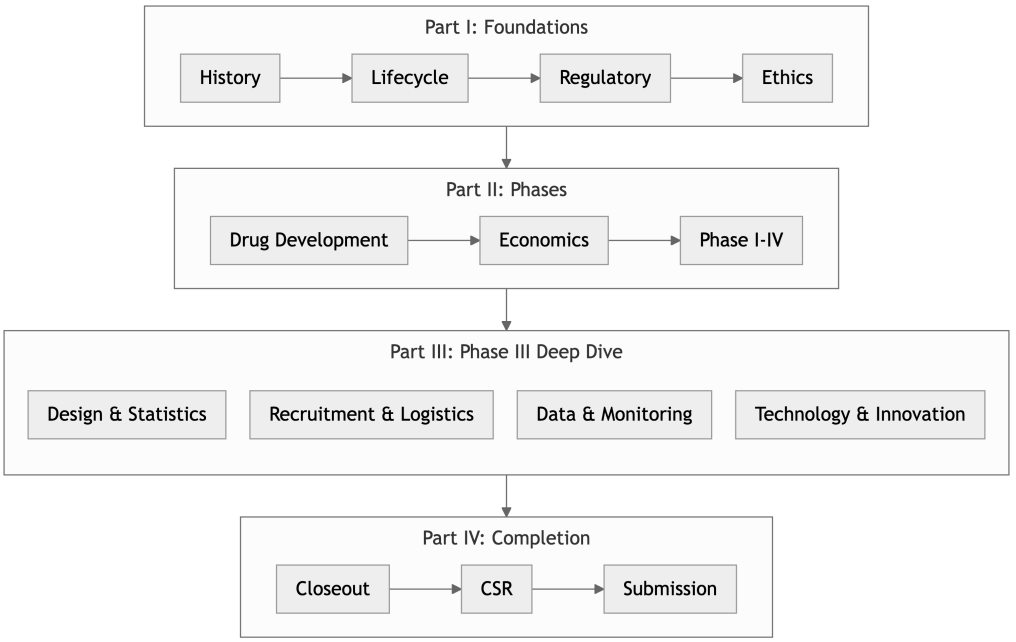


Figure 2.6.: Book structure and chapter relationships

3. Regulatory Framework

An investigational drug does not reach its first human subject in the United States until a sponsor has filed an *Investigational New Drug (IND)* application and either 30 days have passed since FDA received it without a clinical hold or FDA has told the sponsor the study may begin sooner. For most INDs, silence is permission. Much of the framework works that way: obligations fixed in advance, verified afterwards by inspection, with consequences that reach as far as debarment. The focus here is the United States, with the international requirements a multinational program must also satisfy.

The Food and Drug Administration (FDA) regulates drugs, biologics, and medical devices (U.S. Food and Drug Administration 2024h). It sits in the Department of Health and Human Services as part of the Public Health Service, alongside the National Institutes of Health (NIH) and Centers for Disease Control and Prevention (CDC), with a mission to protect public health by ensuring that regulated products are safe, effective, and not adulterated or misbranded.

Which scientific center holds jurisdiction over a product is a developer's first question, because the center sets the requirements (Table 3.1).

The Center for Drug Evaluation and Research (CDER) handles prescription drugs, OTC drugs and generics, and most therapeutic biologics. The Center for Biologics Evaluation and Research (CBER) handles vaccines, blood products, gene and cellular therapies. The Center for Devices and Radiological Health (CDRH) handles medical devices and radiation-emitting products. The Human Foods Program (HFP) absorbed the food functions of the former CFSAN in October 2024, and cosmetics moved to the Office of the Chief Scientist. The Center for Veterinary Medicine (CVM) handles animal drugs and animal food.

Table 3.1.: FDA Scientific Centers

Center	Products Regulated	Key Responsibilities
CDER	Prescription drugs, OTC drugs, generics; many therapeutic biologics (e.g., most monoclonal antibodies and therapeutic proteins)	INDs; NDAs and ANDAs; BLAs for CDER-regulated biologics; biosimilars; post-market drug safety

Center	Products Regulated	Key Responsibilities
CBER	Vaccines, blood and blood products, plasma derivatives, allergenic products, gene therapies, cellular therapies	INDs and BLAs for CBER-regulated biologics; lot-by-lot release for certain products
CDRH	Medical devices (Class I-III), radiation-emitting products	510(k) clearances, De Novo, Premarket Approvals (PMA), device safety
HFP	Foods and dietary supplements (absorbs former CFSAN food functions, effective October 2024; cosmetics moved to the Office of the Chief Scientist)	Food safety standards, nutrition labeling
CVM	Animal drugs, animal food	Veterinary product approval

The agency also works with numerous interagency partners, including the Office for Human Research Protections (OHRP) on clinical trial matters, NIH on scientific investigation, CDC on epidemiological matters, and the Department of Homeland Security on bioterrorism and food safety.

3.1. The Code of Federal Regulations

Clinical research rules are codified in *Title 21 of the Code of Federal Regulations* (21 CFR). It runs to thousands of pages; a few parts carry nearly all the weight for a trial.

Part 50 establishes what informed consent must include and how it must be documented (U.S. Food and Drug Administration 2023b). Potential participants must be told the purpose of the research, the procedures involved, the foreseeable risks and potential benefits, the available alternatives, and how their privacy will be protected. They must also be told that participation is entirely voluntary and that they can withdraw at any time without penalty.

That list is easy to satisfy on paper. Explaining a novel gene therapy to a patient with limited scientific background is not, and neither is obtaining meaningful consent from someone who is seriously ill and may see the trial as their best hope. The framework is in the regulation; the judgment is not.

No FDA-regulated clinical investigation begins until an *Institutional Review Board (IRB)* has reviewed and approved it, apart from narrow exceptions such as the emergency use of a test article, which need only be reported to the IRB within five working days (U.S. Food and Drug Administration 2023c). The IRB is an independent ethics committee of at least five members with diverse backgrounds, including scientific expertise and community representation. Its fundamental question is whether the research balances potential benefits against risks to participants

appropriately; it also examines the equity of subject selection, the adequacy of the consent process, and the safeguards in place. Its authority is broad: it can require design modifications or additional protections, and it can refuse approval entirely.

3.1.1. The Clinical Trial Stakeholder Framework

Figure 3.1 shows how responsibility runs among the parties to a trial.

The sponsor bears ultimate responsibility for the trial's conduct and for regulatory compliance. Tasks can move to a CRO by written transfer; accountability for oversight does not move with them. The principal investigator runs the research at the site, obtaining informed consent and protecting participant safety, under the IRB's independent ethical oversight. Over all of them, the regulatory authority may inspect sponsors, CROs, investigators, and IRBs, and can halt research that fails to protect participants adequately.

3.1.2. The IND Application (21 CFR Part 312)

Part 312 governs the IND, and arguably no part of the title matters more to a development program (U.S. Food and Drug Administration 2024b). Without an IND in effect, there is no first dose.

Three sections carry the application. *Chemistry, manufacturing, and controls* describes what the drug is and how it is made, information critical to ensuring that every batch is consistent and safe. *Pharmacology and toxicology* summarizes the nonclinical studies (typically in animals) that support the safety of proceeding to human trials. The *clinical protocol* details exactly how the proposed study will be conducted.

Thirty days is the review window; absent a clinical hold, the sponsor proceeds. The arrangement trades a measure of regulatory control for the urgency of bringing new therapies to patients. Planned emergency research conducted without prospective informed consent under 21 CFR 50.24 is an exception: it needs a separate IND and may not start without FDA's prior written authorization (U.S. Food and Drug Administration 2024b, 2023b).

3.1.3. Electronic Records and Signatures (21 CFR Part 11)

Part 11 sets the requirements for electronic records and signatures, with one goal: an electronic record should be as trustworthy as a paper one (U.S. Food and Drug Administration 2023a). That means validated computer systems, secure audit trails that track every change to the data, controlled access, and electronic signatures that reliably identify the signer.

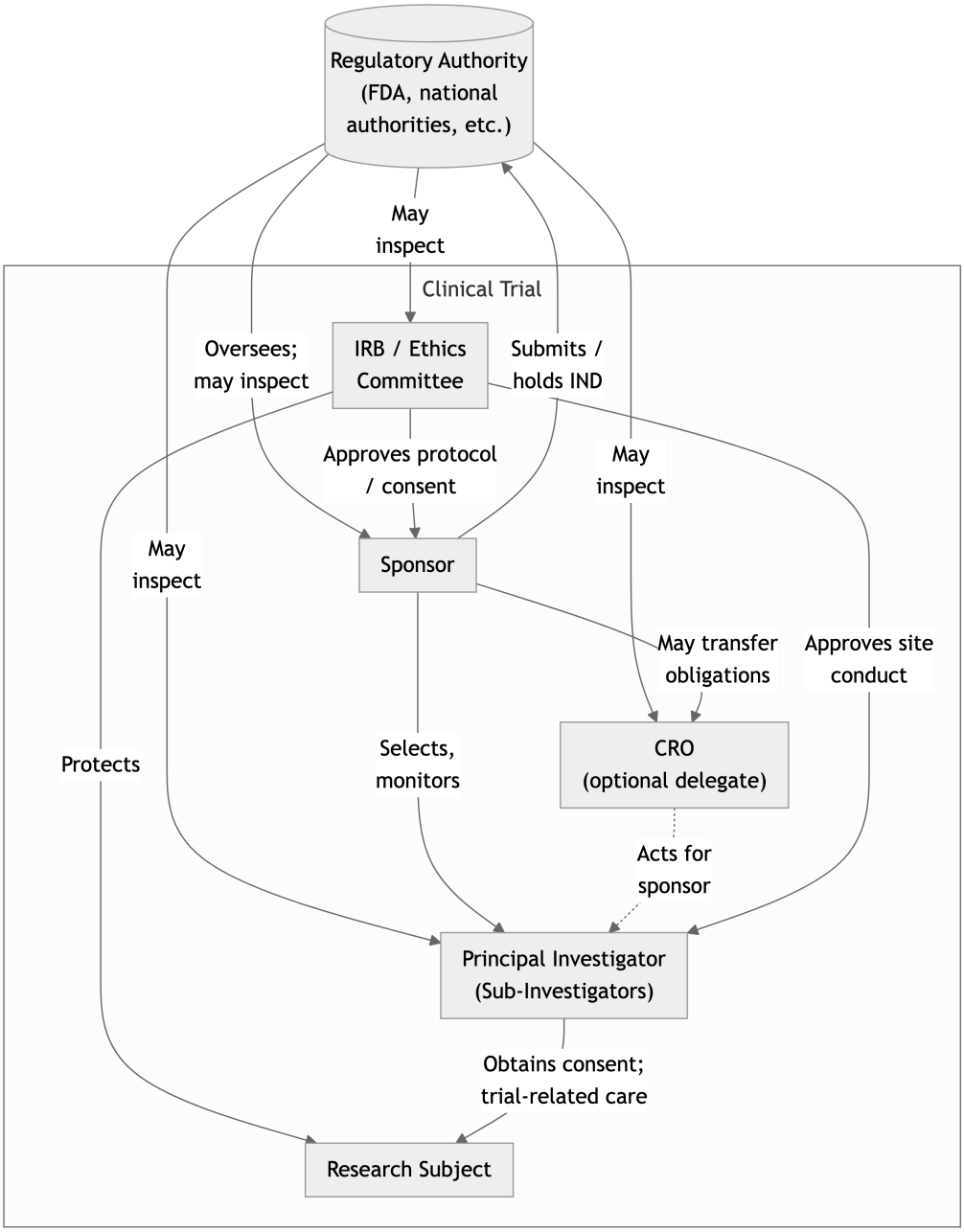


Figure 3.1.: Clinical Trial Stakeholder Relationships

> **Part 11 in Practice**
>
> Part 11 dates from 1997, but the FDA's current approach focuses on the controls most directly related to data integrity. Not every electronic system requires the same level of validation; a risk-based approach guides compliance decisions.

3.2. AI and Automation in a Regulated Trial

The logic behind Part 11 (data integrity, traceability, accountability) now shapes how sponsors deploy AI and automation in clinical development. Most of that activity does not involve a model making an approval decision; it automates *processes* while keeping the evidence chain auditable.

Many high-volume regulatory deliverables are already partly automated through workflow systems: eTMF filing against the TMF Reference Model, query management, and risk-based quality management (RBQM) signals that support targeted monitoring. These can be rules engines plus machine-learning classifiers (e.g., document type classification, completeness checks) that cut manual effort while producing an inspectable audit trail (TMF Reference Model Initiative 2024; Association of Clinical Research Organizations 2025). This matters because inspection findings are frequently about process, not statistical methods: missing essential documents, inconsistent data corrections, inadequate oversight. Automation can reduce *variance* in execution, but only when it is controlled, validated, and monitored like any other regulated computerized system (U.S. Food and Drug Administration 2023a).

Generative AI increasingly drafts narratives, summarizes deviations, and prepares sections of reports and submission documents. The regulatory risk is fluency itself: text that reads well can obscure uncertainty or introduce untraceable statements that do not faithfully reflect the source record (Messeri and Crockett 2024). Empirical work in clinical research document generation has highlighted compliance and factuality issues requiring governance, source linking, and review workflows (Z. Wang et al. 2025).

Governance for AI-enabled workflows mirrors general risk management frameworks: define intended use, identify hazards (hallucinated facts, biased triage, missed signals), implement controls (retrieval with provenance, checklists, dual-review, monitoring), and document residual risk and responsibilities (National Institute of Standards and Technology 2023). For AI/ML built into *products* such as medical devices, regulators and partner agencies have articulated Good Machine Learning Practice (GMLP) principles emphasizing lifecycle management and transparency, which translate well to internal clinical operations tooling (U.S. Food and Drug Administration, Health Canada, and UK Medicines and Healthcare products Regulatory Agency 2021).

Regulators do not require sponsors to avoid AI; they require sponsors to manage risk. The posture is less "ban or bless" than "show your work": keep the decision trace, keep the data lineage, and make it clear where humans remain accountable.

3.3. The IND Process in Practice

The filing is often not the first contact: a Pre-IND meeting commonly precedes it, where sponsors align development plans with FDA expectations (Figure 3.2).

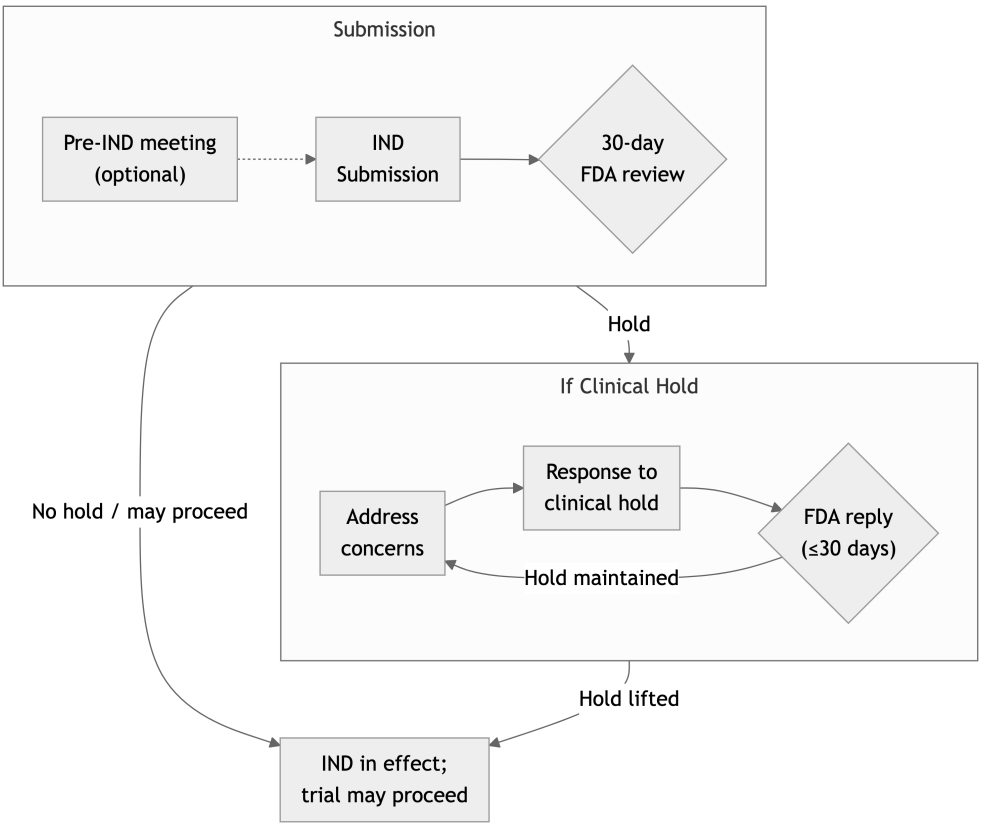


Figure 3.2.: The IND Application Process

Not every IND has the same purpose. A Commercial IND supports research aimed at marketing approval; a Research IND typically comes from an academic institution for non-commercial study. An Emergency IND permits use of an investigational drug when time is of the essence, and a Treatment IND gives broad pre-approval access to promising therapies for patients with serious conditions and no alternative options.

The Food and Drug Omnibus Reform Act (FDORA), enacted in 2022, directs the FDA to require Diversity Action Plans for pivotal and Phase III studies that begin

enrolling more than 180 days after its final implementing guidance is published (U.S. Congress 2022); the FDA's draft implementing guidance called on sponsors to set enrollment goals for underrepresented populations (defined by race, ethnicity, sex, and age) and to describe how those goals would be met. That implementing guidance remains unsettled: the June 2024 Diversity Action Plans draft was withdrawn after a January 2025 executive order, restored by court order weeks later, and was still a draft when its statutory deadline for finalization, June 26, 2025, passed (Grossi 2025; U.S. Food and Drug Administration 2024e). A separate FDA guidance, the December 2025 final *Enhancing Participation in Clinical Trials* (a revision of the agency's earlier eligibility-criteria guidance, issued under a different statutory mandate), retains recommendations for representative enrollment while dropping the earlier "diversity" framing (U.S. Food and Drug Administration 2025c). The statutory mandate for Diversity Action Plans stands, though it takes effect only once that guidance is final, and the specifics of compliance, particularly sex- and gender-based enrollment goals, became genuinely contested in 2025, leaving diverse recruitment somewhere between an ethical best practice and a firm regulatory requirement.

The 21st Century Cures Act (2016) directed the FDA to establish a framework for using real-world evidence (RWE) (U.S. Food and Drug Administration 2018a); subsequent guidance has expanded the role of data from electronic health records, registries, and wearables in supporting new indications and post-marketing safety studies, and, in some rare-disease programs, external or synthetic control arms.

3.4. Investigator-Initiated Trials and the Sponsor-Investigator

Most of this book describes trials that a company designs, pays for and sponsors, with hospitals and clinics working as contracted sites. A large share of clinical research runs the other way. A physician at a university or hospital has a question, writes the protocol, finds the money and runs the study. These are *investigator-initiated trials* (IITs), also called investigator-sponsored or academic trials. They test new uses of approved drugs, compare standard treatments, and run the early pediatric and rare-disease studies that companies seldom start.

"Investigator-initiated" describes who had the idea and who controls the design. It is not a regulatory category. FDA regulations know three roles. A *sponsor* "means a person who takes responsibility for and initiates a clinical investigation". An *investigator* "means an individual who actually conducts a clinical investigation". A *sponsor-investigator* "means an individual who both initiates and conducts an investigation", and it is limited to an individual (U.S. Food and Drug Administration 2024b). The device regulation has the same three roles: a device sponsor initiates but does not conduct the investigation, and a sponsor-investigator initiates and "actually conducts, alone or with others" (U.S. Food and Drug Administration 2026b). An IIT run under the physician's own IND makes that physician a sponsor-

investigator. An IIT that needs no IND still has to name the person or institution that carries the sponsor's responsibilities under GCP.

No registry field marks an investigator-initiated trial, so the share can be bounded but not counted. We queried ClinicalTrials.gov on 20 October 2026 for interventional studies with a US site that started from 2015 through 2024 (79,375 studies) (U.S. National Library of Medicine 2026). Industry was the lead sponsor of 31.8% of them. The other 68.2% had a lead sponsor outside industry, the broadest reading of academic research, and it includes NCI network and other government trials and studies that need no IND, such as behavioral and device studies. A narrower reading keeps studies whose lead sponsor is a university, hospital or individual, with no industry collaborator, testing a drug or biologic: 16.8% of all studies. The narrowest reading keeps studies whose registrant names the sponsor-investigator as the responsible party: 4.9%. None of the three is the share of trials run under a sponsor-investigator IND, because the registry does not record INDs, and registrants fill in the responsible-party field inconsistently. We started in 2015 because registration was less complete in earlier years.

Table 3.2.: Three Bounds on the Share of Investigator-Initiated Trials in ClinicalTrials.gov

Studies started 2015 to 2024, US site	Studies	Industry lead	Non-industry lead	Academic drug or biologic, no industry collaborator	Sponsor-investigator
All interventional	79,375	31.8%	68.2%	16.8%	4.9%
Phase 1 and phase 1/2	14,326	57.3%	42.7%	24.1%	6.1%
Phase 1 cancer studies that enroll children	545	29.4%	70.6%	39.4%	8.4%

The share depends on the kind of trial. Companies led most phase 1 studies, but in phase 1 cancer studies that enroll children they led under a third, and the middle reading reaches 39.4% (215 of 545 studies). The pediatric row counts any study that includes children, so it includes studies of both children and adults, and the condition search matches the registry's own cancer synonyms. Early pediatric oncology is one of the fields where academic sponsors carry much of the work that companies seldom start.

3.4.1. Does the Study Need an IND?

The first regulatory question for an IIT with a marketed drug is whether an IND is needed at all. Under 21 CFR 312.2(b)(1), a clinical investigation of a lawfully marketed drug is exempt only if all five conditions hold (U.S. Food and Drug Administration 2024b):

1. It is not intended to be reported to FDA as a well-controlled study in support of a new indication or any other significant change in labeling.
2. It is not intended to support a significant change in the advertising of a prescription drug.
3. It does not involve a route of administration, dosage level, use in a patient population or other factor that significantly increases the risks, or decreases the acceptability of the risks, associated with the drug.
4. It complies with the IRB requirements of part 56 and the informed consent requirements of part 50.
5. It complies with 21 CFR 312.7, which bars promotion and commercialization of investigational drugs.

The third condition decides most cases. A study that gives an approved oncology drug to children when it is labeled for adults, or at a higher dose than labeled, can fail it, and then the investigator needs an IND. FDA will not accept an IND for a study that meets all five conditions (U.S. Food and Drug Administration 2024b), so the investigator, the IRB and the institution's regulatory office have to make the call together and document it.

Device studies follow a different path. A *nonsignificant risk* device study is treated as having an approved IDE unless FDA says otherwise, and it runs under abbreviated requirements: labeling, IRB approval with the IRB's agreement that the device is nonsignificant risk, informed consent, monitoring, specified records and reports, and the prohibitions of 21 CFR 812.7. A *significant risk* device study needs FDA approval of an IDE application. Some studies are exempt from part 812 altogether, among them certain noninvasive diagnostic devices, consumer preference testing and custom devices (U.S. Food and Drug Administration 2026b).

3.4.2. What the Sponsor-Investigator Takes On

A sponsor-investigator holds both sets of obligations at once. As sponsor, the physician must select qualified investigators, give them the information they need, ensure proper monitoring, keep the IND in effect, and promptly inform FDA and all investigators of significant new adverse effects or risks (21 CFR 312.50). Before each investigator begins, the sponsor collects a signed Form FDA 1572, a statement of qualifications, and financial information for disclosure under part 54 (21 CFR 312.53) (U.S. Food and Drug Administration 2024b). As investigator, the same physician also signs a Form 1572 for his or her own site (U.S. Food and Drug Administration 2015).

FDA wrote a guidance for this role, *Investigational New Drug Applications Prepared and Submitted by Sponsor-Investigators*, issued in May 2015 as a draft. It walks through the IND forms (Form FDA 1571 and Form FDA 3674) and the content of each section. It also makes two practical points. IRB approval is not needed before the IND is submitted, because the 1572 signature is itself a commitment to obtain IRB approval before the study starts. And the guidance reminds the

sponsor-investigator of the annual report, which the regulation itself requires within 60 days of the anniversary of the date the IND went into effect (U.S. Food and Drug Administration 2026a). FDA had not issued a final version as of October 2026, so the document describes FDA's current thinking and binds no one. FDA has also proposed changing the content of IND annual reports (December 2022), and that rule was not final either (U.S. Food and Drug Administration 2022c).

Safety reporting is where the double role is easiest to get wrong. An investigator must report any serious adverse event to the sponsor immediately (21 CFR 312.64(b)), and the sponsor must send a written IND safety report to FDA and all participating investigators within 15 calendar days of determining that a suspected adverse reaction is serious and unexpected, or within 7 calendar days for one that is fatal or life-threatening (21 CFR 312.32) (U.S. Food and Drug Administration 2024a). Chapter 23 describes the reporting chain. In an IIT the sponsor is the physician at the lead site, so the sponsor's clock can start the moment that physician hears of the event, and a multi-site IIT has to have its own system for receiving, assessing and distributing reports, which an industry trial would buy from a pharmacovigilance group.

Table 3.3 lists the main sponsor duties under part 312 and who usually carries them in each model. The right-hand column is the gap an academic sponsor has to fill with its own staff, a coordinating center or purchased services.

Table 3.3.: Sponsor Duties under 21 CFR Part 312 and Who Usually Carries Them (U.S. Food and Drug Administration 2024b, 2024a, 2026a, 2015)

Duty	Source	Industry-sponsored trial	Sponsor-investigator trial
Write the protocol and the IND	21 CFR part 312; FDA 2015 draft guidance	Company clinical, regulatory and medical writing staff	The investigator, with institutional regulatory support if it exists
Select and qualify investigators; collect Form 1572, CVs, financial disclosure	21 CFR 312.53	Company or CRO site start-up team	Lead-site study team or coordinating center
Monitor the investigation	21 CFR 312.50	Company or CRO monitors	Coordinating center, institutional monitoring unit, or a contracted monitor
Report serious and unexpected suspected adverse reactions to FDA and investigators	21 CFR 312.32	Company pharmacovigilance group	The sponsor-investigator, with a safety process written for the trial

Duty	Source	Industry-sponsored trial	Sponsor-investigator trial
Keep drug disposition records	21 CFR 312.62(a)	Site pharmacy under company supply systems	Site pharmacy; the lead site tracks supply across sites
Submit annual reports and keep the IND in effect	21 CFR 312.33	Company regulatory affairs	The sponsor-investigator

Checklist: Before a Sponsor-Investigator Files an IND

- Document the IND decision against all five conditions of 21 CFR 312.2(b)(1), with the institution's regulatory office.
- Name who will monitor each site and how, in a written monitoring plan.
- Write the safety reporting procedure: who receives SAE reports, who judges whether a reaction is serious and unexpected, who files the IND safety report, and who sends it to every site.
- Put the annual report date (60 days after the IND anniversary) and the protocol amendment process on the trial calendar.
- Confirm that the drug supplier's agreement covers supply for every site and the full planned duration.
- Collect a Form 1572 for the sponsor-investigator's own site as well as for every other site.

The operations of a multi-site academic trial (contracts, single IRB review, coordinating centers, cost and accrual) are covered in Section 20.5.

3.5. International Considerations

The *International Council for Harmonisation (ICH)* brings together regulatory authorities and pharmaceutical industries from the United States, European Union, Japan, and beyond to develop harmonized guidelines. Those guidelines are not law and bind nobody by themselves. Member countries adopt them, and that adoption has made them the de facto international standards for an enterprise that now runs trials across many countries and files submissions with agencies around the world.

Key guidelines (Table 3.4) include E6 on Good Clinical Practice, E8 on general considerations for clinical studies, E9 on statistical principles, and M4 on the Common Technical Document format for regulatory submissions.

Table 3.4.: Key ICH Guidelines for Clinical Trials

Guideline	Topic
E6	Good Clinical Practice (R3 updates focus on Quality by Design & Data Governance)
E8	General Considerations for Clinical Studies (R1 introduces Quality by Design & Critical to Quality factors)
E9	Statistical Principles (R1 adds "Estimands" framework)
E2A	Clinical Safety Data Management
E3	Structure and Content of Clinical Study Reports
M4	Common Technical Document (CTD) Format for Regulatory Submissions

Regional regulatory frameworks. In the European Union, clinical trials are governed by the *EU Clinical Trials Regulation 536/2014*, applicable since January 2022. It introduced a single submission portal (the *Clinical Trials Information System*, or CTIS), harmonized assessment timelines across member states, and opened trial information to public access. For multinational sponsors, that replaced a fragmented process requiring a separate submission to each country.

Beyond the EU, major authorities include Japan's Pharmaceuticals and Medical Devices Agency (PMDA), China's National Medical Products Administration (NMPA), Health Canada, and Australia's Therapeutic Goods Administration (TGA). Requirements differ, reflecting different healthcare systems, patient populations, and regulatory traditions, but the ICH guidelines give global programs a common foundation and let one body of clinical data support submissions in several jurisdictions.

3.6. Good Clinical Practice

ICH E6 GCP is the international ethical and scientific quality standard for clinical trials (International Council for Harmonisation 2025b). The 2025 revision (R3) builds on the risk-based quality management that E6(R2) introduced in 2016 (International Council for Harmonisation 2016a) with an explicit *Quality by Design* (QbD) approach, in which quality is built into the trial design from the outset through systematic identification of critical factors, risk assessment, and implementation of appropriate controls.

E6(R3) also adds requirements for *data governance*, since a modern trial draws data from electronic health records, wearables, patient-reported outcomes, and central laboratories, all needing appropriate controls throughout the data lifecycle. It presses for risk-proportionate oversight, letting sponsors match oversight intensity to each trial's complexity and risk, and it addresses technology-enabled designs explicitly: decentralized trials, electronic consent, digital health technologies.

E6 assigns duties to three parties. The *IRB or Independent Ethics Committee* reviews and approves trial documents before research begins, then conducts continuing review and receives reports of safety events and protocol deviations. The *investigator*, typically a physician, conducts the trial according to GCP and the approved protocol, obtains informed consent from each participant, reports adverse events, and maintains accurate records. The *sponsor*, whether a pharmaceutical company, academic institution, or government agency, is responsible for the overall design and conduct of the trial: selecting qualified investigators, providing the investigational product, ensuring proper monitoring, and reporting safety information to regulatory authorities.

Investigators and sponsors must maintain essential documents throughout: the records that demonstrate GCP compliance and let an outsider verify the data independently. They accumulate chronologically: the protocol and investigator's brochure at startup, consent forms and monitoring reports during conduct, final study reports and treatment allocation records at completion.

The Trial Master File

All essential documents should be organized in a *Trial Master File (TMF)*. The sponsor maintains a central file, while each investigator site maintains a site file. These records must be retained for years after the trial concludes. While historically ICH E6(R2) specified a minimum of two years following marketing approval or program discontinuation, the updated ICH E6(R3) (2025) framework defers to applicable regional regulatory requirements for the sponsor's records and asks investigators to keep theirs for the regional period or until the sponsor says they are no longer needed, whichever is longer; in the United States, FDA 21 CFR 312.57(c) requires sponsors to retain records for a minimum of two years. That two-year minimum does not begin at study close: the clock runs from a trigger event (two years after the marketing application is approved, or after investigational shipments cease and FDA is notified), and other regions impose far longer periods: the EU Clinical Trials Regulation requires the trial master file to be archived for at least 25 years after the trial ends (European Parliament and Council of the European Union 2014), so a trial with EU sites keeps its records for decades (see Chapter 24).

3.7. Adaptive and Bayesian Trial Design: Regulatory Guidance

Adaptive and Bayesian designs were once negotiated one product at a time. Over the past decade that case-by-case practice hardened into published guidance defining what a sponsor must provide to use these approaches. Table 3.5 lists the main documents and their status as of November 2026, and the subsections below take them in turn.

Table 3.5.: Main regulatory and reporting documents on adaptive and Bayesian trial design

Document	Status as of November 2026
FDA 2019	Final guidance on adaptive designs for drugs and biologics, issued in November 2019 (U.S. Food and Drug Administration 2019a)
FDA CID	Final guidance on interacting with FDA on complex innovative trial designs, December 2020 (U.S. Food and Drug Administration 2020b)
FDA Bayesian	Draft guidance, January 2026, CDER and CBER, not for implementation (U.S. Food and Drug Administration 2026g)
ICH E20	Step 2b draft, endorsed June 2025; the work plan of 10 July 2026 targets Step 3 and Step 4 sign-off in February 2027 (International Council for Harmonisation 2025a)
EMA 2007	Reflection paper on confirmatory trials with an adaptive design, CHMP/EWP/2459/02, adopted by CHMP in October 2007 (European Medicines Agency, Committee for Medicinal Products for Human Use 2007)
CONSORT ACE	Reporting guideline for adaptive randomised trials, published in the BMJ in 2020 (Dimairo et al. 2020)

3.7.1. FDA Adaptive Design Guidance (2019)

FDA's 2019 guidance on adaptive designs for drugs and biologics sorts adaptive modifications by how much they threaten the Type I error rate (U.S. Food and Drug Administration 2019a). Non-comparative adaptations use pooled data without reference to treatment assignment, as in blinded sample-size re-estimation based on nuisance parameters like the overall variance or event rate; they carry little risk of inflating the false-positive rate and are generally acceptable when pre-specified. Comparative adaptations use interim results broken down by treatment group (even if the groups are labeled only A and B), including dropping doses or treatment arms and response-adaptive randomization, and they directly risk inflating Type I error and biasing the estimated treatment effect, so they demand rigorous, typically simulation-based justification and regulatory agreement in advance. (The "well-understood versus less well-understood" labels sometimes attached to adaptive designs come from the FDA's superseded 2010 draft, not the 2019 guidance.)

Four principles run through the guidance:

1. **Pre-specification.** Adaptation rules must be written into the protocol and statistical analysis plan before enrollment begins. Unplanned changes prompted by comparative interim results make Type I error hard to control and the trial hard to interpret, however beneficial they look, so FDA strongly discourages making them without first meeting with the agency, unless patient safety requires immediate action.

2. **Type I error control.** The overall probability of a false-positive result across all adaptation points must be maintained at the pre-specified level, typically 5% (two-sided) or 2.5% (one-sided). Sponsors must demonstrate that control analytically or through simulation.

3. **Operational integrity.** Comparative interim results must be kept from the people managing and conducting the trial, through firewalled statistical teams and procedures that keep adaptive information away from anyone who can influence trial conduct. FDA generally recommends that no sponsor representative see them, although limited access for a few sponsor staff who take no other part in conducting or managing the trial can be justified, as for a dose-selection decision.

4. **Reliable effect estimation.** Controlling false positives is not enough; the design must also yield treatment-effect estimates and confidence intervals that the adaptation has not systematically biased. Adaptive selection and early stopping distort naive estimates, so the guidance expects sponsors to anticipate that and apply bias-adjusted estimation where needed.

3.7.2. FDA Bayesian Guidance

A Bayesian design raises a question of its own: how much of the evidence may come from outside the current trial. For medical devices, a 2010 guidance applies (U.S. Food and Drug Administration 2010b). For drugs and biologics, FDA issued a draft guidance in January 2026 (U.S. Food and Drug Administration 2026g); before it, the adaptive design guidance and Complex Innovative Design meeting interactions had established the operating norms (U.S. Food and Drug Administration 2019a). The CID paired meeting program, described in FDA's final guidance of December 2020 (U.S. Food and Drug Administration 2020b), gives sponsors of novel designs, including Bayesian ones, additional meetings with FDA to discuss the design and its simulations before the trial starts, and FDA's Bayesian draft cites two CID case studies among its examples (U.S. Food and Drug Administration 2026g). The draft distinguishes Bayesian approaches used *with* Type I error control (calibrating the trial to a pre-specified frequentist error rate, with Bayesian machinery used for computational convenience) from those used *without* it (where an informative prior from prior studies or related evidence reduces the required evidence from the current trial, and the combined posterior probability is the decision criterion).

For the "without control" category, the draft frames the central question as whether the borrowed prior information is *valid, reliable, and interpretable* (U.S. Food and Drug Administration 2026g). Several factors bear on that judgment: the quality and reliability of the contributing data, whether the prior is pre-specified, the relevance of the prior evidence to the current population and question, the rigor of the design of the studies that produced it, and the availability of patient-level data. An informative prior can reduce the evidence the current trial must supply on its own, but the draft is explicit that a design may skip calibration to a fixed Type I error rate only where the sponsor and FDA have agreed to that approach in advance.

Even then, it cautions, the Type I error rate alone will not provide a complete assessment of the prior's influence, which makes sensitivity analyses showing that the trial's conclusions are not driven entirely by the prior essential.

The draft also addresses *discounting*: information borrowed from historical or prior studies should be discounted for the possibility that the historical and current populations differ (U.S. Food and Drug Administration 2026g). Dynamic discounting, which automatically reduces borrowing when the current trial data diverge from the prior, is often chosen over static discounting (a fixed fraction) because it has better operating characteristics, such as less bias, when the populations do differ; the price is more parameters to specify, and the draft leaves the choice of method to case-by-case discussion with FDA.

Success criteria. A trial analyzed with frequentist methods almost always sets its success criteria so that the familywise Type I error rate is no greater than 0.025 one-sided. For a Bayesian primary analysis the draft says that default may not apply, for example when information is borrowed (U.S. Food and Drug Administration 2026g). The criterion is then most often a posterior probability of the form $\Pr(d > a) > c$, and the draft expects both the effect threshold a and the probability c to be justified, for every primary and key secondary estimand and, in a trial that can stop early for efficacy, at each decision point. The three ways the draft describes for choosing a and c are set out in Section 12.5.

Influence of the prior. The draft asks sponsors to document how much the prior drives the result, through the prior mean and the effective sample size (ESS), the number of patients the prior is equivalent to (Section 12.6.1). The ESS should be summarized over the plausible range of outcomes, with its maximum and mean, and for a dynamic prior, whose ESS changes as data accrue, it should be reassessed after the trial. The draft does not recommend using the inflation of the Type I error rate as the measure of borrowing: judging borrowing by what happens when there is no effect is inconsistent with a prior that assumes a nonzero one (U.S. Food and Drug Administration 2026g).

Where FDA says Bayesian methods have been used. The draft lists situations rather than rules (U.S. Food and Drug Administration 2026g): borrowing from earlier trials, augmenting a randomized concurrent control with an external control, pediatric extrapolation (supportive Bayesian analyses in the empagliflozin and linagliptin pediatric type 2 diabetes supplements), borrowing across similar diseases or subtypes (a placebo-controlled trial in epilepsy with myoclonic-atonic seizures, proposed through the CID program, that borrowed from trials in other epilepsies with a hierarchical model), borrowing between subgroups (a hierarchical model for regions in the LEADER liraglutide cardiovascular outcomes trial), and Bayesian dose-finding in oncology.

What FDA Expects in a Bayesian Submission

The draft's documentation section asks for a plan that states the prior, the sources excluded from it, how prior-data conflict will be handled and the planned sensitivity

analyses; the success criteria, including both a and c, at every decision point; how operating characteristics were estimated, with a comprehensive simulation report and its code; a summary of the sampler settings (warmup, iterations, chains, convergence diagnostics) when MCMC is used; and a comparison of the proposed design with alternatives, including simpler designs. It recommends a meeting with FDA before the trial starts for complex designs (U.S. Food and Drug Administration 2026g). The clinical study report then adds the posterior results with credible intervals, sensitivity to other priors, model checking including prior-data conflict, convergence diagnostics, the software used and the seed numbers needed to reproduce MCMC results.

3.7.3. ICH E20: International Harmonization

ICH E20, the first international harmonized guideline specifically addressing adaptive clinical trials, was released as a Step 2b draft in June 2025, and EMA's public consultation on it closed in November 2025 (International Council for Harmonisation 2025a). It proposes common principles for judging adaptive designs across FDA, EMA, and the other ICH regions: pre-specified decision rules, simulation to characterize operating characteristics, appropriate error control. For a global program, E20 alignment should reduce the risk of region-specific redesign. The expert working group's work plan of 12 February 2026 targeted Step 3 and Step 4 sign-off in October 2026; its work plan of 10 July 2026 moved that sign-off to February 2027, after review of the consultation comments by December 2026, so as of November 2026 E20 is still a draft (International Council for Harmonisation 2025a). Until the guideline is final the draft does not confirm regulatory acceptance of its contents, current regional guidance remains valid, and its section on Bayesian methods is not fully harmonized.

Europe's own reference point is older. EMA's reflection paper on methodological issues in confirmatory clinical trials planned with an adaptive design (CHMP/EWP/2459/02) was adopted by the CHMP in October 2007 (European Medicines Agency, Committee for Medicinal Products for Human Use 2007), twelve years before FDA's final adaptive design guidance. On reporting, the CONSORT extension for adaptive designs (CONSORT ACE), published in the BMJ in 2020, adds a checklist for reports of randomised trials that used an adaptive design, so that readers can see what was pre-planned and what changed (Dimairo et al. 2020).

! Regulatory Engagement for Adaptive Designs

Engage regulators before the protocol for any novel adaptive design is final. For FDA that typically means a Type C meeting or similar interaction, at which the agency reviews the proposed adaptation rules, the simulation package, and the decision thresholds. Where the design uses Bayesian methods with informative priors or borrowing, agreement on the prior and discounting scheme should be reached before enrollment. Negotiating adaptive design choices after enrollment has begun is far harder.

3.7.4. Simulation as Regulatory Evidence

Both the FDA adaptive design guidance and ICH E20 treat simulation as the primary tool for demonstrating that a proposed adaptive design has acceptable operating characteristics, and FDA reviewers read a simulation package as they would statistical analysis code: for correctness, completeness, reproducibility. It should characterize four things:

- *Type I error rate*: the probability of a false positive across a range of null scenarios, including those where the adaptation rule might inflate error
- *Power*: the probability of correctly concluding efficacy across a range of true effect sizes and interim trajectories
- *Sample size distribution*: expected and worst-case sample sizes under null and alternative scenarios, including those where adaptation causes substantial expansion
- *Bias*: whether adaptive allocation or early stopping produces estimated effect sizes that systematically differ from the true treatment effect

Bayesian designs not calibrated to the Type I error rate are judged instead by quantities averaged over a prior, such as Bayesian power and expected bias (U.S. Food and Drug Administration 2026g), and platform trials add measures such as arm trajectories and the effect of shared controls (European Medicines Agency 2026b). Section 12.7.5 lists these operating characteristics and Section 12.7.5 describes how to produce them; Section 3.8 gives the platform-specific expectations.

The CID program is one route to agreeing the simulation plan with FDA before the trial starts (U.S. Food and Drug Administration 2020b), and CONSORT ACE governs how the adaptive features and their results are reported afterwards (Dimairo et al. 2020).

What Regulators Expect in a Simulation Report

- **Code and seeds.** Simulation code and seed values sufficient for FDA to reproduce the results independently; FDA's Bayesian draft asks for the code with a detailed description of the simulation design, implementation and results (U.S. Food and Drug Administration 2026g).

- **Scenarios.** Scenarios and assumptions that are pre-specified, comprehensive and plausible (U.S. Food and Drug Administration 2026g).
- **Alternatives.** For complex designs, such as those that borrow external information, a comparison of operating characteristics with alternative designs, including simpler ones (U.S. Food and Drug Administration 2026g).
- **Computation.** FDA requires no particular software, but for complex Bayesian models it may be necessary to check by simulation that the sampling algorithm is reliable before the model is chosen (U.S. Food and Drug Administration 2026g).
- **Placement.** For a confirmatory platform trial, the EMA draft expects the operating characteristics to be characterized in the protocol, with simulations included where needed (European Medicines Agency 2026b).

3.8. Regulatory Guidance for Master and Platform Trials

A master protocol is a single protocol with several substudies that test one or more drugs in one or more diseases (the design types are described in Section 13.6, and their statistics in Section 12.5). Regulators have begun to write guidance specific to these designs, and as of November 2026 two of the main documents are still drafts. This section covers the FDA guidance, the EU documents, where they agree and differ, and what each expects from simulation. It keeps to the statistical and regulatory content; for how a platform trial is run, see Section 20.6.

3.8.1. FDA

FDA's guidance on master protocols for oncology drugs and biologics was finalized in March 2022 (U.S. Food and Drug Administration 2022a). It is specific to cancer trials and deals with matters such as establishing the dose before a drug enters a master protocol, using a common control where feasible and what to submit when substudies are added.

For other therapeutic areas FDA issued a draft guidance, Master Protocols for Drug and Biological Product Development, in December 2023. It was revised and replaced in June 2026 by a draft marked Revision I, which adds recommendations on basket trials and makes small changes on randomization, choice of control and informed consent (U.S. Food and Drug Administration 2026e). As of November 2026 it is a draft, not for implementation, and the points below describe that draft. Its main content is as follows.

Allocation and control. The draft advises considering a randomization scheme that puts more participants on the shared control than on any single drug arm, which raises power for each drug-versus-control comparison at a given total sample size. One example scheme moves the ratio from $\sqrt{2}$:1:1 (control:drug A:drug B) to $\sqrt{3}$:1:1:1 when a third drug enters. If the ratio changes over time, the analysis should account for the periods, for example by stratifying on time or weighting by assignment probability, to avoid bias from temporal trends (U.S. Food and Drug

Administration 2026e). The preferred comparison uses concurrently eligible controls: participants randomized to control while the drug was enrolling who could have been randomized to it. Nonconcurrent controls can raise precision, but the draft allows them only in rare circumstances that the sponsor can justify, and a sponsor planning to use them in a trial meant to support substantial evidence of effectiveness should discuss the rationale with FDA, with bias-correcting methods and sensitivity analyses on concurrent controls alone (U.S. Food and Drug Administration 2026e). The methods it names are adjustment for a function of calendar time and baseline prognostic factors, dynamic borrowing of nonconcurrent control data (for example with a Bayesian hierarchical model), and network meta-analysis combining comparisons between concurrently randomized arms. Promoting a drug tested in the master protocol into the control arm or the background therapy can change whether other drugs are tested for superiority or noninferiority, their sample sizes and how data before and after the change are combined, so the draft asks sponsors to obtain FDA's concurrence before making such a change (U.S. Food and Drug Administration 2026e).

Consent and blinding. Consent should cover all arms, and the form should be updated as drugs enter or leave. The draft warns that consent given to a drug-specific substudy only after randomization can introduce bias if participants who decline differ from those who accept, so a sponsor proposing it needs a strong justification. On blinding, it prefers complete blinding, accepts a distinct matched placebo per drug (partial blinding) with a sensitivity analysis, and allows open-label designs only in rare circumstances (U.S. Food and Drug Administration 2026e).

Multiplicity. FDA generally does not recommend adjusting across drugs compared with a shared control, or across diseases for one drug, because those are separate clinical questions that independent trials would not adjust for, and the expected number of Type I errors is the same either way. Closely related comparisons, such as several doses of the same drug, still need control of the Type I error rate, as do multiple endpoints for one drug. With a shared control the chance of at least one error is lower but the chance of several simultaneous errors is higher than in independent trials, so the distribution of the number of errors should be examined (U.S. Food and Drug Administration 2026e).

Baskets and integrity. For basket trials the draft describes a continuum from separate analyses in each disease, through borrowing of information (for example with a Bayesian hierarchical model), to one combined analysis, with a bias-variance trade-off along it, and it asks for operating characteristics that include scenarios where the drug works in only some populations. It recommends an independent data monitoring committee and a data access plan, because the result for one drug in an event-driven trial can reveal how another is doing (U.S. Food and Drug Administration 2026e).

IND mechanics. Each master protocol should be submitted in a new IND, after a pre-IND meeting, and its substudies are generally conducted under that IND only: a new substudy (a new drug or population) and any change to the master protocol or

a substudy are submitted to it as protocol amendments. Some basket and platform trials may instead use separate substudy INDs held by the same sponsor, for example when the expertise sits in several review divisions; then a new substudy goes to a new substudy IND, and each amendment is filed to the IND it concerns, with notices and cross-references to the master protocol IND and the other substudy INDs (U.S. Food and Drug Administration 2026e). A master protocol sponsor should submit protocol amendments that substantively affect safety, quality or scope, including a new substudy, at least 30 days before starting the change, and notify the regulatory project manager at least 48 hours before submitting such an amendment (U.S. Food and Drug Administration 2026e).

3.8.2. European Union

In the EU, the main document is the Complex clinical trials Q&A from the European Commission, EMA and the Heads of Medicines Agencies, published in 2022 (version of 23 May 2022) (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022). It treats a platform trial as a typical complex trial and describes the protocol as a master part with the common elements plus sub-protocols for each drug or population. It asks that decision rules be pre-specified, that the testing strategy across sub-protocols be set out in the master protocol, and that sub-protocols affected by an interplay (for example through the Type I error) carry the relevant information and be updated together. On Bayesian methods it notes that type I error control is central to frequentist efficacy analyses and says a Bayesian method should allow the corresponding error issues to be evaluated, including by simulation. With the application it asks for prior predictive simulations, simulations of the trial's operating characteristics under a wide range of data scenarios, and simulations showing how the conclusions change under alternative assumptions for the pre-specified parameters and priors; after the trial, sensitivity analyses under those alternative assumptions. It also notes there are few precedents of notable or successful Bayesian approaches for efficacy (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022).

In October 2026 the CHMP adopted for consultation a draft reflection paper on methodological aspects in confirmatory platform trials (EMA/217509/2026); the consultation runs from 19 October 2026 to 31 January 2027 (European Medicines Agency 2026b). It defines a platform trial as a multi-arm trial in a single target population with randomization and comparison to a control, and it leaves operational and clinical trial application aspects aside. Its main positions are:

- **Allocation.** Keep the allocation ratio constant for each comparison of interest. Square-root allocation ($\sqrt{k}$ participants on control for each one on an experimental arm, with k experimental arms) may help in a platform with a fixed number of arms and no adaptations, but not always in an adaptive one, and response-adaptive randomization is "not supported."

- **Controls.** Concurrent controls are strongly preferred. Non-concurrent controls are "usually not acceptable for the primary analysis of the primary endpoint" and are better suited to supplementary analyses or some secondary endpoints, with pre-specified methods against temporal bias. Changing the control treatment for arms that have already started enrolling is discouraged, because the analyses and sample sizes of ongoing arms may need revision and the type I error is hard to control; a treatment that met its endpoint should become the control only once authorised, save in exceptional circumstances, and if the change cannot be avoided a new trial is usually preferable. A different control may be defined for arms not yet started if the standard of care has changed.
- **Multiplicity.** Strong control of the familywise error rate is not required by default across all hypotheses, but is required across hypotheses related to the same overall question and a regulatory decision. Two arms that are doses of one medicine are related; two unrelated new substances are not, if the licensing decisions are independent. A shared control raises the chance of several false positives at once; stricter significance levels or a larger control arm reduce it, but the draft says a second confirmatory trial in the same or a similar setting will usually be the best remedy for multiple false positives and bias.
- **Estimands, integrity and complexity.** Each experimental arm typically has its own estimand, and changes to the estimands during the trial should be documented. Primary analyses run at different times can partly or fully unblind the shared control results, and later adaptations that use that information without pre-specification can inflate the Type I error and bias results. The assessment of confirmatory evidence is holistic: a combination of design features can turn a trial planned as confirmatory into an exploratory one, so simpler designs are preferred where a complex one offers no substantial advantage.

The draft does not mention Bayesian methods.

3.8.3. Where the Agencies Agree and Differ

Read side by side (a comparison neither agency makes itself), the FDA and EMA drafts agree on four points: pre-specify objectives, decision rules and analyses; prefer concurrent controls; protect interim results from the people running the trial; and control multiplicity across closely related hypotheses, such as doses of one drug, but not by default across unrelated drugs. Both also ask for early discussion of a complex design: FDA for nonconcurrent controls and integrity measures (U.S. Food and Drug Administration 2026e), and the EU Q&A for Bayesian methods (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022). The ICH E20 draft is meant to harmonize the broader adaptive-design principles (International Council for Harmonisation 2025a). Table 3.6 sets out the specific points.

Table 3.6.: FDA and EMA drafts on master and platform trials, as of November 2026

Topic	FDA draft (June 2026)	EMA draft (October 2026)
Controls	Concurrently eligible controls preferred	Concurrent controls strongly preferred
Non-concurrent	Rare circumstances, justified, discussed with FDA	Usually not acceptable for the primary analysis of the primary endpoint
Control change	Needs FDA's concurrence	Discouraged for enrolling arms; a new trial usually preferable
Allocation	More to control than to each arm; ratio may change with fixed schemes, analysis must account for it	Constant ratio per comparison; RAR not supported
Multiplicity	Generally no adjustment across drugs or diseases; control for related doses and endpoints	Not by default; strong control across related hypotheses
Simulation	Operating characteristics relative to alternatives, including scenarios where effects differ	Simulations may be required and included in the protocol
Bayesian	Borrowing across baskets described	Not discussed

Even where the conclusions match, the starting points differ, and that matters for borderline cases. FDA starts from separate clinical questions and says adjustment across drugs is generally not recommended, while EMA starts from whether hypotheses serve one regulatory decision and requires strong control within such a family (U.S. Food and Drug Administration 2026e; European Medicines Agency 2026b). Both texts are drafts that may change, so a sponsor should check the current versions and raise the design with each agency early.

3.8.4. What Each Expects from Simulation

For a master protocol the simulation package follows the general rules above and adds the platform-specific items. For basket designs FDA asks that the proposed statistical approach be supported by an evaluation of its operating characteristics against alternatives, including probabilities of false positive and false negative conclusions, bias-variance trade-offs, and scenarios in which the true effects differ across populations; with a shared control it asks that the distribution of the number of Type I errors, and power, be considered; and for adaptive elements it points to its 2019 guidance (U.S. Food and Drug Administration 2026e). EMA expects the operating characteristics, including the type I error rate, the relevant definitions of power (marginal power to reject one hypothesis, disjunctive power to reject at

least one false null hypothesis, conjunctive power to reject all of them, with the targeted one named in the sample size justification), arm trajectories, durations and the effect of shared controls, to be fully characterized in the protocol, with simulations included where they are needed (European Medicines Agency 2026b). The EC, EMA and HMA Q&A likewise points to simulation for evaluating a Bayesian design (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022).

3.9. Inspections and Enforcement

The regulatory framework has teeth. FDA inspections of clinical trial sites verify compliance and data integrity: routine inspections as part of a marketing application review and *for-cause* (also called *directed*) inspections triggered by specific concerns, such as a complaint or a problem identified during application review.

Following an inspection, the FDA classifies findings as *NAI* (no action indicated) when no objectionable conditions or practices are found, *VAI* (voluntary action indicated) when objectionable conditions are found but FDA leaves their correction to the site and recommends no action, or *OAI* (official action indicated) when FDA recommends regulatory or administrative action.

Serious violations can mean rejection of trial data from regulatory submissions, clinical holds on ongoing trials, and disqualification of investigators, which bars them from receiving investigational products and from conducting clinical investigations that support applications to FDA. In extreme cases involving fraud or egregious misconduct, criminal penalties are possible, and a conviction can lead to debarment, which bars the individual from providing services in any capacity to anyone with an approved or pending drug application.

4. Ethical Principles

Clinical research asks people to accept risks and inconveniences not primarily for their own benefit, but so that future patients might have better treatments. The rules governing that arrangement emerged largely in response to abuse: the Nuremberg Code and the Belmont Report arose from the failures described in Chapter 1, while the Declaration of Helsinki built on the Nuremberg Code as the medical profession's own statement of principles. But the absence of abuse is not the same thing as ethical research.

For many people, "ethical research" is synonymous with "informed consent." Consent is necessary in most but not all studies. It is not sufficient: a scientifically worthless study remains unethical even if every participant provides exquisitely informed consent, because it exposes people to risks without any possibility of generating knowledge that could benefit others (Emanuel, Wendler, and Grady 2000).

4.1. Seven Requirements for Ethical Research

Emanuel, Wendler, and Grady proposed seven requirements (Table 4.1) that clinical research must satisfy to be considered ethical (Emanuel, Wendler, and Grady 2000). Satisfying six is not enough.

Table 4.1.: Emanuel's Seven Requirements for Ethical Research

Requirement	Ethical Principle	Key Question	Example Application
1. Value	Research must have social or scientific value	Will this research improve health or increase understanding?	A study duplicating existing knowledge is unethical regardless of consent, as it exposes participants to risk without potential benefit.
2. Scientific Validity	Research must be methodologically rigorous	Is the study designed to answer its research question?	Underpowered studies that cannot detect meaningful effects cannot justify participant risks.
3. Fair Subject Selection	Burdens and benefits should be distributed fairly	Are subjects chosen for scientific reasons, not convenience or vulnerability?	Testing drugs only in young men while marketing to all populations violates this principle.

Requirement	Ethical Principle	Key Question	Example Application
4. Favorable Risk-Benefit Ratio	Risks must be justified by potential benefits	Do potential benefits outweigh risks to participants and society?	High-risk procedures require proportionately important research questions.
5. Independent Review	Conflicts of interest must be managed through oversight	Has the research been reviewed by unaffiliated experts?	IRB approval is mandatory; investigators cannot approve their own studies.
6. Informed Consent	Participants must voluntarily agree with understanding	Do participants understand and freely agree?	Consent forms must be readable; participants can withdraw at any time.
7. Respect for Participants	Researchers have ongoing responsibilities	Are participants' interests protected throughout and after the study?	Privacy must be maintained; participants should be informed of new findings.

4.2. Applying Ethical Principles

The seven requirements are easier to state than to apply. In practice they collide, and honoring one can mean compromising another.

4.2.1. Placebo Controls

Placebo controls are contentious wherever an effective treatment already exists, and most contentious where that treatment works, but imperfectly. A placebo-controlled trial often provides the clearest evidence of efficacy. It can also mean withholding a known-effective treatment from patients who need it.

The Declaration of Helsinki takes a strong position: new interventions must be tested against the best proven intervention, not against placebo. However, placebo controls may be acceptable when no proven intervention exists, or when compelling and scientifically sound methodological reasons require a placebo comparison and participants who receive placebo will not be subject to additional risks of serious or irreversible harm as a result of not receiving the best proven intervention (World Medical Association 2025).

Beneath the debate sits *clinical equipoise*, introduced by Benjamin Freedman in 1987 (Freedman 1987): randomizing patients between two arms is ethical only when the expert medical community is genuinely uncertain which arm is better. That uncertainty is the standard justification for assigning anyone to a control arm, placebo included. It is also perishable. As outcome data accumulate during a trial, equipoise

can erode, which is precisely the situation that response-adaptive randomization exploits by shifting allocation toward the arm that appears to be performing better (Section 12.8). Judging when the accumulating evidence has disturbed equipoise enough to justify reallocating patients, or stopping the trial, is one of the central ethical decisions in adaptive design.

4.2.2. Vulnerable Populations

Vulnerable populations are not protected by being excluded from research: they too deserve therapies developed and tested in relevant populations (Emanuel, Wendler, and Grady 2000). What their vulnerability requires is safeguards matched to it. The Common Rule supplies them, and its subparts on pregnant women and prisoners apply to research that the Department of Health and Human Services conducts or supports (U.S. Department of Health and Human Services 2024).

Children. Children cannot provide legal consent and may not understand the research, so the protections are parental or guardian permission plus the child's assent when the child is capable. The research must involve minimal risk, be justified by direct benefit to the child, or carry only a minor increase over minimal risk and yield knowledge about the child's disorder or condition.

Pregnant women. Risks extend to the fetus. Where scientifically appropriate, preclinical and clinical data on pregnancy risks are required. The research must minimize risk to the fetus, and consent must address fetal risks.

Prisoners. Institutional coercion may compromise voluntariness, so research is limited to the categories in 45 CFR 46.306(a)(2): incarceration and crime, prisons and prisoners, conditions affecting prisoners, or practices likely to benefit the subject. Payment and conditions must not be coercive.

Cognitively impaired participants. Capacity for informed consent is reduced, so consent comes from a legally authorized representative, with the subject's assent when possible. Capacity is assessed on an ongoing basis, and the subject's objection is respected even if the representative consented.

Economically disadvantaged participants. Payment may be unduly influential, so compensation must not be coercive. Payment for time and inconvenience is appropriate. Credit should accrue as the study progresses, with at most a small, non-coercive completion incentive.

Research in resource-limited settings. A trial run in a resource-limited setting by a sponsor based in a wealthier country raises two questions with no universal answer. The first is what the control group gets: the best treatment available globally, or the local standard of care. The second is what happens when the trial ends, and sponsors and researchers must arrange post-trial provisions in advance for all participants who still need an intervention identified as beneficial and reasonably safe in the trial, unless a research ethics committee approves an exception (World Medical Association 2025). Both demand thoughtful engagement, and both come

back to the same requirement: that the research be responsive to the health needs or priorities of the host population and create local social value, not merely exploit the community for data (Council for International Organizations of Medical Sciences (CIOMS) 2016).

4.3. Modern Ethical Challenges

Decentralized trials (DCTs) can collect patient data continuously, via wearables and smartphone apps. GPS location and sleep patterns carry re-identification risk, and they invite "function creep," where data collected for one purpose is repurposed for another. Researchers must weigh the scientific value of that high-frequency record against the participant's right to digital privacy (World Health Organization 2021).

4.3.1. Electronic Informed Consent (eConsent)

Electronic informed consent is now widely used in clinical investigations, and its multimedia tools (videos, quizzes) can improve a participant's understanding. Digitizing the signature is not what makes it ethical. The ethical standard remains comprehension and voluntariness, while the operational standard extends to reliable documentation, auditability, and appropriate protections for privacy and participant autonomy (Emanuel, Wendler, and Grady 2000; U.S. Food and Drug Administration 2016, 2024f).

Remote consent in a decentralized or hybrid trial reduces travel burden and broadens access. It also increases the risk that participants with lower digital literacy, limited broadband access, or limited accessibility accommodations are systematically excluded. That is a *fair subject selection* concern: an operational convenience can become a structural enrollment bias unless alternative pathways are offered (Emanuel, Wendler, and Grady 2000; U.S. Food and Drug Administration 2024n).

Checklist: eConsent Implementation (Ethics-Facing)

- Use plain-language explanations and layered disclosure (short summary plus details), and verify comprehension (e.g., teach-back or comprehension checks). A completed "click-through" is not by itself evidence of understanding (U.S. Food and Drug Administration 2016).
- Provide equivalent non-digital pathways (paper or staff-assisted workflows) to avoid excluding participants who cannot or prefer not to use digital tools (Emanuel, Wendler, and Grady 2000; U.S. Food and Drug Administration 2016).
- Specify identity verification, version control of consent content, and procedures for re-consent when new information emerges (U.S. Food and Drug Administration 2016).

- Ensure documentation and auditability are suitable for regulated electronic records and signatures (e.g., validated systems, audit trails, access control, retention) (U.S. Food and Drug Administration 2023a, 2024f).
- Define privacy-by-design boundaries for any data collected during consent (device identifiers, contact methods, metadata) and ensure participants understand what is collected and why (World Health Organization 2021).

4.3.2. AI and Automation in Clinical Research Ethics

The ethical question about AI in a trial is rarely whether it is allowed. It is whether a particular *use* satisfies the same seven requirements that make any study ethical (Emanuel, Wendler, and Grady 2000). The same system that detects a safety signal earlier, assesses protocol feasibility better, or lowers site burden can also amplify bias, reduce participant understanding, and shift accountability in ways that are hard to justify.

Fair subject selection and justice. Recruitment algorithms trained on historical site performance, claims, or EHR data can reproduce inequities by systematically under-identifying eligible participants in under-resourced health systems or underrepresented groups. Bias auditing should therefore be treated as an ethical safeguard, not an optional "model improvement" step (World Health Organization 2021; Saleiro et al. 2018). Dataset "datasheets" and model "cards" help make these limitations explicit to reviewers and users (Gebru et al. 2021; Mitchell et al. 2019).

Informed consent and understanding. If AI influences eligibility triage, monitoring intensity, or what a coordinator asks a participant to do (e.g., app-based prompts), then "what participants are consenting to" includes the role of automated systems. There is a risk that fluent, confident outputs create an illusion of understanding (among staff and participants) without reliable grounding in the source record or the evidence base (Messeri and Crockett 2024; Bender et al. 2021). Transparency therefore belongs in the consent conversation: what is automated, what is reviewed by humans, and how an error gets corrected.

Respect for participants (privacy, dignity, and autonomy). Continuous sensing and "always-on" monitoring can feel intrusive even when clinically informative. Ethical review should explicitly consider minimization (collect only what is necessary), proportionality (match intrusiveness to expected value), and participant control (pause/opt-out mechanisms) (World Health Organization 2021).

Accountability and governance. Who is answerable when an algorithm misclassifies, misses a safety signal, or increases burden for a subgroup? The World Health Organization's guidance on AI for health holds that AI should enhance human decision-making and empower clinicians rather than replace them, and that humans should remain in full control of medical decisions (World Health Organization 2021). The European Medicines Agency's reflection paper on AI states that it is the responsibility of the clinical trial sponsor, marketing authorization applicant or holder, or manufacturer to ensure that all algorithms, models, datasets, and data

processing pipelines used are fit for purpose (European Medicines Agency 2024b). Governance can be structured with risk management concepts (intended use, hazards, controls, monitoring) rather than marketing claims, as in NIST's AI Risk Management Framework and the good machine learning practice principles written for medical devices (National Institute of Standards and Technology 2023; U.S. Food and Drug Administration, Health Canada, and UK Medicines and Healthcare products Regulatory Agency 2021).

4.4. The IRB in Practice

Independent review is the fifth of Emanuel's seven requirements, and the IRB is where it happens. It does not happen once. A study comes back to the committee at least once a year while it runs, and again whenever a serious or unexpected event raises the question of whether it can continue safely (Table 4.2) (U.S. Food and Drug Administration 2023c):

Table 4.2.: Types of IRB Review

Review Type	Timing	Purpose	IRB Action
Initial Review	Before research begins	Evaluate protocol for ethical compliance	Approve, require modifications, or disapprove research
Continuing Review	At least annually during study	Verify ongoing compliance and assess new information	Reapprove, require changes, suspend, or terminate
Modification Review	When protocol/consent changes proposed	Evaluate impact of changes on risk-benefit	Approve or disapprove modifications
Adverse Event Review	When serious/unexpected events occur	Assess whether study can continue safely	May require protocol changes, additional monitoring, or suspension
Final Review	At study completion	Close out study oversight	Document study completion and any findings

! IRB Authority

The IRB's authority is not advisory: it is decisive. Research cannot proceed without IRB approval. Except for changes necessary to eliminate immediate hazards, all modifications must be approved before implementation.

Part II.

Part II: Clinical Trial Phases

5. Overview of Drug Development

Most compounds that reach human testing never make it to a pharmacy shelf. The ones that do have typically taken 10 to 15 years from laboratory discovery and billions of dollars to get there, and they can transform (and in many cases save) millions of lives.

The constraints accumulate. Biology is uncertain: a mechanism that looks compelling in cell culture may fail in animals, and a signal seen in Phase I or II may disappear in larger, more heterogeneous Phase III populations. Safety is non-negotiable, and rare adverse events can emerge only after thousands of exposures, forcing sponsors to design studies large enough to detect clinically meaningful risks. Manufacturing and quality systems must scale from milligrams in a lab to reproducible commercial production, with tight control of impurities, stability, and batch-to-batch consistency. And every step must be documented in a form that regulators can inspect and independently verify.

So the process is best thought of as a sequence of risk-reduction decisions. Early work asks "*does it plausibly work and is it safe enough to test?*" Later stages ask "*does it work better than existing options, in the patients who will actually use it, with an acceptable benefit-risk profile?*"

5.1. Development Path from Molecule to Medicine

Development proceeds in stages (see Figure 5.1), each designed to answer a specific question and retire some of the uncertainty of working with a novel therapy.

Discovery comes first: researchers identify a biological target (a protein, receptor, or pathway involved in disease) and search for molecules that can interact with it. The search may mean screening thousands of compounds, designing molecules from structural knowledge, or modifying natural products with known activity.

Preclinical research then evaluates a promising compound in laboratory and animal studies. Three questions have to be answered before anyone is dosed: does the compound hit its target and produce the desired biological effect, what are its toxic effects in animals, and how is it absorbed, distributed, metabolized, and eliminated?

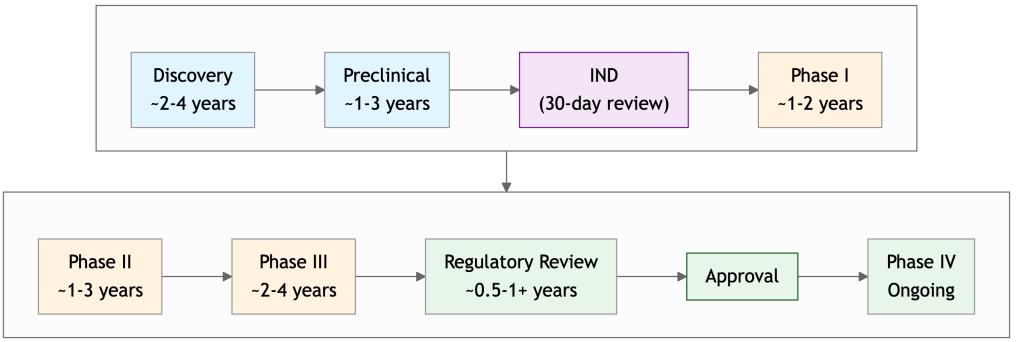


Figure 5.1.: The Drug Development Pipeline from Discovery to Market (typical ranges; varies by modality and indication)

In the United States, the gate between preclinical and clinical development is the *Investigational New Drug (IND)* application to the FDA (see Chapter 3 for the full regulatory framework). The IND contains chemistry, manufacturing, and controls data; preclinical pharmacology and toxicology results; and the clinical protocol. FDA has 30 days to review the IND; if no clinical hold is placed, the sponsor may begin clinical trials (U.S. Food and Drug Administration 2024b).

5.2. Understanding the Clinical Phases

Clinical development is traditionally divided into four phases, numbered I through IV. The designations are useful and a little misleading: the boundaries between phases are not sharp, and different types of studies may occur at various points in development.

The ICH E8(R1) guideline offers a complementary classification based on study objectives rather than timing, and the 2021 revision renamed three of the four categories that the original 1997 guideline had called "therapeutic exploratory," "therapeutic confirmatory," and "therapeutic use" (see Figure 5.2 and Table 5.1) (International Council for Harmonisation 2021). *Human pharmacology* studies assess tolerance, pharmacokinetics, and pharmacodynamics, and increasingly cover immunogenicity, tolerance in patients with renal or hepatic impairment, and cardiac toxicity. *Exploratory* studies explore efficacy, estimate doses, and evaluate biomarkers, patient-reported outcome measures, and adaptive designs. *Confirmatory* studies provide the definitive evidence of efficacy and safety for regulatory approval, and *post-approval* studies refine our understanding of the drug in clinical practice after marketing.

Human pharmacology does not stop at the end of Phase I: it continues throughout development as new formulations are tested, drug interactions are characterized, and effects in special populations are evaluated.

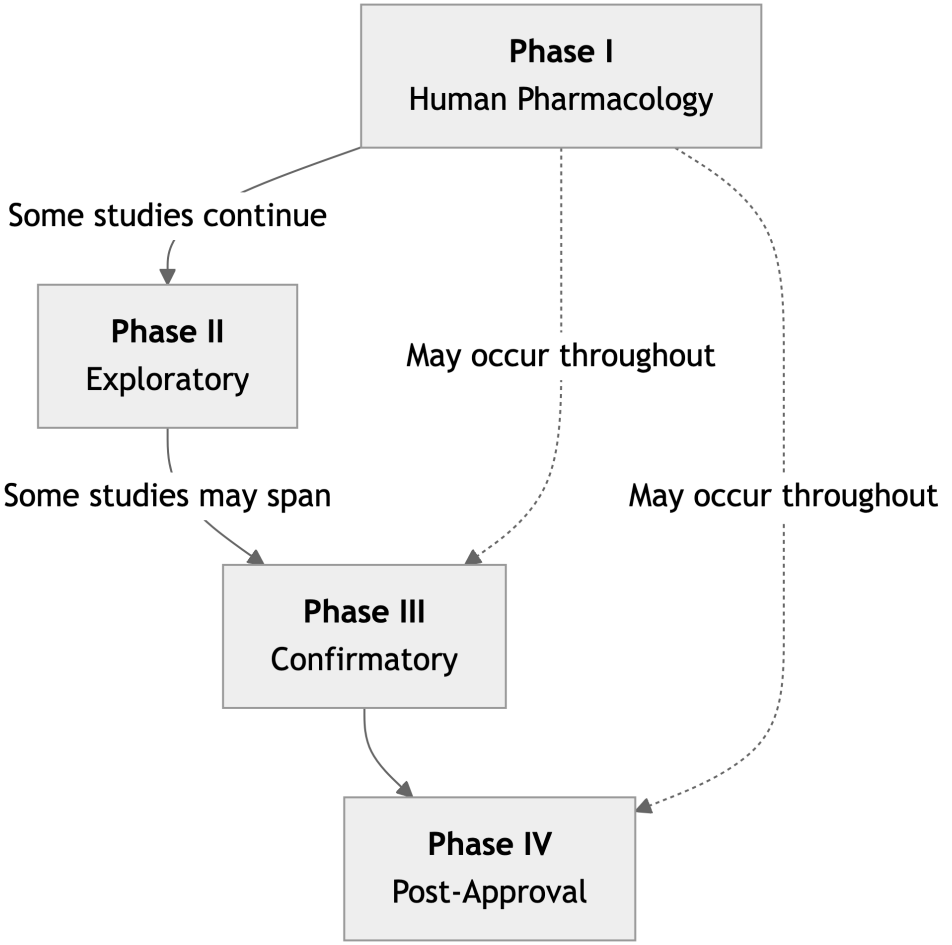


Figure 5.2.: Relationship Between Phases and Study Objectives

ICH E8(R1) also brings *Quality by Design* into clinical studies through *Critical to Quality (CtQ) factors*: those attributes of a study whose integrity is fundamental to protecting participants, to the reliability and interpretability of the results, and to the decisions made on the basis of those results (International Council for Harmonisation 2021). Which attributes those are depends on the study objective. For a Phase I first-in-human study, CtQ factors might include dose escalation safety rules and pharmacokinetic sampling accuracy; for a Phase III confirmatory trial, they typically include endpoint ascertainment, treatment adherence, and minimization of missing data. The guideline also encourages sponsors to build patient input into development planning, endpoint selection, and trial design, so that studies address outcomes meaningful to patients and the burden of participating is justified by the scientific objectives.

Table 5.1.: Classification of Clinical Studies by Objective (ICH E8(R1))

Study Type	Typical Phase	Primary Objectives
Human Pharmacology	Phase I	Assess tolerance, PK/PD, immunogenicity, tolerance in renal or hepatic impairment, and cardiac toxicity; define metabolism and drug interactions
Exploratory	Phase II	Explore use for targeted indication, estimate dosing, evaluate biomarkers and patient-reported outcome measures, provide basis for confirmatory design
Confirmatory	Phase III	Demonstrate/confirm efficacy, establish safety profile, provide adequate basis for marketing approval
Post-Approval	Phase IV	Refine understanding of benefit/risk in general population, identify less common adverse reactions, optimize dosing

5.3. Regulatory Pathways

The standard path runs Phase I, II, and III trials, then a *New Drug Application (NDA)* or *Biologics License Application (BLA)* for FDA review. The agency targets 10 months for a standard review. Not every drug takes that path.

For drugs addressing serious conditions, the FDA offers *expedited programs* that can significantly reduce development time and regulatory burden. Fast Track designation provides more frequent FDA interaction and supports the rolling review of application sections for drugs aimed at an unmet medical need. Breakthrough Therapy status offers more intensive guidance for drugs showing substantial improvement over existing treatments, while Accelerated Approval allows marketing based on surrogate endpoints reasonably likely to predict clinical benefit. Priority

Review compresses the final agency review timeline from ten months to six for drugs representing significant advances in care.

These programs can be combined. Breakthrough Therapy designation already carries all of the Fast Track features, and most drugs holding either designation also receive priority review, significantly accelerating the path to patients.

5.4. The Target Product Profile

The *Target Product Profile (TPP)* is the label a sponsor intends to earn, written down before the data exist: the indication it will treat, the patient population that will use it, the dosing regimen, the claims that can be made about efficacy and safety, and how it will be positioned against competitors. It is not fixed in stone; it evolves as data accumulates and the commercial environment changes. What it buys is alignment. A development team that has agreed on the intended label can check every planned study against the data that label will require.

5.5. The Economics of Attrition

Drug development is an exercise in managing failure. Historic data indicated that only about 10% of drugs entering Phase I reached the market. More recent analysis from 2023-2024 shows overall success rates remaining at 6-7% as a composite probability across all phases, held up by higher stability in Phase I and Phase III and dragged down by Phase II, a formidable "valley of death" where nearly three-quarters of all programs fail (Table 5.2).

Table 5.2.: Clinical Development Success Rates, 2020-2024 Averages (IQVIA Institute for Human Data Science 2025)

Stage	Success Rate (2020-2024)	Key Attrition Drivers
Phase I -> Phase II	~48%	Safety signals, toxicity, strategic portfolio pruning
Phase II -> Phase III	~28%	Primary bottleneck: Lack of efficacy, dose-response uncertainty
Phase III -> Submission	~54%	Failure to hit endpoints, safety signals in large populations
Submission -> Approval	~94%	Regulatory questions, manufacturing (CMC) issues
Overall: Clinic -> Market	~7%	Cumulative probability from Phase I start (composite success rate)

A Phase III failure is the expensive kind. By the time the program fails, the sponsor has typically invested several hundred million dollars and 5 to 7 years of clinical

development (DiMasi, Grabowski, and Hansen 2016), and those losses are ultimately borne by successful products and by investors. Chapter 6 takes up R&D costs and the economics of attrition in detail.

Attrition also falls unevenly across the industry. *Emerging Biopharma (EBP)* companies originate over half of all new drugs, and they face a significantly more difficult path to approval than their larger counterparts: they receive *Complete Response Letters (CRLs)* from the FDA more frequently (IQVIA Institute for Human Data Science 2025). In 2024, 70% of CRLs were issued for non-clinical reasons such as manufacturing quality, product quality, or chemistry data, but the EBP pattern is distinctive: since 2020, 41% of CRLs for EBP sponsors included clinical reasons, while the share of clinical CRLs for non-EBP companies has declined by 81% (IQVIA Institute for Human Data Science 2025). This suggests that EBPs, strong as they are at scientific innovation, often face challenges with both clinical evidence packages and the industrial-scale regulatory and quality infrastructure required to cross the final approval threshold.

Programs most often fail because the drug does not meet its primary outcomes, or because an unacceptable safety signal emerges during larger-scale testing. Failure can also be commercial, when market shifts leave a scientifically successful drug unable to compete with existing or emerging standards of care.

6. Economics of a Trial

Clinical trials are expensive. The pivotal trials behind FDA's 2015-2016 approvals had a median estimated cost of $19 million, and the costliest, a noninferiority trial with clinical-benefit endpoints, about $350 million (Moore et al. 2018). Bringing a new drug to market, counting the trials that fail along the way, requires an estimated $2.6-2.7 billion per approved drug over 10-15 years (Deloitte's 2025 cohort average is $2.7B; the Tufts CSDD study's capitalized estimate is $2.6B) (Deloitte Centre for Health Solutions 2026; DiMasi, Grabowski, and Hansen 2016). Historical success rates ran near 10%. The composite rate was 6% in 2023 and 7% in 2024, the improvement driven by Phase III (IQVIA Institute for Human Data Science 2025).

These economics shape every decision in drug development: which diseases a company pursues, where the trials are run, whether a sponsor builds internal capability or buys it from a CRO, and whether a small biotech partners with a large one or goes it alone. The science does not happen in isolation from the money.

Money enters the system from pharmaceutical and biotechnology sponsors, who fund the vast majority of research and treat each trial as an investment that must yield an approved product and eventually drive revenue. Much of this work is outsourced to Contract Research Organizations (CROs), which provide specialized services ranging from study design to data management in a market Grand View Research put at $59.6 billion in 2025 for healthcare CRO services (Grand View Research 2026), and which are also the largest end users of clinical trial software (Grand View Research 2025). The trials themselves are conducted at clinical sites, hospitals and private practices compensated for their resources and staff time, supported by specialized vendors providing everything from laboratory processing to data capture software. Patients remain the primary participants: often altruistic or hoping for therapeutic benefit, and bearing personal costs (travel expenses, time away from work) that are frequently invisible in formal trial budgets.

6.1. The Cost Structure

In the early 2000s, estimates pegged the total cost of bringing a drug to market at $800 million. Modern analysis paints a starker picture.

6.1.1. Stratospheric R&D Costs

The top 20 biopharma companies spent **$7.7 billion** on terminated trials for assets that failed during the 2024 analysis cycle alone (Deloitte Centre for Health Solutions 2025). In Deloitte's 2025 analysis of its cohort, the average cost to develop a drug from discovery to launch rose to **$2.67 billion** ($2,671 million, up from $2,229 million in 2024), and even that widely cited benchmark often underestimates the full economic burden of modern development (Deloitte Centre for Health Solutions 2026). The Deloitte 2026 report finds a continued rebound in headline productivity, with a forecast **internal rate of return (IRR)** of 7.0% in 2025 (up from 5.9% in 2024 and 4.3% in 2023), driven largely by a few high-value GLP-1 and GLP-1/GIP assets in late-stage pipelines. Without those assets the forecast IRR would be just 2.9% (down from 3.8% in 2024), so the rest of the pipeline weakened even as the headline improved. Average forecast **peak sales** per asset have also increased, to $598 million (from $510 million in 2024), though excluding GLP-1-based assets (mainly obesity and diabetes drugs) would drop this benchmark to $353 million (Deloitte Centre for Health Solutions 2026).

Adding post-approval R&D (new indications, formulations, and post-marketing studies) raises the Tufts CSDD estimate to a **lifecycle cost** of $2.87 billion per approved drug in 2013 dollars (DiMasi, Grabowski, and Hansen 2016). That figure bundles the actual cash spent on successful and failed projects alike, the heavy investment in the approximately 93-94% of compounds that never reach the market (the corollary of composite success rates of 6-7% (IQVIA Institute for Human Data Science 2025)), and the opportunity cost of tying up billions over a 10-15 year development horizon. For the scientific drivers of phase-by-phase attrition, see Chapter 5.

These widely cited cost estimates have been challenged in the academic literature. Avorn (2015) argued that the Tufts CSDD methodology overstates true development costs by including a large opportunity-cost-of-capital adjustment and by relying on confidential, self-reported industry data that cannot be independently verified. An alternative analysis by Prasad and Mailankody (2017), using publicly available data from oncology companies that brought a single drug to market, estimated the median out-of-pocket R&D cost at $648 million ($757 million once capitalized at a 7% cost of capital), less than a third of the Tufts mean, which is capitalized at a higher 10.5% rate. The debate remains unresolved because true R&D expenditures are proprietary, and methodological choices (which costs to include, how to allocate shared overhead, what discount rate to apply) drive large differences in headline figures. Any single cost-to-market number should be interpreted with attention to the assumptions behind it.

A sponsor must fund the successful drug and also provide enough "economic cover" for the roughly 13 to 16 others that failed at today's 6-7% success rates. That is where the extreme pressure on clinical trial timelines and success rates comes from.

Cost rises with protocol complexity and enrollment size. In U.S. data from 2004 to 2012, not adjusted for inflation, the average study cost $1.4-6.6 million in Phase I (safety in small cohorts of 20-80 participants, with high per-patient costs from intensive PK/PD sampling), $7.0-19.6 million in Phase II (20-300 participants), and $11.5-52.9 million in Phase III, depending on the therapeutic area (Sertkaya et al. 2016). Among the pivotal trials behind FDA's 2015-2016 approvals, the median estimated cost was $19 million; the costliest were noninferiority trials with clinical-benefit endpoints and trials that needed large populations to detect small effects or accrue rare events, at up to about $350 million (Moore et al. 2018).

The direct cost of operating a standard Phase II/III trial is approximately $40,000 per day (Z. P. Smith, DiMasi, and Getz 2024). The opportunity cost of a delay is larger still: for a newly launched drug, a lost day of sales now averages roughly $800,000 (about a fifth of the $4-5 million per day long assumed from 1990s blockbusters), rising by therapeutic area to a median of $1.3-1.4 million per day for cardiovascular and hematology products (Z. Smith, DiMasi, and Getz 2024).

6.2. How AI and Automation Can Change Trial Economics

The most expensive failure is the late failure. When Phase III fails, years of work and hundreds of millions of dollars are already sunk, and the "cost of failure" must be recovered by the few programs that succeed (DiMasi, Grabowski, and Hansen 2016). If the economics of development are dominated by timelines and failure, then the economic value of AI is primarily indirect: it comes from shifting *when* and *how confidently* sponsors make decisions, and from reducing high-friction operational work that does not add scientific value. AI cannot eliminate biological uncertainty, but it can support earlier termination of low-probability programs by improving evidence synthesis across preclinical, translational, and early clinical signals, especially when combined with more structured protocols and better data integration. Even modest improvements in decision timing can have large economic consequences because capital is tied up for a decade or more, and because time-to-market directly affects expected returns (Deloitte Centre for Health Solutions 2025).

The second lever is timeline compression. Enrollment delay is the canonical case, because feasibility errors, inefficient patient identification, and retention failures can drive substantial schedule slippage and therefore opportunity cost (Lamberti et al. 2024). Protocol amendments are another: they trigger IRB/ethics resubmissions, re-consent, electronic data capture rebuilds, and supply-chain re-forecasting, and substantial changes can cost hundreds of thousands of dollars once downstream disruption is accounted for (Getz et al. 2016; Tufts Center for the Study of Drug Development 2021). Monitoring and data cleaning can then become a "long tail" of repetitive query generation, verification, and follow-up, which is the argument for risk-based quality management (RBQM) approaches that concentrate effort where it changes outcomes (Association of Clinical Research Organizations 2025).

AI-enabled tools create economic value mainly by automating classification, triage, and routine drafting (document indexing, query suggestion and clustering, standardized narrative scaffolds) so that expert time is shifted from low-information coordination to higher-judgment work. The intended outcome is not the removal of expertise, but a reduction in coordination overhead and in the reconciliation debt that delays interim decisions and final reporting.

Internal discovery vs. external acquisition. Sixty-one percent of the assets in the late-stage pipelines of the top 20 biopharma companies are now **externally sourced** through mergers, acquisitions, and licensing agreements, unchanged from 2024 and up from 59% in 2023 (Deloitte Centre for Health Solutions 2025, 2026). The R&D model at large pharmaceutical companies has shifted decisively toward buying innovation: "bolt-on" acquisitions offset looming patent expiries, and early-stage risk is offshored to smaller, more agile biotech firms. Bought assets carry their own risk. Externally sourced assets are consistently more likely to be terminated in late-stage development than those developed in-house (14% vs 12% termination rate in 2024); Deloitte's prescription is an M&A strategy built around long-term strategic fit rather than short-term pipeline gap-filling (Deloitte Centre for Health Solutions 2025).

The shift toward rare diseases and high-value targets. In 2025, 36% of the late-stage assets in Deloitte's top-20 biopharma cohort were focused on **rare diseases** (orphan indications), holding steady after a significant increase from 31.1% in 2023 to 37.4% in 2024 (Deloitte Centre for Health Solutions 2025, 2026). The pivot is toward precision medicine and high-unmet-need populations, where regulatory pathways may be faster and pricing power is greater. The consequence is extreme concentration: oncology alone accounted for 37% of the cohort's late-stage assets in 2024, which leaves sponsors in intense competition for trial sites and eligible participants (Deloitte Centre for Health Solutions 2025). These categories are not mutually exclusive, since many oncology assets also carry orphan designations.

6.2.1. Shifting the cost structure: labor vs. software and governance

Automation does not "remove" cost; it changes its composition. Some labor costs can decline (less manual reconciliation and administrative follow-up), while software, integration, validation, and oversight costs increase. In regulated settings, sponsors must budget for governance: validation of computerized systems, audit trails, monitoring of model performance, and clear accountability for errors (U.S. Food and Drug Administration 2023a; National Institute of Standards and Technology 2023).

If routine work becomes more software-mediated, CRO margins based on manual throughput are pressured, while differentiated value moves toward execution excellence, specialized expertise, and quality systems that can demonstrate control and traceability. The clinical trial software market, projected to grow more than 10% a year to nearly $29 billion by 2033, reflects this shift toward technology-mediated operations (Grand View Research 2025).

Global R&D spending has risen for two decades, and both *time* and *attrition* dominate what that money buys. Table 6.1 traces the shift.

Table 6.1.: Evolution of Drug Development Economics. The cost row sets the 2003 Tufts CSDD estimate beside Deloitte's 2025 cohort average, which use different methods; the success rows use IQVIA's five-year windows, as labeled in the cells. (Deloitte Centre for Health Solutions 2026; IQVIA Institute for Human Data Science 2025; DiMasi, Grabowski, and Hansen 2016)

Metric	Earlier Estimate	Recent Estimate
Avg Cost to Market	$800 Million (Tufts, 2003)	~$2.67 Billion (Deloitte, 2025)
Phase I Success	~52% (2015-2019)	~48% (2020-2024)
Phase III Success	~59% (2015-2019)	~54% (2020-2024)

Therapeutic area dramatically influences cost. Oncology trials are among the most expensive, driven by complex protocols, specialized imaging and biomarker testing, and extended survival follow-up; neurology and cardiovascular trials are similarly costly, while dermatology and infectious disease trials tend toward the lower end of cost ranges. Table 6.2 projects global pharmaceutical spending by therapeutic area through 2028, and Figure 6.1 ranks the top therapy areas by projected 2028 spending.

Table 6.2.: Global Pharmaceutical Spending by Therapeutic Area, 2024-2028 Projections (IQVIA Institute for Human Data Science 2024b).

Therapeutic Area	2024-2028 Projected Spending	Key Trends
Oncology	$440B by 2028 (14-17% CAGR)	Largest therapeutic area; novel mechanisms (cell/gene therapy, ADCs) now account for **35%** of oncology trials in 2024 (IQVIA Institute for Human Data Science 2025)
Immunology	$192B by 2028 (2-5% CAGR)	Growth slowing due to biosimilar competition (adalimumab from 2023, ustekinumab from 2025)
Diabetes and Obesity	Diabetes $184B (3-6% CAGR) and obesity $74B (24-27% CAGR) by 2028, tracked separately	Massive growth in **obesity** trial starts (**+77%** in 2024 compared to 2023), with 173 medicines in development or marketed (IQVIA Institute for Human Data Science 2025)
Neurology (CNS)	$103B by 2028 (6-9% CAGR)	Continued investment in Alzheimer's, Parkinson's, and rare neurological diseases

Therapeutic Area	2024-2028 Projected Spending	Key Trends
Cardiovascular	$126B by 2028 (2-5% CAGR)	Established market with incremental innovation
Infectious Disease	$75B by 2028 (3-6% CAGR)	Pandemic preparedness and antimicrobial resistance driving investment

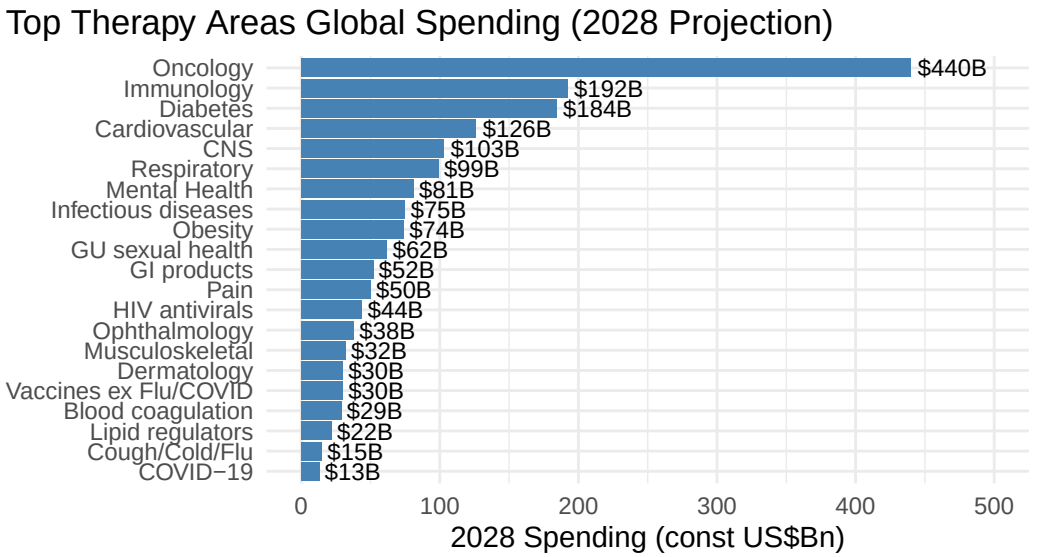


Figure 6.1.: Global Spending by Top Therapy Areas in 2028. Data from IQVIA Institute for Human Data Science (2024b).

6.2.2. The Globalization of Clinical Trials

Geography is a primary driver of cost. Because clinical research costs are driven largely by labor, trials cost far less where physicians, nurses, and study coordinators earn lower salaries, which is why sponsors have moved Phase II and Phase III trials to countries such as India (Glickman et al. 2009).

These cost variations reflect fundamental differences in healthcare systems and medicine utilization patterns. Per capita medicine use varies dramatically by region: Western Europe and Japan average 1,150-1,200 defined daily doses per capita, compared to North America's 800 doses per capita, despite North America having the highest GDP per capita. Eastern Europe demonstrates over three times higher medicine use per capita than China, despite GDP per capita being only 50% higher. These disparities stem from differences in healthcare system structure, reimbursement policies, patient cost-sharing requirements, and disease burden, all of which directly influence both trial site selection and per-patient costs (IQVIA Institute for Human Data Science 2024b).

That differential, combined with the need for large patient pools, drove a massive shift toward global participation. Glickman and colleagues documented the transition in the *New England Journal of Medicine* in 2009, noting that industry-sponsored Phase III trials were increasingly being conducted at sites outside the United States: more than half of all Phase III study sites (13,521 of 24,206) were already located ex-U.S., with approximately one-third of trials conducted solely outside the United States (Glickman et al. 2009). The economic driver was stark: a first-rate academic medical center in India charged approximately $1,500-$2,000 per case report, less than one-tenth the cost at a second-tier U.S. center (Glickman et al. 2009).

Glickman and colleagues also pressed the point on **generalizability**: more than two-thirds of 42 genetic variants associated with pharmacologic response show significant frequency differences between persons of African ancestry and those of European ancestry, and geographically distinct populations can have different genetic profiles that affect drug safety and effectiveness. Differences in medical infrastructure and physician training are often not transparently reported, making it difficult to assess whether standards of care are consistent across regions. Ethically, the "translation" of results is complicated by the fact that many drugs tested in developing countries never become readily available or affordable there after approval, potentially violating the principle that research should be responsive to the health needs of the host community (Glickman et al. 2009).

China's share of global trial country-uses grew from 9% in 2019 to 15% in 2024, a 66% increase that moved it from fifth to fourth place globally, just behind Asia-Pacific as a whole (Figure 6.2).

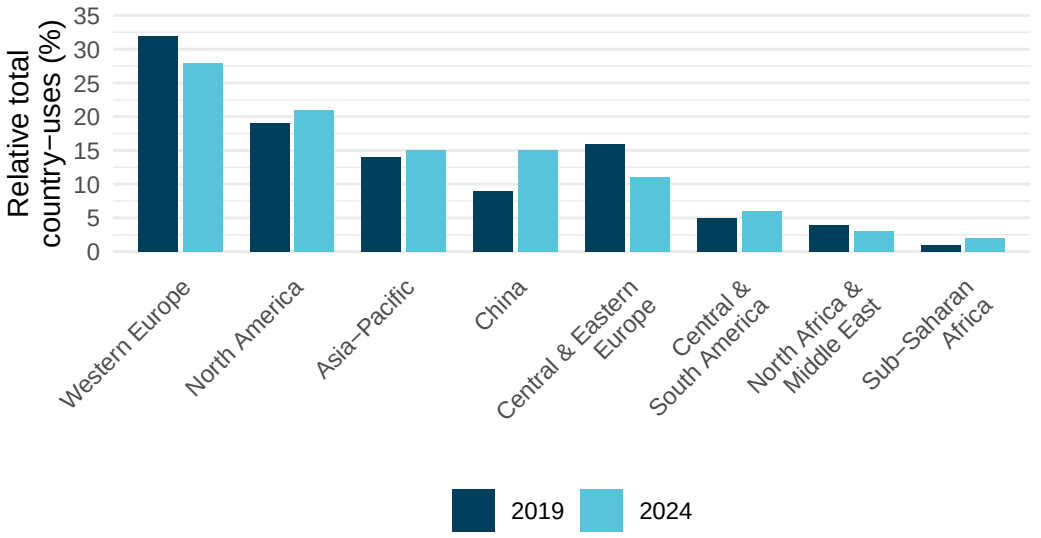


Figure 6.2.: Country Utilization as Percent of Trial Country-Uses, 2019 vs 2024. Data from IQVIA Institute for Human Data Science (2025).

6.2.3. The Regional Pivot: The Rise of China

China now accounts for **30%** of all global trial starts by headquarters location, up from just 1% in 2009 (IQVIA Institute for Human Data Science 2025). Independent data corroborates this trajectory: China-sponsored trial registrations grew from 1,333 cumulative studies in 2009 to over 32,500 by 2018, a 24-fold increase (Xu, Dong, and Liu 2020). China has now approached the United States (35% in 2024) and surpassed Europe (21%) in terms of trial initiation volume, making the innovation ecosystem genuinely multi-polar.

This growth in trial activity reflects substantial capital investment. The market value of publicly listed Chinese biopharma innovation players (innovative biotechs plus CRO/CDMO companies) grew from about $3 billion in 2016 to more than **$380 billion** by July 2021, more than a hundredfold increase in five years (Figure 6.3). China's share of the global innovation pipeline grew from 4.1% in 2015 to 13.9% in 2020, and the number of innovative molecules entering clinical trial applications in China grew by an average of 32% a year, from 114 in 2016 to 344 in 2020, with Chinese biotechs the largest group of submitters by 2020 (Han et al. 2021).

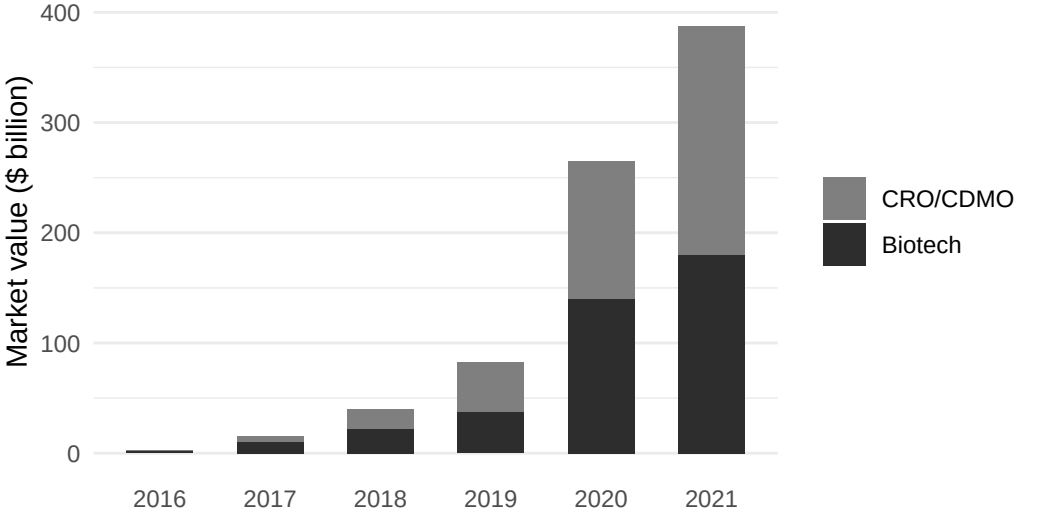


Figure 6.3.: Total Market Value of Listed Innovative China Biotechs and CRO/CDMO Players on Major Stock Exchanges (2021 data as of mid-year). Data from Han et al. (2021).

The composition of that growth matters: CRO/CDMO companies (contract research and manufacturing organizations) grew from a negligible base to represent over half of the total market value by 2021, reflecting both the outsourcing trend and China's emergence as a global services hub for clinical development. Growth in innovative drug development and in clinical services infrastructure together position China as both a source and destination for clinical trial activity.

Where a company is headquartered is not necessarily where its trials are physically conducted. Chinese emerging biopharma companies (EBPs) are prolific, but Chinese companies ran **83% of their trials domestically** between 2020 and 2024 (IQVIA Institute for Human Data Science 2025), while Western sponsors remain heavily tethered to U.S. and European site networks for global regulatory submissions.

Capacity is shifting underneath all of this. As discussed in Chapter 18, WCG's preliminary, partial-year 2024 count of unique U.S. investigators on industry-sponsored interventional trials was 78% below its pandemic-inflated 2020 peak (closer to 73% against the pre-pandemic base), and new trial starts fell sharply from their post-pandemic peak (WCG Clinical 2024). Trial activity is geographically broader, but this suggests that the absolute density of the global investigative environment is under significant strain.

6.2.4. Budgeting Checklist

Budgeting is granular work. Table 6.3 and Table 6.4, adapted from Beal et al. (2004) on industry-sponsored trials, outline the key direct and indirect costs that sites must anticipate.

Table 6.3.: Direct Cost Considerations for Clinical Trial Sites

Category	Direct Cost Considerations
Study Startup	Staff training time, Site initiation visit attendance, IRB preparation & submission fees, Amendment processing
Personnel	PI oversight, Coordinator time, Research assistants, Data entry, AE reporting, Monitor visits
Lab & Procedures	Phlebotomy supplies, Specimen processing/shipping, Dry ice, Biohazard fees, Radiology/Imaging fees, ECGs
Pharmacy	Drug storage, Preparation (compounding), Dispensing, Accountability logs, Destruction/Return
Patient Costs	Stipends/Reasonable reimbursement, Travel/Parking, Meals, Overnight stays (if required)
Administrative	Record storage (long-term), Copying/Scanning supplies, Courier fees, Archiving

Table 6.4.: Indirect Cost Considerations (Institutional Overhead)

Category	Indirect Cost Considerations
Institutional Overhead	Often a fixed % (e.g., 20-30%) covering rent, utilities, building maintenance

Category	Indirect Cost Considerations
Administrative Support	Payroll processing, Accounting/Invoicing, Legal review, Purchasing/Procurement
Infrastructure	Equipment depreciation (freezers, centrifuges), IT support/Network access, Communication (phones/internet)

Budget Line Items for Decentralized Trials and Digital Endpoints

Decentralized elements and digital endpoints change the cost structure as much as they change operations. Sponsors and sites increasingly incur costs for computerized systems, integration work, validation and oversight, and vendor-managed data flows, costs that are distinct from traditional per-visit procedure budgets (U.S. Food and Drug Administration 2024n, 2023h, 2023a). Quality management also remains a material cost driver: Meinert (2013) noted that there is "reason for concern" when quality assurance and control are allocated less than 10% of total direct costs. Quality is purchased through discipline and oversight.

The incremental line items follow the technology. eConsent and participant communication tooling brings platform licensing, configuration, accessibility review, and staff training costs (U.S. Food and Drug Administration 2016, 2024f). DHT and device operations add procurement, provisioning, support, replacement rates, data transmission, and monitoring of the device data itself (U.S. Food and Drug Administration 2023h). Every vendor data pipeline needs interface build, testing, reconciliation, and sustained change control (U.S. Food and Drug Administration 2023f, 2024f). Remote monitoring needs secure remote source access solutions, redaction workflows where applicable, and expanded centralized monitoring analytics (U.S. Food and Drug Administration 2024n; International Council for Harmonisation 2025b). All of it needs governance: computerized system validation and qualification, audit trail review capacity, and inspection readiness drills (U.S. Food and Drug Administration 2023a, 2024f).

6.3. How Money Flows

The sponsor sits at the center of the financial web, paying out to everyone else.

When a sponsor hires a CRO, the fee covers project management, site monitoring, data management, biostatistics, regulatory affairs, medical monitoring, and quality assurance, and it includes a margin over the CRO's direct costs. In fiscal 2025, four large publicly listed CROs reported gross margins (revenue less direct costs, before depreciation) of about 18% to 33%, and operating margins that ranged from a loss at Fortrea to 21% at Medpace, with IQVIA at 13% company-wide and ICON at 5% after impairment charges (IQVIA Holdings Inc. 2026; ICON plc 2026; Medpace Holdings, Inc. 2026; Fortrea Holdings Inc. 2026). Because pass-through costs such as investigator payments are booked as both revenue and direct cost (IQVIA

Holdings Inc. 2026; ICON plc 2026), margins on service revenue alone are higher: Medpace, which reports reimbursed out-of-pocket expenses separately, earned a gross margin of about 51% on revenue net of those expenses (Medpace Holdings, Inc. 2026). Across the itemized costs of U.S. pharmaceutical trials from 2004 to 2012, clinical procedures accounted for 15-22%, administrative staff for 11-29%, and site monitoring for 9-14%, excluding site overhead and sponsors' own costs of monitoring the study (Sertkaya et al. 2016).

The CRO in turn pays its vendors: central laboratories per specimen processed, electronic data capture companies by licensing fee plus per-patient or per-site charges, interactive response technology vendors for randomization and drug supply management. Each service generates revenue for the vendor and represents a pass-through cost to the sponsor.

Sites are paid by sponsors or CROs against negotiated budgets covering investigator time, coordinator time, overhead, and study-specific procedures. One CRO's published cost guide puts the cost of starting a new clinical trial site at $20,000 to $75,000 or more, covering site selection, regulatory submissions, staff training, and infrastructure setup (Andhalkar 2025).

Site payments are typically structured around *per-patient* or *per-visit* fees. A site might receive $3,000 for screening a patient, $500 for each study visit, and defined amounts for specific procedures, plus payments for protocol amendments, additional monitoring burden, and other non-standard activities.

Trial budgets calculate per-subject payments by mapping protocol procedures to specific visits. One-time activities, such as obtaining informed consent or reviewing medical history, are typically billed at the screening visit. Recurring activities (physical exams, vital sign collections, and laboratory tests) are usually billed at every study visit, while specialized assessments like ECGs or pharmacokinetic (PK) sampling occur only at designated intervals. The sponsor also pays professional fees to the investigator (PI) and a coordinator fee to the site for each visit conducted. The per-subject budget is the sum of these procedural and professional components across the screening period, treatment visits, and the final end-of-study assessment.

On top of per-subject costs, sites negotiate *start-up fees* to cover the administrative burden of opening the trial: regulatory document preparation, contract negotiation, pharmacy setup, source document creation, and archiving fees for long-term record storage.

The money that reaches sites must cover personnel time (investigators, coordinators, nurses, regulatory staff), the largest expense, along with administrative and regulatory costs such as IRB submissions, informed consent development, and training. Medical procedures, laboratory costs, and participant reimbursements (meal stipends, travel allowances) are often paid from site budgets as well.

6.4. The CRO-Site Relationship

CROs control payment flow to sites but are paid based on sponsor satisfaction. That asymmetry is where the friction starts. A CRO's business depends on meeting sponsor timelines, controlling costs, and delivering quality data, while sites are focused on patient care and, increasingly, on the operational challenges of running a clinical research enterprise.

Communication breakdowns are common. A 2024 industry column, citing a 2022 ZS Associates report on the voice of sites and patients, reported that 52% of sites were dissatisfied with how communication is handled and that 59% found contracting delayed and inefficient, largely because too many third parties, including CROs, are involved (Bronner 2024). Sites report repeating the same information to multiple contacts from the same sponsor or vendor, dealing with fragmented decision-making, and feeling disconnected from the decisions that shape the trial. The contracting layers (sponsor, CRO, legal departments, institutional offices) slow trial startup, and sites may wait months for executed contracts and first patient payments.

High staff turnover at CROs compounds these problems. When CRO personnel rotate frequently, sites must continuously re-educate their contacts about the trial and their capabilities, and inexperienced CRO staff may not understand protocol requirements, leading to errors that sites must correct.

Sites feel squeezed between CRO cost pressures and the realities of delivering quality research, and that structural tension underlies many operational conflicts.

6.5. Where the Money Actually Goes

In 2023, the biopharmaceutical industry spent over $30 billion directly at clinical trial sites across the United States, generating more than $62 billion in total economic activity (Pharmaceutical Research and Manufacturers of America 2024). These expenditures support jobs (investigators, coordinators, nurses, regulatory staff, administrators) and purchase goods and services from local economies. CROs employ hundreds of thousands of people globally, and the industry supports specialized vendors, technology companies, and service providers. Academic medical centers derive substantial revenue from clinical research, which supports teaching and infrastructure beyond the trials themselves.

A development program that costs $1 billion may never yield an approved product, and the majority do not. The oft-cited figure of $2.6 billion to bring a new drug to market (from a Tufts University study) is an average that already includes capitalized R&D expenditures and the cost of failure, so the out-of-pocket cost of a typical successful program, taken on its own, is lower. This high cost of failure increasingly drives larger pharmaceutical companies to license assets later in the development cycle, shifting much of the early-phase risk, and the majority of trial activity, onto Emerging Biopharma (EBP) companies (IQVIA Institute for Human Data Science 2025).

6.6. Patient Economics

In early-phase oncology trials, nearly half of patients (48%) report monthly out-of-pocket costs of at least $1,000 (Huey et al. 2021). Investigational drugs and many study procedures are covered, but patients face out-of-pocket expenses for travel, parking, meals, and accommodations, and those costs fall disproportionately on patients with lower incomes and those who live far from trial sites.

Time is the other cost. Participation requires clinic visits, sometimes frequent ones: time away from a job for working patients, arranged coverage for patients with caregiving responsibilities. These costs are real even if they do not appear in any budget.

Decentralized trial elements (home visits, remote monitoring, direct-to-patient drug shipment) address some of these burdens but introduce new logistical challenges and costs.

6.7. Funding Interims and Portfolio Management

Interim analyses are usually framed as scientific tools, but they also serve a financial one: deciding whether to commit capital to the next stage of a program.

A *funding interim* (also called an administrative or financial interim analysis) is a pre-specified analysis of accumulating trial data conducted specifically to trigger an investment decision, not to change trial design or stop enrollment. A scientific interim (for futility, efficacy, or dose selection) changes something about the trial. A funding interim changes nothing about the trial at all. It simply answers a binary question for an external funder: "Has the pre-specified threshold been met?"

The trial protocol specifies that an independent statistical group will perform a blinded or unblinded analysis when a defined number of patients have completed follow-up. That group delivers a checkbox (yes/no) to the sponsor and to the funder under a non-disclosure agreement, and the criterion is typically based on predictive probability of trial success, described below. The funder uses the checkbox to decide whether to commit capital to complete the trial; if the threshold is not met, the funder may decline to invest but the trial continues unchanged under its original protocol.

The naming convention is not a semantic nicety. Calling this analysis a "futility" interim creates a regulatory problem, because futility implies that failing the analysis triggers a protocol change. An administrative analysis that changes nothing does not require the same pre-specification, alpha-allocation, and regulatory disclosure as a scientific futility analysis. Sponsors should use "administrative analysis" or "financial analysis" in the protocol and charter, clearly distinguished from any scientific interim planned.

This design solves a structural problem in smaller biotech development. A company that can fund the Phase II portion of a seamless Phase II/III program may need

external capital to fund the Phase III expansion, and a standalone Phase II trial requires a gap between studies (unblinding, data readout, fundraising, regulatory interaction) that adds roughly a year or more to the development timeline and costs additional months of patent life. A seamless design with a built-in funding checkpoint can eliminate that gap, with the funder committing based on the interim checkpoint result.

6.7.1. Predictive Probability of Success as a Funder's Tool

The funding criterion is almost always some form of *predictive probability of success (PPoS)*, the probability that the trial will achieve statistical significance at the planned final analysis, given the data accumulated so far (Marion et al. 2025). This is distinct from power, which is fixed at design time based on an assumed effect size and does not change as data accumulate. PPoS does change with interim data: it rises as the treatment effect looks larger than assumed, falls as it looks smaller, and can swing dramatically depending on what the data show. The full statistical treatment is at Section 12.7.1.

For funders, PPoS is intuitive in a way that p-values and confidence intervals are not. A funder is, in effect, buying a lottery ticket with a stated probability of a payoff. A 70% PPoS at the time of the funding checkpoint means that, given the available data and the model's assumptions, there is a 70% probability the trial will be statistically successful at the final analysis. This is the natural framing for investors who think about expected value and risk-adjusted returns. Even large pharmaceutical companies use internal PPoS calculations as portfolio-management tools for deciding which Phase III programs to continue and which to terminate.

A common communication failure is demanding that PPoS equal power. "We need 90% predictive probability" as a funding criterion means the funder needs to see, at halftime, that the team is 90% likely to win by the required margin. That is an extraordinarily high bar mid-trial, far higher than is typically achievable without an enormous interim effect size. The practical solution is to show funders example data presentations (Kaplan-Meier curves, hazard ratios, mean differences) and work backward to the PPoS associated with each one, building intuition for what a "fundable" interim result looks like.

6.7.2. Assurance and Portfolio Optimization

Assurance (sometimes called Bayesian predictive probability of success) extends the PPoS concept from trial execution to program planning (O'Hagan, Stevens, and Campbell 2005). Where PPoS conditions on the accumulated data at a specific interim time point, assurance is computed before the trial starts: power assumes a specific true effect size and computes the probability of observing a significant result under that assumption, while assurance integrates over a distribution of plausible true effects and computes the average probability of a successful outcome across that distribution.

For portfolio management, assurance is the more relevant quantity. A Phase III program powered at 85% for an assumed hazard ratio of 0.80 sounds high, but if the sponsor's internal evidence places substantial uncertainty on whether the effect is 0.70 or 0.90, the assurance (averaged over that uncertainty) might be 55-60%. Portfolio managers who optimize based on power alone are systematically overestimating the probability of success for their programs; assurance provides a more realistic forecast.

Assurance calculations also make uncertainty transparent. A program team that disagrees about the likely effect size can run assurance under both assumptions and see how dramatically the probability of success changes, which is more productive than arguing about which single effect size to assume for power.

6.8. Platform Trial Economics

The economic case for platform trials (described in detail at Section 13.6) is a reduction in cost per scientific question answered. When sponsors each build separate Phase III programs for drugs targeting the same disease, approximately half of the enrolled patients receive placebo across all programs simultaneously, generating information only about the control-arm trajectory rather than about any experimental treatment. A platform with a shared control arm redirects those patients to experimental arms, substantially reducing the total number of patients required to answer each question while maintaining statistical power (Benjamin R. Saville and Berry 2016).

Patient count is only part of it. Platform infrastructure (site contracts, databases, regulatory submissions, DSMBs, monitoring infrastructure) represents a substantial fixed cost that does not scale with the number of programs run in parallel. Amortizing this fixed cost across multiple arms makes the per-arm cost of subsequent arms a fraction of the cost of a standalone trial. REMAP-CAP investigated many therapies across its multiple treatment domains over its lifespan; each of those questions would have required years of infrastructure development if approached as a standalone trial.

The first-arm problem is the primary economic barrier to platforms. The sponsor of the platform's first arm bears the full fixed cost of building the arena before any amortization benefit materializes. This means platforms in academic or disease-advocacy settings (Healey ALS, I-SPY 2, SNAP) typically require philanthropic or government funding for the first arms; once the platform is proven, commercial sponsors engage at much lower marginal cost. This structural challenge explains why perpetual platforms have succeeded most readily in conditions with strong advocacy communities or NIH investment, while diseases without those resources remain served primarily by standalone trials.

6.9. Strategic Implications

The largest economic levers in a trial are rarely the per-procedure line items. They are the decisions about which programs to fund, when to stop them, and how to share fixed infrastructure across scientific questions.

For sponsors and their funders, this argues for judging programs on assurance instead of power: a program powered at 85% can carry a true probability of success closer to 55-60% once uncertainty in the effect size is taken into account, and stopping such a program early (or structuring a seamless design with an administrative funding checkpoint) protects far more capital than any procedure-level negotiation. For CROs, differentiated value is moving from manual throughput toward execution excellence and the quality systems that can demonstrate control as routine work becomes software-mediated. For sites, financial viability depends on budgeting that captures the indirect and administrative costs (Table 6.4) that per-visit fees routinely miss, so that negotiated compensation actually covers personnel and infrastructure. And for the patients at the center of the enterprise, designs that move cost off the participant are both an ethical obligation and, increasingly, an advantage in recruitment and retention.

7. Phase I: Human Pharmacology

Until the first human dose, everything known about a compound comes from test tubes, cell cultures, and animal models. Phase I is where someone has to convert all of that into a number of milligrams, hand it to a person, and then find out how quickly the drug reaches the bloodstream, how long it stays there, what biological effects it produces, and at what dose the adverse ones become unacceptable.

First-in-human strictly names only the initial study in a development program; Phase I as a category covers every study designed principally to assess safety, tolerability, and pharmacokinetics in humans (International Council for Harmonisation 2021). That work is increasingly the domain of small companies. Emerging Biopharma (EBP) sponsors accounted for 65% of all Phase I trial starts in 2024, more than twice the share of large pharmaceutical companies (24%) (IQVIA Institute for Human Data Science 2025).

The subjects are usually healthy volunteers, since exposing sick people to a drug whose behavior in humans is completely unknown would be inappropriate. Oncology is the standing exception, along with certain other therapeutic areas where the expected toxicity is too significant to put in front of a healthy person.

The starting dose is calculated from preclinical data and set far below anything expected to cause toxicity; Section 7.1 works through that calculation and the alternative framework for high-risk biologics. Escalation follows prespecified rules: each cohort is dosed and observed for signs of toxicity before the next may go higher, and the sequence continues until the *maximum tolerated dose (MTD)* is reached or the desired pharmacological effect is achieved.

> ⚠ Lessons from TGN1412
>
> In 2006, six healthy volunteers who received TGN1412 suffered life-threatening immune reactions. The dose had been calculated by conventional methods. First-in-human design has been under heightened scrutiny ever since, and the guidance that followed pushed sponsors toward target biology and mechanism of action as the basis for dose selection, staggered dosing within cohorts, and extended observation periods before escalation.

7.1. Starting Dose Selection and Risk Mitigation

Two frameworks convert animal data into a first human dose, and for TGN1412 the gap between them was orders of magnitude. Which one governs depends on the compound class and on the plausible mechanisms of adverse effects.

NOAEL-based approach. The *No Observed Adverse Effect Level (NOAEL)* in the most appropriate animal species (by default, the most sensitive) is converted to a human equivalent dose using a body-surface-area correction (HED = NOAEL $\times$ (animal weight/human weight)$^{0.33}$, with FDA's standard 60 kg human) and then divided by a safety factor (10 by default, raised or lowered with a stated rationale) to arrive at the *Maximum Recommended Starting Dose (MRSD)* (U.S. Food and Drug Administration 2005). If the human equivalent of the pharmacologically active dose falls below the MRSD, the starting dose may be lowered further. For small molecules with conventional pharmacology and a well-characterized safety profile in animals, this calculation is generally sufficient.

MABEL approach. The *Minimal Anticipated Biological Effect Level (MABEL)* was developed for compounds whose pharmacological activity in humans begins at doses far below any observable toxicity in animals. It integrates receptor occupancy modeling, in vitro pharmacological activity data, ex vivo assays on human cells, and animal PK/PD to identify the lowest dose expected to produce any measurable biological effect. EMA's 2017 guideline on FIH trials (European Medicines Agency 2017b) calls for MABEL to be estimated alongside the pharmacologically active dose (PAD), and recommends that the starting dose be anchored to the MABEL, the PAD, or the NOAEL according to how much uncertainty remains about the human relevance of the non-clinical findings and about the intended target. In practice the MABEL anchor is the one chosen when a compound is highly potent, acts through a cytokine-mediated mechanism, or engages a target expressed on circulating immune cells. TGN1412 failed precisely because the NOAEL-based MRSD did not capture the potent receptor-mediated cytokine cascade; a MABEL-based approach would have produced a starting dose orders of magnitude lower.

In practice, sponsors calculate both estimates and select the more conservative. A large gap between the MABEL and the NOAEL-derived MRSD is itself a risk signal that calls for protocol-level mitigation.

Sentinel dosing. To limit how many subjects are exposed to a new dose at once, FIH protocols routinely dose sentinels first: within each cohort, one subject receives the investigational drug and one receives placebo, ahead of the rest by an interval set from the compound's half-life and the plausible onset of an adverse reaction (commonly 24 to 48 hours). After that interval, the DSMB or Safety Review Committee reviews the sentinel data before the remaining members are dosed per the cohort's pre-specified active:placebo allocation, which is majority active. EMA's first-in-human guideline endorses staggered dosing within cohorts (European Medicines Agency 2017b), and it is now both regulatory expectation and routine practice for biologics and immunologically active compounds.

Stopping rules. FIH protocols typically prespecify stopping criteria at three levels; the criteria below are common examples, and each rule should state whether the stop is a temporary halt or a final end to dosing (European Medicines Agency 2017b). *Subject-level* rules discontinue one person's dosing immediately for any Grade 3 or higher adverse event, a cytokine elevation above a pre-specified threshold, or a defined change in cardiac rhythm or ECG morphology. *Cohort-level* criteria suspend enrollment pending a safety review when a sentinel subject has a DLT, when two subjects in the cohort have severe adverse reactions (in the same organ system or not), or when any laboratory finding meets a pre-specified threshold of clinical concern. *Study-level* criteria stop dosing across the trial: a potentially drug-related death, a Grade 4 cytokine release syndrome, or, in a healthy-volunteer trial, a single serious adverse reaction in any subject, expected or not.

For immunologically active compounds, cytokine panels (IL-6, IFN-γ, TNF-α) belong in the protocol as early biomarkers of immune dysregulation, checked at prespecified timepoints before clinical symptoms develop.

Every foreseeable safety event should have a pre-written response in the protocol before the first subject is dosed, because the alternative is deciding in real time under clinical pressure.

7.2. Single and Multiple Ascending Dose Studies

Most Phase I programs run both a *single ascending dose (SAD)* study and a *multiple ascending dose (MAD)* study. In the SAD, small cohorts (typically 6 to 8 subjects) receive a single dose each, with placebo typically included in each cohort to help distinguish drug effects from background noise; safety and pharmacokinetics are reviewed and found acceptable before the next cohort receives a higher dose.

The MAD then gives the drug repeatedly over days or weeks, and asks what a single dose cannot: whether the drug accumulates, how long it takes to reach steady state, and whether repeated exposure is still tolerated. It bridges from the artificial world of single doses to the clinical reality of chronic treatment.

7.3. Characterizing Pharmacokinetics

A primary goal of Phase I is to determine how the drug moves through the body: its *pharmacokinetics (PK)* (International Council for Harmonisation 2021). The raw material is a series of concentrations measured in blood (and sometimes other fluids) at multiple time points after dosing; Table 7.1 lists what pharmacokineticists compute from that curve.

Table 7.1.: Key Pharmacokinetic Parameters

PK Parameter	Symbol	Definition	Formula	Clinical Implication
Maximum Concentration	C_{max}	Peak drug concentration achieved	Observed directly from PK curve	Indicates peak exposure; related to tolerability
Time to Maximum	T_{max}	Time to reach peak concentration	Observed directly from PK curve	Related to absorption rate; affects onset of action
Area Under Curve	AUC	Total exposure over time	$\int_0^\infty C(t)\,dt$	Proportional to amount absorbed; key PK/PD metric
Half-life	$t_{1/2}$	Time for concentration to decrease by 50%	$t_{1/2} = \frac{0.693 \times V_d}{CL}$ (one-compartment)	Sets time to steady state (about 4-5 half-lives); guides dosing interval
Clearance	CL	Volume of plasma cleared per unit time	$CL = \frac{Dose}{AUC}$ after an IV dose (CL/F after an oral dose)	Efficiency of drug elimination; guides dosing
Volume of Distribution	V_d	Apparent volume drug distributes into	$V_d = \frac{Dose}{C_0}$ (IV bolus, one-compartment)	Suggests tissue distribution; affects loading doses
Bioavailability	F	Fraction of dose reaching systemic circulation	$F = \frac{AUC_{oral}/Dose_{oral}}{AUC_{IV}/Dose_{IV}}$	Determines oral vs. IV dose equivalence

Half-life is the main guide to the dosing interval: a 4-hour half-life typically requires three or four doses daily, while a 24-hour half-life allows once-daily dosing. Caffeine's is about 5 hours, which is why morning coffee wears off by afternoon; some antidepressants exceed 100 hours, which is why missing a dose matters less and why side effects persist after discontinuation. Formulation and pharmacodynamics can override half-life: extended-release products stretch the interval, and a drug that binds its target irreversibly can act long after it has left the plasma.

Volume of distribution reveals where the drug goes. A V_d close to plasma volume (about 3 liters) suggests the drug stays in the bloodstream, useful for treating blood infections but unable to reach intracellular targets. A V_d of hundreds of liters, far exceeding actual body volume, indicates extensive tissue binding: the drug accumulates in fat, muscle, or specific organs. Chloroquine's V_d exceeds 200 liters per kilogram because it concentrates in tissues, which is why it needs loading doses and why it persists for weeks after the last dose.

Clearance sets the dose needed to hold therapeutic levels: high clearance means rapid elimination and so higher or more frequent doses, low clearance means accumulation

and careful titration to avoid toxicity. Impaired kidney or liver function can reduce clearance, which is why these populations may need dose adjustments; whether they do depends on how much exposure changes and how wide the drug's therapeutic range is.

Bioavailability explains why oral and intravenous doses differ. At 50%, an oral dose must be twice the IV dose to reach the same exposure. Some drugs fall below 10%, making oral administration impractical; others are destroyed by stomach acid or extensively metabolized by the liver before reaching circulation (the "first-pass effect").

Pharmacokinetics tells us what the body does to the drug; *pharmacodynamics (PD)* tells us what the drug does to the body. Phase I studies often include pharmacodynamic endpoints: biomarkers or biological effects that indicate the drug is producing its intended action. If such a biomarker correlates with drug exposure, and if it is a reasonable predictor of clinical effect, *PK/PD modeling* can link the two and guide the choice of dose and regimen for Phase II.

7.4. Special Studies

Beyond first-in-human and dose escalation, Phase I typically carries a set of specialized studies (Table 7.2) (International Council for Harmonisation 2021; U.S. Food and Drug Administration 2024d):

Table 7.2.: Types of Phase I Special Studies

Study Type	Purpose	Design	Regulatory Expectation
Bioavailability	Determine what fraction of dose reaches circulation	Compare oral vs. IV administration (absolute) or different formulations (relative)	Required for oral formulations
Bioequivalence	Show new formulation performs like reference product	Crossover design comparing test vs. reference; assess if 90% CI of AUC/Cmax ratio within 80-125%	Required for generics and major formulation changes; a BCS-based biowaiver can replace the in vivo study for some immediate-release oral products
Food Effect	Assess how meals affect absorption	Crossover comparing fed vs. fasted states	Required; informs label dosing instructions
Drug-Drug Interaction (DDI)	Evaluate interaction with other medications	Test investigational drug with probe substrates/inhibitors/inducers of CYP450 enzymes	Required for commonly co-administered drugs

Study Type	Purpose	Design	Regulatory Expectation
Hepatic Impairment	PK in patients with liver dysfunction	Compare normal vs. mild/moderate/severe hepatic impairment	Required if drug significantly metabolized
Renal Impairment	PK in patients with kidney dysfunction	Compare normal vs. mild/moderate/severe renal impairment	Expected if the drug is likely to be used in patients with impaired renal function, even if not renally eliminated (a reduced design may then suffice) (U.S. Food and Drug Administration 2024j)
Elderly Subjects	PK in aged population (≥ 65 years)	Compare young vs. elderly PK parameters	Geriatric PK expected for drugs used in the elderly; population PK can replace a dedicated study if enough older patients are enrolled
Thorough QT	Assess cardiac repolarization risk	High therapeutic and supratherapeutic doses; measure QTc prolongation	Required unless waived by alternative data

Together they let a sponsor recommend a dose for everyone who might need the drug, accounting for physiological differences and potential interactions. Some of that evidence no longer requires a trial: FDA and EMA now accept physiologically-based pharmacokinetic (PBPK) modeling and model-informed drug development (MIDD) to supplement or waive some dedicated drug-drug interaction and organ-impairment studies, extrapolating exposure to untested populations from validated mechanistic models.

7.5. Phase I in Oncology: Dose Escalation Design

Oncology is the exception to the healthy-volunteer rule. Anticancer drugs often cause significant toxicity, and the people who need them face serious illness with limited treatment options, so Phase I oncology studies are conducted in patients.

The objective has traditionally been the MTD: the highest dose that causes unacceptable toxicity in no more than a defined fraction of patients. *Dose-limiting toxicities (DLTs)* are the pre-specified adverse events (typically Grade 3 or 4 by the CTCAE system) that draw the toxicity boundary, observed over a fixed *observation window* (commonly 21 or 28 days, matching the first treatment cycle) after each dose.

7.5.1. The 3+3 Design and Its Limitations

The *3+3 design* has been the workhorse of oncology Phase I trials for decades. Three patients receive the starting dose; if none experiences a DLT the dose is escalated; if one does, three more are enrolled at that dose, and escalation continues only if none of them has a DLT; if two or more patients at a dose experience DLTs, the dose is declared too high and the next lower dose, usually confirmed in six patients, is the MTD. The design is simple and transparent, requiring no statistical software.

Simulation studies show that the 3+3 often fails to select the correct MTD, and that it tends to err low rather than high (Le Tourneau, Lee, and Siu 2009). The consequences are patients treated below the true MTD and an uncharacterized upper part of the dose-response curve. The cause is structural: the design makes each decision using only the cohort at the current dose, ignoring what the trial has already learned at every other dose, which makes it statistically inefficient (O'Quigley, Pepe, and Fisher 1990).

7.5.2. Modern Dose-Escalation Designs

Two families of escalation designs are the main alternatives to the 3+3 in sponsored oncology trials. Both require specifying a *target toxicity rate* (typically 20-25% in modern practice, lower than the old "highest tolerated dose" convention), and interval designs also use an *acceptable toxicity interval* around it (e.g., 15-35%).

Rule-based "interval" designs (BOIN, mTPI, i3+3). These methods compare the observed DLT rate at the current dose to a target interval and make a simple escalate/stay/de-escalate decision without fitting a dose-toxicity curve across all doses. The *Bayesian Optimal INterval (BOIN)* design, which FDA designated fit-for-purpose for dose finding in December 2021 (U.S. Food and Drug Administration 2025b), is a prominent example: it derives optimal decision boundaries from the target toxicity rate and the acceptable interval through a closed-form calculation, minimizing the probability of incorrect escalation decisions (S. Liu and Yuan 2015). BOIN and related methods (mTPI-2, i3+3) can be pre-computed into a decision table before the trial, which makes them operationally tractable. Their limitation is that each escalation decision uses only the data at the current dose; BOIN pools information across doses only at the end, when isotonic regression selects the MTD (Yuan et al. 2016).

Model-based designs (CRM, BLRM). The *Continual Reassessment Method (CRM)* fits a simple dose-toxicity model (a one-parameter curve in O'Quigley's original formulation) after each patient with complete DLT information, updating the model and assigning the next cohort to the dose estimated to be nearest the target toxicity (O'Quigley, Pepe, and Fisher 1990). The *Bayesian Logistic Regression Model (BLRM)* is a two-parameter extension that provides more realistic uncertainty quantification (Neuenschwander, Branson, and Gsponer 2008). By borrowing information across all tested doses simultaneously, model-based designs use the accumulating data more efficiently than the 3+3, and their advantage over interval

designs is greatest when many dose levels are studied or the true MTD lies at an extreme of the tested range. In published head-to-head simulations, BOIN's average MTD-selection accuracy is comparable to the CRM's (S. Liu and Yuan 2015). The tradeoff is operational complexity: decisions require real-time statistical computation as each patient's DLT outcome becomes available, and the prior distribution for the dose-toxicity curve requires careful elicitation from clinical and historical data.

Escalation with Overdose Control (EWOC) is an additional constraint applied with model-based designs: a dose is allocated only if the posterior probability that its DLT rate exceeds the upper bound (e.g., 35%) is below a threshold (e.g., 25%). Unlike the absolute exclusion rules in interval designs, EWOC exclusion is *temporary*: as the model accumulates data and revises its estimate of a dose's toxicity, a previously excluded dose can be reconsidered (Babb, Rogatko, and Zacks 1998).

Choosing a dose-escalation design

There is no universally superior design. Table 7.3 summarizes the practical tradeoffs.

Table 7.3.: Dose-escalation design comparison

Design	Type	Information borrowing	Operational requirements	Typical use case
3+3	Rule-based	None (current cohort only)	Decision table; no software	Legacy studies; regulatory familiarity
BOIN	Interval rule-based	Escalation uses current dose only; final MTD selection pools doses (isotonic regression)	Pre-computed table; no real-time stats	Most sponsored oncology Phase I; good balance of simplicity and performance
CRM / BLRM	Model-based	Borrows across all tested doses simultaneously	Real-time statistician or software at each decision	Programs with many dose levels; extreme-range MTD; high-potency compounds

Beyond the operational question the table settles, two things push toward a model-based design. The first is the number of doses: with many dose levels, model-based designs gain a larger relative advantage because they can skip doses on the way up and borrow more efficiently. The second is a low target toxicity rate, since at low targets interval methods may be too conservative and model-based designs with EWOC handle the situation more gracefully. Whichever design is chosen, simulate it across a range of plausible dose-toxicity scenarios before the trial begins, assessing MTD-selection accuracy, probability of overdosing, and expected sample size.

7.5.3. From MTD to Optimal Dose: Project Optimus

For decades the regulatory pathway rewarded finding the highest dose patients could tolerate, on the assumption that more was better. Then post hoc analyses of several approved oncology drugs showed that doses below the MTD would have produced equivalent efficacy with substantially less toxicity. FDA's Project Optimus, launched in 2021 for oncology, exists to generate that evidence prospectively (U.S. Food and Drug Administration 2021d, 2024i): the *right* dose is the one that optimizes the balance of efficacy and tolerability, which is not always the maximum patients can survive.

The FDA guidance on dosage optimization for oncology biologics and drugs, issued as a draft in January 2023 and finalized in August 2024, sets out FDA's recommendations for optimizing dosage during development, much of which falls to early-phase programs (U.S. Food and Drug Administration 2024i):

Evidence across multiple candidate doses. Escalating to the MTD and declaring it the recommended Phase 2 dose (RP2D) is no longer enough. Sponsors are expected to characterize at least two candidate RP2Ds through expansion cohorts, seamless Phase I/II designs, or randomized dose-comparison substudies, with the criteria for selecting among those candidates written into the protocol before enrollment begins.

Pharmacodynamic and patient-reported evidence. Dose characterization now includes efficacy biomarkers (receptor occupancy, PD markers mechanistically linked to the drug's activity) and patient-reported outcomes measuring symptom burden and quality of life, alongside the traditional clinical safety assessments. A dose that is more tolerable by PRO metrics is meaningful, even when adverse event grades are equivalent.

Randomized dose comparison. That guidance says explicitly that sponsors should randomize where feasible, in place of uncontrolled parallel-cohort expansion. The randomization need not be powered for formal hypothesis testing: cohorts of, say, 20 to 40 patients per dose can suffice, with pre-specified endpoints covering activity and tolerability and a defined decision rule for RP2D selection. What matters is that the rule exists before the data do.

Biologically active dose. For drugs with clear PD markers, the *biologically active dose (BAD)* has gained regulatory traction: if PK/PD modeling demonstrates that a dose below the MTD achieves full target coverage (e.g., sustained receptor occupancy above 80%), escalating further is unlikely to improve efficacy and may add toxicity; only activity and tolerability data compared across dosages can settle the question. Project Optimus gives sponsors a path to justify selecting the BAD rather than the MTD as the RP2D, provided the exposure-response relationship for the PD marker is reproducible across the tested dose range.

Project Optimus was launched for oncology, but its logic is spreading. Immunology, gene therapy, and other areas where the relationship between dose and effect is

non-linear and toxicity can be severe and irreversible are now having the same dose-finding discussions.

Combination escalation: added dimensions

Oncology increasingly relies on drug combinations, typically pairing a checkpoint inhibitor with a targeted or cytotoxic agent. Combination escalation then has to solve three problems that single-agent escalation never faces.

Two-dimensional dose space. With two drugs each tested at multiple levels, the toxicity surface spans a grid, and exhaustive testing of every combination is infeasible. Escalation designs must prespecify the rules for moving through the grid, although adaptive designs pick each next combination from the accumulating data rather than following a fixed path. A BLRM with an interaction term models how the two drugs' toxicities combine, while model-free designs such as the BOIN combination design and the Product of Independent Beta Probabilities Escalation (PIPE) design estimate each combination's DLT rate directly, without a parametric dose-toxicity model (R. Lin and Yin 2017; Mander and Sweeting 2015). Assuming that the two toxicities arise independently is not the same as assuming they add: two independent 20% risks give a 36% chance of at least one DLT, not 40%.

Scheduling. The sequence and timing of combination dosing often drives tolerability independently of dose: the same total doses may be tolerated when the two drugs are given sequentially but not when they are given simultaneously. Combination protocols must specify schedule as well as dose, and treat a schedule change as a form of dose modification requiring safety review under the protocol.

Attribution. When a toxicity occurs, determining whether it belongs to Drug A, Drug B, or the combination is frequently impossible. DLT definitions and cohort decision rules must prespecify how attribution uncertainty is handled, with the rationale stated in the protocol rather than settled at a case review meeting after the event.

7.6. The Phase I-Phase II Transition

Phase I trials are small by design, and everything after them is sized from what they produce: a safety database characterizing adverse events across a range of doses, pharmacokinetic parameters to guide dosing decisions, pharmacodynamic data linking exposure to biological effects, and, most critically, a *recommended Phase II dose* that balances safety against expected efficacy and initial proof of mechanism. By identifying that dose, Phase I researchers move the drug from laboratory hypotheses into human biology.

The handoff is also one of several *go/no-go decisions* that punctuate drug development. Unexpected toxicity, poor pharmacokinetics, or an absent pharmacodynamic effect can end the program here. Such early terminations, while disappointing, are far preferable to late-stage failures.

The geographic footprint of this work is shifting, with China-headquartered companies accounting for 30% of global trial starts in 2024 (IQVIA Institute for Human Data Science 2025). AI-assisted approaches to trial design and simulation, including enrollment and feasibility modeling and prognostic digital twins for covariate adjustment, are covered in Chapter 26.

Discussion Questions

1. A T-cell engager for relapsed/refractory multiple myeloma has potent cytokine-release activity in ex vivo human blood assays at concentrations well below those toxic in non-human primates. The NOAEL-based MRSD is 10-fold higher than the MABEL-derived starting dose. Which starting dose framework would you recommend to the sponsor, and what protocol-level safeguards would you require to accompany that choice?

2. The 3+3 design selects the correct MTD less reliably in simulation than interval and model-based designs, yet it remains in use. What institutional, regulatory, and operational factors sustain its use, and under what specific circumstances, if any, does its use remain defensible?

3. FDA's dosage-optimization guidance recommends comparing multiple candidate dosages. For a drug whose primary PD marker (target receptor occupancy) is technically feasible but not validated as a predictor of clinical response, how should the sponsor operationalize the dose-selection criteria in the protocol before data are seen?

4. A combination Phase I program pairs a checkpoint inhibitor (approved as monotherapy, with two dose levels under consideration for the combination) with a novel SMAC mimetic (dose-escalating). The development team proposes a simple row-by-row escalation through a dose grid, assuming additive toxicity. What are the statistical and clinical limitations of that assumption, and what design alternatives would you propose?

5. A sentinel stopping rule specifies that dosing of the full cohort is suspended if a DLT occurs in a sentinel subject. A sentinel subject at dose level 3 develops Grade 3 transaminase elevation at hour 36 that resolves completely by hour 72. The medical monitor argues this is manageable and not clinically meaningful. Walk through the considerations for deciding whether to invoke the stopping rule or proceed with cohort dosing.

8. Phase II: Exploratory

Approximately 70% of drugs that enter Phase II never advance to Phase III (Biotechnology Innovation Organization, QLS Advisors, and Informa Pharma Intelligence 2021). Phase I established that the drug can be given to humans with acceptable safety; Phase II must establish that giving it to humans accomplishes something worthwhile. That is the first controlled exploration of whether an investigational drug produces the therapeutic effects that preclinical research suggested it might, and the question shifts from "is this drug safe?" to the more consequential "does this drug actually work?"

The exploration occurs in two stages (see Table 8.1). Early Phase II studies (sometimes called *Phase IIa*) seek *proof of concept*: evidence that the drug produces some measurable therapeutic effect in patients with the target condition. Later Phase II studies (*Phase IIb*) take that signal and turn it into a dose (International Council for Harmonisation 2021):

Table 8.1.: Comparison of Phase IIa and Phase IIb Studies

Aspect	Phase IIa (Proof of Concept)	Phase IIb (Dose Finding)
Primary Objective	Detect therapeutic signal; establish proof of concept	Characterize dose-response; select Phase III dose
Sample Size	Often 20-100 patients	Often 100-300 patients
Design	May be open-label or informally controlled	Randomized, controlled, often with multiple dose arms
Duration	Shorter (weeks to months)	Longer (months)
Endpoints	Exploratory; biomarkers and early efficacy signals	Primary efficacy endpoints; expanded safety data
Key Question	"Does this drug do anything beneficial?"	"What dose balances efficacy and safety?"
Go/No-Go Decision	Proceed to dose-finding or terminate	Proceed to Phase III or terminate

8.1. The Dose-Response Relationship

Understanding how response changes with dose is one of the most critical (and challenging) objectives of Phase II (International Council for Harmonisation 2021). The relationship is rarely simple.

At very low doses, there may be no detectable effect; the drug is present but at concentrations too low to produce meaningful target engagement. As doses increase, response emerges and grows. Eventually, a plateau is reached where higher doses produce little additional benefit. At still higher doses, toxicity may begin to outweigh efficacy.

Dose-finding studies have to characterize that curve well enough to select the optimal dose for Phase III: high enough to produce meaningful efficacy and low enough to minimize toxicity, and sitting where small dose differences do not produce large response differences. Common dose-finding approaches are summarized in Table 8.2.

Table 8.2.: Dose-Finding Approaches in Phase II

Approach	Description	Advantages	Disadvantages
Parallel Dose Groups	Fixed doses compared in separate arms	Straightforward analysis; clear comparisons	Requires larger sample sizes; less efficient
Dose Escalation Within Subjects	Individuals receive increasing doses sequentially	More efficient use of subjects	Effects may reflect cumulative exposure; carryover effects
Adaptive Designs	Allocation modified based on interim data	Concentrates patients in informative dose ranges	More complex; requires careful pre-specification
MCP-Mod	Multiple comparison procedure with modeling	Extracts maximum information about dose-response curve	Requires sophisticated statistical expertise
Bayesian Adaptive	Uses Bayesian updating to allocate to optimal doses	Efficient; incorporates prior knowledge	Complex implementation; regulatory acceptance varies

Whatever the approach, a few principles hold (U.S. Food and Drug Administration 1994). The doses studied should bracket the expected optimal dose, above the target as well as below. Placebo should be included wherever it is ethical, both to quantify the drug's effect over background and to account for the often-substantial placebo response in many indications; without a placebo group, a positive dose-response slope still shows that the drug has an effect, but not how large that effect is. And because exposure-response relationships are often more interpretable than simple dose-response relationships, the analysis should incorporate pharmacokinetic data.

8.1.1. The Emax Model

The most widely used mathematical description of dose-response is the *Emax model*, borrowed directly from pharmacology (MacDougall 2006):

$$E(d) = E_0 + \frac{E_{\max} \cdot d^{\gamma}}{ED_{50}^{\gamma} + d^{\gamma}}$$

where E_0 is the baseline (placebo-level) response, $E_{\max}$ is the maximum drug effect above baseline, ED_{50} is the dose producing half the maximum effect, and γ (the Hill coefficient) controls the steepness of the curve. When $\gamma = 1$ the model reduces to the simple three-parameter Emax model (E_0, $E_{\max}$, ED_{50}); letting γ be a free fourth parameter gives the sigmoid (S-shaped) Emax, or Hill, model, with $\gamma > 1$ producing progressively steeper curves.

For practitioners, the model's value is that it compresses that whole shape into a few interpretable parameters and names the region where Phase III dosing belongs: the plateau, high enough for reliable efficacy but not so high that the incremental toxicity of further dose increases outweighs any additional benefit.

When preclinical or Phase IIa data suggest a plausible range for ED_{50} and $E_{\max}$, those values can serve as Bayesian priors in Phase IIb dose-finding, so the dose-response estimate updates as the trial accumulates data.

8.1.2. MCP-Mod: Combining Significance Testing with Dose-Response Modeling

A purely comparative analysis of dose-response data (testing each dose against placebo with separate t-tests) is statistically inefficient: it cannot interpolate between doses to find the optimal one, and applying multiple comparisons corrections loses power. *MCP-Mod* (Multiple Comparison Procedures combined with Modeling) was developed as a principled solution (Bretz, Pinheiro, and Branson 2005), and it runs in two sequential steps:

1. **Multiple comparison step.** The analyst pre-specifies a family of candidate dose-response models (e.g., Emax, linear, quadratic, logistic, exponential) and, for each one, an optimally weighted contrast test comparing the dose groups against placebo, with the weights chosen to maximize power if that model is correct. The largest of these contrast statistics is compared with a multiplicity-adjusted critical value, so the proof-of-concept test controls Type I error across all candidate models.

2. **Modeling step.** If the test is significant, each candidate model is fitted to the data and the best-fitting model is selected (by the largest contrast test statistic, by an information criterion such as AIC, or by averaging across the candidate models). That model then estimates the dose-response curve and

identifies the target dose: the dose estimated to produce a pre-specified target effect (e.g., 80% of $E_{\max}$, or a clinically meaningful threshold).

For sponsors and CROs, the payoff is a dose recommendation backed by a fitted dose-response curve and a statistically defensible selection procedure, so the Phase III team carries forward a model-based rationale instead of "dose 3 looked best numerically." Both FDA and EMA have recognized MCP-Mod as an acceptable dose-response analysis method: EMA issued a formal qualification opinion in 2014 (EMA Committee for Medicinal Products for Human Use (CHMP) 2014) and FDA designated it fit-for-purpose for dose finding in May 2016 (U.S. Food and Drug Administration 2025b).

8.2. Endpoint Selection

The ideal endpoint would be the clinical outcome we ultimately care about: survival, resolution of disease, or clinically meaningful symptom improvement. But these outcomes often require large samples or long follow-up to observe.

Phase II studies therefore often use *surrogate endpoints*: measurements that are expected to predict clinical outcome but can be observed more quickly and in smaller populations (see examples in Table 8.3) (International Council for Harmonisation 1998b). The ICH E9(R1) estimands framework gives a structured way to define precisely what treatment effect is being estimated (International Council for Harmonisation 2019), including how intercurrent events such as treatment discontinuation, use of rescue medication, or switching to alternative therapy will be handled in the analysis. For example, a Phase II diabetes study using HbA1c as a surrogate must define whether the estimand reflects treatment effect while patients remain on therapy (a "while on treatment" strategy) or the effect regardless of treatment adherence (a "treatment policy" strategy).

FDA's Patient-Focused Drug Development guidance series presses sponsors to build patient perspectives into endpoint selection, particularly through clinical outcome assessments (COAs) that capture treatment benefit as the patient experiences it (U.S. Food and Drug Administration 2023i). The biomarker most convenient to measure is not always the outcome that matters to the patient.

Table 8.3.: Examples of Surrogate Endpoints by Therapeutic Area

Therapeutic Area	Surrogate Endpoint	Clinical Outcome	Validation Status
Cardiovascular	Blood pressure reduction	Myocardial infarction, stroke, cardiovascular death	Well-validated
Oncology	Tumor shrinkage (objective response rate)	Overall survival	Partially validated; varies by cancer type

Therapeutic Area	Surrogate Endpoint	Clinical Outcome	Validation Status
HIV/AIDS	Viral load (HIV RNA)	AIDS progression, death	Well-validated
Diabetes	HbA1c reduction	Diabetic complications (retinopathy, nephropathy)	Well-validated
Alzheimer's	Amyloid plaque reduction (PET imaging)	Cognitive decline	Controversial; under evaluation
Osteoporosis	Bone mineral density	Fracture risk	Validated: FDA qualified total hip BMD (24-month change) in December 2025 (U.S. Food and Drug Administration 2025d)

Surrogates buy speed at a price. Not every change in a surrogate translates to clinical benefit: Drug A might lower blood pressure more than Drug B yet produce no better cardiovascular outcomes. Using surrogates in Phase II is generally accepted, since the goal there is to detect signals and select doses, but the relationship between surrogate and clinical benefit must be established if the surrogate is to support full regulatory approval; a surrogate only *reasonably likely* to predict clinical benefit can support accelerated approval for a serious condition, with confirmation deferred to post-marketing trials.

8.3. Adaptive Designs

Beyond the traditional approach to Phase II (design a study, run it to completion, analyze the results), *adaptive designs* modify the trial as data accumulate. Sample size can be re-estimated to maintain statistical power in the face of unexpected variability, ineffective doses dropped and new ones added to better characterize the dose-response relationship, and randomization shifted toward the better-performing arms (see Section 12.8). At the far end, a seamless Phase II/III design transitions directly into the confirmatory stage on interim evidence (see Section 12.4).

FDA Support for Adaptive Designs

The FDA has issued guidance supporting the use of adaptive designs when appropriately planned (U.S. Food and Drug Administration 2019a): pre-specify the adaptation rules, control Type I error across the adaptation, and maintain the integrity of the trial despite the modifications. For Phase II, advanced Bayesian frameworks are increasingly used to handle these adaptations (including stopping for futility, dropping arms, or response-adaptive randomization), supported by rigorous simulation workflows to maintain statistical integrity (Granholm et al. 2025).

8.4. Stopping for Futility: Predictive Probability of Success

Whether to stop early when interim data are discouraging is one of the hardest calls in a running Phase II trial. Continuing a trial that is unlikely to succeed wastes resources and delays alternative programs; stopping prematurely risks abandoning a drug that would have succeeded with more time.

The classical approach to futility is *conditional power*: given the observed interim data and an assumed true treatment effect (often the interim estimate, sometimes the effect assumed at the design stage), what is the probability of achieving a significant result at the final analysis? A conditional power below 20% is a common (though arbitrary) threshold for stopping.

A more principled alternative is the *predictive probability of success (PPoS)*, described in detail at Section 12.7.1. PPoS integrates over the entire posterior distribution of the treatment effect instead of conditioning on a point estimate: it asks "given everything we've seen, what is the probability that this trial will succeed if we complete it?" That accounts for the statistical uncertainty in the interim estimate, which is often substantial in Phase II when sample sizes are small. A trial that has seen a modest positive trend with wide uncertainty has a meaningfully different PPoS than one where the interim estimate is close to zero with tight confidence.

Futility stopping in practice

A sponsor or DSMB using PPoS for futility monitoring should pre-specify both the stopping threshold (for example, PPoS < 0.05 to 0.20, depending on the indication and the sponsor's willingness to continue for potential upside) and the interim time points at which the analysis is run. Triggering a futility stop mid-trial without pre-specification creates regulatory and operational risk. Simulations across a range of plausible treatment effects should also verify the rule's operating characteristics: that it does not stop trials with genuine efficacy too frequently.

The transition from Phase II to Phase III is the *Phase II decision* (see Figure 8.1). By this point, a sponsor has typically invested tens to hundreds of millions of dollars and several years of development time (DiMasi, Grabowski, and Hansen 2016). Proceeding to Phase III will require several hundred million dollars more. And the probability of success in Phase III, even with positive Phase II data, is far from certain.

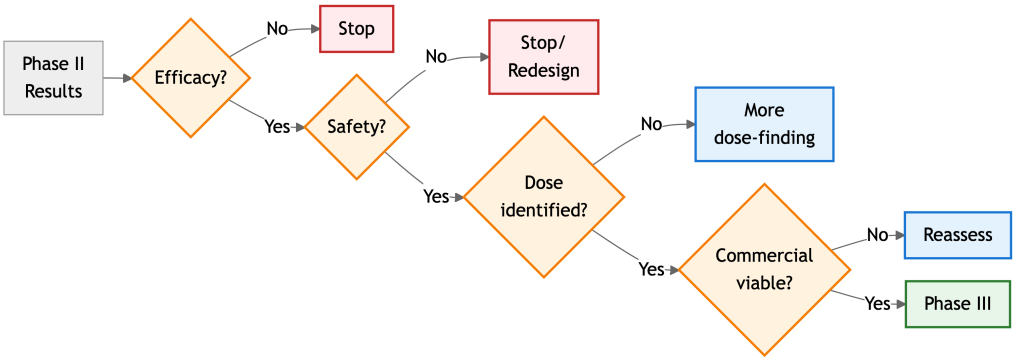


Figure 8.1.: Phase II Go/No-Go Decision Framework

That decision weighs the strength of the efficacy evidence, the acceptability of the safety profile relative to the disease burden, whether an optimal dose for Phase III has been identified, and the drug's commercial viability and competitive position.

Attrition is not uniform. In some therapeutic areas, central nervous system disorders for example, the failure rate runs above the 70% overall rate. Yet the failures serve a clear purpose. A rigorous Phase II program that kills an ineffective drug prevents a far costlier failure in Phase III by learning that the drug does not work before hundreds of millions of additional dollars are spent.

8.5. Biomarkers

Throughout Phase II, *biomarkers* provide mechanistic insight that complements clinical endpoints. The most direct question is target engagement: is the drug hitting its intended target? Others track pathway modulation, whether the biological pathway is being affected as expected, and early disease response, the first signs that a therapeutic effect is appearing.

Biomarkers earn their place when clinical endpoints take time to manifest. Cognitive decline in Alzheimer's disease occurs slowly; biomarkers of amyloid plaque or tau pathology may provide earlier signals that a drug is having biological effect, even if clinical improvement is not yet evident.

Biomarker changes do not always translate to clinical benefit. A drug might successfully engage its target without producing meaningful therapeutic improvement.

The ultimate test remains clinical outcomes, but biomarkers help illuminate the path.

9. Phase III: Confirmatory

The transition from Phase II to Phase III often commits a sponsor to hundreds of millions of dollars on top of 5-7 years of prior work, all of it staked on a therapeutic benefit that Phase II could only suggest. Phase II asks whether a drug might work. Phase III has to demonstrate that it does, with the statistical rigor and methodological quality that regulatory agencies require for approval.

These large, well-controlled *pivotal trials*, classified as *confirmatory* under ICH E8(R1), generate the primary evidence regulators require for marketing approval and establish the safety profile with sufficient precision to support labeling (International Council for Harmonisation 2021).

9.1. Characteristics of Phase III Trials

Table 9.1 sets the two phases side by side (International Council for Harmonisation 2021).

Table 9.1.: Comparison of Phase II and Phase III Trial Characteristics. Phase III trials account for the largest share of clinical trial software spending (Grand View Research 2025).

Feature	Phase II	Phase III
Sample Size	Often 20-300 patients	Hundreds to thousands
Primary Objective	Explore efficacy; find optimal dose	Confirm efficacy with statistical rigor
Design	May use adaptive or exploratory designs	Rigorous randomized controlled trials
Sites	Often single-center or limited multi-center	Multi-center, often multinational
Setting	Specialized research centers	Typical clinical practice settings
Duration	Months	Often years (long-term follow-up)
Control Group	Placebo or active comparator	Placebo (if ethical) or active comparator
Endpoints	May use surrogates	Clinically meaningful outcomes preferred

Feature	Phase II	Phase III
Regulatory Role	Generate "go/no-go" decision	Provide pivotal evidence for approval
Cost	About $7-20 million per study on average (Sertkaya et al. 2016)	About $12-53 million per study on average, and up to about $350 million for the largest pivotal trials (Sertkaya et al. 2016; Moore et al. 2018)

Size is the first difference. Enrolling hundreds to thousands of patients, depending on the indication and endpoints, provides the statistical power to detect clinically meaningful treatment effects with high confidence. Rigor is the second: randomized, controlled designs minimize bias and confounding, and placebo controls (when ethically appropriate) or active comparators allow observed effects to be attributed to the investigational drug and not to other factors.

Phase III also moves closer to routine care. Early studies occur in specialized Phase I units; Phase III takes place in hospitals, clinics, and medical offices where patients receive routine care, under conditions intended to approximate the drug's eventual clinical use. Trials typically run at multiple sites, often across multiple countries, and this multicenter approach helps ensure that the study population is representative of the intended patient population and that results are generalizable across different healthcare settings and geographical regions.

9.2. Trial Design Considerations

The choice of control group is among the most consequential design decisions. Placebo controls provide the clearest evidence of efficacy but may be unethical when effective treatments exist. Active controls demonstrate that the new drug is not worse than existing therapy by more than a prespecified margin (non-inferiority) or that it is better (superiority), though they require larger sample sizes.

The primary endpoint should be clinically meaningful and clearly defined in the protocol, and regulatory agencies increasingly expect trials to demonstrate effects on outcomes that matter to patients: survival, symptom improvement, functional status. When those outcomes require long follow-up, surrogate endpoints that are *reasonably likely to predict clinical benefit* may be acceptable, particularly for serious conditions with unmet medical need.

The ICH E9(R1) estimands framework defines the primary endpoint precisely enough to align the clinical question of interest with the statistical analysis that answers it (International Council for Harmonisation 2019). An estimand specifies five attributes: (1) the treatment being evaluated, (2) the target population, (3) the variable (endpoint) to be measured, (4) how intercurrent events (such as

treatment discontinuation, rescue medication use, or death) will be handled, and (5) the population-level summary measure. For example, a Phase III cardiovascular trial might define its estimand as the effect of Drug X versus placebo on time to cardiovascular death in patients with heart failure, summarized as a hazard ratio, using a treatment policy strategy that counts events regardless of treatment adherence and stating how deaths from other causes are handled.

The statistical analysis plan must be fully specified before unblinding occurs: the primary analysis population, handling of missing data, multiplicity adjustments for multiple endpoints or comparisons, and the specific statistical tests.

Increasingly, sponsors do not stop at the Phase II and Phase III boundary at all. A *seamless design* runs both phases as a single trial: it may begin in Phase II exploratory mode (selecting doses, confirming the target population, generating hypotheses) and then transition without a stop into a confirmatory Phase III stage, with the Phase II data contributing to the final analysis under pre-specified conditions. Designed carefully, it reduces calendar time and can preserve patient data across both stages. The design and operational requirements are covered at Section 12.4.

The *master protocol* goes further still: one overarching design evaluating multiple therapies, disease subtypes, or both under shared infrastructure. Platform trials such as RECOVERY, REMAP-CAP, and I-SPY 2 add and drop treatment arms against a common control as evidence accumulates, and several have produced confirmatory, practice-changing results. Master protocols and platform designs are developed in detail at Section 13.6.

9.3. Interim Analyses and Group-Sequential Designs

Many Phase III trials include one or more pre-planned interim analyses, reviewed by the DSMB. The point is to catch harm early enough to stop the trial or modify it. Two other findings justify an interim look: efficacy convincing enough that continued randomization to the control group becomes unethical, and futility, where accumulating data make a positive outcome at final analysis implausible.

Each interim look gives the trial an additional chance to cross the significance threshold by chance. A trial with one interim analysis and a final analysis, both tested at the conventional $\alpha = 0.05$, has an overall Type I error rate of roughly 8%. *Group-sequential designs* solve this by pre-allocating the significance budget across analysis time points, so the cumulative probability of a false positive never exceeds the target α.

The O'Brien-Fleming boundaries allocate very little alpha to early analyses, requiring near-overwhelming evidence to stop, and more to later ones (O'Brien and Fleming 1979); the conservatism at early time points reflects the greater uncertainty when fewer patients have been followed. Pocock boundaries apply the same nominal threshold at every look, allowing earlier stopping but requiring a more stringent

final threshold (Pocock 1977). The Lan-DeMets generalization allows these boundaries to be expressed as a continuous function of the information fraction, so that unplanned or irregularly spaced interims can still be accommodated without inflating error rates, provided their timing is not chosen in response to the accumulating treatment comparison (Lan and DeMets 1983; U.S. Food and Drug Administration 2019a). The full technical treatment is at Section 11.5.

> **i** What group-sequential analysis does not do
>
> Group-sequential boundaries control Type I error; they do not guarantee that an early stop reflects the true treatment effect. When a trial stops early for efficacy, the observed effect size is typically inflated relative to the true effect because extreme results are more likely to cross a stopping threshold when sample size is small. This *overestimation bias* is relevant to how results from stopped trials should be interpreted and communicated.

Worked example: planning a group-sequential trial

Suppose the primary endpoint is continuous, the assumed treatment effect is 3 units with a standard deviation of 10, the endpoint is measured at 24 weeks, enrollment runs at 5 patients a week, and the trial needs 90% power at one-sided $\alpha = 0.025$, with interims when 40%, 60% and 80% of patients have completed the endpoint. A fixed-sample design needs about 467 patients. With O'Brien-Fleming-type Lan-DeMets spending, the one-sided p-value thresholds at the four looks are about 0.0004, 0.0037, 0.0110 and 0.0211, the cumulative chance of having stopped for efficacy under the assumed effect is about 10%, 45%, 75% and 90%, and the maximum sample size rises only about 2%, to roughly 478. Pocock-type spending sets a threshold of about 0.013 at the first look and about 0.009 at each later one, gives about a 50% chance of stopping at the first look, and raises the maximum sample size by about 18%. (The figures were computed for this book from the standard spending functions; the R packages gsDesign and rpact reproduce them.)

The interims are triggered by *completers*, not by enrolled patients. While enrollment continues, about 120 enrolled patients (24 weeks at 5 a week) are still in follow-up, so the first look (about 191 completers) comes when roughly 311 patients are enrolled, and by the third look (about 382 completers) enrollment is over. A stop at that look saves follow-up but not recruitment, and an expected sample size computed from completers overstates the saving in enrolled patients. In a time-to-event trial the information is the number of *events*: with a hazard ratio of 0.5, 90% power and one-sided $\alpha = 0.025$, a fixed design needs about 88 events and the O'Brien-Fleming-type design about 90, and the interims are scheduled by event count.

Futility stopping asks the opposite question. Efficacy boundaries ask whether the evidence is strong enough to conclude benefit; futility rules ask whether it is weak enough that continuing is not worth the cost. Conditional power and predictive probability of success are the two common metrics, described at Section 12.7.1.

Futility rules are often specified as nonbinding guidance rather than strict triggers, since the costs of a false futility stop (abandoning an effective drug) are high and the statistical uncertainty at interim points can be substantial.

9.4. The Two-Pivotal-Study Paradigm and the Single-Trial Standard

Regulatory agencies have traditionally interpreted *substantial evidence* of effectiveness, as required under the Federal Food, Drug, and Cosmetic Act, to mean at least two adequate and well-controlled clinical trials (U.S. Food and Drug Administration 2019c). A single trial reaching statistical significance could be a false positive. Replication across independent trials, patient populations, and sets of investigators provides meaningful protection against this. Regulators evaluate each pivotal trial at a one-sided $\alpha = 0.025$ (equivalent to a two-sided 0.05), so two independent trials that both succeed have a combined false-positive probability of roughly $0.025 \times 0.025 = 0.000625$, a much higher evidentiary standard.

The FDA Modernization Act of 1997 codified a formal exception (United States Congress 1997): FDA may accept a single pivotal trial when accompanied by confirmatory evidence. The strongest single-trial cases rest on a trial that is large, multicenter, and internally consistent across pre-specified subgroups, with a hard clinical outcome as the primary endpoint (death, major cardiovascular event) rather than a surrogate, a p-value substantially below the conventional 0.05 threshold (typically $p < 0.001$ or even $p < 0.0001$ in recent large outcomes trials), and supportive evidence from mechanistic studies, surrogate endpoint data, or other development programs.

Large cardiovascular outcome trials meeting these criteria have supported single-trial submissions, and some oncology approvals in rare diseases and serious unmet-need settings have relied on single trials with strong effect sizes. For a sponsor, one very large, high-quality trial with a hard primary endpoint can therefore be more regulatory-efficient than two moderately sized studies, provided the effect size and evidence quality are sufficient.

In June 2026, FDA issued a revised draft guidance on demonstrating substantial evidence of effectiveness that moves further in this direction (U.S. Food and Drug Administration 2026c). The revision repositions a single adequate and well-controlled trial supported by confirmatory evidence from an exception toward the default expectation, as FDA's leadership has framed it (Prasad and Makary 2026), and it drops the earlier framing that treated one such trial as equivalent to two independent trials. Because it remains a draft (the public comment period closed on September 22, 2026) rather than final policy, sponsors should read it as a signal of regulatory direction rather than a settled standard when planning pivotal programs.

9.5. Safety Evaluation in Phase III

Phase III substantially expands the safety database. For drugs intended for long-term treatment of non-life-threatening conditions, the ICH E1 guideline recommends a minimum extent of population exposure to characterize the safety profile (Table 9.2) (International Council for Harmonisation 1994b):

Table 9.2.: ICH E1 safety exposure recommendations for drugs intended for long-term treatment

Exposure Metric	ICH E1 Recommendation
Total patients exposed	About 1500, including short-term exposure
Exposed for ≥ 6 months	300-600 patients
Exposed for ≥ 12 months	At least 100 patients

i Detecting Rare Adverse Events

Even with 1500 patients exposed, the safety database is not expected to characterize events occurring in fewer than 1 in 1000 patients (0.1%) (International Council for Harmonisation 1994b): at exactly that rate only 1.5 cases are expected, too few to estimate the risk with any precision, and much rarer events may not appear at all. The "rule of 3" states that if zero events are observed in N patients, the upper 95% confidence limit for the true rate is approximately 3/N. Thus, 1500 patients provides only 95% confidence that the true rate is below 0.2%.

That exposure is necessary to detect adverse events occurring infrequently or developing only after extended use. The events it still misses before approval are why post-marketing surveillance remains important.

9.6. Data Safety Monitoring Boards

Given their size and duration, many Phase III trials are overseen by a *Data Safety Monitoring Board* (DSMB), also known as a Data Monitoring Committee (DMC) or Independent Data Monitoring Committee (IDMC): a group of independent experts who periodically review unblinded data to assess safety and, in some cases, efficacy.

The DSMB may recommend that a trial continue as planned, that it be modified (by changing eligibility criteria, say, or stopping a specific treatment arm), or that it be stopped early for any of the reasons interim analyses exist to detect: harm, clear benefit, or futility.

The operational difficulty is keeping the sponsor and investigators blinded while the DSMB works from unblinded data. A formal charter describes the board's responsibilities, composition, and operating procedures.

9.7. Global Considerations

Phase III trials are increasingly global, with sites across North America, Europe, Asia, Latin America, and elsewhere. Globalization speeds enrollment, gives access to treatment-naive populations, and allows efficacy to be demonstrated across diverse populations. The complications are regulatory and clinical at once: requirements differ across regions and investigators must ensure compliance with all applicable regulations, while differences in medical practice, standard of care, and patient populations may affect results. The ICH E5 guideline addresses the acceptability of foreign clinical data and the concept of *bridging studies* to assess whether results from one region apply to another (International Council for Harmonisation 1998a).

For pivotal trials specifically, representative enrollment is now a U.S. regulatory expectation: under the Food and Drug Omnibus Reform Act of 2022 (FDORA), sponsors are directed to submit Diversity Action Plans for Phase III and other pivotal studies that begin enrolling more than 180 days after FDA publishes final guidance on the plans (U.S. Congress 2022). The Diversity Action Plans implementing guidance has been unsettled since early 2025: FDA briefly withdrew and, under court order, restored its draft, then let the statutory deadline for a final version pass, so the plan requirement has not yet taken effect (Grossi 2025). Separately, FDA's December 2025 final guidance on trial participation (a revision of its earlier eligibility-criteria guidance) keeps the representative-enrollment recommendations while dropping the *diversity* framing (U.S. Food and Drug Administration 2025c). The scientific rationale is unchanged; the operational mechanics of meeting enrollment goals are covered at Chapter 18.

9.8. Example Phase III Protocol Structure

A Phase III protocol follows the structure of earlier phases with greater detail and rigor. Background summarizes earlier clinical development; objectives state the primary and secondary endpoints; the design section describes the randomization scheme, blinding procedures, treatment duration, and visit schedule; subject selection provides the detailed inclusion and exclusion criteria designed to enroll a population representative of the intended use while ensuring patient safety; treatment procedures specify exactly how the investigational product and comparator are to be administered, stored, and accounted for; and the efficacy and safety assessment sections detail how those data are collected and analyzed.

The statistical section carries particular weight: analysis populations, primary analysis method, sample size justification, interim analyses (if planned), and methods for handling multiplicity.

A Phase III that succeeds produces the definitive evidence behind the label, and with it the information prescribers need and the therapy patients eventually receive.

10. Phase IV: Post-Marketing Studies and Surveillance

A drug is approved on evidence from trials that were never large enough or long enough to see everything. Then it meets everyone else. Patients differ from the ones who enrolled, adherence varies, and the drug is taken alongside other medications by people carrying other health conditions. Phase IV is every study run after that point, and it has two main jobs: to learn how the drug performs in the broader population it was intended for, and to keep watching for harm.

10.1. Types of Phase IV Studies

What separates one kind of Phase IV study from another is purpose and, more consequentially, whether anyone can compel the sponsor to finish it (Table 10.1).

Table 10.1.: Types of Phase IV Studies

Study Type	Purpose	Examples	Regulatory Status
Post-Marketing Safety	Detect rare or delayed adverse events	Long-term safety studies, special population studies	May be required or voluntary
Post-Marketing Efficacy	Evaluate real-world effectiveness	Subpopulation analyses, comparative effectiveness	Usually voluntary
Post-Marketing Requirements (PMR)	Satisfy mandated post-approval studies	Confirmatory trials for accelerated approval	Required by statute or regulation
Label Extension	Expand approved uses	Pediatric studies, new indication trials	Required for new indication claims
Outcomes Research	Demonstrate value in real-world practice	Quality of life studies, health economics	Usually voluntary; payer-driven

Post-marketing safety studies look for adverse events the premarketing program did not identify, whether because the events are rare, because they emerge only after

prolonged exposure, or because they fall preferentially on populations the original trials under-enrolled. *Post-marketing efficacy studies* take up what that program had no time or breadth to answer. Does the drug move long-term outcomes? How does it compare with other treatments under real-world conditions? How does it hold up in specific subpopulations?

Post-marketing requirement (PMR) studies are the ones a sponsor is required by statute or regulation to conduct, among them studies of safety in populations of concern and of outcomes not fully evaluated before approval. The heaviest are the trials that confirm efficacy where approval rested on a surrogate endpoint, as in accelerated approval. A *post-marketing commitment (PMC)* is the softer instrument: a study the sponsor agrees to conduct but is not legally required to complete.

The Food and Drug Omnibus Reform Act (FDORA) of 2022 strengthened the FDA's hand over those confirmatory trials: the agency may now require that one be underway before it grants accelerated approval or within a period it specifies after approval, and it can withdraw approval through an expedited procedure when the trial fails to verify clinical benefit or is not conducted with due diligence (U.S. Congress 2022). Melphalan flufenamide (Pepaxto) lost its approval on that logic, effective February 2024, after its confirmatory trial showed worse overall survival.

Sponsors also run studies to support label extensions, seeking approval for new indications, new dosing regimens, or new populations. The historical "pediatric labeling gap" has made that route particularly critical for children. The FDA has tracked pediatric labeling changes made under the Best Pharmaceuticals for Children Act (BPCA), the Pediatric Research Equity Act (PREA), and the Pediatric Rule (U.S. Food and Drug Administration 2025e). Those initiatives have significantly improved pediatric labeling: more than 1,400 changes approved since 1998, carrying evidence-based dosing and safety information for pediatric populations. They have not closed the gap. The "therapeutic orphan" problem (drugs used in children without adequate pediatric studies) remains a primary driver for dedicated Phase IV pediatric programs.

10.2. Pharmacovigilance

Pharmacovigilance is the science of detecting, assessing, and preventing adverse effects of medicines once they are in widespread use. Its first line of detection is *spontaneous reporting*: clinicians and patients voluntarily report suspected adverse reactions to systems such as the FDA Adverse Event Monitoring System (AEMS, launched in March 2026 to replace the FDA Adverse Event Reporting System, FAERS) or, through national competent authorities, the European Union's EudraVigilance, while manufacturers are legally required to forward the reports they receive (in the United States within 15 calendar days for serious, unexpected cases under 21 CFR 314.80; in the EU within 15 days for all serious cases, wherever they occur, and 90 days for non-serious cases occurring in the EU, under GVP Module VI). Spontaneous reports are invaluable for flagging rare or unexpected events that

premarketing trials were too small or too short to detect. The severe liver injury that led to the 2000 US withdrawal of troglitazone (Rezulin) was flagged largely through spontaneous case reports.

Passive reporting does not always get there first. The cardiovascular risk that led to the 2004 withdrawal of rofecoxib (Vioxx) was established instead by randomized trials (VIGOR in 2000 and the halted APPROVe trial in 2004), a case often cited for how slowly passive surveillance can move on its own. Spontaneous reporting's limitations are well known: underreporting, uncertain causality, no denominator data. A signal raised that way therefore often has to be confirmed elsewhere, in epidemiologic studies using healthcare databases, in registries, or in dedicated Phase IV trials.

Active surveillance increasingly complements passive reporting: regulators run standardized queries against large networks of electronic health records and claims data on a routine schedule. The FDA's Sentinel System and the European Medicines Agency's DARWIN EU (Data Analysis and Real World Interrogation Network) apply the approach at scale; Chapter 23 develops the methods of signal detection.

i The Scale of Safety Surveillance

FAERS alone, the drug and biologic database that AEMS is replacing, received more than one million adverse-event reports a year and had accumulated tens of millions over decades (U.S. Food and Drug Administration 2024g, 2026d). Filtering genuine signals out of that volume of noise is the central operational challenge of pharmacovigilance.

10.3. Risk Evaluation and Mitigation Strategies

When significant safety concerns are identified, the FDA may require a *Risk Evaluation and Mitigation Strategy* (REMS) as a condition of approval or continued marketing (U.S. Food and Drug Administration 2024l). A REMS manages known or potential serious risks so that the patients who need the drug can keep getting it.

Table 10.2.: Components of Risk Evaluation and Mitigation Strategies

REMS Component	Description	Example Application
Medication Guide	FDA-approved written information for patients about drug risks/benefits	Required with each dispensing; e.g., NSAIDs cardiovascular risk
Communication Plan	Ensure healthcare providers receive important safety information	Dear Healthcare Provider letters, professional society communications

REMS Component	Description	Example Application
Elements to Assure Safe Use (ETASU)	Specific requirements for prescribing, dispensing, or receiving	Prescriber certification, patient enrollment, restricted distribution
Implementation System	Infrastructure to ensure REMS is executed	Centralized REMS programs with data collection
REMS Assessment	Periodic evaluation of REMS effectiveness	By 18 months, 3 years, and in the 7th year after approval, unless FDA changes the timetable

The elements in Table 10.2 run from a leaflet to a lock on the supply chain. Every REMS includes a timetable for assessments of the strategy (21 U.S.C. § 355-1(d)); the FDA requires the other elements when it determines that they are needed to mitigate a serious risk. At the light end, a *Medication Guide* goes to the patient with every dispensing and a *Communication Plan* pushes safety information to healthcare providers. At the heavy end sit *Elements to Assure Safe Use* (ETASU), the specific actions required for prescribing, dispensing, or receiving the drug, such as prescriber training and certification, patient enrollment in a registry, or dispensing only through certified pharmacies.

10.4. Comparative Effectiveness Research and Real-World Data Sources

Phase IV increasingly turns on *comparative effectiveness research* (CER), which asks what a placebo-controlled efficacy trial run under idealized conditions cannot: among the available treatments for a condition, which does better, and for whom? CER compares the benefits and harms of the different interventions under conditions that approximate routine practice.

CER takes two main forms. Pragmatic clinical trials run in real-world settings with broad inclusion criteria and clinically relevant endpoints. Observational studies compare outcomes among patients receiving different treatments in routine clinical care, drawing on electronic health records, claims databases, or registries.

Patient registries are organized systems that collect uniform observational data on a population defined by a particular disease, condition, or exposure, and they take their names from that focus. Disease registries follow patients with a particular condition to characterize natural history and identify prognostic factors. Product registries, often mandated for drugs with significant safety concerns or accelerated approvals, collect long-term safety and effectiveness data for specific medications. Pregnancy registries are the critical case: for assessing developmental risks they are the primary source of safety information on a population typically excluded from premarketing trials.

10.5. Real-World Evidence

The 21st Century Cures Act of 2016 directed the FDA to evaluate how *real-world evidence* (RWE) could support regulatory decisions, and the agency's subsequent framework defined the terms (U.S. Food and Drug Administration 2018a). RWE is clinical evidence about a drug's usage and its potential benefits or risks, derived from analysis of *real-world data* (RWD): data relating to patient health status or the delivery of health care, routinely collected from electronic health records, claims and billing databases, product and disease registries, and other sources.

The framework and the guidance that followed it describe how RWE may support approval of new indications for already-approved drugs and satisfy post-marketing study requirements (U.S. Food and Drug Administration 2023f). Whether a given body of RWE carries that weight turns on the quality and relevance of the underlying data and on the rigor of the study design and analysis. Chapter 14 develops external control arms and the other real-world evidence methods.

10.6. Health Technology Assessment (HTA) and Payer Value

Regulatory approval answers whether a drug is safe and effective. It does not answer whether anyone should pay for it. That second question belongs to Health Technology Assessment (HTA) bodies and payers, and in many markets approval is only the first hurdle: market access depends on demonstrating value.

Cost-effectiveness analysis asks whether the health benefit, often measured in Quality-Adjusted Life Years, justifies the price. The National Institute for Health and Care Excellence (NICE) in the UK and the Institute for Clinical and Economic Review (ICER) in the US use economic models to answer it. Germany's IQWiG asks something harder: whether the new drug offers a proven advantage over the current standard of care, a higher bar than the placebo-controlled evidence often accepted for regulatory approval.

Phase IV is therefore often where a sponsor generates what payers want to see: resource utilization, comparative effectiveness against active controls, and long-term economic outcomes (National Institute for Health and Care Excellence 2022).

10.7. Lifecycle Management

From the sponsor's side, all of this falls under *lifecycle management*: studies that differentiate the product from competitors, optimize dosing, or expand it into new indications and populations.

Those objectives are commercial, and they have to be held against scientific integrity. The post-marketing period is a continuation of the learning process, not an epilogue to it, and its regulatory stakes have risen. Under FDORA an accelerated approval now stands or falls on whether its confirmatory trial verifies the benefit

that premarketing data only suggested. It is that binding requirement, more than any voluntary commitment, that keeps Phase IV tied to the standard of evidence the earlier phases set out to meet.

Part III.

Part III: Phase III Deep Dive

11. Statistical Methods for Clinical Trials

D.R. Cox, in his foundational *Planning of Experiments* (Cox 1958), describes clinical trials as operating in the domain of "large uncontrolled variations." In physics, once the apparatus is working correctly, closely reproducible results emerge: uncontrolled variation is small relative to the effect being measured. Clinical trials face a different reality. Any given patient's outcome swings with the natural history of disease, environmental factors, the psychological response to treatment, and the supportive care received. The effect of a new drug, often only a modest improvement over existing therapy, has to be distinguished from that underlying variation.

This chapter covers the frequentist core of the toolkit built for that job: hypothesis testing and error control, survival analysis, repeated-measures models, missing data, and group-sequential designs. Two modern extensions whose methods cut across inferential frameworks follow: dynamic treatment regimes (SMART designs) and causal inference for observational and hybrid evidence. Bayesian and adaptive methods are in Chapter 12.

11.1. Hypothesis Testing and Error Control

A five-year survival rate of 40% tells us little on its own; everything depends on what would have happened without the treatment. Clinical trials are built for *comparison*, not absolute measurement, and the quantity they estimate and test is the *treatment effect*: the difference in outcomes between the investigational arm and the control arm.

The frequentist framework structures this comparison around two hypotheses: the *null hypothesis* (H_0), the default assumption of no treatment difference, and the *alternative hypothesis* (H_1), the claim that a difference exists. The analysis produces a *p-value*: the probability of observing results at least as extreme as those seen, assuming H_0 is true. If the p-value falls below a pre-specified threshold α, we reject H_0 and conclude the treatment has an effect.

Two kinds of decision error follow. *Type I error* (false positive) means concluding the drug works when it does not. For confirmatory trials, regulators typically expect this probability to be held at or below $\alpha = 0.025$ one-sided (or 0.05 two-sided): on average, no more than 2.5 false-positive results per 100 trials of ineffective agents.

Type II error (false negative) means failing to detect a drug that genuinely works. The probability of *avoiding* it is *power*, and trials are typically designed for 80% or 90%.

Sample size is determined by four inputs: the minimum clinically important difference (MCID, the smallest effect worth detecting), the variability of the outcome (its standard deviation for a continuous endpoint, or the control-arm event rate for a binary or time-to-event endpoint), the desired power, and the tolerated Type I error. Precision improves only with the square root of sample size: to halve the standard error (and so double the precision), sample size must quadruple. This is why large Phase III trials are expensive and why design efficiency (reducing variability through stratification, using continuous rather than binary endpoints, or fitting a covariate-adjustment model) can meaningfully reduce required enrollment.

i What a p-value does and does not say

A p-value of 0.01 does not mean there is a 99% chance the drug works. It means that if the drug truly had no effect, we would see results this extreme only 1% of the time by chance. The framework controls error rates over many hypothetical repetitions; it does not directly state the probability that *this particular drug* is effective. Surveys consistently show that even experienced researchers misinterpret p-values (Wasserstein and Lazar 2016; Goodman 2008). Bayesian methods (Chapter 12) address this by directly computing posterior probabilities.

11.1.1. The ITT Principle and the Estimands Framework

For decades, the *Intent-to-Treat* (ITT) principle governed trial analysis: analyze every patient according to their randomization assignment, regardless of whether they actually received the treatment or completed the study. A patient randomized to the drug arm who never took a single dose is still counted in the drug arm. Because non-adherence usually pulls the ITT difference toward zero, ITT tends to be conservative in superiority trials but is generally not conservative in non-inferiority or equivalence trials, where arms that look more alike make those conclusions easier to reach (International Council for Harmonisation 1998b). Either way, it preserves the benefits of randomization and blocks manipulation of results by selective exclusion.

ITT does not always answer the clinical question of greatest interest. The ICH E9(R1) addendum introduced the *estimands framework* (International Council for Harmonisation 2019), which requires sponsors to explicitly specify five elements of the treatment comparison: the population, the treatment and control, the endpoint variable, how *intercurrent events* will be handled, and the population-level summary measure. Intercurrent events are the things that happen after randomization and affect interpretation: treatment discontinuation, rescue medication use, crossover.

They cannot be handled silently, because different strategies for handling them answer different clinical questions:

- **Treatment policy strategy** (closest to ITT): measure the outcome regardless of intercurrent events. This estimates the effect of the treatment *policy* ("start on Drug X"), not necessarily of the drug itself.
- **Hypothetical strategy**: estimate what would have happened had the intercurrent event not occurred, which requires modeling assumptions about unobserved outcomes.
- **Principal stratum strategy**: estimate the effect within the subpopulation defined by how patients would respond to the intercurrent event (those who would remain adherent in either arm, for example). The stratum is a well-defined causal subgroup, but membership is not directly observed, so estimation usually requires strong assumptions; some methods infer membership from baseline characteristics.
- **Composite variable strategy**: fold the intercurrent event into the endpoint, so that experiencing it (treatment discontinuation or death, say) counts as a specified, usually unfavorable, outcome. Common when the event is itself a marker of treatment failure.
- **While-on-treatment strategy**: use the outcome only up to the intercurrent event (for death, E9(R1) calls this *while alive*). When the event is stopping treatment, this answers "what is the effect while the patient is actually taking the drug?"

The framework forces that choice before data collection: you cannot retrospectively pick the strategy that makes results look better. It also ties the analysis to the regulatory claim being sought, since different strategies support different label language. Chapter 17 covers the interaction between estimands and endpoint choice.

11.2. Survival Analysis

Time-to-event endpoints are the workhorses of Phase III oncology, cardiovascular, and infectious disease trials. Survival analysis models the time until an event of interest: death, disease progression, hospitalization, or relapse. A binary endpoint records only whether that event had occurred by a fixed date, throwing the timing away.

Censoring is the key technical challenge. A patient still alive at the end of the trial, or lost to follow-up, has not experienced the event, but we know they survived *at least* until their last observation. That partial information has to be used, not discarded.

11.2.1. Kaplan-Meier Estimator

The Kaplan-Meier (KM) estimator (Kaplan and Meier 1958) describes an arm's survival experience by estimating the probability of surviving beyond each observed

event time, accounting for censoring. Its curves are the visual signature of most Phase III oncology and cardiovascular publications.

A KM curve drops only at times when events actually occur, so a censored observation produces a tick mark but no step. Its reliability deteriorates toward the right, where fewer patients remain at risk and the confidence band widens: when only a few patients have been followed for five years, a 5-year survival estimate is much less precise than a 1-year estimate. And comparing two curves by eye can mislead when they cross, because a single summary statistic (the hazard ratio) may hide clinically important early benefit and late harm, or the reverse.

Log-rank test. The log-rank test is the standard significance test for comparing two survival curves: did the two groups experience events at the same rate over time? It is distribution-free (no parametric assumptions about survival times) and maximally sensitive when the proportional hazards assumption holds.

When curves cross or diverge late, the log-rank test can be insensitive. *Weighted log-rank tests* (Fleming-Harrington family) place more or less weight on early versus late events, targeting the test toward the timing of expected benefit. For immunotherapy trials, where effects often emerge slowly, up-weighting late events (equivalently, down-weighting early ones) can improve power.

11.2.2. Cox Proportional Hazards Model

The Cox model (Cox 1972), the most widely used regression model for survival data, estimates the effect of treatment and covariates on the *hazard*: the instantaneous rate of experiencing the event at each point in time, among those who have survived to that point. It produces a *hazard ratio* (HR), the ratio of the hazard in the treatment arm to the hazard in the control arm: an HR of 0.75 means patients on treatment experience the event at 75% of the rate of controls at any given moment.

The *proportional hazards (PH) assumption* states that the HR is constant over time, and it should be checked (Schoenfeld residuals, scaled Schoenfeld plots). When it is violated, the HR is still estimable but represents an average over time rather than a stable effect size. Alternatives for non-proportional hazards include restricted mean survival time (RMST), which compares average time alive up to a horizon t^* without requiring the PH assumption.

Covariate adjustment via the Cox model is standard: pre-specified prognostic variables (age, disease stage, performance status, prior therapy) sharpen the treatment comparison at fixed sample size. The route differs from the linear-model case. Because the hazard ratio is non-collapsible, adjustment shifts the estimand from a marginal to a conditional hazard ratio and can widen the standard error of the log hazard ratio even as it increases power.

Competing risks. A patient in an oncology trial who dies in a car accident is censored in the standard disease-specific survival analysis, and can never experience the event of interest afterward. That censoring is non-informative only if death

from other causes is independent of disease progression, an assumption that may not hold in older or sicker populations, and even when it holds, one minus the Kaplan-Meier curve describes a hypothetical world without competing deaths and overstates the proportion of patients who actually experience the event. *Competing risks* analysis (Fine-Gray subdistribution hazards (Fine and Gray 1999)) models the *cumulative incidence* of each event type explicitly, and for trials with substantial competing events (hematologic cancers, COVID-19, elderly populations) it should be pre-specified as at least a sensitivity analysis.

11.3. Repeated Measures and Longitudinal Data

Trials often measure outcomes repeatedly: pain scores every two weeks, cognitive assessments every three months, quality-of-life questionnaires at each visit. The primary question may be the effect of treatment at a specific timepoint (e.g., change from baseline at week 52), but the repeated measurements contain information that a single-timepoint analysis discards.

ANCOVA for single-timepoint outcomes. Analysis of covariance (ANCOVA) is the standard model when the primary endpoint is a continuous outcome measured at one post-baseline timepoint, with baseline value as a covariate. Its residual variance is always at most as large as the variance of the raw change score, and often substantially smaller. Both the EMA and the FDA endorse adjusting for pre-specified prognostic baseline covariates, including the baseline value of the outcome, because adjustment generally improves efficiency and guards against chance baseline imbalance; FDA's 2023 guidance presents adjustment as an accepted option rather than a requirement, noting that an unadjusted analysis is also acceptable for the primary analysis (U.S. Food and Drug Administration 2023j).

11.3.1. Mixed Model for Repeated Measures (MMRM)

The *Mixed Model for Repeated Measures* (MMRM) has become the de facto standard for longitudinal continuous endpoints in confirmatory trials, particularly in neurology, psychiatry, and metabolic disease. It models all post-baseline measurements simultaneously, with treatment, visit, and treatment-by-visit interaction as fixed effects and an unstructured correlation matrix across visits.

MMRM's advantage over the older *Last Observation Carried Forward* (LOCF) method is in missing data handling. LOCF imputes missing later observations with the last observed value, so a patient who dropped out because they were doing poorly is carried as though they continued at that level indefinitely. MMRM instead uses a likelihood-based approach that is valid under the *Missing at Random* (MAR) assumption: it uses all available data from each patient without imputation, as long as missingness depends only on observed (not unobserved) outcomes. FDA and EMA have increasingly pushed sponsors toward MMRM and away from LOCF for this reason.

The unstructured covariance matrix makes no assumption about the shape of within-patient correlation over time, which suits irregularly spaced visits or an unknown correlation pattern. For trials with many timepoints or complex longitudinal patterns, random-effects models (with random slopes and intercepts) are an alternative that borrows strength across patients.

11.4. Missing Data

Missing data is a design problem that belongs in the protocol and the statistical analysis plan, before enrollment begins.

11.4.1. The MCAR/MAR/MNAR Taxonomy

The inferential consequences of missing data depend on *why* data are missing:

- **Missing Completely at Random (MCAR):** missingness is unrelated to any observed or unobserved data. Rare in practice; a lab machine failure that randomly affects 2% of samples might qualify.
- **Missing at Random (MAR):** missingness depends on observed data but not on the unobserved outcome itself. A patient with worse observed prior scores is more likely to drop out, but conditional on those scores, dropping out does not depend on what the unmeasured future score would have been. MAR is the working assumption for MMRM and multiple imputation.
- **Missing Not at Random (MNAR):** missingness depends on the unobserved outcome itself. A patient stops reporting because they are feeling worse than what their last observed score reflects. This is the problematic case: no analysis fully corrects for MNAR without additional assumptions.

11.4.2. Multiple Imputation and Sensitivity Analyses

Multiple imputation. Multiple imputation (MI) creates several (typically 20-100) complete datasets by drawing from the predictive distribution of missing values given observed data, analyzes each complete dataset separately, and combines results using Rubin's rules (Rubin 1987). Under MAR, MI produces valid estimates. Under MNAR, MI under MAR can be biased in either direction (it overstates the benefit if treated patients who drop out fare worse than similar treated patients who stay, and understates it if placebo patients who drop out do), which is why MNAR sensitivity analyses are required.

Sensitivity analyses. Regulators expect primary analyses that assume MAR to be accompanied by sensitivity analyses that explore MNAR scenarios. A *tipping point analysis* asks how extreme the unobserved outcomes would have to be before the treatment effect stops being significant; if the answer is unrealistically bad outcomes for dropouts in the treatment arm, the MAR result is robust. *Pattern mixture models* specify different outcome distributions for different dropout patterns

(dropouts for adverse events versus dropouts for lack of efficacy, say) and check sensitivity to plausible MNAR assumptions. *Reference-based imputation* imputes missing treatment-arm outcomes using the control arm distribution (the conservative "jump-to-reference" assumption for withdrawals), directly encoding the estimand strategy for treatment discontinuation.

11.5. Group-Sequential Designs and Alpha Allocation

Take a fixed 400-patient trial and add planned interim looks at 200 and 300 patients before the final analysis. You now have three opportunities to declare superiority (or to stop for futility) instead of one, and three opportunities for a false positive: applying the same α threshold (0.025) at each of the three looks raises the experiment-wide false-positive rate above 2.5%. Group-sequential testing (Jennison and Turnbull 2000), the oldest and most widely adopted form of interim flexibility in confirmatory trials, answers this by *allocating* alpha across the looks so the total remains 2.5%. Two boundary families dominate practice:

O'Brien-Fleming boundaries (O'Brien and Fleming 1979) spend very little alpha early, requiring an extremely stringent threshold at interim looks (roughly $p < 0.003$ at the halfway point) while preserving a final threshold close to the unadjusted 0.025. The intuition: a finding that clears a very high early bar is convincing; a finding that merely clears the standard bar at the end requires no sacrifice. That familiar final threshold is why they are common in Phase III trials.

Pocock boundaries (Pocock 1977) distribute alpha more evenly across looks, making early stopping considerably easier but depressing the final threshold. The same nominal threshold applies at every look: crossing it at an interim analysis justifies stopping early for efficacy, and a trial that runs to the end must clear the same stricter threshold, so a final result that would have sufficed in a fixed-sample trial can fall short.

The **Lan-DeMets alpha-spending function** (Lan and DeMets 1983) generalizes both into a single flexible framework: alpha is allocated as a smooth function of the information fraction (the proportion of patients enrolled or events observed), with neither the number nor the timing of interim analyses fixed in advance, provided the timing is not chosen in response to the comparative results. That is why it is now the workhorse of Phase III interim planning. Real interim dates depend on enrollment rate and event accumulation, not on a calendar set at protocol writing.

> **Alpha is distributed, not consumed**
>
> Interim looks are widely believed to "cost alpha" or impose a "penalty." They do not. Alpha is *distributed* across analyses, not spent irrecoverably: you still have 2.5% experiment-wide Type I error, allocated to different timepoints. A final nominal threshold of 0.021 instead of 0.025 records where the rest of it went.
> What requires adjustment is the *action* taken in response to the interim data. Stopping early for superiority spends alpha for that possibility. Stopping early for futility does not: it *deflates* rather than inflates Type I error. A DSMB monitoring a trial for safety without unblinding the sponsor does not touch alpha at all.

Futility rules are pre-specified stopping criteria that end a trial when interim evidence is sufficiently discouraging. Well-designed rules are based on predictive probability (see Section 12.7.1): given the data so far, what is the probability that continuing to the planned sample size would succeed? If that probability falls below a threshold (typically 10-20%), continuing may be difficult to justify. The threshold cuts both ways: an aggressive one may prematurely terminate a trial that would ultimately have succeeded; a conservative one may waste resources. Calibration comes from simulation across plausible effect-size scenarios.

A recurring design trap: a trial reaches an interim with borderline results, and the sponsor considers extending enrollment to "wash out" an unfavorable trend. Without pre-specification, that inflates Type I error. The *promising zone* design (Mehta and Pocock 2011) is a pre-specified version of that extension: it allows sample-size expansion when interim conditional power falls within a defined range, with no additional alpha adjustment. Comparative simulations (Jennison and Turnbull 2015) suggest that a well-built group-sequential design with futility rules can match or exceed the standard promising-zone design's efficiency (higher power at lower average sample size) under many assumptions, though an optimally tuned promising-zone design can narrow the gap. Either way, the flexibility has to be in the design before the interim arrives.

Combination tests and conditional error. The boundaries above assume that the sample size and the timing of the looks are fixed in advance. When the second-stage sample size is chosen from the interim treatment effect, as in the extension just described, the test itself has to keep that choice from inflating the Type I error. A *combination test* computes a p-value from each stage and combines them with a rule fixed before the trial (P. Bauer and Kohne 1994). Cui, Hung and Wang apply the idea to a group-sequential trial by keeping the originally planned stage weights after a sample-size increase; the cost is that patients enrolled after the increase count for less than those enrolled before it (Cui, Hung, and Wang 1999). The *conditional error* principle allows any redesign of the rest of the trial whose Type I error, conditional on the interim data, is no larger than the original

design's (Proschan and Hunsberger 1995; Müller and Schäfer 2001). Both methods are described with the other adaptive designs in Section 12.1.1.

Section 9.3 introduces these concepts in the setting of Phase III planning, and Chapter 12 covers their Bayesian analogs.

11.6. SMART Designs and Dynamic Treatment Regimes

Depression, substance use disorders, cancer, HIV, and chronic pain are not cured by a single fixed treatment. Each requires a *sequence*: what to start with, whether to switch if the initial treatment fails, and what to switch to. A standard RCT comparing Treatment A versus Treatment B answers which starting treatment is better on average. It does not tell you what to do next for the patients who do not respond.

A *Dynamic Treatment Regime* (DTR) is a sequence of decision rules, each mapping a patient's current status (clinical state, response to prior treatment, biomarkers) to a treatment action. DTRs are the formal statistical representation of how clinicians actually practice: "start with drug A; if patient has not responded at week 8, augment with drug B or switch to drug C."

11.6.1. The SMART Design

A *Sequential Multiple Assignment Randomized Trial* (SMART) produces the data for comparing embedded DTRs, with clean randomization at each decision stage (Murphy 2005; Lavori and Dawson 2000). In a common two-stage SMART, patients are first randomized to one of two (or more) initial treatments. After an assessment of response (typically at a clinically meaningful timepoint), patients who responded continue; non-responders are *re-randomized* to one of two (or more) salvage or augmentation options. The result is a tree of sequences: A then augmentation X for non-responders, A then a switch to Y, B then X, and so forth. Each branch defines an *embedded DTR*.

The advantage over a factorial or crossover design is clinical realism: stage-two randomization depends on response status, so each patient is randomized only among options that make clinical sense for them. Non-responders are re-randomized among salvage or augmentation options; responders continue their initial treatment or, when the best maintenance approach is uncertain, are re-randomized among maintenance options, as in the ExTENd trial described below.

Comparing embedded DTRs requires weighted analysis: responders are consistent with more than one embedded regime, because in the design above they are not re-randomized, while each re-randomized non-responder is consistent with only one, so patients follow a given regime with unequal probability (one half for responders, one quarter for non-responders in a prototypical two-stage SMART). Inverse probability weighting (IPW) or doubly-robust estimators correctly account for this. The

estimand is the *mean outcome under the DTR if applied to all patients in the target population*, a well-defined causal quantity.

Sample size for a SMART depends on which comparison is primary, as well as on the effect size and the correlation between stages. A comparison of the initial treatments uses every patient and needs the same enrollment as a standard two-arm RCT, while a comparison of stage-two options uses only the non-responders (in the design above, the only patients re-randomized), so the two-arm enrollment must be divided by the non-response rate (Almirall et al. 2014). A comparison of embedded DTRs needs more patients than a two-arm trial with the same effect size, because each re-randomized non-responder follows only one of the regimes that start with their initial treatment. Murphy (2005) provides the foundational sample size framework for SMART designs (Murphy 2005).

11.6.2. Key Applications

In the ADHD SMART of Pelham et al. (2016), children were randomized to begin with low-dose medication or behavioral intervention, and non-responders were re-randomized to intensify that modality or add the other (Pelham et al. 2016). The trial found that starting with behavioral intervention and augmenting with medication for non-responders produced better outcomes than starting with medication, and did so with less overall medication exposure: a clinically actionable DTR comparison.

The Sequenced Treatment Alternatives to Relieve Depression trial (STAR*D; Rush et al., 2004) was a multi-stage "real-world" depression study in which patients who did not achieve remission at each stage were offered progressively different treatment options (Rush et al. 2004). Sequential randomization was optional at each stage, so it was not a formally pre-specified SMART, but it provided the foundational data motivating the SMART framework in psychiatry and generated the hypothesis that most patients who will ultimately achieve remission do so within two or three medication steps (Rush et al. 2006).

ExTENd (Oslin et al.), the Extending Treatment Effectiveness of Naltrexone study, is a SMART for alcohol use disorder in which all participants began naltrexone and were randomized to a stringent or a lenient definition of early non-response, with non-responders then re-randomized between combined behavioral intervention alone and combined behavioral intervention plus naltrexone, and responders re-randomized between naltrexone alone and naltrexone plus telephone disease management. It demonstrated that the DTR comparison could be operationalized in outpatient substance use settings (Nahum-Shani et al. 2017).

When SMART designs are appropriate. A SMART earns its complexity on four conditions: the clinical uncertainty is about sequencing, not only about which treatment to start with; the re-randomization timepoint corresponds to a meaningful clinical decision window; non-response to the initial treatment is common enough to power stage-two comparisons; and the scientific questions concern how to build

a DTR, whether the primary aim compares initial treatments, stage-two options, or whole embedded DTRs. Where most patients respond to the initial treatment and sequencing rarely arises, a standard RCT is more efficient and the SMART's operational complexity (tracking response status, executing stage-two randomization in real time) buys nothing.

11.7. Causal Inference: From Trials to Real-World Evidence

Randomization balances confounders, measured and unmeasured, in expectation (any imbalance in a particular trial arises by chance, not from how patients were assigned), which is why the randomized trial is the gold standard for causal treatment effects. Where randomization is not possible (rare diseases, historical comparisons, post-marketing effectiveness), causal inference methods attempt to recover valid treatment comparisons from observational data. The vocabulary follows; Chapter 14 has the full regulatory and design framework for using real-world evidence in drug development.

11.7.1. The Potential Outcomes Framework

The potential outcomes (Rubin causal model) framework (Rubin 1974) defines the causal effect of treatment at the individual level as the difference between what would have happened to a patient under treatment and what would have happened under control: two *potential outcomes*, only one of which is ever observed (the *fundamental problem of causal inference*). The average treatment effect (ATE) averages those individual-level differences across the population.

ITT analysis in a randomized trial estimates the ATE under the treatment-policy estimand: it compares what happened to patients *assigned* to treatment with what happened to patients *assigned* to control, averaged over the entire population including non-compliers. When compliance is high, the ITT effect closely approximates the effect of treatment actually received. When non-compliance is substantial, the *complier average causal effect* (CACE, or local average treatment effect, LATE) can be estimated via instrumental variables (IV), using randomization as the instrument.

11.7.2. Propensity Score Methods

In observational data, treatment assignment is not random. *Propensity score* methods (Rosenbaum and Rubin 1983) use the estimated probability of receiving treatment given observed covariates to balance groups, and they come in three forms. *Matching* pairs each treated patient with one or more controls of similar propensity score: balance is verifiable, but the analysis shrinks to the matched pairs and many patients can be discarded. *Inverse probability weighting* (IPW) weights each patient by the inverse of their probability of receiving the treatment they actually received, creating a pseudo-population where treatment is independent of covariates; it retains all patients but is sensitive to extreme weights when propensity scores are near

0 or 1. *Stratification*, or subclassification, estimates treatment effects within strata defined by propensity score quintile, a rougher approach with less bias reduction than matching or IPW.

All three balance only on *observed* covariates. Unmeasured confounding, the persistent concern in any observational analysis, survives propensity scoring untouched, which is why a sensitivity analysis for it (E-values (VanderWeele and Ding 2017), for instance) should accompany any causal claim from a propensity-based analysis. The E-value is the minimum strength of association, on the risk ratio scale, that an unmeasured confounder would need to have with both treatment and outcome to explain away the observed association.

11.7.3. Target Trial Emulation

Target trial emulation (Hernán and Robins 2016) designs an observational analysis to explicitly mimic a hypothetical randomized trial. The analyst specifies the target trial's eligibility criteria, treatment strategies, assignment mechanism, outcomes, follow-up, and estimand, then checks whether these elements can be credibly emulated with the available observational data. That discipline forces analytic decisions (e.g., the time zero for follow-up, which patients qualify as eligible, how to handle time-varying treatment) that are often made implicitly and inconsistently in ad hoc observational analyses. Common biases (immortal time bias, selection bias at time zero, depletion of susceptibles) become visible and correctable when mapped against the target trial template.

It is increasingly used for post-marketing effectiveness studies comparing two approved drugs where no RCT exists, for historical comparisons in rare disease programs, and to inform the design of prospective trials by estimating likely effect sizes and population characteristics from existing registries.

11.7.4. Marginal Structural Models

When treatment is time-varying and affected by prior outcomes (e.g., a patient's dose is adjusted based on their response), standard regression confounds the treatment effect with the confounding pathway that runs through time-varying covariates. *Marginal structural models* (MSMs) (Robins, Hernán, and Brumback 2000), fit using IPW with time-varying weights, handle this correctly by modeling the *marginal* (population-averaged) treatment effect in a population where, by weighting, all patients had equal probability of being on each treatment at each time.

MSMs answer questions like "What would have happened to this patient population if everyone had maintained continuous antiretroviral therapy, compared to a policy of interrupting therapy when CD4 counts exceeded a threshold?" (a question that standard adjusted regression cannot answer because CD4 count is simultaneously a confounder and a mediator).

Every method here serves one goal: valid inference about treatment effects under uncertainty, whether the data come from a randomized trial with missing data, an adaptive trial with pre-specified interim rules, or an observational database where nobody was randomized. Chapter 12 extends the toolkit with Bayesian methods, which update probability estimates as evidence accumulates and can carry prior information about treatment effects.

Discussion Questions

1. A Phase III trial for a novel PD-L1 inhibitor in second-line hepatocellular carcinoma uses overall survival as its primary endpoint and pre-specifies the log-rank test. At the time of analysis, the Kaplan-Meier curves separate at month 6, converge at month 12, and then diverge again in favor of the experimental arm. The log-rank test is not significant ($p = 0.08$). What does this tell you about the proportional hazards assumption, and what alternative analysis approaches (restricted mean survival time, weighted log-rank, landmark analysis) could be pre-specified to handle this pattern without inflating Type I error?

2. A sponsor is designing a 52-week antidepressant trial in treatment-resistant depression with an expected 40% dropout rate. The ICH E9(R1) estimands framework requires explicit handling of treatment discontinuation. Compare the treatment policy strategy and the hypothetical strategy as the primary estimand: what different clinical questions do they answer, what are their respective data requirements, and how does your choice affect the labeling claim you can support?

3. A tipping-point sensitivity analysis for a chronic pain trial shows that the primary MMRM result ($p = 0.03$) would flip to non-significance only if the mean of unobserved outcomes in the treatment arm were 2.8 standard deviations worse than the observed completer mean. How should a clinical reviewer interpret this threshold, and under what circumstances would you consider this result robust versus still uncertain?

4. A SMART design for an anxiety disorder population estimates that 35% of patients will be classified as non-responders at the stage-one re-randomization. The target effect size for the DTR comparison is a 0.35-unit difference on a validated symptom scale, with a standard deviation of 1.2. Why does the non-responder proportion directly determine required enrollment rather than only the stage-two sample size, and what is the role of inverse probability weighting in ensuring a valid DTR comparison?

5. A propensity-score-matched observational study reports that patients who received Drug A had a 22% lower rate of major cardiovascular events than those who received Drug B (HR 0.78, 95% CI 0.67-0.91). The authors report an E-value of 1.9, computed by treating the hazard ratio as a risk ratio. A reviewer argues this is insufficient to rule out unmeasured confounding. Construct the

strongest possible argument that the result is robust and the strongest possible argument that it is not, using the E-value and the study design features.

12. Adaptive and Bayesian Trial Design

As results accumulate showing one treatment superior, continuing to assign patients equally to both arms becomes progressively more *unethical*. For most of the 20th century, traditional methodology demanded exactly that. Trials followed a rigid paradigm: fix the sample size in advance, randomize patients equally to treatment and control, analyze the data once at the end, and never peek at accumulating results. The framework was designed to control the probability of false positive conclusions, it succeeded admirably at that goal, and regulators and statisticians understood it well.

It came at a cost. Patients in trials sometimes continued receiving inferior treatments long after the evidence favored the alternative, because the design could not adapt and information accumulating during the trial was deliberately hidden until the very end. Trials that were clearly failing ran to their planned end, and promising drugs took years to reach patients because trials could not accelerate when results were compelling. The methodology prioritized the conclusions drawn after the trial over the welfare of patients during it.

That ordering had a principled defense. The Belmont Report, published in 1979, established the ethical framework for human subjects research in the United States (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979). It drew a sharp distinction: "The purpose of medical or behavioral practice is to provide diagnosis, preventive treatment or therapy to particular individuals. By contrast, the term 'research' designates an activity designed to test an hypothesis, permit conclusions to be drawn, and thereby to develop or contribute to generalizable knowledge."

This dichotomy shaped clinical trial design for decades. Research was about generalizable knowledge; treatment was about helping patients. The two were separate, and a patient in a clinical trial was participating in science, not necessarily receiving optimal therapy. *Treating a patient in a clinical trial should be as important as treating a patient after the trial*, yet traditional designs ignored this principle.

The past three decades rethought that paradigm with adaptive designs, Bayesian methods, and platform trials, and this chapter traces that evolution from its theoretical foundations to the trials that established it. The frequentist foundations that rethinking builds on (hypothesis testing, Type I and II error control, survival analysis, repeated-measures models, group-sequential designs, and causal inference

methods) are covered in Chapter 11; Berry's review is a practitioner-oriented synthesis of how the adaptive and Bayesian ideas fit together (D. A. Berry 2025).

12.1. Preliminaries

Chapter 11 opened on Cox's domain of "large uncontrolled variations," and everything in this chapter follows from what that variation does to a trial. A patient's condition fluctuates with the natural history of the disease, with environmental factors, with psychological responses to treatment (the placebo effect), and with the supportive care they receive. Even if every patient in a trial received identical treatment, their outcomes would still vary substantially, and the effect of a new drug is usually a modest improvement over existing therapy that has to be detected against that background. Absolute outcomes say little on their own, which is why trials are built for comparison.

The need for concurrent controls (a group of patients receiving placebo or standard therapy, enrolled in the same trial under the same conditions) follows directly from this logic. It is tempting to compare a new treatment's results against historical data: "Patients with this disease used to have 30% survival; now with our drug they have 45% survival." But such comparisons are perilous. Medical practice changes over time. Diagnostic criteria evolve, often detecting disease earlier (lead-time bias). Supportive care improves. Patient populations shift. These "secular changes" can easily produce apparent improvements that have nothing to do with the investigational drug. The classical solution is the concurrent control: with both groups managed under identical conditions, the treatment is the only systematic difference between them.

In the early 20th century, a researcher studying inheritance in rats trained animals to perform a task, then bred from them across multiple generations, measuring how quickly each generation learned the task. He observed steady improvement across generations, apparently supporting a theory that acquired traits could be inherited. But there were no controls: no rats bred under identical conditions from untrained parents. When other researchers repeated the experiment *with* controls, they found that both trained and untrained lineages showed the same improvement over time. The apparent effect was due to secular changes in the health of the rat colony, not inherited learning. Without concurrent controls, this confounding factor was invisible.

Unless a treatment effect is dramatic (what statisticians sometimes call a "penicillin effect," where the improvement is so large it overwhelms any background variation), the comparison must be made against concurrent controls. Except in rare circumstances (such as diseases with 100% mortality and no existing treatment), historical controls cannot provide the certainty needed for regulatory approval.

The structure of a clinical trial follows from its comparative nature. Patients are assigned to *treatment arms* (the groups being compared). The *investigational arm* receives the new therapy being tested. The *control arm* receives either a *placebo*

(an inactive substance designed to be indistinguishable from the active drug) or an *active control* (the current standard-of-care treatment). The *endpoint* is the outcome measured to determine whether the treatment worked, for example, survival time, tumor shrinkage, or improvement on a symptom scale. The *treatment effect* is what we are trying to estimate: the true difference in outcomes between the investigational arm and the control arm.

The *population* is everyone with the condition of interest, all the people who might eventually receive the drug if approved. The *sample* is the much smaller subset actually enrolled in the trial. Because we can only study a sample, our conclusions about the population are always uncertain, and that uncertainty is quantified using *probability*, a number between 0 and 1 (or equivalently, 0% to 100%) running from impossible to certain.

Efficacy is the treatment's ability to produce the desired beneficial effect under the controlled conditions of a trial. *Effectiveness* is how well it works in routine clinical practice, where patients may be older, sicker, or less adherent to the prescribed regimen than those carefully selected for a trial.

A valid clinical trial must satisfy five requirements.

Absence of systematic error (bias). Patients receiving different treatments must not differ systematically in ways that could affect outcomes. If healthier patients are assigned to the investigational arm and sicker patients to the control arm, any observed benefit might be due to patient selection rather than the drug. The difficulty is that such *bias* cannot be removed by statistical analysis after the fact. No calculation of statistical significance can distinguish a treatment effect from a confounding factor: a small p-value says only that the observed difference would be surprising if chance alone were at work, not *what* produced it.

The solution is *randomization*: using a chance mechanism (like a computer-generated random number) to assign each patient to a treatment arm (Cox 2009). (For practical implementation of randomization methods, see Chapter 16.) Randomization serves three purposes: it avoids selection and other biases in a publicly convincing way; it provides a unified basis for estimating statistical error across different trial designs; and it enables formally exact tests of significance. Of these, bias avoidance is largely uncontroversial: randomization ensures that any differences between groups, whether known factors like age and disease severity or unknown factors we have not even thought to measure, are distributed randomly rather than systematically. This allows us to attribute observed differences to the treatment itself with quantifiable confidence.

A related principle: the units assigned to each treatment must respond *independently* of one another. If all patients receiving the new drug are treated at one hospital and all control patients at another, we cannot separate the drug effect from differences between hospitals: the hospital has become the experimental unit, and each arm has only one. The remedy is to randomize patients within each site

or, when the intervention acts on a whole site, to randomize many sites and account for the clustering in the analysis (see Chapter 13).

Precision. Even when systematic error is eliminated through randomization, our estimate of the treatment effect will differ from its true value due to *random error* (the natural fluctuation that occurs when studying a sample rather than the entire population). The *standard error* measures the expected magnitude of this random error. A rough interpretation: in about one case out of three, the estimate will err by more than ± 1 standard error; in about one case out of twenty, by more than ± 2 standard errors; in about one case out of a hundred, by more than ± 2.6 standard errors.

The standard error follows a *square-root law*: it is proportional to $1/\sqrt{n}$, where n is the number of patients per treatment arm (the drug group, placebo group, etc.). To halve the standard error (to double the precision of our estimate) we must quadruple the number of patients in each arm; to reduce it by a factor of ten requires a hundredfold increase in patients. Precision is costly, which is why confirmatory trials (the large Phase III studies required for regulatory approval) are so expensive, and why improved trial design (using techniques like stratification or crossover to reduce variability) can be more efficient than simply enrolling more patients.

Range of validity. The conclusions drawn from a trial strictly apply only to the specific patients and conditions studied. How well the trial's design and conduct protect those conclusions from bias is its *internal validity*. But the scientific and public health value of a trial depends on *external validity* (also called *generalizability*): the extent to which findings can be extrapolated to broader patient populations.

The two are in tension. A trial restricted to young, otherwise healthy patients will have less outcome variability and therefore require fewer subjects to achieve a given precision, but its results may not apply to elderly patients with multiple conditions, the very patients who will receive the drug in routine practice. Broad enrollment criteria reverse the trade: more generalizable results, more patients needed for the same precision because outcomes are more variable.

Simplicity. The trial should be as simple as practical. Complex protocols with frequent treatment changes are difficult for research staff to execute correctly, and operational difficulties can compromise data quality. Nevertheless, some complexity may be advantageous (adaptive designs that modify treatment assignments based on accumulating data, for instance), and it is a matter of judgment how far to go.

Quantifiable uncertainty. It must be possible to calculate the uncertainty in the conclusions, ideally from the trial data themselves. Statistical analysis provides both: an estimate of the treatment effect, and a measure of how confident we can be in it. The use of randomization to eliminate systematic differences automatically makes remaining differences random, which justifies the statistical analysis under weak assumptions.

These five requirements are met, in conventional practice, by the frequentist machinery set out in Chapter 11: the null hypothesis and the p-value, Type I and Type II error, power, and a sample size fixed in advance from the minimum clinically important difference. That chapter also covers the intent-to-treat principle and the ICH E9(R1) estimands framework that now sits alongside it. The name is apt, since the framework reads probability as the frequency of outcomes over many repeated trials, and it has governed regulatory decision-making for decades. What matters here is its shape rather than its details. It does not answer the question sponsors and regulators actually ask, which is how probable it is that this drug works given the data. It answers how improbable the data would be if the drug did nothing.

The framework is rigid by design. Sample size is calculated before the trial begins based on assumptions about the treatment effect, variability, desired power, and acceptable Type I error, and once the trial starts those parameters are fixed. Looking at accumulating data and deciding to enroll more patients if results are borderline (a natural impulse) inflates the Type I error rate; stopping early when results look promising, without pre-specified rules, does the same. The price of controlling error rates is inflexibility.

Abraham Wald laid the mathematical foundations for analyzing sequential decisions with the Sequential Probability Ratio Test, developed during World War II for industrial quality control (Wald 1947), and Albert Shiryaev later formalized them in optimal stopping theory (Shiryaev 1978). Peter Armitage recognized in 1954 that these methods could address an ethical tension in clinical trials (Armitage 1954, 1960): if one treatment is clearly superior, continuing to randomize patients to the inferior arm is unjustifiable. Sequential methods allow trials to stop early while maintaining valid statistical inference, but they require the stopping rules to be specified in advance.

When traditional designs break down. The five requirements for a valid trial (absence of bias, precision, range of validity, simplicity, and quantifiable uncertainty) were developed in an era of relatively homogeneous patient populations and treatments with broad applicability. Modern medicine increasingly operates in a different world (D. A. Berry 2012), and regulators have responded with new guidance frameworks.

A drug may work extraordinarily well in 15% of patients who carry a particular genetic mutation and have no effect in the other 85% who lack it. That pattern is the premise of *precision medicine* (also called *personalized medicine*), where treatments are targeted to patients with specific characteristics (U.S. Food and Drug Administration 2019d; Zhou et al. 2008). The characteristic that marks out the responders is a *biomarker*, a measurable biological indicator (such as a genetic mutation, a protein level in the blood, or a molecular characteristic of a tumor) that can predict whether a patient will respond to a particular treatment. The traditional framework (randomize patients equally, analyze the average treatment effect across all patients) will dramatically underestimate the benefit for the responsive subgroup while exposing non-responders to ineffective treatment.

The FDA's enrichment strategies guidance addresses this challenge by describing *predictive enrichment* (selecting patients likely to respond based on biomarker status) and *prognostic enrichment* (selecting patients at higher risk of the outcome), but these approaches create new tensions. The "range of validity" requirement becomes paradoxical: a trial enrolling all patients has broad generalizability but may miss a genuine effect in the responsive subgroup; a trial restricted to biomarker-positive patients (those with the predictive characteristic) may have insufficient patients for adequate precision. *Adaptive enrichment designs* address this by starting with broad enrollment and using interim data to identify responsive subgroups, restricting subsequent enrollment accordingly (Simon and Simon 2013; Pallmann et al. 2018). Recent advances in *covariate-adaptive randomization* allow trials to balance treatment assignments across patient characteristics more efficiently, with unified frameworks that can handle continuous covariates and complex covariate structures (Atkinson 2002; Ma et al. 2024).

Rare diseases present an even starker challenge (European Medicines Agency 2006; U.S. Food and Drug Administration 2025a). If only 5,000 patients worldwide have a condition, enrolling 1,000 patients for a Phase III trial (a modest size for common diseases) would require recruiting 20% of the entire patient population. The square-root law governing precision becomes an insurmountable barrier: there simply are not enough patients to achieve traditional power levels for modest effect sizes. Recognizing this, regulators have developed more flexible ways to meet the existing evidentiary standard. Under the FDA's proposed Rare Disease Evidence Principles (RDEP) process, one adequate and well-controlled study, possibly single-arm, plus strong confirmatory evidence can generally establish effectiveness; the process is limited to drugs for a known inborn genetic defect in a very small population (generally fewer than 1,000 U.S. patients) whose disease progresses to significant disability or death in a relatively short time and has no adequate therapy that alters its course. The EMA's guideline on small populations explicitly addresses alternative study designs including non-randomized approaches and external controls.

Multiple treatment options compound the difficulty (U.S. Food and Drug Administration 2022a; Redman et al. 2020; Renfro and Sargent 2017). Modern oncology trials may need to evaluate several drugs, each potentially combined with several others, across multiple biomarker-defined subgroups. A traditional approach (separate trials for each combination in each subgroup) would take decades and hundreds of thousands of patients. The requirement for "simplicity" becomes impossible when the clinical reality is inherently complex. The FDA's master protocols guidance describes innovative designs: *basket trials* (studying one drug across multiple tumor types sharing a biomarker), *umbrella trials* (studying multiple drugs within one tumor type based on biomarkers), and *platform trials* (flexible frameworks that can add or remove treatment arms over time). All three put multiple treatments or multiple patient populations inside a single trial infrastructure.

In these settings the traditional design (fixed sample size, equal randomization, single analysis at the end) is not merely suboptimal. It is potentially unethical or

infeasible. Continuing to randomize patients equally to a clearly inferior arm wastes their contribution. Conducting *underpowered* trials (trials with too few patients to reliably detect a true treatment effect) in rare diseases wastes resources and patient willingness to participate, and testing one combination at a time when patients are dying delays potentially life-saving treatments for years.

The case for adaptive designs. These challenges have driven the development of *adaptive trial designs* (trials that modify themselves based on accumulating data according to pre-specified rules) (S. M. Berry et al. 2010). Instead of fixing everything before the first patient enrolls, an adaptive design lets the trial learn and adjust. Two oncology programs show what that buys: the BATTLE trial used Bayesian adaptive randomization to match lung cancer patients to targeted therapies based on their tumor biomarkers (Kim et al. 2011), and the I-SPY 2 platform trial has evaluated more than twenty neoadjuvant breast cancer treatments, graduating effective drugs to Phase III while dropping ineffective ones (H. Wang and Yee 2019).

The rigidity of traditional designs is not inherent to valid inference. If we can prespecify the rules by which adaptations will occur (writing them into the protocol before the trial begins) and use computer simulations to verify that the trial still controls Type I error under various scenarios, we can maintain regulatory standards while gaining flexibility. The adaptation rules become part of the design, not modifications made after seeing the data.

The FDA's 2019 guidance builds the same requirement into its definition: an adaptive design is "a clinical trial design that allows for prospectively planned modifications to one or more aspects of the design based on accumulating data from subjects in the trial" (U.S. Food and Drug Administration 2019a). A change made without such a rule, after someone has seen accumulating results, is a protocol amendment, and it brings back the Type I error and bias problems that pre-specification avoids. Readable overviews of the field are (Pallmann et al. 2018; Thorlund et al. 2018; Burnett et al. 2020).

Adaptive designs can drop ineffective treatment arms early, concentrating resources on promising therapies, and can shift randomization toward better-performing treatments, reducing the number of patients assigned to inferior arms. Sample size can rise if *interim results* (analyses conducted partway through the trial, before all patients have completed) suggest the treatment effect is smaller than anticipated. And where the drug works best in a subgroup defined by biomarkers, the design can find it.

A complementary approach uses *Bayesian* statistical methods, which directly answer the question the frequentist framework does not: "Given the data we have observed, what is the probability that this drug works?" Bayesian methods naturally accommodate accumulating evidence, updating probability estimates as each new patient's data arrives, making them well-suited for adaptive designs. They also provide a coherent framework for *borrowing* information (using data from previous

trials, from other patient subgroups, or from external sources to supplement the current trial's data), which is especially valuable in rare diseases where every patient's data is precious.

Chapter 30 provides definitions of key terms used throughout this chapter.

12.1.1. Types of Adaptive Design at a Glance

The rest of the chapter works through a few adaptive designs in depth. Table 12.1 puts the main types side by side, so that each later section has a place on the map.

Table 12.1.: Common adaptive designs and what each adapts. SSR is sample size re-estimation, MAMS is multi-arm multi-stage, RAR is response-adaptive randomization.

Design	What adapts	Main concern
Group sequential	Stop early for efficacy or futility at planned interim looks	Alpha spent across looks
Blinded SSR	Sample size, from pooled variance or overall event rate	Little effect on Type I error
Unblinded SSR	Sample size, from the interim treatment effect	Needs a combination test or conditional error
Enrichment	Enrollment restricted to a biomarker subgroup	Multiplicity across populations
MAMS	Arms dropped at interim looks	Multiplicity, use of an intermediate outcome
RAR	Allocation probabilities	Efficiency, time trends
Seamless Phase II/III	Dose or arm selection, then confirmation, in one protocol	Selection, combined analysis
Bayesian adaptive	Any of the above, driven by posterior or predictive probabilities	Operating characteristics by simulation

Group-sequential designs stop the trial at planned looks when the evidence crosses a boundary, and they are the oldest and best understood adaptive design (Pocock 1977; O'Brien and Fleming 1979; Lan and DeMets 1983); their alpha spending is covered in Section 11.5 and their use in Phase III in Section 9.3. *Sample size re-estimation* (SSR) changes the planned enrollment. A blinded version recomputes a nuisance parameter such as the variance from the pooled data, uses no information about the treatment difference, and so has little or no effect on the Type I error rate. An unblinded version sets the new sample size from the interim treatment effect or its conditional power, and needs the error-control tools described below; the "promising zone" approach increases the sample size only when the interim effect falls in a range where extra patients would plausibly rescue the trial (Mehta and

Pocock 2011). *Adaptive enrichment* restricts later enrollment to a biomarker-defined subgroup, as sketched earlier in this chapter. The DAWN trial of thrombectomy 6 to 24 hours after stroke used a Bayesian adaptive-enrichment design; enrollment stopped at 31 months, after 206 patients, because of the results of a prespecified interim analysis, and thrombectomy was superior on both coprimary endpoints (posterior probability of superiority above 0.999) (Nogueira et al. 2018). *Multi-arm multi-stage* (MAMS) designs drop arms that fall behind at interim looks, often judging them on an intermediate outcome, and let the remaining arms continue. STAMPEDE in prostate cancer is the standard example: it shared one control arm, added arms over time, and reported a survival benefit for docetaxel and, later, for abiraterone added to hormone therapy (Nicholas D. James et al. 2016; Nicholas D. James et al. 2017). Dose-finding designs such as the continual reassessment method and BOIN adapt the dose of the next cohort (O'Quigley, Pepe, and Fisher 1990; S. Liu and Yuan 2015). Seamless designs, response-adaptive randomization and platform trials have their own sections below (Section 12.4, Section 12.8, Section 12.5), and a Bayesian design can use any of these adaptations, with decisions driven by posterior or predictive probabilities.

Controlling Type I error. A design that looks at the data and then changes its sample size, its population or its arms no longer matches the assumptions of a fixed-sample test, so error control has to be built in. Alpha spending does this for planned looks (Section 11.5). *Combination tests* handle changes that were not fully fixed in advance: the data from each stage give a stage-wise p-value, and the stage-wise results are combined with a rule and weights fixed before the trial, so the combined test keeps its level even if the second-stage sample size is chosen after seeing the first stage (P. Bauer and Kohne 1994; Cui, Hung, and Wang 1999). The *conditional error* principle reaches the same end differently: it records the probability, under the null hypothesis and given the interim data, that the original design would have rejected, and allows any redesign of the remainder whose own conditional Type I error is no larger (Proschan and Hunsberger 1995; Müller and Schäfer 2001). For Bayesian designs the evidence comes from simulation (Section 12.7.5). Twenty-five years of confirmatory adaptive designs and their pitfalls are reviewed in (Peter Bauer et al. 2015).

12.2. Bandit Problems: The Theoretical Foundation

In 1933, statistician W.R. Thompson proposed a radical idea (Thompson 1933). When testing two treatments with unknown efficacy, assign the next patient to treatment A with probability equal to the current probability that A is better than B. Thompson was motivated by ethics: avoiding patients being "sacrificed to the inferior treatment." His proposal predated Austin Bradford Hill's first randomized controlled trial by 15 years.

Thompson had discovered what mathematicians would later call a *multi-armed bandit problem*, a fundamental challenge in decision theory. Imagine slot machines (bandits) with unknown payout rates. You want to maximize total winnings over

many pulls. Each pull gives information (which machine pays better?) and potentially gives a reward (you might win). The dilemma: should you *exploit* by playing the machine that seems best, or *explore* by trying other machines to learn more? This is the *exploration-exploitation tradeoff*.

Clinical trials are bandit problems. Each patient assignment is a "pull." Some pulls give information about treatment efficacy (explore). Some give effective therapy to the patient (exploit). The goal is maximizing successful outcomes across all patients, both those in the trial and those who will be treated after the trial based on its results. The challenges that break traditional designs (precision medicine, rare diseases, multiple treatment options) are instances of this structure, and its formal study provides the theoretical foundation for adaptive trial designs.

Early statisticians recognized that fixed randomization (ignoring accumulating data) was clearly suboptimal. If you know which treatment works better, use it. But in trials, you don't know; you learn as you go. The question becomes: what strategy adapts optimally as evidence accumulates?

Finding the mathematically optimal strategy is computationally demanding, and the difficulty grows explosively as the number of arms and the length of the trial increase. For a small number of arms, though, the optimal strategy can still be computed exactly through *dynamic programming*, working backward from the last patient. For patient N, the decision is easy: choose whichever arm has higher probability of success. For patient N-1, choose the arm that maximizes expected total successes for the last two patients, accounting for what will be learned. Work backward to patient 1.

Consider testing two treatments with unknown success rates in four patients. Starting with uniform prior beliefs (complete uncertainty about both treatments), an optimal adaptive strategy achieves an expected success rate of 56.9%, compared to 50% for any fixed randomization scheme. The benefit grows with sample size: at 100 patients, the adaptive strategy reaches 64.9%; with infinite patients, it approaches 66.7%, a 16.7 percentage point gain representing the *expected value of perfect information*.

Thompson's strategy turns out to be effective in practice, especially when testing many treatments simultaneously. It is not mathematically optimal, but it is intuitive and it performs well. After 70 years of obscurity, Thompson's 1933 paper has seen explosive growth in citations, driven by applications ranging from search engine ad placement to modern platform trials.

12.2.1. Bayesian Bandit Problems: Updating Beliefs

The mathematical foundation of bandit problems rests on *Bayesian updating*, the process of revising probability distributions as evidence accumulates (D. A. Berry 2025). Consider a simple two-armed trial where both treatments have unknown

success probabilities p_1 and p_2. Before observing any data, we might assume complete uncertainty: both success rates follow uniform distributions on [0, 1], which is equivalent to a Beta(1, 1) distribution.

As outcomes are observed, beliefs update according to Bayes' rule:

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

For a binary outcome (success/failure), if a treatment produces s successes and f failures, and we start with a Beta(1,1) prior, the posterior distribution for its success rate p becomes Beta($\alpha_{post}, \beta_{post}$), where:

$$\alpha_{post} = 1 + s, \quad \beta_{post} = 1 + f$$

The posterior mean, representing our updated best estimate of the success rate, is given by

$$\frac{\alpha_{post}}{\alpha_{post} + \beta_{post}}$$

and the variance, representing our uncertainty, is

$$\frac{\alpha_{post}\beta_{post}}{(\alpha_{post} + \beta_{post})^2(\alpha_{post} + \beta_{post} + 1)}$$

This elegant conjugacy enables exact updating without numerical approximation.

Figure 12.1 illustrates this updating process for a four-patient bandit problem, showing how the probability distribution for p_1 evolves while p_2 (with no observations yet) remains uniform:

With uniform priors, a single success shifts the mean from 0.5 to 2/3, while a failure shifts it to 1/3. After one success and one failure, the mean returns to 0.5, but the distribution is no longer uniform. It is now peaked at 0.5 with less variance, reflecting the additional information that the true rate is unlikely to be extreme.

Despite the elegant theory, Bayesian adaptive trials remained curiosities for decades. Biostatisticians in the 1970s and 1980s considered them naive: randomization was sacrosanct, the gold standard of medical research, and varying randomization probabilities based on accumulating data was viewed with deep suspicion.

One investigator remarked that Bayesians were "loosey-goosey" in adhering to rules. The concern was legitimate: if you look at data repeatedly and adjust your trial, don't you inflate Type I error? Can such trials be trusted?

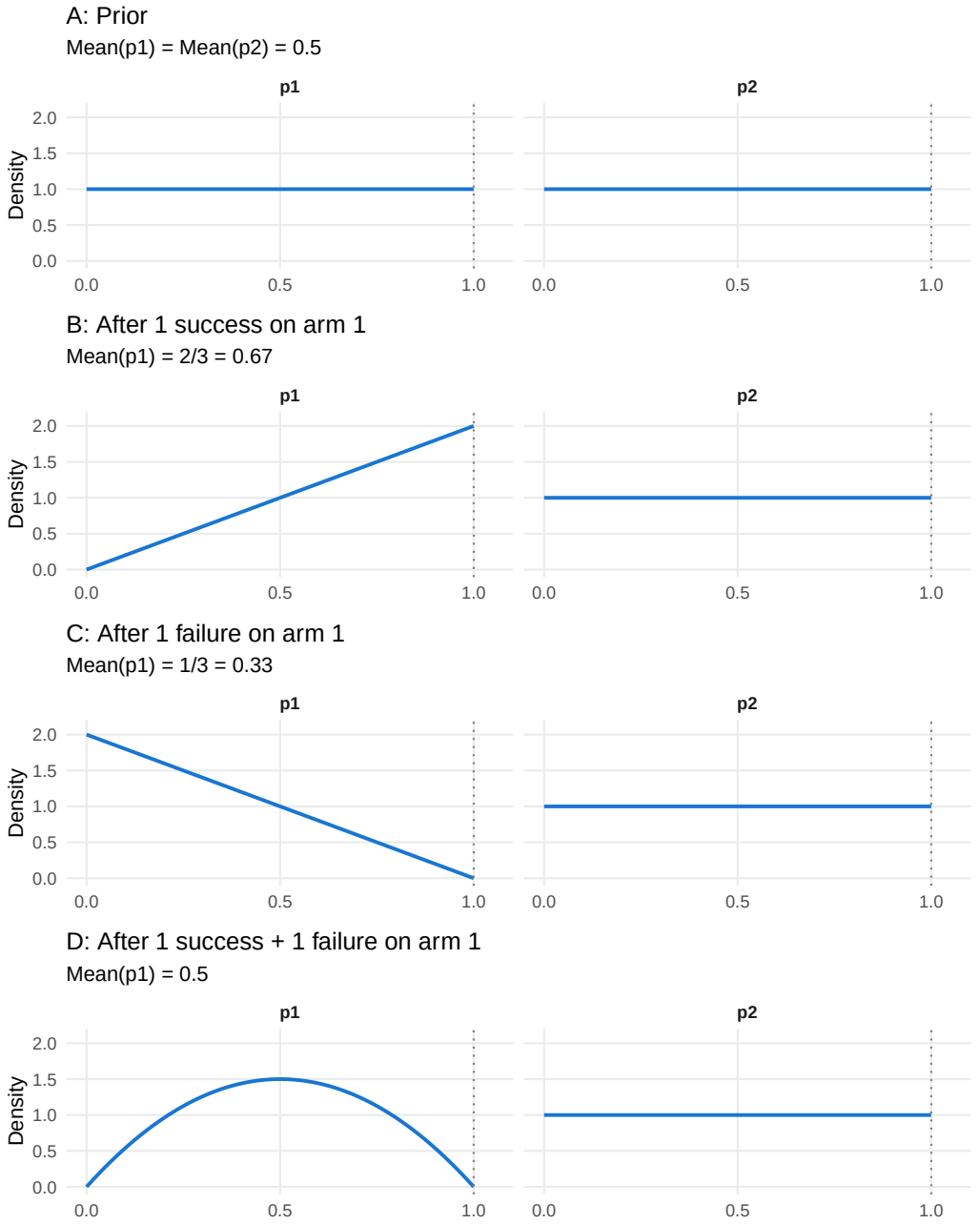


Figure 12.1.: Bayesian updating of success probability distributions in a two-armed bandit. Panel A shows uniform priors for both arms. Panels B-D show how the distribution for p1 evolves after observing a success (B), a failure (C), or one success followed by one failure (D). The distribution for p2 remains uniform throughout since no observations have been made on that arm.

The path forward ran through *traditional* trial design: demonstrate mastery of it first, gain credibility in the clinical research community, then introduce innovations incrementally. It also took advances in computing power and statistical software to make complex calculations feasible, and regulators willing to embrace innovation while maintaining rigorous standards.

By the late 1990s, these pieces were falling into place. The FDA's Center for Devices and Radiological Health (CDRH) led early adoption, commissioning white papers on Bayesian approaches in medical device development. Early successes in late-phase device trials demonstrated that Bayesian methods could work in practice, maintaining control of Type I error while providing decision-makers with more intuitive, actionable information.

12.3. The First Wave of Bayesian Adaptive Trials: Proving the Concept

The first wave of real-world Bayesian adaptive trials (late 1990s through the 2010s) did not set out to replace frequentist practice. They succeeded by embedding Bayesian quantities inside conservative, auditable decision rules that stakeholders already understood: stop or continue enrollment, drop unpromising doses, or graduate a regimen to confirmatory testing. The common thread is operational: these designs turned interim data into actionable forecasts (especially predictive probabilities) while still demonstrating familiar operating characteristics through simulation and error-rate control. The examples below became proof points because they were clinically consequential, computationally feasible, and easy to explain to sponsors and regulators in plain language.

12.3.1. CALGB 49907: Predicting Trial Outcomes

In 1999, a fully Bayesian trial was designed for the Cancer and Leukemia Group B (CALGB) testing whether capecitabine could replace standard chemotherapy in elderly breast cancer patients (Muss et al. 2009). The sponsor insisted on 1800 patients to answer the question adequately. The statistical team estimated this would take 15 years to accrue, by which time the question might no longer be interesting.

The compromise: enroll 1800 patients *only if needed*. Starting at 600 patients, calculate Bayesian *predictive probabilities*: given current data, what is the probability that further follow-up (without additional enrollment) will answer the question conclusively? If this probability exceeds 80%, stop enrollment and continue follow-up.

The trial stopped at the first interim analysis with 633 patients enrolled. Additional 16-month follow-up delivered a compelling answer: capecitabine cannot replace chemotherapy in this population. The trial achieved its scientific goal with one-third the planned sample size.

By Berry's account, this was the first fully Bayesian randomized trial published in the *New England Journal of Medicine* (2009). It demonstrated a key principle: *predictive probabilities* answer the question sponsors and clinicians actually care about ("Will this trial give us a definitive answer?") more directly than traditional power calculations.

12.3.2. ASTIN: Adaptive Dose-Finding in Stroke

Pfizer's ASTIN trial (Acute Stroke Therapy by Inhibition of Neutrophils) tested 15 doses plus placebo of an investigational neutrophil inhibitor in stroke patients (Krams et al. 2003). Allocation probabilities were never fixed upfront. The trial analyzed data daily and used Bayesian modeling to decide which doses to assign the next day.

The algorithm smoothed estimates across doses and used longitudinal models to predict 90-day outcomes from earlier measurements. It began on low doses and placebo, saw little effect, and progressively concentrated on higher doses as evidence accumulated. By the time 500 patients had completed 90-day follow-up, it had concluded the drug was ineffective at every dose and stopped the trial, which randomized 966 patients, well short of the 1300 maximum.

This was dose-finding as a bandit problem: continuously learning which doses were most promising and concentrating resources there, while exploring enough to avoid missing an effective dose.

12.4. Seamless Designs: Combining Phases

Phase II and Phase III decisions are traditionally separated by years and by organizational boundaries: one study ends, a fresh protocol is written, a new trial starts. Seamless Phase II/III designs collapse that gap. Interim learning (often supported by Bayesian modeling and predictive thinking) selects doses or regimens early, and accrual continues into a confirmatory stage under prespecified rules. The motivation is economic and ethical. Cutting the "dead time" between phases delivers decision-grade evidence sooner without sacrificing interpretability, and choosing doses from the trial's own interim data avoids overcommitting to a weak one.

12.4.1. AWARD-5: From Seven Doses to Blockbuster Drug

Eli Lilly's AWARD-5 trial for dulaglutide (Trulicity) in type 2 diabetes began in 2008 with seven doses (0.25 to 3.0 mg) plus two controls, designed in collaboration with the FDA as part of the Critical Path Initiative (D. A. Berry 2025). *Stage 1* used response-adaptive randomization with biweekly interim analyses, scored by a composite *clinical utility index* that combined HbA1c, weight loss, heart rate, and blood pressure, capturing both efficacy and safety in a single metric.

Longitudinal modeling predicted 12-month outcomes from early data, so doses could be selected before any patient reached the primary endpoint. The algorithm picked

two non-adjacent doses (0.75 and 1.50 mg) at the earliest allowed timepoint and shifted directly to *Stage 2*: fixed randomization comparing those doses to controls.

The Data Safety Monitoring Board could depart from the algorithm only on matters of safety (D. A. Berry 2025), and on those grounds, in April 2009, randomization to the 3.0 mg dose was stopped on its recommendation after increases in heart rate and concerns about pancreatic safety (European Medicines Agency 2014a). Its members remarked that they could not otherwise overrule the algorithm's decisions, but that if they could, they would do exactly what the algorithm had done.

Both doses received FDA approval in 2014. Trulicity became Lilly's best-selling drug, reaching peak annual revenue of $7.4 billion in 2022 according to Lilly's financial reporting, and, by Berry's estimate, the seamless design saved at least one year compared to running separate Phase II and III trials (D. A. Berry 2025). The 3.0 mg dose, though dropped in Stage 1, showed a mean weight loss of 3.3 kg (about 7 pounds) in the 15 patients randomized to it, against about 0.5 kg on the control arms, in data up to the dose-selection point (European Medicines Agency 2014a). That signal foreshadowed the modern use of GLP-1 agonists for obesity, even though Trulicity itself was never approved for weight loss.

12.4.2. Lecanemab: From Adaptive Phase II to Phase III

When Eisai designed their Phase II trial (Study 201) for lecanemab (Leqembi) in Alzheimer's disease in 2011, they aimed to avoid a common failure mode: Phase II trials that give false signals, leading to failed Phase III trials (Swanson et al. 2021; Dyck et al. 2023; D. A. Berry 2025).

The trial tested six dose regimens including placebo. Initially, 2/7 of patients were randomized to placebo and 1/7 to each dose; at interim analyses after the first 196 patients and then after every 50 more, up to about 800 (Swanson et al. 2021), the proportion assigned to placebo was adjusted to match the proportion for the most likely Phase III dose. Early data favored the highest doses, and by trial end 48% of patients had received the two highest doses while only 12% received the two lowest. The trial was maximally informative about the regimens of greatest interest.

Berry describes missing data above 30% as typical of Alzheimer's trials. By his account, traditional "last observation carried forward" (LOCF) imputation would have estimated only a 13% reduction in disease progression, probably insufficient to justify Phase III. Bayesian multiple imputation, prospectively specified, learned patterns of missingness: it attributed placebo dropouts largely to the treatment not working, while dropouts on the highest dose were mostly due to a regulatory mandate unrelated to efficacy. The Bayesian model estimated a 27% reduction (D. A. Berry 2025).

Eisai proceeded to Phase III. The confirmatory trial showed 27% less decline than placebo on its primary endpoint, the CDR-SB (Dyck et al. 2023). The match is less exact than it looks: the Phase II 27% was on ADCOMS, and the Phase II Bayesian estimate on the CDR-SB was 33% (Swanson et al. 2021). Both are point

estimates with uncertainty, but the Phase II estimates were accurate enough to right-size the Phase III (D. A. Berry 2025). The FDA granted accelerated approval, then full approval, and lecanemab became the first disease-modifying therapy for Alzheimer's disease with clear efficacy evidence.

12.4.3. Sizing a Seamless Design

In an *inferentially seamless* design the stage-one data count toward the final test, and the analysis has to account for the selection made between the stages. An *operationally seamless* design runs the stages under one protocol but analyzes the final stage separately. For the inferentially seamless case with one arm selected at the end of the first stage, pooling the two stages and using an adjusted final alpha, in the manner of a group-sequential design, outperforms analyzing the second stage alone at full alpha and is slightly better than a combination test (L. R. Berry et al. 2024). The adjusted alpha depends on the number of arms in the first stage and on the share of the final information it supplies. The same paper finds that putting roughly 30 to 50% of the total patients into the first stage is near-optimal across a range of response shapes.

Two cautions apply to sizing. Power computed for the arm that is eventually selected overstates the power of the design, because the first stage sometimes advances the wrong arm; the actual power has to be found by simulating the selection. And when the first-stage endpoint is delayed, patients enrolled while it matures are part of the first stage's cost, so the planned split should count them. Comparisons between stage splits should keep the total sample size fixed.

12.5. Platform Trials

A platform trial is standing infrastructure. Therapies enter and exit over time against a shared control arm, Bayesian hierarchical models can support learning across biomarker-defined subgroups, and pre-specified decision thresholds control false positives. The obvious benefit is efficiency, since a shared control arm reduces redundant randomization. The subtler one is continuity: evidence accumulates in a coherent framework across calendar time while the operational machinery (data pipelines, governance, DSMB processes, endpoint adjudication) stays stable.

12.5.1. I-SPY 2: Prototype for a New Paradigm

From 2010 to 2022, the I-SPY 2 platform in neoadjuvant breast cancer cycled 23 investigational therapies from 18 pharmaceutical companies through a single trial engine (J. W. Park et al. 2016; D. A. Berry 2025). Nine graduated to Phase III on a high Bayesian predictive probability of success ($\geq 85\%$ probability of success in a confirmatory trial). Four have received FDA marketing approval.

Patients were classified into eight molecular subtypes based on hormone receptor status, HER2 status, and genomic markers. Each experimental therapy was evaluated within relevant subtypes, up to ten possible "signatures" (potential regulatory indications). The trial used response-adaptive randomization, assigning 20% to control (or 50% if only one experimental arm was available) and distributing the remainder based on performance.

A common control arm, pooling concurrently randomized patients with historical controls through a "time machine" approach, gave I-SPY 2 a control cohort an order of magnitude larger than a traditional Phase II study. Around it, monthly continuous learning updated probability distributions across all arms and subtypes; longitudinal modeling used tumor volume measurements to predict pathologic complete response well before patients reached their final assessments; and nested factorials within the continuing control arms tested combination therapies simultaneously, isolating the effects of individual components more efficiently than separate trials would permit.

Neratinib graduated in the HER2-positive, hormone-receptor-negative signature. Across all 115 patients assigned to neratinib, pCR rate was 37% versus 20% for controls, a 17% absolute difference. Berry's modeling decomposed that gap (D. A. Berry 2025): 7% was the drug's inherent efficacy, and 10% was due to adaptive randomization concentrating assignments in responsive subtypes. Patients in I-SPY 2 were therefore participating in rigorous research *and* receiving optimized therapy at the same time. The Belmont Report itself allows research and practice to be "carried on together," provided the activity undergoes review for the protection of human subjects (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979).

Pembrolizumab added to standard neoadjuvant chemotherapy graduated in all three biomarker signatures it was tested in, and it was the first of ten agents to graduate in the hormone-receptor-positive, HER2-negative signature (Nanda et al. 2020). The final estimated pCR rates, pembrolizumab versus control, were 44% versus 17% in HER2-negative disease overall, 30% versus 13% in hormone-receptor-positive, HER2-negative disease and 60% versus 22% in triple-negative disease, from 69 patients on pembrolizumab and 181 controls (Nanda et al. 2020). How I-SPY 2 predicts the response of patients still in follow-up is described in Section 12.7.4.

Building on BATTLE, the Bayesian adaptive lung-cancer master protocol that accrued from 2006 to 2009, I-SPY 2 in turn served as prototype for platform trials across diseases: GBM AGILE in glioblastoma, REMAP-CAP in community-acquired pneumonia (which proved critical during COVID-19), and Precision Promise in pancreatic cancer.

12.5.2. REMAP-CAP and GBM AGILE

REMAP-CAP is a Bayesian platform in community-acquired pneumonia, including COVID-19. It is multifactorial: each *domain* is a set of alternative interventions

for one aspect of care, with its own domain-specific appendix, and the trial is embedded in routine care and can switch to a pandemic mode (Angus et al. 2020). It uses response-adaptive randomization, and its decision rules declare efficacy or futility when the posterior probability of superiority over control crosses prespecified thresholds (Angus et al. 2020). Interleukin-6 receptor antagonists (tocilizumab and sarilumab) improved outcomes, including survival, in critically ill patients with COVID-19 (REMAP-CAP Investigators, Gordon, et al. 2021). **GBM AGILE** in glioblastoma is also Bayesian and response-adaptive. It combines a seamless Phase II/III structure within one platform, is subtype-specific, and was designed with registration in mind (Alexander et al. 2018).

What the FDA's Bayesian draft implies for platforms. Three points in the January 2026 draft bear directly on long-running platforms (U.S. Food and Drug Administration 2026g). Borrowing across baskets or sub-studies through a hierarchical model needs a justification of exchangeability, and known effect modifiers such as biomarker status should be modeled explicitly. Each interim decision point needs prespecified success criteria, and the simulations should cover operational parameters such as accrual. And changes in the standard of care, and the recency of earlier data, affect how relevant that data is to current patients, which matters when a platform runs for years.

12.5.3. The Regulatory Evolution

In 2010, FDA's draft guidance allowed that a Bayesian approach to modeling and simulation "might be useful," then added a caution: "Using simulations to demonstrate control of the Type I error rate, however, is controversial and not fully understood" (U.S. Food and Drug Administration 2010a; D. A. Berry 2025). By 2019, the agency had accepted the approach (U.S. Food and Drug Administration 2019a): "The guidance also advises sponsors on the types of information to submit to facilitate FDA evaluation of clinical trials with adaptive designs, including Bayesian adaptive and complex trials that rely on computer simulations for their design."

The 2019 guidance addressed practical issues like *post-simulation*: if operating characteristics were evaluated under certain assumptions (e.g., expected accrual rates) but the trial experienced different conditions, additional simulations after the trial could verify that conclusions remain valid.

The FDA's January 2026 draft guidance on Bayesian methodology in clinical trials (https://www.fda.gov/media/190505/download) spells out what the agency expects when Bayesian methods appear in a submission: Bayesian success criteria and decision thresholds prespecified, operating characteristics evaluated (often through simulation), prior distributions carefully justified (especially when they borrow external information), and sensitivity analyses planned that examine prior-data conflict and other key assumptions. It also stresses documentation and reporting practices sufficient for regulatory review and reproducibility, including clear statements of what was prespecified and how computation was performed (U.S. Food and Drug Administration 2026g).

For Bayesian designs the draft guidance describes the usual success criterion as having the form $\Pr(d > a) > c$, where d summarizes the treatment effect, a is the minimum effect considered beneficial and c is the minimum probability that supports a conclusion of effectiveness (U.S. Food and Drug Administration 2026g). It describes three ways to set a and c. *Calibration to Type I error* suits complex adaptive designs that use Bayesian methods for something other than synthesizing several information sources, and noninformative priors: a is zero for superiority (or based on the noninferiority margin), and c is derived algebraically or approximated by simulation. *Direct interpretation of the posterior probability* applies when the prior accurately summarizes existing belief; a posterior probability of effectiveness of 0.98 then means a probability of 0.02 that the treatment is ineffective, and the question is whether that is small enough in context. *Benefit-risk or decision-theoretic approaches* let the thresholds reflect product risk, disease seriousness and available therapies, for instance through loss functions. The guidance also asks for success criteria at each interim decision point and for every Bayesian primary and key secondary endpoint.

Its operational value is that it turns Bayesian "acceptability" into reviewable components instead of a broad philosophical debate. A sponsor must be able to show how the decision rule maps to error control and interpretability, how prior influence is understood and bounded (including concepts such as effective sample size and discounting when borrowing), how estimands and missing data are treated in a Bayesian setting, and how results can be reproduced, which means software reliability and, when relevant, convergence diagnostics for simulation-based inference (U.S. Food and Drug Administration 2026g).

The draft ICH E20 guideline on adaptive clinical trials adds international harmonization, though its scope centers on pre-specification, simulation-based demonstration of operating characteristics, and Type I error control, and does not reach the full range of Bayesian adaptive applications the FDA Bayesian guidance covers (International Council for Harmonisation 2025a).

ICH E20 and some regulatory commentary have framed Bayesian adaptive trials primarily in terms of borrowing from external data using informative priors. Many Bayesian adaptive trials use *minimally informative priors* that are rapidly overwhelmed by accumulating data; the benefits come from the coherent inferential framework and from the interpretability of decision rules based on posterior and predictive probabilities (U.S. Food and Drug Administration 2019a). Modern Bayesian borrowing, where it does occur, employs *dynamic* approaches in which the degree of borrowing depends on observed consistency between external and new data, reducing the bias risk of static borrowing. The Bayesian framework is well suited for sophisticated borrowing, but borrowing and Bayesian analysis are conceptually separate (International Council for Harmonisation 2025a).

12.5.4. The One-Trial Standard

FDA has grown increasingly open to accepting a single highly stringent trial as substantial evidence of efficacy in place of two independent Phase III trials (U.S. Food and Drug Administration 2019c; United States Congress 1997). The 1997 modernization act codified the exception, and the agency's 2019 guidance describes what strengthens the case for a single trial: a hard clinical endpoint, a substantially lower p-value threshold than the conventional two-trial 0.025, large multicenter enrollment, and internal consistency across pre-specified subgroups. In June 2026, FDA issued a revised draft guidance that repositions the single-trial-plus-confirmatory-evidence approach from a bounded exception toward what FDA's leadership calls the default (Prasad and Makary 2026) and drops the earlier framing that treated one such trial as equivalent to two; because it remains a draft (the public comment period closed on September 22, 2026), it signals regulatory direction rather than settled policy (U.S. Food and Drug Administration 2026c) (see Section 29.7).

The shift fits adaptive designs. A seamless Phase II/III trial with extremely conservative success criteria could deliver both dose optimization and confirmatory evidence in one study, saving years of development time.

It carries costs too. Approvals tied to narrowly defined populations may restrict labeled indications, leaving clinicians to extrapolate off-label or sponsors to run "label expansion" trials after approval. The change could accelerate time to market while shifting weight onto post-approval research and multi-cohort basket architectures.

The optimal evidence standard should arguably depend on context: disease severity, unmet medical need, and patient impact. For fatal rare diseases, a more flexible bar may be justified; for high-prevalence, lower-risk conditions, standards should be stricter. Context-varying standards are good science and hard to implement in regulation.

When Lilly's AWARD-5 data were presented, when device trials using Bayesian predictive probabilities supported marketing approvals, when I-SPY 2 demonstrated that adaptive trials could maintain scientific rigor while improving patient care, resistance gave way to acceptance.

12.6. Mathematical Foundations: Hierarchical Modeling

Modern adaptive trials often employ *Bayesian hierarchical models* (probability distributions of probability distributions), which are well suited to basket trials testing one therapy across multiple tumor types and to platform trials testing multiple therapies across patient subtypes (D. A. Berry 2025).

Consider a therapy tested in K tumor types. For each type k, let θ_k denote the true response rate. A hierarchical model specifies:

$$\theta_k \sim \text{Beta}(\alpha, \beta) \quad \text{for } k = 1, \ldots, K$$

where α and β are themselves unknown parameters with their own prior distributions (hyperpriors). If we observe y_k responders out of n_k patients in tumor type k, the likelihood is:

$$y_k \mid \theta_k \sim \text{Binomial}(n_k, \theta_k)$$

Through the shared hyperparameters, the data from every tumor type inform the posterior for each θ_k. If nine types show 40% response rates and one shows 10%, a typical hierarchical model "shrinks" all estimates toward a common mean, but the observed heterogeneity inflates the estimated variance of the population distribution, so every estimate, the outlier included, is pulled in less than it would be if all ten rates agreed.

Charles Stein proved in the 1950s that when three or more normal means are estimated at once, estimating each separately is "inadmissible" under total squared-error loss: another estimator has lower expected total squared error whatever the true means (Stein 1956). The *James-Stein estimator*, which shrinks every estimate toward a common point, showed that this holds even for unrelated parameters. The gains are largest when the true values lie close together, as related tumor-type response rates may, but the improvement is in the total: a subgroup whose true rate lies far from the rest can be estimated worse.

The amount of borrowing is governed by the estimated between-group variance τ^2. When τ^2 is small (tumor types respond similarly), the model borrows heavily and each type's estimate is pulled toward the overall mean. When τ^2 is large (responses are heterogeneous), the model borrows cautiously and each type's estimate stays closer to its observed rate. The posterior mean for θ_k can be approximated as:

$$\hat{\theta}_k \approx \lambda_k \bar{\theta} + (1 - \lambda_k)\hat{\theta}_k^{\text{MLE}}$$

where $\bar{\theta}$ is the pooled estimate across types, $\hat{\theta}_k^{\text{MLE}} = y_k/n_k$ is the maximum likelihood estimate for type k, and λ_k is a shrinkage weight that decreases as τ^2 grows and decreases as the sample size n_k grows.

By pooling evidence across the full trial, this framework enables basket trials to identify responder populations and support regulatory approvals in molecular subsets even when some subsets have modest sample sizes.

12.6.1. Choosing and Justifying a Prior

The hierarchical model above is one place where a prior does visible work: the amount of borrowing is a prior assumption. More generally, every Bayesian analysis needs a prior, and the choice has to be defensible to a reviewer. FDA's January 2026 draft guidance (a draft as of November 2026) says sponsors should pre-specify

and justify the full details of the proposed prior in the protocol, and that informative priors generally need a greater amount of justification (U.S. Food and Drug Administration 2026g).

Noninformative and minimally informative priors. These priors are meant to express general uncertainty about the parameter. The draft guidance says no single approach is universally preferred. Such a prior can still be influential when data are sparse, and it is not invariant to transformation: a prior that is flat for a response probability implies a prior that is not flat for the log odds, so the induced prior on the scale that drives the decision should be checked (U.S. Food and Drug Administration 2026g). These priors also serve as the reference against which the influence of an informative prior is measured, and when the primary analysis uses an informative prior, an analysis with a minimally informative one helps show how sensitive the result is to that choice.

Skeptical and enthusiastic priors. A *skeptical prior* puts little probability on very large treatment effects. It can be appropriate after related programs have failed or when only an incremental gain over existing therapy is plausible, and the draft guidance notes that such priors have not been standard practice in drug development but could be considered. In an adaptive design a skeptical prior can stand in for a stricter decision rule: it counterbalances early random highs, so stopping early for efficacy requires more evidence. An *enthusiastic prior* (one with some positive belief built in) can play the same role for stopping early for futility (U.S. Food and Drug Administration 2026g).

Effective sample size. The *effective sample size* (ESS) of a prior measures how much information it carries in units of patients. For a Beta(a, b) prior on a response rate, the conjugate update adds observed successes to a and failures to b, so the prior behaves like $a + b$ patients' worth of data; the uniform Beta(1, 1) prior of Figure 12.1 carries the information of two. The guidance describes comparing the prior ESS with the size of the planned trial, or with the total information (prior ESS plus trial size), to say how much of the answer comes from the prior. It also asks for sensitivity analyses, planned in advance, that rerun the analysis under a range of alternative reasonable priors, for example with weaker and stronger borrowing (U.S. Food and Drug Administration 2026g).

Estimands and borrowed data. Borrowing only makes sense when the borrowed data answer the same question as the current trial. Ideally the same primary estimand and estimator are used for both data sources. When they differ, the external data can sometimes be reanalyzed with the approach planned for the new trial, but the guidance warns that reanalyzing completed studies risks overfitting to observed random highs, and it advises discussing the plan with FDA early (U.S. Food and Drug Administration 2026g).

12.7. Practical Implementation Challenges

Decision rules have to be computable fast enough to support interim looks (sometimes daily), robust to missing and delayed outcomes, and transparent enough that a DSMB, sponsor, and regulator can reproduce results from the same locked dataset. Those requirements turn many "simple" theoretical quantities into engineering problems, especially when decisions are driven by predictive probabilities that integrate over uncertainty in both current estimates and future data.

12.7.1. Computing Predictive Probabilities

Predictive probabilities drive adaptive decisions (D. A. Berry 1989). The standard way to compute one is Monte Carlo simulation: for each possible future dataset, draw many random samples, analyze each, and calculate the proportion showing success. Thousands of future scenarios times thousands of Monte Carlo iterations per scenario is where the cost lives.

An approximation skips the simulated future datasets, estimating the predictive probability directly from the current p-value or posterior probability, the information fraction, and the final success threshold (Marion et al. 2025). It shows high concordance with full Monte Carlo methods across diverse settings (dichotomous endpoints, time-to-event analyses, ordinal outcomes, historical borrowing, and longitudinal modeling) while dramatically reducing computational time.

Because each scenario runs much faster, a sponsor can test hundreds of scenarios and estimate operating characteristics precisely across the plausible range of accrual rates, effect size assumptions, heterogeneity, and decision thresholds.

12.7.2. Administrative Analyses for Funding

Sponsors often cannot fund an entire large trial upfront. Venture funds, private equity, and pharmaceutical partners want evidence the drug is working before committing tens of millions more.

Administrative analyses are interim analyses conducted solely for funding decisions, completely firewalled from the trial's scientific conduct. An independent statistical group (often an external CRO) examines interim data against predefined criteria and reports a simple yes/no: has the funding trigger been met? The FDA adaptive design guidance strongly recommends limiting access to comparative interim results to people independent of those conducting or managing the trial (U.S. Food and Drug Administration 2019a), and a funding analysis needs the same firewall to leave the trial's conduct and blinding intact.

A natural metric is Bayesian *predictive probability of trial success*: given current data and all its uncertainty, what is the probability that the final analysis will meet significance (Marion et al. 2025)? It updates at each interim look, unlike traditional *power*, which is fixed at the design stage based on an assumed true effect and never changes as data accumulate. Funding partners find it immediately

relatable, since they routinely use dynamic risk forecasts in their own fields, and a predictive probability at an interim point supports an invest-or-wait decision in familiar terms.

Defining objective criteria often requires simulation. The design team presents realistic datasets and visualizations, asking: "If the results looked like this, would you fund the next phase?" A pattern emerges, and quantitative decision thresholds go into the funding triggers.

The label matters. An administrative analysis should *not* be called a "futility analysis," which risks confusion with the efficacy/futility analyses intended to stop the trial itself.

12.7.3. Estimation After Adaptation

Error-rate control is only half of the inference problem. Adaptation also biases the estimates, and the usual confidence intervals can have the wrong coverage. Two mechanisms matter. *Truncation bias* arises when a trial stops early for efficacy: it stopped because the observed effect crossed a boundary, so the estimate at that moment tends to be too large, and the bias is largest when the trial stops at an early look with little information. *Selection bias* arises when the trial picks the best-looking arm or subgroup, because the winner's observed effect includes some luck.

The remedies depend on the design. For group-sequential trials, estimates based on a stage-wise ordering of the outcomes, median-unbiased estimates and repeated confidence intervals correct for the stopping rule. For designs that select an arm or subgroup, conditional estimators and the uniformly minimum variance conditionally unbiased estimator (UMVCUE) correct for the selection. Shrinkage and Bayesian estimates, of the kind described in Section 12.6, pull extreme observed effects back toward the rest and are a third route. Practical guidance on which estimator to report is given in (Robertson, Choodari-Oskooei, et al. 2023). Whatever the method, the bias and mean squared error of the planned estimator belong among the operating characteristics checked by simulation (Section 12.7.5), and the report should show the size and direction of the adjustment instead of the naive estimate alone.

12.7.4. Learning Speed Versus Enrollment Speed

An adaptive trial has two parts that are both fixed in the protocol: a *learning mechanism* (a model fitted to the accruing data) and the *adaptations* that the learning drives. How much the design gains over a fixed one depends on how fast the trial learns relative to how fast it enrolls. When the endpoint is measured within hours of treatment, each decision rests on nearly complete information for the patients enrolled so far. When the endpoint is measured after 18 months, many patients are enrolled before anything is learned, and an adaptation can arrive too late to affect much.

Patients who have been enrolled but have not yet reached the endpoint are a different kind of missing data. A fixed-sample trial can ignore them because it waits for complete follow-up. An adaptive trial that makes decisions at interim looks has to say something about them, and the usual answer is a longitudinal model, written into the protocol with priors based on historical data, that links early measurements to the final outcome. At each interim the model is fitted, the final outcome is multiply imputed for every patient who has not completed follow-up, the final analysis is run on each completed dataset, and the results are combined into posterior and predictive probabilities (Section 12.7.1). The design work is choosing the early measurement that predicts the final outcome best and checking that decisions stay sound when the model fits poorly, since it runs at every interim without manual refitting. The same logic explains why the simulations should vary the enrollment rate: with a fixed maximum sample size, faster accrual means more patients are enrolled before the model has learned much, so a faster trial can make worse decisions at its interim analyses.

The I-SPY 2 platform shows the method at work. Pathologic complete response is assessed at surgery, months after treatment starts, so a longitudinal model uses serial MRI tumor-volume changes to predict it for patients still in follow-up, and graduation decisions such as those for neratinib and veliparib-carboplatin rested partly on those predictions (J. W. Park et al. 2016; Rugo et al. 2016). ASTIN and AWARD-5 (Section 12.4) relied on the same idea, predicting the final endpoint from early measurements so that doses could be chosen before it had been observed. The model, not the observed outcomes, carries much of the interim evidence in these designs, so its pre-specification and its checking by simulation are part of the trial's credibility.

12.7.5. Simulation and Operating Characteristics

Complex adaptive designs require extensive *clinical trial simulation* before the first patient enrolls. Thousands of scenarios are run across different true effect sizes, dose-response curves, patient accrual rates, and amounts of heterogeneity, and each scenario yields a set of operating characteristics: Type I error rate (probability of false-positive conclusion), power (probability of correct positive conclusion), expected sample size, expected trial duration, probability of selecting the correct dose. The results confirm Type I error control and guide refinement of decision thresholds and sample size caps. Fixed-sample and standard group-sequential designs have those characteristics calculated analytically; more complex adaptive designs usually determine them through simulation, a shift that initially concerned regulators and is now fully accepted.

Why simulate when a formula exists. Simulation also shows things a formula leaves out. An analytic expected sample size often counts the patients who complete follow-up, whereas a simulation reports the patients enrolled by the time the trial stops, and with a delayed endpoint the enrolled patients still in the follow-up pipeline add to the total. Looking at individual simulated trials, not only at averages, is

how designers build intuition and tune rules, for example by examining the early interim results of simulated trials that went on to succeed before fixing a futility threshold.

Type I error in Bayesian trials. Even a Bayesian trial must address Type I error to win regulatory acceptance. The typical approach: simulate the trial under the null hypothesis (no treatment effect) and verify that the probability of a false-positive conclusion is at or below the prespecified level, conventionally 2.5% one-sided for a confirmatory efficacy claim. Bayesian decision rules (e.g., "declare success if the posterior probability of benefit exceeds 97.5%") are then chosen to achieve this frequentist property. The design is Bayesian in its continuous learning and probabilistic reasoning, and it demonstrates traditional operating characteristics.

Some trialists add a disclaimer when reporting Bayesian trials in journals: "All p-values and confidence intervals in this article are descriptive and have no inferential content." The Bayesian posteriors carry the inference; the frequentist measures appear for consistency with convention.

Bayesian operating characteristics. The approach just described is what FDA's January 2026 draft guidance calls calibration to Type I error (Section 12.5.3). For such a design the draft asks for Type I error and power, bias and mean squared error, and interval coverage and width, all by simulation, across the plausible range of assumptions including operational parameters such as accrual rate. A design that is not calibrated to Type I error needs agreement with FDA, and its report adds Bayesian power (power averaged over a prior), the probability of a correct decision (similar to a positive predictive value, because in the Bayesian framework a false positive is defined conditional on a positive conclusion, not on a true null hypothesis), and the expected bias, mean squared error and coverage. The guidance separates the *analysis prior*, which must be pre-specified and drives the decision, from the *design prior* (or sampling prior), which describes the effects assumed when simulating, and it recommends considering design priors with pessimistic assumptions about the treatment effect. Sensitivity to these choices is higher with small samples or early interim analyses (U.S. Food and Drug Administration 2026g).

A Simulation Workflow

Morris, White and Crowther describe any simulation study by five elements, ADEMP: its aims, the data-generating mechanisms, the estimands or targets, the methods compared and the performance measures (Morris, White, and Crowther 2019). Designing a trial by simulation fills in those elements in a fairly standard order, and the steps below are the order we recommend; where FDA's January 2026 draft guidance states an expectation, it is noted.

1. **Frame the decision (aims).** State what the trial is for (screening, dose selection, confirmation) and what counts as a good, a mediocre and an unacceptable effect.

2. **Get an analytic anchor.** Compute a fixed-sample or group-sequential sample size as a starting point.
3. **Write the virtual trial (data-generating mechanism).** Specify the endpoint model and its variability, the time to the endpoint, accrual (including ramp-up and low, medium and high rates), dropout, and for platforms the arrival of arms and any cap on concurrent arms. FDA expects the assumptions to cover the plausible range of statistical parameters, such as the variance or background rate, and of operational ones, such as the accrual rate (U.S. Food and Drug Administration 2026g).
4. **Write the design (estimand and method).** Specify the treatment effect being estimated, the analysis model, the decision quantities (a p-value, a posterior probability or a predictive probability), the timing of interim looks, the success and futility rules, the allocation rule and any selection rule.
5. **Build the scenario grid and simulate (performance measures).** Look at summary operating characteristics and at individual simulated trials.
6. **Iterate.** Tune the boundaries, the number and timing of interims and the split between stages, comparing variants at the same total sample size so that the comparison is fair. The draft guidance describes this iterative adjustment of sample size, priors, success criteria and interim boundaries as the usual practice (U.S. Food and Drug Administration 2026g).
7. **Stress-test.** Add scenarios for drift, slower accrual, a smaller effect and model misspecification.
8. **Freeze and document.** Write a simulation report with the code and a description of the simulation design, implementation and results, with scenarios that are pre-specified, comprehensive and plausible (U.S. Food and Drug Administration 2026g), and confirm that the simulated design is exactly the design in the protocol and the statistical analysis plan.

Scenario design. The minimum grid is the null, the alternative the trial was powered for, and an intermediate effect between them. Multi-arm designs add a global null, one effective arm, a linear dose-response, a peaked response in which a middle dose is best, and several arms near the maximum effect; the best share of patients for a first stage depends on how many arms are near the maximum (L. R. Berry et al. 2024). Operational scenarios vary accrual (low, expected, high), dropout, and for platforms the speed at which arms arrive. These can matter as much as the efficacy assumptions: with fixed-size arms, a cap on concurrent arms and a limit on how long an arm may wait, an effective arm that arrives late may never be tested, and early futility rules for failing arms free the slots. Sampled-effect scenarios draw each arm's true effect from a prior distribution, giving "portfolio" operating characteristics reported by the true class of each arm, not as one power number. Sensitivity scenarios cover time trends in the control rate, effect dilution between stages, heterogeneity and non-proportional hazards.

Simulation results carry Monte Carlo error, which Morris and colleagues recommend reporting alongside every performance measure (Morris, White, and Crowther 2019). A rejection probability near p estimated from n runs has standard error

$\sqrt{p(1-p)/n}$: about 0.0016 for $p = 0.025$ and 10,000 runs, large relative to the gap between 0.025 and a value that would count as inflation, and about 0.0005 for 100,000 runs, which is why runs near the Type I error boundary should number 100,000 or more.

Operating characteristics to report. A complete report includes:

- Type I error for each hypothesis and the family-wise error rate, taken as the worst case over the null space, not only the global null.
- Power or probability of success by scenario, and for selection designs the probability of correct selection (and of selecting the wrong arm).
- Expected sample size and its distribution, counting patients enrolled and patients completing follow-up.
- Expected duration and time to decision.
- Stopping probabilities by interim look and by reason (early success, late success, early or late futility).
- Bias, mean squared error and confidence interval coverage of the effect estimates (Section 12.7.3).
- For platforms, the probability that an arm is never enrolled, the expected number of successful arms, the probability of at least one success, the time to the first success, and results by the true class of the arm.

Software for design simulation. Analytic group-sequential boundaries, power and sample size re-estimation are available in the R packages gsDesign and rpact and in the commercial program East. Bayesian adaptive and response-adaptive designs can be simulated with the R package adaptr, time-to-event designs with non-proportional hazards with simtrial, and the commercial FACTS program covers many of the designs in this chapter. Bespoke platforms and borrowing models are usually simulated with custom code in R, Python or Stan, and then the sponsor owns the verification of that code; regulators can ask for it (U.S. Food and Drug Administration 2019a).

Common Simulation Pitfalls

Four mistakes recur, and each undoes one of the steps above. Comparing design variants at different total sample sizes makes the comparison reflect the sample size, not the design (step 6). Running too few simulations near the Type I error boundary, or omitting the Monte Carlo error, leaves the key number less precise than it looks. Counting futility stops in the null simulations when the monitoring committee may overrule them understates the Type I error, because a trial that continues past a futility boundary has another chance to reject; unless the futility rule is binding, the Type I error should be computed with it switched off. And simulating a design that differs from the one actually run (step 8) is avoided by locking the versions of the design, the simulation code and the protocol together.

12.8. Response-Adaptive Randomization

Response-adaptive randomization (RAR) changes allocation probabilities during the trial in response to accumulating outcome data: arms that appear to be performing better receive a larger share of future enrollments. One common Bayesian mechanism is Thompson sampling (see Section 12.2): allocate the next patient to arm k with probability equal to the current posterior probability that arm k is the best (Thompson 1933; Hu and Rosenberger 2006).

The appeal is straightforward: if interim data suggest that one dose is substantially better than another, equal allocation keeps sending patients to the inferior treatment for the rest of enrollment.

Unlike a group-sequential design, where each interim superiority look spends part of the alpha budget, RAR adds no interim hypothesis tests, so there are no interim looks to spend alpha on. That does not make it automatically safe for Type I error. Because allocation becomes data-dependent, a naive analysis that ignores how patients were assigned can distort the false positive rate, especially when outcomes drift over time (the temporal-trend bias discussed below). Preserving Type I error depends on a pre-specified analysis that accounts for the adaptive allocation, with operating characteristics confirmed by simulation.

RAR's logic has generated a sustained methodological debate about where it pays and where it costs.

Temporal-trend bias. When patients enrolled later in a trial are systematically different from those enrolled earlier (improving standard of care, viral variants, changes in patient mix), RAR can create imbalanced patient characteristics across arms. If, for example, enrollment of healthier patients rises over time and allocation is simultaneously shifting toward the better-looking arm, the arm receiving more late enrollees will appear to perform better than it actually does. The standard rebuttal is to include a time covariate in the primary analysis: because both arms are present throughout the trial, the time effect is estimable and the adjusted treatment comparison remains valid, provided that the treatment effect is approximately additive over time and no treatment-by-time interaction is present. This additivity assumption deserves scrutiny in infectious-disease trials, where pathogen variants and background-care improvements can change the nature of the disease itself during the trial (Korn and Freidlin 2011; Proschan and Evans 2020). The problem is not peculiar to RAR: calendar-time drift (from the sources above, from a changing mix of sites, seasonality or pandemic waves) biases comparisons whenever allocation changes over time, which also happens when arms enter and leave a platform and when enrollment is enriched. Besides the time covariate, the mitigations are to compare only against concurrent controls, to stratify the analysis by period, and to simulate drift scenarios when evaluating the design (Section 12.7.5). Models that borrow non-concurrent controls while smoothing the control response over time, such as the Bayesian time machine (Benjamin R. Saville et al. 2022), are an alternative, and FDA's draft Bayesian guidance cites their proposed use in the

GBM AGILE and Precision Promise platforms (U.S. Food and Drug Administration 2026g) (see Section 12.5); the EMA's October 2026 draft reflection paper says non-concurrent controls are usually not acceptable for the primary analysis of the primary endpoint (European Medicines Agency 2026b).

Power in two-arm trials. When the outcome variance is the same in both arms, equal (1:1) randomization maximizes statistical power for a given total sample size. Unequal allocation, whether 3:1 or 9:1, reduces power. For a standard two-arm superiority trial, RAR therefore requires a non-statistical justification (typically the ethical argument that more patients should receive the better-performing arm) to offset the power cost (Proschan and Evans 2020). Most statisticians advise caution about RAR in two-arm Phase III confirmatory trials. The critique goes further. Simulation studies of two-arm trials report that RAR loses power, produces highly variable sample sizes, in a non-trivial fraction of runs assigns more patients to the inferior arm, and is sensitive to time trends (Thall, Fox, and Wathen 2015; Wathen and Thall 2017), and similar arguments were made earlier by Korn and Freidlin (Korn and Freidlin 2011).

Multi-arm settings. RAR's strongest case is in multi-arm dose-finding or "king of the hill" settings where the goal is to identify the best-performing arm and concentrate learning on it. A Phase II trial with five doses can allocate progressively more patients to the most promising doses, reducing waste on clearly inferior doses while still maintaining the shared control arm (D. A. Berry 2025).

Practical safeguards for RAR

Two design safeguards appear in virtually every well-run RAR trial:

1. **Burn-in period.** Run conventional equal randomization for an initial cohort (for example, the first 10-20% of total patients) before activating the adaptive algorithm. Early outcome data are too sparse to drive allocation reliably; the burn-in ensures the first adaptations are based on reasonable evidence.

2. **Protect the control arm.** Adapting allocation among experimental arms while holding the control at a fixed fraction (e.g., always 25-33% of enrollment) preserves the common comparator that all treatment comparisons depend on. Reducing control allocation degrades the entire trial's inferential base and can lead to misleading arm-level estimates.

Simulation across a range of true effect sizes, enrollment rates, and temporal-trend magnitudes is required to verify that the chosen RAR rule achieves the operating characteristics (power, expected sample size, probability of correct arm selection) claimed in the protocol.

The ethical intuition behind RAR (giving more patients the better-looking arm) is compelling but does not straightforwardly resolve into a design recommendation, because both of its failure modes land on future patients. A less powerful trial

produces less reliable evidence for them. A trial that over-allocates to the winner-so-far and then encounters a temporal trend can produce misleading results. The strongest case for RAR is in settings with many experimental arms, a fast endpoint relative to enrollment rate, and a primary goal of selecting the best arm rather than estimating treatment effects with high precision for a regulatory submission.

A balanced review of the debate separates myths from practical considerations; its recommendations match the safeguards in the callout above (burn-in, limits on extreme allocation, modeling of time trends) and add that the ethical claim should be justified explicitly (Robertson, Lee, et al. 2023). One natural protected control share is $1/(k+1)$ in a trial with k experimental arms, the share equal allocation would give. Platform trials have also compared multi-arm multi-stage designs, which drop arms, with adaptive randomization, which down-weights them (J. Lin and Bunn 2017).

Regulatory positions (as of November 2026). The EMA and FDA differ. The EMA's draft reflection paper on confirmatory platform trials (adopted for consultation in October 2026) reasons that changes in allocation ratios over time can be hard or impossible to adjust for in the analysis, recommends fixed allocation ratios for each comparison of interest, and concludes that RAR is not supported (European Medicines Agency 2026b). FDA's draft master protocols guidance lists modifying the randomization ratio among the adaptive design elements a master protocol may include and applies the principles of the 2019 adaptive design guidance to them; its worked examples of changing ratios are fixed schemes, not driven by outcome data, in which the ratio changes as drugs enter or leave a platform and the comparisons must account for the periods with different ratios (U.S. Food and Drug Administration 2026e).

12.9. The New Standard

Adaptive and Bayesian methods have moved from exotic to mainstream, carried there by two convictions. A trial should *learn from data as soon as it has them*, not years later when it is too late to adjust. And a patient enrolled in a trial deserves effective treatment, not only a part in generating generalizable knowledge.

Traditional methods served well for decades, establishing the principle of rigorous evidence and Type I error control. A better way emerged as computing power grew, statistical methodology advanced, and regulatory understanding deepened.

Platform trials now test dozens of treatments in years instead of decades. Seamless designs combine phases; response-adaptive randomization can improve the care patients receive while the trial is running; hierarchical models extract maximum information from complex data; and predictive probabilities put decisions in the language decision-makers think in naturally.

Berry argues that the Belmont Report, with its nearly half-century-old distinction between research and treatment, is outdated (D. A. Berry 2025). Clinical trials *are*

treatment, and they should be designed accordingly. Adaptive Bayesian methods provide the framework to merge rigorous science with ethical patient care.

13. Study Design

A trial can fail for reasons that have nothing to do with the drug's biology. Define eligibility too broadly, pair it with an endpoint that is noisy or poorly aligned with the mechanism, and the study enrolls a heterogeneous population in which any true benefit is diluted; what comes out is an underpowered "negative" trial that will not persuade a regulator even if a meaningful effect exists in the intended subgroup.

Design is what stands between a clinical question and decision-grade evidence. Every trial begins with the question. Sometimes it is simple: does this drug reduce blood pressure more than placebo? More often it is nuanced: does this drug reduce cardiovascular events more than existing therapy in patients with moderate-to-severe hypertension and high cardiovascular risk who are not adequately controlled on their current regimen? The choices that follow (population, endpoints, control, randomization, blinding, follow-up, and analysis plan) determine whether an observed effect can be credibly attributed to the intervention and interpreted for the intended regulatory and clinical use.

13.1. Randomization

The cornerstone of the modern clinical trial is *randomization*: the random allocation of participants to treatment groups, arguably the most important innovation in the history of clinical research. Age, sex, disease severity, and genetic factors can all influence outcomes, and those can be measured. The case for randomization rests on the rest: the likely hundreds of factors we do not know about or cannot measure. Allocation by chance, performed correctly, balances the groups on known and unknown factors alike. Not in every individual case, but on average across studies.

That balance is what lets us credit an observed difference to the treatment rather than to the groups: if patients receiving the new drug do better than patients receiving placebo, and the groups were comparable at baseline, the most likely explanation is that the drug works.

Simple randomization flips a coin for each patient, which can leave the groups unequal in size. *Block randomization* fixes that by forcing equality after every block of patients (say, 4 or 8). *Stratified randomization* goes further, randomizing separately within subgroups defined by important prognostic factors so that those factors stay balanced.

13.2. Blinding

If randomization protects against bias in how patients are assigned to groups, *blinding* (also called masking) protects against bias in how they are treated and assessed. Table 13.1 summarizes the levels and their applications.

Table 13.1.: Levels of Blinding in Clinical Trials

Blinding Level	Who Is Blinded	Protection Against	Limitations
Open-label	No one	None	Risk of assessment and performance bias
Single-blind	Participants only	Participant expectation effects	Investigator may influence care/assessment
Double-blind	Participants + Investigators	Expectation + Assessment bias	May still be unblinded by side effects
Triple-blind	+ Data analysts	All above + Analytic bias	Maximum protection
Blinded assessment	Outcome assessors only	Assessment bias when other blinding impossible	Limited to specific outcomes

Double-blind is the gold standard for most clinical trials. Triple-blind extends the blind to the statisticians, who learn which group is which only when the analysis plan is executed, so that the analysis itself cannot be manipulated, consciously or otherwise.

Maintaining the blind is not always straightforward. Some drugs have distinctive side effects (certain cancer treatments cause hair loss; some psychiatric medications cause weight gain). Placebo formulations must be indistinguishable from active treatment in appearance, taste, smell, and texture. When blinding is not possible (as in surgical trials or many device studies), alternative designs and analyses must account for the limitations.

13.3. Control Groups

Every trial is a comparison, and the comparator decides what the result can prove and sometimes whether the trial can ethically be run at all.

Placebo controls set the experimental treatment against an inert substance indistinguishable from it. They provide the clearest evidence of efficacy, because any difference between groups can be attributed to the pharmacological effect of the drug rather than to expectations, attention from healthcare providers, or natural fluctuations in disease. But when effective treatments exist, withholding them to

demonstrate that a new drug is better than nothing may be unacceptable. The Declaration of Helsinki requires that new treatments be tested against the best proven intervention, not placebo, except in the circumstances it specifies (World Medical Association 2025); Chapter 4 describes them.

Active controls compare the experimental treatment to an established therapy. That is an ethically appropriate choice when withholding treatment would be harmful, and it comes at a statistical price. Showing *superiority* to an active control requires a larger sample size than showing the drug is better than placebo. Showing *non-inferiority* (not meaningfully worse) requires careful definition of how much difference would be acceptable and raises concerns about trial quality that could mask real differences.

Historical and external controls compare study participants to patients treated in the past or in separate real-world environments. Long viewed with skepticism for pivotal trials, they are increasingly being given regulatory structure in rare diseases and high unmet need areas. In 2025, the MHRA released a draft guideline on the use of external control arms derived from real-world data (Medicines and Healthcare products Regulatory Agency 2025), and EMA published a concept paper outlining plans for a future reflection paper on the same topic (European Medicines Agency 2025), both emphasizing transparent, pre-specified protocols and the fitness for purpose of the external data.

Digital twins push the technique further: prognostic scores generated from high-dimensional baseline data allow for "TwinRCTs" where each patient's outcome is adjusted by their own AI-generated digital counterpart, increasing statistical power in longitudinal studies (Unlearn.AI 2024; Ross et al. 2024). *Bayesian borrowing* instead lets sponsors "borrow" information from historical trials or RWD, discounted for differences between the sources by a fixed weight or dynamically when the data disagree. Discounting does not remove those differences: in a first-line non-small cell lung cancer (NSCLC) case study, a matched real-world external control augmented from a historical trial with a fixed-weight power prior still did not reproduce the survival result of the randomized trial whose control arm it replaced (Struebing et al. 2024).

13.4. Trial Design Types

Table 13.2 compares the major trial design types and the clinical questions each one suits.

Table 13.2.: Comparison of Clinical Trial Design Types

Design	Description	Advantages	Disadvantages	Best Used When
Parallel	Patients randomized to one treatment for study duration	Simple; No carryover concerns; Works for progressive diseases	Larger sample needed; Between-patient variability	Most confirmatory trials; Progressive conditions
Crossover	Each patient receives all treatments in sequence	Smaller sample; Patients serve as own controls	Carryover effects; Requires stable disease	Stable chronic conditions; PK studies
Factorial	Multiple interventions tested simultaneously	Tests interactions; Efficient for 2+ questions	Complex analysis; Interpretation challenges	Testing combinations; Multiple hypotheses
Cluster	Groups (sites, clinics) randomized, not individuals	Practical for system-level interventions	Reduced power; Complex analysis	Community interventions; Educational programs
Adaptive	Design modified based on interim data	Efficient; Smaller samples possible	Complex planning; Implementation challenges	Dose-finding; Rare diseases

The *parallel design*, the default for confirmatory trials, keeps each participant in the group they were randomized to for the duration of the study. A *crossover design* instead gives each participant several treatments in sequence, so that each serves as their own control and the noisier between-patient comparison gives way to a within-patient one, which means smaller sample sizes may suffice. The price is complexity and two requirements: the condition must be stable (the disease should not progress during the study) and treatments must have no lasting effects that carry over from one period to the next.

A *factorial design* tests two or more interventions in the same trial by randomizing each participant to one of the possible combinations (for example, drug A or placebo crossed with drug B or placebo). When the treatments act independently, a single trial delivers the main effect of each intervention, answering two questions for roughly the cost of one; the design can also estimate whether the treatments interact, though detecting an interaction reliably requires a much larger sample than estimating either main effect alone. This same logic reappears later in the chapter in the *multifactorial* platform trials: REMAP-CAP randomizes a single patient within several treatment domains at once, and SNAP layers multiple treatment domains over one shared infrastructure.

In a *cluster-randomized design*, the unit of randomization is a group (a clinic, hospital, or community) rather than an individual. It is the natural choice when the intervention operates at the level of a system (a screening program, or a change in how a ward is staffed) or when treating individuals differently within the same site risks contamination between arms. The cost is statistical: outcomes for patients in the same cluster are correlated, so a cluster trial carries less information than an individually randomized trial of the same size, and the sample size must be inflated by a design effect that grows with the intracluster correlation and the number of patients per cluster. A *stepped-wedge* variant rolls the intervention out to clusters in a randomized sequence until all have crossed over, which suits an intervention expected to do more good than harm, where withholding it from some clusters for the whole trial would be hard to justify.

13.5. Sample Size

How many patients should be enrolled in a trial? Four quantities set the answer: the size of the effect we expect (or the minimum effect we consider clinically meaningful), the variability of the outcome measure, the Type I error rate we are willing to accept (the probability of declaring the drug works when it does not), and the power we require (the probability of detecting a real effect if it exists, commonly 80% or 90%). Smaller effects, noisier outcomes, stricter error rates, and higher power all cost patients.

The calculation is made before the trial begins, from assumptions about effect size and variability drawn from prior studies. If those assumptions prove incorrect (for example, if the drug produces a smaller effect than expected, or if outcomes are more variable), the trial may be underpowered to detect a real effect. Adaptive designs that allow sample size re-estimation based on interim data address this concern.

13.6. Master Protocols and Platform Trials

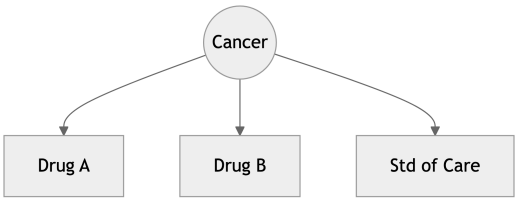
Traditional trials test one treatment in one disease, then shut down and start from scratch for the next question. It is like building a stadium for a single game and then demolishing it: every element (enrollment infrastructure, site contracts, database, regulatory submissions, DSMB) is built and then torn down. *Master protocols* reuse the arena: multiple therapies, or multiple diseases, share a single infrastructure, regulatory submission, database, and operational team (U.S. Food and Drug Administration 2022a).

Three designs operate under the master-protocol umbrella, and they differ in what is held fixed. A **basket trial** fixes the drug and varies the patient types, testing one agent across multiple diseases or subgroups that share a common molecular feature (a BRAF V600E mutation, for example). The design underpinned the first tissue-agnostic FDA approvals, such as pembrolizumab for mismatch-repair-deficient tumors and larotrectinib for NTRK-fusion cancers. Large exploratory baskets such

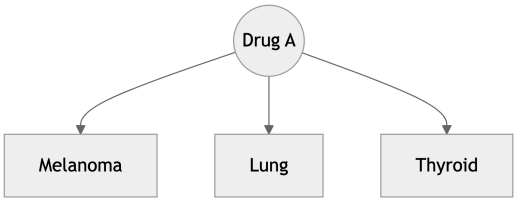
as NCI-MATCH, by contrast, were primarily signal-finding: across dozens of arms only a few showed meaningful activity.

An **umbrella trial** turns that around. The disease is fixed, the multiple is the treatments, and patients are assigned to arms on their biomarker profile. A **platform trial** fixes the disease and the arena and lets the therapies vary over time, in a perpetual or long-running infrastructure where arms enter and exit as data accumulate: arms read out and close while new ones enter, and the shared control arm continues throughout.

The terms are used loosely, and the documents that define them differ in wording. The Adaptive Platform Trials Coalition uses *adaptive platform trial* for a platform trial whose pre-specified adaptations (dropping or graduating arms, response-adaptive randomization, sample-size changes) are governed by decision rules fixed in advance (Coalition et al. 2019). The EC/EMA/HMA question-and-answer document on complex clinical trials defines a platform trial as a shared framework in which several investigational products enter and leave at different times on pre-specified decision rules, possibly in different diseases or conditions (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022). The treatment-specific module added to the master protocol goes by several names: *intervention-specific appendix* (ISA), sub-protocol, arm appendix, and, in REMAP-CAP, domain-specific appendix. Two systematic reviews map how widely basket, umbrella and platform designs have been used (J. J. H. Park et al. 2019; Meyer et al. 2020). What FDA and EMA expect of these designs is set out at Section 3.8.



Umbrella trial design



Basket trial design

13.6.1. Platform Trial Mechanics

Run five separate Phase III trials, each randomizing 80 patients to active and 80 to control, and 400 of the 800 total patients (50%) are on placebo, each serving as the comparator for only one of the five experimental drugs. Put the same five arms in one platform sharing a single control and the arithmetic changes: about 80 patients on active per arm plus 80 shared placebo is 480 total, a 40% reduction in total enrollment, while each active-versus-control comparison retains essentially the same power (Benjamin R. Saville and Berry 2016). That is the *shared control arm*, the key efficiency of the design, and the Healey ALS Platform Trial has demonstrated it at scale, running multiple concurrent active arms against one common placebo group.

A *master protocol* is the governing document that specifies the patient experience, eligibility criteria, endpoints, and the rules by which arms behave (Woodcock and LaVange 2017). It contains nothing specific to any treatment; treatment-specific information enters through modular appendices called substudies or protocol appendices. When a treatment arm closes, the master protocol is unchanged, and a new treatment joins through an amendment that adds its appendix, without rebuilding the rest of the trial. Like other significant amendments (see Section 13.8.1), that amendment needs ethics approval, and whatever regulatory submission or authorization the jurisdiction requires, before the new arm opens (U.S. Food and Drug Administration 2024b, 2022a). In I-SPY 2 (the breast cancer neoadjuvant platform), new agents can be added at any time as agents under test graduate or are dropped, so each enters a standing platform rather than a trial built for it (H. Wang and Yee 2019).

Timing is part of the architecture. FDA's revised draft guidance on master protocols (June 2026) asks the master protocol sponsor to submit amendments that substantively affect safety, quality or scope at least 30 days before the change starts. When all substudies sit under the master protocol IND, a new substudy is therefore submitted at least 30 days before it begins, and the regulatory project manager is notified at least 48 hours before such an amendment is submitted (U.S. Food and Drug Administration 2026e). In the EU, changes not covered by the initial protocol are documented as amendments and submitted as substantial modifications where required, and because a change to one sub-protocol may force changes to others, each sub-protocol should state how far it depends on the rest (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022). As of November 2026 the FDA document is a draft.

Response-adaptive randomization within platforms concentrates patients toward better-performing arms, reducing exposure to inferior treatments during the trial itself; its statistical mechanics and the conditions under which it is most beneficial are described at Section 12.8. The operational safeguard is a burn-in period of equal allocation followed by adaptive weighting governed by the platform's independent statistical center, with the sponsor blinded to comparative results across arms.

Governance follows from that need for separation. Typically a single DSMB oversees all arms simultaneously, an independent statistical center runs the unblinded interim analyses, and a committee approves applications for new arms; the arrangement exists so that no sponsor has visibility into how competitor arms are performing.

13.6.2. Concurrent and Non-Concurrent Controls

A shared control arm raises a question that separate trials never face: which control participants may be compared with a given experimental arm? *Concurrent* controls are those randomized while that arm was open. *Non-concurrent* controls were randomized before the arm entered, and they are tempting because they add patients at no cost. The danger is temporal drift. In a platform that runs for years, standard of care, the mix of participants, trial conduct and sites all change, so an arm that

opens late is compared with controls who were treated under different conditions (Dodd, Freidlin, and Korn 2021).

FDA's June 2026 draft goes one step further than "concurrent". It asks that the primary comparison for a drug use only concurrently eligible controls: participants who met that drug's eligibility criteria and could have been randomized to it but were randomized to control instead. Controls enrolled earlier, and controls enrolled at the same time who could not have received the drug (a kidney-toxic drug excludes some patients, or a site offers only some of the arms), should be left out. The draft states that using non-concurrent data can bias the estimate and change the type I and type II error probabilities even when the analysis tries to model a time trend, and that the shared control keeps most of the efficiency gain even when only concurrent controls are used. A sponsor who still wishes to use non-concurrent data should justify it in advance, weigh the amount of data and the calendar gap, pre-specify the method, and run sensitivity analyses that give the non-concurrent data less weight or none (U.S. Food and Drug Administration 2026e). The EMA draft reflection paper of October 2026 defines concurrent controls as those with a positive probability of being randomized to the arm of interest, states a strong preference for them, and says that non-concurrent controls are usually not acceptable for the primary analysis of the primary endpoint but may support supplementary analyses or rarely observed secondary endpoints (European Medicines Agency 2026b). Both documents are drafts as of November 2026.

Methods for using non-concurrent controls anyway are an active research area. The simplest adds the calendar period as a covariate in the model (Roig et al. 2022). The Bayesian time machine smooths the control response over time and borrows more from nearby periods (Benjamin R. Saville et al. 2022); FDA's draft Bayesian guidance cites its proposed use in the glioblastoma platform GBM AGILE and the pancreatic cancer platform Precision Promise, where the model tries to account for temporal shifts in outcomes such as tumor response (U.S. Food and Drug Administration 2026g); GBM AGILE's design is described at Section 12.5. Simulation studies and reviews examine when the added power is worth the risk of bias (Lee and Wason 2020; Roig et al. 2023; Jiang et al. 2023), and the R package ncc simulates and analyzes platform trials with non-concurrent controls (Krotka et al. 2023). The same Bayesian guidance lists the relevance of borrowed data as recency and changes in standard of care over time, which is the estimand drift a long-running platform has to confront. Where a Bayesian hierarchical model borrows across substudies or subgroups, it should justify exchangeability and include known effect modifiers, such as biomarker status, as covariates rather than treating the groups as interchangeable (U.S. Food and Drug Administration 2026g). Hierarchical borrowing is treated at Section 12.6, and calendar-time drift at Section 12.8.

Allocation to control. Because every arm is compared with the same control, an equal split is not the most powerful one. FDA's draft suggests giving the control arm more participants than any single drug arm, which raises power for each comparison at a given total sample size, while the chance that an individual participant is

assigned to control stays below that in a 1:1 two-arm trial (U.S. Food and Drug Administration 2026e). The EMA draft gives Dunnett's square root rule as one example: with k experimental arms, allocate in the ratio 1 to the square root of k for each experimental arm against the control, which also reduces the correlation between test statistics (European Medicines Agency 2026b). When arms enter and leave, the ratios change over time, and the analysis has to account for that; allocation that responds to outcomes is described at Section 12.8.

13.6.3. Multiplicity Across Arms

The first design question is what counts as the family of hypotheses. Comparisons of unrelated drugs with a shared control answer separate clinical questions that independent trials would not adjust for, so a master protocol does not by default control the error rate across all of its arms. Comparisons that feed one regulatory decision, such as several doses or formulations of one drug, the components of a combination, or several endpoints for one drug, form a family whose error rate the design has to control. The protocol therefore defines the families arm by arm and justifies them in advance; the FDA and EMA positions on where the line falls are set out at Section 3.8.

A shared control changes the shape of the error distribution without changing its mean. Because the test statistics are correlated through the common control, the probability of at least one false positive is lower than in independent trials, while the probability of two or more is higher, so a design should be judged on the distribution of the number of errors and not only on the chance of one. FDA notes that a larger control allocation reduces the risk that a poorly or unusually well performing control produces several correlated erroneous findings. EMA notes that holding the probability of multiple false positives down by tightening each arm's significance level would need thresholds stricter than Bonferroni, and offers alternatives such as more control participants (U.S. Food and Drug Administration 2026e; European Medicines Agency 2026b). The frequentist route to interim arm dropping is the multi-arm multi-stage (MAMS) design, in which arms are dropped for lack of benefit on an intermediate outcome and the familywise error rate is controlled by design (Parmar, Carpenter, and Sydes 2014); it has been compared with adaptive randomization in platform settings (J. Lin and Bunn 2017).

13.6.4. Case Studies

REMAP-CAP (Randomized, Embedded, Multifactorial Adaptive Platform trial for Community-Acquired Pneumonia) is the canonical pandemic-preparedness platform (Angus et al. 2020). Conceived after the 2009 H1N1 influenza pandemic and seeded in 2014 by the EU PREPARE (FP7) network, it enrolled its first non-pandemic CAP patient in April 2018 at a single center (UMC Utrecht) before expanding into a multi-country network, all while carrying a dormant pandemic stratum that could activate when a respiratory pathogen emerged. In February 2020,

the pandemic stratum activated for COVID-19. Over its life it investigated multiple therapies across several treatment domains: corticosteroids, anticoagulation, antivirals, immune modulators, and others. Its primary endpoint for the pandemic stratum was organ-support-free days over 21 days, an ordinal scale with in-hospital death as the worst outcome (REMAP-CAP Investigators, Gordon, et al. 2021). The trial demonstrated that a standing infrastructure built before a crisis can begin randomizing patients within weeks of that crisis. Its domain structure, adaptive allocation and results are described at Section 12.5.

The COVID anticoagulation multi-platform RCT took platform collaboration one step further (ATTACC Investigators et al. 2021; REMAP-CAP Investigators, ACTIV-4a Investigators, et al. 2021). Three independently funded Bayesian platform trials (REMAP-CAP, ATTACC in Canada, and ACTIV-4a at the NIH) asked the same question: therapeutic-dose versus prophylactic-dose anticoagulation in COVID-19. Conflicting results and a post hoc meta-analysis were the obvious risk, so early in the pandemic the three teams harmonized their protocols and statistical analysis plans and combined raw patient data in a single Bayesian hierarchical model with dynamic borrowing across severity strata. The analysis found benefit from therapeutic-dose anticoagulation in moderately ill patients and no benefit in critically ill patients, where the point estimate favored harm, a conclusion that a naive pooled analysis across severity groups could have missed (REMAP-CAP Investigators, ACTIV-4a Investigators, et al. 2021; ATTACC Investigators et al. 2021). The severity strata were specified in advance, so the finding did not rest on a post hoc subgroup analysis.

SNAP (Staphylococcus aureus Network Adaptive Platform) addresses a bacterium that causes hundreds of thousands of bloodstream-infection deaths per year worldwide, with bacteremia mortality that has remained high, around 20 to 30 percent, for decades (Tong et al. 2022). Enrolling across multiple hospitals and countries, SNAP uses three mutually exclusive patient silos by antibiotic susceptibility (penicillin-susceptible S. aureus, penicillin-resistant MSSA, and MRSA) crossed with multiple treatment domains (backbone antibiotic, adjunctive therapy, early oral switch). A single patient can contribute to multiple domains simultaneously, and domains can close, reopen for subpopulations, or add new questions from the network's investigators, including bacteriophage therapy, antiplatelet/anticoagulation, and diagnostic strategies. Each domain reaches its own conclusion inside the shared infrastructure, so SNAP can settle questions such as the preferred first-line agent for MSSA or the value of adjunctive clindamycin, then retire or modify that domain, without launching a standalone trial for each question. The MSSA backbone question illustrates the rationale: by 2018, several large observational cohorts had found cefazolin to give outcomes at least equivalent to flucloxacillin for MSSA, yet, as the authors of the largest stressed, only a randomized trial could convincingly settle the comparison (Davis, Turnidge, and Tong 2018).

PANTHER (the Precision medicine Adaptive Network platform Trial in Hypoxaemic acutE Respiratory failure) illustrates precision-medicine enrichment: patients

are pre-classified by inflammatory subphenotype (hyper- versus hypo-inflammatory) before randomization, addressing the long-standing problem that ARDS encompasses heterogeneous biology with potentially different treatment-response profiles. Funded by the UK's NIHR as a Bayesian adaptive platform, it began with arms testing simvastatin and baricitinib against usual care, the enrichment intended to concentrate each treatment where it is most likely to work (PANTHER Trial Investigators 2024).

The Healey ALS Platform Trial at Massachusetts General Hospital uses two-level nested randomization: patients are randomized to one arm (not blinded across arms) but are blinded to active versus placebo within their assigned arm (Quintana et al. 2023). All patients eligible for multiple arms who receive placebo contribute their data to every eligible arm's primary analysis. The first arms were seeded by fundraising from the ALS community, illustrating a first-arm effect in platform economics: once enrollment and data collection are shown to work, later arms reuse that infrastructure instead of building their own, which lowers the cost and risk of joining for the sponsors behind them.

EPAD (European Prevention of Alzheimer's Dementia) is the canonical platform failure (Ritchie et al. 2016). A 64-million-euro, 5-year IMI-funded project, EPAD built a 2,000-participant readiness cohort with deep pre-characterization (amyloid status, cognitive baselines, longitudinal follow-up) specifically to reduce the very high screen-failure rates that plague Alzheimer's prevention trials. Yet EPAD never enrolled a single treatment arm. By the project's own post-mortem (Saunders et al. 2022), two obstacles proved decisive: intervention owners were ultimately reluctant to hand sponsorship of a critical asset to a university-run platform, and the stakeholders could never agree on a common legal framework usable and acceptable to all parties. When the project's fixed five-year term ended in 2020, the treatment platform had never opened, and the COVID-19 pandemic seriously hampered efforts to secure ongoing funding. The rest of the investment outlived the project: the cohort's data and biobank are freely available to researchers, with follow-on funding from the Alzheimer's Association to maintain access, and the unused platform infrastructure could be reopened or replicated (Saunders et al. 2022). The contrast with Healey ALS is precise: Healey could fund the first arms from philanthropic donations; EPAD could not. The lesson: a platform's first arm must be fundable independently of the commercial pipeline it is meant to serve.

STAMPEDE (Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy) is a UK platform in advanced prostate cancer run from the MRC Clinical Trials Unit at UCL, and an example of the frequentist alternative to the Bayesian platforms above. It is a multi-arm, multi-stage design: experimental arms share one control group on standard hormone therapy, each is assessed at interim looks on an intermediate outcome, and new arms were added as earlier ones reported (N. James 2013; Parmar, Carpenter, and Sydes 2014). Docetaxel improved survival in the first comparisons, and abiraterone with prednisolone, which joined later, reduced the hazard of death in men starting long-term hormone therapy

(hazard ratio 0.63) (Nicholas D. James et al. 2016; Nicholas D. James et al. 2017); Section 12.1.1 places STAMPEDE among the multi-arm multi-stage designs. Its lesson is that a long-lived platform does not need Bayesian machinery: a common control, pre-specified stopping rules and a standing infrastructure were enough.

RECOVERY (Randomised Evaluation of COVID-19 Therapy) is a UK platform in patients hospitalized with COVID-19, built to be large, pragmatic and open-label, with deliberately little data collection beyond what the health service already records and 28-day mortality as the primary outcome. Its first major result was that dexamethasone reduced 28-day mortality, by about one fifth overall (rate ratio 0.83), by more among patients on invasive mechanical ventilation (0.64), and not at all among patients who needed no oxygen (RECOVERY Collaborative Group et al. 2021). It shows what a simple, high-volume platform can do: a clear answer for a cheap, widely available drug within months of the first patient.

PANORAMIC took the same pragmatic approach into UK primary care. The open-label, platform-adaptive trial randomized adults aged 50 or older, or 18 or older with relevant comorbidities, who had been unwell with confirmed COVID-19 for five days or fewer, to molnupiravir plus usual care or usual care alone. Between December 2021 and April 2022 it randomized 26,411 participants, 94% of whom had had at least three vaccine doses. Hospitalization or death within 28 days occurred in about 1% of each group (adjusted odds ratio 1.06), so molnupiravir did not reduce it (Butler et al. 2023). A platform that can enroll that many community patients in under five months can give a clear negative answer on a hard outcome, which is as useful to health systems as a positive one.

Table 13.3 sets the designs in this chapter side by side, with the trade-off each one accepts in exchange for what it buys.

Table 13.3.: Clinical trial design taxonomy: key trade-offs

Design type	Randomization	Key strength	Key limitation	Regulatory precedent
Standard RCT	Concurrent, 1:1 or other fixed ratio	Eliminates confounding; gold standard for causation	Enrollment cost; narrow eligibility limits generalizability	Established across all therapeutic areas
Platform trial	Concurrent; adaptive across arms over time	Shared control arm efficiency; arms enter and exit; perpetual infrastructure	Governance complexity; requires pre-funded first arm; statistical modeling of concurrent arms	REMAP-CAP, Healey ALS, RECOVERY; FDA master protocols guidance 2022

Design type	Randomization	Key strength	Key limitation	Regulatory precedent
Externally controlled trial	None (single arm vs. historical or registry control)	Feasible when concurrent control is unethical or impossible; all patients receive investigational drug	Confounding; population drift; regulatory scrutiny; requires rigorous fit-for-purpose data	FDA 2023 draft ECA guidance; rare pediatric oncology
Hybrid Bayesian (augmented control)	Reduced concurrent control; Bayesian borrowing from external data	Smaller control arm; preserves causal inference of randomization	Prior sensitivity; dynamic borrowing assumptions require pre-specification; inflation risk if historical data mismatch	Pediatric programs (MAP prior); FDA Complex Innovative Trial Design program
Registry-based RCT	Concurrent; executed within registry infrastructure	Broad population; low operational cost; representative outcomes	Outcome ascertainment quality varies; registry must collect primary endpoint reliably	TASTE (NEJM 2013); VALIDATE-SWEDEHEART; Scandinavian cardiac registries

13.7. The Statistical Analysis Plan

Before data from a clinical trial are unblinded, the *statistical analysis plan (SAP)* should be finalized. Modern SAPs must align with the **ICH E9(R1) Estimands Framework** (International Council for Harmonisation 2019), which requires precise definition of the treatment effect of interest *before* data is collected.

The estimand has five attributes: the treatment condition being compared, the population being analyzed, the specific endpoint variable, the strategies for handling intercurrent events (disruptions such as treatment discontinuation, death, or the use of rescue medication), and the population-level summary used to compare the treatments. ICH E9(R1) defines five intercurrent-event strategies. The three most commonly invoked are the treatment policy strategy, which follows the intent-to-treat principle by using the outcome regardless of intercurrent events such as nonadherence; the hypothetical strategy, which estimates the outcome in a specified scenario in which the intercurrent event would not occur (for example, if rescue medication were not available); and the composite strategy, which folds the intercurrent event into the endpoint definition itself. The remaining two are the principal stratum strategy, which restricts the estimate to a causally defined subpopulation, and the

while-on-treatment strategy, which uses the outcome only up to the intercurrent event (*while alive* when the event is death).

A p-value of 0.05 counts as confirmatory evidence only if the test was specified in advance and the Type I error is controlled across the trial's confirmatory tests; an unplanned test can suggest a hypothesis but not confirm one (International Council for Harmonisation 1998b).

13.8. The Protocol

Every design decision lands in the *protocol*, the document that governs how the trial is conducted. A well-written one leaves little room for interpretation: exactly who can participate, what treatments will be given and how, what assessments will be performed and when, and how data will be collected and analyzed.

Writing the protocol is a project in its own right. Development usually starts from a synopsis of the key design choices, detailed enough to seek investigators' views and to request bids from CROs (see Chapter 20), and grows into a full draft through rounds of review by the functions that will have to run the trial. ICH E6(R3) asks for a protocol that is "clear, concise and operationally feasible", designed to minimize unnecessary complexity and developed with input from relevant interested parties (International Council for Harmonisation 2025b). Two public standards shape the document itself. SPIRIT 2025 is a checklist of 34 minimum items that the protocol of a randomized trial should address, accompanied by a diagram of the schedule of enrollment, interventions, and assessments; one of its new items asks how patients and the public will be involved in the trial's design, conduct, and reporting (A.-W. Chan et al. 2025). ICH M11, adopted in November 2025, provides a harmonized protocol template and a technical specification, so that protocols present the same information in the same place and can be exchanged electronically in a format acceptable to regulators (International Council for Harmonisation 2025c). Sign-off by the sponsor's authorized signatories makes it final (International Council for Harmonisation 2025b).

The protocol is read well outside the sponsor. IRBs review it to assess participant protection, regulatory authorities to assess scientific adequacy, and investigators to understand what they are responsible for. Once approved, deviations from it are tracked and reported.

AI tools for protocol authoring, feasibility screening, and design optimization are covered in Chapter 26.

13.8.1. Protocol Amendments

Once the protocol has been approved, it changes only by *protocol amendment*, which ICH E6(R3) defines as "a documented description of a change(s) to a protocol".

Throughout the guideline, "protocol" means the protocol together with its amendments, and each amendment carries its own number and date. New safety information can force an amendment at any point: after an immediate hazard, the sponsor is expected to consider whether the protocol needs amending (International Council for Harmonisation 2025b). Sponsors most often cite regulatory agency requests and changes to study strategy as their reasons (Getz et al. 2024).

Regulation decides which changes need the formal route. For a trial under an IND, FDA requires an amendment for any change to a Phase I protocol that significantly affects participant safety, and for any change to a Phase II or III protocol that significantly affects safety, the scope of the investigation, or the scientific quality of the study. Its examples include raising the dose or the duration of exposure beyond the current protocol, significantly increasing the number of participants, adding or dropping a control group, and adding or dropping a test that monitors safety (U.S. Food and Drug Administration 2024b). The EU sets its threshold at a *substantial modification*, a change "likely to have a substantial impact on the safety or rights of the subjects or on the reliability and robustness of the data generated in the clinical trial" (European Parliament and Council of the European Union 2014).

Any change to approved research needs IRB approval before it is implemented, apart from the exceptions below, although minor changes can go through expedited review by the IRB chair or an experienced member the chair designates (21 CFR 56.110) (U.S. Food and Drug Administration 2023c; International Council for Harmonisation 2025b). The regulator's role differs by jurisdiction. Under an IND, FDA approval is not a precondition: the sponsor may make the change once it has been submitted to FDA and approved by the IRB, in either order, and a new investigator can be added without waiting for FDA, which must be notified within 30 days (U.S. Food and Drug Administration 2024b). In the EU, a substantial modification may be implemented only after it has been approved, and the Regulation spells out that this includes adding a trial site or changing a site's principal investigator (European Parliament and Council of the European Union 2014).

Two kinds of change may go ahead first. A change needed to eliminate an immediate hazard to participants is made at once: the investigator informs the sponsor promptly and reports the hazard, the change, and any proposed amendment to the IRB and, where applicable, to regulators (International Council for Harmonisation 2025b), and under an IND the sponsor then notifies FDA by protocol amendment (U.S. Food and Drug Administration 2024b). A change that involves only logistical or administrative aspects of the trial, such as a new monitor or telephone number, needs no prior IRB approval where the applicable regulations allow it (International Council for Harmonisation 2016a, 2025b).

Approval starts the rollout, and neither step is quick. Each site moves to the new version only when its own approvals are in place, and E6(R3) holds trial systems to the same rule: updates resulting from an amendment are released to a site only after that site has all the approvals it needs. Participants already enrolled should be told promptly of new information that may bear on their willingness

to continue, and where re-consent is needed, the revised consent materials go to the IRB for approval before use (International Council for Harmonisation 2025b; U.S. Food and Drug Administration 2023b). Site staff need training on the new version before they apply it, and trial-specific plans are revised where the change touches them (see Section 20.3.2); the signed amendment and its approvals go into the essential records (see Chapter 24). In Tufts CSDD's 2022 benchmark, the time from identifying the need to amend to the last oversight approval averaged 260 days, and sites worked under different versions of the protocol for a mean of 215 days (Getz et al. 2024). Chapter 20 follows the operational chain through randomization systems, data capture, and drug supply.

Amendments have also become more frequent. In a Tufts CSDD study published in 2016, 57% of protocols had at least one substantial amendment, 45% of those amendments were judged avoidable, and the median direct cost of implementing one was $141,000 for a Phase II protocol and $535,000 for a Phase III protocol (Getz et al. 2016). The 2022 follow-up found at least one amendment in 76% of Phase I-IV protocols and a mean of 3.3 amendments per protocol, up from 2.1, while a much larger share of amendments, 77%, was deemed unavoidable (Getz et al. 2024). Some flexibility can be designed in. E6(R3) suggests acceptable ranges for specific protocol provisions, which can reduce deviations and sometimes remove the need for an amendment, provided the flexibility does not compromise participant safety or scientific validity (International Council for Harmonisation 2025b). Frequent changes also raise questions about the scientific basis of the trial. Chapter 6 takes up the costs, and Chapter 15 the protocol complexity behind many of them.

Discussion Questions

1. A sponsor is developing a gene therapy for a rare pediatric muscle disease affecting approximately 300 patients in the United States annually. No approved therapy exists, and the natural history is well-characterized in a prospective registry spanning 12 years. The FDA has suggested a single-arm trial using the registry as the external control. What properties of the registry data must be evaluated before accepting this design, and what analytic safeguards would you pre-specify to protect against population drift between the historical registry cohort and the current trial population?

2. Platform trials offer substantial efficiency gains but can fail for reasons unrelated to the science. Drawing on the EPAD experience and the contrast with Healey ALS, identify three structural preconditions that must be in place before a platform trial is feasible, and describe one additional failure mode not illustrated by EPAD.

3. Suppose it is 2025 and a consortium is designing an Alzheimer's prevention platform targeting cognitively normal adults with elevated amyloid by PET. Applying the lessons from EPAD's collapse, what funding model, governance structure, and first-arm strategy would you recommend, and how would you

handle screen failure rates given that amyloid PET capacity may be limited outside academic centers?

4. A cardiovascular outcomes trial designed around a time-to-first-event endpoint (composite of MI, stroke, and cardiovascular death) reaches a planned interim analysis at 60% of target events. Observed event rate is 30% lower than assumed in the power calculation. The DSMB must decide whether to recommend continuing to the original event count, revising the target upward, or stopping for futility. What information is needed to make this decision, and what are the statistical and operational consequences of each path?

5. A late-stage diabetic kidney disease trial finds that 45% of participants have stopped randomized treatment by 36 months, substantially higher than the 25% assumed at design, and some of them have also stopped attending study visits. The primary analysis uses a treatment policy estimand (outcomes measured regardless of treatment discontinuation). The sponsor proposes to also report a while-on-treatment analysis as a key secondary. What are the interpretive differences between these two estimands in this setting, and what sensitivity analysis would you require to evaluate the robustness of the primary result?

14. Real-World Evidence and Evidence Integration

Breast cancer in men runs to roughly 2,670 U.S. cases a year. When a drug approved in women with hormone receptor-positive metastatic disease needs a label extension to cover men with the same disease, there are too few of them to power a randomized trial. The randomized controlled trial is the most reliable instrument for establishing that a treatment causes its observed effect. It is not always available, and it is not always enough. A vaccine approved in a clinical trial of healthy volunteers still needs its effectiveness measured across the diverse population receiving it in community pharmacies; a drug studied in a three-year trial needs its long-term safety assessed over a decade; a rare pediatric disease may have only a few hundred patients worldwide, none of whom can ethically be assigned to placebo.

These are not edge cases. They are a large and growing fraction of the evidence questions drug development has to answer, and over fifty years the response has hardened from informal practice into a regulated discipline.

Real-world evidence (RWE) is clinical evidence about the usage and potential benefits or risks of a medical product derived from analysis of *real-world data* (RWD): data collected outside of traditional randomized clinical trials, including electronic health records, insurance claims, disease registries, patient-reported outcomes, and wearable devices. RWD is the raw material; RWE is what rigorous analysis of it produces. Conversation runs the two together, but the analytic step between them is the whole difference.

14.1. The Regulatory Framework

Regulatory acceptance of real-world evidence was built statute by statute and guidance by guidance, and it still varies by jurisdiction.

14.1.1. Statutory Authority and FDA's Program

The modern U.S. basis for RWE is Section 3022 of the 21st Century Cures Act (2016), which directed FDA to evaluate the potential use of RWE to support approval of new indications and to satisfy post-approval study requirements. The mandate was specific: FDA was required to establish the program, publish a draft framework for it within two years of enactment, and issue guidance for industry on the circumstances in which RWE may be used.

FDA published its *Framework for FDA's Real-World Evidence Program* in 2018, establishing two evaluative dimensions for any RWD source. *Relevance*: does the data cover the population, setting, outcomes, and time period pertinent to the regulatory question? *Reliability*: is the data complete, accurate, and consistent, with sufficient provenance and quality control to be trusted? They have organized FDA's RWE reviews since.

Table 14.1 lists the suite that has accumulated since 2018. One of those documents governs this chapter above the rest: the 2023 draft guidance on externally controlled trials, which sets out what FDA recommends a sponsor document when RWD or data from other trials supply the control arm. Under the PDUFA VII (Prescription Drug User Fee Act) commitments, FDA's "Advancing Real-World Evidence Program" targets RWE for *effectiveness* labeling changes, the historically most resistant application.

Table 14.1.: FDA Real-World Evidence Guidance Documents (selected)

Year	Document	Scope
2018	Framework for FDA's RWE Program	Overarching; defines RWD/RWE, relevance/reliability criteria
2022	Submitting Documents Using RWD/RWE to FDA	Procedural: flagging RWD/RWE use in submission cover letters
2023	Externally Controlled Trials for Drug and Biological Products	ECA design, comparability, bias mitigation (draft)
2023	Considerations for Using RWD and RWE	Overarching study design and data quality
2023	Assessing Registries to Support Regulatory Decision-Making	Disease/product registry fit-for-purpose assessment
2024	Assessing EHR and Medical Claims Data	EHR/claims data quality recommendations (finalizes 2021 draft)
2024	Non-Interventional Studies for Drug and Biological Products	Observational study design expectations (draft)
2024	Integrating RCTs Into Routine Clinical Practice	Embedded/pragmatic trial design (draft)

14.1.2. EMA and International Harmonization

The EMA's approach leans infrastructure-centric. Alongside sponsor-facing documents such as its guideline on registry-based studies, EMA established **DARWIN EU** (Data Analysis and Real World Interrogation Network) as a coordination center

for regulatory-led RWD studies, pulling from databases across European healthcare systems: the UK's CPRD (Clinical Practice Research Datalink, >60 million patient records), Spain's BIFAP, Italy's Pedianet, and the Danish national registries (European Medicines Agency 2026a). EMA scientific committees commission the studies and feed the results to national competent authorities and HTA bodies. Sponsors cannot commission one directly.

EMA's April 2024 guide (EMA/152628/2024) describes how regulators can request RWD studies and what study types are available. In parallel, ICH adopted its first harmonization signal on RWE in June 2024: a reflection paper on "Pursuing Opportunities for Harmonisation in Using Real-World Data to Generate Real-World Evidence."

ICH E6(R3) (2025) (International Council for Harmonisation 2025b), the updated Good Clinical Practice guideline, addresses hybrid designs that incorporate EHR and RWD within interventional trials, extending principles of data integrity and traceability to these sources.

Fit for purpose. Both FDA and EMA converge on *fit for purpose* as the evaluative standard, and the weight falls on *purpose*. A registry dataset that serves as an external control for a natural history comparison in a rare pediatric disease may be unfit as the primary comparator for a Phase III efficacy claim in a common chronic condition.

The assessment adds a third test to relevance and reliability: analytic validity, whether the proposed analysis can extract the target estimand from the available data without biases that exceed what is acceptable for the regulatory purpose. Sponsors who engage with FDA early, through pre-submission meetings under the Advancing RWE Program, are encouraged to settle the data source and analysis plan there rather than presenting RWE for the first time in a marketing application.

14.1.3. HTA and Payer Frameworks

Regulatory approval and market access are separate decisions with separate evidence standards. FDA asks whether a drug is safe and effective for the proposed indication; HTA bodies such as NICE (National Institute for Health and Care Excellence, UK) ask whether it is cost-effective relative to standard of care, for a specific patient population in a specific healthcare system. The two questions call for different evidence, and they make different demands on RWE (Table 14.2).

Table 14.2.: Regulatory vs. HTA evidence requirements

Dimension	Regulatory (FDA/EMA)	HTA/Payers (NICE, payers)
Central question	Does it work under trial conditions or real-world use?	Is it cost-effective in this health system?

Dimension	Regulatory (FDA/EMA)	HTA/Payers (NICE, payers)
Comparator	Placebo or trial-specified active comparator	Actual standard of care in that country
Time horizon	Trial duration	Long enough to capture all important cost and outcome differences, often a lifetime extrapolated beyond the trial (National Institute for Health and Care Excellence 2022)
Population	Label indication	Patients who will receive reimbursement
RWE role	Label expansion, post-approval commitments	Fills long-term effectiveness gaps; informs HTA models

NICE's RWE framework (ECD9, June 2022) provides structured guidance for RWE studies submitted in technology appraisals. Outcomes-based *managed entry agreements* (MEAs), reimbursement made conditional on accumulation of real-world evidence confirming effectiveness in routine practice (as distinct from the finance-based MEAs that turn on discounts and price-volume caps), have become common in Europe. Their designs vary. Under coverage with evidence development, the payer covers the drug for a limited period while an agreed study collects data, then maintains, extends, or withdraws coverage, or adjusts the price, according to the results; under payment-by-results schemes, payment or refunds depend on whether each patient reaches a pre-specified response (Wenzl and Chapman 2019). When one study is meant to serve both the regulator and the HTA body, it has to satisfy both sets of evidentiary standards at once.

14.2. Data Sources

A data source is defined as much by what it leaves out as by how many patients it holds.

Electronic Health Records are the richest source of clinical detail: diagnoses, procedures, medications, laboratory values, vital signs, and clinical notes. Their coverage, though, is bounded by the patient's engagement with a particular healthcare system: a patient who sees a cardiologist at a different hospital does not appear in the EHR. U.S. EHR networks of regulatory-grade scale include Flatiron Health (>5 million oncology patients across ~220 practices, with structured abstraction of tumor response and genomic data), Epic Cosmos (>200 million patients across Epic-using health systems), and PCORnet. In the UK, CPRD supplies >60 million patient records from general practice. EHR data are collected for clinical care, not research: coding completeness and measurement fidelity are driven by billing incentives and workflow conventions, not study protocols.

Administrative claims and insurance data capture billing transactions: ICD diagnosis codes, procedure codes, drug dispensing (NDC codes), and enrollment periods. Coverage is population-level for the insured population, billed services are captured throughout enrollment (care paid out of pocket or by another payer is missed), and follow-up can be long. U.S. sources include Medicare (CMS, covering >65 million beneficiaries), Medicaid, and commercial claims (e.g., Optum Clinformatics, Merative MarketScan, and IQVIA). The tradeoff is clinical depth: claims capture what was billed, not what the patient's hemoglobin was or how advanced their tumor was.

Disease and condition registries are purpose-built systems, prospective or retrospective, that collect uniform data on a population defined by a disease, exposure, or condition. On data quality they sit between clinical trials and EHR: entry criteria are defined, data elements standardized, outcomes often adjudicated. The Surveillance, Epidemiology, and End Results (SEER) program covers ~48% of the U.S. population across its core registries. The Scandinavian cardiac registries (SCAAR, SWEDEHEART) have been the foundation for large-scale registry-based randomized trials. Natural history registries (the Pompe Registry, the Cystic Fibrosis Foundation Patient Registry, rare disease-specific databases) supply the comparator population for single-arm trials in rare disease.

Linked databases combine two or more data sources using patient identifiers. SEER-Medicare links cancer diagnoses with Medicare claims for fee-for-service enrollees and encounter records for Medicare Advantage enrollees, which supports studies of clinical outcomes and health service utilization together. Flatiron Health linked its EHR with Foundation Medicine's comprehensive genomic profiling (Flatiron+FMI), enabling studies of genomic markers as predictors of real-world treatment outcomes. Whether a linkage is deterministic (exact identifier match) or probabilistic, and what its error rate is, feeds directly into bias.

Global RWD networks enable federated analysis across many institutions. The FDA's **Sentinel System** maintains over 1.3 billion person-years of mostly U.S. administrative claims data (covering more than 371 million unique patient identifiers as of April 2024; a patient who changes health plans can hold more than one) for active post-market safety surveillance, and is now expanding to some effectiveness analyses (Sentinel Operations Center 2025); DARWIN EU serves the analogous function for EMA. The OHDSI/OMOP network is a global open-science consortium of 544 data sources across 54 countries representing >974 million unique patient records, all standardized to the OMOP Common Data Model (Ryan 2024). Standardizing the vocabulary (SNOMED, RxNorm, LOINC) and the table structure is what lets identical analysis code run across distributed sites without patient data leaving any institution. PCORnet, the U.S. patient-centered clinical research network, links 78 U.S. health systems, with more than 13,000 hospitals and other care sites, on its own Common Data Model, which was originally based on the FDA's Mini-Sentinel Common Data Model (Cohen et al. 2026; PCORnet 2025).

Patient-level data sharing platforms open IPD (individual patient data) from completed clinical trials, the raw material for an external control built from historical trial patients. Vivli hosts >6,000 clinical trials; the YODA Project (Yale Open Data Access) provides third-party review of data sharing requests for J&J and Kenvue datasets; Project Data Sphere focuses on oncology trial control arm data. These data came out of trials, under standardized protocols, unlike EHR or claims.

14.2.1. Outcome Ascertainment: The Critical Quality Issue

A structural weakness of most RWD sources is that outcomes were not designed to be measured. The diagnosis code for "disease progression" in an EHR reflects what a physician documented for billing purposes, not a protocol-specified assessment against RECIST criteria at a pre-specified timepoint. The gap between *documented* and *protocol-adjudicated* outcomes is why real-world endpoints carry their own names: rwOS (real-world overall survival), rwPFS (real-world progression-free survival), and rwTR (real-world tumor response). These are not equivalent to their trial-measured counterparts.

Friends of Cancer Research ran multi-partner pilots that standardized the definitions of these endpoints so that they can be measured the same way across RWD sources; its 2023 considerations call for evidence that the dataset is fit for purpose when RWD support a regulatory submission and urge caution in comparing real-world endpoints with trial endpoints, because the populations and measurements differ (Friends of Cancer Research 2023). Flatiron Health has published validation studies comparing rwOS and rwPFS against clinical trial outcomes across multiple tumor types.

Phenotyping algorithms (computable definitions of conditions, exposures, and outcomes applied to RWD) require validation against a reference standard, often medical record review, sometimes an established external source such as national death records (U.S. Food and Drug Administration 2024k). The positive predictive value of ICD coding varies widely by condition and again across care settings: in one systematic review, it was 86-98% for hip fracture when hospital and physician claims were combined, but below 60% for osteoporosis diagnoses (Hudson et al. 2013). Non-differential outcome misclassification usually biases effect estimates toward the null, although with perfect specificity the risk ratio can be left unbiased; differential misclassification, where the rate differs between treatment groups, can bias in either direction. A serious RWE study reports the algorithm, its validation statistics against the reference standard (PPV, sensitivity, and specificity), and a sensitivity analysis for outcome misclassification.

14.3. Study Design Taxonomy

The designs below run from no randomization at all to randomization executed inside a national registry. The clinical question fixes the estimand (International

Council for Harmonisation 2019); where a study sits on that range determines what must be assumed to estimate it, and those assumptions, together with data quality and study conduct, set how much weight a regulator can give the result (U.S. Food and Drug Administration 2018a).

14.3.1. Observational Comparative Effectiveness Research

In observational CER both treatment groups are drawn from RWD and nothing is randomized: patients who received Drug A in routine clinical practice are compared with patients who received Drug B. The central challenge is confounding: patients and their physicians chose their treatments for reasons that may also predict outcomes. Propensity scores, outcome regression, and doubly robust estimators attempt to control for the measured confounders; the unmeasured ones remain the irreducible limitation.

Observational CER is most credible when the outcome is objective and reliably captured in routine care (overall survival is better than symptom scores), the comparison is between two established alternatives rather than between treated and untreated, which concentrates confounding by indication, and the database is large enough to achieve covariate balance after matching or weighting. Beyond that, sensitivity analyses have to demonstrate that the result is robust to plausible unmeasured confounding.

Natural history and registry studies. Natural history studies characterize disease progression, event rates, and mortality in untreated or standard-of-care populations. They do not compare treatments. In rare disease drug development a well-designed natural history study is often the indispensable prerequisite for a single-arm trial: without a reliable characterization of the untreated disease course, the single arm's outcomes cannot be interpreted against anything. FDA's 2023 guidance on registry assessment sets out the data elements, follow-up duration, and outcome adjudication a registry needs to serve as a regulatory-grade external control comparator.

14.3.2. External Control Arms and Synthetic Control Arms

An **external control arm** (ECA) replaces the concurrent randomized control arm with a population assembled from outside the trial: everyone in the trial receives the investigational treatment, and their outcomes are compared with a pre-specified external cohort drawn from historical trials, registries, EHR, or claims. The FDA's 2023 guidance on externally controlled trials (U.S. Food and Drug Administration 2023e) treats ECAs as most appropriate where concurrent randomization is infeasible or unethical, the disease has a well-characterized natural history, outcomes can be measured reliably in the external data source, and the comparison population is demonstrably similar to the trial population at baseline.

A **synthetic control arm** (SCA) is the version in which patient-level data from historical trials, often proprietary datasets, are statistically matched to the current

trial's population by propensity score or covariate-matching algorithms. Medidata's platform, drawing on >38,000 trials and 12 million patients, is one of the largest commercial SCA systems. In a widely cited example from 2020, Medicenna Therapeutics gained FDA support to use a synthetic control arm within a hybrid Phase 3 registrational design for MDNA55 in recurrent glioblastoma, in which matched external controls supply roughly two-thirds of the pooled control arm, reducing the number of patients randomized to standard of care (Medicenna Therapeutics Corp. 2020; Medidata Solutions 2024d). SCAs drawn from historical RCT patient data are generally stronger than ECAs from EHR or claims because the data were collected under controlled conditions.

14.3.3. Hybrid Randomized Designs and Bayesian Augmented Controls

Hybrid designs randomize a *reduced* concurrent control arm while borrowing information from external data to increase effective sample size. The borrowing runs through Bayesian methods that "discount" the external data according to how consistent it is with the emerging concurrent control data, and three frameworks dominate.

The power prior (Ibrahim and Chen 2000) modifies the likelihood of the historical data by a parameter $\alpha \in [0, 1]$: $\alpha = 0$ borrows nothing, $\alpha = 1$ fully pools historical and current data. A fixed α chosen before the trial controls the degree of borrowing; a data-adaptive α estimated from the data consistency is more robust to historical-current heterogeneity but introduces operational complexity.

The Meta-Analytic-Predictive (MAP) prior (Neuenschwander et al. 2010) derives the prior from a meta-analysis of multiple historical studies, accounting for between-study heterogeneity through a hierarchical model. The resulting prior on the control event rate carries both the central estimate from historical data and the uncertainty about how far the current population may differ. MAP priors are implemented in the `RBesT` R package and have been used in pediatric trials, where adult trial data informs the prior for smaller pediatric randomized comparisons.

Commensurate priors (Hobbs et al. 2011) make the amount of borrowing a function of the observed similarity between historical and current control arm data, updating as current data accumulates: when the two are concordant, borrowing is high; when they diverge, it shrinks. That adaptation is the protection against the principal risk of Bayesian borrowing: bias introduced by over-weighting historical data that does not represent the current population.

Mixture and elastic priors. A mixture prior combines an informative component built from the external data with a vague component, and the mixture weights can be read as the probability that the external data apply to the current trial. When the concurrent data conflict with the external data, the weight moves to the vague component and borrowing falls. The hard part is choosing the weights, which set the strength of borrowing. Elastic priors work in two steps: a congruence measure quantifies how similar the external and current data are, and an elastic function

maps that measure to a degree of borrowing between none and full. FDA's January 2026 draft guidance describes both, alongside the power prior, its commensurate and supervised extensions and Bayesian hierarchical models (U.S. Food and Drug Administration 2026g); the mixture, elastic, commensurate and supervised approaches are dynamic, borrowing less as the data diverge, and the guidance's comparison of static and dynamic discounting is summarized at Chapter 3.

How much effective sample size the borrowed data contribute depends on the discount applied. Regulatory agencies expect pre-specified borrowing parameters, with operating characteristics evaluated through simulation across scenarios of historical-current concordance and discordance. FDA's January 2026 draft guidance on the use of Bayesian methodology in trials of drugs and biological products (U.S. Food and Drug Administration 2026g) speaks directly to this setting, asking that borrowed information be valid, reliable, and interpretable, judged through five factors that include data quality, pre-specification of the borrowing, and the relevance and design of the contributing studies.

14.3.4. Justifying Borrowed Information

The general rules for choosing a prior (noninformative and skeptical priors, effective sample size, sensitivity analyses) are at Section 12.6.1. Borrowing external data adds a burden of its own. FDA's January 2026 draft guidance (a draft as of November 2026) asks for a strong justification for an informative prior that borrows external information, one that considers the feasibility of approaches that do not borrow and the relevance of the available information; it names pediatrics and rare diseases as the areas where such priors have most often been proposed (U.S. Food and Drug Administration 2026g). A sponsor should show that the external information is relevant (a similar population, endpoint, treatment conditions and handling of intercurrent events, comparable measurement, recent data and no important change in standard of care), that it was not cherry-picked (inclusion standards set before a systematic search, and a documented check that favorable studies were not preferentially selected) and that it rests on sound designs (borrowing from randomized comparisons relies on fewer and more plausible assumptions than borrowing a historical control arm). The guidance says the information should typically come from patient-level data, which allow the relevance of the external data to be examined and covariates to be adjusted for when eligibility criteria do not match.

The model then has to state how the external and current data relate. The guidance describes four kinds of assumption: equal relevance (all patients in one trial analyzed together, which needs scrutiny when adolescents are enrolled with adults), exchangeability (as in the hierarchical models of Section 12.6), an assumed bias parameter, and functional dependence (for example a pharmacometric model that predicts pediatric response from adult studies). Some discounting is usually needed on top. In pediatric extrapolation, undiscounted adult results would often meet the success criterion on their own and overwhelm the pediatric data whatever they showed, so the guidance suggests a prior centered on a benefit similar to the adult

one but with greater variability (U.S. Food and Drug Administration 2026g). The simulations that support the design should include prior-data conflict, down to no effect in the target population, and the influence of the prior is reported as its mean and effective sample size in the way described at Chapter 3.

14.3.5. Registry-Based Randomized Controlled Trials

A **registry-based RCT** (RRCT) conducts randomization within an existing clinical registry and uses it for enrollment, baseline data, and some or all outcomes; some RRCTs rely on registries alone for outcomes, while others add trial-specific follow-up and adjudication. The TASTE trial (2010-2013) randomized 7,244 patients with ST-elevation myocardial infarction to thrombus aspiration versus conventional PCI within the SCAAR Swedish cardiac registry; outcomes were collected from registry records without source data verification. The VALIDATE-SWEDEHEART trial compared bivalirudin versus heparin in myocardial infarction within the same registry infrastructure but added nurse telephone follow-up at 7 and 180 days and blinded central adjudication of the primary end point (Erlinge et al. 2017, 2016). Both appeared in the *New England Journal of Medicine*, combining the operational efficiency of registry data collection with the inferential validity of randomization.

RRCTs are most powerful where an existing registry covers the relevant disease, collects the primary outcome reliably enough to carry the comparison, and can randomize at the point of care through its own interface. The Scandinavian registries (supported by universal personal identification numbers enabling linkage across data sources) have been the paradigmatic setting, but the methodology travels wherever a well-maintained registry exists.

14.3.6. Platform Trials and External Data

Platform trials (multi-arm adaptive protocols that can add and remove treatment arms over time) are treated in Section 13.6 and Section 12.5. Three of their features belong here.

External control arms in biomarker-selected platforms. Basket trials run one drug across tumor types sharing a molecular alteration; umbrella trials run several drugs within one tumor type, stratified by biomarker. Either way a biomarker-defined cohort can be too small to randomize internally, and the platform reaches for an external control arm, historical or registry-based, for that subgroup. The FDA master protocols guidance (U.S. Food and Drug Administration 2022a) recommends a single common control arm wherever randomization is feasible and addresses non-randomized, activity-estimating substudies where it is not; the design requirements for the external comparator itself, pre-specification of the external population and demonstrated baseline comparability, come from FDA's externally controlled trials guidance (U.S. Food and Drug Administration 2023e). Oncology platforms such as NCI-MATCH and LUNG-MAP have assessed molecularly selected cohorts against historical or external benchmarks.

Streamlined outcome collection in platform designs. The RECOVERY trial (Randomised Evaluation of COVID-19 Therapy) enrolled >40,000 patients across NHS hospitals in the UK with minimal trial data collection: coordinators enroll electronically, treatment is assigned, a single online follow-up form is completed at discharge, death, or day 28, and routine health care and registry data supplement it (RECOVERY Collaborative Group et al. 2021). The operational saving is dramatic. The REMAP-CAP trial (Randomized, Embedded, Multifactorial Adaptive Platform trial for Community-Acquired Pneumonia) embedded enrollment and randomization in intensive care routines but still recorded outcomes on trial case report forms, which one U.S. health system populated largely automatically from its EHR (Angus et al. 2020; King et al. 2024). Both are embedded pragmatic designs at platform scale, running a multi-arm adaptive protocol inside routine care.

Bayesian borrowing inside a platform. In multi-domain platforms (REMAP-CAP ran immunological, anticoagulation, and antiviral domains at once), one analysis model includes patients from every domain, but not as a shared control arm. Patients randomized in other domains sharpen the estimates of covariate effects such as site, age, and calendar time, while each intervention's effect is estimated from patients who could have been randomized to it within its own domain. The borrowing runs between related treatment effects: REMAP-CAP nested tocilizumab and sarilumab under a common interleukin-6 receptor antagonist effect and shrank each estimate toward it when the two agreed (REMAP-CAP Investigators, Gordon, et al. 2021). It is the hierarchical logic of the MAP prior described above, applied to concurrent randomized comparisons inside one protocol instead of to historical controls.

14.3.7. Embedded and Pragmatic Trials

Pragmatic trials assess treatments under conditions approximating routine clinical practice: broad eligibility criteria, usual care comparators, real-world outcomes, and heterogeneous patient populations. *Embedded pragmatic trials* (EPTs) go further, living inside healthcare system infrastructure and using the EHR or claims for recruitment, randomization, and outcome collection. IMPACT-AFib is the canonical U.S. example: claims flowing through FDA's Sentinel distributed database identified patients with atrial fibrillation in five health plans, and patients (not practices) were individually randomized to receive or not receive a mailed educational outreach. The primary analysis covered those with no claims evidence of oral anticoagulant use in the prior 12 months, and its outcome, anticoagulation initiation, came from claims (Pokorney et al. 2022). Claims replaced dedicated data collection, not research staff: two coordinating centers and investigators and programmers at the five participating health plans ran the distributed queries, checked refreshed data extracts, and arranged the mailings (Garcia et al. 2020).

FDA's 2024 draft guidance on integrating randomized trials into routine clinical practice sets out the agency's recommendations for EHR-based recruitment and

outcome collection in an EPT. PCORnet and the NIH Health Care Systems Research Collaboratory have built the infrastructure enabling EPTs at scale.

14.4. Pharmacoepidemiology Design Principles

Observational RWE studies are subject to a set of biases that have no analog in randomized trials. These biases arise from the way treatment is assigned in clinical practice and from the structure of RWD sources. Understanding them is prerequisite to critically reading or designing an RWE study.

Confounding by indication is the most pervasive. Physicians assign treatments based on patient characteristics that may also predict outcomes: sicker patients get more aggressive treatment; frail patients get less. If those characteristics are not fully captured in the RWD, the treatment comparison is contaminated by the indication. The *active comparator, new user* (ACNU) design is the principal design-level solution.

The active comparator design draws the comparator group from patients who received a different active treatment for the same indication at the same time, not from patients who received no treatment at all. If both drugs are used for the same condition in the same clinical context, much of the confounding by indication cancels: the choice between Drug A and Drug B is less driven by disease severity than the choice to prescribe *any* drug.

The new user design restricts the cohort to patients who newly initiated treatment (no use in a washout period, typically 6-12 months). That aligns cohort entry with the start of treatment, so baseline covariates can be measured before exposure, mirroring a randomized trial that enrolls patients not already on the drug (a finite washout cannot show that they never used it). It also avoids *prevalent user bias*: prevalent users have already survived early adverse events and been selected as tolerating the drug, a systematically different population from incident users. The two-drug version of that problem is *depletion of susceptibles*. If early users of Drug A who experienced adverse events switched to Drug B, the remaining Drug A user population is enriched with tolerators and depleted of susceptibles, creating an apparent safety advantage for Drug A. Applying the restriction to both exposure groups removes that selection at cohort entry.

Immortal time bias occurs when the interval between cohort entry and first treatment exposure is incorrectly classified as follow-up time in the treated group (Suissa 2008). During that interval the patient could not have died: they are still alive to receive the treatment. Counting it as follow-up hands the treated group an artificial survival advantage. Target trial emulation (see Chapter 11) prevents this by construction, aligning the eligibility date with the treatment start date.

Index date alignment requires that baseline covariates be measured *before* the index date (treatment initiation) and outcomes *after*. Adjusting for a covariate that could itself have been caused by treatment, such as a post-initiation lab value, can

block part of the effect being estimated or induce collider bias, depending on where it sits in the causal structure. The rule sounds trivial. It is routinely violated when analysts reach for "all available" data without strict time-ordering.

14.5. Statistical Methods for Real-World Evidence

The foundational methods are in Chapter 11. What observational causal inference adds begins before any of them, with the causal structure of the problem itself.

14.5.1. Causal Diagrams as the Starting Point

Three kinds of variable have to be told apart. Confounders are shared causes of treatment and outcome, and must be controlled. Mediators sit on the causal pathway from treatment to outcome, so controlling them blocks the very effect being estimated. Colliders are common effects, whether of treatment and outcome themselves or of their causes, and controlling one opens a spurious association. *Directed Acyclic Graphs* (DAGs) provide the formal language for the distinction (Pearl 2009): a DAG specifies the minimum sufficient adjustment set for the causal estimand, and just as precisely what to leave alone, mediators in many contexts and colliders unless the path they would open is blocked elsewhere.

DAGs are widely taught in pharmacoepidemiology and give an RWE analysis an explicit, reviewable rationale for its choice of confounders, and DAGitty makes the construction and analysis accessible without formal training in graphical models.

14.5.2. Propensity Score and Doubly Robust Methods

Propensity score (PS) methods (covered in Chapter 11) are the workhorses of observational RWE. The PS model predicts treatment assignment from baseline covariates measured before the index date, and variable selection follows its own rules: include the causes of the outcome, exclude instruments, and be cautious with variables associated only with treatment. Balance is demonstrated, never assumed. Report standardized mean differences (SMD) before and after adjustment, with SMD < 0.1 the conventional threshold for acceptable.

Doubly robust estimators (AIPW, g-computation with IPW, TMLE) are more resistant to model misspecification than PS-only or outcome-model-only approaches, and should be preferred for high-stakes primary analyses. Targeted Maximum Likelihood Estimation (TMLE) combines machine learning-based estimation of both the PS and the outcome model with a targeted bias-correction step that restores valid inference when data-adaptive models are used, provided the usual regularity conditions hold (adequate positivity, sufficiently accurate nuisance estimation, and cross-fitting or a comparable safeguard against overfitting).

14.5.3. Instrumental Variables

When unmeasured confounding is suspected, *instrumental variables* (IV) offer an alternative identification strategy. An instrument is a variable that: (1) affects treatment; (2) affects the outcome only through its effect on treatment; (3) shares no common cause with the outcome, so that it is unassociated with the unmeasured confounders of the treatment-outcome relationship. In pharmacoepidemiology, instruments include physician prescribing preference: some physicians systematically prefer Drug A, independent of patient characteristics. Formulary changes that shift prescribing toward one drug based on calendar time, and geographic discontinuities in access, are used the same way.

The three conditions alone only bound the effect. In the standard case of a binary instrument and a binary treatment, adding *monotonicity* (no *defiers*, patients who would do the opposite of whatever the instrument encourages) lets IV identify the *local average treatment effect* (LATE), the effect in the subpopulation whose treatment status is changed by the instrument, not the average treatment effect in the full population (Baiocchi, Cheng, and Small 2014). So IV earns its place when a strong instrument exists and the LATE is clinically meaningful. Weak instruments produce imprecise estimates.

14.5.4. Difference-in-Differences and Regression Discontinuity

Difference-in-differences (DiD) compares the pre-to-post change in outcomes in a treated group with the pre-to-post change in an untreated group, differencing out unmeasured confounders whose effects do not change over time. The identifying assumption is *parallel trends*: absent the treatment, the two groups would have followed the same trajectory. Because it concerns the untreated trajectory after treatment starts, it cannot be tested directly. Parallel pre-treatment trends make it more plausible and belong in the report, but a pre-trend test can lack the power to detect a violation large enough to matter (Roth 2022). DiD is common in policy evaluation and increasingly used in pharmacoepidemiology for studies of formulary changes, guideline implementations, or drug market entries.

Regression discontinuity (RD) exploits threshold-based treatment assignment: patients scoring just above a clinical cutoff receive treatment, patients just below do not, and near the cutoff the two groups are approximately comparable. RD identifies a local treatment effect at the threshold, useful wherever clinical practice sets one: CD4 count thresholds for starting antiretroviral therapy (Bor et al. 2014), lipid level thresholds for statin prescribing, age-based eligibility cutoffs.

14.5.5. Unmeasured Confounding Assessment

No observational method fully eliminates unmeasured confounding, so the response is to quantify how strong an unmeasured confounder would have to be to explain the observed result away. That quantity is the **E-value** (VanderWeele and Ding 2017), covered in Chapter 11: a large one means only a very strong unmeasured

confounder could explain the result away. It addresses unmeasured confounding only; selection bias and outcome misclassification need their own checks.

Negative control outcomes and exposures are, respectively, outcomes the treatment under study cannot cause and exposures that cannot cause the study outcome, chosen to share the confounding and measurement problems of the main comparison. If the same analysis finds an association with a negative control that chance does not explain, the method is producing spurious results (systematic bias), and the primary estimate is suspect to the extent that the control shares its sources of bias (Lipsitch, Tchetgen Tchetgen, and Cohen 2010). It is a powerful validity check.

Quantitative bias analysis (QBA) goes further, into probabilistic sensitivity analysis: Monte Carlo simulation over plausible values of unmeasured confounder prevalence and effect size. What comes out is a bias-adjusted distribution for the effect estimate, in place of a single point estimate that assumes no unmeasured confounding.

14.6. Landmark Applications

Three cases carried real-world evidence from method into regulatory and clinical practice.

14.6.1. Oncology: Ibrance for Male Breast Cancer

Male breast cancer occurs in approximately 2,670 U.S. patients annually, and only a fraction of them have the hormone receptor-positive, HER2-negative advanced disease palbociclib treats, so a randomized trial in men would be slow to enroll; men had also been excluded from the trials behind palbociclib's earlier approvals (U.S. Food and Drug Administration 2019b). In April 2019 FDA approved Pfizer's supplemental NDA for palbociclib (Ibrance) to add male breast cancer to the indication, the canonical landmark case for RWE-supported label expansion. The evidence came from three sources: Flatiron Health's oncology EHR supplied real-world tumor response and treatment patterns; IQVIA insurance data added treatment patterns and concomitant medications; Pfizer's global post-marketing safety database carried the adverse event profiles in male patients.

FDA treated this package as supportive rather than primary. The approval rested on palbociclib's efficacy in women, established in randomized trials, and on a strong mechanistic rationale for the same effect in men; the Flatiron data, 3 responses among the 12 men on palbociclib whose tumors were assessed, added modest support, and FDA left the IQVIA results out of its benefit-risk assessment because of their limitations (U.S. Food and Drug Administration 2019b). The label's statement about men concerns safety: "Based on limited data from postmarketing reports and electronic health records, the safety profile for men treated with IBRANCE is consistent with the safety profile in women treated with IBRANCE." FDA never called

the real-world data equivalent to a randomized trial; it accepted the evidence as adequate for one specific regulatory question, whether the drug's pharmacology and clinical profile in men are consistent with benefit in women, where randomized evidence exists.

14.6.2. COVID-19 Vaccine Effectiveness at Scale

Vaccines were authorized under Emergency Use Authorization on clinical trial data in 2020-2021, and real-world effectiveness still had to be measured across diverse populations, viral variants, and healthcare settings, questions the trials could not answer in time. What followed was a rapid, large-scale deployment of RWE.

The dominant design was the *test-negative design* (TND): among people with symptoms who were tested for COVID-19 by PCR, compare vaccination status between those who tested positive and those who tested negative. Restricting to people who sought testing helps control for differences in healthcare-seeking behavior, and the comparison estimates vaccine effectiveness against symptomatic disease. Lopez Bernal and colleagues applied it to England's national surveillance data to estimate BNT162b2 effectiveness against the Delta variant (Lopez Bernal et al. 2021). Dickerman and colleagues took a different route, emulating a target trial in U.S. veterans data to compare BNT162b2 against mRNA-1273 directly, at continental scale (Dickerman et al. 2022).

Observational studies also documented protection waning with time since vaccination, and that evidence informed booster policy. The COVID-19 experience showed that RWE can provide actionable regulatory-grade evidence in a compressed timeframe when the methodological infrastructure is in place.

14.6.3. Registry-Based Trials: TASTE and SWEDEHEART

TASTE (Fröbert et al. 2013) and VALIDATE-SWEDEHEART (Erlinge et al. 2017), both described above, randomized thousands of patients for a fraction of traditional Phase III cost (Erlinge 2021), in a population more representative of clinical practice than the carefully selected cohorts conventional trials assemble. TASTE took its endpoints from national registries. VALIDATE-SWEDEHEART added its telephone follow-up and adjudication because the registry did not capture bleeding complications well enough. Where a suitable registry already exists, randomization can be built into its interface cheaply, but an endpoint the registry does not capture reliably still needs trial-specific follow-up and adjudication, which adds complexity and cost (Erlinge 2021). The evidence carries the inferential advantages of randomization with the population representativeness of observational data. The design is now being replicated internationally.

14.7. Infrastructure: OMOP, OHDSI, and Federated Analytics

Running any of this at scale takes computational infrastructure, and the OHDSI (Observational Health Data Sciences and Informatics) community has built it as an open-science ecosystem. The **OMOP Common Data Model** standardizes both the table structure and the vocabulary of observational databases: SNOMED CT for conditions, RxNorm for drugs, LOINC for measurements. Analysis code written once can then be run across the network's standardized data sources, which in 2024 numbered 544 in 54 countries and together held more than 974 million unique patient records (Ryan 2024). Patient data never leave the institution; only aggregate results are shared.

The **HADES** toolkit (Health Analytics Data-to-Evidence Suite) comprised 37 R packages in 2024 for running standardized epidemiological analyses on OMOP-format data: cohort construction, propensity score estimation, outcome modeling, negative control calibration, and diagnostics (Ryan 2024). PhenotypeLibrary maintains a version-controlled repository of cohort definitions (phenotype algorithms) that undergo voluntary open peer review, with diagnostics run on volunteer data partners to check that each definition executes across the network (Rao 2026).

The DARWIN EU network applies OMOP-compatible standards across its European data partners, so the same standardized analytic code that runs on OMOP databases elsewhere can be applied to European data within a regulatory-grade framework.

Where a model has to be trained on patient-level data that cannot be pooled, *federated learning* trains it across the distributed sites, with model updates rather than patient data crossing institutional boundaries. It costs more complexity than distributed analytics and buys model-level analysis, but the updates are aggregates too and can leak information about the patients behind them, so privacy-sensitive settings (rare diseases, small populations, genomic data) call for safeguards such as differential privacy or secure aggregation (Ogier du Terrail et al. 2025). Owkin has applied it to construct external control arms from real-world data distributed across hospital networks (see Chapter 26).

14.8. Generalizability and Transportability

Even a perfectly unconfounded RWE study estimates the treatment effect in the study population. Whether that estimate carries to the target population, the patients who will actually receive the drug, is a separate question with its own vocabulary: *generalizability* runs from the study sample to the study population, *transportability* from the study population to a different target population.

Dahabreh and colleagues built a formal framework, combining evidence from randomized trials and observational data to estimate treatment effects in a target population beyond the trial's eligibility criteria (Dahabreh et al. 2019, 2020). Measure

and model the factors that make the trial population differ from the target, and inverse probability of sampling weights re-weight the trial evidence to represent that target. Weighting works only if those measured factors account for every difference between the populations that bears on the treatment effect, and if every covariate pattern in the target also occurs among trial participants; a group the eligibility criteria excluded outright, on a factor that modifies the effect, cannot be recovered by weighting (Dahabreh et al. 2020).

Trial eligibility criteria systematically exclude older patients, those with comorbidities, and those on concomitant medications, and those are precisely the patients who will receive the drug after approval. Within those limits, transportability methods estimate whether the trial effect size survives that translation, which bears on labeling claims and post-market commitments. The same machinery does the cross-border work: when a drug approved in one country is taken to another for approval or HTA, the approval trial's evidence may not transport to the target country's population, and MAIC (Matching-Adjusted Indirect Comparison) is the widely used HTA implementation of the idea.

14.9. When Does RWE Belong in a Development Program?

The conditions that make RWE credible share one logic: confounding is the irreducible threat, and each condition either limits its impact or keeps other biases from adding to it.

The strongest protection comes when the outcome is not subject to differential ascertainment. Overall survival qualifies because death is recorded regardless of which arm the patient was in; hospitalization is a billing event, generated the same way whichever treatment the patient received; treatment dispensing is a pharmacy record. None of the three is complete by default: U.S. EHR and claims data generally record a death only when it occurs under medical care, unless they are linked to vital statistics, and care or prescriptions obtained outside the contributing health system or insurance plan are missed (U.S. Food and Drug Administration 2024k). Completeness has to be established for each source, often through linkage. Symptom scores, imaging-defined tumor progression, and patient-reported outcomes fare worse, because they depend on protocol-mandated assessments that routine care does not perform. A completely captured objective outcome leaves confounding by treatment selection as the problem to solve. A subjectively ascertained one adds outcome measurement bias on top of it.

Rare diseases and rare populations are a different case: a randomized comparison may not be feasible, and a comparison against natural history can still persuade when the untreated course is well characterized and predictable and the drug's effect is large (U.S. Food and Drug Administration 2023e). Rarity does not remove confounding. Treated patients can differ from natural-history patients in disease severity, prognosis, comorbidities, and prior treatment, and a historical comparison

also absorbs changes in diagnosis and standard of care over time. Pediatric oncology and rare hereditary diseases, where the natural history is the only available comparison, are where regulatory agencies have been most receptive.

An RCT demonstrating efficacy, supplemented by RWE demonstrating real-world effectiveness, is substantially more compelling than RWE as the primary evidence. FDA's endorsement of emulation benchmarking studies, which test whether RWE analyses can replicate the findings of completed RCTs in the same setting, follows that logic: where concordance has been established, regulatory confidence in RWE for similar questions increases.

All of this depends on pre-specification. FDA and EMA expect the external control population, the analytic methods, the confounders to be controlled, and the sensitivity analyses to be specified before any unblinding of the internal trial data. An analysis decided after the result is known is much weaker evidence.

The Ibrance precedent, the COVID-19 vaccine experience, and the growing body of emulation benchmarking studies are collectively moving the evidentiary floor, most clearly where RWE extends randomized evidence rather than replacing it. That movement is what the Advancing RWE Program is built to convert into effectiveness labeling changes, the application that has historically resisted RWE hardest.

Real-world evidence does not replace randomized trials. It answers the questions trials cannot: what happens to a broader population over a longer period under the conditions of clinical practice, using endpoints that routine care can measure. The methodological and regulatory infrastructure for generating credible RWE is now substantial, and its role already runs from informing trial design through supporting regulatory submissions to satisfying post-market commitments and HTA requirements. It will keep expanding as that infrastructure matures.

Discussion Questions

1. A sponsor is preparing a supplemental NDA for a targeted therapy in a rare solid tumor (approximately 800 annual U.S. cases) and plans to use Flatiron Health oncology EHR data as the primary external control. FDA's fit-for-purpose framework requires evaluation of data relevance and reliability. Walk through the specific questions FDA would ask about this data source for the proposed regulatory question, and identify the two most likely reasons the submission could be rejected on data-quality grounds.

2. A retrospective study comparing a newer oral anticoagulant to warfarin uses a prevalent-user cohort: patients enter at their first database capture, by which time they have been on treatment for a median of 14 months, but their follow-up is counted from treatment initiation. Explain why prevalent user bias and immortal time bias are present in this design, what determines the direction in which each would bias the effect estimate, and what design changes would correct both.

3. An analyst emulating a hypothetical RCT comparing two antihypertensive drug classes using claims data sets the index date as the date of the first dispensing of the study drug. A reviewer notes that some patients have lab values recorded between the first prescription date and the first dispensing date, which the analyst included as baseline covariates. Does this necessarily violate the target trial emulation framework's requirement for correct time-zero alignment? What would you need to know about when treatment actually begins, and what bias would follow if some patients had started treatment before their first recorded dispensing?

4. A hybrid Bayesian design for a rare hematologic malignancy with a binary primary endpoint uses a MAP prior derived from three historical trials, with a historical mean control-arm event rate of 30% and an effective sample size of 25 patients. The concurrent control arm enrolls 30 patients. At the interim analysis, the observed control-arm event rate is 42%, 40% higher than the historical mean. Describe what happens to the posterior estimate of the control arm event rate, how commensurate versus fixed power priors would respond differently to this discordance, and what you would recommend the DSMB do with this information.

5. NICE is evaluating a KRAS G12C inhibitor for second-line non-small cell lung cancer under a managed entry agreement. The sponsor must design a real-world evidence study to confirm effectiveness in routine NHS practice within 24 months of conditional reimbursement. What study design would you propose (specifying data source, population, comparator, outcome, and analytic method), and what are the two greatest threats to the study's ability to produce a result that NICE will accept as confirmatory?

15. Protocol Design Benchmarks

The average Phase III protocol runs 116 pages, specifies 266 procedures, and collects more than 3.4 million datapoints across roughly 82 investigative sites in 14 countries (Table 15.1). Roughly a quarter of those procedures support no primary or key secondary endpoint. Protocols have grown steadily more complex since the 1980s, and benchmarking consistently finds that the added complexity correlates with worse performance.

15.1. The Trend Toward Complexity

The pursuit of blockbuster therapies in the 1980s expanded the number of assessments packed into a trial, particularly in Phase III, along with the investigators engaged and the patients enrolled. Cost containment in the 1990s then increased Phase II complexity, paradoxically, as sponsors tried to gather more information before committing to expensive Phase III programs.

Globalization followed in the 2000s. Sponsors went abroad for treatment-naive populations and for experienced investigators at lower cost, and to support simultaneous international submissions, while quality by design principles and enhanced risk evaluation drove growth in safety procedures and data collection. Between 2010 and 2020 sponsors piloted novel designs (adaptive trials, master protocols), and the proportion of programs targeting rare diseases and narrowly defined patient subpopulations increased dramatically, supported by growth in biomarker and genetic data collection.

Nearly all of these changes have been additive: each new requirement layers atop the ones already in place. Tufts Center for the Study of Drug Development (Tufts CSDD) benchmarking consistently finds that no design element has yet shown a downward trend (Getz, Smith, and Kravet 2023).

15.2. Current Benchmarks by Phase

These benchmarks come from protocols completed just before the COVID-19 pandemic, so read them as a floor and not a ceiling: subsequent benchmarking has found complexity still climbing across the same design dimensions. Even Phase I, focused on safety and pharmacokinetics in small populations, collects more than 330,000 datapoints per protocol, and Phase II and III expand most dimensions substantially from there (see Table 15.1).

Table 15.1.: Clinical Trial Protocol Benchmarks by Phase (Getz, Smith, and Kravet 2023)

Metric	Phase I	Phase II	Phase III
Endpoints	16	21	19
Eligibility Criteria	32	31	30
Distinct Procedures	29	34	35
Total Procedures	169	260	266
Protocol Pages	87	108	116
Countries	<2	~6	~14
Investigative Sites	~5	~35	~82
Datapoints Collected	>330,000	>2 million	>3.4 million

15.3. Oncology and Rare Disease Protocols

Oncology protocols run longer and impose heavier procedural burdens than non-oncology protocols while enrolling similar numbers of patients (Table 15.2).

Table 15.2.: Oncology vs. Non-Oncology Protocol Benchmarks (Getz, Smith, and Kravet 2023)

Metric	Oncology	Non-Oncology
Total Procedures	315	243
Planned Visits	29	16
Countries	13	8
Follow-up Duration	331 days	77 days
Completion Rate	31%	80%

One figure stands out: across the benchmarked Phase II and III oncology protocols, the mean completion rate (completers as a share of enrolled patients, averaged across protocols) was only 31%, which the authors suggest may partly reflect disease progression leading to early discontinuation (Getz, Smith, and Kravet 2023). Not completing the protocol is not the same as dropping out of the analysis, though. Progression or death is often the endpoint itself, and patients who stop treatment can still be followed for it, so sample size planning allows for the expected loss of outcome follow-up, not for the completion rate as such (International Council for Harmonisation 2019).

For rare disease protocols the strain falls hardest on recruitment and site activation (Table 15.3).

Table 15.3.: Rare vs. Non-Rare Disease Protocol Benchmarks (Getz, Smith, and Kravet 2023)

Metric	Rare Disease	Non-Rare Disease
Sample Size	224	425
Sites	27	64
Distinct Procedures	38	33
CRF Pages	244	159
Enrollment Duration	603 days	444 days
Study Initiation	173 days	141 days

Finding eligible patients takes 603 days against 444 for non-rare trials, despite much smaller target sample sizes.

15.4. The Complexity-Performance Relationship

Protocols carrying relatively more endpoints, eligibility criteria, and procedures tend to perform worse across a range of operational measures: physicians refer fewer patients, sites shoulder a heavier procedure-administration burden, prospective volunteers are less willing to enroll, recruitment and retention rates fall, dose adherence declines, data volume grows, and protocol deviations and amendments become more frequent (Getz, Smith, and Kravet 2023).

Those outcomes contribute to higher failure rates, longer cycle times, poorer data quality, and greater costs. Tufts CSDD found significant correlations among the design elements: the number of endpoints correlates with the number of eligibility criteria and with total datapoints collected, and protocols with more investigative sites and countries take longer to complete (Getz, Smith, and Kravet 2023).

The average Phase III protocol takes 1,328 days from protocol finalization to database lock (Getz, Smith, and Kravet 2023). Study initiation, from protocol approval to first patient first visit, averages 166 days. Enrollment, from that first visit to last patient first visit, averages 508 days. Treatment, from that same first visit to last patient last visit, averages 1,064 days. Oncology runs longer still: across Phase II and III, oncology protocols average 1,599 days from protocol finalization to database lock, about 1.5 times the 1,081-day non-oncology average (roughly 48% longer) (Getz, Smith, and Kravet 2023). Among the component periods, the widest gap comes after enrollment closes: from last patient first visit to last patient last visit, oncology protocols average 659 days against 335, nearly twice as long, while enrollment itself runs about a third longer (575 days against 437).

A **core procedure** is one tied to a primary or key secondary endpoint; **non-core** procedures support supplemental, tertiary, or exploratory objectives. Across Phase II and III protocols, nearly a quarter (24-25%) of all procedures are non-core, which suggests a substantial opportunity for simplification. Rare disease trials hold the share to 14%, reflecting a tighter focus on core assessments, while in non-rare disease protocols the figure often exceeds 25%. Every non-core procedure adds cumulative burden to sites and patients without directly supporting the primary approval decision, so each one has to justify its incremental cost and complexity.

15.5. Drivers and Implications

Several interlocking forces drive the persistent trend toward higher complexity. Ambitious development programs often target more challenging diseases, which requires biomarker stratification to understand differences between patient subgroups, and regulatory demand for detailed safety data necessitates more frequent procedures and longer follow-up periods to satisfy risk evaluation and mitigation requirements. Enrollment targets then pull trials into more countries and more sites, each of which brings logistical and coordination overhead. Running underneath all of it is a competitive urge to "collect everything" that often leads sponsors to believe larger datasets will improve their regulatory and commercial positioning.

The strong correlation between executional variables (the number of sites and countries) and longer trial durations suggests that efficiency gains are possible through better design. Prioritizing site quality over quantity, and questioning whether every endpoint and non-core procedure adds value, lets designers cut complexity without sacrificing the integrity or enrollment speed of the trial. That discipline is now a regulatory expectation as much as a practical one: ICH E6(R3) (International Council for Harmonisation 2025b) directs sponsors to identify the factors critical to a trial's quality and to keep trial processes proportionate to the risks that matter, avoiding unnecessary complexity, procedures, and data collection (see Chapter 3).

Decentralized clinical trial (DCT) elements (telemedicine, wearables, home visits, direct-to-patient shipment) may reduce enrollment duration and protocol amendments through more flexible execution. They also introduce new data sources, vendors, and operational practices that can increase short-term protocol complexity. As trials move toward wherever patients can most conveniently participate, the balance between executional flexibility and protocol simplicity remains an open question. For detailed treatment of DCT logistics and implementation, see Chapter 20.

16. Randomization and Blinding

For decades, hormone replacement therapy was thought to prevent heart disease, on the strength of observational studies showing that women who took hormones had lower rates of heart attacks. When randomized trials were finally conducted, they showed the opposite: combined hormone therapy increased the risk of coronary heart disease by roughly 29% and stroke by roughly 41% (Rossouw et al. 2002). The observational studies had been confounded. Women who chose to take hormones were healthier to begin with.

The same confounding arises at the bedside. A physician enthusiastic about a new treatment prescribes it to patients who seem likely to benefit and withholds it from those who seem too sick to respond. The treated patients do better. Nothing in the data can say whether that is the treatment or the selection.

The methodological history of medicine is full of treatments that looked effective in uncontrolled studies and proved worthless, or even harmful, when properly tested. Randomization and blinding are what stand between a trial and that history. Chapter 13 treated both as design-level choices; what follows is the mechanics.

Randomization eliminates the confounding (Cox 2009). A coin flip (or its statistical equivalent) decides who receives treatment and who receives control, so the groups are comparable at baseline and differences that emerge afterwards can be attributed to the treatment itself. Cox identifies three broad purposes for randomization: to avoid selection and other biases in a publicly convincing way, to provide a unified approach to estimating error in standard designs, and to provide a basis for exact tests of significance. Of these, bias avoidance is the most compelling in clinical trials, since failure to randomize appropriately may fatally compromise an investigation, and it is also what convinces those not directly involved in the trial that the comparison is fair.

16.1. Types of Randomization

The methods in Table 16.1 differ in how tightly they control group sizes and in how easily the next assignment can be guessed, and tighter control usually costs some unpredictability.

Table 16.1.: Randomization method comparison

Method	Information needed at randomization	Balance guarantee	Risk of prediction	Best suited for
Simple	None	None (probabilistic only)	Low	Large trials (n > 200); unpredictability valued
Block	Block size	Balance after each complete block	Moderate (last slot in block predictable)	Most confirmatory trials; standard practice
Stratified block	Key prognostic strata for each patient	Balance within each stratum after each block	Moderate (same as block, within stratum)	Trials with 1-3 strong prognostic factors; multi-site
Minimization	Current enrollment totals by covariate level	Near-perfect balance on all included covariates	Low-moderate (deterministic rule creates some predictability)	Multi-center trials with many prognostic covariates; moderate sample sizes
Response-adaptive (RAR)	Accruing outcome data from enrolled patients	None; allocation shifts dynamically toward better arm	Low to moderate (single assignments stay probabilistic, but a shifting allocation ratio can leak interim performance)	Multi-arm early-phase or platform trials; rarely appropriate for two-arm Phase III
Cluster	Group (site, clinic, community) membership	None inherent; set by how clusters are allocated (simple, matched, or stratified), and few clusters make chance imbalance likely	Low	Community or system-level interventions; implementation science

Simple randomization is a coin flip for each participant. In a large trial that works well enough; in a trial randomizing 20 patients, chance can produce 13 in one group and 7 in the other. Bradley Efron addressed the problem in 1971 with the *biased coin design*, which increases the probability of assigning to the underrepresented

group when imbalance develops (Efron 1971).

16.1.1. Block Randomization

Block randomization attacks the same imbalance by randomizing within blocks of fixed size (Figure 16.1).

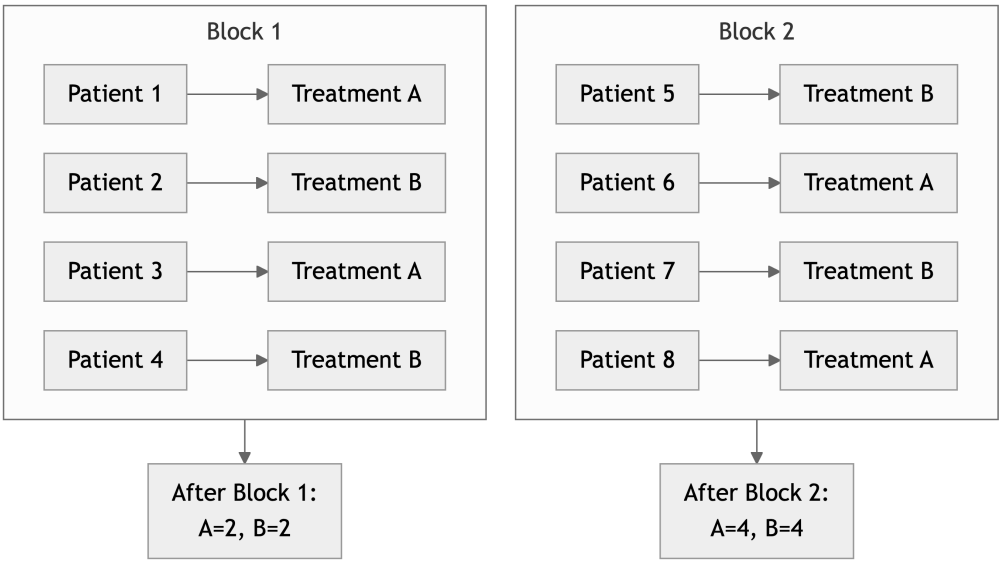


Figure 16.1.: Block randomization ensures equal group sizes after each complete block. This example shows a block size of 4.

With blocks of 4, a block might run ABAB, ABBA, BABA, or any of the six arrangements with two A's and two B's. After every complete block, the groups are guaranteed to be equal in size.

Stratified randomization goes further, running a separate randomization sequence inside each subgroup defined by an important prognostic factor. If disease severity predicts outcome, patients might be stratified as mild, moderate, or severe and randomized independently within each stratum, so the groups stay balanced on severity even when overall enrollment skews toward one level.

Minimization (or *dynamic allocation*) gives up randomizing each patient independently (Taves 1974; Pocock and Simon 1975). The algorithm reads the current imbalance among enrolled patients and assigns each new one to reduce it: if the treatment group has more older patients, the next older patient may be somewhat more likely to go to control. Balance can hold on several factors at once, and simulation studies show that minimization gives better-balanced groups than restricted or unrestricted randomization (Scott et al. 2002). A 2023 tutorial found minimization particularly beneficial when numerous major prognostic factors are known, when

many centers of varying sizes recruit patients, or when the sample size is moderate (Coart et al. 2023).

16.2. Maintaining Randomization Integrity

Investigators have held sealed envelopes up to the light to read the assignment inside (Schulz 1995). That gesture is the whole problem: if the next allocation can be predicted or influenced, randomization has already stopped protecting against bias.

Allocation concealment is the mechanism that prevents it, and numbered envelopes were its old, sometimes inadequate form. Modern trials use *Interactive Response Technology (IRT)*: computer or telephone systems that release the assignment only after the investigator has committed to enrolling a specific patient.

Once randomization occurs, the assignment should be followed. *Treatment crossover*, where patients switch from their assigned group to another, can blur the distinction between groups and complicate analysis. Sometimes it is unavoidable, as when patients on placebo deteriorate and ethical obligations require access to active treatment, and the analysis then has to account for it.

The 2025 revision of ICH's Good Clinical Practice guideline, E6(R3), frames randomization and blinding as factors critical to trial quality and builds them into its quality-by-design expectations (International Council for Harmonisation 2025b). A dedicated provision, *Safeguard Blinding in Data Governance*, directs sponsors to control access to unblinded data and to document any unblinding, planned or unplanned. They must also guard against inadvertent unblinding by the vendors and service providers who handle trial data.

16.3. The Power of Blinding

If randomization protects against bias in allocation, blinding protects against bias in everything that follows.

Suppose the investigators know which patients are receiving the experimental drug. Consciously or not, they might monitor those patients more closely, interpret a symptom more charitably, or encourage them to continue despite side effects. Patients who know they are receiving active treatment might report more improvement due to expectations. Where the outcome is subjective (pain, mood, quality of life), these biases can be substantial. Even a supposedly objective outcome is exposed: a radiologist who knows a patient is on active treatment might interpret a borderline scan as showing improvement. Blinding addresses these concerns by keeping treatment assignment unknown to those who might be influenced by it.

16.4. Levels of Blinding

Open-label trials make no attempt at blinding. They are common when the interventions are too different to disguise, surgery versus medication for instance, and they are susceptible to observer bias. Single-blind trials conceal the assignment from patients, which reduces participant expectations. Double-blind trials, the industry standard, conceal it from investigators as well. Triple-blind designs, the most rigorous, withhold it from those performing the statistical analysis too, until the study is formally concluded.

16.5. Technical Aspects of Blinding

Treatment and placebo must be *indistinguishable* in appearance (color, size, shape), taste, smell, and texture. A capsule and a tablet may both need over-encapsulation to match.

Unblinding procedures must be in place for emergencies. When a participant has a medical crisis and the treating physician needs the treatment assignment, the blind has to break for that one individual and hold for everyone else.

Side effects can compromise blinding on their own. A distinctive adverse effect (dry mouth, dizziness, skin discoloration) can let participants and investigators guess the assignment, which is why trials sometimes include *perception of assignment* assessments to evaluate whether blinding was maintained. When true blinding is not possible at all, *blinded assessment* salvages part of it: an assessor whose only contact with patients is the evaluation itself may remain blind when other study staff cannot, and imaging can be read by radiologists with no knowledge of assignment.

16.6. The Challenge of Device and Behavioral Trials

How do you blind a patient to whether they received surgery? How do you blind a therapist to whether they are delivering cognitive behavioral therapy or a control intervention? Devices, surgical procedures, and behavioral interventions are where blinding is hardest.

Sham procedures are one answer. Control patients in a surgical trial may receive anesthesia and incisions but not the actual procedure. This is ethically controversial, since exposing patients to surgical risk without potential for direct benefit strikes some as unacceptable, and it may still be the only way to separate the effects of the procedure from placebo effects and expectation.

Active comparator designs sidestep the question. Both groups receive an intervention, and the trial asks which one is better.

Decisions about randomization and blinding shape the rest of the trial architecture. Each additional treatment arm complicates stratification and balance. The planned

duration of blinding has to cover cross-over periods and long-term follow-up. Sponsors must also decide how to handle treatment discontinuations: a patient who stops the study medication still needs follow-up and data collection, or the randomized comparison loses its integrity.

16.7. Advanced Topics

Two randomization approaches extend beyond the fixed-probability designs above: response-adaptive randomization, which updates allocation probabilities as outcome data accumulate, and covariate-adaptive randomization, which conditions allocation on patient characteristics measured at enrollment.

16.7.1. Response-Adaptive Randomization

Response-adaptive randomization (RAR) shifts allocation probabilities during the trial in response to accruing outcome data, sending more patients to whichever arm is performing better. The ethical motivation is direct: if one treatment is proving superior, fewer patients should be assigned to the inferior arm during the trial itself.

The *randomized play-the-winner* rule of Wei and Durham (L. J. Wei and Durham 1978) is among the most transparent RAR designs. A success on one arm adds a ball of that treatment's color to an urn; a failure adds a ball of the other treatment's color. Each new patient draws from the urn, so the accumulating success rates steer allocation toward the better arm. Later Bayesian formulations implement the same intuition via Thompson sampling: at each allocation, compute the posterior probability that each arm is best and assign the next patient with that probability (see Section 12.8 for the statistical mechanics).

The most influential and contested application of play-the-winner ended with one patient in the control arm. Bartlett and colleagues were evaluating extracorporeal membrane oxygenation for newborns with severe respiratory failure, a condition with near-certain mortality untreated (Bartlett et al. 1985). In the 1985 trial that one infant died, and the eleven assigned to ECMO all survived. The ethical appeal was obvious: very few infants were exposed to the inferior arm. The evidentiary problem was equally clear. A trial with one control patient cannot estimate the control-arm survival rate with any precision, and the result was vigorously contested. A subsequent UK collaborative ECMO trial using conventional equal randomization confirmed the benefit (UK Collaborative ECMO Trial Group 1996).

ECMO remains the reference case for the central tension in RAR. Full treatment of the statistical debate, including temporal-trend bias, the Proschan objection, and the conditions under which RAR is and is not appropriate, is at Section 12.8.

When RAR is useful and when it is not

RAR is most defensible in multi-arm settings (three or more arms) where the goal is to efficiently identify the best arm, particularly in early-phase or platform trials. In two-arm Phase III confirmatory trials, the power loss and potential for bias under RAR are rarely worth the patient-protection benefit, and conventional equal randomization is generally preferred (Korn and Freidlin 2011; Proschan and Evans 2020). Where RAR is used, two mitigations are practical: a burn-in period (equal allocation for the first N patients before adaptation begins) and a protect-control safeguard (never reducing the control arm below a minimum proportion).

16.7.2. Covariate-Adaptive Randomization

Covariate-adaptive randomization extends minimization, folding patient characteristics into allocation in more sophisticated ways (Atkinson 2002; Y. Lin, Zhu, and Su 2015). Randomizing one patient at a time creates its own exposure: whoever knows the previous allocations may be able to predict the next assignment, which could enable selection bias. Atkinson carefully compared procedures that trade covariate balance against unpredictability (Cox 2009). FDA's 2023 guidance on covariate adjustment recommends that sponsors specify the covariate-adjusted analysis prospectively, before unblinding, and include any stratification factors in that analysis model; complex covariate-adaptive randomization designs should be discussed with the review division in advance (U.S. Food and Drug Administration 2023j).

For the mathematical foundations of randomization theory and practice, see Rosenberger and Lachin (Rosenberger and Lachin 2016); for response-adaptive methods specifically, Hu and Rosenberger (Hu and Rosenberger 2006).

Discussion Questions

1. Minimization is sometimes described as "not truly random" because the allocation of each patient is at least partly deterministic given the current enrollment counts. Some regulatory agencies have expressed concern about this. What is the substantive statistical concern (selection bias, inference validity, or both), how serious is it in practice, and what design modifications (e.g., a probabilistic minimization rule with a biased coin) address the concern?

2. A Phase III sponsor using response-adaptive randomization in a two-arm cardiovascular outcomes trial argues the design is justified because the trial will run for four years and it would be unethical to continue assigning patients to a placebo if the drug is clearly superior at year two. A statistician objects that temporal confounding could bias the estimate in a multi-year trial. Construct the statistician's argument precisely, and propose the minimum safeguards that would make RAR defensible in this specific setting.

3. A trial investigator states: "We have allocation concealment, so blinding is not necessary for our open-label surgical trial." Explain why allocation concealment and blinding protect against different sources of bias, using a concrete example from a surgical or device trial context to illustrate each.

4. A sponsor planning a 120-patient Phase II trial proposes to stratify by six prognostic factors (each binary), generating 64 possible strata. Most strata will have 0-2 patients. What specific statistical and operational problems arise from over-stratification at this sample size, and what would you recommend instead?

5. A placebo-controlled antipsychotic trial achieves double-blind status for capsule appearance and taste, but 68% of patients correctly guess their treatment assignment at the end-of-study perception survey, predominantly due to sedation and weight gain in the active arm. The primary endpoint is a subjective symptom rating scale. What are the implications for interpreting the primary efficacy result, what additional analyses could help bound the bias, and how should this be reported in the publication?

17. Endpoints and Outcomes

In 1987, the FDA approved zidovudine (AZT) for HIV based on a trial stopped early for improved survival after 8 to 24 weeks of follow-up (Fischl et al. 1987). In 2011, the FDA revoked accelerated approval for bevacizumab in breast cancer after confirmatory trials found a much smaller effect on progression-free survival, the endpoint behind the approval, and no significant gain in overall survival to offset serious toxicity. In 2021, the approval of aducanumab for Alzheimer's disease sparked controversy because the drug reduced amyloid plaques (the surrogate) but did not clearly improve cognition (the clinical outcome).

All three turn on one question: what should a trial measure to show that a drug works? The answer drives the design, the sample size, and the regulatory strategy, and in the end decides whether patients gain access at all.

An *endpoint* is the defined outcome a trial is designed to measure, the variable that answers the central question: did the treatment work? Some are assessed at a fixed time, such as the end of treatment or follow-up; others, such as the time to a first event, are captured as they occur.

Endpoints must be specified before the trial begins. The *primary endpoint* is the one the trial will be judged on: a Phase III trial with survival as its primary endpoint succeeds if the treatment group lives significantly longer than the control group, and fails if it does not. Secondary endpoints add information but do not determine the trial's overall success.

Choosing the right endpoint means balancing what matters clinically (does the patient feel better? live longer?) against what is practical (can we measure it reliably, in a reasonable timeframe, with an affordable sample size?). That tension runs through everything that follows.

17.1. The Endpoint Hierarchy

Table 17.1.: Clinical Trial Endpoint Hierarchy

Level	Definition	Examples	Regulatory Acceptance	Trial Implications
Clinical Endpoints	Direct patient outcomes	Survival, symptomatic improvement, functional status	Gold standard for approval	Large trials, long follow-up
Surrogate Endpoints	Predict clinical outcomes	Blood pressure -> CV events; HbA1c -> diabetes complications	Validated or reasonably likely	Smaller trials, faster readout
Biomarkers	Biological process indicators	Drug concentration, receptor occupancy, gene expression	Generally not for approval	Early development, dose selection

At the apex of the hierarchy (Table 17.1) sits what we ultimately care about: whether patients live longer, feel better, or function more effectively. These *clinical endpoints* (sometimes called *hard endpoints*) measure patient outcomes directly. Survival is the paradigm: either the patient is alive or they are not. There is no ambiguity, no measurement error, no room for interpretation.

Below them sit *surrogate endpoints*: measurements that are not themselves clinical outcomes but are reasonably expected to predict them. Blood pressure stands in for stroke and heart attack, viral load for AIDS progression, tumor shrinkage for cancer survival.

Further down are *biomarkers*: measurable characteristics that indicate biological processes, disease states, or responses to intervention. A biomarker can be a surrogate, but one that reliably reflects drug activity may still fail to predict clinical benefit.

17.2. Clinical Endpoints

In oncology, *overall survival* (OS) is the gold standard: the most reliable way to determine whether a cancer treatment extends life. In heart failure, the composite of *cardiovascular death and heart failure hospitalization* captures both mortality and major morbidity.

Progression-free survival (PFS), the time from randomization to objective tumor progression or death, is often used as a primary oncology endpoint when waiting for OS would make a trial impractically long. FDA accepts PFS as a basis for approval in many settings, but treats it as a judgment call, not an automatic substitute for clinical benefit: the magnitude of the delay, whether tumor-related symptoms improve, and the drug's toxicity all matter, because a longer time to progression

does not by itself establish that patients live longer or feel better (U.S. Food and Drug Administration 2018d).

That reliability comes at a price in trial size and duration. Events can be relatively rare and outcomes slow to develop, so a survival endpoint in a slowly progressing cancer can mean thousands of patients followed for years, with everything that implies for cost, time to approval, and patient access to therapy.

Patient-reported outcomes (PROs) ask the patient directly. How do you feel? What is your quality of life? Can you perform daily activities? Standardized questionnaires capture a perspective that may differ from what objective measurements suggest.

Regulators and payers increasingly want that perspective on record: a drug that moves a biomarker without making patients feel better is worth less than one that relieves symptoms. But responses can be influenced by expectations, mood, and cultural factors, so the instrument itself has to be shown to work.

Unlike a blood test, a questionnaire cannot be calibrated against a physical standard, so PROs are validated psychometrically instead, to prove they measure what they claim to. Content validity, usually established through qualitative interviews with patients, shows that the instrument covers the relevant symptoms; reliability makes the measure stable and reproducible; construct validity checks the score against other known measures of the disease; and sensitivity to change confirms the instrument can detect meaningful improvements. Regulators require this evidence before approving a labeling claim based on a PRO (U.S. Food and Drug Administration 2023i).

Smartphones and wearables have produced a newer category: endpoints derived from Digital Health Technologies (DHTs), which promise to replace the snapshot of a brief clinical assessment with continuous real-world monitoring. Digital biomarkers such as subtle changes in gait or voice may predict disease progression.

Validation is the hurdle. Regulators must be convinced that the device measures the concept accurately, that the measurement correlates with clinical status, and that the data is secure and attributable (U.S. Food and Drug Administration 2023h). The FDA's Digital Health Center of Excellence exists to guide these tools into regulatory decision-making.

17.3. Surrogate Endpoints

A surrogate can be measured sooner and in fewer patients, which can accelerate development and cut cost. The reasoning is clean: if intervention A lowers blood pressure more than intervention B, and lower blood pressure reduces cardiovascular events, then A should reduce cardiovascular events more than B.

But the logic can fail. The relationship between surrogate and outcome may not hold under all circumstances: an intervention might move the surrogate through

a pathway that does not influence clinical outcomes, or affect clinical outcomes through pathways the surrogate never captures.

The CAST trial of the late 1980s tested antiarrhythmic drugs in post-heart attack patients. The drugs effectively suppressed the abnormal heart rhythms they were designed to treat, and the surrogate looked excellent. The trial was stopped early because treated patients were *more* likely to die than those on placebo. The trial report called the mechanism of the excess deaths unknown (Echt et al. 1991): the rhythm abnormalities may have been a marker of underlying heart damage rather than a cause of death, and the drugs that suppressed them may have added cardiac risks of their own.

Hormone replacement therapy improved lipid profiles and was widely prescribed for cardiovascular prevention on the strength of surrogate reasoning and observational data. Then the Women's Health Initiative randomized trial showed that, despite the favorable lipid effects, hormone therapy increased cardiovascular events (Rossouw et al. 2002). The surrogate had predicted the wrong outcome.

The FDA distinguishes between *validated surrogates* (where the relationship to clinical outcome is well established) and *reasonably likely surrogates* (where the relationship is plausible but not fully established). The latter can support *accelerated approval* for serious conditions, but sponsors must then confirm the clinical benefit in post-marketing studies.

Molecular surrogates are moving through exactly this pathway now. In blood cancers, *minimal residual disease (MRD)*, the small population of cancer cells that highly sensitive assays can detect after treatment, has emerged as a candidate surrogate for long-term survival. In April 2024, the FDA's Oncologic Drugs Advisory Committee voted unanimously (12-0) that MRD is an appropriate endpoint to support accelerated approval in multiple myeloma, where confirming a progression-free or overall survival benefit can take many years (Landgren and Devlin 2025). In solid tumors, *circulating tumor DNA (ctDNA)* clearance is being studied as an analogous early signal in adjuvant trials. These molecular markers promise dramatically smaller and faster trials, but they carry the same obligation as every surrogate before them: the effect on the marker must ultimately be confirmed to predict the effect on how patients live and feel.

17.4. Primary and Secondary Endpoints

Table 17.2.: Endpoint Categories and Their Roles

Category	Role	Statistical Handling	Interpretive Rules
Primary	Main outcome for trial success/failure	Fully powered; Controls Type I error	Trial succeeds or fails on this

Category	Role	Statistical Handling	Interpretive Rules
Key Secondary	Pre-specified important outcomes	Part of testing hierarchy; Multiplicity-adjusted	Interpret only if primary positive
Secondary	Additional efficacy/safety info	May be multiplicity-adjusted	Supportive; Interpret with caution
Exploratory	Hypothesis-generating	No multiplicity adjustment	Signals for future studies only

The *primary endpoint* is the outcome the trial is designed and powered to assess, the basis for the primary analysis and usually the endpoint regulatory decisions rest upon. *Secondary endpoints* fill in around it: other aspects of efficacy, safety, or tolerability, outcomes in subpopulations, the durability of effects over time. Table 17.2 sets out how each category is analyzed and interpreted.

What a secondary endpoint is worth depends on the primary. If the primary is negative (no statistically significant difference between treatment groups), positive secondaries are difficult to interpret: they may represent chance findings from multiple comparisons. If the primary is positive, they confirm and characterize the result.

The problem behind those chance findings is *multiplicity*. With a conventional significance threshold of 0.05, one in twenty tests will be "positive" by chance alone, so the more endpoints a trial tests, the higher its overall probability of a false positive finding. *Multiplicity adjustment* (such as hierarchical testing, gatekeeping procedures, or graphical approaches) controls the inflation.

17.5. Composite Endpoints

A *composite endpoint* combines multiple outcomes into a single measure. A participant might be counted as having an event if they experience any one of: death, heart attack, stroke, or hospitalization for unstable angina. In the standard time-to-first-event analysis, each participant contributes only their first component event, and all components carry equal weight regardless of clinical severity (a hospitalization counts the same as a death). Because death precludes any later event, it acts as a competing risk for the non-fatal components.

Composites offer statistical efficiency: events are more common, since many participants will have at least one component, so smaller sample sizes may suffice. They also give a fuller picture of disease burden, capturing multiple manifestations of benefit or harm.

The cost is interpretive. If the composite is positive overall, which components drove it? Is a treatment that reduces hospitalizations but has no effect on death meaningfully beneficial, and should a drug that moves only the minor events be approved?

Components should be clinically related and of roughly similar importance; death and mild headache are not.

17.6. Time-to-Event Endpoints

A *time-to-event* endpoint records how long until a participant experiences the outcome, which, given long enough follow-up, lets the analysis separate a treatment that reduces eventual event rates from one that only delays them.

Survival analysis methods (Kaplan-Meier curves, log-rank tests, Cox proportional hazards models) are designed for this kind of data. They accommodate participants who have not yet experienced an event (*censored* at the time of analysis) and those who drop out before one occurs.

Many of these analyses assume *proportional hazards*: the hazard ratio is constant over time. If a treatment lowers the hazard (the event rate among patients still event-free) by 30% at month 6, it should also lower it by 30% at month 12; the relative reduction in cumulative risk is smaller than 30% and shrinks as events accumulate. When this assumption is violated (as when a treatment takes time to work, or when its effects wane over time) alternative methods may be needed.

17.7. Ordinal Endpoints

Many outcomes that appear binary are actually ordered. A patient with heart failure may die, be hospitalized, remain symptomatic but outpatient, or be nearly asymptomatic; collapsing that spectrum into "event" or "no event" discards clinically meaningful information.

Ordinal endpoints keep the ordering, treating the outcome as a ranked scale. The *modified Rankin Scale (mRS)* in stroke trials is the canonical example: it runs from 0 (no symptoms) through 6 (death), with each step representing a clinically distinct level of functional disability (Saver 2007). The WHO Clinical Progression Scale used in COVID-19 trials similarly ranked patients from "uninfected" (score 0) to "dead" (score 10), with intermediate steps for degrees of oxygen support and hospitalization (WHO Working Group on the Clinical Characterisation and Management of COVID-19 infection 2020).

The standard analysis for ordinal outcomes is the *proportional-odds (PO) model*, a generalization of logistic regression (Agresti 2010). It assumes the treatment odds ratio is the same at every cut-point of the ordinal scale (on treatment, the odds of being in a better category than the reference are OR times higher, regardless of where the cut-point is placed). When that assumption holds, the model uses all the information in the ordinal scale instead of dichotomizing at an arbitrary threshold. The PO assumption can be checked with a score test and, if it is violated, the model replaced by a proportional-hazards model on discrete time or by non-parametric rank tests.

For sponsors the benefit is power: at a given sample size, an ordinal analysis is typically more powerful than a binary analysis of the same data, because the ordinal outcome carries more information. For regulators the benefit is interpretability, since an effect that shifts the entire distribution toward better categories says more about clinical benefit than a count of patients crossing one threshold.

17.8. The Win Ratio and DOOR

A composite of "death or hospitalization" counts a 2-day readmission the same as death, which misrepresents clinical priority. Two related methods restore the ranking a composite flattens.

The *win ratio* adjudicates every possible pair of patients (one from the treatment arm, one from the control) against a hierarchy of outcomes: check the most severe outcome first, death, and if one patient wins on it, that patient's arm wins the pair; if the pair is tied on death, check the next outcome, hospitalization, and so on (Pocock et al. 2012). The statistic is the number of pairs the treatment patient wins divided by the number the control patient wins, with tied pairs excluded. Investigators set the hierarchy before the trial, which makes the clinical value judgment explicit. Win ratio analyses have been used in heart failure, oncology, and COVID-19 trials.

Desirability of Outcome Ranking (DOOR) extends the same idea to continuous outcomes and patient-reported data (Evans et al. 2015). Every patient receives a score (their DOOR) from a pre-specified combination of clinical outcomes, adverse events, and, optionally, patient preferences. Patients are ranked from worst to best, and the summary statistic is the probability that a randomly selected treatment patient has a better DOOR than a randomly selected control patient, with ties counted as half ($P[T > C] + 0.5\,P[T = C]$); no difference between arms gives 0.5, and a value above 0.5 means a treated patient is more likely to do better than worse, across the entire outcome spectrum (Evans and Follmann 2016). DOOR is most valuable when benefit and harm have to be summarized together, as in antibiotic trials where one measure should capture cure, adverse events, and resistance emergence.

17.9. Estimands and Death as an Intercurrent Event

The ICH E9(R1) estimands framework (International Council for Harmonisation 2019) formalized a question that had long been implicit in trial design: what treatment effect, precisely, is the trial trying to estimate? An *estimand* is the complete specification of the treatment comparison: the population, the treatment and control, the endpoint variable, how intercurrent events are handled, and the population-level summary measure.

Intercurrent events are events that occur after treatment initiation and affect either the interpretation or the observability of the endpoint. Treatment discontinuation, rescue medication use, and crossover to the other arm are common examples. Death is the hardest case.

When the primary endpoint is a non-fatal outcome (symptom score, functional status, quality of life), death is an intercurrent event that makes the outcome unobservable: a dead patient has no quality-of-life score. ICH E9(R1) defines five strategies for handling intercurrent events, and the choice among them for death fundamentally changes the estimand:

- **Composite (variable) strategy:** treat death as a component of the endpoint itself, typically by assigning it the worst possible score. This estimates the effect on the composite of survival and the non-fatal outcome.
- **Hypothetical strategy:** estimate what the outcome would have been if the patient had survived. This is a counterfactual that requires modeling assumptions.
- **Principal stratum strategy:** restrict the estimate to the subpopulation of patients who would survive under either treatment assignment (the survivor average causal effect).
- **While on treatment strategy:** use the outcome only up to the intercurrent event; for death, E9(R1) calls this *while alive*, and the estimand concerns symptoms or function while the patient is alive. Treatment discontinuation, a different intercurrent event, needs its own strategy.
- **Treatment policy strategy:** use the endpoint value regardless of whether the patient discontinued treatment or took rescue medication. Unlike the strategies above, it offers no way to handle death, because a non-fatal outcome such as quality of life simply does not exist once the patient has died.

The practical consequence for trial teams is that an endpoint cannot be chosen without specifying its estimand at the same time: a quality-of-life endpoint and a symptom-improvement endpoint can be the same measurement and still be different estimands, depending on how death is handled. Pre-specifying it prevents post hoc ambiguity about what the trial demonstrated and keeps the analysis aligned with the clinical question that motivated the trial.

17.10. Endpoint Selection in Practice

Regulators generally favor well-validated clinical outcomes, but sponsors must also confirm that the endpoint captures benefit patients and payers will recognize. Operationally, the endpoint has to be measurable with enough precision to maintain statistical power, and timed to avoid unnecessary delay while still providing a reliable prediction of long-term benefit. For patient-reported outcomes, established and validated instruments are the most direct route to findings that are reproducible and credible to external stakeholders. AI-based endpoint assessment, including automated image scoring, wearable-derived digital biomarkers, and NLP extraction of outcomes from clinical notes, is covered in Chapter 26.

18. Site Selection and Feasibility

A clinical trial exists only on paper until it enrolls patients, and patients are enrolled at clinical sites: hospitals, academic medical centers, private practices, or specialty clinics where investigators conduct the research. Industry surveys consistently show that patient recruitment is among the most common reasons for trial delays, and the wrong sites are where slow enrollment and poor data quality can begin. Historically, a Tufts CSDD benchmark of more than 150 trials run between 2008 and 2010 found that roughly 11% of investigative sites fail to enroll a single patient, with an additional 37% under-enrolling relative to their targets (Tufts Center for the Study of Drug Development 2013; Getz 2012). For a significant portion of activated sites, the substantial investment in startup and training yields little to no scientific data. More recent Tufts CSDD benchmarking is somewhat more encouraging: comparing site-activation and enrollment metrics across 2012, 2019, and 2023, Lamberti and colleagues found that actual enrollment exceeded planned enrollment for a majority of studies and that timelines were often shorter than expected, though achievement varied by region and site type (Lamberti et al. 2024).

For a Phase III trial requiring 1,000 patients, a sponsor might need 100 or more sites across multiple countries, each capable of identifying eligible patients, conducting the study procedures competently, maintaining regulatory compliance, and producing high-quality data. Finding sites that meet all these criteria, and that are genuinely committed to the trial, is harder than it might appear.

18.1. Site Identification

The search for sites begins with investigators who have access to the target patient population. A rheumatoid arthritis trial needs rheumatologists. An advanced lung cancer trial needs oncologists at centers that see late-stage disease. A rare disease trial may need the handful of specialists worldwide who see enough patients to contribute meaningfully.

Access to patients matters most: a distinguished investigator who rarely sees the target population will not enroll patients. *Geographic distribution* matters for global trials, where regulatory authorities may require data from relevant populations and where sponsors want to understand drug effects across regions, and *prior experience* with the sponsor, the therapeutic area, or clinical research generally can indicate capability.

Databases of investigators, clinical trial registries, publication records, and prior trial performance all inform site identification, as do sponsor relationships with key opinion leaders and clinical networks. What those sources turn up ranges from large academic medical centers (AMCs) and integrated health systems to small, independent physician practices and site networks (see Figure 18.1).

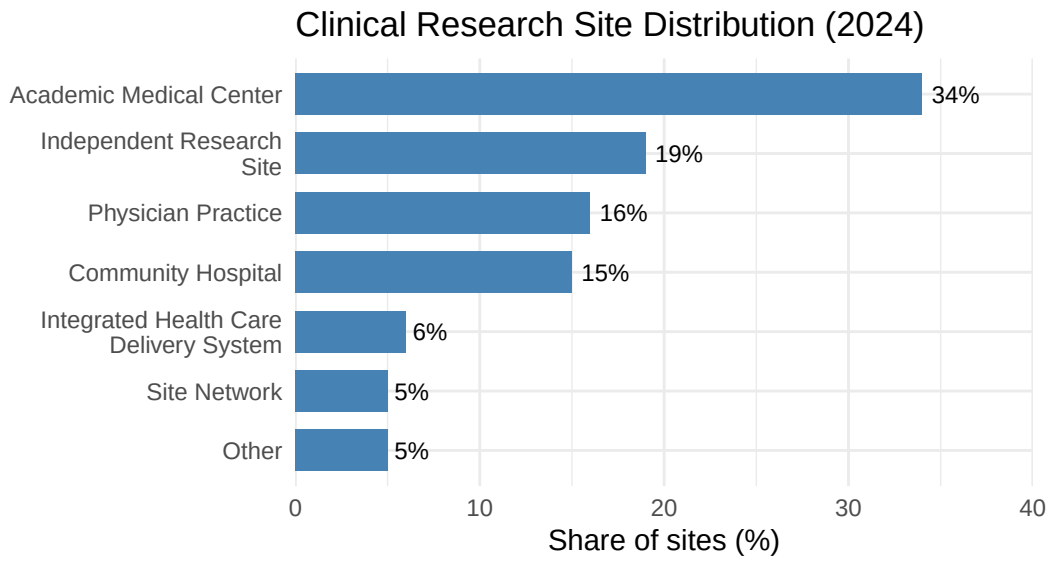


Figure 18.1.: Distribution of Clinical Research Site Types (2024). Data from WCG Clinical (2024).

The pool of available investigators is shrinking. A 2004 survey found that about 13% of physicians actively served as clinical investigators (Harris Interactive 2004); the 2024 WCG Clinical Research Site Challenges Report documents the number of unique U.S. investigators falling roughly 78%, from a pandemic-inflated peak of over 28,000 in 2020 to approximately 6,200 in 2024, alongside a 57% decrease in U.S. trial starts over the same period (WCG Clinical 2024). That headline overstates the structural trend: measured from the pre-pandemic level of about 23,000 investigators, the decline is closer to 73%, still a steep erosion of the base. The 2024 investigator and trial-start counts are also preliminary, drawn from a partial-year ClinSphere snapshot that exaggerates the single-year drop from 2023; WCG's more recent site-challenges reporting has continued to show the same downward direction of travel.

The two trends in Figure 18.2 are distinct. The investigator pool surged with the pandemic in 2020 and then crashed, while new trial starts held relatively stable, peaking near 2,000 in 2021, before falling to fewer than 800 in 2024. Several forces drive that decline. COVID-19-related clinical activity, which peaked in 2021, has dropped substantially. Biopharma funding slowed sharply in 2022 and only partially rebounded in 2023, remaining well below its 2020-21 pandemic peak and constraining emerging-biopharma companies (IQVIA Institute for Human Data Sci-

ence 2024a). With the cost of bringing a single novel asset to launch, failures included, now estimated at $2.6-2.7 billion (Deloitte Centre for Health Solutions 2026; DiMasi, Grabowski, and Hansen 2016), sponsors have grown risk-averse, concentrating capital on existing late-stage pipelines instead of new early-phase starts. Mounting regulatory hurdles and administrative burdens, meanwhile, have pushed study start-up past 60 days at most sites, discouraging new starts during periods of economic uncertainty (WCG Clinical 2024).

Other pressures are structural. Increasing clinical demands leave physicians with limited bandwidth for research, while research team staffing shortages further strain site capacity. This has exacerbated the **"one and done" phenomenon**, where approximately 66% of physicians who participate in a clinical trial do so only once before exiting the research ecosystem (WCG Clinical 2024). The consequences of this investigator attrition for recruitment and retention are discussed further in Chapter 19.

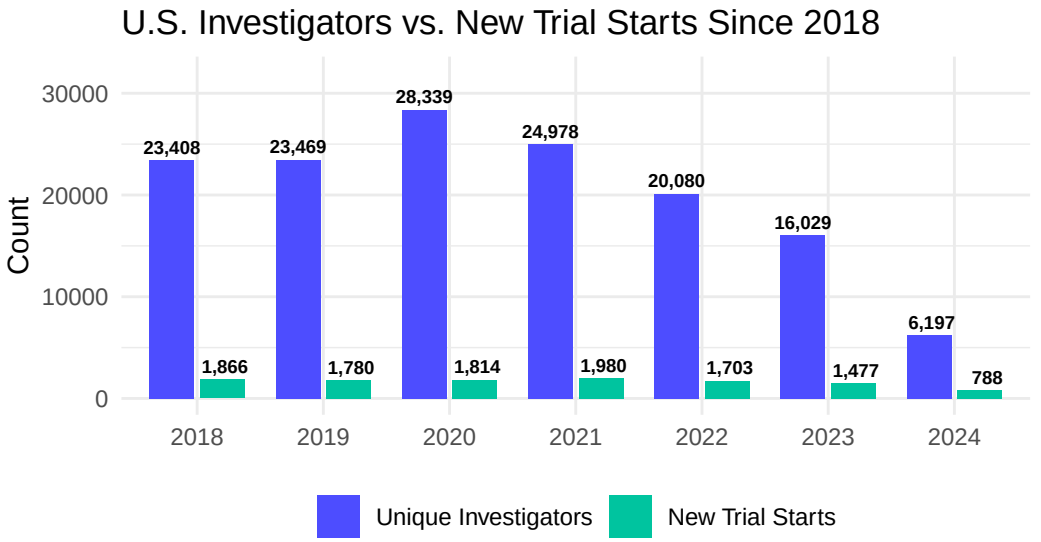


Figure 18.2.: U.S. Investigators Participating in Clinical Trials Compared to New Trial Starts (2018-2024). Data from WCG Clinical (2024).

In that same 2004 survey, among physicians not involved in clinical trials, 38% cited lack of opportunity as the main reason for not serving as a Principal or Sub-Investigator, while 32% indicated that the time commitment is prohibitive (Harris Interactive 2004). Those constraints, compounded by the current exit of experienced investigators, make site identification a search for a rare combination: expertise, patient access, and operational bandwidth.

Historically, clinical trials have often failed to represent the diverse populations that will ultimately use the investigated therapies, which compromises the external validity of the results and can obscure important differences in safety or efficacy across

demographic groups. The Food and Drug Omnibus Reform Act of 2022 (FDORA) requires sponsors to submit Diversity Action Plans for Phase III trials (and certain other pivotal studies) to the FDA, but only for studies that begin enrolling more than 180 days after FDA publishes final guidance on the plans (U.S. Congress 2022). That statutory requirement stands, but its implementing guidance has been in flux: FDA's June 2024 draft Diversity Action Plans guidance was withdrawn following a January 2025 executive order on diversity, equity, and inclusion, then restored under court order weeks later, and its final version was not issued by the statutory deadline of June 26, 2025 (Grossi 2025; U.S. Food and Drug Administration 2024e). A separate December 2025 final guidance, *Enhancing Participation in Clinical Trials*, kept the emphasis on representative enrollment while dropping the "diversity" framing (U.S. Food and Drug Administration 2025c). Until the final Diversity Action Plans guidance appears, the submission requirement has not taken effect, so sponsors should confirm the current posture rather than treat the mandate as settled.

Site selection is the primary operational lever for achieving diversity goals, which means engaging sites embedded in underrepresented communities: community hospitals, federally qualified health centers (FQHCs), or practices in geographic areas with high demographic density of the target populations. Those sites may lack the dedicated research infrastructure (study coordinators, -80°C freezers, secure storage) found in major academic centers, so the assessment must weigh a site's potential capability with appropriate support alongside its current capability. The goal is a site network that geographically and demographically mirrors the epidemiology of the disease.

18.2. Feasibility Assessment

Once potential sites are identified, a *feasibility assessment* evaluates whether each site can realistically conduct the study. It typically involves questionnaires completed by site staff and may be supplemented by in-person or virtual visits. Table 18.1 summarizes the questions it asks and the answers that count as red flags.

Table 18.1.: Site Feasibility Evaluation Criteria

Category	Key Questions	Red Flags
Patient Access	How many target patients per year? Competing trials?	<10 eligible patients/year; Multiple competing studies
Investigator	Prior trial experience? Enrollment track record?	No research experience; History of poor enrollment
Infrastructure	Dedicated research staff? Equipment? Data systems?	No study coordinator; Inadequate facilities
Regulatory	IRB timeline? Institutional requirements?	IRB backlog >3 months; Complex approval layers

Category	Key Questions	Red Flags
Commitment	PI availability? Staff turnover?	Part-time PI; High coordinator turnover

Each row calls for judgment. A site's count of eligible patients has to be checked against the trials it is already running; investigator review covers licensure as well as enrollment track record; infrastructure means space for study procedures and laboratory capabilities along with dedicated research staff; and regulatory readiness turns on the efficiency of the local IRB and on any institutional requirements that could delay startup. Commitment is gauged from PI availability and coordinator turnover history, since a part-time PI or high turnover can undermine execution even at an otherwise well-resourced site.

18.3. The Enrollment Prediction Problem

Sites are notoriously over-optimistic about how many patients they will enroll. An investigator who claims to see 50 eligible patients per month and predicts enrolling 30 may ultimately enroll 3.

The gap has several sources. Investigators may not accurately know how many of their patients meet eligibility criteria: the criteria are often more restrictive than general clinical practice. Patients who appear eligible on paper may decline participation or prove ineligible upon screening, competing trials may draw away others, and staff turnover or other operational issues may limit enrollment capacity.

Experienced sponsors treat site predictions with skepticism and routinely discount them, often steeply (Getz 2008). Historical performance in prior trials is a more reliable guide than investigator optimism.

18.4. Financial and Contractual Negotiations

Between a qualified site and an activated one lies the negotiation of the Clinical Trial Agreement (CTA) and the site budget, and it is a consistent bottleneck: industry benchmarks show that the cycle time from site selection to site initiation is driven more often by contract execution than by regulatory approval (WCG Clinical 2024).

The CTA is a legally binding contract governing the relationship between sponsor and institution, addressing liability, indemnification, intellectual property rights, data ownership, and publication policy. Negotiations can be protracted, particularly with large academic medical centers that adhere to rigid institutional policies regarding indemnification and subject injury. For global trials, these agreements must also align with local laws regarding data privacy and anti-bribery regulations.

The site budget requires a granular accounting of both per-patient costs and overhead fees. A well-constructed one covers per-subject costs for clinical procedures, investigator time, and pharmacy services; startup fees for the administrative effort of regulatory submission and staff training; pass-through costs for variable expenses such as IRB fees and participant travel; and an institutional overhead surcharge applied to direct costs.

A related exercise, especially for U.S. sites, is the *coverage analysis*. Before a trial opens, the site reviews each protocol-required procedure to determine which costs may be billed to a participant's insurance as routine clinical care and which must be paid by the sponsor as research costs. In the United States this determination rests on the Medicare Clinical Trial Policy (National Coverage Determination 310.1, "Routine Costs in Clinical Trials") and, for commercial plans, the Affordable Care Act's requirement that the routine patient costs of qualifying clinical trials be covered. Getting it wrong exposes the institution to double-billing and false-claims liability, so it demands specialized revenue-cycle expertise. Coverage analysis is one component of the study start-up work (alongside budgets and contracts) that sites most often identify as their leading source of start-up delay (WCG Clinical 2024).

18.4.1. Operational Bottlenecks by Site Type

The administrative burden of study start-up falls unevenly across the research ecosystem. According to the 2024 WCG Clinical Research Site Challenges Report, "Study Start-up" remains the single largest operational obstacle for larger institutions (academic medical centers, health systems, and community hospitals), with 43% of these sites identifying it as a primary challenge (WCG Clinical 2024). In contrast, smaller independent sites, physician practices, and site networks report start-up as a top challenge only 24% of the time, focusing instead on recruitment and retention as their primary burden (39%).

The timelines diverge accordingly. Approximately 60% of independent sites and physician practices report being able to initiate studies in under 60 days; 77% of academic medical centers, health systems, and community hospitals report start-up timelines exceeding 60 days, often because of the complexity of managing multiple internal stakeholders, the lack of centralized research administration, and bureaucratic regulatory layers (WCG Clinical 2024). Across all site types in the United States, "Budgets and Contracts" remain the single largest bottleneck, cited by 77% of sites as the primary contributor to start-up delays (see Figure 18.3).

Sponsors design their selection and activation strategy around these profiles, often favoring independent networks for rapid enrollment and academic centers for complex therapeutic areas or highly specialized patient populations.

Budget negotiation requires balancing fiscal discipline with the reality that sites must cover their costs to remain viable. Underfunding a site invariably shows up later as slow data entry and as deprioritization against better-funded trials.

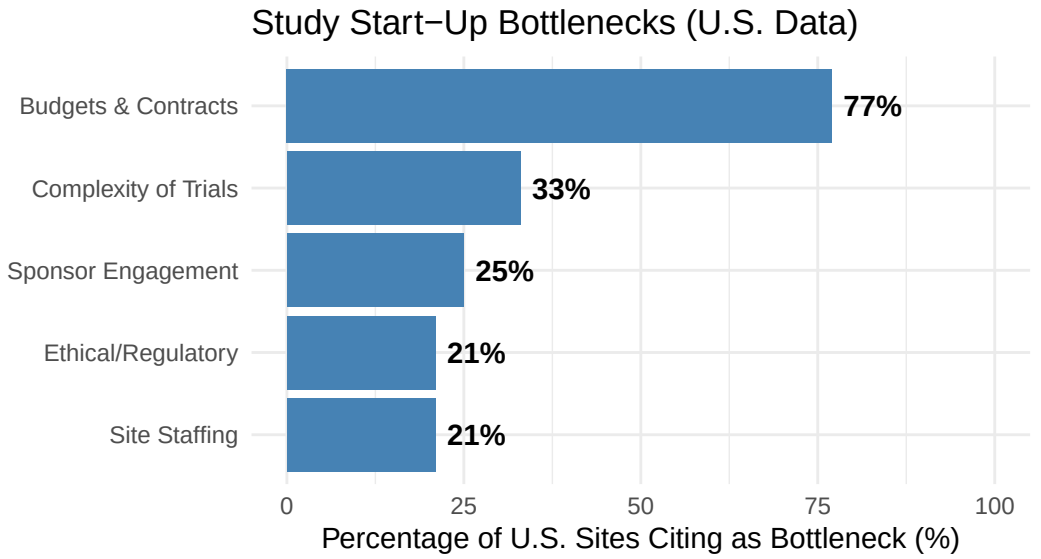


Figure 18.3.: Largest Contributors to Study Start-Up Timelines in the U.S. (2024). Data from WCG Clinical (2024).

18.5. Site Initiation

Once a site is selected and regulatory approvals are obtained, it must be *initiated*: prepared to enroll patients and conduct the study. Figure 18.4 shows the full site lifecycle from identification through closeout.

The *Site Initiation Visit (SIV)* formalizes training. During the SIV, study personnel review the protocol in detail, walk through study procedures, demonstrate equipment, set up data systems, and ensure that essential supplies are in place. The investigator and key site staff must participate; their engagement (or lack thereof) often signals future performance.

Following the SIV, the site is authorized to begin screening and enrolling patients. The time from site selection to the point at which a site is cleared to enroll (the *site activation time*), and from there to the first patient enrolled, can range from weeks to months depending on regulatory complexity, contract negotiations, and site responsiveness.

A global network needs training delivered consistently across languages and time zones, usually through a combination of e-learning platforms and regional investigator meetings. Communication has to strike a balance: enough information to keep sites aligned, without overwhelming staff with unnecessary updates. Performance management systems track real-time metrics (enrollment rates, data quality, query resolution times) to identify the sites that need additional support or, in extreme cases, closure. Sponsors treat high-performing sites as long-term partners,

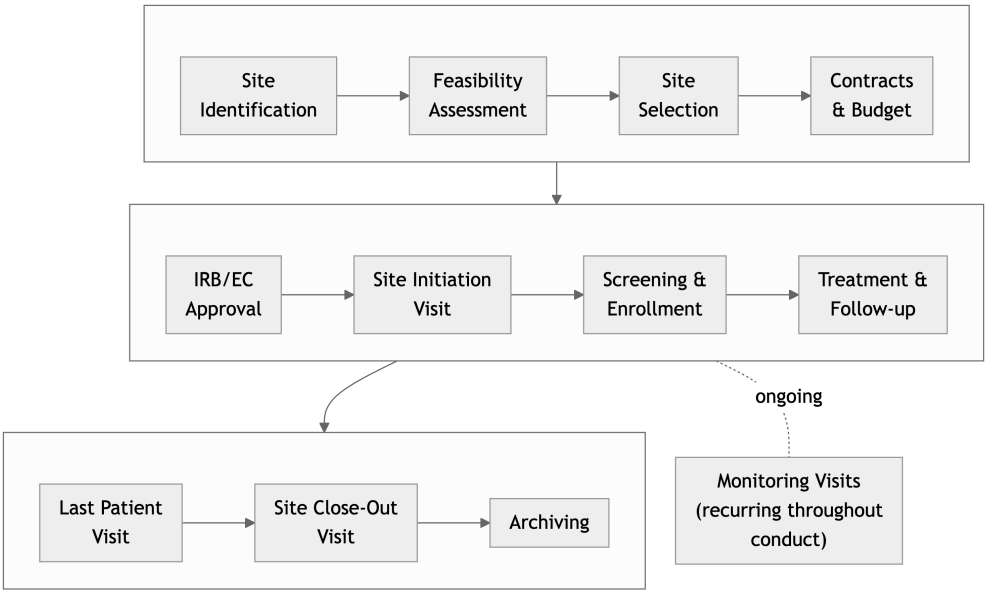


Figure 18.4.: Clinical trial site lifecycle from identification through study closeout (monitoring visits recur throughout active conduct)

and investment in those relationships through transparent communication and professional support improves enrollment, data quality, and site retention across the drug development program.

Global trials add complications of their own. Regulatory requirements differ by country, and language barriers affect training and documentation. Cultural factors influence patient willingness to participate in research, expectations about investigator relationships, and approaches to adverse event reporting. Standard of care may differ across regions, affecting choice of comparator and interpretation of results, while patient populations may differ in disease characteristics, comorbidities, and concomitant medications. Some regions may have limited research infrastructure requiring more intensive support, while others may have mature research environments but intense competition for investigators and patients. All of it feeds back into trial design and analysis.

18.6. Sites New to Research

Table 18.1 lists “no research experience” as a red flag, and for a sponsor that needs fast enrollment it is one. For the trial population it is often the opposite. Patients who live far from academic centers can join a trial only if a site near them takes part, and many such sites have never run one. This section covers what it takes to bring a hospital or clinic with no trial experience into a study: what it lacks, who supplies it, and how a sponsor oversees it.

The literature has no standard term or definition. Industry writing uses "research-naive" or "trial-naive" loosely for community hospitals and clinics with little or no trial history. "Rural" is no better defined: a scoping review of pediatric trials in rural and community settings found no common definition of rurality, and about a third of the articles described their setting too briefly to tell (Watson et al. 2022).

18.6.1. Where the Gap Is

In cancer, the size of the gap is known. Using Commission on Cancer accreditation data for 2013 to 2017, Unger and colleagues estimated that 7.1% of adult cancer patients took part in a treatment trial and at least 21.9% in some kind of research study. Treatment trial participation was 21.6% at NCI-designated comprehensive cancer centers, 5.4% at academic programs without NCI designation, 5.7% at integrated network programs and 4.1% at community programs. The programs outside NCI designation made up 96.4% of the accredited sites, and the authors point to the infrastructure and staff support that NCI centers receive and other programs lack (Unger et al. 2024).

The facilities to close the gap already exist. Chapter 19 cites the estimate that nearly 17% of U.S. residents aged 35 and older live more than 100 miles from an NCI-funded trial site. The same study found that counting the Commission on Cancer accredited programs as well cuts that share to 1.6%; for people below the poverty line, the figures are 20.2% and 1.9%. Over 97% of NCI cancer center and NCORP facilities lie in urban core areas, against 85% of accredited programs, and only three NCI-funded sites lie outside urban cores, against 179 accredited programs (Shriver et al. 2025). Most of the hospitals that could bring trials to rural patients treat cancer already. What they lack is trial capacity.

For children, the gap shows up as distance to a subspecialist. In 2019, depending on the subspecialty, between 2% and 53% of US children lived 80 or more miles in a straight line from a pediatric subspecialist, measured from the subspecialist's address to the child's zip code (Turner, Ricketts, and Leslie 2020). A 2025 erratum corrected the paper's sampling and coding errors, and the corrected range is the one quoted here.

18.6.2. What a First-Time Site Lacks

Direct evidence on first-time sites is thin. The most detailed assessment comes from a network that was building capacity on purpose. When the ECHO IDeA States Pediatric Clinical Trials Network assessed its 17 sites and 7 affiliates while the network was being built, every principal investigator reported prior trial experience, but only 38% to 42% of study coordinators reported experience in the relevant domains. Most sites had an on-site pharmacy (93%) and a research pharmacy for investigational agents (79%), and 92% to 96% could store and ship biosamples, yet about 17% did not use an electronic medical record. The barriers the sites named were knowledge gaps among the regional affiliates about local study conduct and

follow-up, little protected research time, grant budgeting and contracting, and the need for trusted community champions (Annett et al. 2021). These were sites of an organized network with academic leads, so a hospital starting alone is likely to lack more.

The pediatric scoping review found the same leading barrier from the other side. Across 27 articles covering 31 U.S. studies, insufficient staff and resources was the barrier cited most often, followed by family distrust, lack of family time, burdensome or frightening procedures, and transportation. Only six of the studies were predominantly or exclusively rural, only two examined rural retention, and none compared strategies head to head or measured their cost-effectiveness (Watson et al. 2022).

One study measured what first-time sites had to build. In 2017 the same network joined a pharmacokinetic study run by the Pediatric Trials Network. Of its 18 sites, 4 had been active before the network existed and 14 were activated between April 2018 and May 2019, and together they recruited 382 participants by the study's close in September 2019. Of the 20 principal investigators, 7 (35%) were early-stage investigators and 7 (35%) were new to pharmacokinetic research, as were 8 of the 18 coordinators. To take part, 10 sites (56%) added study staff, 7 (39%) added equipment or space, 12 (67%) built new specimen-processing workflows, 5 (28%) wrote new SOPs, and 16 (89%) built new EHR queries to find eligible patients. Sites rated the effect on their research team's confidence at 8.1 on a 0 to 10 scale (Knight et al. 2022). The paper reports activation dates and enrollment, not comparisons with experienced sites, and does not report how many participants were rural.

From these sources and from the start-up steps described earlier in this chapter, a site's first trial typically lacks five things: a coordinator who knows GCP and the sponsor's systems; a regulatory file and the habit of keeping it; a contract and budget office that has negotiated a clinical trial agreement before; a billing process that can separate research costs from routine care under a coverage analysis; and a pharmacy that can receive, store and account for investigational product. Each of these can be supplied from outside the site, but someone has to be named to supply it.

18.6.3. Who Does What

Table 18.2 sets out a division of labor that puts the specialized work on the party that already has the expertise, and leaves the site the tasks only it can do. It is a planning template, not a description of measured practice.

Table 18.2.: Division of Labor for a Site New to Research

Task	Sponsor, CRO or coordinating center	Site investigator and coordinator	Host institution
Readiness assessment	Visits or calls the site; checks staff, space, pharmacy, records, internet access	Answers honestly, including what is missing	Confirms support from administration
Regulatory file	Supplies a pre-built file structure and templates (delegation log, training log, Form 1572)	Keeps the file current from the first day	Provides secure storage
Training	GCP, protocol, EDC and safety reporting training before activation; repeated after amendments	Completes training and documents it	Protects research time for staff
IRB	Arranges reliance on a single IRB where the trial allows it	Completes local paperwork	Signs the reliance agreement
Contract and budget	Offers a standard template and a budget that covers start-up and coordinator time	Reviews feasibility of the budget	Signs; sets up the research account
Coverage analysis and billing	Provides or funds the coverage analysis	Flags research-only procedures at each visit	Bills routine care and research correctly
Investigational product	Ships, sets up accountability logs, trains on storage	Dispenses and records	Pharmacy receives and stores
First enrolled patient	Reviews the first consent and first visits promptly, remotely or on site	Calls the monitor before and after the first visit	
Ongoing oversight	Central review of data and deviations; visits targeted to the site's risks	Answers queries quickly	

18.6.4. Oversight in Proportion to Risk

ICH E6(R3) asks sponsors to scale oversight to the risks that matter for participant safety and the reliability of results (International Council for Harmonisation 2025b), and site inexperience is a risk factor in its own right. In practice that argues for more attention early (review of the first consents, the first eligibility decisions and the first safety reports) and less once the site has shown it can do the work. Central monitoring of the kind described in Chapter 22 can follow a small site between visits, because the data reach the sponsor whether or not a monitor travels.

Decentralized elements can let a small site take part without carrying every procedure. FDA's 2024 guidance on decentralized elements describes trial visits conducted by telehealth, in participants' homes, or with local health care providers (U.S. Food and Drug Administration 2024n), so a rural clinic can handle routine visits close to home while the specialized procedures stay at a larger center. Chapter 19 describes what those elements do for patients' travel.

18.6.5. What Capacity Programs Have Measured

Several programs have tried to bring trials to community and rural settings. Table 18.3 lists those with published figures, and what each figure does and does not show.

Table 18.3.: Published Measures from Community and Rural Trial Capacity Programs

Program or study	What was measured	What it does not show
NCI Community Oncology Research Program (NCORP)	Launched 2014; in 2019, 7 research bases, 32 community sites, 14 minority/underserved sites (at least 30% minority or rural patients) and more than 1,000 affiliates (National Cancer Institute 2019). NCI's NCORP director reported 71,184 accruals from August 2014 to April 2022 (McCaskill-Stevens 2022)	Accrual per newly added affiliate; rural share of accrual
Mayo Clinic, three campuses, 2016 to 2017	731 of 2,313 trial participants (32%) were rural, against 19% of the population of the nine states served; rural participants traveled 103 miles on average, urban 68 (Bharucha et al. 2021)	Whether regional sites, rather than referral to campuses, drove rural enrollment
Mayo Clinic decentralized trial program, 2024 to 2025	Across 765 trials and 7,469 participants, the rural share rose from 16.6% to 24.9% between the first quarters of 2024 and 2025 (Haddad et al. 2026)	Cause: a descriptive study, and the authors do not attribute the change to the program
Academic molecular tumor board for rural community cancer centers	Of 151 patients reviewed, 27 were matched to a trial and 1 enrolled; distance to the trial center was the barrier cited most (70%) (Kumar et al. 2024)	Expertise without a local trial did not produce enrollment
ASCO and ACCC site self-assessment pilot	62 sites; 82% identified opportunities for improvement (Guerra et al. 2023)	Any change in enrollment after the assessment

Program or study	What was measured	What it does not show
Iowa shared care of children with cancer, about 15 years	137 community primary care physicians shared management of 226 children on oncology protocols; survival did not differ from 240 children treated by pediatric oncologists alone (Kisker et al. 1997)	Retrospective comparison, decades old
SWOG Pragmatica-Lung, 2024 update	About 20% of accrued patients lived in a rural location, with accrual spread across NCORP sites, academic sites and other rostered sites (National Cancer Advisory Board 2024)	Rural definitions varied; a single trial

Two kinds of program have no outcome data at all. We found no peer-reviewed report of Project ECHO tele-mentoring producing trial activations or enrollment at community sites; the one curriculum aimed at trial access that we found reported preliminary survey results only (National Association of Chronic Disease Directors 2026). We also found no citable trial-access outcomes for networks serving American Indian and Alaska Native communities. Evidence can also be lost: a 2023 retrospective of an NCI program that placed radiation oncology trials in six institutions serving underserved patients could not collect patient data so long after the program ended, and relied on a survey of its principal investigators (Coleman et al. 2023).

18.6.6. What the Evidence Still Does Not Show

No study we found compares the activation time, accrual or protocol deviation rates of first-time sites with those of experienced sites, and none compares ways of supporting them. The ISPCTN pharmacokinetic study reports what new sites built and when they opened, not how they performed against others. Figures for NCORP's share of national accrual and its minority accrual by site type appear in secondary sources that we could not check against the original evaluation. The advice above rests on the regulatory requirements, the barriers sites report, and the start-up steps every site must complete. A sponsor that brings new sites into a trial should record their start-up times, deviations and enrollment, because at present that is the only way anyone will know what works.

Checklist: Activating a Site with No Trial Experience

- Name one person at the sponsor, CRO or coordinating center as the site's contact for the whole start-up.
- Assess readiness in person or by video, including pharmacy, records access and internet connection.
- Provide a pre-built regulatory file and walk the coordinator through it.
- Train on GCP, the protocol, EDC and safety reporting before the initiation visit, and record the training.
- Fund the coordinator's time and the coverage analysis in the budget.
- Use a single IRB and a standard contract where the trial allows.
- Review the first consent, eligibility decision and safety report within days, not at the next routine visit.
- Track start-up time, deviations and enrollment for the site, and compare them with the trial's experienced sites.

19. Patient Recruitment and Retention

A trial cannot succeed without patients. Getting enough of them, quickly enough, is one of the most persistent problems in clinical research. Enrollment periods projected to take months stretch into years. Sites expected to contribute dozens of patients contribute single digits. Some trials are abandoned altogether when enrollment proves impossible.

The Forum on Drug Discovery, Development, and Translation identifies participant recruitment as one of the most "onerous" and high-risk steps in the entire development process, characterized by significant cost, time, and likelihood of failure (Wagner et al. 2018a). Industry surveys have historically identified recruitment and staffing as the primary bottlenecks, but by 2024 the "Complexity of Clinical Trials" (38%) had overtaken recruitment and retention (36%), study start-up (35%), and site staffing (31%) as the top challenge facing research sites (Figure 19.1) (WCG Clinical 2024). Nearly half (47%) of all sites (and 50% of U.S. sites) reported that these operational challenges forced them to reduce the number of studies they agreed to participate in during the preceding year (WCG Clinical 2024).

Sites respond by adding capacity and by shedding work. Hiring additional staff (42%) and prioritizing training (36%) are the common answers, but many sites also elect to participate in fewer studies (27%) or to select trials with less complex designs (21%) to stay operationally viable Figure 19.2.

The pressure has been building for a long time. Trials have grown larger, target populations more specialized, and competition for the same subjects more intense; decades of Tufts CSDD benchmarking document steadily rising protocol complexity and enrollment targets, with more procedures and stricter eligibility criteria per study (Getz, Smith, and Kravet 2023; Tufts Center for the Study of Drug Development 2021). Yet only a small fraction of eligible patients ultimately enroll, even though public-perception surveys consistently find that most people say they would be willing to consider a trial if asked. That gap between stated willingness and actual enrollment is what recruitment strategy exists to close.

Slow recruitment has cascading consequences. Patients wait longer for therapies that might help them, sites stay open and under oversight while the budget runs, and every month of delay shortens the effective patent life of a drug that works. A prolonged enrollment period may also compromise scientific validity by introducing population heterogeneity.

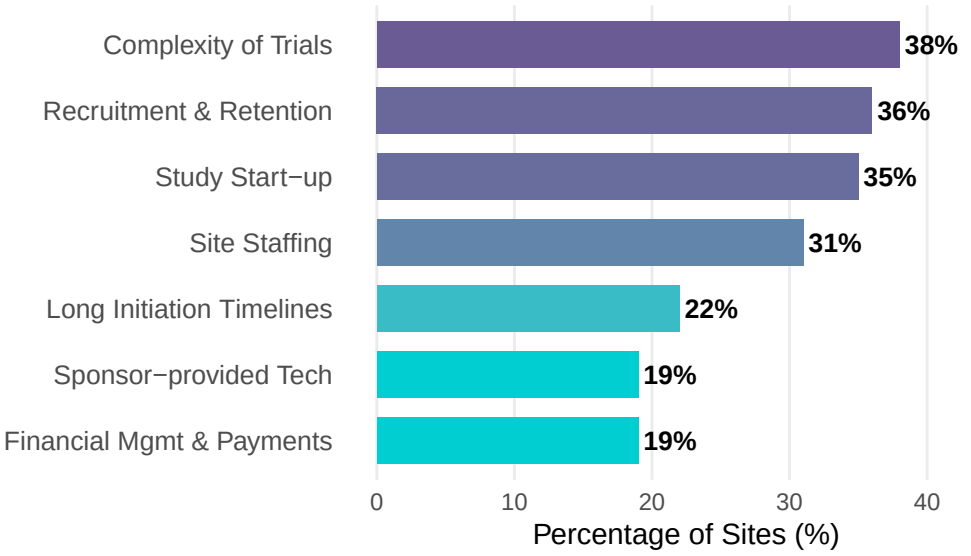


Figure 19.1.: Top issues facing clinical research sites in 2024. Data from WCG Clinical (2024).

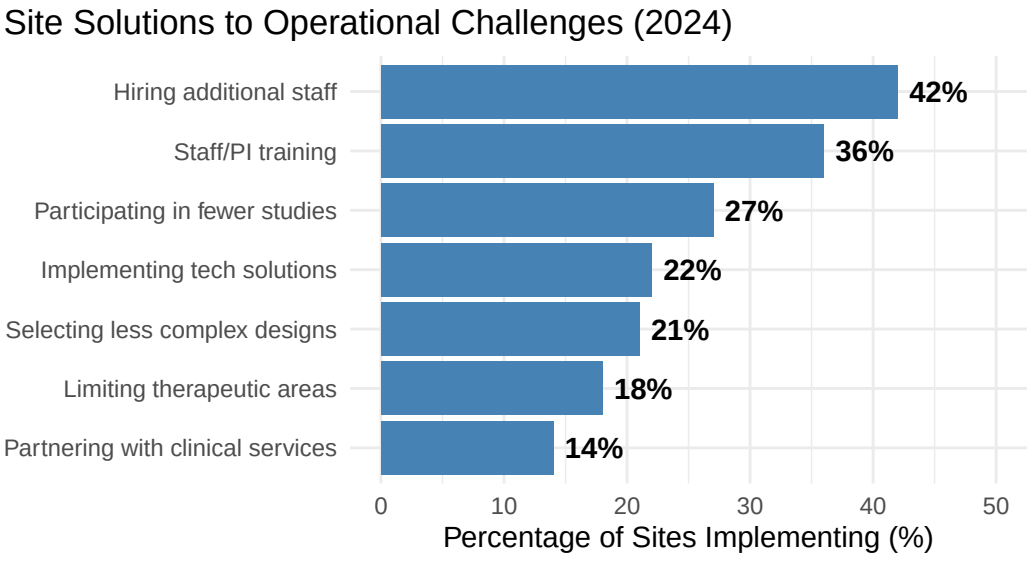


Figure 19.2.: Solutions implemented by clinical research sites to combat operational challenges (2024). Data from WCG Clinical (2024).

Start with the calculus each prospective participant faces, weighing perceived risks against potential benefits. Patients with serious illness may see a trial as their most promising treatment option; patients whose condition is already well managed weigh uncertain medical risks against the practical burden of frequent visits and travel. Altruism enters the calculation, and so does the patient's trust in the healthcare system, a trust that a community's historical experience with clinical research can shape.

19.1. Strategies for Recruitment

Recruitment is an attrition problem. Figure 19.3 shows the shape: between the target population and the patients who actually enroll, a different barrier removes people at every stage. Reaching enough of them means working several channels at once (Table 19.1).

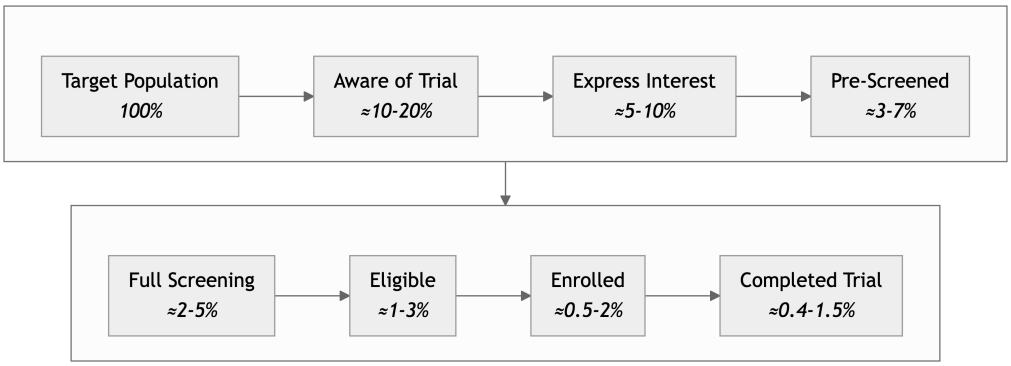


Figure 19.3.: Illustrative patient recruitment funnel (order-of-magnitude ranges; varies by indication and protocol)

Table 19.1.: Recruitment Strategies Comparison

Strategy	Description	Best For	Typical Cost
Provider Referral	Physicians identify/refer eligible patients	Specialty conditions; Established relationships	Low-Medium
Community Outreach	Patient groups, advocacy organizations, support groups	Rare diseases; Close-knit patient communities	Low-Medium
Direct Advertising	TV, radio, print, digital media	Large trials; Common conditions	High
Social Media	Targeted campaigns on Facebook, Instagram, patient forums	Younger populations; Specific demographics	Medium

Strategy	Description	Best For	Typical Cost
Registry Databases	Pre-screened patient lists from research networks	Rare diseases; Longitudinal studies	Medium
Decentralized Elements	Remote consent, telemedicine, home visits	Geographic barriers; Mobility-limited patients	Medium-High

Provider referral depends on a physician or other clinician spotting an eligible patient and raising the option. Because most patients are not actively seeking trials, provider engagement is a primary driver of enrollment; because busy clinicians may not think to mention a trial to an appropriate patient, education and reminders are part of the work. *Direct-to-patient advertising* goes around the clinic, using television and radio advertisements, online banner ads, social media campaigns, and search engine marketing, all of it subject to regulations governing what claims such advertising can make. *Decentralized trial elements* work on burden instead of awareness, bringing the trial to patients who cannot easily travel to a traditional research site.

19.2. AI/ML-Enabled Recruitment and Patient Engagement

AI and machine learning can help recruitment along two routes: making eligibility and outreach *computable*, so that organizations can find potentially eligible participants in real-world data with less manual effort, and supporting scalable, consistent communication that reduces friction and helps sustain retention. Both are sociotechnical systems whose value depends on data quality, workflow design, and human oversight, and both can introduce new risks if deployed without governance.

19.2.1. Making eligibility computable: feasibility and EHR-based identification

Before outreach begins, sponsors and sites have to decide whether the protocol is operationally plausible at all: how many patients are likely to qualify at a site or network, and which criteria will drive the screen failures. Data-driven feasibility answers those questions from historical EHR and claims data, estimating cohort size and anticipating operational constraints such as visit cadence, laboratory requirements, and competing studies. That shifts planning from informal judgment toward testable assumptions, though the validity of any estimate depends on how well eligibility criteria can be represented in data and how representative those data are of the intended trial population (U.S. Food and Drug Administration 2024k; Hripcsak and Albers 2013).

Automated identification from EHRs is a pipeline: translate protocol criteria into a computable phenotype, execute it against clinical data, then put the candidate list in front of a clinician or coordinator before anyone is contacted or screened. Many criteria map cleanly to *structured* elements (diagnosis codes, medications, laboratory values, procedures, and timestamps); others reside in *unstructured* notes (symptom narratives, imaging interpretations, nuanced exclusions), which is why natural language processing complements rules and statistical models when candidate lists are assembled (Hripcsak and Albers 2013).

Standardization through common data models and cohort tooling, exemplified by OHDSI and the OMOP Common Data Model, lets a cohort definition execute consistently across sites in a federated manner (Hripcsak et al. 2015; Overhage et al. 2012). Standards-based interoperability supports the same workflows through APIs: SMART on FHIR supports substitutable applications that retrieve patient-level data, and the FHIR Bulk Data Access specification supports population-level exports for governed research workflows (Mandel et al. 2016; Health Level Seven International 2024). Because EHR data were not originally collected for research, the practical constraints are often provenance, missingness, and fitness-for-purpose, not model choice alone; sponsors should align their use of EHR data with regulatory expectations for documentation, traceability, and data integrity in clinical investigations (U.S. Food and Drug Administration 2018c).

19.2.2. Engagement, retention, and equity in AI-mediated workflows

Communication technologies can answer common questions, deliver consistent study education, collect structured pre-screening information, send scheduling reminders, and route complex concerns to a human. The design challenge is to reduce coordinator load and participant burden without creating misleading medical advice or undue influence, which puts human-AI interaction design and clear escalation paths at the center (Amershi et al. 2019). The evidence base for conversational agents in health remains mixed and context-dependent, with systematic reviews noting that many evaluations are early-stage and heterogeneous, and often lack robust safety assessment even when user satisfaction is high (Laranjo et al. 2018). In trials, electronic informed consent and remote communication can reduce friction, but they must preserve comprehension, voluntariness, and proper documentation (U.S. Food and Drug Administration 2016).

A model that prioritizes outreach or determines who is contacted first is deciding who gets access to research, which amplifies privacy, bias, and accountability risks; reliance on proxies such as cost or utilization can encode structural inequities even when overall accuracy appears high (Obermeyer et al. 2019). A defensible approach therefore includes explicit inclusion goals, audit trails for cohort logic and model versions, subgroup performance checks, and clear procedures for human review and overrides (National Institute of Standards and Technology 2023; World Health Organization 2021; U.S. Food and Drug Administration 2024e, 2025c). Recruitment-facing tools should be documented with enough transparency for review and repli-

cation, and AI-enabled protocols and reports can be structured using emerging extensions such as SPIRIT-AI, CONSORT-AI, and TRIPOD+AI, depending on whether the system is a trial intervention, a reporting artifact, or an operational prediction model (Rivera et al. 2020; X. Liu et al. 2020; Collins et al. 2024).

19.3. Barriers to Recruitment

Lack of awareness remains a primary obstacle: many eligible patients never learn that a trial for their condition exists, or how to reach one. Those who do learn are often excluded by restrictive eligibility criteria, disqualified by a common comorbidity or a routine medication. Those who qualify may still find the logistics impossible, since intensive testing schedules and dietary constraints can be incompatible with daily work and family life. An informed consent process misaligned with a patient's culture or language can also feel intimidating or exclusionary.

19.3.1. Geographic Access and Rural Populations

Between 2008 and 2022, only 42.4% of U.S. counties hosted at least one Phase 1-3 cancer clinical trial, and rural counties were significantly less likely to have a trial site (Gu et al. 2024). In 2022, 86% of nonmetropolitan counties had no active cancer treatment trials, compared to 44% of metropolitan counties (Unger, McAneny, and Osarogiagbon 2025). Research sites concentrate in urban academic medical centers, systematically excluding rural populations.

Nearly 17% of the U.S. population over age 35 lives more than 100 miles from an NCI-funded trial site, and that burden falls disproportionately on low-income individuals (Shriver et al. 2025). Travel distance, time away from work, and unreliable transportation reinforce one another, and for many rural patients the combination rules out participation entirely.

Rural populations experience higher cancer mortality rates and different disease patterns, yet remain systematically underrepresented in the trials that generate treatment evidence. Outcomes data suggest that the treatment disparity may be largely attributable to access rather than biology: an analysis of 36,995 patients in 44 SWOG trials found that rural and urban patients achieved similar survival outcomes when given uniform access to care through clinical trial participation (Unger et al. 2018).

Community-based trial networks can narrow that gap. The NCI Community Oncology Research Program (NCORP) is one, and in the SWOG network-group trials described above, which enrolled patients from 1986 to 2012, 19.4% of participants came from rural areas, the same as the rural share of the U.S. population with cancer (Unger et al. 2018; Unger, McAneny, and Osarogiagbon 2025).

Decentralized elements attack the travel itself. Telemedicine, remote monitoring, home visits by nurses, remote consent, virtual study visits, and direct-to-patient drug shipment reduce travel burden and can maintain data quality while expanding

access; a small pilot comparing decentralized and conventional site models reported that 89% of patients in the decentralized arm completed the study versus 60% in the conventional arm (Sommer et al. 2018), though such comparisons may reflect selection effects, since the sites and patients willing to adopt decentralized elements may differ systematically. Mobile clinics go one step further, taking trial and screening activities directly into rural communities while maintaining protocol compliance: the Ohio State University Comprehensive Cancer Center coordinates a mobile mammography program that brings breast cancer screening to women in rural areas of Appalachia (Unger, McAneny, and Osarogiagbon 2025).

Operationally, geographic barriers are a site-selection problem: they call for the same deliberate reach beyond high-volume academic centers, and the same investment in building site capability where it is thin, that Section 18.6 describes.

19.4. The Importance of Retention

Enrollment is the first half of the problem. High dropout rates can severely reduce a trial's statistical power and introduce bias, particularly if the attrition is related to treatment response or side effects. Retention is won in the details of participant experience: flexible scheduling, timely compensation, and clear, empathetic communication from study staff. Central to all of it is the relationship between the investigator and the participant, since patients who feel personally connected to the research team are more likely to continue through to the study's conclusion.

19.4.1. Physician Participation and the "One and Done" Phenomenon

The same attrition runs on the investigator side, and it deepens the recruitment crisis. The base of active U.S. investigators has shrunk sharply, and most physicians who serve on a trial never do so again, the "one and done" phenomenon documented in Chapter 18. Among the reasons they leave are the complexity of modern protocols, the mounting technology burden, and financial structures that fail to account for the heavy time investment required relative to routine clinical care. Each departure costs the industry expertise it must then replace by onboarding new, less-experienced sites, which slows development further.

The attrition extends to the research staff as well. High turnover among clinical research coordinators (CRCs) and nurses disrupts continuity and increases the risk of documentation errors, and while turnover rates have stabilized somewhat in 2024, a significant portion of the industry still faces high churn, complicating the long-term viability of sites (see Figure 19.4).

19.4.2. Strategic Enrollment and Generalizability

An accrual target can be met with the wrong patients. Enrollment has to reflect the populations that will ultimately use the treatment, and the reason is scientific

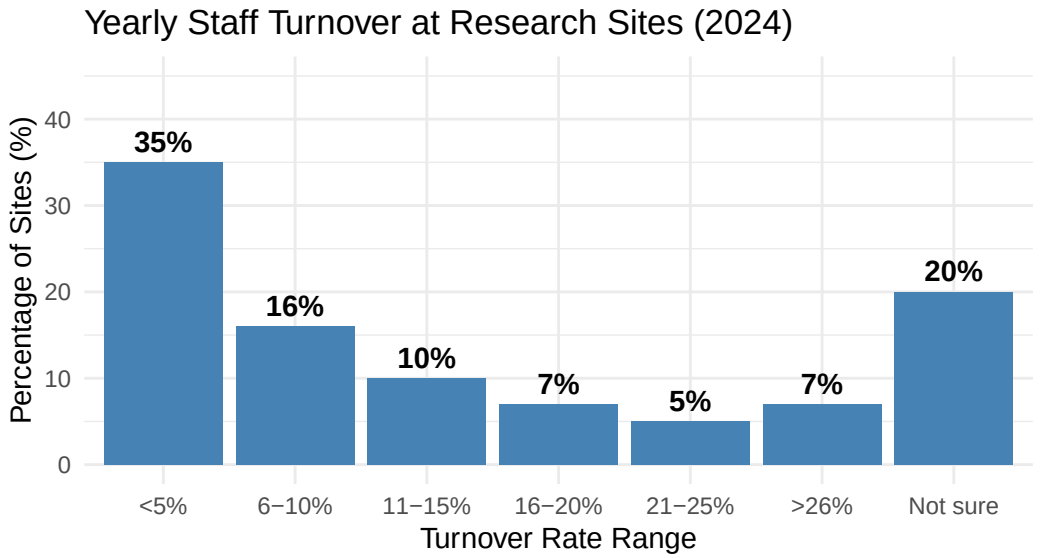


Figure 19.4.: Yearly staff turnover rates at clinical research sites (2024). Data from WCG Clinical (2024).

before it is regulatory: drug metabolism, disease expression, and treatment response vary systematically across age, sex, genetic factors, and comorbidities.

Pharmacokinetics differ substantially by age, driven by changes in renal and hepatic function, body composition, and drug-drug interactions from polypharmacy. Sex affects drug distribution through body fat percentage, metabolism through CYP enzyme activity, and disease pathways through hormonal influence. Genetic polymorphisms in drug-metabolizing enzymes such as CYP2D6 and CYP2C19 vary in frequency across populations and can alter therapeutic response and toxicity profiles. The disease phenotype itself may differ: hypertension, diabetes, and cardiovascular disease show distinct clinical presentations and progression patterns across demographic groups.

If trial enrollment systematically excludes the elderly, women, or specific genetic backgrounds, the resulting efficacy and safety data may not generalize to real-world use: a treatment optimized for a narrow trial population may be suboptimal, or even unsafe, when prescribed more broadly. Pediatric dosing cannot be reliably extrapolated from adult trials, and elderly patients with multiple comorbidities may experience different benefit-risk profiles than younger, healthier trial participants.

Representative enrollment requires intentional design: site selection that reaches diverse geographies and care settings, eligibility criteria that do not unnecessarily exclude common comorbidities, and recruitment strategies that address logistical barriers (transportation, language, visit burden). Regulatory guidance increasingly expects sponsors to justify enrollment targets based on the intended-use population and to monitor accrual across relevant subgroups during the trial (U.S. Food and

Drug Administration 2024e). That guidance and its recent turbulence, including the withdrawn-then-restored 2024 Diversity Action Plans draft and the December 2025 final *Enhancing Participation in Clinical Trials* guidance that dropped the explicit "diversity" framing, are detailed in Chapter 18. The statutory basis for representative enrollment and its scientific rationale are unchanged.

19.4.3. Digital retention toolkit: engagement beyond enrollment

Traditional retention frameworks addressed dropout and loss to follow-up through clinic visits, phone calls, and mailed reminders (Meinert 2013). In the DCT era, retention increasingly depends on digital engagement: app-based reminders, SMS check-ins, telehealth visits, and patient portals that reduce the friction of continued participation (U.S. Food and Drug Administration 2024n).

Digital engagement requires infrastructure and continuing attention. Executed poorly, it can feel intrusive or impersonal and may increase dropout rather than reduce it.

Checklist: Digital Retention Toolkit

- *Platform selection*: validated ePRO/eCOA and patient communication platforms with accessibility features (screen reader compatibility, multilingual support, offline functionality where relevant).
- *Message cadence and content*: prespecified reminder schedules; personalized content where feasible; clear escalation paths for non-response.
- *Telehealth integration*: defined workflows for remote visits (scheduling, consent for recording if applicable, documentation, safety escalation).
- *Help desk support*: accessible support for technical issues, with defined response times and escalation to site coordinators for clinical questions.
- *Retention monitoring*: dashboards tracking visit completion, ePRO compliance, and engagement metrics; predefined thresholds for coordinator outreach.
- *Participant feedback loop*: mechanism for participants to report burden concerns or technical issues, with documented response and protocol amendment pathways if patterns emerge.

19.5. Metrics and Monitoring

Enrollment problems show up in the metrics before they show up in the timeline (Table 19.2).

Table 19.2.: Key Recruitment and Retention Metrics. Targets shown are typical planning values, not industry benchmarks; achievable rates vary substantially by indication, phase, and protocol burden.

Metric	Definition	Target	Action if Off Target
Screening Rate	Patients screened per site per week	Protocol-specific	Increase outreach; Boost advertising
Screen Failure Rate	% screened who are ineligible	<50%	Review eligibility criteria; Improve pre-screening
Enrollment Rate	Patients enrolled per site per month	2-4 typically	Additional training; Site support; Consider closing site
Retention Rate	% enrolled completing study	>85%	Reduce protocol burden; Improve patient engagement
Time to First Patient	Days from site activation to first enrollment	<30 days	Expedite startup; Ensure materials available

Low screening rates often indicate a need for broader outreach or increased advertising, while high screen-failure rates may suggest that eligibility criteria are overly restrictive or that pre-screening processes need refinement. Poor enrollment and retention rates should prompt a review of the protocol's practical burden on patients and an investigation into site-level training or engagement strategies. Tracked against projections, these numbers catch an emerging shortfall while it is still correctable. AI agent applications across the full development lifecycle, including site identification, feasibility prediction, and decentralized trial infrastructure, are covered in Chapter 26.

20. Logistics and Project Management

A large Phase III trial might involve 200 clinical sites across 25 countries, 3,000 patients, half a dozen specialized vendors, and a timeline of 4 years. Each patient visit generates data that must be captured, cleaned, and locked. Each site requires contracts, regulatory approvals, training, monitoring, and closeout, and each country applies its own regulatory requirements, language considerations, and import procedures for investigational drugs. Temperature-sensitive biologics must be shipped in validated cold chains that hold -70°C across continents, and randomization systems must be available 24/7 to assign treatments the moment a patient qualifies. All of it has to happen without compromising the scientific integrity that makes the trial's results meaningful.

This is the domain of clinical trial logistics: the practical coordination of people, documents, supplies, and processes that converts a protocol on paper into executed research. A brilliant study design means nothing if the drug never reaches the patient, if the data are lost in transmission, or if the site closes before enrollment completes.

20.1. Organizational Framework and Roles

A clinical trial runs on delegated work. Accountability does not travel with it. Sponsors hand work to CROs and specialized vendors, principal investigators hand tasks to coordinators and sub-investigators, and each transfer moves the work without moving the ultimate responsibility for it.

20.1.1. The Key Players

The *sponsor* initiates, manages, and finances the trial. Pharmaceutical companies, biotech firms, academic institutions, and government agencies all serve as sponsors. The sponsor owns the trial and bears ultimate responsibility for its conduct, including investigator selection, monitoring, and regulatory compliance (21 CFR 312.50).

The *investigator*, usually a physician designated as the *Principal Investigator (PI)*, takes responsibility for the trial at a specific site. The PI supervises the research team, ensures patient safety, obtains informed consent, and guarantees that the protocol is followed faithfully. Though the PI may delegate tasks to coordinators

and sub-investigators, regulatory accountability cannot be delegated: the PI who signs FDA Form 1572 is held responsible for all trial conduct at that site.

Contract Research Organizations (CROs) are specialized service providers that sponsors hire to conduct trial activities. CROs may provide anything from a single service (such as monitoring or data management) to full end-to-end trial management. Grand View Research estimated the global healthcare CRO market at $59.6 billion in 2025 (Grand View Research 2026).

The *study coordinator*, also called the Clinical Research Coordinator (CRC), works at the site under the PI's supervision. Coordinators are the workhorses of clinical research, managing patient visits, collecting data, maintaining regulatory documents, and serving as the primary point of contact between the site and the sponsor or CRO.

Vendors provide specialized services: central laboratories analyze specimens, electronic data capture companies provide software platforms, imaging core labs review scans, and interactive response technology vendors manage randomization and drug supply.

20.1.2. Delegation, Accountability, and Partner Selection

Delegation follows specific rules. Sponsors may transfer obligations to CROs through written agreements (21 CFR 312.52); once transferred, the CRO assumes regulatory responsibility for those tasks, while the sponsor must still oversee the CRO's performance, review monitoring reports, and verify that delegated tasks are executed properly. As industry observers note, the sponsor remains captain of the ship even when the CRO handles navigation: the sponsor is still responsible for reaching the destination safely. At the site, a *delegation log* documents which coordinators, sub-investigators, and nurses perform which tasks, and the PI remains accountable for all of it.

Selecting the right partner is therefore a critical project management task, and it typically follows a path defined by internal SOPs. Before contacting vendors, the sponsor must define exactly what is being outsourced, ranging from a specific functional service (such as data management alone) to a full-service model where the CRO handles everything from site selection to the final clinical study report. The sponsor then issues a *Request for Proposal (RFP)* outlining study specifications (protocol synopsis, number of patients, number of sites, timeline) and asking vendors to provide detailed budgets and proposals. Vague requirements produce vague, incomparable bids.

Top candidates are then invited to a *bid defense* meeting. Price is on the table, but so is chemistry: the sponsor meets the team who will run the project, not the salespeople, and assesses their experience in the therapeutic area, their available staffing, and whether their systems (EDC, CTMS) are compatible with the sponsor's infrastructure.

The contract that follows formalizes a scope of work, and managing that scope is a primary duty of the project manager: work outside it triggers a change order to adjust budget and timeline, and uncontrolled scope creep is a common cause of budget overruns.

20.2. Biospecimen Logistics

In many modern trials, particularly in oncology and rare diseases, the collection, processing, and transport of **biospecimens** (blood, tissue, urine, or other biological materials) is a primary logistical thread. Clinical data can be captured electronically; biospecimens are physical assets that require sophisticated chain-of-custody and stability controls.

The chain has several links, and each of them can fail. Sites collect under strict protocol-defined procedures that fix the timing relative to dosing and the collection tubes or kits to be used. Many specimens then require immediate processing at the site (centrifugation, aliquoting) before freezing. Storage means calibrated -20°C or -80°C freezers with alarm systems, since an unnoticed temperature excursion loses the sample. Specialized couriers move the material in dry ice or liquid nitrogen to central laboratories or biorepositories. Throughout, a "Biospecimen Log" or specialized software tracks each sample from the patient to the final analysis, maintaining the link between the participant and their biological data while preserving confidentiality.

A missed collection window or a temperature excursion during transit can lose critical exploratory or primary endpoint data, which makes biospecimen management a high-risk component of the trial's operational design.

20.3. Trial Governance and Documentation

A trial is governed by a pyramid of documents, from the protocol at the top to the daily records of every patient interaction and drug shipment, and by the standing procedures of the organizations that run it.

20.3.1. The Protocol as Governing Document

The protocol is the constitution of a clinical trial. It specifies everything: the scientific rationale, the objectives, the study design, the patient population, the treatments, the assessments, the endpoints, and the statistical analysis.

The sponsor develops the protocol, often with input from investigators and CROs. Once finalized, the protocol needs ethics approval, and whatever regulatory submission or authorization the jurisdiction requires, before any patient can be enrolled. Any later change requires a formal amendment. Most amendments need ethics approval before they are implemented, and the more significant ones must also

be submitted to or authorized by regulators first; changes needed to eliminate an immediate hazard to participants are among the exceptions (see Section 13.8.1).

The PI and site agree to follow the protocol by signing it (or a separate agreement). ICH E6(R3) section 2.5 (Compliance with Protocol) directs the investigator to comply with the protocol, GCP, and applicable regulatory requirements (International Council for Harmonisation 2025b). Deviations are documented and reported; systematic deviations trigger regulatory scrutiny.

20.3.2. The Protocol and the SOPs

A monitor on a site visit works from two sets of instructions. The protocol says what this trial does to its participants and when, which is also what makes any departure from it a deviation. The sponsor's *standard operating procedures* (SOPs) say how every monitoring visit is prepared, conducted, reported, and followed up, whatever the trial. ICH E6(R3) defines SOPs as "detailed, documented instructions to achieve uniformity of the performance of a specific activity" and keeps them next to the protocol as a yardstick for conduct. Monitors follow "the sponsor's monitoring plan and applicable monitoring procedures", audits check activities against "the protocol, applicable standard operating procedures (SOPs), Good Clinical Practice (GCP) and the applicable regulatory requirement(s)", and noncompliance with the protocol or the SOPs calls for proportionate action by the sponsor (International Council for Harmonisation 2025b).

Keeping such procedures is a sponsor duty. E6(R2) made the sponsor responsible for quality assurance and quality control systems "with written SOPs" (International Council for Harmonisation 2016a). E6(R3) asks for "documented procedures" and names several that a sponsor needs: for auditing, for the content and frequency of monitoring reports, for unblinding, for reporting incidents such as security breaches, and for developing, maintaining, and using computerized systems (International Council for Harmonisation 2025b). The other parties keep procedures too. FDA requires every IRB to follow written procedures, including ones that keep changes to approved research from starting before the IRB has approved them, except to eliminate apparent immediate hazards (U.S. Food and Drug Administration 2023c). CROs and sites have SOPs for the work they do, and Chapter 22 describes the families a clinical research operation keeps. A single trial therefore runs under several organizations' procedures at once.

Table 20.1 sets the two documents side by side. In practice they differ most in who approves a change and when it takes effect. A change to the protocol is an amendment. It usually needs IRB approval before it is implemented, and a significant change must also go to regulators first; each site switches to the new version only once its own approvals are in place, so sites can work under different versions of the same protocol for months (Getz et al. 2024). An SOP changes by internal revision, approved by the organization's line management and quality assurance, and no IRB or regulator approves it. From its effective date the new version governs everyone who performs the activity, on every trial. Outsiders see an SOP when

a sponsor assesses a vendor, an auditor examines a process, or an inspector asks which procedure was in force (International Council for Harmonisation 2025b; U.S. Environmental Protection Agency 2007).

Table 20.1.: How a Protocol and an SOP Differ (International Council for Harmonisation 2025b; U.S. Environmental Protection Agency 2007)

Aspect	Protocol	SOP
Scope	One trial	One activity, across every trial the organization runs
Written by	The sponsor, with clinical, statistical, regulatory, and operational input	The organization that performs the activity: sponsor, CRO, site, or IRB
Signed or approved by	The sponsor's authorized signatories and each investigator; approved by IRBs; submitted to or authorized by regulators	The organization's line management and quality assurance
Changed by	Protocol amendment, approved before implementation except in narrow cases	A new version under the organization's change control
Takes effect	Site by site, as each site's approvals arrive	From the version's effective date, for everyone who performs the activity
Afterward	Kept with the trial's essential records	Superseded versions archived out of use but available for review

An SOP usually opens with its purpose and scope, defines its terms, assigns qualifications and responsibilities, and sets out the steps in order, with the checklists and forms that document each completed action attached where they are used. Each page generally carries the document's ID, revision number, and date (U.S. Environmental Protection Agency 2007). The completed forms are how a trial later shows that a procedure was followed. E6(R3)'s example is a database lock checklist produced by following the data management SOPs, which belongs among the trial's essential records; the SOPs themselves are essential records too, kept outside any single trial's files alongside master services agreements and validation records (International Council for Harmonisation 2025b).

The organization's procedure for writing procedures, the "SOP on SOPs", sets the lifecycle. People who do the work write the draft, in enough detail that someone with a basic grasp of the work but little experience of the procedure could carry it out unsupervised. Others with experience of the process review it and ideally try it out, and line management and quality assurance approve it. The SOP is revised and re-approved whenever the process changes, reviewed at set intervals even when it has not, and withdrawn when the process is no longer followed. Superseded and withdrawn versions are archived where they cannot be used by mistake but can

still be consulted. That matters because an inspection may come years after the work, and the revision number and date establish which procedure governed a task on the day it was done (U.S. Environmental Protection Agency 2007). Staff are trained on each new version, and their training records are part of the evidence that the people doing trial work were qualified for it (International Council for Harmonisation 2025b).

What varies from trial to trial is carried by a layer between the two documents: trial-specific plans for risk management, monitoring, safety, data management, data validation, and statistical analysis. E6(R3) lists them among a trial's essential records and expects them to be as clear, concise, and operationally feasible as the protocol they carry out. The monitoring plan shows how the layers divide the work. It is tailored to the risks of one protocol, focuses on the aspects critical to quality, describes the monitoring strategy, methods, and tools with the rationale for them, and references the sponsor's applicable procedures. Committees are set up the same way: a DSMB or an endpoint adjudication committee operates under written procedures such as a charter (International Council for Harmonisation 2025b). The SOPs can stay stable while each trial adds what its design needs, and the plans, like the consent form and the case report forms, have to follow the protocol whenever it changes.

Two situations test whose procedures apply. Regulatory requirements differ by country (Table 20.3), so a sponsor's global SOPs either meet the stricter national rule where one applies or are supplemented by country-level procedures that do. And when a sponsor hands obligations to a CRO, FDA requires the transfer to be described in writing; any obligation the description leaves out stays with the sponsor (U.S. Food and Drug Administration 2024b). E6(R3) keeps ultimate responsibility with the sponsor in any case and expects it to have access to a service provider's SOPs to select and oversee it (International Council for Harmonisation 2025b). A sponsor that outsources monitoring therefore has to know whose procedures the CRO's monitors work to, and to check that the work matches them.

The protocol's own life runs through SOPs as well. A sponsor's procedures typically set how a protocol is drafted, reviewed, signed, and amended, and E6(R3) asks that the protocol name the persons authorized to sign it and its amendments for the sponsor, and that every amendment carry its own number and date (International Council for Harmonisation 2025b). Chapter 13 follows a protocol from synopsis to signature, and Section 13.8.1 follows an amendment from the decision to change to the last site that adopts it.

20.3.3. The Flow of Documents

The *Trial Master File (TMF)* is the sponsor's collection of essential documents, defined by ICH GCP as the archive demonstrating that the trial was conducted according to GCP and regulatory requirements. It includes the protocol and its amendments, regulatory approvals, ethics committee correspondence, investigator agreements, monitoring reports, and safety data.

The *Investigator Site File (ISF)* is the site's equivalent, maintained at each clinical trial site, and includes the site's copy of the protocol, regulatory approvals, IRB correspondence, delegation logs, training records, and source document worksheets.

The pathways between them are regulated. The sponsor provides the *Investigator's Brochure* containing preclinical and clinical information about the investigational product; the site returns signed agreements and regulatory forms. Informed consent forms, developed by the sponsor, are reviewed and approved by each site's IRB before use. Safety reports flow from sites to sponsors, and from sponsors to regulators and to every participating site.

ICH E6(R2) organized the *essential documents* into three categories, and TMF practice still follows that temporal structure even though E6(R3) replaced R2's minimum list, which could already be supplemented or reduced where justified, with a risk-proportionate essential-records table (see Chapter 24): protocol, approvals, and CVs before trial initiation; monitoring reports, updated consents, and serious adverse event reports during the trial; final reports, final reconciliations, and archiving records after completion.

20.4. Scale, Timelines, and Global Execution

The protocol and the essential documents are a framework on paper. Execution means holding that framework across immense geographies and multi-year timelines.

20.4.1. Development Timelines and Success Rates

Nearly half of all clinical research sites reported that operational challenges forced them to reduce the number of studies they participated in during the past year, a pattern Chapter 19 examines in detail (WCG Clinical 2024). Long durations and high attrition rates compound the logistical challenge (Table 20.2). Total clinical trial starts declined from the 2021 peak, returning to 2019 levels by 2024, driven largely by the fading of COVID-19-related research (which declined 62% in 2024) but also by lower non-COVID trial starts from both larger companies and Emerging Biopharma (EBP) firms, concentrated in 2022 and 2023, with Phase I starts rebounding for both segments in 2024 (IQVIA Institute for Human Data Science 2025).

Table 20.2.: Current Drug Development Timeline and Success Metrics (IQVIA Institute for Human Data Science 2025; U.S. Food and Drug Administration 2018b)

Metric	Current Data (2023-2024)	Context
Total development time	10-15 years	Discovery through FDA approval

Metric	Current Data (2023-2024)	Context
Clinical phase duration	9.3 years (2024; sum of phase medians)	Phase I through NDA approval
Inter-trial intervals	17 months (2024)	Gaps between research phases; recovered from pandemic peak of 32 months (IQVIA Institute for Human Data Science 2025)
Overall clinical success	6-7%	Phase I to market launch (composite success rate)
Phase I success	47% (2024)	Safety and dosing assessment
Phase II success	**25%** (2024)	Proof-of-concept; lowest success rate
Phase III success	59% (2024)	Confirmatory efficacy trials
NDA/BLA approval rate	98% (2024)	Post-submission to approval
First-cycle approval	74% (2024)	Approved without major deficiencies
Expedited programs	79% (2024)	Launches using fast track, breakthrough, priority review, or accelerated approval

In Deloitte's 2024 analysis, total cycle time from Phase I through regulatory filing had reached 100.4 months, over 8.3 years, with all stages of development lengthening over the preceding five years (Deloitte Centre for Health Solutions 2025). Phase III had expanded the most, by 12.0% over those same five years, reflecting the rising complexity of late-stage global trials and stricter regulatory requirements.

Total development time also reflects **inter-trial intervals**: the gaps from the end of treatment in one trial to the start of enrollment in the next (or to regulatory submission), where data are analyzed, funding is secured, and subsequent protocols are finalized. These intervals contributed 17 months to overall medicine development time in 2024, a recovery from the pandemic-driven peak of 32 months in 2022, which makes further interval reduction one of the most significant opportunities for accelerating patient access (IQVIA Institute for Human Data Science 2025).

China, India, and Eastern Europe offer large patient pools at the cost of varying regulatory timelines, language barriers, and complex global supply chains. The standard a sponsor has to hold is that a site initiation in Brazil is as rigorous as one in Boston.

20.4.2. Multi-Regional Trials and Regulatory Framework

Late-stage programs are increasingly executed as *multi-regional clinical trials (MRCTs)*, conducted across multiple countries and regulatory jurisdictions. This

geographic scope is driven by practical and regulatory considerations: access to larger patient populations, faster enrollment through diversification, and regulatory expectations that evidence should be relevant to the populations in which a drug will be used (Z. P. Smith, DiMasi, and Getz 2024; International Council for Harmonisation 2017).

ICH E5 (Ethnic Factors) addresses when clinical data generated in one region can support approval in another. It distinguishes between *intrinsic* ethnic factors (genetics, body weight, organ function) and *extrinsic* factors (medical practice, diet, environment). When these factors are likely to affect drug response, regulators may require local data or *bridging studies* to extrapolate from foreign trials (International Council for Harmonisation 1998a).

ICH E17 (Multi-Regional Clinical Trials) provides fuller guidance on designing a single trial to support approval across multiple regions simultaneously. In practice, it pushes teams to treat "global" as a prespecified design choice: the trial should be structured so that, if the pooled result is positive, regulators can assess whether the evidence is sufficiently consistent and interpretable across regions (even when individual regions are not powered for standalone hypothesis tests). It also emphasizes planning regional representation prospectively and prespecifying regional analyses and decision-relevant summaries rather than discovering "regional stories" after the fact (International Council for Harmonisation 2017).

Although ICH harmonization has aligned much of clinical trial regulation, important differences remain. In the European Union, for example, the Clinical Trials Regulation (CTR) and its CTIS portal standardize much of the submission and coordination process across member states, while still leaving meaningful national operational differences in how ethics and competent-authority review are executed (European Medicines Agency 2022).

Submission pathways, ethics committee structures, inspection practices, and expectations for local participation can differ across regions in ways that directly affect startup timelines and monitoring models. "One global trial" therefore behaves like a portfolio of region-specific startup and oversight workflows, and its operational controls have to absorb local requirements without fragmenting data standards or protocol interpretation (International Council for Harmonisation 2017). Table 20.3 summarizes the differences across major regulatory regions.

Table 20.3.: Regional Regulatory Differences in Multi-Regional Trials. *Note: Regulatory frameworks evolve; verify current requirements with applicable authorities.*

Aspect	FDA (US)	EMA (EU)	PMDA (Japan)	NMPA (China)
Application type	IND	CTA (national or centralized)	CTN	IND

Aspect	FDA (US)	EMA (EU)	PMDA (Japan)	NMPA (China)
GCP standard	ICH E6(R3)	ICH E6(R3) + local annexes	J-GCP (ICH E6 + local)	China GCP (ICH E6 + local)
Local data requirement	Flexible	Flexible	Bridging often required	Local patients required for registration
IRB/EC structure	Central IRB common	National EC per member state	Central + local	Local EC per site
Inspection authority	FDA	National competent authorities	PMDA	NMPA

A multi-regional trial may require separate regulatory submissions in each country, with different timelines and documentation requirements. The United States permits central IRBs that cover multiple sites; many other countries require local ethics approval at each institution.

20.4.3. Operational Friction and the Activation Bottleneck

Those administrative layers land on study start-up, which remains a primary logistical bottleneck. Start-up timelines vary dramatically by site type (Chapter 18): 77% of academic medical centers, health systems, and community hospitals require more than 60 days, while 60% of independent sites can initiate in under 60 days, and 77% of U.S. sites cite "Budgets and Contracts" as the dominant bottleneck (WCG Clinical 2024). Ex-U.S. sites more frequently cite ethical/regulatory review (58%) as the primary obstacle.

Technology adds friction of its own. In one industry survey, 60% of sites reported using more than 20 technology systems per day to conduct research (WCG Clinical 2024). This "portal proliferation" imposes an immense cognitive and administrative load on site staff, often leading to data entry errors and protocol deviations that further extend timelines, and sponsor-mandated proprietary software frequently conflicts with the site's existing technology infrastructure, forcing sites into manual workarounds.

Sites respond by hiring additional staff, prioritizing training, and (for 27% of them) participating in fewer studies as a deliberate strategy to maintain quality (see Chapter 19 for detailed breakdowns of site solutions and Figure 19.2). Staff turnover has improved modestly: the proportion of sites reporting low turnover (less than 5% annually) rose from 22% in 2023 to 35% in 2024 (see Figure 19.4 in Chapter 19) (WCG Clinical 2024).

Figure 20.1 visualizes these process delays across site types.

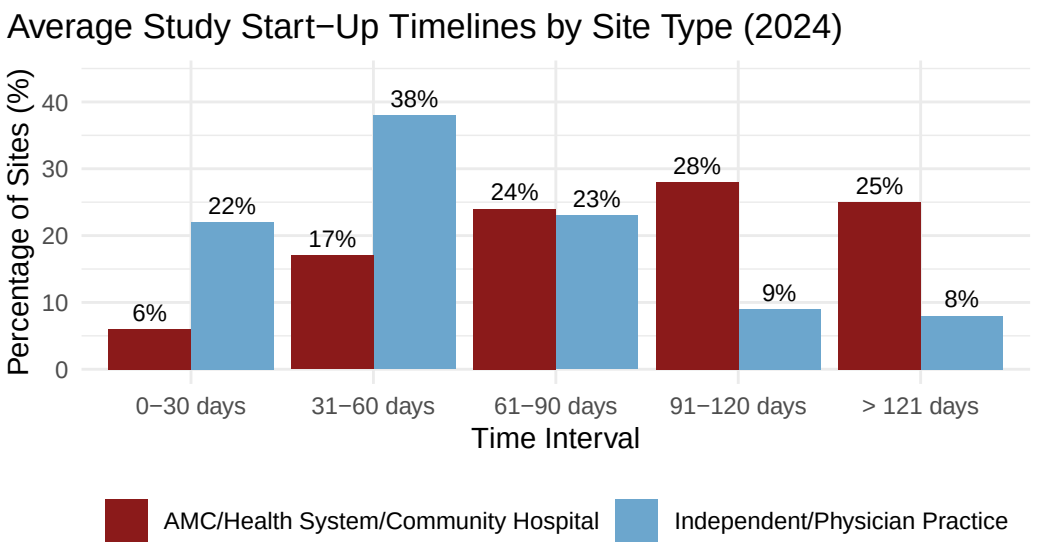


Figure 20.1.: Average clinical trial site start-up durations (days) by institution type. Data from WCG Clinical (2024).

20.4.4. Operational Challenges and Execution Models

Language requirements extend far beyond translation accuracy: informed consent forms, patient materials, and training manuals must be culturally adapted and medically equivalent across all participating regions. Global teams spanning a dozen or more time zones rely on structured asynchronous communication, reserving carefully scheduled real-time calls for critical decisions. Investigational product has to clear each country's own customs and import regulations, which is hardest for temperature-sensitive biologics that require validated cold-chain logistics and rigorous contingency planning for border delays. Data privacy law is fragmented as well, and GDPR and PIPL often require regional hosting or complex pseudonymization protocols. The financial logistics of managing payments across dozens of currencies and banking systems demand meticulous attention to fair market value assessments and anti-corruption compliance.

Three organizational models recur. A centralized model runs the entire trial from a single global operations center to maximize consistency, sometimes at the cost of local expertise and responsiveness. A regional hub model delegates authority to specialized leads in Europe, Asia-Pacific, or the Americas, balancing global protocols against regional nuance. Partnering with local CROs buys deep expertise in specific emerging markets and accelerates startup, at the price of a heavier vendor management burden.

The choice depends on the sponsor's organizational structure, the complexity of the protocol, and the specific countries involved. Whichever model is chosen, decision

rights have to be explicit: when a site in Japan raises a protocol interpretation question, who has authority to answer?

The logistical footprint of a trial is often characterized by its *executional design characteristics*, the countable elements that define the scope of the operation. Benchmarks from the Tufts Center for the Study of Drug Development (Tufts CSDD) put a typical Phase III trial at an average of 14 countries and 82 investigative sites (Getz, Smith, and Kravet 2023).

Oncology and rare disease trials are often dispersed further while enrolling relatively few patients. A rare disease trial might enroll only 200 patients globally and still require 27 sites across 10 countries to find them. Oncology trials involve about 60% more countries than non-oncology trials (13 versus 8) and require significantly more patient visits (29 versus 16), each visit adding to the logistical burden on both sites and patients (Getz, Smith, and Kravet 2023).

Vendor counts have grown too, to an average of 4 to 6 specialized providers per trial (Getz, Smith, and Kravet 2023): central labs, imaging core labs, EDC providers, and logistics firms, each requiring disciplined coordination so that data flow correctly and samples are processed within required windows.

20.5. Academic and Investigator-Initiated Multi-Site Trials

The site start-up figures above come from industry trials, where a company or CRO runs start-up for every site at once and pays for it. When an academic investigator sponsors a multi-site trial (Section 3.4), the lead institution does that work with a grant budget, and each participating institution runs its own contract office, IRB process and pharmacy. The steps are the same as in an industry trial. What differs is who does them, how much money there is to do them, and how many separate institutions must agree.

20.5.1. Where the Time Goes

The most detailed counts come from cancer research. Dilts and colleagues mapped the development of NCI cooperative group phase III trials and found at least 769 processing steps and up to 36 groups or individuals who had to approve a trial before it opened. The median time from formal concept submission to the first enrolled patient was about 920 days, with a range of 456 to 2,440 days, and at one cancer center the time to open an already approved group trial ranged from 21 to 836 days (Dilts et al. 2010). An earlier study at one cancer center counted more than 110 steps to open a trial when the process was mapped in fine detail, and judged about half of them to add no value (Dilts and Sandler 2006).

Those steps can be shortened. At MD Anderson, a department that reorganized itself as a dedicated clinical trials unit took a median of 4.3 months (interquartile range 3.0 to 6.0) from submission to activation. For institution-sponsored trials matched against other departments, the unit took 2.9 months rather than 5.1 from

IRB approval to activation, and 0.5 months rather than 1.2 from activation to the first patient (Tang et al. 2017).

Contracts and IRB review are the two longest single items for the participating sites. When 29 sites of the NIH Clinical and Translational Science Award (CTSA) program were activated for one multi-site study, final contract negotiation took a median of 39 calendar days, full contract execution 91 days, and IRB approval 53 days. Contract work ran alone for a mean of 33 days and IRB review alone for 48 days, and the two overlapped for only 22 days, so most sites handled them one after the other rather than in parallel (Kiriakis et al. 2013). Table 20.4 collects these measured intervals.

Table 20.4.: Measured Start-Up Intervals for Academic Multi-Site Trials

Interval	Measured value	Setting	Source
Concept submission to first patient, phase III group trial	Median about 920 days (range 456 to 2,440)	NCI cooperative groups	Dilts et al. (2010)
Time to open an approved group trial at one cancer center	21 to 836 days	One NCI cancer center	Dilts et al. (2010)
Submission to activation, dedicated trials unit	Median 4.3 months (IQR 3.0 to 6.0)	MD Anderson	Tang et al. (2017)
Final contract negotiation	Median 39 days	29 CTSA sites, one study	Kiriakis et al. (2013)
Full contract execution	Median 91 days	29 CTSA sites, one study	Kiriakis et al. (2013)
IRB approval	Median 53 days	29 CTSA sites, one study	Kiriakis et al. (2013)
Contract negotiation, ad hoc terms versus master agreement	Mean 55 versus 22 days	One academic medical center, 2010 baseline	Lawrence et al. (2023)

The table has gaps. We found no public benchmark that splits an academic multi-site trial's start-up into protocol finalization, budget negotiation, contract, IRB, regulatory submission and database build. No source reports how long budget negotiation takes, and no public figure exists for the time or cost of building an EDC database for an academic trial. The best sources are also old: the CTSA contract data date from 2013 and the cooperative group mapping from 2010.

20.5.2. Standard Agreements and Single IRB Review

Two reforms target the two longest items. The first is the standard contract template. Two are in wide academic use: the Accelerated Clinical Trial Agreement

(ACTA) and the FDP-CTSA subaward template used by the NIH Trial Innovation Network. At one academic medical center, negotiation of a clinical trial agreement took a mean of 55 days in a 2010 baseline and 22 days with a master agreement. Across surveys, ACTA saved an average of 48 days and the FDP-CTSA template 57 days. These counts cover contract negotiation only; they exclude IRB review and budget negotiation (Lawrence et al. 2023).

The second is single IRB review. NIH policy requires a single IRB of record for non-exempt human subjects research carried out at more than one U.S. site under the same protocol, for competing grant applications due on or after January 25, 2018. Exceptions are rare, the cost of the single IRB is not accepted as a reason for one, applicants must put single IRB costs in the budget, and the single IRB is named at the Just-in-Time stage before award (National Institutes of Health 2018). The revised Common Rule added the same requirement for federally funded cooperative research approved on or after January 20, 2020 (National Institutes of Health 2018; U.S. Department of Health and Human Services 2024). FDA regulations allow but do not require a single IRB for FDA-regulated multi-site research. FDA proposed a requirement in September 2022 (U.S. Food and Drug Administration 2022b); as of October 2026 no final rule had been published.

Single review moves the work rather than removing it. Each participating institution still has to sign a reliance agreement and complete its own local checks before its site can open. In the first year of the INVESTED trial at 15 PCORnet sites, which used the SMART IRB master reliance agreement, the mean time from receipt of the protocol to IRB approval was 81 days at the 13 sites that ceded review, against 121 days at the one site that kept its own review; time to site activation was 116 days against 142 (Vardeny et al. 2019). The comparison group was a single site, so the result shows feasibility more than size of effect.

20.5.3. Cost, Funding and Failure to Accrue

The only detailed cost study of investigator-sponsored trials is European. Griessbach and colleagues reconstructed the resource use of 93 randomized trials sponsored by investigators in Switzerland, Germany and the United Kingdom. The median total cost was $645,824 (interquartile range $269,846 to $1,577,924). Planning took a median 27.5% of the cost, conduct 57.3% and finalization 12.7%; protocol writing (7.2%) and data management (5.0%) were the largest single items. Of the 66 trials with an original budget, 46 (69.7%) overran it by more than 50%. Using routinely collected data for the primary outcome was associated with lower cost (Griessbach et al. 2024). No equivalent distribution has been published for U.S. investigator-initiated trials.

Industry support for an IIT can take the form of money, of drug or device supplied without money, or both. We found no federal rule that governs these agreements; their terms are negotiated with each institution's contract office. When a funder changes during a trial, every agreement written for the original funder (the clinical trial agreement, subawards, reliance agreements) has to be reviewed against the

new funder's terms. We found no published guidance or study on how often that happens or how long it takes.

Accrual is the most common way academic trials fail. Among 1,017 randomized trials whose protocols were approved by six research ethics committees in 2000 to 2003, 24.9% were discontinued and poor recruitment was the most frequent reason. Discontinuation for poor recruitment was 26.5% for investigator-sponsored trials against 8.4% for industry-sponsored ones (Kasenda et al. 2014). In the ClinicalTrials.gov results database, insufficient accrual was the leading reason for terminating trials that stopped for reasons other than their data (57%) (Williams et al. 2015). Among 787 adult phase II and III trials launched by the NCI cooperative groups from 2000 to 2011, 18% closed with low accrual or were still below half their target three or more years after opening (Bennette et al. 2016). These studies count trials that opened. We found no study with a denominator of academic trials that were approved but never opened.

20.5.4. Coordinating Centers and Consortia

Networks that run many academic trials concentrate the sponsor's work in a central office. Table 20.5 compares several U.S. networks whose public pages describe how they are funded and organized.

Table 20.5.: Coordinating Structures of Selected Academic Trial Networks (National Cancer Institute, Division of Cancer Treatment and Diagnosis 2026; Pediatric Brain Tumor Consortium 2026; Duke Clinical Research Institute 2026; NIH Environmental influences on Child Health Outcomes (ECHO) Program 2026; National Center for Advancing Translational Sciences 2026; Pediatric Oncology Experimental Therapeutics Investigators' Consortium 2026a, 2026e)

Network	Scope	Funding	Central office	Sites (as stated on its pages)
NCI Pediatric Early Phase Clinical Trials Network (PEP-CTN)	Pediatric cancer phase 1 and pilot trials, established 2019	NCI cooperative agreement	Operations and statistics center inside the Children's Oncology Group's; central monitoring for all trials	21 core U.S. sites in all trials; 21 non-core sites in selected trials
Pediatric Brain Tumor Consortium (PBTC)	Phase 1 and 2 trials in childhood CNS tumors, formed by NCI in 1999	NCI; no further five-year award beyond March 2026	Not described on the pages read	15 to 16 U.S. and Canadian centers

Network	Scope	Funding	Central office	Sites (as stated on its pages)
Pediatric Trials Network (PTN)	Pediatric drug studies under the Best Pharmaceuticals for Children Act, first awarded 2010	NICHD contract, renewed March 2026	Duke Clinical Research Institute as clinical coordinating center	More than 100 sites; 54 studies; 26 FDA pediatric label changes
ECHO IDeA States Pediatric Clinical Trials Network (ISPCTN)	Pediatric trials in IDeA states	NIH ECHO program	Data coordinating and operations center at the University of Arkansas for Medical Sciences	18 sites in 18 states
Rare Diseases Clinical Research Network (RDCRN)	More than 280 rare diseases	NCATS and ten other NIH entities	One data management and coordinating center for all consortia	Hundreds of sites
POETIC	Pediatric cancer phase 1 and 2 trials, founded 2003	Member institutions, philanthropy and trial-specific funders	Research Development and Management Center	14 academic member institutions; non-members may join trials

The NCI page stated in 2026 that PBTC trials will be completed where feasible, through close-out support or transfer to PEP-CTN, and the PBTC's own page listed no trials enrolling (National Cancer Institute, Division of Cancer Treatment and Diagnosis 2026; Pediatric Brain Tumor Consortium 2026). Networks end, and their trials need somewhere to go when they do. For NCI trials, standard common data elements and template case report forms held in the caDSR registry are used across trials (see Chapter 21) (National Cancer Institute, Center for Biomedical Informatics and Information Technology 2026).

Small numbers per site are normal in these trials. In arm E of the NCI-COG Pediatric MATCH trial, which tested selumetinib in children with tumors carrying MAPK pathway alterations, 58 matched patients came from 42 sites, and 20 patients started treatment (Eckstein et al. 2022). On average each site contributed fewer than two matched patients to this arm. Numbers like these are the economic case for a coordinating center: the fixed cost of opening a site is paid once by the network rather than again for every trial.

20.5.5. Worked Example: An Early-Phase Pediatric Oncology Consortium

The Pediatric Oncology Experimental Therapeutics Investigators' Consortium (POETIC) shows how a network without a standing federal award runs small early-phase trials. Its public pages, trial registrations and one archived protocol describe the mechanics; the account below uses only those sources, as read in October 2026.

Origin. Investigators at Memorial Sloan-Kettering Cancer Center and the University of Colorado founded POETIC in February 2003 to widen access to new therapies for children whose cancer had recurred (Pediatric Oncology Experimental Therapeutics Investigators' Consortium 2026a). Its first trial, of the NCI-supplied agent 17-AAG in children with recurrent cancers, opened on September 20, 2004 at nine founding centers (UF Health 2004).

Coordinating center. The Research Development and Management Center ran from Memorial Sloan-Kettering for 16 years and moved to Stanford University in 2020. Its stated functions are program management and quality, trial design and management, oversight of all regulatory activities, data tools, analysis and publication support, contract negotiation, consortium finances, and mentoring of junior investigators (Pediatric Oncology Experimental Therapeutics Investigators' Consortium 2026a, 2026f).

Members and money. The members page lists 14 member institutions, all academic cancer centers or children's hospitals, including one in Canada and one in Israel, and the consortium can include non-member institutions in its studies. Members contribute staff, money and time; philanthropy and trial-specific funders (companies, foundations and, for one trial, the Department of Defense) pay for individual studies, and the drug has come from NCI or the manufacturer (Pediatric Oncology Experimental Therapeutics Investigators' Consortium 2026e, 2026a, 2026c).

Sponsorship. On ClinicalTrials.gov the lead sponsor of each trial is an institution or an investigator, with the consortium as collaborator or coordinating network (Pediatric Oncology Experimental Therapeutics Investigators' Consortium 2026c). Each trial is therefore an investigator-initiated trial in the sense of Section 3.4, with the coordinating center doing much of the sponsor's work.

Scale. In March 2026 the home page reported 3 active trials, 16 completed, and 429 children enrolled since the start, mostly in phase 1 and phase 1/2 trials (Pediatric Oncology Experimental Therapeutics Investigators' Consortium 2026d). Among the registered trials we reviewed, enrollment ranges from 14 patients (a neratinib trial) to 70 (the first 17-AAG trial), and the largest active trial plans 90 (Pediatric Oncology Experimental Therapeutics Investigators' Consortium 2026c).

How a trial runs. A 2014 protocol amendment from the Memorial Sloan-Kettering era sets out the mechanics. A site could not receive investigational drug until all site and study regulatory documents had reached the consortium's data and coordinating center, which then notified the site of activation and cleared it to order drug.

The center assigned a research study assistant to each study, and sites sent case report forms and source documents weekly or one week after each treatment cycle. Each project's steering committee met weekly by conference call, the center and the study chair spoke every two weeks, and the center issued queries and audited sites, either from source documents sent to it or on site. The lead institution's data and safety monitoring committee oversaw the trial (Pediatric Oncology Experimental Therapeutics Investigators' Consortium 2014). The consortium also meets in person twice a year (Pediatric Oncology Experimental Therapeutics Investigators' Consortium 2026b).

What is not public. No public page or paper reports the consortium's activation times, accrual per site, use of a single IRB, or contract templates, and we found no peer-reviewed paper on its operating model. The example shows the structure; it does not show how well the structure performs.

In this example the coordinating center holds activation as a gate, keeps regulatory files and data review central, and holds the study team to a fixed meeting schedule, while the sites keep consent, treatment and source records. That division is close to the one Table 18.2 in Chapter 18 proposes for sites new to research.

Checklist: Start-Up for an Academic Multi-Site Trial

- Send the contract and the IRB reliance packet to each site on the same day, so that the two run in parallel.
- Offer a standard template (ACTA or the FDP-CTSA subaward) and record which sites accept it unchanged.
- For NIH-funded work, name the single IRB at Just-in-Time and put its fees in the budget.
- Build the case report forms from existing common data elements before writing new ones.
- Put a central safety reporting process and a monitoring plan in place before the first site opens (Table 3.3).
- Write an accrual plan with a date and a threshold at which the steering committee decides whether to add sites, amend eligibility or close.
- Keep a list of every agreement tied to each funder, so that a change of funder can be handled agreement by agreement.

20.6. Operating an Adaptive or Platform Trial

The statistics of adaptive and platform designs are set out at Section 12.5 and Section 12.7.5, and the design questions at Section 13.6. This section covers what those designs ask of the people who run them. A conventional trial has one protocol, one randomization list and one drug supply plan, and they change rarely. In an adaptive or platform trial the decision rules act on interim data, so the trial's operations have to deliver clean data on a schedule, pass the resulting decisions to

the randomization system and the depots, and do so without letting the study team learn what the data show.

20.6.1. Interim Data Cuts

An interim analysis runs on a *data cut*, a snapshot of the database taken at a stated cut-off date. The steps are the same whatever the design. The protocol or statistical analysis plan names the information that triggers the look (for example the number of participants who have completed the primary endpoint, or the number of events in a time-to-event trial). The data management team agrees which variables must be clean for the look (the primary endpoint and any early predictor of it, randomization, and exposure) and runs an expedited cleaning pass on them, with reconciliation of serious adverse events against the safety database. The cut is then frozen, together with the analysis code, so that the result can be reproduced and filed in the trial master file, and the independent statistical group runs pre-validated programs on it for the data monitoring committee. How long this takes depends on the endpoint, the data flow and the design, so the plan should set the interval from a dry run on the trial's own data rather than assume one.

Two timing decisions deserve explicit planning. The first is whether enrollment continues while the interim is run. Pausing protects the participants who would otherwise be assigned under rules that the interim may change, but costs calendar time and risks losing sites; continuing keeps the trial moving but means more participants are enrolled than the decision can influence. Platform designs that update allocation or arms on a frequent schedule need data that are clean to the same standard at each update, which is why they tend to rely on a small core data set collected in the same way at every site, and on data captured in routine care where possible (RECOVERY and REMAP-CAP are the best-known examples, Section 13.6). The second is the information measure: interims are triggered by completers or events, not by enrolled participants, and the gap between the two can be large when the endpoint is delayed (see Section 9.3).

20.6.2. The Firewall

The data monitoring committee and the independent statistical group see comparative interim results, and the study team does not; FDA's adaptive design guidance recommends limiting access to those results to people independent of the ones conducting or managing the trial (U.S. Food and Drug Administration 2019a) (the committee's role is described at Section 9.6). The firewall is more than the committee charter, because adaptations leak information. A dropped arm, a changed sample size or a shifted allocation ratio can each tell an observer which way the data point. The EMA draft reflection paper on confirmatory platform trials (October 2026, a draft for consultation as of November 2026) asks that as many evaluations and decisions as possible remain blinded to participants, site staff and the sponsor, that an independent monitoring committee and an adequate firewall shield the decision makers in the trial team, and that maintaining full blinding until all

concurrently recruiting experimental arms are completed be considered (European Medicines Agency 2026b). It suggests that different analysis teams, or different study teams, for different interventions may limit leakage. It also warns that in an event-driven platform trial even the pooled count of events for one intervention and its concurrent control can unblind the study teams when several arms run at once, and suggests reporting only projected analysis times to the teams while external parties monitor event accrual. Where leakage cannot be excluded, it mentions running the analyses of all concurrently running arms at the same time, even if that means postponing some of them. The same paper acknowledges that integrity cannot always be kept: results are not realistically secret if part of a trial is submitted for marketing authorization or published while other parts continue.

20.6.3. Allocation, Supply, Contracts and Amendments

When allocation changes, the change has to travel from the independent statistical group to the interactive response technology (IRT, also called randomization and trial supply management, RTSM) system without passing through blinded staff. The system then has to be validated for the change and keep an audit trail of it. In a platform trial with arms that open and close, each participant may be eligible for only some of the open arms, so the randomization list or algorithm, the eligibility checks at screening and the kit assignment all depend on which arms are currently open. Drug supply is exposed to the same volatility: shifting allocation and arm dropping make forecasts unreliable, so supply should be simulated under the design's scenarios, with buffer stock and resupply triggers planned in advance. The simulation approach used for operating characteristics at Section 12.7.5 can be extended to supply for this purpose.

Contracts and approvals follow the design. A common practice in multi-sponsor platforms is a master clinical trial agreement with each site, so that adding an arm needs only an amendment to the site budget rather than a new contract. The submission rules (FDA's 30-day lead time for a new substudy, the EU's substantial modifications) are set out at Section 13.6 and Section 3.8; three further points bear on planning. The EC, EMA and HMA question-and-answer document on complex clinical trials expects the master protocol to be submitted with at least one sub-protocol in the initial application, and warns that a change to the master protocol may need substantial modifications of every sub-protocol when it substantially affects participants' safety or rights or the reliability of the data (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022). It also anticipates that, once the Clinical Trials Information System supports it, a sponsor studying the same product in several sub-protocols run as separate trials will be able to submit such modifications in parallel as a single application. FDA's draft guidance on master protocols (Revision I, June 2026, a draft as of November 2026) asks that the cover letter for a major change, such as a new substudy or an amended key design element, give the status of every substudy: which are open or complete, how enrollment stands, and the substudy IND number where it differs (U.S. Food and Drug Administration 2026e).

Each amendment then has operational consequences that a manager should plan for as one package: the ethics committee or IRB review, revised consent documents and re-consent of enrolled participants where the change affects what they were told, case report form changes, an IRT update and a supply re-forecast (the general handling of amendments is at Section 13.8.1). The EMA reflection paper adds that amendments made after some results are available need a strong justification so that the ongoing parts of the platform are not compromised (European Medicines Agency 2026b). A platform that expects to change often can reduce the burden through the structure of its protocol. REMAP-CAP uses a core protocol with domain-specific and region-specific appendices, so that an update requires only the relevant appendix to be added or modified, and it carried a dormant pandemic stratum that it activated for COVID-19 in February 2020 (Angus et al. 2020).

20.7. Project Management and Execution Flow

The project management of a clinical trial focuses on the transition from planning to execution, ensuring that sites are activated, staff are trained, and data flows reliably from the clinic to the sponsor.

20.7.1. Site Initiation and Monitoring

Before a site can enroll patients, it must be formally initiated. The *Site Initiation Visit (SIV)* marks this transition. Typically conducted by CRO personnel, the SIV ensures that all regulatory approvals are in place, the investigational product has arrived and is properly stored, site staff understand the protocol and procedures, and the site's essential documents are in order. Benchmarks show that study initiation (from protocol approval to first patient first visit) averaged 166 days for Phase III protocols completed just before the pandemic (Getz, Smith, and Kravet 2023).

Checklist: Remote/Hybrid Site Initiation Visit (SIV)

The increasing use of decentralized elements and remote work has shifted many "initiation" activities from a single in-person meeting toward a hybrid model. FDA guidance emphasizes that decentralized elements must preserve participant safety and data integrity; operationally, this implies that initiation must explicitly cover remote access, training, and computerized-system controls rather than assuming that in-person presence resolves them (U.S. Food and Drug Administration 2024n, 2024f).

- *Regulatory and essential documents readiness*: confirm approvals are current; confirm controlled document versions are distributed; confirm delegation and training records are complete and attributable (International Council for Harmonisation 1996, 2025b).
- *Consent readiness*: confirm the correct consent versions, procedures for re-consent, and (if applicable) eConsent workflows and documentation controls (U.S. Food and Drug Administration 2016, 2024f).

- *Computerized systems readiness*: confirm access provisioning, role-based permissions, and audit trail availability for EDC/ePRO/IRT and any remote source-review workflows (U.S. Food and Drug Administration 2023a, 2024f).
- *Remote/hybrid visit workflows*: define how remote visits will be conducted, documented, and supported (telehealth procedures, home health vendor roles, escalation paths) (U.S. Food and Drug Administration 2024n).
- *Investigational product logistics*: confirm chain-of-custody processes for site and (if applicable) direct-to-participant shipments; confirm temperature excursion procedures and reconciliation responsibilities (U.S. Food and Drug Administration 2024n).
- *Data flow and reconciliation*: confirm expected data sources (central labs, devices, ePRO), transfer cadence, and responsibility for reconciliation and query resolution (U.S. Food and Drug Administration 2023h).

Monitors (whether employed by the sponsor or CRO) visit sites periodically to verify compliance. They review source documents, check informed consent processes, reconcile investigational product, and identify any issues requiring correction. Monitoring reports document these visits and any corrective actions required.

Beyond routine monitoring, sponsors and CROs may conduct audits, which are formal examinations of trial processes and documents to verify compliance with protocols, SOPs, GCP, and regulations. Regulators may also conduct inspections, particularly of pivotal trials submitted to support marketing applications.

20.7.2. Site Team Roles and Relationships

At the site level, the investigator shoulders substantial regulatory obligations. *Patient safety* comes first: the investigator must ensure that qualified staff are available, that appropriate medical care is provided, and that participants are protected from unnecessary risks. The investigator is responsible for all trial-related medical decisions. *Informed consent* must be obtained before any trial procedures, following approved procedures and forms, with participants informed of any new information that might affect their willingness to continue. *Protocol compliance* means conducting the study exactly as specified, with every deviation documented. *Regulatory maintenance* requires keeping the ISF current, reporting to the IRB, and cooperating with monitors and auditors. Investigators must retain trial records for specified periods (often 15 years or longer) and be available for regulatory inspection even years after a trial concludes.

Study coordinators occupy a unique position: they are on the ground at the site, managing the daily logistics that determine whether a trial succeeds or fails. Coordinators screen potential participants against eligibility criteria, schedule and conduct study visits, administer questionnaires and assessments, collect and enter data, and manage investigational product accountability. They maintain the ISF, respond to monitor queries, and prepare for monitoring visits. They also manage relationships with patients (who need coordination for visits and follow-up), with investigators

(who rely on coordinators to keep the trial running smoothly), and with sponsors and CROs (who depend on coordinators for data and compliance). The coordinator role is often underappreciated. Experienced coordinators are among the strongest determinants of site performance, and sites with high coordinator turnover struggle with enrollment and data quality. Sites themselves rank staffing and retention among their top challenges, and finding research professionals with enough experience to deliver good, accurate trial data remains a persistent difficulty (WCG Clinical 2024). The best sponsors and CROs invest in coordinator relationships, recognizing that these individuals make or break trial execution.

The three-way relationship among sponsors, CROs, and sites can be complicated. CROs are accountable to sponsors: their contracts and revenue depend on sponsor satisfaction. Sites depend on sponsors (often through CROs) for trial opportunities and funding. This creates potential tension: CROs may prioritize sponsor timelines over site needs; sites may feel caught between competing demands. Communication breakdowns are common. A 2024 industry column, citing a 2022 ZS Associates report, said that over half of sites were dissatisfied with how communication is handled and that sites often repeat the same information to multiple contacts (Bronner 2024). High staff turnover at CROs exacerbates the problem: sites must repeatedly educate new monitors about local circumstances. Best practices emphasize clear role definitions, consistent contacts, and responsive communication.

20.8. Modern Logistics and Innovation

The logistics of clinical research are being transformed by new technologies and methodologies that address the friction of global operations and the shift toward more patient-centric designs.

20.8.1. AI and Automation in Trial Logistics

A trial that once shipped temperature-controlled product to a few dozen sites may now ship it to thousands of homes. Add SOPs harmonized across 20+ countries with distinct regulatory requirements, plus the protocol deviations that multiply as complex protocols meet site staff overburdened with multiple systems, and the friction exceeds what manual processes can absorb.

The industry's answer has been IT and AI, treated in detail in Chapter 25 and Chapter 26, but autonomy is the wrong frame here. The practical contribution is *coordination at scale*: converting streams of operational events (shipments, visits, queries, deviations) into timely decisions and documented actions. The ethical and regulatory constraints do not change (patient safety, data integrity, and accountability), so successful automation must produce auditable traces and preserve human oversight (U.S. Food and Drug Administration 2023a; National Institute of Standards and Technology 2023).

Decentralized elements and direct-to-patient (DtP) distribution multiply the endpoints, and many shipments require strict temperature control (IQVIA 2024a). Here AI earns its place by forecasting resupply from enrollment trajectories, visit schedules, and dispensing patterns, and by pushing shipments with early warning signals (temperature excursions, customs delays, address failures) to the top of the queue. Reconciliation is the other half: sensor data, courier scans, and dispensing logs linked into a single, reviewable chain-of-custody record suitable for quality oversight and inspection.

Many of the longest delays sponsors experience as "logistics problems" are document and workflow problems. Feasibility questionnaires, contracts, training records, IRB/EC packets, and essential document completeness determine whether sites can activate on time and whether trial conduct remains inspection-ready. Workflow automation and AI-assisted classification can reduce filing and quality-control backlogs, but only when the system retains provenance and supports review; one that silently "auto-files" errors into an eTMF structure has bought speed at the cost of traceability (TMF Reference Model Initiative 2024; Amershi et al. 2019).

Protocol deviations follow the same logic: they are often downstream of cognitive overload, from complex schedules, many systems, and inconsistent site communication. Risk-based quality management (RBQM) formalizes the idea that oversight should follow risk signals rather than fixed calendars (Association of Clinical Research Organizations 2025). An AI layer that detects anomalous patterns (repeated out-of-window visits at a site, say) and directs monitoring attention and corrective action accordingly is most defensible when it also produces a structured explanation of why the record or site was flagged, one a reviewer can challenge and revise as part of an auditable process.

Automation can fail, ethically and operationally, if it simply shifts work onto sites (more alerts, more portals, more passwords) without reducing burden. The baseline is already heavy: in a 2022 Society for Clinical Research Sites poll reported in WCG's 2024 site-challenges survey, 60% of the sites surveyed were using more than 20 different technology systems each day (WCG Clinical 2024). While 43% of sponsors view their own software's adoption as a key site selection criterion, approximately 38% of sites find these sponsor-provided technologies inadequate for their workflows, and 42% now stress the need for sponsor acceptance of the site's own technology infrastructure as a condition for study participation (WCG Clinical 2024).

Coordinators already sit at the center of execution, and systems that add friction can worsen data quality rather than improve it, however advanced they look: sites report the sprawl of sponsor-provided tools as a top challenge in its own right, and the same survey's prescription is integration that cuts data entry and administrative burden (WCG Clinical 2024). Treat coordinator time as a constrained resource: automation should remove repetitive work and reduce context switching, and give exceptions a clear escalation path with a visible owner for the next decision.

20.8.2. Direct-to-Patient and Decentralized Trials

Driven by patient demand for convenience and enabled by regulatory guidance (U.S. Food and Drug Administration 2024n), logistics providers now routinely ship investigational products directly to participants' homes. That shift to *decentralized clinical trials (DCTs)* and a direct-to-patient supply chain is the most significant logistical evolution of the 2020s.

Many of these shipments require strictly controlled temperatures, which means smart packaging with IoT sensors that track stability in real time (IQVIA 2024a). Home health nurses administer infusions and draw blood in the patient's home, extending the trial's reach to those unable to travel to sites. Centralized *ePharmacies* dispense trial medications for simpler protocols, bypassing the site-based pharmacy altogether. Each of these moves burden off the patient and onto the sponsor, who now runs a distributed supply chain.

AI can cut the coordination overhead that would otherwise overwhelm sites and sponsors: reminders, instructions, and scheduling handled automatically, remote data capture standardized, exceptions (missed visits, sensor failures, shipping issues) routed to humans with enough context to act. Medable positions a unified DCT platform that bundles eConsent, eCOA/ePRO, remote monitoring, and study build workflows (Medable 2026). The pattern matters more than the vendor: decentralized trials become practical when tooling collapses distributed coordination tasks into auditable workflows that can be monitored and corrected in near real time, consistent with regulatory expectations for decentralized elements (U.S. Food and Drug Administration 2024n).

Early reports suggest that DCT elements may shorten enrollment durations, which currently average over 500 days for Phase III trials. The offset, at least in the short term, is that non-standard datasets and novel practices may add protocol complexity and demand more specialized logistical management.

20.8.3. Project Management Fundamentals

A clinical trial is a *project* in the formal sense: a temporary endeavor with a defined beginning, end, and scope, typically running from protocol finalization through database lock and study reporting, or even through regulatory submission. The *project manager* (or clinical project manager, or clinical operations lead) coordinates activities across functions, monitors progress against timelines and budgets, identifies risks, and drives issues to resolution.

Four constraints define the job. *Scope* is what the trial will accomplish: the protocol establishes scientific scope, and the project plan translates that into operational scope. *Schedule* is when activities must occur, anchored on the milestones every trial shares: first site initiated, first patient enrolled, last patient enrolled, last patient last visit, database lock, final study report. *Budget* runs to detailed line items: site costs, CRO fees, laboratory expenses, vendor costs, internal labor. *Quality* means

conducting the trial according to the protocol, GCP, and regulatory requirements while generating clean, analyzable data. Table 20.6 gives typical timelines and budgets by development phase.

Table 20.6.: Typical Clinical Development Timelines and Budgets (DiMasi, Grabowski, and Hansen 2016; IQVIA Institute for Human Data Science 2024a; U.S. Food and Drug Administration 2026f)

Phase	Duration	Patients	Trial Cost	Key Milestones	Common Challenges
Phase I	6-12 months	20-100	$1-5M	FIH, MTD, PK	Dose selection; Safety stops
Phase II	1-2 years	100-500	$10-30M	PoC, Dose-response	Go/no-go decision; Endpoint selection
Phase III	2-4 years	300-3000+	$50-200M+	LPLV, DB Lock, CSR	Enrollment delays; Data quality
NDA/BLA	10-12 months	N/A	$4.6M FDA fee	Submission, Review, Approval	Complete response; Advisory committee
Phase IV	Ongoing	Varies	Varies	REMS, PMRs	Safety signal management

[!NOTE] FIH = First-in-Human; MTD = Maximum Tolerated Dose; PoC = Proof of Concept; LPLV = Last Patient Last Visit; PMRs = Post-Marketing Requirements

The NDA/BLA cost is FDA's application fee for fiscal year 2027 (an application requiring clinical data); the sponsor's own cost of preparing the dossier is not included.

Planning precedes the first patient. The *study plan* (or clinical operations plan) documents how the trial will be executed: site selection criteria, recruitment strategy, monitoring approach, data management processes, safety reporting procedures, and vendor management.

Timelines are built backward from the required endpoints. If the sponsor needs a completed CSR by a certain date, when must database lock occur? When must the last patient complete the last visit? Backward planning reveals the critical path; Table 20.7 sets out its key activities and their dependencies.

Table 20.7.: Typical Critical Path Dependencies

Activity	Key Steps	Dependencies
Protocol Development	Draft protocol, Investigator Review, Final Approval	Precedes all site activities
Regulatory	IND/CTA Submission, Ethics Approval, Green Light	Precedes investigational product shipment
Site Activation	Feasibility, Contract Negotiation, SIV, Green Light	Precedes Enrollment
Enrollment	Screen patients, Enroll, Randomize, Treat	Dependent on Site Activation
Data Management	Database Build, UAT, Data Entry, Cleaning, Lock	Runs parallel to Enrollment
Reporting	Analysis, TLF Generation, CSR Writing	Dependent on Database Lock

Resource planning identifies what people, equipment, and services are needed when: how many monitors at peak enrollment, what laboratory capacity. *Risk assessment* anticipates what might go wrong, and the common answers are familiar: slow enrollment, poor data quality, site non-compliance, vendor underperformance, regulatory delays.

Once the trial is underway, the work becomes tracking and correction. *Status tracking* follows the metrics that indicate trial health: sites activated, patients enrolled, data cleaned, queries resolved, protocol deviations, serious adverse events, surfaced through dashboards and reports. *Variance analysis* compares actual performance against plan; if enrollment is running 30% behind projection, what is causing the gap? Issues are logged, assigned to owners, and tracked to resolution. Regular team meetings, status reports, and steering committee updates keep sponsors, CROs, and key sites current.

Change control governs modifications to scope, schedule, or budget. Site additions, vendor changes, and budget increases require formal approval and documentation; protocol amendments are the disruptive case (see Section 13.8.1), and by 2022 protocols averaged 3.3 amendments each (Getz et al. 2024). Operationalizing an amendment requires submission to every IRB/EC and, where the change calls for it, to regulators, re-consenting patients if risks or procedures change, updating EDC databases and IVRS/IRT randomization logic, and revising supply chain forecasts. A single substantial amendment can cost $500,000 or more in direct and indirect fees (Getz et al. 2016).

Coordination runs across specialized functions, each with its own deliverables and interdependencies: clinical operations manages sites and monitors, data management handles collection and cleaning, biostatistics performs the analyses, regulatory

affairs manages submissions, pharmacovigilance monitors safety, medical monitoring provides oversight, and CMC/supply chain ensures product availability. The project manager owns the handoffs between them.

Clinical Trial Management Systems (CTMS) track site and patient status, *Electronic Data Capture (EDC)* platforms enable remote data review, collaboration platforms carry communication across distributed teams, and analytics dashboards feed risk-based monitoring.

Problems arrive anyway. Enrollment shortfalls are the most common, answered by adding sites, extending timelines, or intensifying recruitment. Data quality issues may require site retraining, safety signals may require protocol amendments, and budget overruns force difficult decisions about scope or resources. Experienced project managers expect all of it and build margin into their plans, keeping contingency reserves and the relationships that let a problem be solved quickly.

20.9. Publication and Data Sharing Policies

How and when trial results become public is now as much a matter of policy as of science.

20.9.1. Publication policy: GPP 2022 and modern transparency expectations

Classic operational handbooks addressed publication policy in terms of authorship criteria and sponsor review rights (Meinert 2013). The 2022 update to Good Publication Practice (GPP 2022) widens that frame (DeTora et al. 2022): it encourages plain-language summaries as enhanced content, asking that publications support the communication of scientific information to lay audiences through such summaries and other accessible formats. The guidelines explicitly address patients as potential authors or contributors, recognizing that patient expertise may be relevant to interpreting outcomes, burden, or real-world applicability. They also emphasize transparency about medical writing support and statistical analysis, requiring disclosure of any sponsor involvement in study design, data access, and manuscript preparation.

Publication planning should begin early in the trial, ideally during protocol development, so that data collection supports the planned analyses and authorship expectations are clear before results emerge.

Template: Publication Policy (GPP 2022-Aligned)

- *Authorship criteria*: ICMJE criteria for authorship; explicit documentation of contributions; prohibition of ghost authorship.
- *Plain-language summaries*: commitment to producing accessible summaries for each primary publication; process for patient review of draft summaries.

- *Patient contributors*: defined process for including patient authors or acknowledged contributors where their input meaningfully shaped interpretation or presentation.
- *Sponsor review rights*: timeline for sponsor review (typically 30-60 days, sometimes up to 90, with a further limited delay if the sponsor needs to file a patent) (University-Industry Demonstration Partnership 2014; Mello, Clarridge, and Studdert 2005); explicit prohibition of sponsor suppression of unfavorable results; documentation of any sponsor-requested changes and author responses.
- *Data access and transparency*: statement of investigator access to study data; commitment to registration and results reporting per the Food and Drug Administration Amendments Act (FDAAA) and NIH policies.
- *Publication timing*: planned timeline for primary results publication relative to database lock; coordination with regulatory submissions.

20.9.2. Ancillary study policy: alignment with NIH Data Management and Sharing requirements

Traditional ancillary study policies addressed resource neutrality, Steering Committee review, and restrictions on treatment-group analyses until main trial completion (Meinert 2013). The 2023 Data Management and Sharing (DMS) Policy adds a requirement for NIH-funded research: ancillary studies must inherit or establish their own data sharing plans, and the relationship between parent and ancillary data must be documented (National Institutes of Health 2023).

An ancillary study proposal should therefore address how the data generated will be managed, retained, and shared, alongside the usual scientific merit and resource impact. For specimens or data collected under the parent trial's consent, investigators must verify that consent language supports the proposed ancillary use, which argues for broad consent language in the original protocol.

Template: Ancillary Study Policy (NIH DMS-Aligned)

- *Proposal requirements*: scientific rationale; resource impact assessment; timeline relative to parent trial milestones; investigator qualifications.
- *Review and approval*: Steering Committee or designated subcommittee review; sponsor concurrence for industry-funded trials; documentation of approval.
- *Data management plan*: ancillary study-specific DMS plan or explicit adoption of parent trial plan; identification of repositories for data and specimens.
- *Consent alignment*: verification that parent trial consent supports proposed ancillary use; process for obtaining additional consent if required.
- *Embargo and coordination*: restrictions on ancillary publications that could compromise parent trial integrity or unblinding; timeline for lifting embargoes relative to primary results.
- *Data sharing timeline*: commitment to share ancillary study data per NIH DMS timelines (generally at time of publication or end of award, whichever comes first).

21. Data Management

A nurse draws blood, a patient completes a questionnaire, a physician conducts an examination. At that moment the trial's evidence is a tube of blood, marks on paper, a clinical impression. By the time a regulator reads it, the same evidence has to be a number in a locked database, traceable back to the moment it was observed and sitting alongside the medical histories, laboratory values, imaging results, patient-reported outcomes, adverse event records, and protocol deviations that a single trial accumulates. Converting those raw observations into reliable, analyzable data is the province of clinical data management.

Regulatory decisions affecting millions of patients rest on whether those numbers can be trusted, and the entire edifice of evidence-based medicine assumes they reflect what actually happened to patients. Figure 21.1 shows the pipeline that produces them.

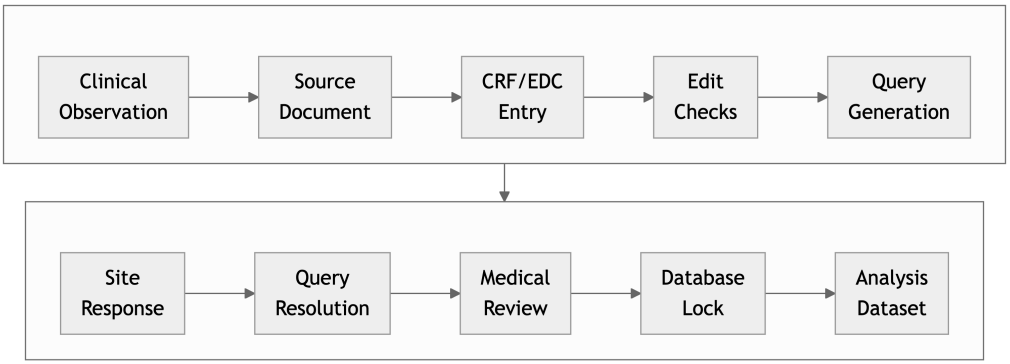


Figure 21.1.: Clinical trial data flow from point of collection through database lock

Observations are first *captured*, entered into the trial's data collection system. Historically that meant transcribing information onto paper *case report forms (CRFs)*; today, *electronic data capture (EDC)* systems are standard, with site staff entering data directly into validated computer systems.

Captured data is then *cleaned*: reviewed for errors, inconsistencies, missing values, and implausibilities, with queries issued to sites until the data is judged acceptably clean. Only then is the database *locked* and the data *extracted* for statistical analysis.

21.1. Electronic Data Capture

Type a systolic blood pressure of 12 where 120 belongs, or a birth date that would make the patient 200 years old, and the system flags the discrepancy immediately. That is what the *edit checks* programmed into EDC systems are for: catching errors at the point of entry. On paper, the same mistake might go undiscovered for months, which is why this *front-end validation* improves data quality.

An EDC system is a validated software platform whose electronic forms mirror protocol requirements, capturing each data point as site staff enter it. That gives sponsors *real-time visibility* into trial data: they can monitor enrollment, track data completeness, and identify sites with quality issues much more rapidly than was possible with paper systems.

Audit trails automatically log every entry and modification along with timestamps and user identities. The resulting record of how the data evolved, protected from modification, is required for demonstrating compliance with regulatory requirements, and it is what gets consulted when a question arises later.

21.2. Data Flow Beyond EDC

A contemporary trial's data no longer arrives only through a site's EDC screen. Central laboratories, imaging core labs, ePRO/eCOA platforms, wearable and sensor-derived endpoints, and real-world data sources each deliver their own stream, with its own provenance, latency, and error modes (U.S. Food and Drug Administration 2023h, 2023f, 2024k). The traditional operational framing of *data flow* still helps, particularly the contrast between bolus flow (data sent from the sites to a processing site in batches) and continuous flow (data sent as they are keyed or shortly after) and the emphasis on timeliness for monitoring and trial control (Meinert 2013), but it now has to describe several channels at once.

Ingestion alone is not the central difficulty. The quality and regulatory problem is *traceable control* across systems: documented interfaces, controlled transformations, and auditability sufficient to support reconstruction of what was collected, when, by whom or by what system, and how it was transformed into analysis-ready datasets (U.S. Food and Drug Administration 2023a, 2024f).

Checklist: Data Transfer Specification (DTS)

For each external source, a defensible DTS should specify:

- *Purpose and scope*: what data are transferred, at what cadence, and how they support protocol endpoints and monitoring (U.S. Food and Drug Administration 2023h).
- *Provenance and identifiers*: subject identifiers and pseudonymization rules; site identifiers; visit/timepoint mapping; timezone conventions (U.S. Food and Drug Administration 2024k).

- *Format and standards*: file formats, data dictionaries, controlled terminology, and mapping plans into SDTM/ADaM where applicable (U.S. Food and Drug Administration 2023f).
- *Validation and reconciliation*: checksums, record counts, missingness expectations, range and logic checks, and discrepancy handling, including "source of truth" rules (U.S. Food and Drug Administration 2024f).
- *Security and access control*: encryption in transit and at rest, least-privilege access, role-based audit trails (U.S. Food and Drug Administration 2023a, 2024f).
- *Change control*: versioning of schemas and code, deprecation policies, documented impact assessment for mid-trial changes (U.S. Food and Drug Administration 2024f).
- *Retention and inspection readiness*: how raw transfers, transformation logs, and audit trails will be retained and retrieved under inspection timelines (U.S. Food and Drug Administration 2023a, 2024f).

When an endpoint depends on a measurement from a *digital health technology (DHT)*, the data management plan should address device selection and performance, participant usability and training, data completeness monitoring, and the integrity of transmission and storage processes. These are endpoint-validity controls, not merely IT hygiene (U.S. Food and Drug Administration 2023h).

When EHR or claims-derived datasets are used to support regulatory decision-making, sponsors should document data provenance, capture processes, missingness and measurement error, and the relationship between clinical care documentation and research variables. The governing question is whether the dataset is fit for the stated regulatory purpose, with traceability and transparency sufficient for review (U.S. Food and Drug Administration 2024k, 2023f).

21.2.1. Common Data Elements

Meinert's handbook emphasized bespoke form design, tailored to each trial's specific requirements (Meinert 2013). Protocol-specific customization is still necessary, but the trend is toward *Common Data Elements (CDEs)*: standardized variable definitions, value sets, and collection instruments shared across studies.

The NIH CDE Repository provides curated element definitions across therapeutic areas (National Library of Medicine 2024), so researchers can adopt a pre-validated instrument instead of reinventing a measurement. CDISC standards (CDASH for collection, SDTM for tabulation, ADaM for analysis) provide the structural framework for regulatory submissions: FDA and PMDA require the tabulation and analysis standards, while CDASH is encouraged but not required (Clinical Data Interchange Standards Consortium 2024; U.S. Food and Drug Administration 2024m). The benefit extends well beyond compliance. A trial built on CDEs can feed meta-analyses, external control arms, and AI-driven signal detection, capabilities that custom-designed data structures often preclude.

Checklist: CDE Selection and Implementation

- *Identify applicable repositories*: NIH CDE Repository, CDISC Therapeutic Area User Guides, disease-specific consortia standards.
- *Map protocol endpoints to existing CDEs*: prioritize validated instruments for primary and key secondary endpoints, and document the rationale for any custom element.
- *Specify controlled terminology*: standard dictionaries (MedDRA for adverse events, WHODrug for medications, SNOMED CT or ICD for diagnoses) with explicit version control.
- *Plan CDISC mapping early*: define CDASH annotation during CRF design rather than retrofitting SDTM mapping at database lock.
- *Document deviations*: where protocol requirements preclude a standard CDE, record the deviation and keep a crosswalk to related standards for future interoperability.

21.3. Data Quality Standards

What makes data "good"? *Accuracy* is the first demand: recorded data reflects what actually happened, so a blood pressure of 140/90 was actually 140/90, not misread and not mistranscribed. *Completeness* asks that all required data be collected, though some gaps are unavoidable: a patient may miss a visit, and what matters then is that the gap is documented and explained.

Consistency means the data makes sense internally. A subject recorded as male on one form and female on another tells you something is wrong without telling you which form is wrong. *Timeliness* means data is entered promptly, because data entered long after collection is more likely to carry the errors of memory lapse and lost source documents. *Legibility* and *traceability* close the set: for electronic data they mean clear audit trails, and for paper documents legible handwriting and proper corrections (single-line strikethrough with initials and date, not obliteration).

The *ALCOA* family of principles captures these standards; Table 21.1 gives the base set.

Table 21.1.: ALCOA Data Integrity Principles

Principle	Definition	Examples	Common Failures
Attributable	Who recorded it?	User ID, signature, initials	Shared login; Unsigned entries
Legible	Can it be read?	Clear handwriting; Readable font	Illegible corrections; Truncated fields
Contemporaneous	Recorded when observed	Timestamp matches activity	Backdated entries; Batched data entry

Principle	Definition	Examples	Common Failures
Original	First record of data	Source document; Certified copy	Transcribed from informal notes
Accurate	Truthful and correct	Matches source document	Transcription errors; Rounding

Modern standards extend ALCOA to *ALCOA+*, adding *Complete* (nothing missing), *Consistent* (defined and handled the same way throughout the data life cycle), *Enduring* (readable throughout retention), and *Available* (retrievable for inspection). A further refinement, *ALCOA++*, adds *Traceable* (every value can be followed through its full lifecycle, including audit trails of any changes) (European Medicines Agency 2023).

21.4. Data Cleaning and Query Management

Edit checks at the point of entry do not catch everything. Data management personnel run programmed checks against the whole database: cross-field validations (is the hospitalization date after the enrollment date?), range checks (is this laboratory value physiologically plausible?), and protocol logic checks (if the patient was randomized to arm A, should they have received arm B at this visit?).

Anything that fails becomes a *query*: a formal request to the site naming the discrepancy and asking staff to verify or correct the value. Site staff go back to their source documents, the original medical records, and either correct the data or explain why it is correct as recorded. *Query resolution* is an iterative dialogue, some of it closing immediately and some of it requiring investigation, and the number of open queries and the time to resolution are key data management metrics.

21.5. Source Data and Verification

The trial database is not the ultimate source of truth; the *source documents* are. Source documents are the original records where clinical observations first appear: medical records, laboratory printouts, imaging reports, patient diaries.

That framing assumes a paper or pre-existing record stands behind every database value, and it no longer holds universally. Under *eSource* and *direct data capture (DDC)*, data are entered directly into a validated electronic system at the moment of observation, with no antecedent paper record; the electronic entry is itself the source. What defines source data is that it is the first, original record of an observation, in whatever medium. When that record is electronic, its trustworthiness rests on the same controls of attribution, contemporaneity, and audit trail that apply to any electronic record in a clinical investigation (U.S. Food and Drug Administration 2024f, 2013a).

Source data verification (SDV) is the monitor's side of this: comparing the trial database against source documents to ensure that transcription has been accurate. Historically, SDV was exhaustive: 100% of key variables for 100% of patients. More recently, *risk-based monitoring* approaches target SDV toward high-risk data and sites, recognizing that exhaustive verification is expensive and that some data points are more important than others.

The whole effort culminates in the database lock. All queries must be resolved, programmed cleaning must be finalized, and medical reviews must confirm the accuracy of safety coding and adverse event records. Once the database is locked, it is frozen. Reopening it takes a documented, authorized unlock and relock, reserved for errors in safety or efficacy data that could change the analysis; lesser errors found afterward are documented, for example in the study report, rather than corrected in the data (Society for Clinical Data Management 2023). After unblinding, ICH E6(R3) recommends limiting data changes to exceptional circumstances, each justified, authorized by the investigator, captured in the audit trail, and reported in the clinical study report (International Council for Harmonisation 2025b). The locked data is then transferred to biostatisticians for analysis, and that transfer must itself be validated, because the final clinical study report rests on the dataset being complete, accurate, and protected from uncontrolled change.

21.6. Data Standards and Traceability

Regulatory agencies require submission data in standardized formats that enable efficient review and cross-study comparisons (U.S. Food and Drug Administration 2024m), which in practice means the international standards developed by the Clinical Data Interchange Standards Consortium (CDISC) (Clinical Data Interchange Standards Consortium 2024), now required for the study data in marketing applications and commercial INDs submitted to FDA and in new drug applications submitted to PMDA (U.S. Food and Drug Administration 2021e; Ministry of Health, Labour and Welfare 2024). The Study Data Tabulation Model (SDTM) provides the foundation by organizing raw clinical data into standardized domains (such as demographics, adverse events, and laboratory results) using a uniform structure and variable naming convention, so that any reviewer can interpret a blood pressure measurement without consulting study-specific custom documentation.

While SDTM tabularizes the data as collected, the Analysis Data Model (ADaM) prepares it for statistical analysis. ADaM datasets include derived variables, analysis flags, and complex calculations like time to event, providing the direct basis for the primary efficacy and safety analyses. The define.xml file is the reviewer's map: machine-readable metadata specifying the origin and derivation of every dataset and variable. Underneath all of these sits controlled terminology, which mandates standardized codes for categorical values like adverse event severity or dosing routes, so that data can be pooled and compared automatically across an entire research portfolio.

21.6.1. The Traceability Chain

Regulators expect to trace any number in a submission back to its source (U.S. Food and Drug Administration 2024m). This *traceability chain* links:

1. **Source documents** (medical records, lab reports)
2. **CRF/EDC data** (as collected)
3. **SDTM datasets** (tabulated for submission)
4. **ADaM datasets** (derived for analysis)
5. **Tables, listings, and figures (TLFs)** (in the CSR)
6. **eCTD submission** (regulatory dossier)

At each step, the transformation should be documented and reproducible. A reviewer who questions an efficacy p-value should be able to walk that chain back, link by link, to the source document.

The chain breaks in recognizable ways: a derivation left undocumented, so nobody can say how "baseline" was defined; mapping documentation missing, so how EDC variable X became SDTM variable Y is guesswork; corrections applied ad hoc, outside the validated systems; dataset versions drifting out of step across analysis steps.

21.6.2. The eCTD Submission Structure

Submissions to major regulatory agencies use the *electronic Common Technical Document (eCTD)* format, which organizes the dossier into a navigable electronic structure (Table 21.2).

Table 21.2.: eCTD Module Structure

Module	Content	Data Management Relevance
Module 1	Regional administrative information	Country-specific forms and labels
Module 2	Summaries and overviews	Clinical summaries synthesizing data
Module 3	Quality (CMC)	Manufacturing; not directly DM
Module 4	Nonclinical	Preclinical studies; not directly DM
Module 5	Clinical	CSRs, SDTM/ADaM datasets, define.xml

Module 5 is where clinical data management work lands, and how well that work was done directly affects how efficiently reviewers can navigate and verify the submission.

21.7. Regulatory Compliance

Several requirements overlap here. *21 CFR Part 11* governs electronic records and electronic signatures, addressing system validation, audit trails, access controls, and the signatures themselves (U.S. Food and Drug Administration 2023a). *Computer System Validation (CSV)* is the process of validating EDC and other clinical trial systems for their intended purpose before use, with documentation that the systems function as specified. *Good Clinical Practice (GCP)* attaches to the data itself, which must be accurately recorded, attributable, contemporaneous, and available for inspection.

Records retention requirements mandate that trial records be maintained for specified periods and be available for regulatory inspection. ICH E6(R3) (2025) (International Council for Harmonisation 2025b) defers to applicable regional regulations rather than specifying a single retention period; in practice, FDA 21 CFR 312.57(c) requires sponsors to retain records for two years after marketing approval or, if the drug is not approved, after investigational shipment and delivery stop and FDA is notified (U.S. Food and Drug Administration 2024b), while the EU Clinical Trials Regulation requires the trial master file to be archived for at least 25 years after the trial ends (European Parliament and Council of the European Union 2014).

AI tools for automated data cleaning, discrepancy detection, and natural language processing of source documents are covered in Chapter 26.

22. Clinical Monitoring

A clinical trial is not self-executing. The protocol may be brilliant, the sites carefully selected, the data capture systems impeccably designed, and the trial can still drift away from its intended course. Good Clinical Practice requires sponsors to "implement a system to manage quality throughout all stages of the trial process" (International Council for Harmonisation 2016a), and clinical monitoring is a central component of that system: the systematic oversight that keeps a trial's conduct safe, competent, and compliant with GCP.

Monitoring protects research participants by identifying safety issues, protocol violations, and inadequate consent processes. It protects data integrity by identifying errors, inconsistencies, and potential fraud. It also protects the sponsor's investment: a problem caught early can still be corrected, while the same problem caught late may already have contaminated the data or triggered regulatory action.

The person doing this is the monitor, formally the *Clinical Research Associate (CRA)*, the sponsor's representative at the investigator site. CRAs are typically experienced professionals with clinical or scientific backgrounds, trained in GCP, the protocol's requirements, and monitoring procedures. A single CRA may be responsible for several sites, visiting each on a regular schedule.

22.1. Types of Monitoring Visits

Four kinds of visit punctuate a site's life in the trial (Table 22.1).

A site that looks promising on paper but lacks the caseload or infrastructure to deliver is far cheaper to identify before activation than after. That is the work of the *Pre-Study Qualification Visit*, or site selection visit, which takes place while the sponsor is still deciding where to place the trial. The monitor assesses the site's interest and its access to the target patient population, the qualifications and availability of the investigator and staff, and whether the facilities and equipment can support the protocol's procedures.

The *Site Initiation Visit (SIV)* comes before the site enrolls its first patient. The monitor confirms that regulatory approvals are in place, that staff are trained on the protocol and study procedures, that investigational product (IP) has arrived and is properly stored, and that the essential documents are in the Trial Master File. It is also where the working relationship between monitor and site is established.

Interim Monitoring Visits (IMVs) recur throughout the trial, monthly, quarterly, or at whatever interval the monitoring plan sets. During IMVs the monitor reviews enrollment, verifies source data, checks protocol compliance, reviews adverse event documentation, reconciles investigational product inventory, and addresses whatever has gone wrong since the last visit.

The *Closeout Visit* follows the end of enrollment and follow-up at a site. Data queries must be resolved, investigational product accounted for (returned, destroyed, or documented as dispensed), and site files completed and ready for long-term retention.

Table 22.1.: Types of Monitoring Visits

Visit Type	Timing	Key Objectives
Pre-Study Qualification	Prior to selection	Assess site interest, capability, facilities, and staff availability
Site Initiation (SIV)	Prior to activation	Train staff on protocol/EDC, verify TMF/ISF, confirm IP receipt
Interim Monitoring (IMV)	Throughout trial	Verify source data, check protocol compliance, manage IP accountability, review safety reporting
Close-Out	After last patient last visit (LPLV)	Resolve data queries, return/destroy IP, reconcile files, submit final report

22.2. On-Site Monitoring Activities

Source Data Verification (SDV) is the core of an on-site visit: the monitor compares data entered in the trial database against the original source documents (medical records, lab reports, imaging results) to confirm that the two agree. Historical standards often demanded 100% verification; risk-based approaches now aim SDV at the most critical data points. Source Document Review (SDR) asks a broader question of the same records: are they complete, are they consistent, and do they suggest faithful adherence to the protocol?

The monitor must also verify that every participant gave valid informed consent, on the correct approved version of the form, before any study procedure began. Investigational product accountability reconciles dispensing records against physical inventory and storage conditions against the protocol. The Investigator Site File (ISF) has to be current and every staff member's qualifications documented. The part least amenable to a checklist is conversation: the monitor talks to the site staff, gauging team capability, surfacing training needs, and catching operational bottlenecks before they escalate into systemic issues.

22.3. The Monitoring Report

After each visit the monitor prepares a *monitoring report* documenting findings, follow-up items, and any corrective actions required. It becomes part of the Trial Master File and stands as the contemporaneous record of what the site looked like on that day.

Monitoring reports should be objective, factual, and specific. "Informed consent was adequate" tells a later reader nothing. "100% of consents reviewed (n=15); all documented appropriately; 2 consents were signed on the same day as randomization (acceptable per protocol)" tells them what was examined, what was found, and why it passed.

Follow-up items are tracked until resolution. A site that is not properly documenting adverse events gets the finding recorded, the corrective action specified, and the fix verified at the subsequent visit.

22.4. Risk-Based Quality Management (RBQM)

A 2025 report by the Association of Clinical Research Organizations (ACRO) found that 96% of the 3,758 outsourced studies reported by seven CROs for 2024 incorporated at least one risk-based monitoring (RBM) or RBQM component, such as centralized statistical monitoring (CSM) or targeted SDV, up from 53% in 2019 (Association of Clinical Research Organizations 2025). Traditional monitoring, built on frequent on-site visits and 100% SDV, is now an outdated model. *Risk-Based Quality Management (RBQM)*, called for by ICH E6(R2) and reinforced in E6(R3), focuses oversight on the risks that matter most to patient safety and data reliability (Figure 22.1).

The gains are in detection and in economics. Statistical algorithms can find fabricated data, such as "perfect" blood pressure readings, that a human reviewer would miss; monitors visit sites less frequently, going where the central signals point them; and the budget that once bought travel buys data analytics instead. None of this is less monitoring. It is smarter monitoring, and technology enables oversight that would be impossible through on-site visits alone.

RBM starts with *risk assessment*: which aspects of this particular trial pose the greatest risk to patient safety or data integrity. A narrow safety margin may pull resources toward safety monitoring, subjective patient-reported endpoints toward endpoint verification, a therapeutic area with a history of data integrity problems toward detecting potential fraud. The resulting *risk-based monitoring plan* may send low-risk data to remote review or reduced SDV; high-risk data receives more intensive oversight.

Risk-based practice has settled on two named instruments, only the first of which is explicit in the ICH text (International Council for Harmonisation 2016a; U.S.

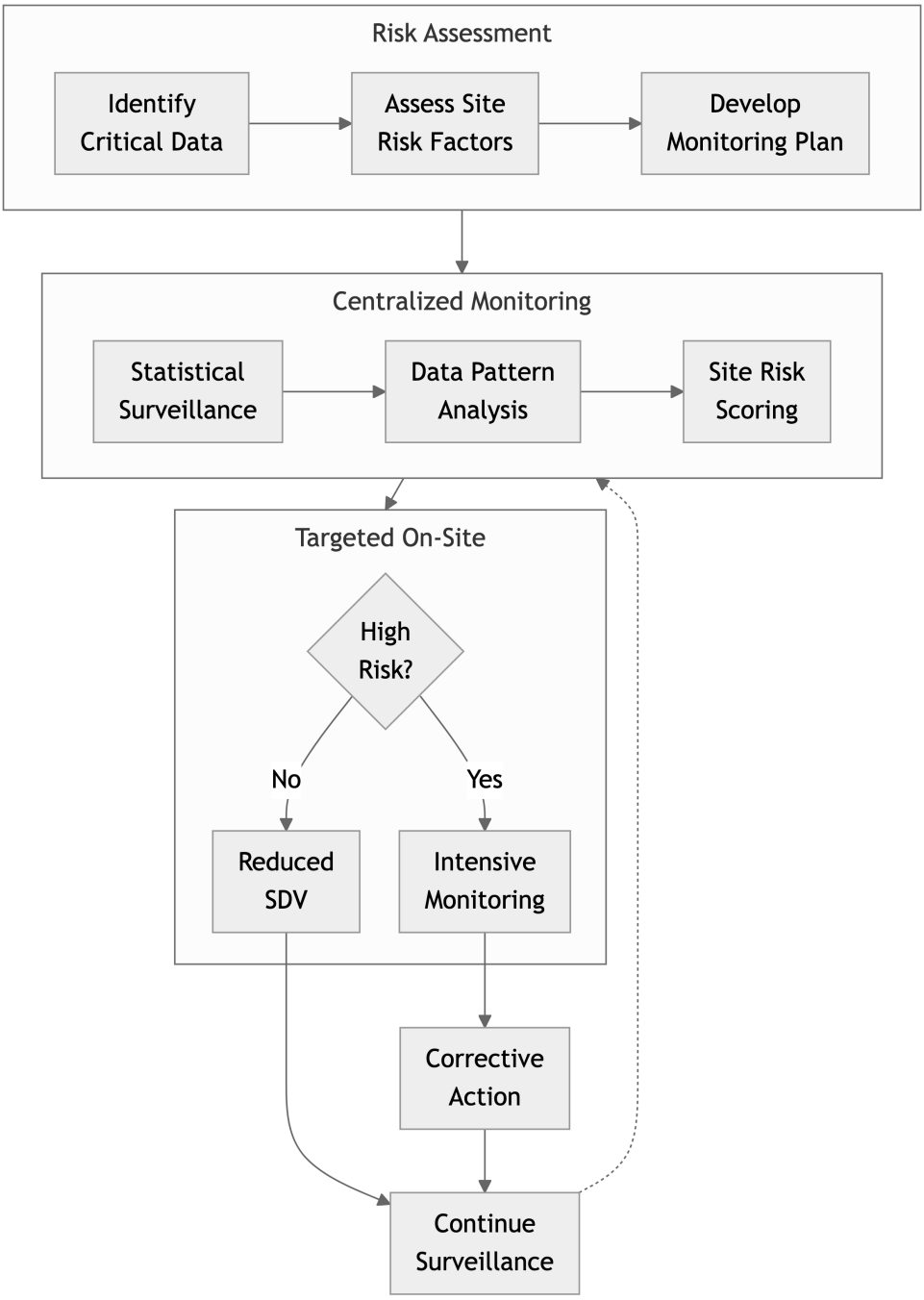


Figure 22.1.: Risk-based monitoring workflow combining centralized surveillance with targeted on-site oversight

Food and Drug Administration 2013b). *Quality Tolerance Limits (QTLs)* are pre-specified thresholds on parameters critical to participant safety and data reliability, set at the protocol level (for example, an acceptable ceiling on the rate of a key eligibility deviation across the whole trial). A breach does not by itself indicate misconduct; it triggers evaluation of whether a systematic problem is present. *Key Risk Indicators (KRIs)* are site-level metrics (enrollment pace, query rates, adverse-event reporting frequency, protocol-deviation counts) tracked centrally to flag sites whose patterns diverge from expectations and warrant closer scrutiny.

Centralized monitoring complements on-site visits, analyzing trial data for patterns suggestive of problems: unusual distributions, unexpected correlations, digit preferences that suggest data fabrication, enrollment patterns inconsistent with site capacity.

The COVID-19 pandemic accelerated adoption of *remote monitoring*, conducted through remote access to electronic health records, video calls with site staff, and remote review of scanned source documents. Some work still requires a physical presence: a video link is a poor substitute for checking pharmacy storage conditions or seeing how a site actually operates. Many other monitoring functions can be performed effectively without a site visit, which is how *hybrid monitoring* settled into its present shape: periodic on-site visits wrapped in more frequent remote oversight.

22.5. Quality Metrics and Ongoing Oversight

Monitoring sits inside a larger quality system whose three parts are easy to confuse (Figure 22.2).

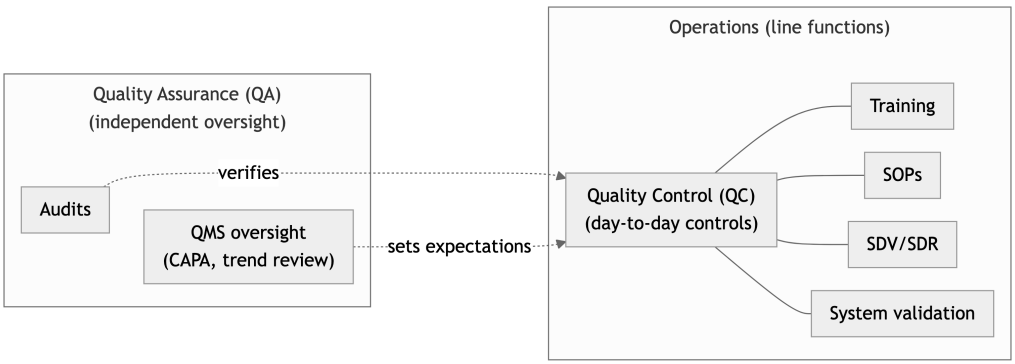


Figure 22.2.: Relationship between Quality Assurance, Quality Control, and Audit Functions

Quality Control is the day-to-day operational work that builds quality into the trial: training, SOPs, source data verification, template development. Quality Assurance is the overarching system meant to keep that work reliable. Audits are how QA finds

out whether that work is achieving its intended outcomes, conducted separately from line operations so that the people judging QC are not the people doing it.

The metrics that indicate site and trial quality are, in risk-based terms, the KRIs described earlier. They work when reviewed regularly and compared across sites, which is how patterns and outliers become visible. A site that consistently under-enrolls may require intervention or closure. A high protocol deviation rate suggests a training need, or a protocol too complex to follow.

Query rates point at data entry problems: a site generating many more queries than its comparators may have inadequate training or inadequate attention to data quality. Adverse event reporting patterns can identify sites over- or under-reporting relative to others, and both directions warrant investigation. Time to issue resolution indicates site responsiveness. A site that leaves queries unresolved or corrective action items unaddressed may need more intensive oversight.

Standard operating procedures (SOPs). SOPs are written instructions intended to achieve uniformity in the performance of specific functions. One common way to organize the ones a full clinical research operation needs is in two families. Administrative and general SOPs govern the operational framework: the "SOP on SOPs" that defines how procedures are written and approved, along with training documentation, records management (including archiving and destruction), and crisis management plans. Regulatory SOPs cover the ground where a lapse becomes a compliance failure: obtaining informed consent, communicating with the IRB, handling safety reports, managing investigational product from receipt to destruction, and screening and enrolling subjects to ensure eligibility. Section 20.3.2 sets these procedures beside the protocol and follows an SOP from drafting to retirement.

22.6. Audits and Inspections

Monitoring is the sponsor's own line oversight. Audits and inspections are checks from outside the operating team, and the two differ enough in authority and consequence that conflating them leaves a team under-prepared for the one carrying real regulatory risk.

Sponsor audits are systematic, internal quality assurance activities conducted by the sponsor's QA group or by independent contractors. Where monitoring is continuous, audits land at strategic intervals: early in the trial to verify initial setup, mid-trial to assess ongoing compliance, or near closeout to confirm inspection readiness. They compare what the site and the sponsor actually did against the protocol, the SOPs, and GCP.

Regulatory inspections are conducted by government authorities such as the FDA, EU national inspectorates (with EMA coordinating those for centralized applications), or Japan's PMDA, and the stakes are different in kind. An inspection may be routine surveillance, for-cause and triggered by a specific concern, or pre-approval,

conducted before a new drug is cleared for market. Findings can produce warning letters, data rejection, or a delayed marketing approval.

22.6.1. Preparing for Inspection

Inspection readiness is a standing operational condition: on the day an inspector arrives, a team either has it or does not. Keeping it means that the TMF and ISF stay current, documents version-controlled and signed; that site and sponsor staff understand their roles and the GCP requirements well enough to answer questions clearly and factually; that retrieval systems can surface a requested record within hours; and that audit trails attribute and explain every data modification. Unexplained deletions and late entries in an audit trail are immediate red flags to any regulatory inspector.

Checklist: Digital Inspection Readiness (Computerized Systems)

Inspection readiness now extends to the computerized systems used for trial conduct and data handling. Expectations for electronic systems emphasize validation commensurate with intended use, reliable audit trails, appropriate access controls, and the ability to reconstruct trial-relevant records and signatures (U.S. Food and Drug Administration 2023a, 2024f; International Council for Harmonisation 2025b).

- *System inventory*: list all computerized systems used for trial conduct and data (EDC, ePRO/eCOA, IRT/RTSM, eTMF, QMS, safety database, device platforms), including vendor responsibilities and interfaces (International Council for Harmonisation 2025b).
- *Validation and intended use*: maintain documentation of validation/qualification aligned to intended use, including change control evidence for material releases (U.S. Food and Drug Administration 2024f).
- *Audit trails*: ensure audit trails are enabled, reviewable, and practically reviewable under inspection timelines; define when routine review is expected (e.g., consent workflows, key efficacy/safety data, late changes) (U.S. Food and Drug Administration 2024f).
- *Access control and identity*: role-based access, unique user identities, and controlled electronic signature processes (including account provisioning and deprovisioning) (U.S. Food and Drug Administration 2023a, 2024f).
- *Record retrieval drills*: demonstrate rapid retrieval of critical records (consent documentation, safety reporting, protocol deviation documentation, monitoring reports, essential documents) (International Council for Harmonisation 2025b).
- *Data transfer traceability*: maintain transfer specifications and reconciliation evidence for transfers between systems, including vendor-managed pipelines (U.S. Food and Drug Administration 2024f).

During the inspection. The sponsor typically designates a *back room* team to support document retrieval and a *front room* presence to escort inspectors and

manage interactions. Daily close-out meetings between inspectors and the sponsor allow clarification of findings before they become formal observations.

Inspectors may request additional documents, interview staff, or tour facilities. Answers should be factual and concise; speculation and defensiveness rarely help. A question that cannot be answered on the spot is better met with a commitment to follow up than with a guess.

22.6.2. Inspection Findings and Responses

Findings are typically graded by severity, and the grade shapes what happens next (Table 22.2).

Table 22.2.: Inspection Observation Categories

Category	Description	Examples	Typical Consequence
Critical	Conditions that pose serious risk to participant safety or data integrity	Fabricated data; Unreported deaths; Systemic GCP violations	Data rejection; Clinical hold; Warning letter
Major	Significant deviations from GCP or protocol	Missing consent signatures; Unqualified investigators; Poor AE documentation	Regulatory correspondence; Required corrective action
Minor	Administrative issues unlikely to affect safety or data integrity	Filing errors; Minor documentation gaps	Noted for improvement

In the United States, FDA inspectors who observe conditions that may violate the regulations list them on a *Form 483*, issued at the close of the inspection. The inspected sponsor or investigator may respond in writing within 15 business days, typically with a *Corrective and Preventive Action (CAPA)* plan (U.S. Food and Drug Administration 2021a, 2020a), which should address both the immediate finding and the systemic cause that produced it.

Findings tend to cluster around a few constant issues. Informed consent produces a family of them: outdated versions, signatures obtained after procedures have begun, signatures missing altogether. Protocol deviations become findings when they were neither documented nor reported through the appropriate channels, data integrity findings when the trial database and the original medical records disagree, investigational product findings when dispensing, returns, or reconciliation were inadequately documented. On the sponsor's side, the recurring finding is oversight itself: monitoring activities inadequately documented.

22.7. Trial Governance Committees

Clinical trials, and large Phase III studies especially, also run under specialized governance committees supplying independent expertise and accountability.

22.7.1. Data Safety Monitoring Board (DSMB)

The *Data Safety Monitoring Board* (DSMB; see Section 9.6 for alternate names) provides independent oversight of accumulating safety and efficacy data. It sees unblinded comparative results that are normally withheld from the sponsor and the investigators, which is what allows it to recommend stopping a trial early for safety, futility, or overwhelming efficacy.

Composition typically runs to 3-7 members: clinical experts in the disease area, a biostatistician experienced in interim analyses, and sometimes an ethicist or patient representative. Members must be independent, free from conflicts of interest with the sponsor and uninvolved in the trial's conduct.

The DSMB operates under a *charter* that settles in advance what it will examine and what it may do: scope of review (safety only, or safety and efficacy), meeting frequency and the triggers for ad hoc meetings, statistical stopping rules and decision thresholds, the process for communicating recommendations to the sponsor, and confidentiality requirements.

DSMB meetings typically include an *open session* (where the sponsor presents trial status without unblinding) and a *closed session* (where the unblinded statistician presents interim analyses to DSMB members only). The DSMB may recommend: continue as planned, modify the protocol, stop enrollment, stop treatment, or stop the trial entirely. Recommendations go to the sponsor, who retains authority for final decisions but must document the rationale if departing from DSMB advice.

> **Note: Committee Documentation as an Inspection Artifact**
>
> Committee governance is a documentation problem: committees matter operationally only insofar as their processes are specified, followed, and evidenced. Meinert's emphasis on formal committee structures and procedures remains relevant even as meeting modalities shift (e.g., secure virtual or hybrid meetings) (Meinert 2013). From a GCP and inspection standpoint, the relevant artifacts include charters, membership independence and conflict management, meeting minutes, recommendation letters, and controlled distribution of any unblinded information (International Council for Harmonisation 2025b).

22.7.2. DSMBs for Platform and Adaptive Trials

A platform DSMB may be overseeing six, eight, or ten treatment arms at once, each at a different stage of enrollment and maturity. The SNAP platform (S. aureus bacteremia) had a single DSMB reviewing multiple treatment domains at once.

A DSMB member handed a standard two-column comparison table can form a judgment quickly. The platform equivalent requires holding in mind which arms are still enrolling, which have closed, and which interim result from one domain might change conduct in another.

The board has to be constituted for that complexity. Not every member can be an expert in every arm's therapy, so domain-specific consultants are sometimes added for particular reviews. Conflicts of interest are harder to manage too: where several companies' drugs are under evaluation at once, members with no relationship to any sponsor are difficult to find, and disclosure and recusal procedures must be detailed in the charter.

When a single arm fails, the recommendation is arm closure, not trial termination, and the DSMB has to say so precisely: "close domain B for futility in the MSSA population; continue all other domains as specified." A partial stop of that kind requires a charter that explicitly anticipates arm-level action and specifies how the remaining platform continues without the closed arm. Spillover needs anticipating as well. If domain A's success changes the standard of care and with it the control event rate for domain B, that has to be surfaced.

Complex Bayesian adaptive platforms also require real-time statistical computation for interim analyses that a DSMB's own biostatistician may not be equipped to run independently. The standard solution is a separate *unblinded statistical team* (sometimes called the independent statistical center) that prepares the interim analysis package for the DSMB. This team is firewalled from the sponsor's statisticians and has full access to unblinded data. The charter must specify exactly what it prepares, what forms of results reach the DSMB (for example, which summaries the sponsor may see in the open session, and in what aggregated form), and how results are archived to support audit.

Adjusting for temporal trends, to account for changes in background treatment or disease severity over the trial's lifespan, creates a presentation problem of its own: should the DSMB see raw observed data or model-adjusted comparisons? Raw safety data from different time periods may not be directly comparable without the adjustment, and DSMB members without statistical modeling backgrounds may struggle to evaluate model-adjusted estimates. For safety reviews in particular, where model-adjusted data may require machinery that is not readily available, a DSMB may see raw event counts alongside model-adjusted efficacy summaries. The unblinded statistician should be prepared to explain what the model is doing and what a DSMB member would see in the raw data, so the board can calibrate its judgment.

The COVID anticoagulation multi-platform RCT, pooling REMAP-CAP, ATTACC, and ACTIV-4a, used a novel governance structure: each platform kept its own DSMB, but all three reviewed the same combined analyses, produced under a common statistical analysis plan, and coordinated their recommendations. Each platform could then satisfy its own regulatory obligations (each had its own sponsor,

protocol, and regulatory and ethics approvals) while the adaptive stopping decisions were made on the full combined evidence. The coordination required detailed prespecification of how stopping decisions would be communicated across platforms.

22.7.3. Endpoint Adjudication and Steering Committees

Endpoint adjudication committees. Where endpoints are subject to interpretation (cardiovascular events, disease progression, cause of death), a *Clinical Events Committee* (CEC) or *Endpoint Adjudication Committee* provides independent, blinded review. Adjudication is what keeps endpoint determinations consistent across sites and investigators: a myocardial infarction diagnosed at one hospital should meet the same criteria as one diagnosed at another. The CEC reviews source documents (ECGs, troponin values, imaging) and applies protocol-defined criteria to classify events.

Operationally, adjudication adds timeline pressure. Cases must be prepared, reviewed, and adjudicated before database lock, and large trials with thousands of potential events may require dedicated adjudication infrastructure and close coordination with data management.

Steering committees. A *Steering Committee* (SC) provides scientific oversight and guidance on protocol interpretation, drawing its membership from the principal investigators, sponsor representatives, and sometimes independent experts. Its authority is whatever the sponsor delegates to it, from advice alone to primary responsibility for the study's design and the quality of its conduct (U.S. Food and Drug Administration 2006). It may recommend protocol amendments, interpret eligibility questions, and guide publication strategy. It should remain blind to outcomes by treatment arm, and the distinction between scientific guidance (SC) and safety oversight (DSMB) must be maintained to preserve the DSMB's independence.

AI tools for remote monitoring, risk-signal prioritization, and automated source data review within risk-based quality management frameworks are covered in Chapter 26.

23. Safety Monitoring and Reporting

A patient in a trial has a heart attack. From the moment the investigator learns of it a clock runs: the report must reach the sponsor immediately (most protocols write that as within 24 hours), and if it turns out to be a life-threatening reaction nobody expected, the sponsor then has 7 calendar days to reach regulators in every country where the study is running. However thoroughly a drug has been studied in laboratories and animal models, the first patients to receive it are pioneers, and pioneers face risks. Safety reporting is how the rest of the trial finds out in time to act.

Each category below carries its own reporting obligation (Table 23.1).

Table 23.1.: Adverse Event Classification Hierarchy

Category	Definition	Examples	Reporting Timeline
Adverse Event (AE)	Any untoward medical occurrence	Cold, headache, unrelated hospitalization	Recorded in the CRF per protocol; generally not expedited to regulators
Serious AE (SAE)	Death, life-threatening, hospitalization, disability, congenital anomaly, or medically important	Heart attack, stroke, cancer diagnosis	Immediately to sponsor (protocols typically specify 24 hours)
Adverse Drug Reaction (ADR)	AE judged to be drug-related	Rash after drug administration	Per protocol
SUSAR	Serious, unexpected ADR	Novel reaction not in Investigator's Brochure	7 days (fatal/life-threat); 15 days (other)

An *adverse event (AE)* is any untoward medical occurrence in a trial participant, caused by the investigational drug or not. A patient who develops a cold during a trial has experienced an adverse event. So has a patient hospitalized for a condition entirely unrelated to the study drug. The net is cast this wide to keep a potential

safety signal from being missed, though rare or delayed ones may surface only after the drug reaches a larger, more diverse population (Chapter 10).

A *serious adverse event (SAE)* meets one of six criteria: it results in death, is life-threatening, requires hospitalization (or prolongs existing hospitalization), causes persistent or significant disability, is a congenital anomaly, or is otherwise medically important. Seriousness says nothing about causality (an SAE may have nothing to do with the study drug), but it triggers expedited reporting and heightened scrutiny.

Seriousness is a regulatory classification driven by an event's outcome (the six criteria above). *Severity* describes intensity, most often graded on the Common Terminology Criteria for Adverse Events (CTCAE) scale that runs from Grade 1 (mild) through Grade 2 (moderate), Grade 3 (severe), and Grade 4 (life-threatening) to Grade 5 (death). The two move independently. A high-grade but asymptomatic laboratory abnormality that meets no seriousness criterion is severe without being serious; a mild event that lands the patient in hospital is serious whatever its grade.

Adverse drug reactions (ADRs) are adverse events thought to be caused by the investigational product, and deciding which events qualify is one of the most consequential clinical judgments in a trial. The evidence is circumstantial: timing, biological plausibility, and whether anything else explains the event. Investigators and sponsors both grade their certainty, though many scales are in use and none is an international standard (International Council for Harmonisation 1994a). On one common five-level scale, "Definite" requires a clear temporal link, a positive response when the drug is withdrawn or reintroduced, and no alternative explanation. As the evidence weakens, events fall to "Probable" or "Possible," depending on the strength of the temporal link and the plausibility of other causes. "Unlikely" and "Unrelated" sit at the far end, where the timing is improbable or an extraneous cause (a pre-existing condition, a traumatic accident) supplies a clear alternative.

Suspected unexpected serious adverse reactions (SUSARs) are serious adverse reactions whose nature or severity is not consistent with the drug's reference safety information, the list of expected reactions in the Investigator's Brochure or an alternative document such as an approved drug's label (International Council for Harmonisation 1994a, 2010, 2025b). They trigger the most expedited reporting, because they are new safety information that may change how the drug is viewed. A reaction the brochure mentions only as a class effect, or a listed one in a more specific or more severe form (hepatic necrosis where only hepatitis is listed), still counts as unexpected (U.S. Food and Drug Administration 2024a).

Cutting across this hierarchy is the *adverse event of special interest (AESI)*: an event, serious or non-serious, that the protocol or safety plan singles out for heightened, often accelerated monitoring because of a specific concern about the drug or its class. An AESI is tracked and reported on a protocol-specified timeline regardless of where it otherwise falls among the categories above. Drug-induced liver injury,

suicidal ideation, and, for immunomodulatory therapies, cytokine release syndrome are common examples.

23.1. Investigator Responsibilities

Immediate SAE reporting, and everything upstream of it, is a site obligation: investigators bear primary responsibility for protecting the participants in front of them. The duty begins with recognizing an adverse event as one. From there it must be documented thoroughly and reported on the timelines the protocol and the regulations set (serious events that are themselves study endpoints, such as deaths in a mortality trial, may follow the protocol's schedule unless there is evidence suggesting the drug caused them); the SAE window is tight so that the sponsor can assess a potential signal promptly and act if warranted.

The participant still has to be treated. A drug may need to be interrupted, a dose adjusted, a concomitant medication prescribed. Clinical judgment guides those decisions, though protocols typically provide guidance on dose modification and stopping rules.

23.2. Sponsor Responsibilities

A sponsor sees what no single site can: safety information from every site and every participant lands in one place. Figure 23.1 traces the path from an event at the site to a report at the regulator.

The sponsor must decide whether an incoming SAE report triggers expedited reporting. Fatal or life-threatening SUSARs must be reported within 7 calendar days, with as complete a report as possible following within a further 8 days (15 days in total); other SUSARs must be reported within 15 calendar days (International Council for Harmonisation 1994a). They go to regulatory authorities in every country where the study is being conducted.

Information also has to travel back down. A newly identified risk, a pattern of adverse events, or a change in the benefit-risk assessment prompts a *safety report* that goes to every investigator on the study.

Once a year the aggregate picture is compiled into a *Development Safety Update Report (DSUR)* and submitted to regulatory authorities (International Council for Harmonisation 2010), covering the drug's safety profile across all ongoing studies.

23.3. Safety Data Management and Analysis

Every SAE lives in two places: the clinical database, which holds all trial data, and the safety database, which holds distinct pharmacovigilance records for SAEs. Reconciling the two is a critical quality control step, and Figure 23.2 shows how it runs.

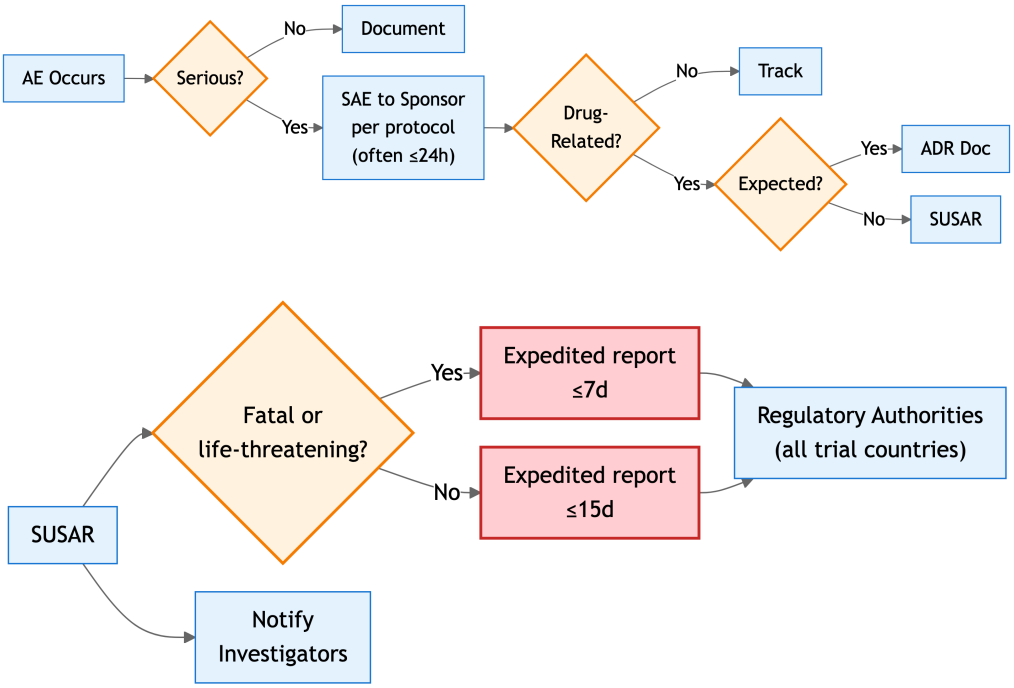


Figure 23.1.: Adverse event reporting workflow from site identification through regulatory notification. The lower panel continues from the SUSAR node of the upper panel.

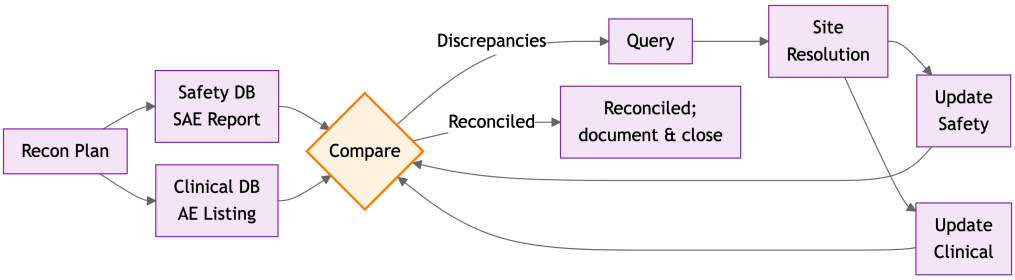


Figure 23.2.: Reconciliation Process between Clinical and Safety Databases

Analyzing laboratory safety data. A laboratory adverse event is often just a number moving, with no symptom to announce it. The analysis has to go looking. *Treatment-Emergent Abnormal Values (TEAVs)* pick out patients who moved from a normal baseline to an abnormal post-treatment state. *Shift tables* (typically 3x3 matrices of patient status, e.g., Low, Normal, High, before versus after treatment) show where a population moved as a whole. *Clinically significant changes* flag values past a defined threshold, such as >3x Upper Limit of Normal for liver enzymes, where the concern is organ toxicity.

23.4. Signal Detection

A safety signal is a potential drug-safety risk that warrants further investigation. The simplest way to find one is to count: if headache occurs in 30% of patients on drug and 10% on placebo, this suggests a drug effect. But many adverse events are tracked at once, so some of those gaps open by chance: *frequency-based methods* treat an unadjusted difference as a flag for closer review rather than proof of harm, and while adjusting for multiple comparisons limits false alarms, the greater concern in safety analysis is usually missing a real signal (International Council for Harmonisation 1998b).

Counting alone misses a good deal. *Pattern recognition* looks past raw frequencies at the shape of the data: a cluster of similar events, events concentrated at a particular dose, or events emerging only after prolonged treatment may suggest a drug effect when no individual rate is statistically significant. *Disproportionality analysis*, run on databases of spontaneous reports, asks whether an event makes up a larger share of one drug's reports than of the reports for all other drugs: if liver injury is reported disproportionately often for a drug, that warrants attention, though a reporting ratio is not an incidence rate (European Medicines Agency 2017a). A single case can also carry the argument: a rare event occurring shortly after drug administration in an otherwise healthy patient may be drug-related on its own, which is what *individual case assessment* is for.

Modern pharmacovigilance increasingly applies data mining and machine learning to large databases, though human judgment remains essential for assessing clinical significance.

For trials involving serious conditions, significant risks, or large participant populations, an independent *Data Safety Monitoring Board (DSMB)* reviews unblinded accumulating data at predetermined intervals and may recommend stopping a trial early for safety, futility, or overwhelming efficacy. Its composition, charter, and operating procedure are covered in Section 9.6 and Chapter 22.

23.5. Regulatory Reporting Requirements

The obligations are largely harmonized across major markets, but each jurisdiction has its own legal instrument and reporting channel. In the United States, expedited

IND safety reports are governed by 21 CFR 312.32 (U.S. Food and Drug Administration 2024a): sponsors must send an IND safety report to the FDA and all participating investigators as soon as possible and no later than 15 calendar days after determining that the information qualifies, which covers SUSARs but also findings from other studies or from animal or in vitro testing that suggest a significant risk, and any clinically important increase in the rate of a serious suspected adverse reaction. An unexpected fatal or life-threatening suspected adverse reaction must also reach the FDA within 7 calendar days of the sponsor's initial receipt of the information, and relevant follow-up information is submitted as soon as it is available. The European Union imposes parallel obligations through the Clinical Trials Regulation and its associated guidelines (European Medicines Agency 2022), with the EudraVigilance database centralizing safety reporting across member states. Beneath both regimes sit the ICH E2A and E2B guidelines (International Council for Harmonisation 1994a, 2013), which supply the harmonized safety definitions and the electronic transmission standards that let a single report move between systems.

Inadequate or late safety reporting can trigger clinical holds and warning letters, and, for egregious violations, criminal penalties.

23.6. The Benefit-Risk Calculation

A drug for a minor condition that causes serious adverse events may be unacceptable. The same drug for a life-threatening condition with no alternatives may be essential. The benefit-risk calculation interprets safety data in the context of the disease being treated and the benefits on offer, a considered judgment made by regulators but also by individual patients and their physicians.

Under its Prescription Drug User Fee Act (PDUFA) V commitments the FDA adopted a structured *Benefit-Risk Framework*, now used to organize the review of every new drug and biologic across four dimensions: Analysis of Condition, Current Treatment Options, Benefit, and Risk and Risk Management (U.S. Food and Drug Administration 2023d). The framework does not reduce the decision to a score; it disciplines the reasoning, making explicit what is known, what is uncertain, and how identified risks will be managed. The European Medicines Agency pursues the same end through its PrOACT-URL approach (Problem, Objectives, Alternatives, Consequences, Trade-offs, Uncertainty, Risk attitude, Linked decisions) and an effects table that lays favorable and unfavorable effects side by side.

The calculation is never final. As more patients receive a drug and follow-up lengthens, a risk that seemed acceptable early in development may prove unacceptable once the magnitude or nature of the harm is better understood. AI approaches to adverse event case processing and signal detection are covered in Chapter 26.

24. Essential Documents and the Trial Master File

When an FDA inspector arrives at a clinical trial site, they do not re-run the study. They interview patients, who have long since completed their participation, generally only when they suspect serious problems (U.S. Food and Drug Administration 2020a). They cannot observe the blood draws, the physical exams, or the conversations between investigators and participants. What they can do is examine the paper trail: the carefully maintained documentation that reconstructs how the trial was conducted.

This is why documentation matters: in clinical research, the record *is* the trial. A consent form signed and dated before procedures began documents that informed consent was obtained. A delegation log with training records shows that the person drawing blood was qualified, and a temperature log that the investigational product was stored correctly. Without these documents, there is no evidence that the trial was conducted properly. Without that evidence, the data cannot be trusted.

ICH E6(R2) Section 8.1 defines essential documents as those that "individually and collectively permit evaluation of the conduct of a trial and the quality of the data produced" (International Council for Harmonisation 2016a). They serve operational purposes during the trial and evidentiary purposes afterward, spanning a lifecycle that runs from pre-trial preparations through active data recording and safety monitoring to final reconciliation and archiving.

In 2025, ICH E6(R3) restructured this material as *essential records* in Appendix C. Where E6(R2) Section 8 prescribed a minimum list, which could be supplemented or, where justified before the trial began, reduced (International Council for Harmonisation 2016a), E6(R3) defines essential records more broadly, as the documents and data, in any format, that permit evaluation of the trial's conduct and the reliability of its results, and asks sponsors and investigators to judge on risk-proportionate grounds which records are essential for a given trial. The individual files keep their names and purposes; what changes is the emphasis, as E6(R3) ties the essential record set more explicitly to the trial's design and risks (International Council for Harmonisation 2025b).

The sponsor manages the Trial Master File (TMF), the repository of the sponsor's study-wide and site-specific essential documents, now increasingly electronic for real-time tracking and secure global access. Each clinical site keeps its own Investigator

Site File (ISF) covering its conduct of the trial. Sponsors increasingly offer electronic site files (eISF) to streamline this, but the investigator remains personally responsible for the ISF's completeness and accuracy, whether digital or physical.

24.1. Critical Essential Documents

Table 24.1 lists documents that draw particular scrutiny during inspections, where each is held, and what inspectors look for.

Table 24.1.: Critical TMF/ISF Documents for Regulatory Inspection

Document	Location	Purpose	Inspection Focus
Protocol + Amendments	TMF + ISF	Governing rules of trial	Version control; Approval dates
Investigator's Brochure (IB)	TMF + ISF	Safety information for investigators	Current version; Receipt acknowledgment
Informed Consent Forms	ISF	Record of ethical participation	Dates; Signatures; Correct versions
FDA Form 1572	TMF + ISF (IND studies)	PI commitment to comply	Signature; Sites, labs, and subinvestigators listed
Delegation of Authority Log	ISF	Staff responsibilities	Training documentation; Dated signatures
IRB/EC Approvals	TMF + ISF	Ethics approval	Correspondence; Amendment approvals
Financial Disclosure	TMF	Conflict of interest documentation	Completeness; Accuracy
Lab Certifications	TMF + ISF	Lab qualifications	Currency; Scope of certification
Monitoring Reports	TMF	Record of sponsor oversight	Timeliness; Issue follow-up
Drug Accountability	ISF	IP receipt, dispensing, return	Reconciliation; Temperature logs

Under 21 CFR Part 54 (U.S. Food and Drug Administration 2024c), sponsors must collect and, in the marketing application, disclose (or certify the absence of) the relevant financial interests of each clinical investigator in a covered clinical study, including those of a spouse or dependent child: compensation tied to the study outcome, a proprietary interest in the tested product, significant equity (more than $50,000 in a publicly traded sponsor, or any equity in a privately held one), and other significant payments exceeding $25,000 beyond the costs of conducting the study, tracked during the trial and for one year after it ends. The requirement does

not forbid such interests; it makes them visible, so that reviewers can judge whether a conflict might have biased the trial's conduct or interpretation.

24.2. The Quality Standard: ALCOA++

A blood pressure reading without a signature and date is useless; an adverse event narrative without an author cannot be verified. ALCOA is the shorthand for the properties that make a clinical trial record trustworthy. Chapter 21 tabulates the base ALCOA principles in Table 21.1 and describes the ALCOA+ and ALCOA++ extensions.

Each record must be *attributable*: it must be clear who created it, when, and under what circumstances. Records must be *legible*, readable and intelligible years later to anyone who needs to review them. They must be *contemporaneous*, created at the time the activity occurred rather than reconstructed from memory days or weeks afterward, and *original*, the first capture of the data, or a certified copy that preserves its content and context, rather than a transcription that might introduce errors (U.S. Food and Drug Administration 2024f). And they must be *accurate*, truthfully reflecting what actually happened.

ALCOA+ and then ALCOA++ added requirements aimed at electronic systems and long-term data integrity. Records must be *complete*, with no missing fields, unexplained gaps, or deleted entries; *consistent*, following the same format and logic across the trial; and *available*, retrievable for review or inspection when needed. They must also be *enduring*, readable throughout the required retention period, which is a challenge when file formats and storage media become obsolete. The second plus adds *traceable*: every value can be followed through its full lifecycle, with audit trails documenting any change.

24.3. TMF Management and Oversight

A trial generates documents continuously (protocols are amended, investigators added, IRB approvals renewed, monitoring visits conducted, safety events reported), and each must reach the right file within a reasonable timeframe. Targets vary by document type and organization; a sponsor SOP might call for filing within 5-10 business days of receipt or creation, a target that is easily missed when trial teams are stretched thin.

An inspector asks to see "the informed consent form for patient 101-005 at the time of their third visit" and expects an answer within minutes. A well-organized ISF can produce the signed form, and the TMF the version the ethics committee had approved by then. The *TMF Reference Model* provides a standardized taxonomy for organizing TMF documents, defining zones (such as Trial Management, Regulatory, and Site Management), sections within each zone, and specific artifact types within each section. Originally convened under the Drug Information Association, the

model has been stewarded by CDISC since 2022 (TMF Reference Model Initiative 2024).

TMF health metrics track whether the file is complete and current: typically filing lag (average time from document creation to filing), completeness rate (percentage of expected documents actually present), and quality score (percentage of documents passing quality review without issues). Sponsors set targets, perhaps 95% completeness and 90% quality. Sites that fall behind receive additional attention, and persistent deficiencies may trigger for-cause audits.

During site visits, monitors verify that the Investigator Site File holds every required document in its current version, signed and dated where required. They also reconcile the site file against the sponsor's central TMF. Deficiencies noted in a monitoring report become action items that must be resolved before the next visit.

Missing essential documents can hold up database lock when the gaps bear on participant safety or the reliability of the data, a problem that can delay regulatory submissions by months when discovered late. Inspectors who find disorganized or incomplete TMFs may question the overall quality of trial conduct, and in extreme cases regulators can refuse to rely on a trial's data when the TMF cannot demonstrate that it was conducted according to GCP.

24.4. Archiving and Retention

The commitment to documentation does not end when the last patient leaves or the final report is signed, and retention periods are set by regional regulation rather than by any single global rule. In the United States, FDA 21 CFR 312.57(c) requires sponsors to keep records for two years after a marketing application for the drug is approved or, if none is approved, for two years after investigational shipment and delivery stop and FDA is notified, while 312.62(c) runs an investigator's two years from approval for the indication under investigation or, failing that, from discontinuation of the investigation with FDA notified; the EU Clinical Trials Regulation (536/2014, Article 58) requires the trial master file to be archived for at least 25 years after the trial ends; and ICH E6(R3) defers to the applicable regional requirement, adding that investigators should keep their records until the sponsor says they are no longer needed, if that is later. Because a global trial must satisfy the strictest jurisdiction, sponsors commonly adopt a single retention policy sized to the longest applicable period, which for a trial with EU sites means at least 25 years (U.S. Food and Drug Administration 2024b; European Parliament and Council of the European Union 2014; European Medicines Agency 2022; International Council for Harmonisation 2025b).

The investigator is responsible for site record retention, the sponsor for the overall TMF. Investigators have faced regulatory action years after their trials concluded, when inspectors discovered that site files had been improperly destroyed or were incomplete.

A large Phase III trial generates tens of thousands of pages across dozens of sites. Monitors spend substantial portions of their visits reviewing regulatory binders, and study coordinators give hours each week to filing, organizing, and reconciling documents. Electronic systems have improved efficiency without eliminating the burden: documents still must be created, reviewed, signed, filed, and retained.

The principle underneath is simple: in clinical research, the documentation is the proof. If a procedure was performed but not documented, regulators will question whether it occurred. If consent was obtained but the form is unsigned, the enrollment is problematic. Contemporaneous, complete, and accurate documentation is the foundation on which the credibility of clinical evidence rests.

25. Clinical Trial IT Infrastructure

A decade ago, sponsors managed clinical data through a patchwork of vendor systems that rarely communicated: EDC databases that could not talk to randomization systems, trial master files stored in SharePoint folders with manual indexing, and clinical trial management systems that required spreadsheet reconciliation to produce accurate enrollment counts. Data flowed through exports, imports, and emails, and every handoff added latency, transcription errors, and audit risk.

What replaced it is the subject of this chapter: the four systems that now form the transactional backbone of a trial, Electronic Data Capture (EDC), Clinical Trial Management Systems (CTMS), electronic Trial Master Files (eTMF), and Randomization and Trial Supply Management (RTSM), plus the platform market shaping them and the regulatory framework (21 CFR Part 11, EMA's guideline on computerized systems in clinical trials, GMP Annex 11, and ICH E6) that governs computerized systems in clinical research.

These platforms determine what data a trial can capture and integrate, and whether their records hang together decides whether it is inspection-ready. They are also the substrate for the AI and automation tools discussed in Chapter 26: without validated, well-integrated core systems, advanced analytics and agentic automation have nothing reliable to operate on. Technology can reduce friction and surface risks earlier. What makes it defensible in an inspection is validation, governance, and oversight.

25.1. The Modern Clinical Trial Ecosystem

Today's leading platforms aspire to unified architectures where patient data, operational metrics, essential documents, and supply chain signals flow through shared data models. In the best-integrated environments, particularly single-vendor platforms like Veeva Vault, a single site action can cascade through the whole stack: randomize a patient and enrollment counts update in CTMS dashboards, treatment-specific CRF pages unlock in the EDC, drug shipment requests trigger in the supply management system, and expected document checklists populate in the eTMF. Many sponsors do not work in that environment. They still run heterogeneous stacks with systems from multiple vendors, connected through middleware and custom integrations that require configuration, maintenance, and periodic reconciliation. How far the integration goes varies widely from one organization to the next,

but the direction of travel is the same: fewer manual handoffs, and the shared data infrastructure on which analytics and automation depend.

25.1.1. The Technology Ecosystem

The four platforms each serve a distinct but complementary function, and Figure 25.1 shows how they interconnect (Medidata Solutions 2024a).

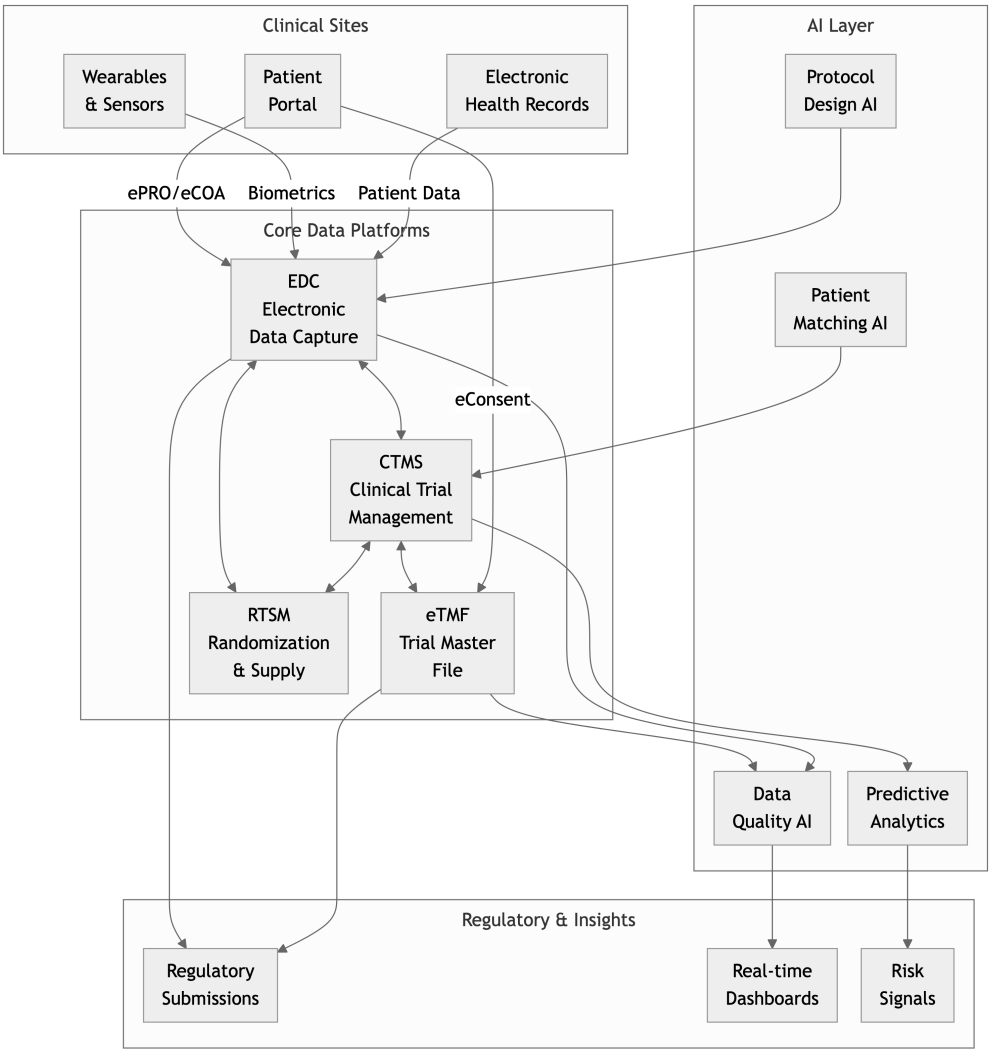


Figure 25.1.: The Modern Clinical Trial Technology Stack

Electronic Data Capture (EDC) replaces paper case report forms with validated electronic forms that capture patient data at the point of care. When a coordinator records a blood pressure reading, administers a questionnaire, or documents an adverse event, that data flows into the EDC. Built-in edit checks flag impossible

values (a heart rate of 500?) or logical inconsistencies (an adverse event dated before the patient enrolled) in real-time, catching errors before they propagate. Every efficacy and safety analysis in the submission traces back to data captured here (U.S. Food and Drug Administration 2023a).

Clinical Trial Management System (CTMS) is the operational command center. Where EDC holds patient data, CTMS holds the state of the trial: which sites are open, how many patients each has enrolled, when the next milestone or monitoring visit falls due, what the budget burn rate looks like. It manages contracts and payments, and it generates the operational metrics sponsors use to assess trial health (Grand View Research 2025). It is what keeps a 50-site, 14-country trial from descending into chaos.

Electronic Trial Master File (eTMF) is the regulatory archive. A single trial generates thousands of documents: the protocol and its amendments, informed consent forms, IRB approvals, investigator CVs, monitoring reports, safety letters, correspondence. Regulators require sponsors to maintain a complete Trial Master File as evidence that the trial was conducted properly, and eTMF systems organize it according to the TMF Reference Model (originally convened under DIA, now stewarded by CDISC), tracking document completeness and inspection readiness (TMF Reference Model Initiative 2024). When an FDA inspector arrives, the eTMF is the first artifact they examine.

Randomization and Trial Supply Management (RTSM), sometimes called Interactive Response Technology (IRT), handles treatment assignment and drug supply. When a patient is eligible for randomization, the RTSM system assigns them to a treatment arm according to the randomization scheme, maintaining the blind while ensuring balanced allocation. At the same time it tracks investigational product inventory at each site and triggers resupply shipments. For trials with temperature-sensitive biologics or personalized therapies, RTSM is mission-critical (Clinical Leader 2024).

The four exchange data continuously. Complete a monitoring visit and the report is filed in eTMF while the visit status updates in CTMS. That constant traffic is why unified platforms like Veeva Vault, which house all four systems in a single architecture, have gained so much traction (Veeva Systems 2024b).

25.2. The Foundational "Backbone": Platform Wars

A handful of enterprise players dominate the clinical trial platform market; Table 25.1 compares four of them (Medidata Solutions 2024a).

Table 25.1.: Comparison of Major Clinical Trial Platforms

Vendor	Primary Strengths	Market Position	Cloud Model	AI Capabilities
Medidata (Dassault)	Industry-standard EDC (Rave), 25-year track record, 38,000+ studies	Market leader in EDC	Cloud/SaaS	AI-powered signal detection, synthetic control arms
Veeva Systems	Unified Vault platform (eTMF, CTMS, EDC), life sciences focus	Fast-growing challenger	Cloud-native	TMF Intake Agent, Quality Check Agent
Oracle	Enterprise scale, Siebel CTMS and Oracle Clinical One, regulatory expertise	Established incumbent	Cloud/On-prem	ML-based safety analytics
IQVIA	Real-world data integration, global CRO services	CRO-integrated platform	Cloud/SaaS	Intelligent eTMF, predictive enrollment

Cloud Delivery

All four platforms in Table 25.1 are delivered as cloud services, and Oracle is the only one of the four that also offers on-premise installation. Grand View Research lists data privacy and security concerns, especially with cloud-based and decentralized platforms, among the restraints on the clinical trial software market (Grand View Research 2025).

25.2.1. Major Platform Vendors

Medidata (acquired by Dassault Systemes in 2019) remains the dominant EDC platform. The 2025 ISR Benchmarking Report ranked its *Rave EDC* the top-preferred EDC system in independent sponsor evaluations (Industry Standard Research (ISR) 2025). Medidata's scale is substantial: over 700,000 certified site users, 1.8 million EDC users, and more than 38,000 studies managed across all phases and therapeutic areas (Medidata Solutions 2024b).

Medidata's AI offerings include *Acorn AI*, which provides synthetic control arms using historical patient data to reduce or eliminate placebo groups in certain trial designs (Medidata Solutions 2024d). Their *Sensor Cloud* integrates wearable device data directly into the EDC, enabling continuous physiological monitoring without manual data entry (Medidata Solutions 2024c).

Veeva has gained market share fast with a unified *Vault* platform that puts eTMF, CTMS, and EDC in one system, against Medidata's historically modular approach.

Veeva serves only life sciences, unlike Oracle or Salesforce, which operate across multiple industries, and that concentration shows in a platform built around clinical and regulatory workflows. It is also deploying specialized *AI Agents* aimed at the most tedious parts of clinical operations; Figure 25.2 follows a site document through two of them.

The *TMF Intake Agent* automatically classifies documents uploaded by sites, extracting metadata such as investigator name and document date to route files to the correct TMF binder. The *Quality Check Agent* then reviews documents for errors (missing signatures, wrong versions, incomplete forms) before a human ever sees them (Veeva Systems 2024a). Separately, Veeva reports a decrease of more than 80% in TMF migration preparation time for its eTMF product, a product-level claim rather than a measured effect of either agent (Veeva Systems 2026).

The eTMF market alone is worth *$1.4 billion* and growing at 12.8% annually (MarketsandMarkets 2024). Table 25.2 compares three of the leading eTMF platforms:

Table 25.2.: eTMF Platform Feature Comparison

Feature	Veeva eTMF	IQVIA eTMF	Phlexglobal eTMF
Auto-Classification	AI-powered TMF mapping	ML-based indexing	Intelligent auto-filing
Completeness Prediction	Expected document lists	Milestone-based gaps	Risk-based prioritization
Inspection Readiness	Real-time dashboards	Inspection-ready reports	Audit trail analytics
Site Integration	SiteVault connected	Site-facing portal	Sponsor-site bridge
Unique Strength	Unified Vault ecosystem	RWD integration	eTMF-specialist focus

All three apply AI in the same two places. *Auto-indexing* uses machine learning models to classify unorganized scans into the TMF Reference Model structure. *Completeness prediction* algorithms identify missing documents based on study milestones, flagging that "Site 101 has initialized but is missing a financial disclosure form" (IQVIA 2024b; Phlexglobal 2024).

Medable took a different path, building for the decentralized clinical trial (DCT) from the outset; as hybrid and virtual trials became mainstream, its modular platform let patients participate from home (see Table 25.3) (Medable 2024a):

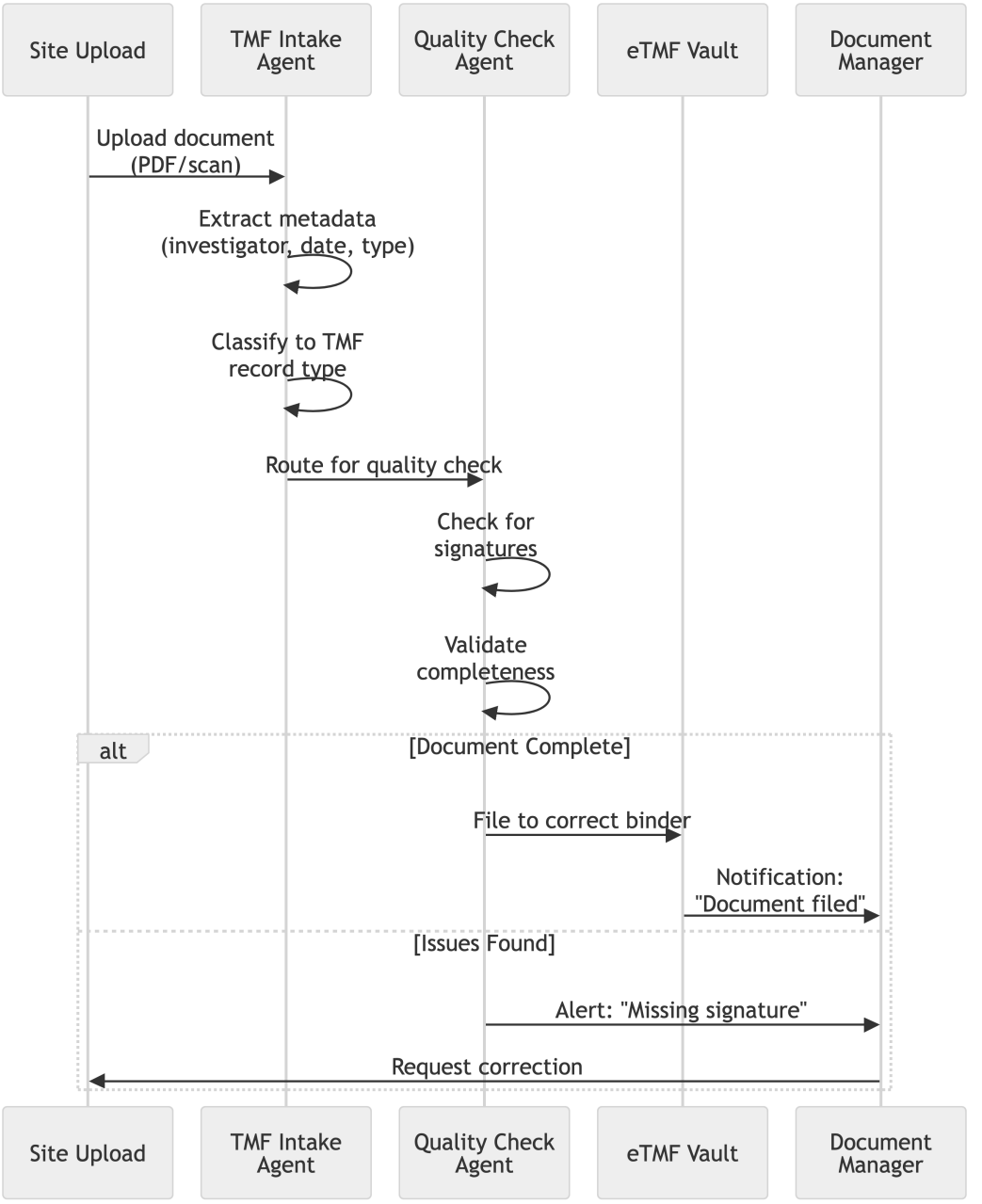


Figure 25.2.: How Veeva's AI Agents Process TMF Documents

Table 25.3.: Medable DCT Capabilities (impacts are implementation- and protocol-dependent) (Medable 2024b)

Capability	What It Does	Impact
TeleVisit	Video conferencing for remote assessments	Reduced travel for suitable protocols (implementation-dependent)
eConsent	Multimedia-rich digital consent	Improved comprehension and workflow consistency (context-dependent)
Medable AI	Generates digital eCOA from paper protocols	Faster digitization and reuse of instruments (vendor-reported)
TMF Automation	Processes DCT-generated document flood	Helps manage higher document volume (protocol-dependent)

25.3. Regulatory Framework for Computerized Systems

Every computerized system in a trial, from a validated EDC platform to an AI-enabled document classifier, operates inside a regulatory framework that dictates how it must be validated, documented, and overseen. What an inspector expects to see as evidence depends on which tier of that framework a requirement sits in.

25.3.1. Enforceable Standards vs. Recommendations

Failure to comply with an enforceable requirement can result in warning letters, clinical holds, or rejection of a submission. Guidance documents represent FDA or EMA "current thinking" and are generally recommendations, not mandates, though departing from them without justification invites scrutiny. An exception is FDA guidance issued under section 745A(a) of the Federal Food, Drug, and Cosmetic Act, such as the standardized study data guidance, which binds sponsors to the electronic submission formats it specifies (U.S. Food and Drug Administration 2021e). Table 25.4 sorts the main obligations into those tiers.

Table 25.4.: Regulatory Hierarchy for AI in Clinical Trials

Category	Examples	Consequence of Non-Compliance
Enforceable Law/Regulation	21 CFR Part 11 (electronic records), 21 CFR Part 312 (INDs), EU Clinical Trials Regulation (which makes GCP compliance legally binding in the EU)	Warning letters, clinical holds, application rejection, criminal liability

Category	Examples	Consequence of Non-Compliance
GxP Standards (legal force varies by jurisdiction)	ICH E6(R3) computerized systems provisions (nonbinding FDA guidance; the EU Clinical Trials Regulation requires appropriate account of ICH GCP), EMA computerized-systems guideline (EU trials), Annex 11 (EU GMP), data integrity requirements	Inspection findings, Form 483 observations, regulatory action
Regulatory Guidance	FDA draft guidance on AI for regulatory decisions (Jan 2025), EMA reflection paper on AI (Sept 2024)	Increased scrutiny, requests for additional information, delays

25.3.2. Validation and Documentation Requirements

AI systems and computerized systems used in clinical trials must meet the same validation requirements as any other regulated software. Which standards apply depends on the regulatory jurisdiction and the system's role: 21 CFR Part 11 in the United States; in the EU, EMA's 2023 guideline on computerized systems and electronic data in clinical trials, with GMP Annex 11 governing systems used in manufacturing (see Table 25.5). Both jurisdictions require that computerized systems generating, modifying, or storing electronic records that FDA or EU regulations require sponsors to maintain or submit meet validation and control standards (U.S. Food and Drug Administration 2023a, 2024f; European Medicines Agency 2023; European Commission 2011). In practice that means documented evidence that the software performs as intended, AI model verification included; computer-generated, time-stamped audit trails recording every system action that creates, modifies, or deletes an electronic record; unique user identification, authentication, and role-based permissions; controls that keep data attributable, legible, contemporaneous, original, and accurate (ALCOA); and documented procedures for system use, maintenance, and change control.

ICH E6(R3), adopted in January 2025, adds computerized systems requirements that bear directly on AI (International Council for Harmonisation 2025b). A system must be validated as fit for the specific use it is put to in the trial, and the validation effort is expected to be proportional to the system's impact on patient safety and data reliability. A dedicated data governance section requires documented data and records management, and the guideline recognizes data from wearables, sensors, and automated systems as primary source data.

Table 25.5.: Computerized Systems Validation Requirements for AI

Requirement	21 CFR Part 11	EU (EMA 2023; Annex 11)	ICH E6(R3)
System validation	Required	Required	Required (risk-based)
Audit trails	Required	Required	Required
Access controls	Required	Required	Required
Change control	Required	Required	Required
Data backup/recovery	Required	Required	Required
Training documentation	Required	Required	Required
Supplier qualification	N/A	Required	Required

The requirements are deliberately technology-neutral. The same expectations for validation, audit trails, access control, and data integrity apply whether a record is created by a coordinator typing into an EDC field or by a model classifying a document, which is what lets AI and automation enter the workflow without a separate rulebook, provided they are held to the same standard of evidence. Chapter 26 turns to those tools, and to what it takes to keep them defensible in an inspection once they are layered onto this infrastructure.

26. AI Agents in Clinical Development

Agents can draft text. Whether one runs inside a trial turns on something else: can it operate within a regulated environment at all, preserving data integrity, producing audit-ready traces, and remaining under accountable human oversight? The *2024 EMA reflection paper* on AI and risk-based approaches for AI use in drug development formalizes that expectation (Unlearn.AI 2025; European Medicines Agency 2024b; National Institute of Standards and Technology 2023; Amershi et al. 2019).

Clinical trials run on a layered digital infrastructure: the core transactional systems the previous chapter described (EDC, CTMS, eTMF, RTSM), integration middleware, and the analytics and automation services that continuously convert operational events into risk signals and decisions. Cloud adoption and regulatory acceptance of technology-enabled trial conduct, including decentralized elements and digital health technologies for remote data acquisition, have accelerated the shift toward that model (U.S. Food and Drug Administration 2024n, 2023g). This chapter takes up that top layer, along with the methodological approaches changing how trials are designed and how evidence is generated.

The transition is conceptual as much as technological: from *automation as a feature* (dashboards, rules engines, isolated machine-learning models) to *automation as an operating model*, in which software agents can plan, execute, and verify multi-step workflows across the clinical operations stack. This shifts the trial from a linear "pipeline" towards a dynamic, electronic map of activities, where information infrastructure identifies bottlenecks and navigates the "diverse web of failure-prone iterative learning loops" that characterize modern drug development (Wagner et al. 2018a). "Agentic AI" here is a design pattern: "compound" systems that combine models with retrieval, tools, and control logic, grounded in computer science work on "generative agents" that simulate human-like behavior and goal-directed action (J. S. Park et al. 2023; Berkeley Artificial Intelligence Research 2024). Preliminary findings from 2024-2025 benchmarks suggest agentic systems can improve clinical task performance by up to 60 percentage points over base language models in specific tasks such as evidence retrieval and task planning, though most improvements are more modest and results vary considerably by task and study design (Abou Ali, Dornaika, and Charafeddine 2026). The same shift converges with *Digital Twin "Moonshots"*, which aim to integrate personalized digital twins directly into medi-

cal records to optimize both clinical care and trial participation (Duke Center for Virtual Imaging Trials 2024).

The money has moved accordingly. The clinical trial software market was estimated at about USD 11 billion in 2024 and is projected to grow at a compound annual rate of about 11% through 2033 (Grand View Research 2025), with AI applications growing rapidly within it (Fortune Business Insights 2024). None of that is evidence of clinical benefit on its own, but it explains why sponsors, CROs, and platform vendors are reorganizing workflows around automation and AI-mediated operations.

26.1. Agentic AI in Clinical Workflows

Pull the *right* protocol amendment from an eTMF. Verify whether a site has an updated IRB approval. Check that a safety narrative is consistent with the underlying case data. Each of these is a retrieval and verification problem under audit constraints, not a "creative writing" task, and none of them is settled by calling a model once.

26.1.1. Building Blocks and System Architecture

The recent resurgence of "agents" is a systems shift. Agentic systems orchestrate multiple calls, state, and tools to complete a task, decomposing goals, retrieving context, taking actions, and iterating on feedback (Cheng et al. 2024; L. Wang et al. 2024). On the Berkeley AI Research account of "compound AI systems," reliability gains often come from engineered compositions (retrievers, checkers, constrained tool calls, repeated sampling) rather than from a single monolithic model invocation (Berkeley Artificial Intelligence Research 2024).

Three technical ideas recur. Structured intermediate reasoning, such as chain-of-thought prompting, can improve performance on complex tasks, though it does not guarantee correctness in the presence of missing or stale information (J. Wei et al. 2022). Retrieval-augmented generation (RAG) externalizes "memory" into a maintained knowledge base and, when paired with citations to retrieved sources, can supply provenance (Lewis et al. 2020; Douze et al. 2024). Coupling reasoning with tool use, where a model decides when to query, calculate, or fetch, also yields more controllable trajectories than purely generative text (Yao et al. 2023).

Agents also need consistent interfaces to external systems, and tool calling standards such as the Model Context Protocol (MCP) aim to standardize how AI applications connect to data sources and actions, turning "integration" into a first-class design surface (Anthropic 2025). In regulated contexts the payoff is not novelty; it is enforceable access controls, a log of every tool invocation, and agent actions a reviewer can follow.

Clinical workflows are naturally multi-role (data management, monitoring, safety, regulatory, sites), and multi-agent frameworks formalize that by allocating roles and

coordinating conversations among specialized agents (Wu et al. 2023). Empirical analyses, however, show that such systems fail in predictable ways: specification gaps, coordination failures, and weak verification loops (Cemri et al. 2025). "Agent quality" therefore has to be measured on cost, robustness, and reproducibility alongside task accuracy (Kapoor et al. 2024), which is where evaluation is heading: benchmarks now emphasize realistic multi-step work with tool use and resource constraints, and propose clearer reporting of cost and reproducibility (J. S. Chan et al. 2024; Cappello et al. 2025). Human-in-the-loop debugging and steering tools are emerging so that multi-step traces can be inspected and edited (Epperson et al. 2025).

Run agents concurrently across several workflows and the operational concerns become first-order: context windows turn into a managed resource, tools must be scheduled and access-controlled, and traces must be stored in an auditable way. Emerging "agent runtime" proposals make this explicit by separating agent applications from shared services such as scheduling, context management, and access control (Mei et al. 2024). Pipeline frameworks that treat agent workflows as composable graphs add systematic optimization and regression testing across versions, which matters in any environment where changes must be validated rather than "shipped and hoped" (Khattab et al. 2023).

One risk remains whatever the architecture. Agentic systems can create an illusion of competence, in which fluent outputs are mistaken for validated decisions: a known socio-technical risk in scientific work, where productivity gains may coexist with a decline in genuine understanding and critical scrutiny (Messeri and Crockett 2024). Human-AI interaction guidelines answer it by making uncertainty visible, supporting oversight and correction, and ensuring users understand system limitations, principles that map directly to quality management and inspection readiness (Amershi et al. 2019).

26.2. Regulatory Framework for AI in Clinical Trials

Most of what FDA and EMA have said about AI in drug development is guidance: influential, expected of sponsors, and not legally binding. One AI-specific instrument in this section is law, and it is not a medicines regulation at all. The validation and documentation requirements for computerized systems (21 CFR Part 11, EMA's computerized-systems guideline, Annex 11, ICH E6) are covered in Chapter 25.

26.2.1. The 10 Guiding Principles for Good AI Practice

In January 2026, FDA (CDER and CBER) and EMA jointly published *Guiding Principles of Good AI Practice in Drug Development*, the first joint regulatory statement establishing foundational expectations for AI across the drug product lifecycle (U.S. Food and Drug Administration and European Medicines Agency 2026). The ten principles represent international consensus on what "good practice" means here.

Joint FDA-EMA Guiding Principles (January 2026)

The following is quoted verbatim from the joint FDA-EMA document:

1. **Human-centric by design** The development and use of AI technologies align with ethical and human-centric values.
2. **Risk-based approach** The development and use of AI technologies follow a risk-based approach with proportionate validation, risk mitigation, and oversight based on the context of use and determined model risk.
3. **Adherence to standards** AI technologies adhere to relevant legal, ethical, technical, scientific, cybersecurity, and regulatory standards, including Good Practices (GxP).
4. **Clear context of use** AI technologies have a well-defined context of use (role and scope for why it is being used).
5. **Multidisciplinary expertise** Multidisciplinary expertise covering both the AI technology and its context of use are integrated throughout the technology's life cycle.
6. **Data governance and documentation** Data source provenance, processing steps, and analytical decisions are documented in a detailed, traceable, and verifiable manner, in line with GxP requirements. Appropriate governance, including privacy and protection for sensitive data, is maintained throughout the technology's life cycle.
7. **Model design and development practices** The development of AI technologies follows best practices in model and system design and software engineering and leverages data that is fit-for-use, considering interpretability, explainability, and predictive performance. Good model and system development promotes transparency, reliability, generalizability, and robustness for AI technologies contributing to patient safety.
8. **Risk-based performance assessment** Risk-based performance assessments evaluate the complete system including human-AI interactions, using fit-for-use data and metrics appropriate for the intended context of use, supported by validation of predictive performance through appropriately designed testing and evaluation methods.
9. **Life cycle management** Risk-based quality management systems are implemented throughout the AI technologies' life cycles, including to support capturing, assessing, and addressing issues. The AI technologies undergo scheduled monitoring and periodic re-evaluation to ensure adequate performance (e.g., to address data drift).
10. **Clear, essential information** Plain language is used to present clear, accessible, and contextually relevant information to the intended audience, including users and patients, regarding the AI technology's context of use, performance, limitations, underlying data, updates, and interpretability or explainability.

All ten operationalize one expectation: AI in drug development must be validated, documented, and maintained with the same rigor as any other regulated activity,

plus extra attention to what makes AI different, its data-dependency, its potential opacity, and its tendency to drift over time.

26.2.2. The FDA Risk-Based Credibility Framework

FDA's January 2025 draft guidance *"Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products"* (U.S. Food and Drug Administration 2025f) sets out a seven-step risk-based framework and, though still a draft, represents FDA's expected approach for evaluating AI-generated evidence.

> **!** Scope of FDA AI Guidance
>
> The FDA guidance applies when AI is used to *produce information or data intended to support regulatory decision-making* regarding safety, effectiveness, or quality. It does *not* apply to AI used in drug discovery (the in-scope drug product life cycle begins at nonclinical development, so not all pre-IND work is excluded), nor to AI used purely for operational efficiencies such as internal workflows and resource allocation that do not impact patient safety, drug quality, or study reliability.

The Seven-Step Credibility Assessment Process:

1. *Define the Question of Interest*: What specific question, decision, or concern is the AI model addressing?
2. *Define the Context of Use (COU)*: What is the specific role and scope of the AI model? Will other evidence be used alongside it?
3. *Assess Model Risk*: Combine *model influence* (contribution of AI evidence relative to other evidence) with *decision consequence* (significance of adverse outcomes from incorrect decisions). See Figure 26.1 for how common AI applications map onto this two-dimensional risk space.
4. *Develop a Credibility Assessment Plan*: Document model architecture, training data, evaluation methods, and performance metrics, with rigor proportional to model risk.
5. *Execute the Plan*: Implement the credibility assessment activities.
6. *Document Results*: Prepare a credibility assessment report with any deviations from the plan.
7. *Determine Adequacy*: Evaluate whether model credibility is sufficient for the COU.

26.2.3. What You CAN and CANNOT Do with AI in Clinical Trials

Table 26.1 synthesizes the enforceable requirements and regulatory expectations from FDA guidance, EMA reflection papers, and ICH E6(R3), ordered from the applications a sponsor can deploy today to the one it cannot deploy at all.

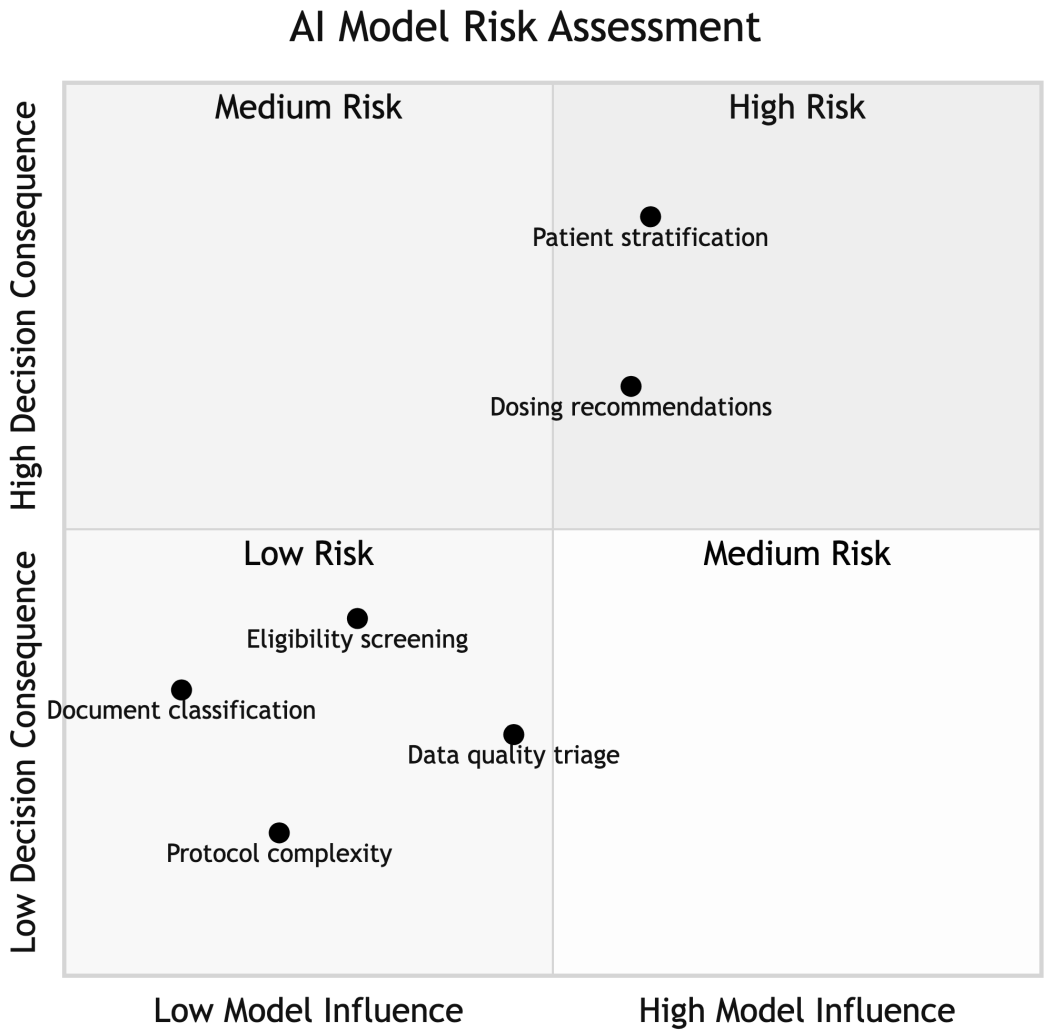


Figure 26.1.: FDA AI Model Risk Matrix: Risk Increases with Model Influence and Decision Consequence

Table 26.1.: AI Applications in Clinical Trials: Regulatory Status and Requirements

AI Application	Regulatory Status	Requirements	Limitations
Document classification and TMF filing	Permitted with oversight	Validate classification accuracy; maintain human review for inspection-critical documents	Cannot replace human accountability for TMF completeness
Data quality screening and query generation	Permitted with oversight	Document AI logic; validate query generation; apply risk-based human review of generated queries	No sponsor changes to investigator- or participant-entered data without the investigator's advance agreement; automated query closure needs a validated, documented process
Patient matching/eligibility pre-screening	Permitted as decision support	Validate against eligibility criteria; investigator makes final determination	Cannot make final eligibility decisions; investigator responsibility under GCP
Protocol complexity scoring	Permitted	Document methodology; validate predictions against historical data	Operational tool only; no regulatory submission required
Site selection and feasibility	Permitted	Document data sources and model logic	Operational tool; sponsor retains responsibility for site qualification
Adverse event case processing	Permitted with lifecycle monitoring	Monitor model performance; human review of serious/unexpected cases	Cannot replace pharmacovigilance qualified person oversight
Statistical analysis endpoints	Permitted with pre-specification	Pre-specify in SAP; in pivotal trials, freeze model before database lock; prospective validation required for high-impact uses	Non-prespecified model changes after unblinding make the analysis post hoc, not confirmatory
Primary endpoint assessment	Conditional: high regulatory scrutiny	EMA: prospective testing with newly acquired data required; FDA: credibility assessment proportional to risk	Model must be frozen and fully documented in SAP

AI Application	Regulatory Status	Requirements	Limitations
Dosing/treatment assignment decisions	Conditional: requires early FDA/EMA engagement	Full credibility assessment; human-in-the-loop required; extensive safety monitoring	Cannot be sole determinant without validated safety controls
Replacing clinical judgment	Not permitted	N/A	AI supports decisions; investigators and physicians retain accountability under GCP

⚠ Critical Principle: Human Accountability Cannot Be Delegated to AI

Under ICH E6 GCP and FDA regulations, the investigator is responsible for medical decisions affecting trial participants, and the sponsor is responsible for trial conduct and data integrity. AI can support these decisions but cannot replace the accountable human. An investigator cannot defend a protocol deviation by stating "the AI said it was acceptable."

26.2.4. AI Model Lifecycle Maintenance

Unlike static software, AI models may degrade or drift over time as input data distributions change. Both FDA and EMA guidance therefore expect *lifecycle maintenance* for models deployed over extended periods: in manufacturing (quality control, process optimization), in pharmacovigilance (case classification, signal detection), and in any AI system whose performance may change as new data arrives. Five activities make it up.

1. *Performance monitoring*: Define metrics and thresholds; monitor on risk-based frequency.
2. *Drift detection*: Identify when input data diverges from training data distribution.
3. *Revalidation triggers*: Pre-define conditions requiring model retesting or retraining.
4. *Change management*: Evaluate model changes through the applicable change-control process (for manufacturing models, the pharmaceutical quality system).
5. *Regulatory notification*: Report changes impacting model performance per applicable requirements (e.g., 21 CFR 314.70 for manufacturing changes under an approved NDA).

EMA on Incremental Learning

For pharmacovigilance applications, EMA permits a "more flexible approach" where incremental learning can continuously enhance models. However, the MAH retains responsibility to validate, monitor, and document model performance as part of the pharmacovigilance system.

26.2.5. When to Engage Regulators

Early engagement with FDA or EMA is strongly recommended for AI applications with *high regulatory impact* or *high patient risk*; Table 26.2 summarizes the pathways.

Table 26.2.: Regulatory Engagement Pathways for AI in Clinical Trials

AI Use Case	Recommended Engagement	FDA Contact	EMA Contact
Novel clinical trial design using AI	CDER C3TI or CID Meeting Program	CDERclinicaltrialinnovation@fda.hhs.gov	Innovation Task Force (ITF)
AI for endpoint evaluation	Drug Development Tools (DDT) or ISTAND	CDER-BiomarkerQualificationProgram@fda.hhs.gov (biomarkers; clinical outcome assessments and ISTAND have their own contacts)	SAWP qualification
AI-enabled digital health technology	DHT Program	DHTsforDrugDevelopment@fda.hhs.gov	N/A
AI in pharmacovigilance	Emerging Drug Safety Technology Program	AIMLforDrugDevelopment@fda.hhs.gov	PRAC interaction
AI in manufacturing	Emerging Technology Program (CDER ETP)	CDER-ETT@fda.hhs.gov	N/A
Model-informed drug development	MIDD Paired Meeting Program	MIDD@fda.hhs.gov	SAWP scientific advice

26.2.6. EMA-Specific Considerations

The EMA reflection paper (September 2024) introduces terminology that differs somewhat from FDA's. It sorts applications by *high patient risk*, meaning AI uses affecting patient safety such as dosing or treatment assignment, and by *high regulatory impact*, meaning AI uses substantially affecting regulatory decisions such

as primary endpoint analysis. The rigor of the credibility assessment should be proportional to both. Five positions follow from that framing.

1. *Transparent models preferred*: "The use of transparent models is preferred" to strengthen accountability. Black box models may be acceptable if transparent models show unsatisfactory performance, with additional documentation and monitoring requirements.
2. *Frozen models for pivotal trials*: "Prior to the database lock and subsequent unblinding…the data pre-processing pipeline and all models should be frozen and documented in a traceable manner in the statistical analysis plan."
3. *Prospective validation for high-impact uses*: "Prior to model deployment in high regulatory impact settings, such as in relation to the primary endpoint, performance should be tested with prospectively generated data (future calendar time) that is acquired in a setting or population representative of the intended context of use."
4. *No incremental learning in pivotal trials*: "Incremental learning approaches are not accepted, and any modification of the model during the trial requires a regulatory interaction."
5. *Human-in-the-loop for precision medicine*: AI-driven indication or posology recommendations are "high patient risk as well as high regulatory impact" and require "fall-back treatment strategies in cases of technical failure."

26.2.7. The EU AI Act

This is the instrument that is law. The European Union's *AI Act* (Regulation (EU) 2024/1689) (European Parliament and Council of the European Union 2024) is the first comprehensive, binding AI statute, and it applies horizontally across sectors rather than being specific to medicines. Having entered into force in August 2024, it applies in phases: prohibitions on a narrow set of unacceptable-risk practices from February 2025, obligations for general-purpose AI models from August 2025, and the core obligations for high-risk systems originally due from August 2026, with a further tranche tied to regulated products (via the Medical Device Regulation and In Vitro Diagnostic Regulation) from August 2027. The Digital Omnibus on AI, provisionally agreed in May 2026 and in force from 27 July 2026, deferred both high-risk tranches: standalone (Annex III) systems now apply from 2 December 2027 and product-embedded (Annex I) systems from 2 August 2028. The Article 50 transparency obligations kept their 2 August 2026 start, except that providers of generative AI systems already on the market before that date have until 2 December 2026 to mark their outputs, in machine-readable form, as artificially generated (European Parliament and Council of the European Union 2026).

Most AI used in trial operations (document filing, query triage, feasibility scoring) is not "high-risk" under the Act's definitions, so its obligations are light: providers and deployers must take measures to support their staff's AI literacy, and Article

50's transparency duties apply only to particular functions, such as systems that interact directly with people or generate synthetic content (European Parliament and Council of the European Union 2024, 2026). AI that functions as, or is a safety component of, a regulated medical device can fall into the high-risk tier, inheriting obligations layered on top of the device regulations. The Act sits alongside, not in place of, the EMA reflection paper and the Clinical Trials Regulation, so sponsors running EU trials should assess where each AI application falls under its classification: a system treated as "decision support" under GCP may still trigger binding obligations under the Act.

26.2.8. Practical Implications: A Decision Framework

Figure 26.2 turns all of this into a path a sponsor can walk before committing to a deployment.

26.2.9. Summary: The Regulatory Bottom Line

Five things hold throughout. Human accountability for medical decisions and regulatory compliance cannot be delegated to AI; investigators, sponsors, and MAHs retain responsibility. Rigor of validation, documentation, and oversight should scale with model risk, which combines decision consequence and model influence. In pivotal trials, EMA expects the data pipeline and every model used for confirmatory analysis to be frozen and documented in the statistical analysis plan before database lock and unblinding; non-prespecified changes made after the data are opened make the results post hoc, unsuitable as confirmatory evidence (European Medicines Agency 2024b). Models that keep operating over time, in manufacturing and pharmacovigilance, need performance monitoring, drift detection, and change management. For high-risk applications, early consultation with FDA or EMA aligns expectations before a late-stage objection becomes expensive.

26.3. AI across the Development Lifecycle

AI now touches every phase of clinical development. What follows marks where automation delivers measurable value and where human oversight remains essential.

26.3.1. AI in Protocol Design

Benchmarking studies suggest that most clinical trials fail to meet planned enrollment timelines, often because of design choices baked in before the first patient is screened (Lamberti et al. 2024). The tools in Table 26.3 attack the problem where it originates, at the design stage, following the workflow in Figure 26.3.

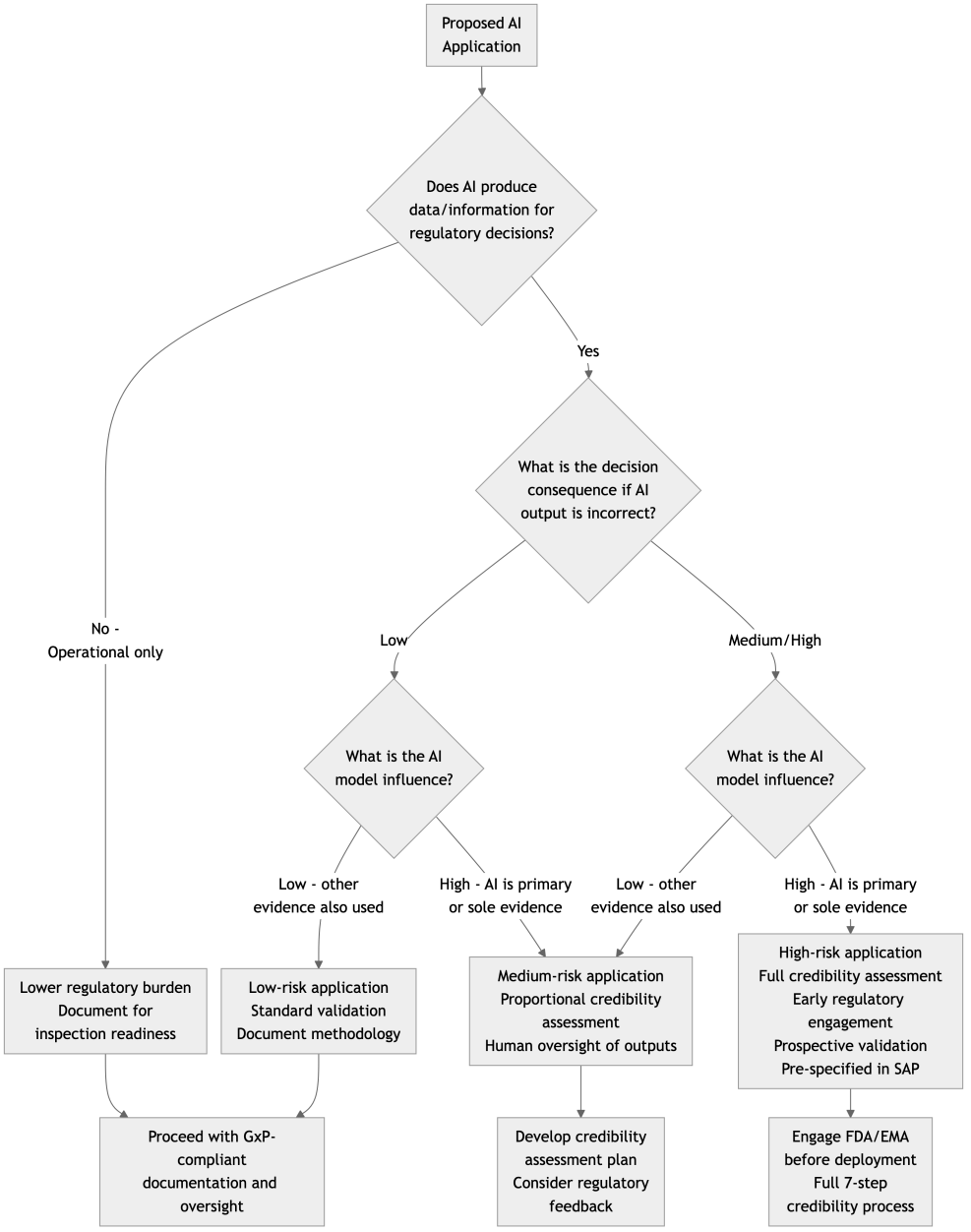


Figure 26.2.: Decision Framework for AI Deployment in Clinical Trials

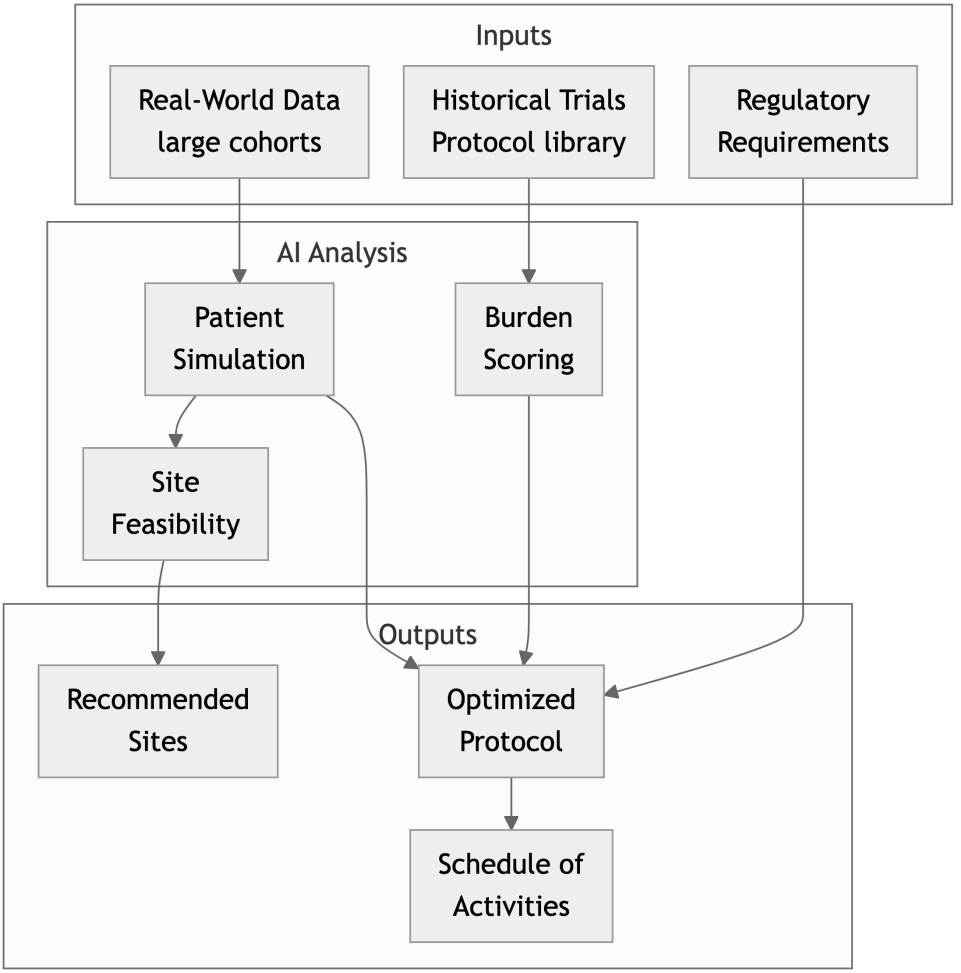


Figure 26.3.: AI-Driven Protocol Design Workflow

Table 26.3.: Protocol Design AI Tools Comparison

Tool	Data Source	Primary Use Case	Key Metric
Faro Health	Protocol library + RWD	Operational burden prediction	Complexity score
Phesi	100M+ patient profiles	Enrollment simulation	Patient availability
Medidata AI	Historical trial data	Protocol optimization	Predicted enrollment rate
Formation Bio (formerly TrialSpark)	Site network data	Site selection	Per-site enrollment probability

Faro Health exemplifies the approach. Instead of writing a static Word document, study teams design the trial in a structured cloud platform. The AI scores the complexity of the schedule of assessments against real-world data to predict operational burden, shows the team which visits pile too many procedures onto one patient, and then generates the protocol document and Schedule of Activities from the structured design (Faro Health 2024).

Phesi takes simulation further, drawing on data from over 100 million patients to model trial outcomes before the protocol is finalized. Their *Digital Patient Profile* reduces the likelihood of the "zero-enrollment site" problem mentioned in Chapter 18.

26.3.2. From Structured Protocol to Data Collection

The schedule of activities in a protocol is usually rebuilt by hand: a data manager reads it and builds the case report forms, visit structure and edit checks in the EDC system (Chapter 21). Where sites or vendors read the same schedule differently, the differences surface later as data that are hard to pool. Two developments aim at the step between protocol and database, one a data standard and one a set of AI methods, and they are at very different stages.

The standard side is further along. ICH M11 gives the protocol a harmonized structure and an electronic exchange format (Chapter 13). CDISC's *Unified Study Definitions Model* (USDM), developed with TransCelerate under the Digital Data Flow initiative, describes the content of a study definition (objectives, design, arms, visits, activities) as data. Version 4.0 was released on June 3, 2025, with a logical model, an API specification, controlled terminology, an implementation guide and conformance rules, and CDISC describes it as aligned as closely as possible with M11. USDM reuses CDASH, the collection standard, rather than replacing it. CDISC names EDC systems as downstream users of study definitions, but it states that the project will not create standard case report forms, and that how study definitions

will interact with each company's own eCRFs is still under discussion (CDISC 2025). A structured protocol can therefore supply the visits and activities an EDC build needs, but the forms themselves are still designed by people.

The AI side has less evidence than the attention it gets suggests. The strongest published study found is on extraction, not generation. Babaeipour and colleagues built a retrieval-augmented system that extracted information, including schedule-of-events content, from trial protocols. It reached 89.0% accuracy against expert-supported reference answers, compared with 62.6% for standalone language models with tuned prompts, and coordinators doing simulated tasks with the system finished them 40% faster (Babaeipour, Charest, and Wright 2026). The workflows were simulated, not live trials. We found no peer-reviewed study that measures how accurately a language model detects inconsistencies within a protocol, or that builds case report forms or edit checks from a protocol and measures the result. Claims that AI can turn a protocol into a database build should be read with that in mind.

For a multi-site trial, the practical gain from either route is the same: one definition of each data item, used at every site. The common data elements in Chapter 21 already deliver much of that without any AI. If a tool is used to draft forms or a schedule from the protocol, its output is a draft for the data manager and needs the same checks as a hand-built database.

Checklist: Reviewing a Database Build Drafted from a Protocol

- Compare every visit and every activity in the draft against the protocol's schedule of activities, cell by cell.
- Check that each primary and key secondary endpoint has the fields needed to derive it.
- Check that each eligibility criterion can be confirmed from a field or a documented source.
- Check visit windows and edit check limits against the protocol text, not against the tool's summary of it.
- Map each field to a CDASH variable or a named common data element, or record why it is custom.
- Keep the tool's output, the reviewer's changes and the approved version, so the build can be reconstructed at inspection.

Whether such a tool falls under FDA's draft credibility framework turns on whether its output affects participant safety, drug quality or study reliability (see "Scope of FDA AI Guidance" earlier in this chapter). Either way, the database it helps build is a computerized system that must be validated like any other (Section 25.3).

26.3.3. AI in Recruitment and Site Selection

Finding the right patients remains the perennial bottleneck. Benchmarking studies suggest that recruitment and retention often take substantially longer and cost more than planned, and that each day of delay can be economically material for sponsors in some therapeutic areas (Lamberti et al. 2024; Deloitte Centre for Health Solutions 2025). AI tools now scan electronic health records for "needle in the haystack" candidates and, when integrated into workflow, can reduce the manual effort of identifying potentially eligible participants. Figure 26.4 shows the typical pipeline; Table 26.4 profiles the leading vendors.

Table 26.4.: Patient Recruitment AI Vendors

Vendor	Technology	Data Assets	Best For
NextTrial.ai	NLP + ML matching	EHR integration	Complex I/E criteria
H1	KOL mapping, investigator analytics	Publications, claims, trials	Site selection, investigator finding
Deep 6 AI	Real-time EHR search	Health system partnerships	Oncology, rare disease
TriNetX	Federated network	300M+ patients (2026)	Global feasibility
Komodo Health	Healthcare map	Claims + RWD	Patient path analysis

NextTrial.ai uses natural language processing to bridge the gap between protocol criteria and patient records. The platform ingests unstructured data (clinician notes, pathology reports, and genetic profiles), flags the patients who meet complex inclusion/exclusion criteria, and predicts which investigator sites hold the highest density of them (NextTrial 2024).

H1 maps the global network of Key Opinion Leaders and investigators instead. Analyzing large volumes of data from publications, claims records, and clinical trial databases, it helps sponsors find investigators who actively treat the target patient population, not only those who are academically prominent (H1 2024).

26.3.4. AI in Clinical Operations and Data Quality

Once data starts flowing, it must be cleaned, and wearables generate 1,000+ data points per patient per day. At that volume manual query resolution stops being sustainable. The emerging approach layers several AI techniques across the data pipeline, each suited to a different kind of quality problem; Table 26.5 profiles the leading tools.

Rule-based checks remain the foundation: programmed validations that flag impossible values (negative ages, dates in the future) or logical inconsistencies (randomization before consent). They are deterministic, fast, auditable, and well-understood.

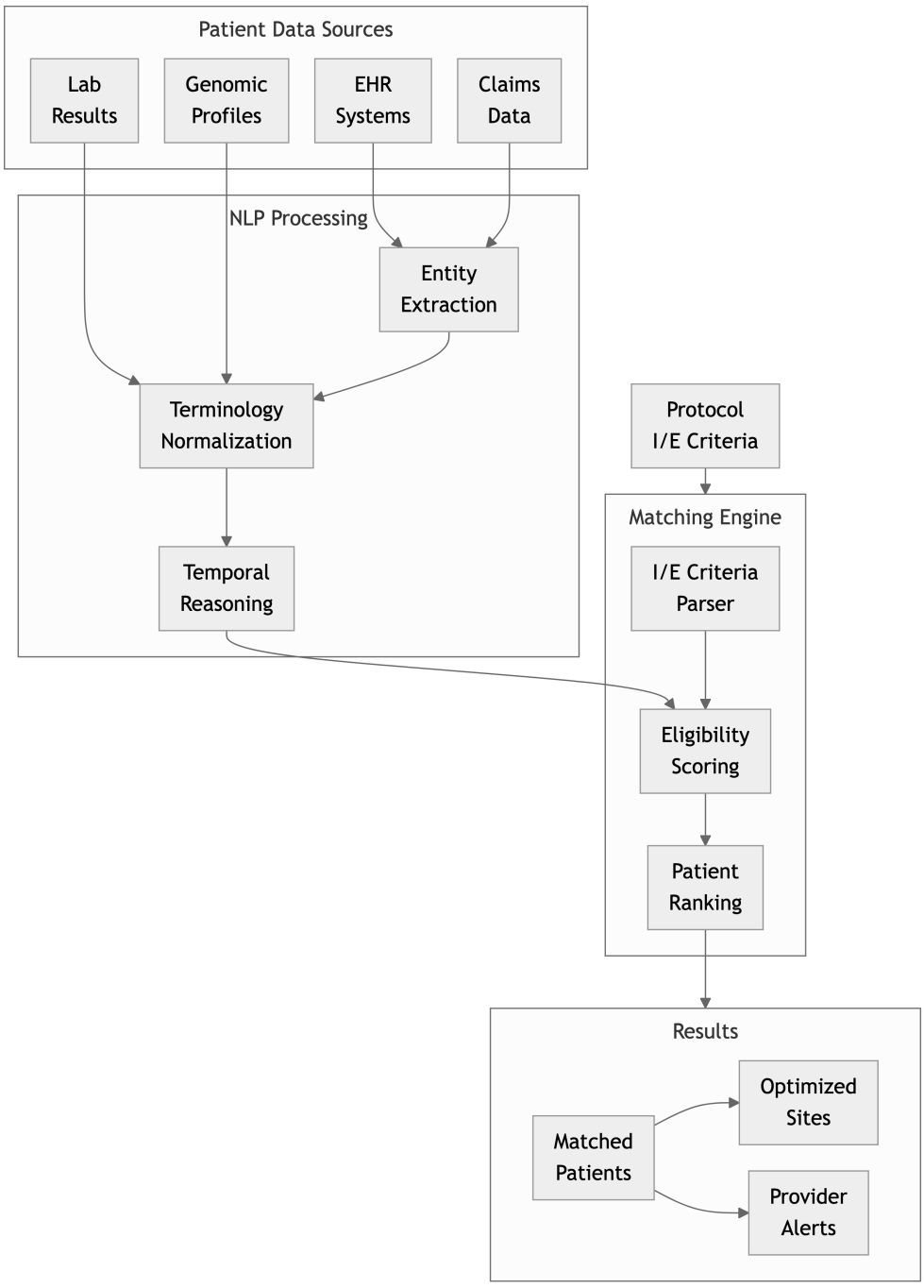


Figure 26.4.: AI-Powered Patient Matching Pipeline

Machine learning anomaly detection adds statistical pattern recognition. Models trained on historical trial data can identify unusual distributions, digit preferences suggestive of fabrication, or sites whose data patterns diverge from comparators, which means they surface problems nobody anticipated during study design.

Large language models (LLMs) parse the free-text fields, adverse event narratives, medical history, concomitant medication entries, for inconsistencies invisible to structured checks: a narrative describing "chest pain radiating to left arm" against a coded adverse event of "headache."

Cross-domain validation checks EDC entries against central lab results, device uploads against visit schedules, and ePRO responses against clinical observations; discrepancies across domains often indicate transcription errors or protocol deviations.

The output is a set of *automated actions*: auto-generated queries with draft text for site response, risk flags that prioritize monitoring attention, trend alerts that surface site-level quality patterns, and quality reports for oversight review (Figure 26.5). The goal is not to displace data management staff but to move their time off low-value triage and onto judgment-intensive resolution.

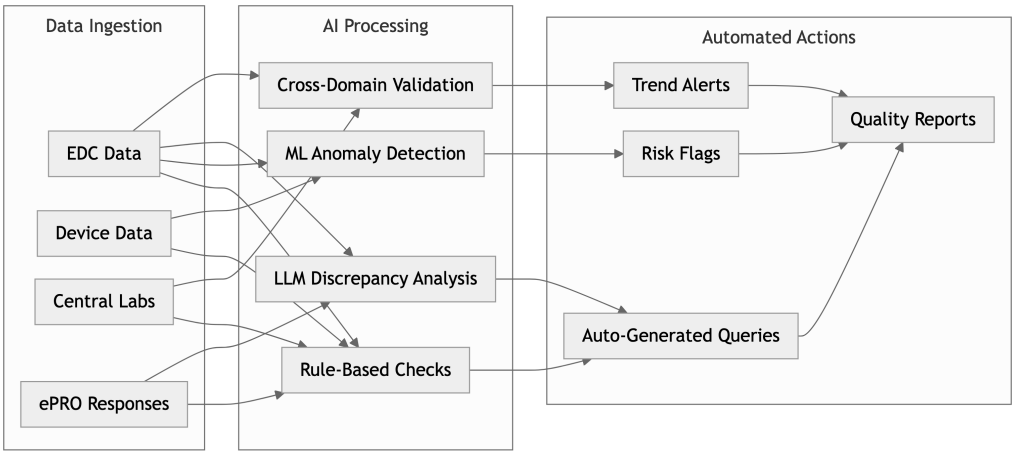


Figure 26.5.: AI-Powered Data Quality Management

Table 26.5.: Data Quality AI Tools

Tool	Primary Approach	Best Feature	Integration Depth
Saama	Clinical Command Center	Unified EDC/CTMS/eTMF view	Deep multi-system
Octozi	LLM-based discrepancy detection	Natural language queries	EDC-focused

Tool	Primary Approach	Best Feature	Integration Depth
Veeva CDB	Vault-native data management	Single-platform simplicity	Veeva ecosystem
Medidata Detect	Signal detection algorithms	Safety signal identification	Rave ecosystem

Saama provides an AI-driven Clinical Command Center, unifying data from EDC, CTMS, and eTMF into an integrated view of trial health. Its models, trained on large proprietary datasets from clinical trial operations, predict *site non-compliance* and *enrollment delays*, enabling intervention before problems escalate (Saama Technologies 2024).

Octozi applies Large Language Models to data review, scanning for inconsistencies such as "male patient listed as pregnant" or "adverse event date before informed consent." By surfacing these issues immediately, Octozi narrows the line-by-line manual review that traditionally consumed data management resources, leaving reviewers to adjudicate the flagged records rather than scan every listing (Octozi 2024).

26.4. Emerging Methodologies and Operational Solutions

Innovation in clinical trials is not limited to software; it includes new methodological frameworks that use real-world data, federated learning, and digital simulations to rethink the evidence-generation process.

26.4.1. Solving Logistical Challenges with IT

Modern trials face significant logistical friction (see Section 20.8.1). The challenges are structural: supply chains that must deliver temperature-sensitive products to thousands of endpoints, sites overwhelmed by redundant data entry across disconnected systems, protocols too complex for manual compliance tracking, and global operations fragmented across time zones and organizations. These are coordination problems, and coordination problems respond to better information infrastructure, as Table 26.6 illustrates.

Supply chain complexity has intensified with decentralized trials and direct-to-patient distribution. Traditional approaches relied on spreadsheets and email to track inventory and shipments; delays were discovered after the fact. Modern systems integrate IoT sensors that monitor temperature, location, and chain of custody in real time. AI-based demand forecasting uses enrollment trajectories and visit schedules to predict resupply needs before stockouts occur. The result is reduced waste (fewer expired products) and fewer patient visits disrupted by supply failures.

Site burden accumulates when coordinators must enter the same data into multiple systems: pulling information from the electronic health record, transcribing it to the EDC, and reconciling discrepancies later. EHR-to-EDC integration automates the flow of structured data (lab values, vital signs, demographics) from the source system to the trial database, reducing transcription errors and freeing coordinator time for patient-facing work. This integration requires careful validation and mapping, but when implemented well, it addresses one of the most persistent complaints from clinical sites.

Protocol deviations often result from complexity: too many visits, too many procedures, too many eligibility criteria for staff to track manually. Real-time nudge systems monitor visit windows and upcoming assessments, alerting coordinators before a deviation occurs rather than flagging it retrospectively. Detection systems analyze patterns across sites to identify systematic compliance failures that may indicate training gaps or protocol design problems.

Global operations compound all of these challenges. When a trial spans 25 countries, 80 sites, and multiple CRO partners, fragmented regional teams operating from different data sources create reconciliation overhead and conflicting reports. Unified command centers aggregate operational data into a single platform, providing consistent metrics, standardized escalation pathways, and a shared view of trial status across all stakeholders.

Table 26.6.: IT Solutions for Logistical Challenges (reported impacts vary by implementation)

Challenge	Traditional Approach	Modern IT Solution	Reported Impact
Supply Chain	Spreadsheets, email	IoT sensors, AI forecasting	Reduced waste, fewer stockouts
Site Burden	Redundant data entry	EHR-to-EDC integration	Less transcription time
Protocol Deviations	Retrospective monitoring	Real-time nudge engines	Fewer major deviations
Global Operations	Fragmented regional teams	Unified command centers	Single source of truth

26.4.2. Novel Methodological Approaches

This section covers a set of approaches that challenge traditional assumptions about how evidence is generated and compared: using observational data to support decisions (RWE), using historical or synthetic comparators to reduce concurrent control enrollment, and using models to predict counterfactual outcomes. All remain subjects of active regulatory and scientific debate.

Clinical research data are fragmented across institutions, each with privacy regulations, competitive interests, and technical barriers to sharing. *Federated learning* addresses this by training models collaboratively without centralizing data: algorithms travel to data sources, train locally, and share only model updates (gradients or weights) rather than patient records. Owkin has applied federated learning to construct external control arms from real-world data distributed across hospital networks. Their 2025 publication in *Nature Communications* described the method, FedECA, and deployed it across cancer centers in France, Spain, and the United States without pooling patient data; the authors note that the shared updates could in theory still be exploited to recover patient information, and they tested differential privacy as a safeguard (Ogier du Terrail et al. 2025). The practical value is clearest in rare diseases and oncology, where recruiting concurrent controls may be infeasible or ethically questionable. However, the approach requires sophisticated infrastructure, standardized data formats across institutions, and careful attention to the comparability of patient populations.

Digital twin is borrowed from engineering, where virtual models of physical systems enable simulation and optimization (Laubenbacher, Sluka, and Glazier 2021). In clinical research the term means a computational model that predicts an individual patient's disease trajectory under control conditions, based on baseline characteristics and historical data from similar patients (Sel et al. 2025; Ross et al. 2024).

Unlearn.AI's *PROCOVA* methodology is the worked example. Models trained on historical patient data generate, for each enrolled patient, a prognostic score forecasting that patient's likely outcome on the control arm; the score then enters the primary analysis as a baseline covariate, reducing residual variance and allowing smaller control groups while preserving unbiased treatment effect estimates and Type I error control (European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP) 2022; Ross et al. 2024). The European Medicines Agency issued a favorable qualification opinion in September 2022 (European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP) 2022). FDA has not qualified the method; Unlearn reports that the agency's Center for Drug Evaluation and Research, responding to its submission, found the approach already covered by FDA's guidance on covariate adjustment in randomized trials (Fisher 2024).

What EMA qualified was not "twin replaces a randomized patient." It qualified PROCOVA as a *covariate adjustment method*, analogous to using baseline HbA1c or age as a covariate in an ANCOVA, except that the covariate is itself the forecast of a deep neural network trained on historical data. The randomized control arm is still required; what the method saves is sample size, not concurrent randomized controls.

A team evaluating PROCOVA therefore has one question to answer. Standard covariate adjustment using well-chosen baseline predictors (baseline disease score, amyloid level, MMSE) can achieve much of the same variance reduction without a machine learning model, so the value of PROCOVA turns on whether the deep

neural network predicts the 18-month outcome better than those covariates do. Indication by indication, that is not settled.

> **!** AI in regulated trials: proprietary models require transparency
>
> If the prognostic score driving the analysis comes from a proprietary black-box model, FDA and EMA cannot independently verify the model's behavior, and sponsors cannot audit whether the prediction is stable across the training data and population. Any AI or machine learning model used as part of a primary statistical analysis in a regulated trial should be written down in enough detail that a third party can reprogram it from the specification and improve it as new data emerge. A score delivered without the model behind it is a governance risk even when regulators have accepted the method class, because acceptance of a method class is not acceptance of any specific model implementation.

26.4.3. Real-World Evidence and Regulatory Considerations

The randomized controlled trial remains the gold standard for establishing causal treatment effects, but *real-world evidence* (RWE) is playing an expanding role in drug development and regulatory decision-making. *Real-world data* (RWD) is data on patient health and health care delivery collected routinely rather than for a traditional trial: electronic health records, claims databases, disease registries, wearable devices, and patient-generated data. Analyzed appropriately, in observational studies or within randomized designs such as pragmatic trials, it yields evidence about drug safety, effectiveness, and utilization patterns, and the distinction matters: RWD is the raw data, RWE the clinical evidence derived from it through rigorous analysis (U.S. Food and Drug Administration 2018a).

For *safety*, RWE has long been used in post-marketing surveillance, detecting rare adverse events that trials were not powered to identify. For *effectiveness*, FDA acceptance is more cautious but expanding: the 21st Century Cures Act directed the agency to evaluate RWE for approving new indications for existing drugs and for satisfying post-marketing study requirements, and FDA's 2023 final guidance on RWE for regulatory decisions emphasized that data relevance, reliability, and analytic rigor determine acceptability, not the mere existence of a large dataset (U.S. Food and Drug Administration 2023f).

Single-arm trials with external controls compare treated patients against matched historical or concurrent observational cohorts; they are most defensible in diseases with well-characterized natural history and no effective standard of care. *Hybrid designs* randomize a reduced control arm and borrow information from external data to increase statistical precision. *Post-marketing effectiveness studies* use RWD to ask whether efficacy demonstrated under controlled conditions survives contact with real-world populations carrying the comorbidities and concomitant medications that pivotal trials excluded.

The core challenge is confounding: in observational data, treatment selection is not random. Patients who receive a therapy may differ systematically from those who do not, and these differences, rather than the treatment itself, may explain observed outcomes. Propensity score methods, instrumental variables, and target trial emulation frameworks attempt to address confounding, but none can fully substitute for randomization. Unmeasured confounders remain a fundamental limitation.

Nonrandomized RWE is most credible when the outcome is objective and reliably captured in routine care, the comparison is against natural history rather than an active comparator, the patient population in the RWD source is demonstrably similar to the trial population, and sensitivity analyses show robustness to plausible unmeasured confounding. Even then, regulators typically treat such evidence as supportive rather than dispositive for efficacy claims: it strengthens a submission that already contains randomized evidence.

FDA's 2023 draft guidance on externally controlled trials sets out how to build an external control arm from patient-level data from other clinical trials or from real-world data, addressing data quality, comparability, and bias mitigation (U.S. Food and Drug Administration 2023e). It emphasizes that external controls are most appropriate when concurrent randomization is infeasible or unethical, and when the disease has a well-characterized natural history with reliable outcome measurement in routine care.

In 2020, FDA accepted a hybrid *synthetic control arm* (SCA) design for Medicenna Therapeutics' planned Phase III trial in recurrent glioblastoma, a precedent-setting acceptance of a hybrid external control in a registrational trial for an indication that had previously relied on conventional randomized controls. In that design, matched external controls make up two-thirds of the pooled control arm, cutting prospective control enrollment by approximately two-thirds relative to conventional 1:1 randomization (Medicenna Therapeutics Corp. 2020; Medidata Solutions 2020).

An SCA constructs the external comparator from patient-level data in historical trials rather than recruiting new control patients. Medidata's platform draws on a database spanning over 38,000 trials and 12 million patients to statistically match historical controls to current trial populations. The approach is most applicable in rare or life-threatening diseases with inadequate standard-of-care, where historical control data are plentiful and disease progression is well-characterized; hybrid designs, which add historical controls to a smaller concurrent control arm, are one way to extend it to later-stage trials.

Legitimate skepticism attaches to all of this, and it is specific. *Verification, validation, and uncertainty quantification (VVUQ)* remain incompletely standardized. A 2025 review in *npj Digital Medicine* called VVUQ frameworks essential for bringing digital twins into clinical practice, argued that VVUQ processes should be standardized as far as possible for quality assurance and risk management, and noted that continually updated digital twins pose a regulatory challenge that FDA's existing medical-device framework does not fully address (Sel et al. 2025). Uncertainty is

the piece most often underemphasized: the same review observes that uncertainty quantification is often lacking in modeling work, even though it is what attaches confidence bounds to a model's predictions (Sel et al. 2025). A single predicted outcome per patient is therefore inadequate; a digital twin used as a counterfactual should report a distribution of plausible outcomes, not a point estimate.

Model opacity carries a cost that is easy to overlook. PROCOVA is a special case of classical ANCOVA: the adjustment covariate is a prognostic score from a model trained on independent historical data, in Unlearn's implementation a proprietary machine-learning model (Ross et al. 2024; European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP) 2022). EMA qualified the adjustment procedure but not any procedure for building that model, which leaves each sponsor to develop and validate the prognostic model for its own trial, and it expects sponsors to justify the advantage PROCOVA can attain over ANCOVA adjusting for individual covariates (European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP) 2022). When model details are withheld, sponsors and regulators cannot interrogate, replicate, or improve the methodology, and EMA's reflection paper on artificial intelligence in the medicinal product lifecycle expects models that influence regulatory decision-making to be fully documented, traceable, and open to independent assessment (European Medicines Agency 2024b).

Data quality and completeness constrain all approaches. In oncology, the data needed to model tumor dynamics are often noisy, incomplete, and subject to collection burden (Venkatesh, Raza, and Kvedar 2022), and statistical models layer their own uncertainty on top, particularly when predictions must generalize across populations and disease subtypes.

Comparability assumptions underpin every external control approach. If the historical population differs systematically from the current trial population (because standard of care, patient selection, or measurement practices have changed), treatment effect estimates may be biased regardless of the sophistication of the matching algorithm.

In a *JAMA* Special Communication published online in October 2024 (January 2025 issue), then-FDA Commissioner Robert Califf and senior FDA officials listed digital twins that generate simulated clinical records to predict placebo-group response among the current and potential uses of AI in clinical research, alongside matching individuals to trials and harmonizing data cleaning, and called for flexible, risk-based regulatory schemes (Warraich, Tazbaz, and Califf 2025). FDA's January 2025 draft guidance on AI in drug development makes that risk basis concrete: the oversight and credibility evidence expected for a model scale with its influence on a regulatory decision and the consequence of an incorrect decision, and operational uses that do not affect patient safety, drug quality, or the reliability of study results fall outside its scope (U.S. Food and Drug Administration 2025f).

Opaque, proprietary methods may therefore face higher evidentiary bars than interpretable, well-documented ones, with sponsors expected to show statistical valid-

ity, model transparency, reproducibility, and appropriate characterization of uncertainty. Table 26.7 summarizes where each approach currently stands.

Table 26.7.: Emerging Clinical Trial Technologies: Status and Limitations

Approach	How It Works	Regulatory Status	Key Limitation
Federated Learning	Models train locally; only updates shared	FDA/EMA engagement; no formal qualification	Infrastructure complexity
Digital Twins (PROCOVA)	Prognostic scores as covariates	EMA qualified (2022) (European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP) 2022); FDA: covered by existing guidance, per Unlearn (2024) (Fisher 2024)	Model opacity; uncertainty
Synthetic Control Arms	Historical data matched to trial	FDA accepted hybrid Phase III design (2020)	Comparability assumptions

26.5. Future Impact and Economics

The technologies in this chapter bear on the cost drivers of clinical development: shorter cycle times, smaller samples, and less coordination labor, set against new operational complexity.

26.5.1. AI Impact on Trial Operations

Recall the economics from Chapter 6: *$2.7 billion* to bring a drug to market (Deloitte's 2025 cohort), *$40,000 per day* to operate a Phase II/III trial, and, on average, roughly *$800,000 per day* in opportunity cost from delays, rising to $1.3-1.4 million per day in the most affected therapeutic areas (Z. P. Smith, DiMasi, and Getz 2024; Z. Smith, DiMasi, and Getz 2024; Deloitte Centre for Health Solutions 2026). AI-enabled tools are being applied to each of these cost drivers (see Table 26.8), though the realized impact depends on implementation quality, governance, and the specific operational context.

Table 26.8.: Potential AI Impact on Clinical Trial Cost Structure (industry projections; actual results vary)

Current Cost Structure	AI Interventions	Impact Estimates
≈ $2-3B to market (est.)	Protocol Optimization AI	Reduced Development Cost
High attrition (≈ 90%)	Digital Twins & Synthetic Arms	Smaller Trials, Similar Power
High daily burn (≈ $40K/day)	Autonomous Agents	Lower coordination overhead
CRO gross margins (≈ 18-33% of revenue)	Direct Automation	Margin Pressure
Time to first patient (≈ 166d avg)	Parallel Processing AI	Faster Timelines

Cost to market decomposes into a small set of dominant drivers: late-stage attrition, enrollment delays, protocol amendments, monitoring intensity, and the long tail of data cleaning. Automation cannot eliminate scientific uncertainty; the question is whether it can shift those operational drivers enough to change the capitalized cost of development in a meaningful way. Table 26.9 sets the main levers discussed in this chapter against the drivers they target.

Table 26.9.: Potential AI Impact on Major Trial Cost Drivers (estimates based on industry reports; actual results depend on implementation) (Lamberti et al. 2024; Getz et al. 2016; Anagnostopoulos et al. 2023; Sertkaya et al. 2016)

Cost Driver	Current State	AI Approach	Potential Impact
Failed Programs	~90% of drugs do not reach market	Protocol simulation, patient selection AI	Earlier go/no-go decisions; potentially fewer late-stage failures
Enrollment Delays	Most trials miss enrollment targets	NLP-based patient matching, predictive site selection	Faster enrollment (magnitude varies by indication)
Data Cleaning	Manual query resolution at $50-100/query	ML-assisted triage and draft resolutions	Reduced query burden
Monitoring Costs	Site monitoring ≈ 9-14% of itemized per-study costs	Risk-based + AI-detected anomalies	Reduced on-site visit frequency

Cost Driver	Current State	AI Approach	Potential Impact
Protocol Amendments	$141K (Phase II) to $535K (Phase III) median direct cost per substantial amendment	Design simulation before finalization	Potentially fewer amendments

The largest listed CROs reported fiscal 2025 gross margins of roughly 18-33% of revenue (revenue running about 23-50% above direct costs) and operating margins ranging from a loss to about 21% (see Chapter 6). AI-enabled tools are increasingly capable of performing tasks that have traditionally justified those margins: document processing, query management, routine monitoring, and medical coding. As sponsors develop their own AI capabilities, they may demand either lower CRO fees or differentiated services that require domain expertise AI cannot replicate. The speed and extent of this shift remain uncertain.

The "daily burn" of operating a trial is largely a labor-and-coordination cost: monitors reviewing data, coordinators entering and reconciling information, medical writers preparing narrative sections, and project managers coordinating activities across teams and vendors. Some of that work is amenable to partial automation, such as drafting or triage; the rest is inherently judgment- and accountability-driven. Table 26.10 decomposes the spend by role and task category.

Table 26.10.: Clinical Operations Roles and Illustrative AI Automation Potential (the percentages and timelines are the authors' illustrative judgments, not measured benchmarks)

Role	Current Cost Contribution	AI Automation Potential	Timeline
Data Manager	Query generation, cleaning	80% automatable	Now - 2027
Clinical Monitor	Source verification, oversight	50% automatable (remote SDV)	2025 - 2028
Medical Writer	Narratives, CSR sections	60% draft automation	Now - 2026
Project Manager	Status tracking, reporting	40% automatable	2026 - 2029
Regulatory Affairs	Submission compilation	70% automatable	2025 - 2027

Where automation reduces repetitive coordination work and shortens the reconciliation tail, the operating cost per day could decline, by a margin that depends on protocol complexity, data sources, and the quality of implementation and oversight.

Delays rarely arise from a single missing document or a single late shipment. They arise from cascades: a contract delay postpones site activation, which shifts the enrollment curve, which changes the drug demand forecast, which raises the risk of stockouts or wastage. That is what makes the logistics challenges described in Chapter 20 (multi-country execution, dozens of sites, long startup timelines, and tightly constrained supply chains) coordination problems in a precise sense: progress depends on many interdependent tasks distributed across organizations, time zones, and systems of record. Oversight imposes its own constraints, since actions must be traceable, exceptions must be reviewable, and accountability cannot be delegated to an opaque process.

In that environment, "agentic" automation is most defensible when it behaves like structured orchestration rather than free-form autonomy (Figure 26.6). Its practical contributions are to maintain state across workflows, to triage and route exceptions, to generate standardized artifacts (draft correspondence, document metadata), and to propose next actions that a responsible owner can approve. Used this way, orchestration turns scattered operational signals into a prioritized work queue with clear provenance, while high-stakes decisions (changes that affect participant safety, protocol interpretation, or regulatory commitments) stay under explicit human control.

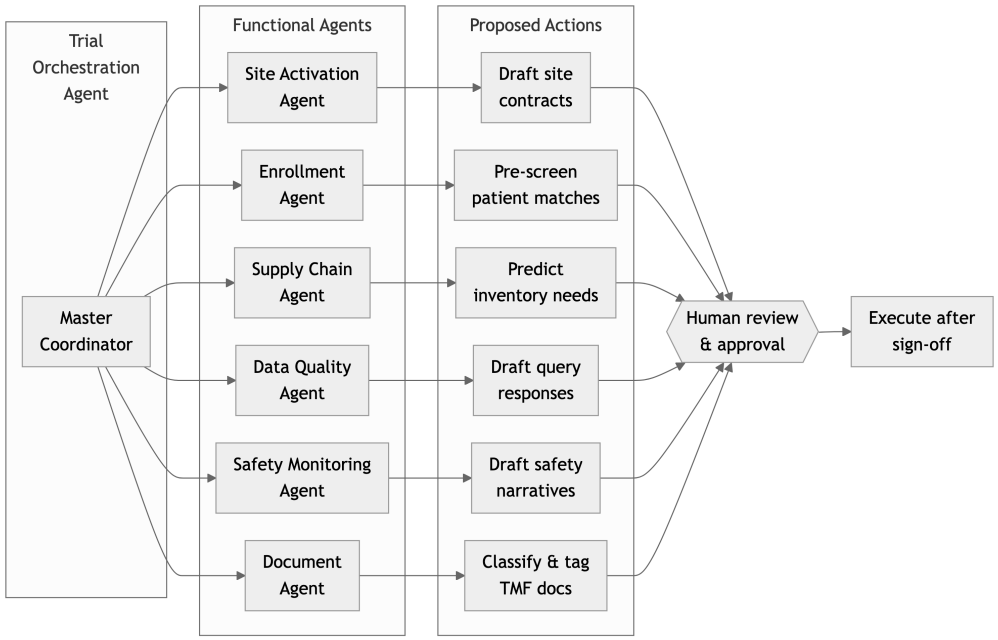


Figure 26.6.: Structured Agent Orchestration in Clinical Trials (Human-in-the-Loop)

Study start-up (from final protocol to first patient first visit) averaged 166 days for Phase III protocols in the Tufts CSDD benchmark (Getz, Smith, and Kravet 2023).

The delay is distributed across feasibility assessment, contract negotiation, IRB/EC submission, site training, and investigational product shipment. Each step has its own bottleneck, and the steps are typically processed one after another.

AI-enabled tools attack each bottleneck. Predictive site scoring trained on historical enrollment data can replace the manual feasibility questionnaire, contract generation from standardized templates can shorten legal negotiation cycles, and auto-populated submission packages can accelerate IRB preparation. Adaptive e-learning modules can replace in-person training for routine content, and predictive supply chain systems can coordinate IP shipment with site readiness.

The bigger shift is from sequential to parallel processing. Software can execute *some* activation workstreams simultaneously (assembling submission-ready document sets, preparing templated contract packets) while review, signature, and site-specific customization still run in series. Parallelization can compress elapsed time, but the magnitude depends on integration quality, local ethics processes, contracting norms, and the extent to which sites and sponsors adopt common platforms.

Decentralized trials (DCTs) create a logistics challenge of much greater scale (see Table 26.11): instead of shipping investigational product to a limited number of sites, sponsors may ship to *thousands of patient homes* and coordinate services across a wider set of vendors and geographies (U.S. Food and Drug Administration 2024n). Routine coordination and exception handling at that volume can overwhelm operational teams, which is what pushes organizations toward greater automation and analytics, with the same division of labor: standardized, traceable workflows for the routine components, accountable human review for anything safety- or compliance-critical.

Table 26.11.: AI Solutions for Decentralized Trial Logistics

DCT Challenge	Traditional Approach	AI-Enabled Approach
Cold-Chain Monitoring	Periodic temperature logs	Real-time IoT + predictive intervention
Home Nursing Coordination	Manual scheduling	AI-assisted routing and scheduling
Patient Adherence	Site calls, diaries	Wearable + app AI detecting non-adherence patterns
Document Collection	Email reminders, faxes	TMF agents that chase, collect, and file automatically
Regulatory Compliance	Per-country manual review	Multi-jurisdiction compliance AI

26.5.2. Economics and Business Models

Sponsors face a strategic choice: build AI capabilities internally, buy them from vendors, or partner with AI-native CROs. Table 26.12 sets out what each path costs, what it risks, and which sponsors it suits.

Table 26.12.: AI Adoption Strategies and Economic Trade-offs (illustrative author estimates; actual budgets vary widely by sponsor size and scope)

Strategy	Upfront Investment	Ongoing Cost	Risk	Best For
Build Internal AI	\$10-50M+	High (talent, compute)	Technology obsolescence	Top 20 pharma
Buy Platform AI	\$1-5M licensing	Medium (per-trial fees)	Vendor lock-in	Mid-size sponsors
Partner with AI-Native CRO	Minimal	Higher per-trial cost	Dependency on partner	Biotech, small sponsors
Hybrid Model	\$5-20M	Medium	Complexity	Most large sponsors

26.5.3. The 2030 Trial: A Vision

By 2030, a Phase III trial *could* look different in organizations that successfully integrate automation into validated workflows. Decision-making does not become "fully autonomous." What changes is that a larger fraction of routine coordination work becomes standardized, traceable, and partially automated: documents assembled from structured sources instead of rewritten, operational signals reconciled across systems of record rather than re-keyed into decks, exceptions surfaced with clear provenance instead of discovered late through periodic reconciliation.

That operating model needs three things at once: data capture and integration mature enough that key events (screen failures, visit-window risk, temperature excursions, missing essential documents) arrive as machine-readable signals; validation, access controls, and audit trails good enough that automated steps stay inspectable and accountable under GCP and 21 CFR Part 11 expectations; and sites and vendors that actually adopt the workflow, so that automation reduces burden instead of shifting it into more portals and alerts. The scenario in Figure 26.7 and the metrics in Table 26.13 are aspirational on purpose: they show where cycle time and coordination cost might fall, not a single inevitable trajectory.

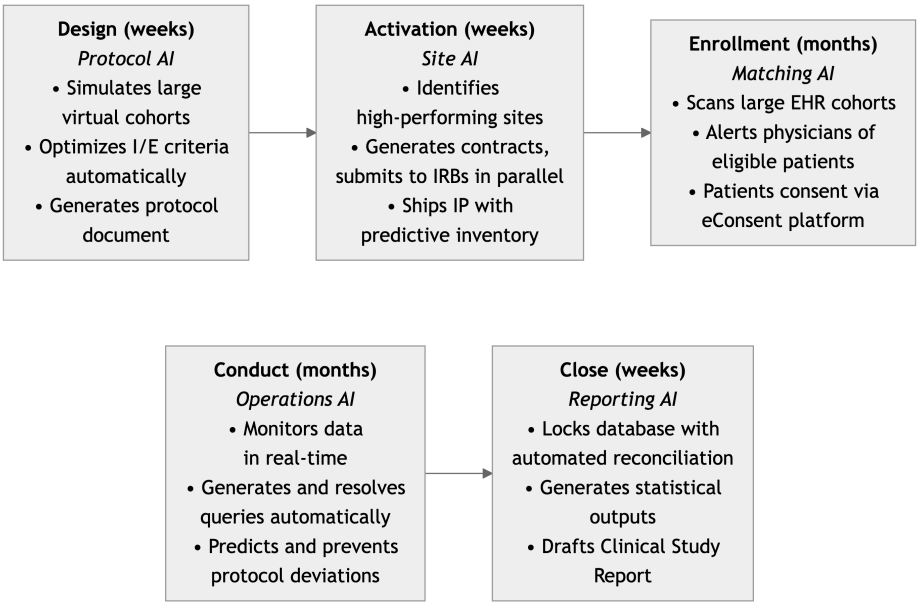


Figure 26.7.: The AI-Native Clinical Trial of 2030

Table 26.13.: Scenario trial metrics in an AI-native operating model (illustrative; not a forecast)

Metric	2025 Benchmark	2030 Projection	Change
Time to First Patient	166 days	60 days	-64%
Enrollment Duration	500+ days	200 days	-60%
Daily Operational Cost	$40,000	$18,000	-55%
Total Phase III Cost	$100M	$45M	-55%
Human FTEs per Trial	50-100	15-30	-70%
Data Quality Issues	15-20% query rate	3-5% query rate	-75%

26.5.4. Industry and Workforce Implications

The same efficiency gain is a saving for one party and a revenue loss for another. Costs, responsibilities, and bargaining power are unevenly distributed across the clinical trial enterprise: organizations bearing the largest absolute operational costs capture most of the direct savings, while organizations earning revenue from the labor-intensive activities that automation targets find the same improvement economically disruptive. Regulatory exposure cuts the same way: actors responsible for

compliance, inspection readiness, and patient protection face higher requirements for validation, audit trails, and accountability, which can slow adoption and increase the fixed costs of implementation.

Where automation is deployed matters too (site workflows, sponsor oversight, CRO service delivery, or platform infrastructure), as does whether a new tool reduces burden or simply shifts it across an organizational boundary. Table 26.14 summarizes the resulting incentives.

Table 26.14.: Potential AI Transition Impact by Stakeholder (author assessment)

Stakeholder	Likely Position	Rationale
Large Sponsors	Positioned to benefit	Can afford AI investment; may capture efficiency gains if governance is effective
Biotech	Mixed	May benefit from lower trial costs, but less capital to invest in AI infrastructure
Traditional CROs	Facing pressure	Core services may be commoditized; business models may need to evolve
AI-Native CROs	Potentially advantaged	Built for automation; lower cost structure if quality can be demonstrated
Clinical Sites	Mixed	Less manual work, but also potentially less revenue per patient
Patients	Potentially benefit	May see faster access to trials, less burden, more decentralized options
Regulators	Adapting	Must develop AI validation frameworks while maintaining safety standards

The clinical research workforce (coordinators, monitors, data managers) faces significant disruption. Automation may reduce some routine tasks while work shifts toward system implementation, validation, and oversight; preserving domain expertise through that transition will take deliberate reskilling and role redesign.

The clinical trial technology ecosystem is moving in two directions at once. Core systems are consolidating around a small number of platforms that can support regulated, end-to-end execution across EDC, CTMS, eTMF, and supply, while sponsors experiment with a growing set of analytics and automation capabilities that sit "on top" of those systems and reshape day-to-day work in feasibility, monitoring, data cleaning, and documentation. That is a gradual reallocation of investment rather than a single wave of disruption: stability in the transactional backbone, faster iteration at the analytics and workflow layer (Table 26.15) (Tufts Center for the Study of Drug Development 2024; Fortune Business Insights 2024):

Table 26.15.: Clinical Trial Technology Market Projections (analyst estimates; subject to revision; the AI 2030 value is interpolated from the source's 2034 forecast at its stated CAGR) (Tufts Center for the Study of Drug Development 2024; Fortune Business Insights 2024; MarketsandMarkets 2024)

Segment	2025 Market Size	2030 Projection	CAGR	Key Trend
CTMS Platforms	\$2.4B	\$4.9B	15%	Cloud consolidation
eTMF Systems	\$1.4B	\$2.5B	13%	AI-powered automation
AI in Clinical Trials	\$3.8B	\$20B	39%	Rapid adoption
EDC Systems	\$1.2B	\$2.1B	12%	EHR integration

The IT stack has evolved from a passive repository to an active intelligence layer. Sponsors who integrate design, recruitment, and operations data in near real time can often detect operational risks earlier, and they may be able to run trials faster and at lower operational cost than fragmented legacy workflows allow, by a margin set by disease area, protocol complexity, and the quality of implementation and governance.

Three questions cut through most of what gets called "innovation" in this space. *What decision does it change*, and what evidence shows that the change improves patient safety, data integrity, or development efficiency in the target context? *What are the failure modes*, and how are they detected, documented, and corrected under GCP and inspection expectations? *How does it integrate into real workflows*, including incentives across sponsors, CROs, and sites, so that it reduces burden instead of shifting it? Tools that answer those questions well are more likely to be adopted, defended, and sustained in regulated clinical research.

Part IV.

Part IV: Trial Completion and Beyond

27. Trial Closeout and Analysis

A brilliantly designed and flawlessly executed trial can still fail if the closeout is mishandled or the analysis is flawed. Closeout converges on one moment, *database lock*: the formal point at which the trial database is frozen and, barring a documented and exceptional unlock, no further changes are made. Before it, the data can still be corrected. After it, the trial is whatever the locked database says it is.

Four streams run at once, as Figure 27.1 shows. Data finalization runs from the Last Patient Last Visit (LPLV) through query resolution and data review to database lock; unblinding then triggers the statistical analysis and the Clinical Study Report (CSR). Sites reconcile drug supplies and archive records in parallel, and on a pivotal program the regulatory team prepares the NDA or BLA submission.

Getting to a clean database takes a final push. Outstanding queries are resolved, sites are contacted for missing data or clarification of discrepancies, medical coding of adverse events is completed and reviewed, and protocol deviations are adjudicated. Anything feeding the primary analysis is scrutinized for accuracy.

Before lock, the *statistical analysis plan (SAP)* must be finalized. Usually developed early and refined as specifics become clearer, it specifies exactly how the data will be analyzed: the analysis populations, the statistical methods, the handling of missing data, and the sensitivity analyses. The sequencing is essential to the integrity of the analysis. If analysts could try different approaches until they found one that produced favorable results, the statistical properties of the tests would be meaningless, so the SAP is closed before anyone sees unblinded data.

27.1. Unblinding

With the database locked and the SAP finalized, the trial *unblinds*, though rarely to everyone at once. A small group, often just the biostatistician running the analysis and a designated medical officer, sees the treatment assignments first, and results flow outward in a controlled manner so that nothing leaks before the analysis is complete and the findings are properly interpreted.

The analysis then proceeds according to the SAP. Programs already tested on blinded or simulated data are run on the real thing: primary endpoints first, then secondary endpoints, subgroup analyses, and safety summaries. The analysis feeds the *clinical study report (CSR)*, which documents the design, conduct, and results and stands as both the definitive record of the trial and a primary component of the regulatory submission. Chapter 28 covers its structure.

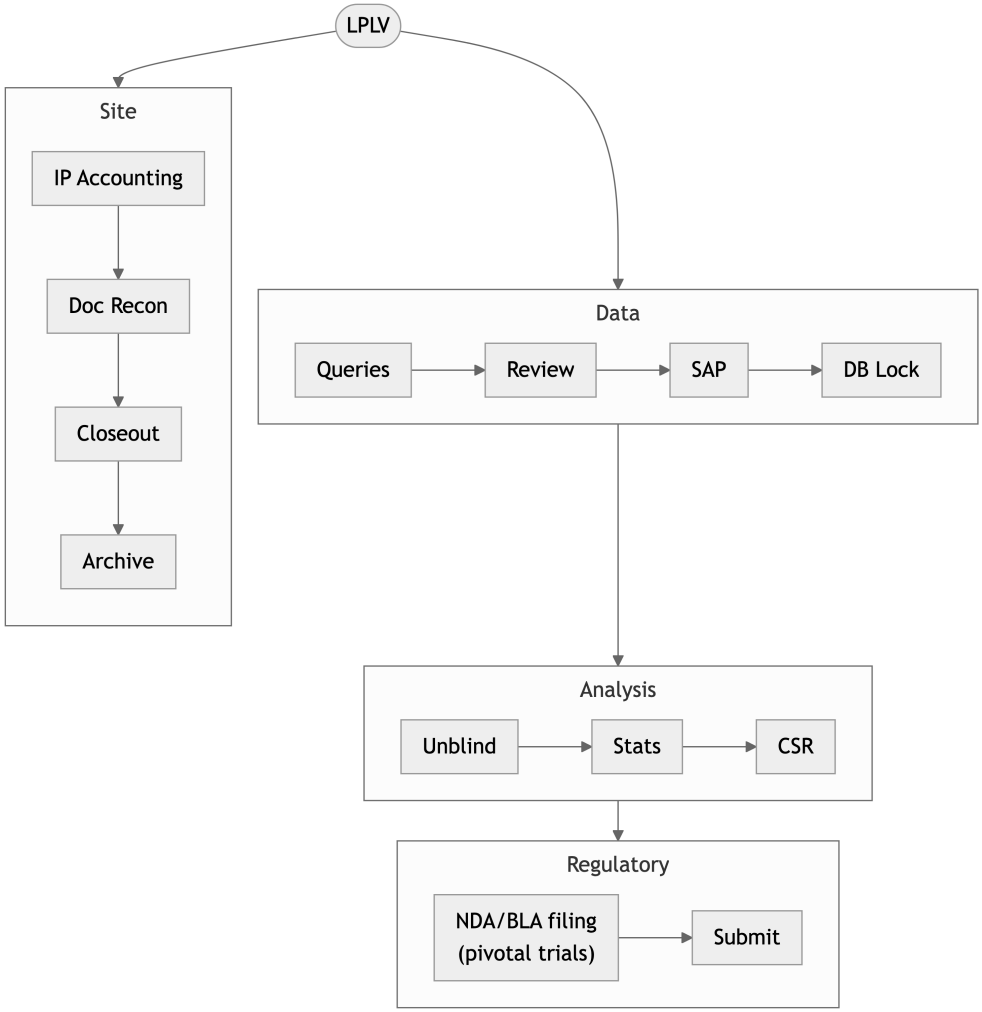


Figure 27.1.: Clinical trial closeout process from last patient visit through archiving

27.2. Interpreting the Results

The analysis produces numbers: p-values, confidence intervals, hazard ratios, differences in means. Numbers are not interpretations, and understanding what the trial found means integrating statistical results with clinical judgment. A trial that meets its primary endpoint is typically counted a success, but how large is the effect? Is it clinically meaningful, or merely statistically detectable? Is it consistent across subgroups, and do the secondary endpoints support the primary finding?

A miss poses different questions. An ineffective drug, or an inadequately powered trial? A control group that performed unexpectedly well, or a treatment group that performed unexpectedly poorly? Any suggestion of benefit in subpopulations that might justify further study?

27.2.1. Results as Probability, Not Binary Verdict

A study showing $p = 0.049$ and a study showing $p = 0.051$ contain nearly identical information about the treatment's effect, yet under a binary decision rule the first is "positive" and the second is "negative." Evidence is continuous; the verdict is not. "The trial succeeded" and "the trial failed" are useful shorthand and a misleading model of what a trial actually produces.

A more accurate framing is probabilistic: after the trial, what is our probability that the drug produces a meaningful benefit, given everything we know, including the trial result, prior evidence, and biological plausibility? A "positive" trial at $p = 0.04$ in a field with roughly two decades of consecutive disease-modifying failures (as was the case in Alzheimer's disease by the mid-2010s) carries much less certainty than a "positive" trial at $p = 0.001$ in a well-characterized mechanism where Phase II was convincingly positive. The p-value is a property of the data; the interpretation is a property of the evidence base.

For closeout teams this lands on the write-up. A negative result should specify the effect size observed (not just the p-value), the confidence interval, and where that result falls relative to the minimum clinically important difference. A trial pre-specified to detect a 30% improvement but observing 15% with $p = 0.07$ is not "no evidence of benefit"; it is modest evidence of a smaller-than-assumed benefit, and that distinction bears on how the result combines with other evidence and on whether there is a next study at all.

The COVID anticoagulation multi-platform RCT makes the cost of a binary verdict concrete. Therapeutic-dose anticoagulation helped one severity group but not the other: in non-critically-ill (moderate) patients it improved organ-support-free days (adjusted odds ratio 1.27, 95% credible interval 1.03 to 1.58, with a 98.6% posterior probability of benefit), whereas in critically-ill patients it did not, with a point estimate favoring harm (adjusted odds ratio 0.83) (ATTACC Investigators et al. 2021; REMAP-CAP Investigators, ACTIV-4a Investigators, et al. 2021). Pooled across the whole population, those opposing effects could have averaged toward

a null and been reported as "no effect of therapeutic anticoagulation," a verdict that would have been wrong for moderately ill patients because the aggregation was at the wrong level of analysis. The hierarchical model that accounted for the differential severity effect preserved the reversal, and the conclusions reshaped clinical practice.

27.2.2. Case Study: Aducanumab and the Limits of Binary Verdict

Aducanumab (Biogen/Eisai) is an anti-amyloid antibody targeting the amyloid plaques that characterize Alzheimer's disease, and its program is a widely cited example of what a binary success/failure framework does to discordant evidence. Two large Phase III trials (EMERGE and ENGAGE) enrolled patients with early Alzheimer's disease and were halted at an interim analysis in March 2019 when a futility analysis suggested they were unlikely to succeed. With more data from patients already enrolled, Biogen reanalyzed: EMERGE appeared positive at the high dose (a statistically significant 22% reduction in decline on the primary global clinical composite, the Clinical Dementia Rating-Sum of Boxes), while ENGAGE did not show the same benefit (Budd Haeberlein et al. 2022).

On 6 November 2020, FDA's Peripheral and Central Nervous System Drugs Advisory Committee voted ten "no," zero "yes," and one "uncertain," largely on the grounds that EMERGE alone, with ENGAGE non-confirmatory, did not constitute reliable evidence of clinical benefit. FDA nonetheless granted accelerated approval in June 2021, based on the drug's effect on amyloid plaque reduction as a surrogate endpoint reasonably likely to predict clinical benefit.

Two trials with opposite results resist a binary verdict: one "win" and one "loss" is genuinely ambiguous evidence. Nor was the surrogate secure, since the adequacy of amyloid plaque reduction as a stand-in for cognitive benefit had been contested for years, because earlier failed anti-amyloid antibodies had reduced amyloid without improving cognition. The discordance might also reflect dosing rather than drug failure: the two trials were identical in design, but EMERGE started later, so more of its patients enrolled after the protocol amendments that let more high-dose patients reach 10 mg/kg (29% of patients in EMERGE and 22% in ENGAGE received all 14 possible 10 mg/kg doses) (Budd Haeberlein et al. 2022).

Subsequent programs in the same mechanism produced clearer evidence. Lecanemab received full FDA approval in 2023 based on a pre-specified Phase III trial showing a 27% slowing of clinical decline on the primary global cognitive and functional composite (CDR-Sum of Boxes) (Dyck et al. 2023), suggesting that the amyloid hypothesis was correct but that the evidence from aducanumab alone was not sufficient to support it.

For closeout and reporting teams the lesson is to report what was observed (effect sizes, confidence intervals, consistency across doses and subgroups) and let the verdict wait. In a field with high prior uncertainty, a discordant two-trial result may

be exactly the correct outcome of an experiment, carrying the information needed to interpret what happened once the subsequent program is in view.

27.3. Site Closeout

At the *site closeout visit*, the question a regulatory authority will eventually ask is simple: can every dose be accounted for? Investigational product is returned to the sponsor, destroyed at the site with documentation, or otherwise reconciled against dispensing records. Queries are resolved or documented as unresolvable so that the database reflects the final state of the data, and the site's trial file is reviewed for completeness: all required documents, properly organized and ready for retention.

The investigator's obligations do not end with the visit. Trial records must be retained for the period that regional regulation and the protocol require (Chapter 24 gives the periods; for a trial with EU sites the trial master file is kept for at least 25 years), the sponsor must be notified before any record destruction, and the investigator must remain available for future regulatory inspections. Participants may have questions of their own: their individual treatment assignment (particularly after a long blind), and what to do if they experience late adverse effects.

27.3.1. Closeout checklist for computerized systems and external data streams

Closeout now reaches past site binders and investigational product into a network of computerized systems and external data flows. Because trial conclusions depend on traceability from source through submission, teams should close out systems with the same discipline applied to physical records: controlled access, retention, and retrievability for inspection (U.S. Food and Drug Administration 2023a, 2024f; International Council for Harmonisation 2025b).

- *Finalize data transfers*: confirm the last scheduled transfers have occurred (central labs, imaging, ePRO, device streams, vendor databases) and reconcile expected against received counts and timestamps (U.S. Food and Drug Administration 2023h).
- *Preserve audit trails*: ensure audit trails remain available and reviewable after database lock, and document any post-lock administrative correction as a controlled deviation with rationale (U.S. Food and Drug Administration 2024f).
- *Access changes*: deprovision non-essential access after closeout, while preserving the sponsor and regulatory access needed for inspection and reconstruction (U.S. Food and Drug Administration 2023a).
- *Retention plan*: document where raw data extracts, transfer logs, validation evidence, and final locked datasets are retained, and how they can be retrieved within inspection timelines (International Council for Harmonisation 2025b; U.S. Food and Drug Administration 2024b).

- *Vendor/system handover*: for vendor-managed systems, document responsibilities for long-term hosting, export formats, and contract clauses supporting retrieval during inspection windows (U.S. Food and Drug Administration 2024f).

27.4. What Happens to the Data

Some of what happens next is not optional. Under FDAAA 801 and its Final Rule, applicable trials must post summary results to ClinicalTrials.gov no later than one year after the primary completion date, and noncompliance carries civil monetary penalties (United States Congress 2007). Data sharing is voluntary by contrast, and modern initiatives increasingly make anonymized datasets available to the external scientific community for transparency and independent scrutiny. Researchers conduct post hoc analyses to explore new hypotheses suggested by the findings, and pooled analyses across trials supply the statistical power to detect rare safety events that single studies might miss. Long-term retention of trial records (the period is set by regional rules and is at least 25 years for the trial master file of a trial with EU sites; see Chapter 24) keeps the study accessible for future regulatory inspections or scientific inquiries.

Regulatory pathways. A successful Phase III program ends in a regulatory submission: an NDA or BLA in the United States, or an MAA in Europe. Chapter 29 covers how these applications are built, submitted, and reviewed.

27.5. When Trials Fail

Trials fail in three broad ways. Efficacy failures occur when a treatment produces no meaningful clinical benefit, sending researchers back to the underlying mechanism or the trial's design. Safety failures are more consequential, occurring when unacceptable toxicity emerges either immediately or through the accumulation of data. Operational failures can derail a program through poor enrollment or through data quality that undermines the scientific validity of the findings. Even then, the data informs future decisions and prevents other researchers from repeating unsuccessful approaches.

27.5.1. Common failure modes (and why they show up late)

Many "failed" trials are a chain of small mismatches between the scientific question, the protocol, and real-world execution, and some links in the chain become obvious only near closeout, when the sponsor tries to lock the database, assemble the Trial Master File (TMF), and write a defensible clinical study report.

Some failures reflect underpowered or mis-specified effects: the true effect is smaller than expected, variability is larger than assumed, the endpoint is noisier than planned, or intercurrent events (treatment discontinuation, treatment switching, rescue medication) complicate interpretation. The result can be an apparently

negative trial that is not decision-grade for regulators or clinicians. Others arise from endpoints and assessment processes poorly aligned with mechanism, subject to measurement drift across sites, or sensitive to missing data and visit windows; here adjudication delays or inconsistent rater training may only become obvious when datasets are reconciled late.

Inclusion and exclusion criteria can yield a cohort that is too heterogeneous, diluting signal, or too narrow, slowing enrollment, or simply not representative of intended use. When protocol deviations accumulate on top of that, the realized population may drift from the intended estimand, leaving the sponsor with an analytically clean plan and a pragmatically messy dataset.

Protocol complexity often expresses itself as amendment churn. Each amendment disrupts sites, triggers retraining, and creates opportunities for version confusion; near closeout, teams sometimes discover that determining *which version applied to which participant* is itself a data-quality problem. Enrollment shortfalls build the same way: poor feasibility assumptions, competing studies, strict criteria, and screening bottlenecks can slow accrual, and slow accrual is a scientific risk as well as a timeline delay, because standard of care may change mid-study, introducing interpretability challenges.

Site performance variation amplifies all of it, since a subset of sites can drive a disproportionate share of protocol deviations, missing data, and query burden, and the damage may stay latent until the sponsor confronts the reconciliation backlog that prevents database lock. The blockers that surface then are familiar: reconciliation debt from late or incomplete entry, coding backlogs for medical history and adverse events, and inconsistencies across EDC, ePRO, central labs and safety databases; vendor and supply-chain problems (drug supply interruptions, temperature excursions, device failures, shipping problems) that create missingness that is not random; and regulatory-readiness gaps such as incomplete documentation, missing signatures, unclear audit trails, inconsistent versions or gaps in training records, which surface when someone starts preparing for inspection or submission.

27.5.2. The cost of failure and delay

Closeout is where the cost of failure becomes concrete. The direct costs are visible in the monthly burn: site payments, CRO and vendor fees, monitoring, data management, safety operations, and internal labor, and a delayed database lock or clinical study report extends every one of them past last patient last visit. The opportunity cost is substantial too. Time on a stalled program is time not spent on the next study, the next indication, or the next molecule, and delay can compress the effective patent-protected commercial window, reducing expected return even when a drug ultimately succeeds.

The scientific cost surfaces here as well. When data are incomplete or inconsistently collected, the outcome may be *inconclusive* rather than negative, forcing additional studies or post hoc salvage analyses that are difficult to interpret and harder to

defend. There is a human cost as well. Participants, investigators, and coordinators invest effort and accept risk; when a program fails late, what is lost is more than money, and the loss deepens when results are not disseminated clearly or cannot be translated into actionable knowledge.

27.5.3. How AI can reduce delays and failure risk

In most programs the useful contribution of AI is operational rather than predictive: it reduces coordination overhead and reconciliation debt (see Chapter 26 for the underlying tools and governance model). Analytics can track real-time screen-failure reasons and accrual drift, flagging problems before timelines slip irreversibly. Automation can shorten the long tail of closeout by triaging queries, reconciling data across disparate systems, checking Trial Master File readiness through signature and version checks, flagging outlier raters, and speeding safety case processing, all without adding administrative burden on site staff and with final sign-off still resting with trained professionals.

Trial "failure," then, is often the visible outcome of earlier, correctable operational and design problems, and AI can improve timelines and reduce failure rates by making those problems legible sooner and by shrinking the manual burden of reconciliation and documentation at closeout. That holds only under rigorous governance and traceability: every AI-assisted output that influences a trial must be attributable to a specific dataset, instruction, and accountable reviewer, and the system must be designed for auditability, oversight, and continuous monitoring rather than autonomous decision-making.

28. The Clinical Study Report

When a trial ends, everything it produced has to be distilled into one document: the Clinical Study Report, or CSR, which accounts for what was planned, what was done, and what was found. A regulatory reviewer reads it with a critical eye, looking for any evidence of bias, error, or inadequate methodology, and has to decide from it whether the trial was conducted properly and whether the data are reliable enough to support the conclusions drawn from them. On that judgment rests the approval of a new therapy.

Regulatory reviewers are not the only readers. Inside the sponsor company the CSR sets development strategy: a positive report may trigger investment in further development, a negative one may end the program. The medical and scientific community may eventually see it, or publications based on it, so it has to meet standards of scientific rigor and transparency. Nor does it expire. Preserved as part of the permanent record, a CSR may be examined years or decades later in response to safety questions, litigation, or regulatory inquiries.

28.1. The ICH E3 Structure

CSR structure follows international guidelines, primarily ICH E3, which establishes a standardized format for presenting clinical study results. Table 28.1 shows the major sections, with indicative lengths. Note the proportions. The conclusions run one to two pages; the appendices behind them, which carry the protocol, the statistical analysis plan, the patient data listings, and other supporting materials, can run past a thousand.

Table 28.1.: Clinical Study Report Structure after ICH E3, with Typical Lengths

Section	Content	Typical Length	Key Elements
Title Page/Synopsis	Summary of entire study	2-3 pages	Design, key results, conclusions
Introduction	Background and rationale	2-5 pages	Disease context, drug mechanism, program fit
Study Objectives	What trial was designed to determine	1-2 pages	Primary, secondary, exploratory objectives
Investigational Plan	Study design details	10-30 pages	Randomization, blinding, dosing, procedures

Section	Content	Typical Length	Key Elements
Study Patients	Who enrolled	10-20 pages	Disposition, demographics, baseline characteristics
Efficacy Results	Primary and secondary analyses	30-100 pages	Endpoint analyses, subgroups, sensitivity
Safety Results	AEs, labs, vitals	20-50 pages	AE tables, SAE narratives, lab shifts
Discussion	Interpretation of results	5-10 pages	Clinical significance, limitations, context
Conclusions	Key findings	1-2 pages	Summary statements
Appendices	Supporting materials	100-1000+ pages	Protocol, SAP, TLFs, patient listings

The opening sections orient the reader. A title page and synopsis condense the whole study into two or three pages for anyone who needs a quick overview. The introduction supplies context, and study objectives specify exactly what the trial was designed to determine, with efficacy questions usually primary and safety, pharmacokinetics, or other endpoints secondary. The investigational plan then describes the design in detail: the type of trial (randomized, controlled, blinded), the treatment groups, the dosing regimen, the duration of treatment and follow-up, and the inclusion and exclusion criteria that defined the population on paper.

The study patients section reports who actually enrolled: how many were screened, randomized, completed, and discontinued, with reasons. Demographic and baseline characteristics follow, and their job is to establish that the population the trial enrolled is the population it intended to enroll. Efficacy results then present the primary and secondary endpoint analyses together with the sensitivity and subgroup analyses that stress them; figures and tables display the data, and the text interprets it. Safety results present adverse events, serious adverse events, laboratory abnormalities, vital signs, and any other safety parameters collected, tabulated by treatment group, with notable events discussed in narrative form. The discussion interprets all of it for the investigational product, the disease, and the development program; the conclusions summarize. Everything else is appendix.

28.2. The Writing Process

A CSR takes several months and many hands. Medical writers sit at the center, synthesizing the locked database, the statistical analysis plan (SAP), and the tables, listings, and figures (TLFs) generated by the biostatistics team. Clinical operations contributes context on trial execution, site audits, and the operational challenges

that may have affected the data. Medical and scientific reviewers keep the clinical narrative coherent, and regulatory affairs confirms that the document meets submission requirements and supports the intended marketing application.

The work runs in phases, and experienced teams start it early. A CSR "shell" or outline is often built months before the trial concludes, with the introduction, study objectives, and investigational plan drafted while the trial is still active. That front-loading is what keeps the post-lock period manageable: it allows early internal review of the study's background and methodology, and leaves only the results and discussion for intensive drafting once the final data sets arrive. Results go in as statistical outputs become available, and the draft then goes through multiple rounds of review and quality control before finalization for regulatory submission. Increasingly, medical writers draft narrative sections with the help of large language model tools that generate first-draft text directly from the structured TLF and SAP inputs; medical judgment, discrepancy resolution, and final sign-off remain human-owned (Chapter 26 covers these tools and their current limits).

Not every clinical study warrants a full technical report. Under ICH E3 guidelines, abbreviated CSRs (using summarized data or with some sections deleted) may be acceptable for uncontrolled studies or other studies not designed to establish efficacy, for seriously flawed or aborted studies, or for controlled studies examining conditions clearly unrelated to the claim; a controlled safety study, however, is still reported in full (International Council for Harmonisation 1995). These criteria are keyed to a study's design and role, not to the direction of its result: a negative but well-conducted controlled pivotal trial still requires a full CSR. Where an abbreviation applies, narrative sections such as detailed medical background or extensive investigator biographies may be summarized or omitted, which preserves the trial's core results and safety data within the Trial Master File without producing a full publication-quality report.

28.3. Quality Standards

Every figure and data point in the report must match the underlying analysis datasets, because any discrepancy can erode reviewer confidence. Results sections have to align with stated objectives, terminology has to remain uniform across the document, and a missing regulatory element can trigger delays and supplemental inquiries. Transparency asks for something harder: an honest appraisal of the trial's limitations and unexpected outcomes. Any attempt to minimize safety signals or obscure unfavorable findings will likely be discovered by inspectors, and it is the sponsor's long-term credibility that pays (International Council for Harmonisation 1995).

The era of the CSR as a purely confidential document is ending. Driven by demands for scientific transparency and by ethical obligations to trial participants, regulators increasingly require public disclosure of clinical data. The EU Clinical

Trials Regulation (CTR) 536/2014 requires sponsors to provide a summary of results for laypersons (European Medicines Agency 2022): plain language suitable for the general public, typically covering the trial's rationale, who participated, what happened, and the side effects and benefits observed. Writing one well means simplifying complex medical terminology without losing scientific accuracy.

EMA Policy 0070 (European Medicines Agency 2014b) goes further, requiring proactive publication of the clinical reports themselves once the regulatory decision is made, which hands the sponsor a redaction and anonymization problem: protect participant privacy while retaining data utility. Sponsors must document a re-identification risk assessment in an anonymization report before these documents are posted to public portals; EMA prefers quantitative methods (a qualitative approach has to be justified as a deviation) and expects the residual risk to sit at or below a pre-set target threshold, commonly 0.09.

What full CSR access is worth became concrete in the Tamiflu (oseltamivir) episode. For years, public claims that the drug prevented serious flu complications rested largely on journal articles and summaries. When independent reviewers obtained the clinical study reports themselves, a systematic re-analysis found that treatment shortened symptoms in adults by about 17 hours, from seven days to 6.3 days, and did not clearly support the larger public-health benefits that had justified government stockpiling (Jefferson et al. 2014). The CSR that regulators receive is increasingly the document that independent science will eventually read, so it must stand on its own, without unpublished context that only the sponsor holds.

28.4. Statistical Communication and Bayesian Reporting

A clinical study report has to make its numbers mean something to readers who range from trained statisticians to prescribing physicians.

28.4.1. Reporting Against the Pre-Specified Estimand

Since ICH E9(R1), the efficacy section of a CSR is expected to be organized around the trial's pre-specified estimand, not around a bare treatment comparison. Each primary and key secondary estimand should be restated in full: the population, the treatment condition, the endpoint variable, the strategy for each intercurrent event, and the population-level summary measure. The report then confirms that the primary analysis estimates that estimand, and presents analyses built on alternative intercurrent-event strategies as labeled supplementary analyses, not as competing headline results: a different strategy defines a different estimand, whereas a sensitivity analysis keeps the estimand and tests the main estimator's assumptions. E9(R1) is explicit that main, sensitivity, and supplementary results should be reported systematically, so a reviewer can judge how robust the primary conclusion is to the assumptions embedded in the chosen strategy. The estimands framework

and the five intercurrent-event strategies are developed in Chapter 11 and Chapter 17; the CSR is where those design-stage choices are shown to have been honored (International Council for Harmonisation 2019).

28.4.2. From p-values to Probability Statements

The standard CSR statistical narrative reports a test statistic, a p-value, and a confidence interval. This format answers the question "Is the observed difference statistically significant?" It does not directly answer the question most relevant to regulators and clinicians: "How probable is it that this drug provides a clinically meaningful benefit?"

When a trial uses conventional frequentist methods and the p-value falls clearly below or above the threshold, the distinction between these two questions rarely matters for regulatory decisions. It matters in two cases. The first is a result near the threshold: a p-value of 0.04 in one trial and 0.06 in another carry different narrative weight despite containing similar evidence. The second is a trial using Bayesian methods, whose natural output is a probability statement, not a test statistic.

For Bayesian trials, the CSR should report the primary result in the same language the protocol and SAP specified: the posterior probability that the treatment is superior to control, the posterior probability of a benefit exceeding a clinically meaningful threshold, and the credible interval for the treatment effect estimate. The FDA review of REBYOTA (an approved fecal microbiota product) reported a 99.1% Bayesian posterior probability that the product was superior to placebo rather than a p-value, evaluating that probability against two pre-specified success thresholds of differing stringency: the result cleared the less stringent bar but not the more stringent one (US Food and Drug Administration, Center for Biologics Evaluation and Research 2022).

For frequentist trials, the effect size and its confidence interval are as important as the p-value and should appear in the synopsis, the summary of efficacy findings, and the conclusions. A result at $p = 0.03$ whose confidence interval reaches a hazard ratio of 0.99, nearly no clinical effect at one end, is meaningfully different from a result at $p = 0.003$ with an interval entirely below 0.85. Both are "significant." They do not mean the same thing clinically, which is why the CSR narrative should discuss both the lower and the upper bound of the interval.

28.4.3. Communicating Uncertainty

Statistical reviewers work through the statistical appendix, but many readers of a CSR, clinicians in particular, may read little beyond the synopsis and the conclusions. A CSR that gives them only point estimates and binary pass/fail verdicts underserves the readers whose decisions it exists to inform. Those sections should communicate the effect size in clinically interpretable terms (what a hazard ratio of 0.78 means for a patient's risk over a year), the range of plausible effects in the same

terms, and the consistency of the primary result across pre-specified subgroups and secondary endpoints. For Bayesian analyses, they should also report the sensitivity of the conclusion to the prior specification, showing that the conclusion holds across alternative priors.

The aducanumab program (discussed in Chapter 27), whose two Phase III trials reached discordant results, shows why a report must address such discordance directly. When a result is unexpected, the report should say explicitly whether it is consistent with prior evidence, what alternative interpretations exist, and what additional information would be needed to resolve the uncertainty.

28.4.4. Reporting Adaptive Trial Results

The standard CSR template was designed for fixed-sample trials. An adaptive design has to document more than that template asks for: the adaptation rules pre-specified in the protocol and SAP, which adaptations were actually triggered, and against which dates and data cuts. It also reports the final sample size against the original plan, with the reason for any deviation. Unblinded interim analysis reports usually go in as appendices, with access controlled per the charter. The report also has to show how the alpha-spending or error-control method keeps the reported p-value valid despite the adaptations.

For platform trials, a CSR covering a single arm must make clear that its primary analysis used a shared control, and describe the concurrent controls (and the non-concurrent controls, if any) that contributed to the primary comparison, with the dates of contribution and the time-adjustment method applied.

28.5. Integration with Regulatory Submissions

In the *Common Technical Document (CTD)* format used for major regulatory submissions, CSRs appear in Module 5 (Clinical Study Reports); the CTD also includes summaries and overviews that synthesize information across multiple studies (International Council for Harmonisation 2016b).

For pivotal studies, the CSR will be read closely: reviewers examine methodology, verify that the analysis followed the pre-specified plan, and assess whether the conclusions are supported by the data. A supportive study may draw a less intensive review, but its CSR must still give a reviewer enough to understand what was done and what was found. Either way the document outlives the review and remains the trial's authoritative record long after the trial concludes.

29. Regulatory Submission

Years of research, billions of dollars in investment (see Chapter 6), and thousands of patients' participation arrive at FDA as a dossier, and the agency's first decision about it is not yet about whether the drug works. For 60 days it asks only whether the application is *complete*, whether it contains enough to permit substantive review at all. One deemed incomplete may be refused to file and returned without review. What must clear that threshold is the sponsor's case: the accumulated evidence from preclinical studies and clinical trials that, they hope, demonstrates the product is safe and effective.

Different products and circumstances call for different applications. The *New Drug Application (NDA)* is the primary vehicle for small-molecule drugs in the United States, carrying complete data on chemistry, manufacturing, preclinical studies, and clinical trials; the *Biologics License Application (BLA)* follows a similar structure for biological products such as proteins, vaccines, and cellular therapies, with additional manufacturing detail specific to biologics. The European Union equivalent is the *Marketing Authorization Application (MAA)*, submitted centrally or through national procedures. Three further routes rest on work already done: the *Abbreviated New Drug Application (ANDA)* for generics, which relies on the reference product; biosimilar applications, which demonstrate similarity to an approved reference biologic; and supplements and variations, which modify an existing approval for a new indication or a manufacturing change.

Harmonization gave them all a common container: the Common Technical Document (CTD), which organizes a submission into the five modules of Table 29.1.

Table 29.1.: Common Technical Document (CTD) Structure

Module	Name	Content Description
1	Regional Administrative	Forms, labeling, and jurisdiction-specific materials
2	Summaries & Overviews	Integrated quality, nonclinical, and clinical summaries
3	Quality (CMC)	Chemistry, Manufacturing, and Controls documentation
4	Nonclinical	Reports from toxicology and pharmacology studies

Module	Name	Content Description
5	Clinical	Clinical study reports and related data materials

Modules 2 through 5 form a single "core dossier" that can go to regulatory agencies around the world simultaneously; Module 1 carries each region's own material, and the core modules have only a few regional slots, such as section 3.2.R (Regional Information) in Module 3. That is what makes it practical to bring a new therapy to several markets at once.

29.1. Building the Submission

A major application is assembled by several specialized functions at once. Regulatory affairs leads, owning the submission strategy and the direct communication with agencies that keeps every legal and technical requirement in view. Clinical development compiles and interprets the clinical data; CMC specialists document the manufacturing and quality controls that guarantee product stability; nonclinical researchers supply the foundational toxicology. Medical writers turn those separate streams into summaries clear enough for a reviewer facing thousands of pages. The finished package is transmitted securely to authorities worldwide as an electronic CTD (eCTD).

29.2. Pre-Submission Interactions

Discovering a major issue *before* submission is far better than discovering it during review. That asymmetry is why sponsors engage with regulatory agencies throughout development, and heavily in the run-up to a major filing.

Pre-IND meetings, held before first-in-human studies, clarify the data needed to support clinical development. The end-of-Phase 2 meeting is a key checkpoint: the sponsor puts its Phase III design and evidentiary standards in front of the agency for feedback. Pre-NDA/BLA meetings (Pre-MAA in Europe) come last, confirming that the dossier is complete and free of critical unresolved issues. The exchanges are formal, in writing or in person, and each type carries its own timeline and objective (Table 29.3).

29.3. Expedited Regulatory Pathways

The pathways regulatory agencies built for serious conditions are designed to accelerate development and review. They do not lower the evidentiary standard for safety or efficacy. What they change is the intensity of the interaction between sponsor and regulator, and in some cases they let a surrogate endpoint reasonably likely to predict clinical benefit carry the case.

Fast Track designation, for drugs that treat serious conditions and fill an unmet medical need, buys more frequent meetings with the FDA and the possibility of rolling review, where completed sections of the application go in before the entire dossier is finalized. Breakthrough Therapy is more intensive, reserved for drugs whose preliminary clinical evidence indicates they may demonstrate substantial improvement over available therapy on a clinically significant endpoint. Accelerated Approval rests on a surrogate or intermediate clinical endpoint reasonably likely to predict clinical benefit, and the Food and Drug Omnibus Reform Act of 2022 (FDORA) tightened it: FDA may now require that a confirmatory trial be underway before accelerated approval is granted or within a period it specifies after approval, and it can use expedited procedures to withdraw an approval when the confirmatory trial fails to verify benefit or is not conducted with due diligence (U.S. Congress 2022). Priority Review commits the FDA to complete review within six months of filing (about eight months from submission for new molecular entities and original biologics) for drugs offering significant improvements in safety or effectiveness (U.S. Food and Drug Administration 2021c).

A single drug program may hold several of these at once: a Breakthrough Therapy can also receive Accelerated Approval and Priority Review. Table 29.2 contrasts the mechanisms.

Table 29.2.: FDA Expedited Programs for Serious Conditions (U.S. Food and Drug Administration 2014)

Designation	Criteria	Key Benefit
Fast Track	Serious condition + unmet need	Rolling review, frequent communication
Breakthrough	Serious condition + substantial improvement	Intensive guidance, organizational commitment
Accelerated Approval	Serious condition + surrogate endpoint	Approval based on surrogate (requires confirmation)
Priority Review	Significant improvement	6-month review clock (vs 10 months)

Table 29.3.: FDA Meeting Types and Timelines (U.S. Food and Drug Administration 2021c)

Meeting Type	Purpose	Examples	FDA Scheduling Goal
INTERACT	Early advice, before pre-IND	Novel technology or complex manufacturing questions	75 days

Meeting Type	Purpose	Examples	FDA Scheduling Goal
Type A	Critical path issues; stalled development	Dispute resolution; clinical hold discussion	30 days
Type B	Key development milestones	Pre-IND, Pre-NDA/BLA	60 days
Type B (EOP)	End-of-phase milestones	End-of-Phase 1, End-of-Phase 2	70 days
Type C	General guidance	Protocol design review; new indication discussion	75 days
Type D	Narrow, focused issues	Follow-up on one or two tightly related topics	50 days

29.4. The Review Process

Figure 29.1 traces the path from submission to action letter.

An application that survives filing enters a review queue governed by *user fee goals*, performance targets established under the Prescription Drug User Fee Act (PDUFA). For new molecular entities and original biologics, standard review targets 10 months from the 60-day filing date, about 12 months from the original submission, and priority review targets 6 months from filing; for other original NDAs the clock starts at receipt (U.S. Food and Drug Administration 2014, 2021c). Review is multidisciplinary: medical officers evaluate clinical efficacy and safety; pharmacologists assess pharmacokinetics; chemists review manufacturing; statisticians verify the analyses; toxicologists examine nonclinical safety.

Reviewers may issue *information requests*, which require the sponsor to clarify or supplement the submission, and a sponsor that responds slowly delays its own review.

For novel or complex applications, FDA may convene an *Advisory Committee*, a panel of outside experts who discuss the application in a public meeting and vote on approval recommendations. The vote is not binding. FDA usually follows it.

29.5. Bioresearch Monitoring (BIMO) Inspections

Positive data are worthless if the agency cannot trust that they were collected accurately and ethically. In parallel with the scientific review, therefore, the FDA conducts on-site inspections through its Bioresearch Monitoring (BIMO) program, verifying data integrity and compliance with regulations.

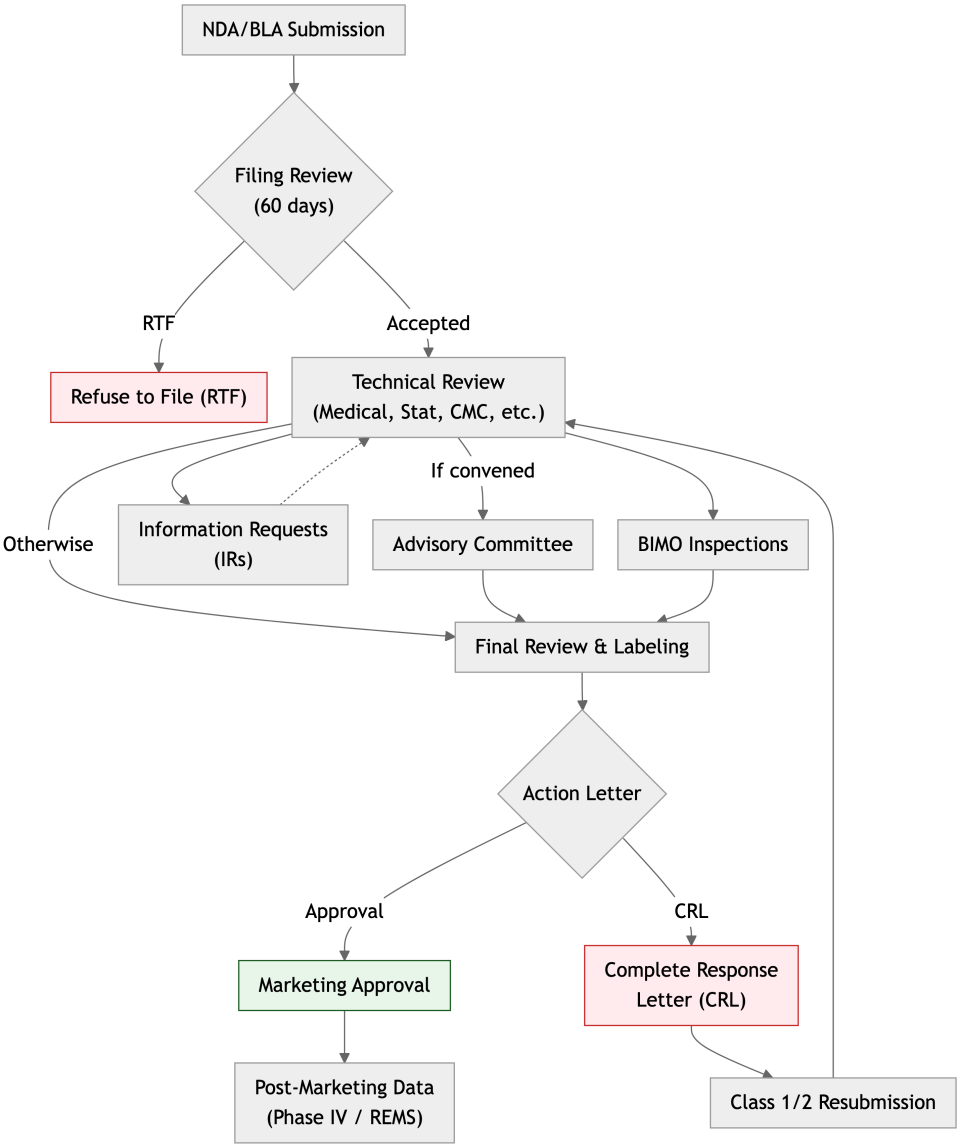


Figure 29.1.: The NDA/BLA Regulatory Review Cycle

At a clinical site, inspectors compare source documents against the data submitted in the case report forms, looking for discrepancies or evidence of fraud. An inspection of the sponsor examines oversight instead: monitoring reports, data management systems, safety reporting processes. Where the Institutional Review Boards involved are inspected, the question is whether ethical oversight was maintained appropriately throughout the study.

Significant findings during a BIMO inspection can result in a Form 483 (listing observations) or a Warning Letter. In severe cases, the FDA may invoke the "Application Integrity Policy," pausing the review until the data's reliability can be re-established. Inspection readiness is therefore continuous work for clinical operations teams.

29.6. The Decision

At the conclusion of review, FDA issues an action letter. An *approval letter* authorizes marketing: the product may now be sold, usually with specific labeling, potentially with post-marketing requirements or commitments, and subject to ongoing safety monitoring.

A *complete response letter (CRL)* indicates that the application cannot be approved in its current form and specifies the deficiencies that must be addressed. The sponsor may respond, addressing each one, and multiple review cycles may be needed before approval. Or the sponsor may ultimately abandon the product.

29.7. Single-Trial Substantial Evidence

The traditional interpretation of "substantial evidence" (introduced at Chapter 9) requires replication across at least two adequate and well-controlled trials. But the statistical basis for the two-trial rule is less absolute than it appears, and in practice FDA has regularly approved drugs on a single trial.

29.7.1. The Statistical Argument

If two independent trials each require a one-sided p-value below 0.025 to succeed, a drug with no effect clears both with probability $0.025^2 = 0.000625$. That is the false-positive standard the two-trial rule actually imposes on a sponsor who runs two trials, each tested at $p < 0.025$. A sponsor who designs a single trial to achieve $p < 0.000625$ matches the combined false-positive probability of a two-trial program, at potentially lower overall sample size.

The efficiency follows. Two trials each powered at 90% give the development program $0.90 \times 0.90 = 81\%$ power; a single trial designed to the combined threshold can achieve equivalent program power with approximately 80% of the total sample size of the two-trial program (assuming normally distributed test statistics and the same

true effect and variance in every trial), because the experiment is unified instead of split.

The two-trial rule also creates a classification problem. A program where trial 1 achieves $p = 0.001$ and trial 2 achieves $p = 0.026$ technically "wins one and loses one," while two trials that each achieve $p = 0.024$ both win. Yet the first program carries the stronger evidence. Fisher's method, a standard way to combine independent p-values (a raw product of p-values is not itself a p-value), puts the first program at about 0.0003 and the second at about 0.005; held to the two-trial rule's false-positive rate of 0.000625, the first would pass and the second would not. Pool the evidence instead of scoring each trial pass or fail, and the ordering reverses.

29.7.2. When FDA Accepts a Single Trial

The single-trial exception is codified in statute (the Food and Drug Administration Modernization Act (FDAMA) of 1997, 21 U.S.C. § 355(d) (United States Congress 1997)) and elaborated in FDA draft guidance (U.S. Food and Drug Administration 2019c). Many recent approvals have been based on a single trial together with confirmatory evidence in place of two fully independent replications.

Five features support such a submission. The strongest case has a hard endpoint: survival, cardiovascular death, or another endpoint whose biological connection to patient outcome is unambiguous, since surrogate endpoints carry more risk of false-positive results that would not replicate. The result should be statistically very persuasive, but there is no universal p-value threshold: the 0.000625 derived above is one benchmark, and the June 2026 revised draft discussed below ties the appropriate significance level to how likely the drug was to be effective before the trial, so the level should be agreed with FDA in advance (U.S. Food and Drug Administration 2026c). Enrollment is large and multicenter, which demonstrates generalizability and reduces the risk that results are site-specific. Pre-specified subgroup analyses should show that the overall result is not driven by one atypical subgroup, supplying within one trial the replication two trials provided across trials. Confirmatory mechanistic or surrogate evidence, biological plausibility plus supportive Phase II data, strengthens the case further.

As this book goes to press, the regulatory ground beneath this section is shifting. In June 2026 FDA issued a revised draft of its substantial-evidence guidance (U.S. Food and Drug Administration 2026c) that repositions a single adequate and well-controlled trial plus confirmatory evidence from a bounded exception toward a default, presented as applicable across disease areas instead of reserved for special cases. The draft itself does not use the word "default"; the framing comes from FDA's leadership, Vinay Prasad and Martin Makary, who wrote in February 2026 that "a one-trial requirement will be the FDA's new default standard" (Prasad and Makary 2026). The revision is one piece of a broader HHS initiative (Operation TrialBlazer) aimed at accelerating U.S. drug development. The draft also steps back from framing a single trial as the formal equivalent of two, emphasizing instead the strength and quantity of the confirmatory evidence a program can marshal.

At the time of writing (September 2026) it remains a draft; the public comment period closed on September 22, 2026, and until FDA finalizes it, the earlier guidance remains the reference point. But the direction of travel is clear, and sponsors designing pivotal programs now should track where it lands.

29.7.3. The Breadth Tradeoff

The genuine statistical cost of the one-trial approach is breadth. When sponsors run two trials in different but related patient populations, approval covers both, and labeling reflects evidence in each. A single trial in one population generally supports a label only for that population, with off-label use carrying no safety or efficacy guarantee.

That argues for enrolling a broader population within the single trial and prespecifying subgroup analyses that function like a basket, showing the effect is consistent across the range of patients who will use the drug in practice. When a single pivotal trial supports the first approval, the initial label may be narrow, with expansion trials to follow after approval. The breadth question bears directly on labeling negotiations with FDA at the pre-BLA meeting, so population scope should be settled before the pivotal trial is designed.

29.8. International Regulatory Pathways

The U.S. NDA/BLA process is one system among several. A sponsor pursuing global registration runs parallel pathways in the other major markets, each with its own clock and its own expectations about local data.

European Medicines Agency (EMA). The *centralized procedure* turns a single Marketing Authorization Application (MAA) into a marketing authorization valid in all EU/EEA member states. It is mandatory for certain product categories (biotechnology-derived products, orphan medicines, advanced therapy medicinal products) and optional for other innovative therapies. The Committee for Medicinal Products for Human Use (CHMP) conducts the scientific assessment, with a standard timeline of 210 active review days (European Parliament and Council of the European Union 2004). Sponsors may also use decentralized or mutual-recognition procedures for products not requiring centralized authorization. Europe's decision instruments are the marketing authorization itself and the request for supplementary information.

Pharmaceuticals and Medical Devices Agency (PMDA, Japan). Japan's pathway runs through the *J-NDA* (new drug application), submitted to PMDA, which conducts both the scientific review and GMP inspections. Japan historically required local clinical data, and *bridging studies* under ICH E5 (International Council for Harmonisation 1998a) remain common to demonstrate that foreign data apply to the Japanese population. Japan's participation in ICH and acceptance of

multi-regional clinical trial data under ICH E17 has increasingly enabled global submissions.

National Medical Products Administration (NMPA, China). China's regulatory framework has undergone substantial reform since 2017, including acceptance of international multi-center clinical trial data for registration, removal of the requirement that an imported drug be approved or already under study overseas before Chinese trials could begin, and alignment with ICH guidelines (China became an ICH member in 2017). These reforms have contributed to China's emergence as a major clinical trial market (see Chapter 6). The NMPA still generally requires Chinese patients in pivotal trials, and local regulatory expertise remains essential as requirements evolve.

ICH harmonization and multi-regional submissions. The core dossier is portable across ICH member regions because ICH maintains the CTD format itself, under M4, and its electronic version, the eCTD, under M8. ICH E17 supplies the companion piece: the scientific framework for designing multi-regional clinical trials that can simultaneously support regulatory decisions in multiple jurisdictions, reducing duplicative regional studies while preserving the ability to assess consistency of treatment effects across populations (International Council for Harmonisation 2017, 2016b, 2015).

Concurrent oncology review. Two FDA Oncology Center of Excellence programs compress cancer timelines. Real-Time Oncology Review (RTOR), piloted in 2018, lets sponsors submit key data components before the complete application, so reviewers can begin their analysis earlier (U.S. Food and Drug Administration, Oncology Center of Excellence 2024b). Project Orbis, launched in 2019, frames concurrent submission and review of oncology products among international partner regulators, so that a single dossier can support near-simultaneous decisions in several countries (U.S. Food and Drug Administration, Oncology Center of Excellence 2024a).

Approval ends the submission, not the obligations. Sponsors are often required to conduct post-marketing requirements (PMRs) or agree to post-marketing commitments (PMCs), which can involve additional safety studies or trials in specialized populations. For therapies with significant identified risks, a Risk Evaluation and Mitigation Strategy (REMS) may be mandated to ensure safe use through communication plans or restricted distribution (U.S. Food and Drug Administration 2024l). Throughout the marketed life of the product, sponsors must also monitor safety and submit periodic reports to regulators confirming that the benefit-risk balance remains favorable as real-world experience accumulates.

Part V.

Appendices

30. Glossary and Abbreviations

Abbreviations first (Table 30.1), then the statistical and clinical trial terms used throughout the book (Table 30.2).

30.1. Abbreviations

Table 30.1.: Common abbreviations in clinical development

Abbreviation	Full Term
ADaM	Analysis Data Model; a CDISC standard for deriving analysis-ready datasets from SDTM tabulations
ADR	Adverse Drug Reaction; a noxious and unintended response to a drug for which a causal relationship to the drug is at least a reasonable possibility (ICH E2A); distinguished from an adverse event, which need not be drug-related
AE	Adverse Event; any untoward medical occurrence in a trial participant, whether or not related to the investigational product
AI	Artificial Intelligence; computational systems that perform tasks typically requiring human intelligence, such as pattern recognition or decision support
ALCOA	Attributable, Legible, Contemporaneous, Original, and Accurate; the five principles governing data integrity in clinical trials, extended by ALCOA+ (adding Complete, Consistent, Enduring, Available) and ALCOA++ (adding Traceable)
ANDA	Abbreviated New Drug Application; a regulatory submission to FDA for approval of a generic drug product, referencing the innovator's safety and efficacy data
API	Application Programming Interface; a set of protocols enabling software systems to exchange data
BIMO	Bioresearch Monitoring; the FDA program that inspects clinical investigators, sponsors, IRBs, and contract laboratories to verify data integrity and regulatory compliance
BLA	Biologics License Application; the regulatory submission to FDA for marketing approval of a biologic product

Abbreviation	Full Term
BTD	Breakthrough Therapy Designation; an FDA expedited program for drugs intended to treat serious conditions where preliminary clinical evidence indicates the drug may demonstrate substantial improvement over existing therapies on one or more clinically significant endpoints
CAPA	Corrective and Preventive Action; a systematic process for identifying, investigating, and resolving quality issues to prevent recurrence
CBER	Center for Biologics Evaluation and Research; the FDA center responsible for regulating biological products
CDASH	Clinical Data Acquisition Standards Harmonization; a CDISC standard defining recommended fields for CRF design
CDER	Center for Drug Evaluation and Research; the FDA center responsible for regulating drugs
CDISC	Clinical Data Interchange Standards Consortium; the organization that develops data standards (SDTM, ADaM, CDASH) used in regulatory submissions
CEC	Clinical Events Committee; an independent committee that adjudicates clinical endpoints to ensure consistent classification across sites
CFR	Code of Federal Regulations; the codification of US federal rules, including 21 CFR governing FDA-regulated activities
CHMP	Committee for Medicinal Products for Human Use; the EMA committee responsible for scientific assessment of marketing authorization applications through the centralized procedure
CIOMS	Council for International Organizations of Medical Sciences; an international organization that develops guidelines for safety reporting, including the CIOMS form for individual case safety reports
CMC	Chemistry, Manufacturing, and Controls; the body of data covering a drug product's composition, manufacturing process, and quality specifications (CTD Module 3)
COU	Context of Use; the specific role and scope for which a drug development tool or AI model is applied
CRA	Clinical Research Associate; a monitor who conducts on-site and remote oversight of investigator sites on behalf of the sponsor
CRF	Case Report Form; the document (paper or electronic) used to record trial data for each participant
CRL	Complete Response Letter; an FDA communication indicating that an application cannot be approved in its current form and specifying deficiencies to address
CRO	Contract Research Organization; a company hired by a sponsor to perform one or more trial-related functions such as monitoring, data management, or regulatory submissions
CSR	Clinical Study Report; the comprehensive document describing the design, conduct, results, and interpretation of a completed clinical trial

Abbreviation	Full Term
CSV	Computer System Validation; the documented process of ensuring that a computerized system performs as intended for its regulated purpose
CTA	(1) Clinical Trial Authorization, the regulatory approval required in certain jurisdictions (e.g., EU) before a clinical trial may begin; (2) Clinical Trial Agreement, the legally binding sponsor-site contract covering budget, indemnification, intellectual property, data ownership, and publication rights (see Chapter 18)
CTD	Common Technical Document; the internationally harmonized format for organizing regulatory submissions into five modules
CTIS	Clinical Trials Information System; the EU portal for submitting and managing clinical trial applications under the Clinical Trials Regulation
CTMS	Clinical Trial Management System; software used to manage the operational aspects of clinical trials, including site management, visit tracking, and milestone reporting
CTR	Clinical Trials Regulation; the EU regulation (536/2014) governing clinical trial conduct in member states
DCT	Decentralized Clinical Trial; a trial design where some or all activities occur at locations other than a traditional clinical site, such as the participant's home
DDT	Drug Development Tool; a method, material, or measure with the potential to facilitate drug development; FDA's DDT Qualification Programs formally qualify a tool for a specified context of use
DHT	Digital Health Technology; electronic systems that capture health-related data from participants, including wearable sensors, mobile apps, and remote monitoring devices
DLT	Dose-Limiting Toxicity; a pre-defined adverse event in dose-escalation studies that signals the dose has reached an unacceptable level of toxicity
DMC	Data Monitoring Committee; see DSMB
DMP	Data Management Plan; the document specifying procedures for data collection, cleaning, coding, transfer, and quality control throughout a trial
DSMB	Data Safety Monitoring Board; an independent group of experts who periodically review unblinded safety and efficacy data and may recommend stopping or modifying a trial
DSUR	Development Safety Update Report; an annual safety report submitted to regulators summarizing the safety profile of an investigational product across all ongoing trials
eCOA	Electronic Clinical Outcome Assessment; electronic systems for capturing patient-reported, clinician-reported, or observer-reported outcomes
eConsent	Electronic Informed Consent; the use of electronic systems (multimedia, interactive tools) to present and document the informed consent process

Abbreviation	Full Term
eCRF	Electronic Case Report Form; the electronic version of a CRF used within an EDC system to capture trial data
eCTD	Electronic Common Technical Document; the electronic format for submitting the CTD to regulatory agencies
EDC	Electronic Data Capture; a computerized system for collecting clinical trial data, replacing paper CRFs
EHR	Electronic Health Record; a digital version of a patient's medical record maintained by healthcare providers
EMA	European Medicines Agency; the EU regulatory authority responsible for evaluating and supervising medicinal products
EOP	End-of-Phase; the FDA meeting held at the end of a phase of clinical investigation (End-of-Phase 1 or End-of-Phase 2) to agree the plan for the next phase
ePRO	Electronic Patient-Reported Outcome; patient-reported outcome data collected via electronic devices such as smartphones, tablets, or web portals
eTMF	Electronic Trial Master File; an electronic system for storing and managing essential trial documents required by GCP
FDA	Food and Drug Administration; the US regulatory agency responsible for protecting public health through oversight of drugs, biologics, and medical devices
FDAAA	Food and Drug Administration Amendments Act; 2007 US law whose Section 801 requires registration and results reporting for applicable clinical trials on ClinicalTrials.gov
FDAMA	Food and Drug Administration Modernization Act; 1997 US law that, among other provisions, allows approval on one adequate and well-controlled trial together with confirmatory evidence
FDORA	Food and Drug Omnibus Reform Act; 2022 US legislation that modernized FDA authorities, including provisions for decentralized trials and diversity action plans
FIH	First-in-Human; the initial clinical study in which an investigational product is administered to human subjects
GAMP	Good Automated Manufacturing Practice; a framework (GAMP 5) providing guidance on computer system validation in regulated industries
GCP	Good Clinical Practice; the international ethical and scientific quality standard for designing, conducting, recording, and reporting clinical trials (ICH E6)
GDPR	General Data Protection Regulation; the EU regulation governing the processing and protection of personal data
GLP	Good Laboratory Practice; the quality system governing non-clinical laboratory studies intended to support regulatory submissions

Abbreviation	Full Term
GMP	Good Manufacturing Practice; the quality system governing the manufacture of investigational and commercial pharmaceutical products
GxP	Good Practice; a collective term for quality guidelines and regulations (e.g., GCP, GMP, GLP) in the pharmaceutical industry
HIPAA	Health Insurance Portability and Accountability Act; US legislation that establishes standards for protecting patient health information
HTA	Health Technology Assessment; the payer-side evaluation of a medicine's cost-effectiveness and value (e.g., by NICE) that informs reimbursement decisions, distinct from a regulator's assessment of safety and efficacy
IB	Investigator's Brochure; the document providing investigators with clinical and non-clinical data on the investigational product relevant to the study
ICF	Informed Consent Form; the document that participants sign to confirm they understand and voluntarily agree to participate in a clinical trial
ICH	International Council for Harmonisation; the organization that brings together regulators and industry to develop harmonized guidelines (e.g., E6 GCP, E8, E9) for pharmaceutical development
IDMC	Independent Data Monitoring Committee; see DSMB
IEC	Independent Ethics Committee; a body that reviews clinical trial protocols to protect the rights, safety, and well-being of participants; the EU equivalent of an IRB
IND	Investigational New Drug; the FDA submission that must go into effect (i.e., 30 days after FDA receives it unless FDA imposes a clinical hold, or sooner if FDA notifies the sponsor that the study may begin; 21 CFR 312.40) before an investigational product may be administered to humans in the US
IP	Investigational Product; the drug, biologic, or device being tested in a clinical trial, including placebo or active comparator
IRB	Institutional Review Board; a committee that reviews and approves clinical research to protect the rights and welfare of human subjects; the US equivalent of an IEC
IRT	Interactive Response Technology; a system that manages randomization and drug supply in clinical trials via web or telephone interfaces
ISF	Investigator Site File; the collection of essential documents maintained at each investigator site for regulatory inspection readiness
ITT	Intent-to-Treat; an analysis population that includes all randomized subjects regardless of protocol adherence
IWRS	Interactive Web Response System; a web-based system for managing randomization and trial supply; a specific type of IRT
J-NDA	Japanese New Drug Application; the regulatory submission to PMDA for marketing approval of a new drug in Japan

Abbreviation	Full Term
KRI	Key Risk Indicator; a metric used in risk-based quality management to monitor trial conduct and trigger risk mitigation actions
LLM	Large Language Model; an AI model trained on large text corpora that can generate, summarize, and analyze text
LOA	Likelihood of Approval; a metric estimating the probability that a drug in development will ultimately receive regulatory approval
LPLV	Last Patient Last Visit; the final protocol-specified visit by the last enrolled participant, marking the end of data collection
MAA	Marketing Authorization Application; the regulatory submission to EMA for marketing approval of a medicinal product in the EU
MCID	Minimum Clinically Important Difference; the smallest change in an outcome that patients or clinicians would consider meaningful, and the usual basis for the effect a trial is sized to detect
MCP	Model Context Protocol; a standard for connecting AI applications to external data sources and tools in a controlled, auditable manner
MedDRA	Medical Dictionary for Regulatory Activities; the standardized medical terminology used for coding adverse events and medical history in clinical trials and pharmacovigilance
ML	Machine Learning; a subset of AI in which algorithms learn patterns from data without being explicitly programmed for each task
MTD	Maximum Tolerated Dose; the highest dose of an investigational product that can be administered without unacceptable toxicity
NDA	New Drug Application; the regulatory submission to FDA for marketing approval of a new drug product
NLP	Natural Language Processing; AI techniques for interpreting and generating human language
NMPA	National Medical Products Administration; the regulatory authority in China responsible for drug, device, and cosmetic oversight
NOAEL	No Observed Adverse Effect Level; the highest dose in preclinical studies at which no adverse effects are observed; used to calculate the starting dose for first-in-human trials
PD	Pharmacodynamics; the study of what a drug does to the body, including its mechanism of action and biological effects
PDUFA	Prescription Drug User Fee Act; US legislation authorizing FDA to collect fees from drug manufacturers to fund the drug review process
PI	Principal Investigator; the physician or qualified individual at a trial site who is responsible for the conduct of the trial at that site
PK	Pharmacokinetics; the study of how the body absorbs, distributes, metabolizes, and excretes a drug
PMC	Post-Marketing Commitment; a study or clinical trial that a sponsor agrees to conduct after approval, not required by statute or regulation

Abbreviation	Full Term
PMDA	Pharmaceuticals and Medical Devices Agency; the Japanese agency that conducts the scientific review of drug and medical device applications; formal marketing approval is granted by the Ministry of Health, Labour and Welfare (MHLW)
PMR	Post-Marketing Requirement; a study or clinical trial that a sponsor is required by statute or regulation to conduct after approval
PRO	Patient-Reported Outcome; a measurement of any aspect of a patient's health status that comes directly from the patient, without interpretation by a clinician
QA	Quality Assurance; the overarching system of planned activities ensuring that trial processes comply with GCP and applicable regulations
QC	Quality Control; the operational techniques and activities undertaken to verify that data and processes meet quality requirements
QMS	Quality Management System; the organizational structure, processes, and resources for implementing quality management across trial activities
RAG	Retrieval-Augmented Generation; an AI technique that grounds language model outputs by retrieving relevant documents before generating a response
RAR	Response-Adaptive Randomization; a design in which allocation probabilities shift during the trial toward better-performing arms based on accumulating outcome data (see Section 12.8)
RBQM	Risk-Based Quality Management; a systematic approach to trial quality that focuses oversight on the risks most likely to affect participant safety and data reliability (ICH E6 R2/R3)
RDEP	Rare Disease Evidence Principles; regulatory principles guiding the use of innovative evidence approaches in rare disease drug development
REMS	Risk Evaluation and Mitigation Strategy; an FDA-required safety plan for certain drugs with known or potential serious risks, designed to ensure benefits outweigh risks
RP2D	Recommended Phase 2 Dose; the dose selected from Phase I studies for further evaluation in Phase II efficacy trials
RTF	Refuse to File; an FDA decision that a submitted application is insufficiently complete to permit substantive review
RTOR	Real-Time Oncology Review; an FDA Oncology Center of Excellence program in which sponsors submit key components of an oncology application before the complete application so that review can begin earlier
RTSM	Randomization and Trial Supply Management; a system that integrates randomization with drug supply logistics, ensuring correct treatment allocation and supply forecasting

Abbreviation	Full Term
RWD	Real-World Data; data relating to patient health status or healthcare delivery collected outside of traditional clinical trials (e.g., from EHRs, claims, registries)
RWE	Real-World Evidence; clinical evidence derived from analysis of real-world data
SAE	Serious Adverse Event; an adverse event that results in death, is life-threatening, requires or prolongs hospitalization, causes persistent or significant disability, is a congenital anomaly, or is another medically important event (ICH E2A)
SAP	Statistical Analysis Plan; the document specifying the statistical methods, populations, endpoints, and handling of missing data for a clinical trial, finalized before unblinding
SDTM	Study Data Tabulation Model; a CDISC standard for organizing and formatting clinical trial data for regulatory submission
SDV	Source Data Verification; the process of comparing data entered in the CRF/EDC against original source documents to ensure accuracy
SIV	Site Initiation Visit; the sponsor or CRO visit to a trial site before enrollment begins, to train staff, review the protocol, and verify site readiness
SOP	Standard Operating Procedure; documented step-by-step instructions for carrying out routine operations in compliance with regulations and quality standards
SUSAR	Suspected Unexpected Serious Adverse Reaction; a serious adverse event that is both suspected to be related to the investigational product and not consistent with known risk information; requires expedited reporting to regulators
TLF	Tables, Listings, and Figures; the standard set of statistical outputs generated from clinical trial data for inclusion in the CSR and regulatory submission
TMF	Trial Master File; the collection of essential documents that individually and collectively permit evaluation of the conduct of a trial and the quality of data produced (ICH E6)
VVUQ	Verification, Validation, and Uncertainty Quantification; the framework for assessing whether computational models are correctly implemented and fit for their intended use
WMA	World Medical Association; the international organization that authored the Declaration of Helsinki, establishing ethical principles for medical research involving human subjects

30.2. Statistical Terms

Table 30.2.: Key statistical terms for clinical trials

Term	Definition
Active control	The current standard-of-care treatment used as a comparator
Adaptive design	A trial that modifies itself based on accumulating data according to pre-specified rules
Alpha (α)	The maximum acceptable probability of Type I error; typically 2.5% or 5%
Basket trial	A master protocol that tests one therapy across multiple diseases (e.g., tumor types) sharing a common biomarker
Bayesian	Statistical approach that directly estimates the probability a treatment works, given observed data; naturally updates as evidence accumulates
Bias (systematic error)	A flaw causing treatment groups to differ in ways unrelated to treatment, distorting results
Biomarker	A measurable biological indicator (e.g., genetic mutation, protein level) that can predict treatment response
Confidence interval	A range of values, computed from the data, that would contain the true treatment effect in a stated percentage (e.g., 95%) of repeated trials; narrower intervals indicate more precise estimates. It describes the reliability of the procedure, not the probability that one particular interval contains the true value
Control arm	The comparison group receiving placebo or standard of care
Effectiveness	How well a treatment works in routine clinical practice
Efficacy	The treatment's ability to produce beneficial effects under controlled trial conditions
Endpoint	The outcome measured to determine treatment effect (e.g., survival, tumor response, symptom improvement)
Estimand	A precise definition of the treatment effect being estimated, including how to handle intercurrent events
External control arm (ECA)	A comparator group built from data outside the trial (historical trials, registries, EHRs, or claims) in place of a concurrent randomized control, used when randomization is infeasible or unethical
External validity (generalizability)	Whether conclusions apply to broader patient populations
Fixed design	A traditional trial where sample size is set before the trial begins and cannot change
Frequentist	Statistical approach that interprets probability as the frequency of outcomes over many repeated trials; focuses on controlling error rates

Term	Definition
Hazard ratio	The ratio of event rates between treatment and control groups in a time-to-event analysis; a hazard ratio of 0.75 means the treatment group experiences events at 75% the rate of the control group
Intent-to-Treat (ITT)	Analyzing all patients according to their original randomization, regardless of protocol adherence
Intercurrent events	Events occurring after treatment initiation that affect the interpretation or the existence of the endpoint measurement (e.g., discontinuation, rescue medication)
Interim analysis	An analysis conducted partway through a trial, before all patients have completed
Internal validity	Whether the trial's conclusions are correct for the patients actually studied
Investigational arm	The group receiving the new therapy being tested
Master protocol	A single design that evaluates multiple therapies, disease subtypes, or both under shared infrastructure; basket, umbrella, and platform trials are its principal forms
Minimum clinically important difference (MCID)	The smallest change in an outcome that patients or clinicians would consider meaningful, and the usual basis for the effect a trial is sized to detect
Null hypothesis	The default assumption that there is no treatment difference
p-value	The probability of observing results as extreme as those seen, assuming the null hypothesis is true
Placebo	An inactive substance given to the control group
Platform trial	A master protocol in which treatment arms enter and exit over time against a shared control arm, enabling continuous evaluation of multiple therapies
Population	The entire universe of patients with the condition of interest who might receive the drug if approved
Power	The probability of correctly detecting a true treatment effect; typically set at 80% or 90%
Precision	The degree of certainty in an estimate; higher precision means smaller standard error
Precision medicine	Treatments targeted to patients with specific biological characteristics
Prior and posterior probability	In Bayesian analysis, the prior is the probability of a hypothesis or parameter before observing the current data; the posterior is the updated probability after combining the prior with the observed data via Bayes' theorem
Probability	A number between 0 and 1 (or 0%-100%) quantifying how likely something is to occur; 0 = impossible, 1 = certain

Term	Definition
Random error	Natural fluctuation in results due to studying a sample rather than the entire population
Randomization	Using a chance mechanism to assign patients to treatment arms, eliminating systematic differences
Sample	The subset of patients actually enrolled in the trial
Sample size	The number of patients enrolled; balances Type I error, power, and detectable effect size
Standard error	A measure of the expected magnitude of random error in an estimate
Statistical significance	A result is "significant" when the p-value falls below the α threshold, suggesting the effect is unlikely due to chance
Treatment arms	The groups being compared in a trial
Treatment effect	The true difference in outcomes between investigational and control treatments
Type I error (false positive)	Concluding an ineffective drug works; regulators limit this to $\leq 2.5\%$ (one-sided) or $\leq 5\%$ (two-sided)
Type II error (false negative)	Failing to detect that an effective drug works
Umbrella trial	A master protocol that tests multiple therapies within a single disease (e.g., one tumor type), assigning patients to arms by biomarker
Underpowered trial	A trial with too few patients to reliably detect a true treatment effect

References

Abou Ali, Mohamad, Fadi Dornaika, and Jinan Charafeddine. 2026. "Agentic AI: A Comprehensive Survey of Architectures, Applications, and Future Directions." *Artificial Intelligence Review* 59: 11. https://doi.org/10.1007/s10462-025-11422-4.

Agresti, Alan. 2010. *Analysis of Ordinal Categorical Data*. 2nd ed. Wiley. https://doi.org/10.1002/9780470594001.

Alexander, Brian M., Sujuan Ba, Mitchel S. Berger, Donald A. Berry, Webster K. Cavenee, Susan M. Chang, Timothy F. Cloughesy, et al. 2018. "Adaptive Global Innovative Learning Environment for Glioblastoma: GBM AGILE." *Clinical Cancer Research* 24 (4): 737–43. https://doi.org/10.1158/1078-0432.CCR-17-0764.

Almirall, Daniel, Inbal Nahum-Shani, Nancy E. Sherwood, and Susan A. Murphy. 2014. "Introduction to SMART Designs for the Development of Adaptive Interventions: With Application to Weight Loss Research." *Translational Behavioral Medicine* 4 (3): 260–74. https://doi.org/10.1007/s13142-014-0265-0.

Amershi, Saleema, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, et al. 2019. "Guidelines for Human-AI Interaction." In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–13. https://doi.org/10.1145/3290605.3300233.

Anagnostopoulos, Chris, David Champagne, Thomas Devenyns, Alex Devereson, and Heikki Tarkkila. 2023. "How Artificial Intelligence Can Power Clinical Development." McKinsey & Company. https://www.mckinsey.com/industries/life-sciences/our-insights/how-artificial-intelligence-can-power-clinical-development.

Andhalkar, Niranjan. 2025. "Phase-by-Phase Clinical Trial Costs Guide for Sponsors." ProRelix Research blog. https://prorelixresearch.com/phase-by-phase-clinical-trial-costs-guide/.

Angus, Derek C., Scott Berry, Roger J. Lewis, Farah Al-Beidh, Yaseen Arabi, Wilma van Bentum-Puijk, Zahra Bhimani, et al. 2020. "The REMAP-CAP (Randomized Embedded Multifactorial Adaptive Platform for Community-Acquired Pneumonia) Study: Rationale and Design." *Annals of the American Thoracic Society* 17 (7): 879–91. https://doi.org/10.1513/AnnalsATS.202003-192SD.

Annett, Robert D., Scott Bickel, John C. Carlson, Kelly Cowan, Sara Cox, Mark J. Fisher, J. Dean Jarvis, Alberta S. Kong, et al. 2021. "Capacity Building for a New Multicenter Network Within the ECHO IDeA States Pediatric Clinical Trials Network." *Frontiers in Pediatrics* 9: 679516. https://doi.org/10.3389/

fped.2021.679516.

Anthropic. 2025. "Model Context Protocol: Introduction." https://modelcontextprotocol.io/docs/getting-started/intro.

Armitage, Peter. 1954. "Sequential Tests in Prophylactic and Therapeutic Trials." *Quarterly Journal of Medicine* 23 (91): 255–74.

———. 1960. *Sequential Medical Trials*. Oxford: Blackwell Scientific Publications.

Association of Clinical Research Organizations. 2025. "2025 Risk-Based Quality Management (RBQM) Summary Report." https://www.acrohealth.org/wp-content/uploads/2025/06/ACRO_2025-RBQM-Report_Final-062425.pdf.

Atkinson, A. C. 2002. "The Comparison of Designs for Sequential Clinical Trials with Covariate Information." *Journal of the Royal Statistical Society: Series A* 165 (2): 349–73. https://doi.org/10.1111/1467-985X.00564.

ATTACC Investigators, ACTIV-4a Investigators, REMAP-CAP Investigators, Patrick R. Lawler, Ewan C. Goligher, Jeffrey S. Berger, et al. 2021. "Therapeutic Anticoagulation with Heparin in Noncritically Ill Patients with Covid-19." *New England Journal of Medicine* 385 (9): 790–802. https://doi.org/10.1056/NEJMoa2105911.

Avorn, Jerry. 2015. "The $2.6 Billion Pill — Methodologic and Policy Considerations." *New England Journal of Medicine* 372 (20): 1877–79. https://doi.org/10.1056/NEJMp1500848.

Babaeipour, Ramtin, François Charest, and Madison Wright. 2026. "AI-Assisted Protocol Information Extraction for Improved Accuracy and Efficiency in Clinical Trial Workflows." *Journal of Biomedical Informatics* 179: 105036. https://doi.org/10.1016/j.jbi.2026.105036.

Babb, James, Andre Rogatko, and Shelemyahu Zacks. 1998. "Cancer Phase I Clinical Trials: Efficient Dose Escalation with Overdose Control." *Statistics in Medicine* 17 (10): 1103–20. https://doi.org/10.1002/(SICI)1097-0258(19980530)17:10%3C1103::AID-SIM793%3E3.0.CO;2-9.

Baiocchi, Michael, Jing Cheng, and Dylan S. Small. 2014. "Instrumental Variable Methods for Causal Inference." *Statistics in Medicine* 33 (13): 2297–2340. https://doi.org/10.1002/sim.6128.

Ballentine, Carol. 1981. "Taste of Raspberries, Taste of Death: The 1937 Elixir Sulfanilamide Incident." FDA Consumer. https://www.fda.gov/files/about%20fda/published/The-Sulfanilamide-Disaster.pdf.

Bartlett, Robert H., Donald W. Roloff, Robert G. Cornell, Arthur F. Andrews, Patrick W. Dillon, and Joseph B. Zwischenberger. 1985. "Extracorporeal Circulation in Neonatal Respiratory Failure: A Prospective Randomized Study." *Pediatrics* 76 (4): 479–87. https://doi.org/10.1542/peds.76.4.479.

Bauer, Peter, Frank Bretz, Vladimir Dragalin, Franz König, and Gernot Wassmer. 2015. "Twenty-five years of confirmatory adaptive designs: opportunities and pitfalls." *Statistics in Medicine* 35 (3): 325–47. https://doi.org/10.1002/sim.6472.

Bauer, P., and K. Kohne. 1994. "Evaluation of Experiments with Adaptive Interim Analyses." *Biometrics* 50 (4): 1029. https://doi.org/10.2307/2533441.

Beal, Kristy, Juliette Dean, James Chen, Elena Dragaon, Ann Saulino, and Charles

D. Collard. 2004. "Budget Negotiation for Industry-Sponsored Clinical Trials." *Anesthesia & Analgesia* 99 (1): 173–76. https://doi.org/10.1213/01.ANE.0000122633.97424.76.

Bender, Emily M., Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?" In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–23. https://doi.org/10.1145/3442188.3445922.

Bennette, Caroline S., Scott D. Ramsey, Cara L. McDermott, Josh J. Carlson, Anirban Basu, and David L. Veenstra. 2016. "Predicting Low Accrual in the National Cancer Institute's Cooperative Group Clinical Trials." *Journal of the National Cancer Institute* 108 (2): djv324. https://doi.org/10.1093/jnci/djv324.

Berkeley Artificial Intelligence Research. 2024. "The Shift from Models to Compound AI Systems." https://bair.berkeley.edu/blog/2024/02/18/compound-ai-systems/.

Berry, Donald A. 1989. "Monitoring Accumulating Data in a Clinical Trial." *Biometrics* 45 (4): 1197–1211. https://doi.org/10.2307/2531771.

———. 2012. "Adaptive Clinical Trials in Oncology." *Nature Reviews Clinical Oncology* 9 (4): 199–207. https://doi.org/10.1038/nrclinonc.2011.165.

———. 2025. "Adaptive Bayesian Clinical Trials: The Past, Present, and Future of Clinical Research." *Journal of Clinical Medicine* 14 (15): 5267. https://doi.org/10.3390/jcm14155267.

Berry, Lindsay R., Joe Marion, Scott M. Berry, and Kert Viele. 2024. "Optimal sample size division in two-stage seamless designs." *Pharmaceutical Statistics* 23 (6): 854–63. https://doi.org/10.1002/pst.2394.

Berry, Scott M., Bradley P. Carlin, J. Jack Lee, and Peter Müller. 2010. *Bayesian Adaptive Methods for Clinical Trials.* Chapman & Hall/CRC Biostatistics Series. Boca Raton, FL: CRC Press. https://doi.org/10.1201/EBK1439825488.

Bharucha, Adil E., Chung Il Wi, Sushmitha Grama Srinivasan, Hyuckjae Choi, et al. 2021. "Participation of Rural Patients in Clinical Trials at a Multisite Academic Medical Center." *Journal of Clinical and Translational Science* 5 (1): e190. https://doi.org/10.1017/cts.2021.813.

Biotechnology Innovation Organization, QLS Advisors, and Informa Pharma Intelligence. 2021. "Clinical Development Success Rates 2011-2020." https://go.bio.org/rs/490-EHZ-999/images/ClinicalDevelopmentSuccessRates2011_2020.pdf.

Bor, Jacob, Ellen Moscoe, Portia Mutevedzi, Marie-Louise Newell, and Till Bärnighausen. 2014. "Regression Discontinuity Designs in Epidemiology: Causal Inference Without Randomized Trials." *Epidemiology* 25 (5): 729–37. https://doi.org/10.1097/EDE.0000000000000138.

Bretz, Frank, Jose C. Pinheiro, and Michael Branson. 2005. "Combining Multiple Comparisons and Modeling Techniques in Dose-Response Studies." *Biometrics* 61 (3): 738–48. https://doi.org/10.1111/j.1541-0420.2005.00344.x.

Bronner, Denise. 2024. "Why Site-CRO Relationships Are so Strained, and How to Fix Them." Clinical Leader, guest column. https://www.clinicalleader.com/doc/why-site-cro-relationships-are-so-strained-and-how-to-fix-them-0001.

Budd Haeberlein, Samantha, Paul S. Aisen, Frederik Barkhof, Spyridon Chalkias,

Tianle Chen, Sandra Cohen, Gary Dent, et al. 2022. "Two Randomized Phase 3 Studies of Aducanumab in Early Alzheimer's Disease." *Journal of Prevention of Alzheimer's Disease* 9 (2): 197–210. https://doi.org/10.14283/jpad.2022.30.

Burnett, Thomas, Pavel Mozgunov, Philip Pallmann, Sofia S. Villar, Graham M. Wheeler, and Thomas Jaki. 2020. "Adding flexibility to clinical trial designs: an example-based guide to the practical use of adaptive designs." *BMC Medicine* 18 (1). https://doi.org/10.1186/s12916-020-01808-2.

Butler, Christopher C, F D Richard Hobbs, Oghenekome A Gbinigie, Najib M Rahman, Gail Hayward, Duncan B Richards, Jienchi Dorward, et al. 2023. "Molnupiravir plus usual care versus usual care alone as early treatment for adults with COVID-19 at increased risk of adverse outcomes (PANORAMIC): an open-label, platform-adaptive randomised controlled trial." *The Lancet* 401 (10373): 281–93. https://doi.org/10.1016/S0140-6736(22)02597-1.

Cappello, Franck, Sandeep Madireddy, Robert Underwood, Neil Getty, Nicholas Lee-Ping Chia, Nesar Ramachandra, et al. 2025. "EAIRA: Establishing a Methodology for Evaluating AI Models as Scientific Research Assistants." *arXiv Preprint arXiv:2502.20309*. https://arxiv.org/abs/2502.20309.

CDISC. 2025. "Digital Data Flow and the Unified Study Definitions Model (USDM)." https://www.cdisc.org/ddf.

Cemri, Mert, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, et al. 2025. "Why Do Multi-Agent LLM Systems Fail?" *arXiv Preprint arXiv:2503.13657*. https://arxiv.org/abs/2503.13657.

Chan, An-Wen, Isabelle Boutron, Sally Hopewell, David Moher, Kenneth F. Schulz, Gary S. Collins, et al. 2025. "SPIRIT 2025 Statement: Updated Guideline for Protocols of Randomised Trials." *BMJ* 389: e081477. https://doi.org/10.1136/bmj-2024-081477.

Chan, Jun Shern, Neil Chowdhury, Oliver Jaffe, James Aung, Dane Sherburn, Evan Mays, Giulio Starace, et al. 2024. "MLE-Bench: Evaluating Machine Learning Agents on Machine Learning Engineering." *arXiv Preprint arXiv:2410.07095*. https://arxiv.org/abs/2410.07095.

Cheng, Yuheng, Ceyao Zhang, Zhengwen Zhang, Xiangrui Meng, Sirui Hong, Wenhao Li, Zihao Wang, et al. 2024. "Exploring Large Language Model Based Intelligent Agents: Definitions, Methods, and Prospects." *arXiv Preprint arXiv:2401.03428*. https://arxiv.org/abs/2401.03428.

Clinical Data Interchange Standards Consortium. 2024. "CDISC Foundational Standards: SDTM, ADaM, and CDASH." https://www.cdisc.org/standards/foundational.

Clinical Leader. 2024. "Best Practices for Randomization and Trial Supply Management (RTSM)." https://www.clinicalleader.com/doc/best-practices-for-randomization-and-trial-supply-management-rtsm-0001.

Coalition, The Adaptive Platform Trials, Derek C. Angus, Brian M. Alexander, Scott Berry, Meredith Buxton, Roger Lewis, Melissa Paoloni, et al. 2019. "Adaptive platform trials: definition, design, conduct and reporting considerations." *Nature Reviews Drug Discovery* 18 (10): 797–807. https://doi.org/10.1038/s41573-019-0034-3.

Coart, Elisabeth, Perrine Bamps, Emmanuel Quinaux, Geneviève Sturbois, Everardo D. Saad, Tomasz Burzykowski, and Marc Buyse. 2023. "Minimization in Randomized Clinical Trials." *Statistics in Medicine* 42 (28): 5285–5311. https://doi.org/10.1002/sim.9916.

Cohen, Lauren W., Rowena J. Dolor, Mark Weiner, Nadine Zemon, Charles Bailey, Thomas Carton, Erika Cottrell, et al. 2026. "PCORnet: 10 Years of Research Innovation." *Medical Care* 64: S185–90. https://doi.org/10.1097/MLR.0000000000002269.

Coleman, C. Norman, Rosemary Wong, Daniel G. Petereit, Patrick D. Maguire, et al. 2023. "The National Cancer Institute's Cancer Disparities Research Partnership Program: A Unique Funding Model 20 Years Later." *Journal of the National Cancer Institute* 115 (12): 1465–74. https://doi.org/10.1093/jnci/djad173.

Collins, Gary S. et al. 2024. "TRIPOD+AI Statement: Updated Guidance for Reporting Clinical Prediction Models That Use Regression or Machine Learning Methods." *BMJ* 385: e078378. https://www.bmj.com/content/385/bmj-2023-078378.

Council for International Organizations of Medical Sciences (CIOMS). 2016. *International Ethical Guidelines for Health-Related Research Involving Humans*. Fourth. Geneva: CIOMS. https://cioms.ch/publications/product/international-ethical-guidelines-for-health-related-research-involving-humans/.

Cox, D. R. 1958. *Planning of Experiments*. New York: John Wiley & Sons.

———. 1972. "Regression Models and Life-Tables." *Journal of the Royal Statistical Society: Series B (Methodological)* 34 (2): 187–220. https://doi.org/10.1111/j.2517-6161.1972.tb00899.x.

———. 2009. "Randomization in the Design of Experiments." *International Statistical Review* 77 (3): 415–29. https://doi.org/10.1111/j.1751-5823.2009.00084.x.

Cui, Lu, H. M. James Hung, and Sue-Jane Wang. 1999. "Modification of Sample Size in Group Sequential Clinical Trials." *Biometrics* 55 (3): 853–57. https://doi.org/10.1111/j.0006-341X.1999.00853.x.

Dahabreh, Issa J., Sarah E. Robertson, Jon A. Steingrimsson, Elizabeth A. Stuart, and Miguel A. Hernan. 2020. "Extending Inferences from a Randomized Trial to a New Target Population." *Statistics in Medicine* 39 (14): 1999–2014. https://doi.org/10.1002/sim.8426.

Dahabreh, Issa J., Sarah E. Robertson, Eric J. Tchetgen Tchetgen, Elizabeth A. Stuart, and Miguel A. Hernan. 2019. "Generalizing Causal Inferences from Individuals in Randomized Trials to All Trial-Eligible Individuals." *Biometrics* 75 (2): 685–94. https://doi.org/10.1111/biom.13009.

Davis, Joshua S., John Turnidge, and Steven Y. C. Tong. 2018. "A Large Retrospective Cohort Study of Cefazolin Compared with Flucloxacillin for Methicillin-Susceptible Staphylococcus aureus Bacteraemia." *International Journal of Antimicrobial Agents* 52 (2): 297–300. https://doi.org/10.1016/j.ijantimicag.2018.02.013.

Deloitte Centre for Health Solutions. 2025. "Be Brave, Be Bold: Measuring the Return from Pharmaceutical Innovation." https://www.deloitte.com/content/dam/assets-shared/docs/industries/life-sciences-health-care/2025/deloitte-

measuring-return-from-pharmaceutical-innovation-2025.pdf.

———. 2026. "Navigating the GLP-1 Boom: Measuring the Return from Pharmaceutical Innovation." https://www.deloitte.com/content/dam/assets-shared/docs/industries/life-sciences-health-care/2026/measuring-the-return-from-pharmaceutical-innovation.pdf.

DeTora, Lisa M., Dikran Toroser, Angela Sykes, Christine Vanderlinden, Fiona J. Plunkett, Trevor Lane, Eline Hanekamp, et al. 2022. "Good Publication Practice (GPP) Guidelines for Company-Sponsored Biomedical Research: 2022 Update." *Annals of Internal Medicine* 175 (9): 1298–1304. https://doi.org/10.7326/M22-1460.

Dickerman, Barbra A., Hanna Gerlovin, Arin L. Madenci, Katherine E. Kurgansky, Brian R. Ferolito, Mariantonia Figueroa Munõz, David R. Gagnon, et al. 2022. "Comparative Effectiveness of BNT162b2 and mRNA-1273 Vaccines in U.S. Veterans." *New England Journal of Medicine* 386 (2): 105–15. https://doi.org/10.1056/NEJMoa2115463.

Dilts, David M., Steven K. Cheng, Joshua S. Crites, Alan B. Sandler, and James H. Doroshow. 2010. "Phase III Clinical Trial Development: A Process of Chutes and Ladders." *Clinical Cancer Research* 16 (22): 5381–89. https://doi.org/10.1158/1078-0432.CCR-10-1273.

Dilts, David M., and Alan B. Sandler. 2006. "Invisible Barriers to Clinical Trials: The Impact of Structural, Infrastructural, and Procedural Barriers to Opening Oncology Clinical Trials." *Journal of Clinical Oncology* 24 (28): 4545–52. https://doi.org/10.1200/JCO.2005.05.0104.

Dimairo, Munyaradzi, Philip Pallmann, James Wason, Susan Todd, Thomas Jaki, Steven A Julious, Adrian P Mander, et al. 2020. "The Adaptive designs CONSORT Extension (ACE) statement: a checklist with explanation and elaboration guideline for reporting randomised trials that use an adaptive design." *BMJ*, m115. https://doi.org/10.1136/bmj.m115.

DiMasi, Joseph A., Henry G. Grabowski, and Ronald W. Hansen. 2016. "Innovation in the Pharmaceutical Industry: New Estimates of R&D Costs." *Journal of Health Economics* 47: 20–33. https://doi.org/10.1016/j.jhealeco.2016.01.012.

Dodd, Lori E., Boris Freidlin, and Edward L. Korn. 2021. "Platform Trials , Beware the Noncomparable Control Group." *New England Journal of Medicine* 384 (16): 1572–73. https://doi.org/10.1056/NEJMc2102446.

Douze, Matthijs, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, and Hervé Jégou. 2024. "The Faiss Library." *arXiv Preprint arXiv:2401.08281*. https://arxiv.org/abs/2401.08281.

Duke Center for Virtual Imaging Trials. 2024. "Virtual Imaging Trials in Medicine 2024 Summit." https://cvit.duke.edu/vitm24/.

Duke Clinical Research Institute. 2026. "DCRI Secures Third Award for Pediatric Trials Network." https://dcri.org/news/dcri-secures-third-award-pediatric-trials-network.

Dyck, Christopher H. van, Chad J. Swanson, Paul Aisen, Randall J. Bateman, Christopher Chen, Michelle Gee, Michio Kanekiyo, et al. 2023. "Lecanemab in

Early Alzheimer's Disease." *New England Journal of Medicine* 388 (1): 9–21. https://doi.org/10.1056/NEJMoa2212948.

Echt, Debra S., Philip R. Liebson, L. Brent Mitchell, Robert W. Peters, Dulce Obias-Manno, Allan H. Barker, Daniel Arensberg, et al. 1991. "Mortality and Morbidity in Patients Receiving Encainide, Flecainide, or Placebo: The Cardiac Arrhythmia Suppression Trial." *New England Journal of Medicine* 324 (12): 781–88. https://doi.org/10.1056/NEJM199103213241201.

Eckstein, Olive S., Carl E. Allen, P. Mickey Williams, Sinchita Roy-Chowdhuri, David R. Patton, Brent Coffey, Joel M. Reid, Jin Piao, et al. 2022. "Phase II Study of Selumetinib in Children and Young Adults with Tumors Harboring Activating Mitogen-Activated Protein Kinase Pathway Genetic Alterations: Arm E of the NCI-COG Pediatric MATCH Trial." *Journal of Clinical Oncology* 40 (20): 2235–45. https://doi.org/10.1200/JCO.21.02840.

Efron, Bradley. 1971. "Forcing a Sequential Experiment to Be Balanced." *Biometrika* 58 (3): 403–17. https://doi.org/10.1093/biomet/58.3.403.

EMA Committee for Medicinal Products for Human Use (CHMP). 2014. "Qualification Opinion of MCP-Mod as an Efficient Statistical Methodology for Model-Based Design and Analysis of Phase II Dose Finding Studies Under Model Uncertainty." European Medicines Agency. https://www.ema.europa.eu/en/documents/scientific-guideline/qualification-opinion-mcp-mod-efficient-statistical-methodology-model-based-design-analysis-phase-ii_en.pdf.

Emanuel, Ezekiel J., David Wendler, and Christine Grady. 2000. "What Makes Clinical Research Ethical?" *JAMA* 283 (20): 2701–11. https://doi.org/10.1001/jama.283.20.2701.

Epperson, Will, Gagan Bansal, Victor Dibia, Adam Fourney, Jack Gerrits, Erkang Zhu, and Saleema Amershi. 2025. "Interactive Debugging and Steering of Multi-Agent AI Systems." *arXiv Preprint arXiv:2503.02068*. https://arxiv.org/abs/2503.02068.

Erlinge, David. 2021. "Registry-Based Randomized Clinical Trials in Cardiology: Opportunities and Challenges." *REC: Interventional Cardiology* 3: 157–60. https://doi.org/10.24875/RECICE.M21000205.

Erlinge, David, Sasha Koul, Peter Eriksson, Fredrik Scherstén, Elmir Omerovic, Rikard Linder, Olof Petter Östlund, Lars Wallentin, Ole Fröbert, and Stefan James. 2016. "Bivalirudin Versus Heparin in Non-ST and ST-Segment Elevation Myocardial Infarction: A Registry-Based Randomized Clinical Trial in the SWEDEHEART Registry (the VALIDATE-SWEDEHEART Trial)." *American Heart Journal* 175: 36–46. https://doi.org/10.1016/j.ahj.2016.02.007.

Erlinge, David, Elmir Omerovic, Ole Fröbert, Rikard Linder, Mikael Danielewicz, Mehmet Hamid, Eva Swahn, et al. 2017. "Bivalirudin Versus Heparin Monotherapy in Myocardial Infarction." *New England Journal of Medicine* 377 (12): 1132–42. https://doi.org/10.1056/NEJMoa1706443.

European Commission. 2011. "EudraLex Volume 4, EU Guidelines for Good Manufacturing Practice for Medicinal Products for Human and Veterinary Use, Annex 11: Computerised Systems." https://health.ec.europa.eu/document/download/

e63bde49-af31-4d1d-8a2e-1f3f8e5f6d1c_en.

European Commission and European Medicines Agency and Heads of Medicines Agencies. 2022. "Complex Clinical Trials: Questions and Answers." https://health.ec.europa.eu/system/files/2022-06/medicinal_qa_complex_clinical-trials_en.pdf.

European Medicines Agency. 2006. "Guideline on Clinical Trials in Small Populations." https://www.ema.europa.eu/en/clinical-trials-small-populations-scientific-guideline.

———. 2014a. "Assessment Report: Trulicity (Dulaglutide)." EMA/CHMP/524604/2014, Procedure No. EMEA/H/C/002825/0000. https://www.ema.europa.eu/en/documents/assessment-report/trulicity-epar-public-assessment-report_en.pdf.

———. 2014b. "European Medicines Agency Policy on Publication of Clinical Data for Medicinal Products for Human Use (Policy 0070)." https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/clinical-data-publication.

———. 2017a. "Guideline on Good Pharmacovigilance Practices (GVP) Module IX Addendum I: Methodological Aspects of Signal Detection from Spontaneous Reports of Suspected Adverse Reactions." https://www.ema.europa.eu/en/documents/scientific-guideline/guideline-good-pharmacovigilance-practices-gvp-module-ix-addendum-i-methodological-aspects-signal-detection-spontaneous-reports-suspected-adverse-reactions_en.pdf.

———. 2017b. "Guideline on Strategies to Identify and Mitigate Risks for First-in-Human and Early Clinical Trials with Investigational Medicinal Products." EMEA/CHMP/SWP/28367/07 Rev. 1. https://www.ema.europa.eu/en/documents/scientific-guideline/guideline-strategies-identify-mitigate-risks-first-human-early-clinical-trials-investigational_en.pdf.

———. 2022. "Clinical Trials Regulation (EU) No 536/2014 and Clinical Trials Information System (CTIS)." https://www.ema.europa.eu/en/human-regulatory/research-development/clinical-trials/clinical-trials-regulation.

———. 2023. "Guideline on Computerised Systems and Electronic Data in Clinical Trials." https://www.ema.europa.eu/en/documents/regulatory-procedural-guideline/guideline-computerised-systems-and-electronic-data-clinical-trials_en.pdf.

———. 2024a. "ICH E6(R3) Good Clinical Practice Guidelines." https://www.ema.europa.eu/en/ich-e6-good-clinical-practice-scientific-guideline.

———. 2024b. "Reflection Paper on the Use of Artificial Intelligence (AI) in the Lifecycle of Medicinal Products." https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf.

———. 2025. "Draft Concept Paper on the Development of a Reflection Paper on the Use of External Controls for Evidence Generation in Regulatory Decision-Making." https://www.ema.europa.eu/en/documents/scientific-guideline/draft-concept-paper-development-reflection-paper-use-external-controls-evidence-generation-regulatory-decision-making_en.pdf.

———. 2026a. "DARWIN EU Data Partners Onboarded in Year I, II, III, IV,

and V." https://www.ema.europa.eu/en/documents/other/darwin-eu-data-partners-onboarded-phases-i-ii-iii_en.pdf.

———. 2026b. "Reflection Paper on Methodological Aspects in Confirmatory Platform Trials." https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-methodological-aspects-confirmatory-platform-trials_en.pdf.

European Medicines Agency, Committee for Medicinal Products for Human Use. 2007. "Reflection Paper on Methodological Issues in Confirmatory Clinical Trials Planned with an Adaptive Design (CHMP/EWP/2459/02)." European Medicines Agency. https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-methodological-issues-confirmatory-clinical-trials-planned-adaptive-design_en.pdf.

European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP). 2022. "Qualification Opinion for Prognostic Covariate Adjustment (PROCOVA)." European Medicines Agency. https://www.ema.europa.eu/en/documents/regulatory-procedural-guideline/qualification-opinion-prognostic-covariate-adjustment-procovatm_en.pdf.

European Parliament and Council of the European Union. 2004. "Regulation (EC) No 726/2004 Laying down Community Procedures for the Authorisation and Supervision of Medicinal Products for Human and Veterinary Use and Establishing a European Medicines Agency." https://eur-lex.europa.eu/eli/reg/2004/726/oj/eng.

———. 2014. "Regulation (EU) No 536/2014 on Clinical Trials on Medicinal Products for Human Use, and Repealing Directive 2001/20/EC." https://eur-lex.europa.eu/eli/reg/2014/536/oj/eng.

———. 2024. "Regulation (EU) 2024/1689 Laying down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)." https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng.

———. 2026. "Regulation (EU) 2026/1744 Amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as Regards the Simplification of the Implementation of Harmonised Rules on Artificial Intelligence (Digital Omnibus on AI)." http://data.europa.eu/eli/reg/2026/1744/oj.

Evans, Scott R., and Dean Follmann. 2016. "Using Outcomes to Analyze Patients Rather Than Patients to Analyze Outcomes: A Step Toward Pragmatism in Benefit:risk Evaluation." *Statistics in Biopharmaceutical Research* 8 (4): 386–93. https://doi.org/10.1080/19466315.2016.1207561.

Evans, Scott R., David Rubin, Dean Follmann, Gene Pennello, W. Charles Huskins, John H. Powers, Jerry Rhorer, et al. 2015. "Desirability of Outcome Ranking (DOOR) and Response Adjusted for Duration of Antibiotic Risk (RADAR)." *Clinical Infectious Diseases* 61 (5): 800–806. https://doi.org/10.1093/cid/civ495.

Faro Health. 2024. "Faro Health: AI-Powered Protocol Design." https://farohealth.com.

Fine, Jason P., and Robert J. Gray. 1999. "A Proportional Hazards Model for the Subdistribution of a Competing Risk." *Journal of the American Statistical Association* 94 (446): 496–509. https://doi.org/10.1080/01621459.1999.10474144.

Fischl, Margaret A., Douglas D. Richman, Michael H. Grieco, Michael S. Gottlieb, Paul A. Volberding, Oscar L. Laskin, John M. Leedom, et al. 1987. "The Efficacy of Azidothymidine (AZT) in the Treatment of Patients with AIDS and AIDS-Related Complex: A Double-Blind, Placebo-Controlled Trial." *New England Journal of Medicine* 317 (4): 185–91. https://doi.org/10.1056/NEJM198707233170401.

Fisher, Charles K. 2024. "US FDA Comments on Unlearn's PROCOVA Methodology." Unlearn.AI blog. https://www.unlearn.ai/blog/us-fda-comments-on-unlearns-procova-methodology.

Fortrea Holdings Inc. 2026. "Fortrea Reports Fourth Quarter and Full-Year 2025 Results; Issues Full-Year 2026 Guidance." https://ir.fortrea.com/news-releases/news-release-details/fortrea-reports-fourth-quarter-and-full-year-2025-results-issues.

Fortune Business Insights. 2024. "AI in Clinical Trials Market Size, Share & Industry Analysis." https://www.fortunebusinessinsights.com/ai-in-clinical-trials-market-114081.

Freedman, Benjamin. 1987. "Equipoise and the Ethics of Clinical Research." *New England Journal of Medicine* 317 (3): 141–45. https://doi.org/10.1056/NEJM198707163170304.

Friends of Cancer Research. 2023. "Considerations for Leveraging Real-World Endpoints in Oncology Drug Development." https://friendsofcancerresearch.org/wp-content/uploads/Considerations_Leveraging_Real-World_Endpoints_Oncology.pdf.

Fröbert, Ole, Bo Lagerqvist, Göran K. Olivecrona, Elmir Omerovic, Thorarinn Gudnason, Michael Maeng, Mikael Aasa, et al. 2013. "Thrombus Aspiration During ST-Segment Elevation Myocardial Infarction." *New England Journal of Medicine* 369 (17): 1587–97. https://doi.org/10.1056/NEJMoa1308789.

Garcia, Crystal J., Kevin Haynes, Sean D. Pokorney, Nancy D. Lin, Cheryl McMahill-Walraven, Vinit Nair, Lauren Parlett, et al. 2020. "Practical Challenges in the Conduct of Pragmatic Trials Embedded in Health Plans: Lessons of IMPACT-AFib, an FDA-Catalyst Trial." *Clinical Trials* 17 (4): 360–67. https://doi.org/10.1177/1740774520928426.

Gebru, Timnit, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. "Datasheets for Datasets." *Communications of the ACM* 64 (12): 86–92. https://doi.org/10.1145/3458723.

Getz, Kenneth A. 2008. "Is Investigative Site Feasibility Feasible?" *Applied Clinical Trials*. https://www.appliedclinicaltrialsonline.com/view/investigative-site-feasibility-feasible.

———. 2012. "Enrollment Performance: Weighing the 'Facts'." *Applied Clinical Trials* 21 (5). https://www.appliedclinicaltrialsonline.com/view/enrollment-performance-weighing-facts.

Getz, Kenneth A., Zachary P. Smith, and Marcy Kravet. 2023. "Protocol Design and Performance Benchmarks by Phase and by Oncology and Rare Disease Subgroups." *Therapeutic Innovation & Regulatory Science* 57 (1): 49–56. https:

//doi.org/10.1007/s43441-022-00438-5.

Getz, Kenneth A., Zachary Smith, Emily Botto, Elisabeth Murphy, and Arnaud Dauchy. 2024. "New Benchmarks on Protocol Amendment Practices, Trends and Their Impact on Clinical Trial Performance." *Therapeutic Innovation & Regulatory Science* 58 (3): 539–48. https://doi.org/10.1007/s43441-024-00622-9.

Getz, Kenneth A., Stella Stergiopoulos, Mary Short, Leon Surgeon, Randy Krauss, Sybrand Pretorius, Julian Desmond, and Derek Dunn. 2016. "The Impact of Protocol Amendments on Clinical Trial Performance and Cost." *Therapeutic Innovation & Regulatory Science* 50 (4): 436–41. https://doi.org/10.1177/2168479016632271.

Glickman, Seth W., John G. McHutchison, Eric D. Peterson, Charles B. Cairns, Robert A. Harrington, Robert M. Califf, and Kevin A. Schulman. 2009. "Ethical and Scientific Implications of the Globalization of Clinical Research." *New England Journal of Medicine* 360 (8): 816–23. https://doi.org/10.1056/NEJMsb0803929.

Goodman, Steven. 2008. "A Dirty Dozen: Twelve *P*-Value Misconceptions." *Seminars in Hematology* 45 (3): 135–40. https://doi.org/10.1053/j.seminhematol.2008.04.003.

Grand View Research. 2025. "Clinical Trials Support Software Solutions Market Size, Share & Trends Analysis Report." https://www.grandviewresearch.com/industry-analysis/clinical-trials-support-software-solutions-market-report.

———. 2026. "Healthcare Contract Research Organization Market Report, 2026-2033." https://www.grandviewresearch.com/industry-analysis/healthcare-contract-research-outsourcing-market.

Granholm, Anders, Aksel Karl Georg Jensen, Theis Lange, Anders Perner, Morten Hylander Møller, and Benjamin Skov Kaas-Hansen. 2025. "Designing and Evaluating Bayesian Advanced Adaptive Randomised Clinical Trials: A Practical Guide." *Pharmaceutical Statistics* 24 (6). https://doi.org/10.1002/pst.70042.

Griessbach, Alexandra, Benjamin Speich, Alain Amstutz, et al. 2024. "Resource Use and Costs of Investigator-Sponsored Randomized Clinical Trials in Switzerland, Germany, and the United Kingdom: A Metaresearch Study." *Journal of Clinical Epidemiology* 176: 111536. https://doi.org/10.1016/j.jclinepi.2024.111536.

Grossi, Giuliana. 2025. "FDA Quietly Removes Draft Guidance on Diversity in Clinical Trials Following Executive Order on DEI." The American Journal of Managed Care (AJMC). https://www.ajmc.com/view/fda-quietly-removes-draft-guidance-on-diversity-in-clinical-trials-following-executive-order-on-dei.

Gu, Ningzhou, Zizi Elsisi, Ryan Suk, and Meng Li. 2024. "Geographic Disparity in the Distribution of Cancer Clinical Trials in the United States and the Associated Factors." *Journal of Managed Care & Specialty Pharmacy* 30 (4): 376–85. https://doi.org/10.18553/jmcp.2024.30.4.376.

Guerra, Carmen, Alice Pressman, Patricia Hurley, Elizabeth Garrett-Mayer, et al. 2023. "Increasing Racial and Ethnic Equity, Diversity, and Inclusion in Cancer Treatment Trials: Evaluation of an ASCO-Association of Community Cancer

Centers Site Self-Assessment." *JCO Oncology Practice* 19 (4): e581–88. https://doi.org/10.1200/OP.22.00560.

H1. 2024. "H1: Healthcare Provider Intelligence Platform." https://h1.co.

Haddad, Tufia C., Lydia A. Mercado, Jennifer G. Le-Rademacher, David A. Enarson, et al. 2026. "A Multiregional Decentralized Clinical Trial Program to Improve Access." *JAMA Network Open* 9 (6): e2617204. https://doi.org/10.1001/jamanetworkopen.2026.17204.

Han, Kiki, Franck Le Deu, Fangning Zhang, and Josie Zhou. 2021. "The Dawn of China Biopharma Innovation." https://www.mckinsey.com/industries/life-sciences/our-insights/the-dawn-of-china-biopharma-innovation.

Harris Interactive. 2004. "Most Physicians Do Not Participate in Clinical Trials Because of Lack of Opportunity, Time, Personnel Support and Resources." https://www.biospace.com/b-harris-interactive-b-release-most-physicians-do-not-participate-in-clinical-trials-because-of-lack-of-opportunity-time-personnel-support-and-r.

Health Level Seven International. 2024. "HL7 FHIR Bulk Data Access (Flat FHIR)." https://hl7.org/fhir/uv/bulkdata/.

Hernán, Miguel A., and James M. Robins. 2016. "Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available." *American Journal of Epidemiology* 183 (8): 758–64. https://doi.org/10.1093/aje/kwv254.

Hobbs, Brian P., Bradley P. Carlin, Sumithra J. Mandrekar, and Daniel J. Sargent. 2011. "Hierarchical Commensurate and Power Prior Models for Adaptive Incorporation of Historical Information in Clinical Trials." *Biometrics* 67 (3): 1047–56. https://doi.org/10.1111/j.1541-0420.2011.01564.x.

Hripcsak, George, and David J. Albers. 2013. "Next-Generation Phenotyping of Electronic Health Records." *Journal of the American Medical Informatics Association* 20 (1): 117–21. https://pmc.ncbi.nlm.nih.gov/articles/PMC3555337/.

Hripcsak, George, Jon D. Duke, Nigam H. Shah, Christian G. Reich, Vojtech Huser, Martijn J. Schuemie, Marc A. Suchard, et al. 2015. "Observational Health Data Sciences and Informatics (OHDSI): Opportunities for Observational Researchers." *Studies in Health Technology and Informatics*. https://pmc.ncbi.nlm.nih.gov/articles/PMC4815923/.

Hu, Feifang, and William F. Rosenberger. 2006. *The Theory of Response-Adaptive Randomization in Clinical Trials*. Hoboken, NJ: John Wiley & Sons. https://doi.org/10.1002/047005588X.

Hudson, Marie, Antonio Avina-Zubieta, Diane Lacaille, Sasha Bernatsky, Lisa Lix, and Sonia Jean. 2013. "The Validity of Administrative Data to Identify Hip Fractures Is High: A Systematic Review." *Journal of Clinical Epidemiology* 66 (3): 278–85. https://doi.org/10.1016/j.jclinepi.2012.10.004.

Huey, Ryan W., Goldy C. George, Penny Phillips, Revenda White, Siqing Fu, Filip Janku, Daniel D. Karp, et al. 2021. "Patient-Reported Out-of-Pocket Costs and Financial Toxicity During Early-Phase Oncology Clinical Trials." *The Oncologist* 26 (7): 588–96. https://doi.org/10.1002/onco.13767.

Ibrahim, Joseph G., and Ming-Hui Chen. 2000. "Power Prior Distributions for Regression Models." *Statistical Science* 15 (1): 46–60. https://doi.org/10.1214/

ss/1009212673.

ICON plc. 2026. "Annual Report on Form 20-F for the Fiscal Year Ended December 31, 2025." https://investor.iconplc.com/static-files/ce9930d9-1163-4cbe-907e-540cb4a46d01.

Industry Standard Research (ISR). 2025. "EDC Benchmarking and Market Dynamics (6th Edition)." https://isrreports.com/reports/2025-edc-benchmarking/.

International Council for Harmonisation. 1994a. "ICH Harmonised Tripartite Guideline: Clinical Safety Data Management: Definitions and Standards for Expedited Reporting E2A." https://database.ich.org/sites/default/files/E2A_Guideline.pdf.

———. 1994b. "ICH Harmonised Tripartite Guideline: The Extent of Population Exposure to Assess Clinical Safety for Drugs Intended for Long-Term Treatment of Non-Life-Threatening Conditions E1." https://database.ich.org/sites/default/files/E1_Guideline.pdf.

———. 1995. "ICH Harmonised Tripartite Guideline: Structure and Content of Clinical Study Reports E3." https://database.ich.org/sites/default/files/E3_Guideline.pdf.

———. 1996. "ICH Harmonised Tripartite Guideline: Guideline for Good Clinical Practice E6(R1)." https://www.ema.europa.eu/en/documents/scientific-guideline/ich-e6-r1-guideline-good-clinical-practice_en.pdf.

———. 1998a. "ICH Harmonised Tripartite Guideline: Ethnic Factors in the Acceptability of Foreign Clinical Data E5(R1)." https://www.ema.europa.eu/en/documents/scientific-guideline/ich-e-5-r1-ethnic-factors-acceptability-foreign-clinical-data-step-5_en.pdf.

———. 1998b. "ICH Harmonised Tripartite Guideline: Statistical Principles for Clinical Trials E9." https://database.ich.org/sites/default/files/E9_Guideline.pdf.

———. 2010. "ICH Harmonised Tripartite Guideline: Development Safety Update Report E2F." https://database.ich.org/sites/default/files/E2F_Guideline.pdf.

———. 2013. "ICH E2B(R3): Electronic Transmission of Individual Case Safety Reports (ICSRs): Data Elements and Message Specification." https://ich.org/page/e2br3-individual-case-safety-report-icsr-specification-and-related-files.

———. 2015. "ICH M8: Electronic Common Technical Document (eCTD), Version 4.0." https://www.ich.org/page/ich-electronic-common-technical-document-ectd-v40.

———. 2016a. "ICH E6(R2): Integrated Addendum to ICH E6(R1) - Guideline for Good Clinical Practice." https://database.ich.org/sites/default/files/E6_R2_Addendum.pdf.

———. 2016b. "ICH M4: Organisation of the Common Technical Document for the Registration of Pharmaceuticals for Human Use." https://www.ich.org/page/ctd.

———. 2017. "ICH Harmonised Guideline: General Principles for Planning and Design of Multi-Regional Clinical Trials E17." https://www.ema.europa.eu/en/documents/scientific-guideline/ich-guideline-e17-general-principles-planning-and-design-multi-regional-clinical-trials-step-5-first-version_en.pdf.

———. 2019. "ICH E9(R1): Addendum on Estimands and Sensitivity Analysis in Clinical Trials to the Guideline on Statistical Principles for Clinical Trials." https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf.

———. 2021. "ICH E8(R1): General Considerations for Clinical Studies." https://database.ich.org/sites/default/files/ICH_E8-R1_Guideline_Step4_2021_1006.pdf.

———. 2025a. "E20 Adaptive Designs for Clinical Trials." https://www.ema.europa.eu/en/ich-e20-adaptive-designs-clinical-trials-scientific-guideline.

———. 2025b. "ICH E6(R3): Guideline for Good Clinical Practice (GCP)." https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf.

———. 2025c. "ICH M11: Clinical Electronic Structured Harmonised Protocol (CeSHarP)." https://database.ich.org/sites/default/files/ICH_Step4_M11_Final_Guideline_2025_1119.pdf.

IQVIA. 2024a. "Decentralized Clinical Trials Solutions." https://www.iqvia.com/solutions/research-and-development/decentralized-trials.

———. 2024b. "IQVIA Intelligent eTMF." https://www.iqvia.com/library/fact-sheets/iqvia-intelligent-etmf.

IQVIA Holdings Inc. 2026. "Annual Report on Form 10-K for the Fiscal Year Ended December 31, 2025." https://s201.q4cdn.com/580005511/files/doc_financials/2025/q4/IQV-2025-12-31-10K_Filed-with-exhibits.pdf.

IQVIA Institute for Human Data Science. 2024a. "Global Trends in R&D 2024: Activity, Productivity, and Enablers." https://www.iqvia.com/insights/the-iqvia-institute/reports-and-publications/reports/global-trends-in-r-and-d-2024-activity-productivity-and-enablers.

———. 2024b. "Global Use of Medicines 2024: Outlook to 2028." https://www.iqvia.com/insights/the-iqvia-institute/reports-and-publications/reports/the-global-use-of-medicines-2024-outlook-to-2028.

———. 2025. "Global Trends in R&D 2025." https://iqvia.com/insights/the-iqvia-institute/reports-and-publications/reports/global-trends-in-r-and-d-2025.

James, Nicholas. 2013. "Systemic Therapy in Advanced or Metastatic Prostate cancer: Evaluation of Drug Efficacy." https://doi.org/10.1186/ISRCTN78818544.

James, Nicholas D., Johann S. de Bono, Melissa R. Spears, Noel W. Clarke, Malcolm D. Mason, David P. Dearnaley, Alastair W. S. Ritchie, et al. 2017. "Abiraterone for Prostate Cancer Not Previously Treated with Hormone Therapy." *New England Journal of Medicine* 377 (4): 338–51. https://doi.org/10.1056/NEJMoa1702900.

James, Nicholas D, Matthew R Sydes, Noel W Clarke, Malcolm D Mason, David P Dearnaley, Melissa R Spears, Alastair W S Ritchie, et al. 2016. "Addition of docetaxel, zoledronic acid, or both to first-line long-term hormone therapy in prostate cancer (STAMPEDE): survival results from an adaptive, multiarm, multistage, platform randomised controlled trial." *The Lancet* 387 (10024): 1163–77. https://doi.org/10.1016/S0140-6736(15)01037-5.

Jefferson, Tom, Mark Jones, Peter Doshi, Elizabeth A. Spencer, Igho Onakpoya,

and Carl J. Heneghan. 2014. "Oseltamivir for Influenza in Adults and Children: Systematic Review of Clinical Study Reports and Summary of Regulatory Comments." *BMJ* 348: g2545. https://doi.org/10.1136/bmj.g2545.

Jennison, Christopher, and Bruce W. Turnbull. 2000. *Group Sequential Methods with Applications to Clinical Trials*. Boca Raton, FL: Chapman & Hall/CRC.

———. 2015. "Adaptive Sample Size Modification in Clinical Trials: Start Small Then Ask for More?" *Statistics in Medicine* 34 (29): 3793–3810. https://doi.org/10.1002/sim.6575.

Jiang, Ziren, Chengxing (Cindy) Lu, Jialing Liu, Satrajit Roychoudhury, Daniel Meyer, Bo Huang, and Haitao Chu. 2023. "Nonconcurrent Controls in Platform Trials: Can We Borrow Their Concurrent Observation Data?" *Statistics in Biopharmaceutical Research* 17 (1): 17–26. https://doi.org/10.1080/19466315.2023.2267502.

Junod, Suzanne White. 2008. "FDA and Clinical Drug Trials: A Short History." U.S. Food and Drug Administration. https://www.fda.gov/media/110437/download.

Kaplan, E. L., and Paul Meier. 1958. "Nonparametric Estimation from Incomplete Observations." *Journal of the American Statistical Association* 53 (282): 457–81. https://doi.org/10.1080/01621459.1958.10501452.

Kapoor, Sayash, Benedikt Stroebl, Zachary S. Siegel, Nitya Nadgir, and Arvind Narayanan. 2024. "AI Agents That Matter." *arXiv Preprint arXiv:2407.01502*. https://arxiv.org/abs/2407.01502.

Kasenda, Benjamin, Erik von Elm, John You, Anette Blümle, Yuki Tomonaga, Ramon Saccilotto, Alain Amstutz, et al. 2014. "Prevalence, Characteristics, and Publication of Discontinued Randomized Trials." *JAMA* 311 (10): 1045–51. https://doi.org/10.1001/jama.2014.1361.

Khattab, Omar, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, et al. 2023. "DSPy: Compiling Declarative Language Model Calls into Self-Improving Pipelines." *arXiv Preprint arXiv:2310.03714*. https://arxiv.org/abs/2310.03714.

Kim, Edward S., Roy S. Herbst, Ignacio I. Wistuba, J. Jack Lee, George R. Blumenschein, Anne Tsao, David J. Stewart, et al. 2011. "The BATTLE Trial: Personalizing Therapy for Lung Cancer." *Cancer Discovery* 1 (1): 44–53. https://doi.org/10.1158/2159-8274.CD-10-0010.

King, Andrew J., Lisa Higgins, Carly Au, Salim Malakouti, Edvin Music, Kyle Kalchthaler, et al. 2024. "Automatic Population of the Case Report Forms for an International Multifactorial Adaptive Platform Trial Amid the COVID-19 Pandemic." *AMIA Joint Summits on Translational Science Proceedings* 2024: 276–84. https://pmc.ncbi.nlm.nih.gov/articles/PMC11141839/.

Kiriakis, James, Nicholas Gaich, S. Claiborne Johnston, Darlene Kitterman, Daniel Rosenblum, Libby Salberg, and Adam Rifkind. 2013. "Observational Study of Contracts Processing at 29 CTSA Sites." *Clinical and Translational Science* 6 (4): 279–85. https://doi.org/10.1111/cts.12073.

Kisker, C. Thomas et al. 1997. "Shared Management of Children with Cancer." *Archives of Pediatrics & Adolescent Medicine* 151 (10): 1008–13. https://doi.

org/10.1001/archpedi.1997.02170470042008.

Knight, Lisa, Barbara A. Pahud, Margaret Scheffler, Joshua C. Euteneuer, et al. 2022. "Capacity Building in a New Clinical Trials Network Through Inter-Network Collaboration." *The Journal of Pediatrics* 240: 5–8. https://doi.org/10.1016/j.jpeds.2021.04.062.

Korn, Edward L., and Boris Freidlin. 2011. "Outcome-Adaptive Randomization: Is It Useful?" *Journal of Clinical Oncology* 29 (6): 771–76. https://doi.org/10.1200/JCO.2010.31.1423.

Krams, Michael, Kennedy R. Lees, Werner Hacke, Andrew P. Grieve, Jean-Marc Orgogozo, and Gary A. Ford. 2003. "Acute Stroke Therapy by Inhibition of Neutrophils (ASTIN): An Adaptive Dose-Response Study of UK-279,276 in Acute Ischemic Stroke." *Stroke* 34 (11): 2543–48. https://doi.org/10.1161/01.STR.0000092527.33910.89.

Krotka, Pavla, Marta Bofill Roig, Katharina Hees, Peter Jacko, and Dominic Magirr. 2023. "NCC: Simulation and Analysis of Platform Trials with Non-Concurrent Controls." https://doi.org/10.32614/CRAN.package.ncc.

Kumar, Anivarya, Jennifer R. Owen, Nicholette T. Sloat, Elizabeth Maynard, et al. 2024. "Expansion of an Academic Molecular Tumor Board to Enhance Access to Biomarker-Driven Trials and Therapies in the Rural Southeastern United States." *Current Oncology* 31 (11): 7244–57. https://doi.org/10.3390/curroncol31110534.

Lamberti, Mary Jo, Abigail Dirks, Nicholas Kikuchi, Neha Patel Cervantes, and Kenneth Getz. 2024. "Benchmarking Site Activation and Patient Enrollment." *Therapeutic Innovation & Regulatory Science* 58 (4): 696–703. https://doi.org/10.1007/s43441-024-00638-1.

Lan, K. K. Gordon, and David L. DeMets. 1983. "Discrete Sequential Boundaries for Clinical Trials." *Biometrika* 70 (3): 659–63. https://doi.org/10.1093/biomet/70.3.659.

Landgren, Ola, and Sean M. Devlin. 2025. "Minimal Residual Disease as an Early Endpoint for Accelerated Drug Approval in Myeloma: A Roadmap." *Blood Cancer Discovery* 6 (1): 13–22. https://doi.org/10.1158/2643-3230.BCD-24-0292.

Laranjo, Liliana, Adam G. Dunn, Huong Ly Tong, A. Baki Kocaballi, Jessica Chen, Rabia Bashir, Didi Surian, et al. 2018. "Conversational Agents in Healthcare: A Systematic Review." *Journal of the American Medical Informatics Association* 25 (9): 1248–58. https://pmc.ncbi.nlm.nih.gov/articles/PMC6118869/.

Laubenbacher, Reinhard, James P. Sluka, and James A. Glazier. 2021. "Using Digital Twins in Viral Infection." *Science* 371 (6534): 1105–6. https://doi.org/10.1126/science.abf3370.

Lavori, Philip W., and Ree Dawson. 2000. "A Design for Testing Clinical Strategies: Biased Adaptive Within-Subject Randomization." *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 163 (1): 29–38. https://doi.org/10.1111/1467-985X.00154.

Lawrence, Colleen E., Virginia M. Bruce, Libby D. Salberg, Terri Edwards, Casi Morales, Marisha Palm, and Gordon R. Bernard. 2023. "Quantitative Assessment of the Impact of Standard Agreement Templates on Multisite Clinical

Trial Start up Time." *Journal of Clinical and Translational Science* 7 (1): e204. https://doi.org/10.1017/cts.2023.622.

Le Tourneau, Christophe, J. Jack Lee, and Lillian L. Siu. 2009. "Dose Escalation Methods in Phase i Cancer Clinical Trials." *Journal of the National Cancer Institute* 101 (10): 708–20. https://doi.org/10.1093/jnci/djp079.

Lee, Kim May, and James Wason. 2020. "Including non-concurrent control patients in the analysis of platform trials: is it worth it?" *BMC Medical Research Methodology* 20 (1). https://doi.org/10.1186/s12874-020-01043-6.

Lewis, Patrick, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, et al. 2020. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." *Advances in Neural Information Processing Systems* 33: 9459–74. https://arxiv.org/abs/2005.11401.

Lin, Jianchang, and Veronica Bunn. 2017. "Comparison of multi-arm multi-stage design and adaptive randomization in platform clinical trials." *Contemporary Clinical Trials* 54: 48–59. https://doi.org/10.1016/j.cct.2017.01.003.

Lin, Ruitao, and Guosheng Yin. 2017. "Bayesian Optimal Interval Design for Dose Finding in Drug-Combination Trials." *Statistical Methods in Medical Research* 26 (5): 2155–67. https://doi.org/10.1177/0962280215594494.

Lin, Yunzhi, Mumu Zhu, and Zheng Su. 2015. "The Pursuit of Balance: An Overview of Covariate-Adaptive Randomization Techniques in Clinical Trials." *Contemporary Clinical Trials* 45: 21–25. https://doi.org/10.1016/j.cct.2015.07.011.

Lipsitch, Marc, Eric Tchetgen Tchetgen, and Ted Cohen. 2010. "Negative Controls: A Tool for Detecting Confounding and Bias in Observational Studies." *Epidemiology* 21 (3): 383–88. https://doi.org/10.1097/EDE.0b013e3181d61eeb.

Liu, Suyu, and Ying Yuan. 2015. "Bayesian Optimal Interval Designs for Phase I Clinical Trials." *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 64 (3): 507–23. https://doi.org/10.1111/rssc.12089.

Liu, Xiaoxuan et al. 2020. "CONSORT-AI Extension: Reporting Guidelines for Clinical Trials Evaluating Artificial Intelligence Interventions." *BMJ* 370: m3164. https://www.bmj.com/content/370/bmj.m3164.

Lopez Bernal, Jamie, Nick Andrews, Charlotte Gower, Eileen Gallagher, Ruth Simmons, Simon Thelwall, Julia Stowe, et al. 2021. "Effectiveness of Covid-19 Vaccines Against the B.1.617.2 (Delta) Variant." *New England Journal of Medicine* 385 (7): 585–94. https://doi.org/10.1056/NEJMoa2108891.

Ma, Wei, Ping Li, Li-Xin Zhang, and Feifang Hu. 2024. "A New and Unified Family of Covariate Adaptive Randomization Procedures and Their Properties." *Journal of the American Statistical Association* 119 (545): 151–62. https://doi.org/10.1080/01621459.2022.2102986.

MacDougall, James. 2006. "Analysis of Dose-Response Studies: Emax Model." In *Dose Finding in Drug Development*, edited by Naitee Ting, 127–45. Springer. https://doi.org/10.1007/0-387-33706-7_9.

Mandel, Joshua C., David A. Kreda, Kenneth D. Mandl, Isaac S. Kohane, and Rachel B. Ramoni. 2016. "SMART on FHIR: A Standards-Based, Interoperable Apps Platform for Electronic Health Records." *Journal of the American Medical*

Informatics Association 23 (5): 899–908. https://pmc.ncbi.nlm.nih.gov/articles/PMC4997036/.

Mander, Adrian P., and Michael J. Sweeting. 2015. "A Product of Independent Beta Probabilities Dose Escalation Design for Dual-Agent Phase I Trials." *Statistics in Medicine* 34 (8): 1261–76. https://doi.org/10.1002/sim.6434.

Marion, Joe, Elizabeth Lorenzi, Cora Allen-Savietta, Scott Berry, and Kert Viele. 2025. "Predictive Probabilities Made Simple: A Fast and Accurate Method for Clinical Trial Decision-Making." *Statistics in Medicine* 44 (13-14): e70120. https://doi.org/10.1002/sim.70120.

MarketsandMarkets. 2024. "Electronic Trial Master File System Market." https://www.marketsandmarkets.com/Market-Reports/electronic-trial-master-file-system-market-94357456.html.

McCaskill-Stevens, Worta. 2022. "Lessons Learned from the National Cancer Institute's Community-Based Clinical Trials." Curtis Meinert lecture, Society for Clinical Trials annual meeting, May 16, 2022. https://sctweb.org/presentations/2022/Curtis_Meinert_2022.pdf.

Medable. 2024a. "Medable Releases New AI Agent to Automate Trial Master File Processing." https://www.biopharmatrend.com/news/medable-releases-new-ai-agent-to-automate-trial-master-file-processing-1447/.

———. 2024b. "Medable: Decentralized Clinical Trial Platform." https://www.medable.com.

———. 2026. "Medable Clinical Trial Platform." https://www.medable.com.

Medicenna Therapeutics Corp. 2020. "Medicenna Provides MDNA55 rGBM Clinical Program Update Following Positive End of Phase 2 Meeting with the U.S. Food and Drug Administration (FDA)." Press release, October 15, 2020. https://ir.medicenna.com/news-releases/news-release-details/medicenna-provides-mdna55-rgbm-clinical-program-update-following/.

Medicines and Healthcare products Regulatory Agency. 2025. "MHRA Draft Guideline on the Use of External Control Arms Based on Real-World Data to Support Regulatory Decisions." https://www.gov.uk/government/consultations/mhra-draft-guideline-on-the-use-of-external-control-arms-based-on-real-world-data-to-support-regulatory-decisions.

Medidata Solutions. 2020. "Medidata Synthetic Control Arm Supported by the US Food and Drug Administration (FDA) for Use in Medicenna Therapeutics, Corp. Phase 3 Registrational Trial in Recurrent Glioblastoma." Press release (Business Wire), October 28, 2020. https://www.businesswire.com/news/home/20201028005277/en/Medidata-Synthetic-Control-Arm-Supported-by-the-US-Food-and-Drug-Administration-FDA-for-Use-in-Medicenna-Therapeutics-Corp.-Phase-3-Registrational-Trial-in-Recurrent-Glioblastoma.

———. 2024a. "Everest Group EDC PEAK Matrix Assessment." https://www.medidata.com/en/everest-group-edc-peak-matrix/.

———. 2024b. "Medidata Rave EDC." https://www.medidata.com/en/clinical-trial-products/clinical-data-management/edc-systems.

———. 2024c. "Medidata Sensor Cloud: Wearable Sensors." https://www.medidata.com/en/patient-experience/wearable-sensors/.

———. 2024d. "Medidata Synthetic Control Arm." https://medidata.com/en/clinical-trial-products/medidata-ai/real-world-data/synthetic-control-arm.

Medpace Holdings, Inc. 2026. "Medpace Holdings, Inc. Reports Fourth Quarter and Full Year 2025 Results." https://investor.medpace.com/news-releases/news-release-details/medpace-holdings-inc-reports-fourth-quarter-and-full-year-2025/.

Mehta, Cyrus R., and Stuart J. Pocock. 2011. "Adaptive Increase in Sample Size When Interim Results Are Promising: A Practical Guide with Examples." *Statistics in Medicine* 30 (28): 3267–84. https://doi.org/10.1002/sim.4102.

Mei, Kai, Xi Zhu, Wujiang Xu, Wenyue Hua, Mingyu Jin, Zelong Li, Shuyuan Xu, Ruosong Ye, Yingqiang Ge, and Yongfeng Zhang. 2024. "AIOS: LLM Agent Operating System." *arXiv Preprint arXiv:2403.16971*. https://arxiv.org/abs/2403.16971.

Meinert, Curtis L. 2013. *Clinical Trials Handbook: Design and Conduct*. Hoboken, NJ: John Wiley & Sons.

Mello, Michelle M., Brian R. Clarridge, and David M. Studdert. 2005. "Academic Medical Centers' Standards for Clinical-Trial Agreements with Industry." *New England Journal of Medicine* 352 (21): 2202–10. https://doi.org/10.1056/NEJMsa044115.

Messeri, Lisa, and M. J. Crockett. 2024. "Artificial Intelligence and Illusions of Understanding in Scientific Research." *Nature*. https://www.nature.com/articles/s41586-024-07146-0.

Meyer, Elias Laurin, Peter Mesenbrink, Cornelia Dunger-Baldauf, Hans-Jürgen Fülle, Ekkehard Glimm, Yuhan Li, Martin Posch, and Franz König. 2020. "The Evolution of Master Protocol Clinical Trial Designs: A Systematic Literature Review." *Clinical Therapeutics* 42 (7): 1330–60. https://doi.org/10.1016/j.clinthera.2020.05.010.

Ministry of Health, Labour and Welfare. 2024. "Revision of Notification on Handling of Submission of Electronic Study Data for New Drug Applications." https://www.pmda.go.jp/files/000267942.pdf.

Mitchell, Margaret, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. 2019. "Model Cards for Model Reporting." In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 220–29. https://doi.org/10.1145/3287560.3287596.

Moore, Thomas J., Hanzhe Zhang, Gerard Anderson, and G. Caleb Alexander. 2018. "Estimated Costs of Pivotal Trials for Novel Therapeutic Agents Approved by the US Food and Drug Administration, 2015-2016." *JAMA Internal Medicine* 178 (11): 1451–57. https://doi.org/10.1001/jamainternmed.2018.3931.

Morris, Tim P., Ian R. White, and Michael J. Crowther. 2019. "Using simulation studies to evaluate statistical methods." *Statistics in Medicine* 38 (11): 2074–2102. https://doi.org/10.1002/sim.8086.

Müller, Hans-Helge, and Helmut Schäfer. 2001. "Adaptive Group Sequential Designs for Clinical Trials: Combining the Advantages of Adaptive and of Classical Group Sequential Approaches." *Biometrics* 57 (3): 886–91. https:

//doi.org/10.1111/j.0006-341X.2001.00886.x.

Murphy, Susan A. 2005. "An Experimental Design for the Development of Adaptive Treatment Strategies." *Statistics in Medicine* 24 (10): 1455–81. https://doi.org/10.1002/sim.2022.

Muss, Hyman B., Donald A. Berry, Constance T. Cirrincione, Maria Theodoulou, Ann M. Mauer, Alice B. Kornblith, Ann H. Partridge, et al. 2009. "Adjuvant Chemotherapy in Older Women with Early-Stage Breast Cancer." *New England Journal of Medicine* 360 (20): 2055–65. https://doi.org/10.1056/NEJMoa0810266.

Nahum-Shani, Inbal, Ashkan Ertefaie, Xi Lu, Kevin G. Lynch, James R. McKay, David W. Oslin, and Daniel Almirall. 2017. "A SMART Data Analysis Method for Constructing Adaptive Treatment Strategies for Substance Use Disorders." *Addiction* 112 (5): 901–9. https://doi.org/10.1111/add.13743.

Nanda, Rita, Minetta C. Liu, Christina Yau, Rebecca Shatsky, Lajos Pusztai, Anne Wallace, A. Jo Chien, et al. 2020. "Effect of Pembrolizumab Plus Neoadjuvant Chemotherapy on Pathologic Complete Response in Women With Early-Stage Breast Cancer." *JAMA Oncology* 6 (5): 676. https://doi.org/10.1001/jamaoncol.2019.6650.

National Association of Chronic Disease Directors. 2026. "Using Project ECHO Cancer Clinical Trials to Increase Participation and Access to Cancer Clinical Trials in Pennsylvania." https://chronicdisease.org/success-story/using-project-echo-cancer-clinical-trials-increase-participation-and-access-to-cancer-clinical-trials-in-pennsylvania-2/.

National Cancer Advisory Board. 2024. "Clinical Investigations Subcommittee Meeting Summary, June 2024." https://www.cancer.gov/about-nci/advisory-boards/ncab/subcommittees/ncab-clinical-investigations-subcom-summary-june-2024.pdf.

National Cancer Institute. 2019. "NCI Community Oncology Research Program (NCORP)." https://www.cancer.gov/research/infrastructure/clinical-trials/ncorp.

National Cancer Institute, Center for Biomedical Informatics and Information Technology. 2026. "Data Elements for Cancer Research." https://www.cancer.gov/about-nci/organization/cbiit/vocabulary/data-elements-for-cancer-research.

National Cancer Institute, Division of Cancer Treatment and Diagnosis. 2026. "Pediatric Networks." https://dctd.cancer.gov/research/networks/pediatric.

National Center for Advancing Translational Sciences. 2026. "Rare Diseases Clinical Research Network (RDCRN)." https://ncats.nih.gov/research/research-activities/rdcrn.

National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. 1979. "The Belmont Report: Ethical Principles and Guidelines for the Protection of Human Subjects of Research." https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/read-the-belmont-report/index.html.

National Institute for Health and Care Excellence. 2022. "NICE Technology Appraisal and Highly Specialised Technologies Guidance: The Manual." https:

//www.nice.org.uk/process/pmg36.

National Institute of Standards and Technology. 2023. "Artificial Intelligence Risk Management Framework (AI RMF 1.0)." https://www.nist.gov/itl/ai-risk-management-framework.

National Institutes of Health. 2018. "Single IRB Policy for Multi-Site Research." https://grants.nih.gov/policy/humansubjects/single-irb-policy-multi-site-research.htm.

———. 2023. "NIH Policy for Data Management and Sharing." https://sharing.nih.gov/data-management-and-sharing-policy/about-data-management-and-sharing-policies.

National Library of Medicine. 2024. "NIH Common Data Elements (CDE) Repository." https://cde.nlm.nih.gov/.

Neuenschwander, Beat, Michael Branson, and Thomas Gsponer. 2008. "Critical Aspects of the Bayesian Approach to Phase I Cancer Trials." *Statistics in Medicine* 27 (13): 2420–39. https://doi.org/10.1002/sim.3230.

Neuenschwander, Beat, Gorana Capkun-Niggli, Michael Branson, and David J. Spiegelhalter. 2010. "Summarizing Historical Information on Controls in Clinical Trials." *Clinical Trials* 7 (1): 5–18. https://doi.org/10.1177/1740774509356002.

NextTrial. 2024. "NextTrial AI: Clinical Trial Intelligence." https://www.nexttrial.ai.

NIH Environmental influences on Child Health Outcomes (ECHO) Program. 2026. "IDeA States Pediatric Clinical Trials Network." https://echochildren.org/idea-states-pediatric-clinical-trials-network/.

Nogueira, Raul G., Ashutosh P. Jadhav, Diogo C. Haussen, Alain Bonafe, Ronald F. Budzik, Parita Bhuva, Dileep R. Yavagal, et al. 2018. "Thrombectomy 6 to 24 Hours after Stroke with a Mismatch between Deficit and Infarct." *New England Journal of Medicine* 378 (1): 11–21. https://doi.org/10.1056/NEJMoa1706442.

Nuremberg Military Tribunal. 1949. "The Nuremberg Code." https://collections.nlm.nih.gov/catalog/nlm:nlmuid-01130400RX2-mvpart.

O'Brien, Peter C., and Thomas R. Fleming. 1979. "A Multiple Testing Procedure for Clinical Trials." *Biometrics* 35 (3): 549–56. https://doi.org/10.2307/2530245.

O'Hagan, Anthony, John W. Stevens, and Michael J. Campbell. 2005. "Assurance in Clinical Trial Design." *Pharmaceutical Statistics* 4 (3): 187–201. https://doi.org/10.1002/pst.175.

O'Quigley, John, Margaret Pepe, and Lloyd Fisher. 1990. "Continual Reassessment Method: A Practical Design for Phase 1 Clinical Trials in Cancer." *Biometrics* 46 (1): 33–48. https://doi.org/10.2307/2531628.

Obermeyer, Ziad, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. 2019. "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations." *Science* 366 (6464): 447–53. https://doi.org/10.1126/science.aax2342.

Octozi. 2024. "Octozi: AI for Clinical Operations." https://www.octozi.com.

Ogier du Terrail, Jean, Quentin Klopfenstein, Honghao Li, Imke Mayer, Nicolas Loiseau, Mohammad Hallal, Michael Debouver, Thibault Camalon, et al. 2025.

"FedECA: Federated External Control Arms for Causal Inference with Time-to-Event Data in Distributed Settings." *Nature Communications* 16 (1): 7496. https://doi.org/10.1038/s41467-025-62525-z.

Overhage, J. Marc, Patrick B. Ryan, Christian G. Reich, Abraham G. Hartzema, and Paul E. Stang. 2012. "Validation of a Common Data Model for Active Safety Surveillance Research." *Journal of the American Medical Informatics Association* 19 (1): 54–60. https://doi.org/10.1136/amiajnl-2011-000376.

Pallmann, Philip, Alun W. Bedding, Babak Choodari-Oskooei, Munyaradzi Dimairo, Laura Flight, Lisa V. Hampson, Jane Holmes, et al. 2018. "Adaptive Designs in Clinical Trials: Why Use Them, and How to Run and Report Them." *BMC Medicine* 16 (1): 29. https://doi.org/10.1186/s12916-018-1017-7.

PANTHER Trial Investigators. 2024. "PANTHER: Precision Medicine Adaptive Network Platform Trial in Hypoxaemic Acute Respiratory Failure." https://www.panthertrial.org.

Park, Jay J. H, Ellie Siden, Michael J. Zoratti, Louis Dron, Ofir Harari, Joel Singer, Richard T. Lester, Kristian Thorlund, and Edward J. Mills. 2019. "Systematic review of basket trials, umbrella trials, and platform trials: a landscape analysis of master protocols." *Trials* 20 (1). https://doi.org/10.1186/s13063-019-3664-1.

Park, John W., Minetta C. Liu, Douglas Yee, Christina Yau, Laura J. van 't Veer, W. Fraser Symmans, Melissa Paoloni, et al. 2016. "Adaptive Randomization of Neratinib in Early Breast Cancer." *New England Journal of Medicine* 375 (1): 11–22. https://doi.org/10.1056/NEJMoa1513750.

Park, Joon Sung, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. "Generative Agents: Interactive Simulacra of Human Behavior." In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. https://doi.org/10.1145/3586183.3606763.

Parmar, Mahesh K B, James Carpenter, and Matthew R Sydes. 2014. "More multiarm randomised trials of superiority are needed." *The Lancet* 384 (9940): 283–84. https://doi.org/10.1016/S0140-6736(14)61122-3.

PCORnet. 2025. "PCORnet Common Data Model (CDM) Specification, Version 7.0." https://pcornet.org/wp-content/uploads/2025/05/PCORnet_Common_Data_Model_v70_2025_05_01.pdf.

Pearl, Judea. 2009. *Causality: Models, Reasoning, and Inference*. 2nd ed. Cambridge: Cambridge University Press.

Pediatric Brain Tumor Consortium. 2026. "Pediatric Brain Tumor Consortium." https://www.pbtc.org/.

Pediatric Oncology Experimental Therapeutics Investigators' Consortium. 2014. "Protocol POE10-02: Vorinostat with Etoposide in Pediatric Patients (Amended March 25, 2014)." Protocol posted on ClinicalTrials.gov, NCT01294670. https://cdn.clinicaltrials.gov/large-docs/70/NCT01294670/Prot_SAP_000.pdf.

———. 2026a. "About POETIC." https://poeticphase1.org/about/.

———. 2026b. "POETIC Bi-Annual Meeting." https://poeticphase1.org/bi-annual-meeting/.

———. 2026c. "POETIC Clinical Trials." https://poeticphase1.org/programs/

clinical-trials/.

———. 2026d. "POETIC Home Page." https://poeticphase1.org/.

———. 2026e. "POETIC Member Institutions." https://poeticphase1.org/members/.

———. 2026f. "Research Development and Management Center (RDMC)." https://poeticphase1.org/rdmc/.

Pelham, William E., Gregory A. Fabiano, James G. Waxmonsky, Andrew R. Greiner, Elizabeth M. Gnagy, et al. 2016. "Treatment Sequencing for Childhood ADHD: A Multiple-Randomization Study of Adaptive Medication and Behavioral Interventions." *Journal of Clinical Child & Adolescent Psychology* 45 (4): 396–415. https://doi.org/10.1080/15374416.2015.1105138.

Pharmaceutical Research and Manufacturers of America. 2024. "The Economic Impact of the u.s. Biopharmaceutical Industry." https://phrma.org/resource-center/topics/economic-impact/industry-economic-impact.

Phlexglobal. 2024. "Phlexglobal eTMF Software." https://www.phlexglobal.com/etmf-software.

Pocock, Stuart J. 1977. "Group Sequential Methods in the Design and Analysis of Clinical Trials." *Biometrika* 64 (2): 191–99. https://doi.org/10.1093/biomet/64.2.191.

Pocock, Stuart J., Cono A. Ariti, Timothy J. Collier, and Duolao Wang. 2012. "The Win Ratio: A New Approach to the Analysis of Composite Endpoints in Clinical Trials Based on Clinical Priorities." *European Heart Journal* 33 (2): 176–82. https://doi.org/10.1093/eurheartj/ehr352.

Pocock, Stuart J., and Richard Simon. 1975. "Sequential Treatment Assignment with Balancing for Prognostic Factors in the Controlled Clinical Trial." *Biometrics* 31 (1): 103–15. https://doi.org/10.2307/2529712.

Pokorney, Sean D., Noelle Cocoros, Hussein R. Al-Khalidi, Kevin Haynes, Shuang Li, Sana M. Al-Khatib, et al. 2022. "Effect of Mailing Educational Material to Patients with Atrial Fibrillation and Their Clinicians on Use of Oral Anticoagulants: A Randomized Clinical Trial." *JAMA Network Open* 5 (5): e2214321. https://doi.org/10.1001/jamanetworkopen.2022.14321.

Prasad, Vinay, and Sham Mailankody. 2017. "Research and Development Spending to Bring a Single Cancer Drug to Market and Revenues After Approval." *JAMA Internal Medicine* 177 (11): 1569–75. https://doi.org/10.1001/jamainternmed.2017.3601.

Prasad, Vinay, and Martin A. Makary. 2026. "One Pivotal Trial, the New Default Option for FDA Approval: Ending the Two-Trial Dogma." *New England Journal of Medicine* 394 (8): 815–17. https://doi.org/10.1056/NEJMsb2517623.

Proschan, Michael A., and Scott R. Evans. 2020. "Resist the Temptation of Response-Adaptive Randomization." *Clinical Infectious Diseases* 71 (11): 3002–4. https://doi.org/10.1093/cid/ciaa334.

Proschan, Michael A., and Sally A. Hunsberger. 1995. "Designed Extension of Studies Based on Conditional Power." *Biometrics* 51 (4): 1315. https://doi.org/10.2307/2533262.

Quintana, Melanie, Benjamin R. Saville, Matteo Vestrucci, Michelle A. Detry, Lori

Chibnik, Jeremy Shefner, James D. Berry, et al. 2023. "Design and Statistical Innovations in a Platform Trial for Amyotrophic Lateral Sclerosis." *Annals of Neurology* 94 (3): 547–60. https://doi.org/10.1002/ana.26714.

Rao, Gowtham A. 2026. "The OHDSI Phenotype Library." R package and GitHub repository, version 3.37.0, part of HADES. https://github.com/OHDSI/PhenotypeLibrary.

RECOVERY Collaborative Group, Peter Horby, Wei Shen Lim, Jonathan R. Emberson, Marion Mafham, Jennifer L. Bell, et al. 2021. "Dexamethasone in Hospitalized Patients with Covid-19." *New England Journal of Medicine* 384 (8): 693–704. https://doi.org/10.1056/NEJMoa2021436.

Redman, Mary W., Vassiliki A. Papadimitrakopoulou, Glenwood D. Goss, et al. 2020. "Biomarker-Driven Therapies for Previously Treated Squamous Non-Small-Cell Lung Cancer (Lung-MAP SWOG S1400): A Biomarker-Driven Master Protocol." *The Lancet Oncology* 21 (12): 1589–1601. https://doi.org/10.1016/S1470-2045(20)30475-7.

REMAP-CAP Investigators, ACTIV-4a Investigators, ATTACC Investigators, Ewan C. Goligher, Corrinne A. Bradbury, Bryan J. McVerry, et al. 2021. "Therapeutic Anticoagulation with Heparin in Critically Ill Patients with Covid-19." *New England Journal of Medicine* 385 (9): 777–89. https://doi.org/10.1056/NEJMoa2103417.

REMAP-CAP Investigators, Anthony C. Gordon, Paul R. Mouncey, Farah Al-Beidh, Kathryn M. Rowan, Alistair D. Nichol, et al. 2021. "Interleukin-6 Receptor Antagonists in Critically Ill Patients with Covid-19." *New England Journal of Medicine* 384 (16): 1491–1502. https://doi.org/10.1056/NEJMoa2100433.

Renfro, Lindsay A., and Daniel J. Sargent. 2017. "Statistical Controversies in Clinical Research: Basket Trials, Umbrella Trials, and Other Master Protocols: A Review and Examples." *Annals of Oncology* 28 (1): 34–43. https://doi.org/10.1093/annonc/mdw413.

Ritchie, Craig W., José Luis Molinuevo, Luc Truyen, Andrew Satlin, Serge Van der Geyten, Simon Lovestone, and EPAD Consortium. 2016. "Development of Interventions for the Secondary Prevention of Alzheimer's Dementia: The European Prevention of Alzheimer's Dementia (EPAD) Project." *Lancet Psychiatry* 3 (2): 179–86. https://doi.org/10.1016/S2215-0366(15)00454-X.

Rivera, Samantha Cruz et al. 2020. "SPIRIT-AI Extension: Guidance for Clinical Trial Protocols for Interventions Involving Artificial Intelligence." *BMJ* 370: m3210. https://www.bmj.com/content/370/bmj.m3210.

Robertson, David S., Babak Choodari-Oskooei, Munya Dimairo, Laura Flight, Philip Pallmann, and Thomas Jaki. 2023. "Point estimation for adaptive trial designs II: Practical considerations and guidance." *Statistics in Medicine* 42 (14): 2496–2520. https://doi.org/10.1002/sim.9734.

Robertson, David S., Kim May Lee, Boryana C. López-Kolkovska, and Sofía S. Villar. 2023. "Response-Adaptive Randomization in Clinical Trials: From Myths to Practical Considerations." *Statistical Science* 38 (2): 185–208. https://doi.org/10.1214/22-STS865.

Robins, James M., Miguel A. Hernán, and Babette Brumback. 2000. "Marginal

Structural Models and Causal Inference in Epidemiology." *Epidemiology* 11 (5): 550–60. https://doi.org/10.1097/00001648-200009000-00011.

Roig, Marta Bofill, Cora Burgwinkel, Ursula Garczarek, Franz Koenig, Martin Posch, Quynh Nguyen, and Katharina Hees. 2023. "On the use of non-concurrent controls in platform trials: a scoping review." *Trials* 24 (1). https://doi.org/10.1186/s13063-023-07398-7.

Roig, Marta Bofill, Pavla Krotka, Carl-Fredrik Burman, Ekkehard Glimm, Stefan M. Gold, Katharina Hees, Peter Jacko, et al. 2022. "On model-based time trend adjustments in platform trials with non-concurrent controls." *BMC Medical Research Methodology* 22 (1). https://doi.org/10.1186/s12874-022-01683-w.

Rosenbaum, Paul R., and Donald B. Rubin. 1983. "The Central Role of the Propensity Score in Observational Studies for Causal Effects." *Biometrika* 70 (1): 41–55. https://doi.org/10.1093/biomet/70.1.41.

Rosenberger, William F., and John M. Lachin. 2016. *Randomization in Clinical Trials: Theory and Practice*. 2nd ed. Hoboken, NJ: John Wiley & Sons.

Ross, Jessica L., Arman Sabbaghi, Run Zhuang, and Daniele Bertolini. 2024. "Enhancing Longitudinal Clinical Trial Efficiency with Digital Twins and Prognostic Covariate-Adjusted Mixed Models for Repeated Measures (PROCOVA-MMRM)." *arXiv Preprint arXiv:2404.17576*. https://arxiv.org/abs/2404.17576.

Rossouw, Jacques E., Garnet L. Anderson, Ross L. Prentice, Andrea Z. LaCroix, Charles Kooperberg, Marcia L. Stefanick, Rebecca D. Jackson, et al. 2002. "Risks and Benefits of Estrogen Plus Progestin in Healthy Postmenopausal Women: Principal Results from the Women's Health Initiative Randomized Controlled Trial." *JAMA* 288 (3): 321–33. https://doi.org/10.1001/jama.288.3.321.

Roth, Jonathan. 2022. "Pretest with Caution: Event-Study Estimates After Testing for Parallel Trends." *American Economic Review: Insights* 4 (3): 305–22. https://doi.org/10.1257/aeri.20210236.

Rubin, Donald B. 1974. "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies." *Journal of Educational Psychology* 66 (5): 688–701. https://doi.org/10.1037/h0037350.

———. 1987. *Multiple Imputation for Nonresponse in Surveys*. New York: John Wiley & Sons. https://doi.org/10.1002/9780470316696.

Rugo, Hope S., Olufunmilayo I. Olopade, Angela DeMichele, Christina Yau, Laura J. van 't Veer, Meredith B. Buxton, Michael Hogarth, et al. 2016. "Adaptive Randomization of Veliparib-Carboplatin Treatment in Breast Cancer." *New England Journal of Medicine* 375 (1): 23–34. https://doi.org/10.1056/NEJMoa1513749.

Rush, A. John, Maurizio Fava, Stephen R. Wisniewski, Philip W. Lavori, Madhukar H. Trivedi, Harold A. Sackeim, Michael E. Thase, et al. 2004. "Sequenced Treatment Alternatives to Relieve Depression (STAR*D): Rationale and Design." *Controlled Clinical Trials* 25 (1): 119–42. https://doi.org/10.1016/S0197-2456(03)00112-0.

Rush, A. John, Madhukar H. Trivedi, Stephen R. Wisniewski, Andrew A. Nierenberg, Jonathan W. Stewart, Diane Warden, George Niederehe, et al. 2006. "Acute and Longer-Term Outcomes in Depressed Outpatients Requiring One or Several Treatment Steps: A STAR*d Report." *American Journal of Psychiatry*

163 (11): 1905–17. https://doi.org/10.1176/ajp.2006.163.11.1905.

Ryan, Patrick B. 2024. "OHDSI for Real World Evidence (RWE)." Presentation slides, Observational Health Data Sciences and Informatics. https://www.ohdsi.org/wp-content/uploads/2024/12/2.-OHDSI-for-Real-World-Evidence-RWE-Patrick-Ryan.pdf.

Saama Technologies. 2024. "Saama: Life Sciences Analytics Cloud." https://www.saama.com.

Saleiro, Pedro, Benedict Kuester, Loren Hinkson, Jesse London, Abby Stevens, Ari Anisfeld, Kit T. Rodolfa, and Rayid Ghani. 2018. "Aequitas: A Bias and Fairness Audit Toolkit." *arXiv Preprint arXiv:1811.05577*. https://arxiv.org/abs/1811.05577.

Saunders, Stina, Sarah Gregory, Matthew H. S. Clement, Cindy Birck, Serge Van Der Geyten, and Craig W. Ritchie. 2022. "The European Prevention of Alzheimer's Dementia Programme: An Innovative Medicines Initiative-Funded Partnership to Facilitate Secondary Prevention of Alzheimer's Disease Dementia." *Frontiers in Neurology* 13: 1051543. https://doi.org/10.3389/fneur.2022.1051543.

Saver, Jeffrey L. 2007. "Novel End Point Analytic Techniques and Interpreting Shifts Across the Entire Range of Outcome Scales in Acute Stroke Trials." *Stroke* 38 (11): 3055–62. https://doi.org/10.1161/STROKEAHA.107.488536.

Saville, Benjamin R., and Scott M. Berry. 2016. "Efficiencies of Platform Clinical Trials: A Vision of the Future." *Clinical Trials* 13 (3): 358–66. https://doi.org/10.1177/1740774515626362.

Saville, Benjamin R, Donald A Berry, Nicholas S Berry, Kert Viele, and Scott M Berry. 2022. "The Bayesian Time Machine: Accounting for temporal drift in multi-arm platform trials." *Clinical Trials* 19 (5): 490–501. https://doi.org/10.1177/17407745221112013.

Schulz, Kenneth F. 1995. "Subverting Randomization in Controlled Trials." *JAMA* 274 (18): 1456–58. https://doi.org/10.1001/jama.1995.03530180050029.

Scott, Neil W., Gladys C. McPherson, Craig R. Ramsay, and Marion K. Campbell. 2002. "The Method of Minimization for Allocation to Clinical Trials: A Review." *Controlled Clinical Trials* 23 (6): 662–74. https://doi.org/10.1016/S0197-2456(02)00242-8.

Sel, Kaan, Andrea Hawkins-Daarud, Anirban Chaudhuri, Deen Osman, Ahmad Bahai, David Paydarfar, Karen Willcox, Caroline Chung, and Roozbeh Jafari. 2025. "Survey and Perspective on Verification, Validation, and Uncertainty Quantification of Digital Twins for Precision Medicine." *Npj Digital Medicine* 8: 40. https://doi.org/10.1038/s41746-025-01447-y.

Sentinel Operations Center. 2025. "An Assessment of the Sentinel System (2022 to 2024)." U.S. Food and Drug Administration. https://www.sentinelinitiative.org/sites/default/files/documents/PDUFA-VII-Sentinel-System-Report_An-Assessment-of-the-Sentinel-System_2022-to-2024.pdf.

Sertkaya, Aylin, Hui-Hsing Wong, Amber Jessup, and Trinidad Beleche. 2016. "Key Cost Drivers of Pharmaceutical Clinical Trials in the United States." *Clinical Trials* 13 (2): 117–26. https://doi.org/10.1177/1740774515625964.

Shiryaev, Albert N. 1978. *Optimal Stopping Rules*. Berlin: Springer-Verlag.

Shriver, Sharon P., Liora Sahar, Vanhvilai L. Douangchai Wills, Devon V. Adams, and Mark E. Fleury. 2025. "Assessing Populations with Access to National Cancer Institute-Funded Sites Using Local Distance-Based Service Areas." *Journal of Clinical and Translational Science* 9 (1): e218. https://doi.org/10.1017/cts.2025.10148.

Simon, Noah, and Richard Simon. 2013. "Adaptive Enrichment Designs for Clinical Trials." *Biostatistics* 14 (4): 613–25. https://doi.org/10.1093/biostatistics/kxt010.

Smith, Zachary P., Joseph A. DiMasi, and Kenneth A. Getz. 2024. "New Estimates on the Cost of a Delay Day in Drug Development." *Therapeutic Innovation & Regulatory Science* 58 (5): 855–62. https://doi.org/10.1007/s43441-024-00667-w.

Smith, Zachary, Joseph A. DiMasi, and Kenneth A. Getz. 2024. "Quantifying the Value of a Day of Delay in Drug Development." White Paper. Tufts Center for the Study of Drug Development. https://csdd.tufts.edu/sites/default/files/2025-02/Aug2024%20Day%20of%20Delay%20White%20Paper%20Final.pdf.

Society for Clinical Data Management. 2023. "Database Closure." Good Clinical Data Management Practices chapter, Journal of the Society for Clinical Data Management. https://scdm.org/wp-content/uploads/2024/07/Database-Closure.pdf.

Sommer, C., D. Zuccolin, V. Arnera, N. Schmitz, P. Adolfsson, N. Colombo, R. Gilg, and B. McDowell. 2018. "Building Clinical Trials Around Patients: Evaluation and Comparison of Decentralized and Conventional Site Models in Patients with Low Back Pain." *Contemporary Clinical Trials Communications* 11: 120–26. https://doi.org/10.1016/j.conctc.2018.06.008.

Stein, Charles. 1956. "Inadmissibility of the Usual Estimator for the Mean of a Multivariate Normal Distribution." In *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, 1:197–206. https://www.degruyter.com/document/doi/10.1525/9780520313880-018/html.

Struebing, Alessandria, Chelsea McKibbon, Haoyao Ruan, Emma Mackay, Natalie Dennis, Russanthy Velummailum, Philip He, et al. 2024. "Augmenting External Control Arms Using Bayesian Borrowing: A Case Study in First-Line Non-Small Cell Lung Cancer." *Journal of Comparative Effectiveness Research* 13 (5). https://doi.org/10.57264/cer-2023-0175.

Suissa, Samy. 2008. "Immortal Time Bias in Pharmacoepidemiology." *American Journal of Epidemiology* 167 (4): 492–99. https://doi.org/10.1093/aje/kwm324.

Swanson, Chad J., Yong Zhang, Shobha Dhadda, Jianing Wang, June Kaplow, Robert Y. K. Lai, Lars Lannfelt, et al. 2021. "A Randomized, Double-Blind, Phase 2b Proof-of-Concept Clinical Trial in Early Alzheimer's Disease with Lecanemab, an Anti-aβ Protofibril Antibody." *Alzheimer's Research and Therapy* 13 (1): 80. https://doi.org/10.1186/s13195-021-00813-8.

Tang, Chad, Kenneth R. Hess, Dwana Sanders, Suzanne E. Davis, Aman U. Buzdar, Razelle Kurzrock, J. Jack Lee, Funda Meric-Bernstam, et al. 2017. "Modifying the Clinical Research Infrastructure at a Dedicated Clinical Trials Unit: Assess-

ment of Trial Development, Activation, and Participant Accrual." *Clinical Cancer Research* 23 (6): 1407–13. https://doi.org/10.1158/1078-0432.CCR-16-1936.

Taves, Donald R. 1974. "Minimization: A New Method of Assigning Patients to Treatment and Control Groups." *Clinical Pharmacology & Therapeutics* 15 (5): 443–53. https://doi.org/10.1002/cpt1974155443.

Thall, P., P. Fox, and J. Wathen. 2015. "Statistical controversies in clinical research: scientific and ethical problems with adaptive randomization in comparative clinical trials." *Annals of Oncology* 26 (8): 1621–28. https://doi.org/10.1093/annonc/mdv238.

Thompson, William R. 1933. "On the Likelihood That One Unknown Probability Exceeds Another in View of the Evidence of Two Samples." *Biometrika* 25 (3/4): 285–94. https://doi.org/10.2307/2332286.

Thorlund, Kristian, Jonas Haggstrom, Jay JH Park, and Edward J Mills. 2018. "Key design considerations for adaptive clinical trials: a primer for clinicians." *BMJ*, k698. https://doi.org/10.1136/bmj.k698.

TMF Reference Model Initiative. 2024. "Trial Master File Reference Model." https://tmfrefmodel.com/.

Tong, Steven Y. C., Jocelyn Mora, Asha C. Bowen, Matthew P. Cheng, Nick Daneman, Anna L. Goodman, George S. Heriot, et al. 2022. "The Staphylococcus aureus Network Adaptive Platform Trial Protocol: New Tools for an Old Foe." *Clinical Infectious Diseases* 75 (11): 2027–34. https://doi.org/10.1093/cid/ciac476.

Tufts Center for the Study of Drug Development. 2013. "89% of Trials Meet Enrollment, but Timelines Slip, Half of Sites Under-Enroll." Tufts CSDD Impact Report 15(1), January/February 2013. https://csdd.tufts.edu/publications/impact-reports.

———. 2021. "Tufts Center for the Study of Drug Development Impact Reports." https://csdd.tufts.edu/publications/impact-reports.

———. 2024. "Clinical Trial Management System Benchmarks." https://csdd.tufts.edu/publications/impact-reports.

Turner, Ann, Thomas Ricketts, and Laurel K. Leslie. 2020. "Comparison of Number and Geographic Distribution of Pediatric Subspecialists and Patient Proximity to Specialized Care in the US Between 2003 and 2019." *JAMA Pediatrics* 174 (9): 852–60. https://doi.org/10.1001/jamapediatrics.2020.1124.

UF Health. 2004. "Group Forms to Test New Cancer Drugs in Children." https://ufhealth.org/news/2004/group-forms-test-new-cancer-drugs-children.

UK Collaborative ECMO Trial Group. 1996. "UK Collaborative Randomised Trial of Neonatal Extracorporeal Membrane Oxygenation." *The Lancet* 348 (9020): 75–82. https://doi.org/10.1016/S0140-6736(96)04100-1.

Unger, Joseph M., Barbara L. McAneny, and Raymond U. Osarogiagbon. 2025. "Cancer in Rural America: Improving Access to Clinical Trials and Quality of Oncologic Care." *CA: A Cancer Journal for Clinicians* 75 (4): 341–61. https://doi.org/10.3322/caac.70006.

Unger, Joseph M., Adrienne Moseley, Beulah Symington, Mariana Chavez-MacGregor, Scott D. Ramsey, and Dawn L. Hershman. 2018. "Ge-

ographic Distribution and Survival Outcomes for Rural Patients with Cancer Treated in Clinical Trials." *JAMA Network Open* 1 (4): e181235. https://doi.org/10.1001/jamanetworkopen.2018.1235.

Unger, Joseph M., Lawrence N. Shulman, Matthew A. Facktor, Heidi Nelson, and Mark E. Fleury. 2024. "National Estimates of the Participation of Patients with Cancer in Clinical Research Studies Based on Commission on Cancer Accreditation Data." *Journal of Clinical Oncology* 42 (18): 2139–48. https://doi.org/10.1200/JCO.23.01030.

United States Congress. 1951. "An Act to Amend Section 503(b) of the Federal Food, Drug, and Cosmetic Act (Durham-Humphrey Amendment)." https://www.govinfo.gov/content/pkg/STATUTE-65/pdf/STATUTE-65-Pg648.pdf.

———. 1997. "Food and Drug Administration Modernization Act of 1997."

———. 2007. "Food and Drug Administration Amendments Act of 2007 (FDAAA), Public Law 110-85, Section 801." https://clinicaltrials.gov/policy/fdaaa-801-final-rule.

University of North Carolina at Chapel Hill. 2024. "Nuremberg Code." https://research.unc.edu/human-research-ethics/resources/ccm3_019064/.

University-Industry Demonstration Partnership. 2014. "Accelerated Clinical Trial Agreement (ACTA)." https://uidp.org/our-work/acta/.

Unlearn.AI. 2024. "Unlearn.AI: Digital Twins for Clinical Trials." https://www.unlearn.ai.

———. 2025. "A Risk-Based Approach for Leveraging AI in Clinical Trials." https://www.unlearn.ai/blog/how-unlearn-boosts-trial-power-using-the-fdas-ai-framework.

U.S. Congress. 2022. "Food and Drug Omnibus Reform Act (FDORA) of 2022." https://www.fda.gov/regulatory-information/selected-amendments-fdc-act/food-and-drug-omnibus-reform-act-fdora-2022.

U.S. Department of Agriculture. 1937. "Report of the Secretary of Agriculture on Deaths Due to Elixir Sulfanilamide-Massengill."

U.S. Department of Health and Human Services. 2024. "45 CFR Part 46: Protection of Human Subjects (Common Rule)." https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-A/part-46.

U.S. Environmental Protection Agency. 2007. "Guidance for Preparing Standard Operating Procedures (SOPs) (EPA QA/G-6)." EPA/600/B-07/001. https://www.epa.gov/sites/default/files/2015-06/documents/g6-final.pdf.

U.S. Food and Drug Administration. 1994. "Dose-Response Information to Support Drug Registration." U.S. Food and Drug Administration. https://www.fda.gov/media/71279/download.

———. 2005. "Estimating the Maximum Safe Starting Dose in Initial Clinical Trials for Therapeutics in Adult Healthy Volunteers." Guidance for Industry, Center for Drug Evaluation and Research. https://www.fda.gov/media/72309/download.

———. 2006. "Guidance for Clinical Trial Sponsors: Establishment and Operation of Clinical Trial Data Monitoring Committees." https://www.fda.gov/media/75398/download.

———. 2010a. "Adaptive Design Clinical Trials for Drugs and Biologics: Draft Guidance for Industry." https://web.archive.org/web/2012/http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/UCM201790.pdf.

———. 2010b. "Guidance for the Use of Bayesian Statistics in Medical Device Clinical Trials." U.S. Food and Drug Administration. https://www.fda.gov/media/71512/download.

———. 2013a. "Electronic Source Data in Clinical Investigations: Guidance for Industry." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/electronic-source-data-clinical-investigations.

———. 2013b. "Oversight of Clinical Investigations: A Risk-Based Approach to Monitoring." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/oversight-clinical-investigations-risk-based-approach-monitoring.

———. 2014. "Expedited Programs for Serious Conditions – Drugs and Biologics: Guidance for Industry." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/expedited-programs-serious-conditions-drugs-and-biologics.

———. 2015. "Investigational New Drug Applications Prepared and Submitted by Sponsor-Investigators: Guidance for Industry (Draft)." https://www.fda.gov/files/drugs/published/Investigational-New-Drug-Applications-Prepared-and-Submitted-by-Sponsor-Investigators.pdf.

———. 2016. "Use of Electronic Informed Consent in Clinical Investigations: Questions and Answers." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/use-electronic-informed-consent-clinical-investigations-questions-and-answers.

———. 2018a. "Framework for FDA's Real-World Evidence Program." https://www.fda.gov/media/120060/download.

———. 2018b. "The Drug Development Process." https://www.fda.gov/patients/learn-about-drug-and-device-approvals/drug-development-process.

———. 2018c. "Use of Electronic Health Record Data in Clinical Investigations: Guidance for Industry." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/use-electronic-health-record-data-clinical-investigations-guidance-industry.

———. 2018d. "Clinical Trial Endpoints for the Approval of Cancer Drugs and Biologics." https://www.fda.gov/media/71195/download.

———. 2019a. "Adaptive Designs for Clinical Trials of Drugs and Biologics: Guidance for Industry." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/adaptive-design-clinical-trials-drugs-and-biologics-guidance-industry.

———. 2019b. "Approval Package for Application Number 207103Orig1s008: IBRANCE (Palbociclib)." https://www.accessdata.fda.gov/drugsatfda_docs/nda/2019/207103Orig1s008.pdf.

———. 2019c. "Demonstrating Substantial Evidence of Effectiveness for Human Drug and Biological Products." U.S. Food and Drug Administration. https:

//www.fda.gov/media/133660/download.

———. 2019d. "Enrichment Strategies for Clinical Trials to Support Approval of Human Drugs and Biological Products." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/enrichment-strategies-clinical-trials-support-approval-human-drugs-and-biological-products.

———. 2020a. "Compliance Program 7348.811: Clinical Investigators and Sponsor-Investigators." https://www.fda.gov/media/75927/download.

———. 2020b. "Interacting with the FDA on Complex Innovative Trial Designs for Drugs and Biological Products: Guidance for Industry." U.S. Food and Drug Administration. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/interacting-fda-complex-innovative-trial-designs-drugs-and-biological-products.

———. 2021a. "Compliance Program 7348.810: Sponsors and Contract Research Organizations." https://www.fda.gov/media/75916/download.

———. 2021b. "Conduct of Clinical Trials of Medical Products During the COVID-19 Public Health Emergency: Guidance for Industry, Investigators, and Institutional Review Boards." https://www.fda.gov/media/137496/download.

———. 2021c. "PDUFA Reauthorization Performance Goals and Procedures, Fiscal Years 2023 Through 2027." https://www.fda.gov/media/151712/download.

———. 2021d. "Project Optimus: Reforming the Dose Optimization and Dose Selection Paradigm in Oncology." FDA Oncology Center of Excellence. https://www.fda.gov/about-fda/oncology-center-excellence/project-optimus.

———. 2021e. "Providing Regulatory Submissions in Electronic Format: Standardized Study Data. Guidance for Industry." https://www.fda.gov/media/82716/download.

———. 2022a. "Master Protocols: Efficient Clinical Trial Design Strategies to Expedite Development of Oncology Drugs and Biologics." U.S. Food and Drug Administration. https://www.fda.gov/media/120721/download.

———. 2022b. "Institutional Review Boards; Cooperative Research (Proposed Rule)." Federal Register 87 FR 58752. https://www.govinfo.gov/content/pkg/FR-2022-09-28/html/2022-21089.htm.

———. 2022c. "Investigational New Drug Application Annual Reporting (Proposed Rule)." https://www.federalregister.gov/documents/2022/12/09/2022-26731/investigational-new-drug-application-annual-reporting.

———. 2023a. "21 CFR Part 11: Electronic Records; Electronic Signatures." https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-11.

———. 2023b. "21 CFR Part 50: Protection of Human Subjects." https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-50.

———. 2023c. "21 CFR Part 56: Institutional Review Boards." https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-56.

———. 2023d. "Benefit-Risk Assessment for New Drug and Biological Products." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/benefit-risk-assessment-new-drug-and-biological-products.

———. 2023e. "Considerations for the Design and Conduct of Externally Controlled Trials for Drug and Biological Products." https://www.fda.gov/

regulatory-information/search-fda-guidance-documents/considerations-design-and-conduct-externally-controlled-trials-drug-and-biological-products.

———. 2023f. "Considerations for the Use of Real-World Data and Real-World Evidence to Support Regulatory Decision-Making for Drug and Biological Products." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/considerations-use-real-world-data-and-real-world-evidence-support-regulatory-decision-making-drug.

———. 2023g. "Digital Health Technologies for Remote Data Acquisition in Clinical Investigations." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/digital-health-technologies-remote-data-acquisition-clinical-investigations.

———. 2023h. "Digital Health Technologies for Remote Data Acquisition in Clinical Investigations: Guidance for Industry, Investigators, and Other Stakeholders." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/digital-health-technologies-remote-data-acquisition-clinical-investigations.

———. 2023i. "Patient-Focused Drug Development: Incorporating Clinical Outcome Assessments into Endpoints for Regulatory Decision-Making." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/patient-focused-drug-development-incorporating-clinical-outcome-assessments-endpoints-regulatory.

———. 2023j. "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products: Guidance for Industry." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/adjusting-covariates-randomized-clinical-trials-drugs-and-biological-products.

———. 2024a. "21 CFR 312.32: IND Safety Reporting." https://www.ecfr.gov/current/title-21/chapter-I/subchapter-D/part-312/subpart-B/section-312.32.

———. 2024b. "21 CFR Part 312: Investigational New Drug Application." https://www.ecfr.gov/current/title-21/chapter-I/subchapter-D/part-312.

———. 2024c. "21 CFR Part 54: Financial Disclosure by Clinical Investigators." https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-54.

———. 2024d. "Clinical Pharmacology Guidance Documents." https://www.fda.gov/drugs/guidance-compliance-regulatory-information/guidances-drugs.

———. 2024e. "Diversity Action Plans to Improve Enrollment of Participants from Underrepresented Populations in Clinical Studies: Guidance for Industry." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/diversity-action-plans-improve-enrollment-participants-underrepresented-populations-clinical-studies.

———. 2024f. "Electronic Systems, Electronic Records, and Electronic Signatures in Clinical Investigations: Questions and Answers." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/electronic-systems-electronic-records-and-electronic-signatures-clinical-investigations-questions.

———. 2024g. "FDA Adverse Event Reporting System (FAERS) Public Dashboard." https://www.fda.gov/drugs/drug-approvals-and-databases/fda-adverse-event-reporting-system-faers-database.

———. 2024h. "FDA Organization." https://www.fda.gov/about-fda/fda-organization.

———. 2024i. "Optimizing the Dosage of Human Prescription Drugs and Biological Products for the Treatment of Oncologic Diseases." Guidance for Industry, Center for Drug Evaluation and Research and Center for Biologics Evaluation and Research. https://www.fda.gov/media/164555/download.

———. 2024j. "Pharmacokinetics in Patients with Impaired Renal Function: Study Design, Data Analysis, and Impact on Dosing. Guidance for Industry." https://www.fda.gov/media/78573/download.

———. 2024k. "Real-World Data: Assessing Electronic Health Records and Medical Claims Data to Support Regulatory Decision-Making for Drug and Biological Products." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/real-world-data-assessing-electronic-health-records-and-medical-claims-data-support-regulatory.

———. 2024l. "Risk Evaluation and Mitigation Strategies (REMS)." https://www.fda.gov/drugs/drug-safety-and-availability/risk-evaluation-and-mitigation-strategies-rems.

———. 2024m. "Study Data Technical Conformance Guide." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/study-data-technical-conformance-guide-technical-specifications-document.

———. 2024n. "Conducting Clinical Trials with Decentralized Elements." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/conducting-clinical-trials-decentralized-elements.

———. 2025a. "CDER/CBER Rare Disease Evidence Principles (RDEP)." https://www.fda.gov/industry/fda-rare-disease-innovation-hub/cdercber-rare-disease-evidence-principles-rdep.

———. 2025b. "Drug Development Tools: Fit-for-Purpose Initiative." https://www.fda.gov/drugs/development-approval-process-drugs/drug-development-tools-fit-purpose-initiative.

———. 2025c. "Enhancing Participation in Clinical Trials: Eligibility Criteria, Enrollment Practices, and Trial Designs." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/enhancing-participation-clinical-trials-eligibility-criteria-enrollment-practices-and-trial-designs.

———. 2025d. "FDA Qualifies Total Hip Bone Mineral Density (BMD) as Surrogate Endpoint for Osteoporosis Drug Development." https://www.fda.gov/drugs/drug-safety-and-availability/fda-qualifies-total-hip-bone-mineral-density-bmd-surrogate-endpoint-osteoporosis-drug-development.

———. 2025e. "Pediatric Labeling Changes." https://www.fda.gov/science-research/pediatrics/pediatric-labeling-changes.

———. 2025f. "Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/considerations-use-artificial-intelligence-support-regulatory-decision-making-drug-and-biological.

———. 2026a. "21 CFR 312.33: Annual Reports." https://www.ecfr.gov/current/

title-21/chapter-I/subchapter-D/part-312/subpart-B/section-312.33.

———. 2026b. "21 CFR Part 812: Investigational Device Exemptions." https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-812.

———. 2026c. "Demonstrating Substantial Evidence of Effectiveness for Human Drug and Biological Products." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/demonstrating-substantial-evidence-effectiveness-human-drug-and-biological-products.

———. 2026d. "FDA Launches New Adverse Event Look-Up Tool." https://www.fda.gov/news-events/press-announcements/fda-launches-new-adverse-event-look-tool.

———. 2026e. "Master Protocols for Drug and Biological Product Development: Guidance for Industry." https://www.fda.gov/regulatory-information/search-fda-guidance-documents/master-protocols-drug-and-biological-product-development.

———. 2026f. "Prescription Drug User Fee Rates for Fiscal Year 2027." Federal Register 91 FR 48160, 30 July 2026. https://www.federalregister.gov/documents/2026/07/30/2026-15334/prescription-drug-user-fee-rates-for-fiscal-year-2027.

———. 2026g. "Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products." U.S. Food and Drug Administration. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/use-bayesian-methodology-clinical-trials-drug-and-biological-products.

U.S. Food and Drug Administration, Oncology Center of Excellence. 2024a. "Project Orbis." https://www.fda.gov/about-fda/oncology-center-excellence/project-orbis.

———. 2024b. "Real-Time Oncology Review (RTOR)." https://www.fda.gov/about-fda/oncology-center-excellence/real-time-oncology-review.

U.S. Food and Drug Administration, and European Medicines Agency. 2026. "Guiding Principles of Good AI Practice in Drug Development." https://www.fda.gov/media/189581/download.

U.S. Food and Drug Administration, Health Canada, and UK Medicines and Healthcare products Regulatory Agency. 2021. "Good Machine Learning Practice for Medical Device Development: Guiding Principles." https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles.

U.S. National Library of Medicine. 2026. "ClinicalTrials.gov API Version 2: Study Counts by Lead Sponsor Class and Responsible Party." https://clinicaltrials.gov/data-api/api.

US Food and Drug Administration, Center for Biologics Evaluation and Research. 2022. "Clinical Review Memorandum: REBYOTA (Fecal Microbiota, Live-jslm), BLA 125739." US Food and Drug Administration. https://www.fda.gov/media/164040/download.

VanderWeele, Tyler J., and Peng Ding. 2017. "Sensitivity Analysis in Observational Research: Introducing the E-Value." *Annals of Internal Medicine* 167 (4): 268–74. https://doi.org/10.7326/M16-2607.

Vardeny, Orly, Adrian F. Hernandez, Lauren W. Cohen, Amy Franklin, Mina Baqai, Sarah Palmer, Barbara E. Bierer, and Nichelle Cobb. 2019. "Transitioning to the National Institutes of Health Single Institutional Review Board Model: Piloting the Use of the Streamlined, Multi-Site, Accelerated Resources for Trials IRB Reliance." *Clinical Trials* 16 (3): 290–96. https://doi.org/10.1177/1740774519832911.

Veeva Systems. 2024a. "Veeva TMF Intake Agent and Quality Check Agent." https://www.veeva.com/resources/tmf-intake-agent-and-quality-check-agent/.

———. 2024b. "Veeva Vault Clinical Suite." https://www.veeva.com/products/vault-clinical/.

———. 2026. "Veeva eTMF." https://www.veeva.com/products/veeva-etmf/.

Venkatesh, Kaushik P., Marium M. Raza, and Joseph C. Kvedar. 2022. "Health Digital Twins as Tools for Precision Medicine: Considerations for Computation, Implementation, and Regulation." *Npj Digital Medicine* 5: 150. https://doi.org/10.1038/s41746-022-00694-7.

Wagner, John, Andrew M. Dahlem, Lynn D. Hudson, Sharon F. Terry, Russ B. Altman, C. Taylor Gilliland, Christopher DeFeo, and Christopher P. Austin. 2018a. "A Dynamic Map for Learning, Communicating, Navigating and Improving Therapeutic Development." *Nature Reviews Drug Discovery* 17 (2): 150. https://doi.org/10.1038/nrd.2017.217.

———. 2018b. "Application of a Dynamic Map for Learning, Communicating, Navigating, and Improving Therapeutic Development." *Clinical and Translational Science* 11 (2): 166–74. https://doi.org/10.1111/cts.12531.

Wald, Abraham. 1947. *Sequential Analysis*. New York: John Wiley & Sons.

Wang, Haiyun, and Douglas Yee. 2019. "I-SPY 2: A Neoadjuvant Adaptive Clinical Trial Designed to Improve Outcomes in High-Risk Breast Cancer." *Current Breast Cancer Reports* 11 (4): 303–10. https://doi.org/10.1007/s12609-019-00334-2.

Wang, Lei, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, et al. 2024. "A Survey on Large Language Model Based Autonomous Agents." *Frontiers of Computer Science* 18 (6): 186345. https://arxiv.org/abs/2308.11432.

Wang, Zifeng, Junyi Gao, Benjamin Danek, Brandon Theodorou, Ruba Shaik, Shivashankar Thati, Seunghyun Won, and Jimeng Sun. 2025. "Compliance and Factuality of Large Language Models for Clinical Research Document Generation." *Journal of the American Medical Informatics Association*. https://doi.org/10.1093/jamia/ocaf174.

Warraich, Haider J., Troy Tazbaz, and Robert M. Califf. 2025. "FDA Perspective on the Regulation of Artificial Intelligence in Health Care and Biomedicine." *JAMA* 333 (3): 241–47. https://doi.org/10.1001/jama.2024.21451.

Wasserstein, Ronald L., and Nicole A. Lazar. 2016. "The ASA Statement on *p*-Values: Context, Process, and Purpose." *The American Statistician* 70 (2): 129–33. https://doi.org/10.1080/00031305.2016.1154108.

Wathen, J Kyle, and Peter F Thall. 2017. "A simulation study of outcome adaptive randomization in multi-arm clinical trials." *Clinical Trials* 14 (5): 432–40. https:

//doi.org/10.1177/1740774517692302.

Watson, Sara E., Paul Smith, Jessica Snowden, Vida Vaughn, Lesley Cottrell, Christi A. Madden, Alberta S. Kong, Russell McCulloh, et al. 2022. "Facilitators and Barriers to Pediatric Clinical Trial Recruitment and Retention in Rural and Community Settings: A Scoping Review of the Literature." *Clinical and Translational Science* 15 (4): 838–53. https://doi.org/10.1111/cts.13220.

WCG Clinical. 2024. "2024 Clinical Research Site Challenges Report: Data-Driven Insights on Current Site Challenges and Strategic Recommendations to Overcome Barriers and Boost Clinical Trial Efficiency." https://www.wcgclinical.com/wp-content/uploads/2024/10/WCG_2024_Clinical_Research_Site_Challenges_Report.pdf.

Wei, Jason, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models." *Advances in Neural Information Processing Systems* 35: 24824–37. https://arxiv.org/abs/2201.11903.

Wei, L. J., and S. Durham. 1978. "The Randomized Play-the-Winner Rule in Medical Trials." *Journal of the American Statistical Association* 73 (364): 840–43. https://doi.org/10.1080/01621459.1978.10480109.

Wenzl, Martin, and Suzannah Chapman. 2019. "Performance-Based Managed Entry Agreements for New Medicines in OECD Countries and EU Member States: How They Work and Possible Improvements Going Forward." OECD Health Working Paper 115. Paris: Organisation for Economic Co-operation and Development. https://doi.org/10.1787/6e5e4c0f-en.

WHO Working Group on the Clinical Characterisation and Management of COVID-19 infection. 2020. "A Minimal Common Outcome Measure Set for COVID-19 Clinical Research." *Lancet Infectious Diseases* 20 (8): e192–97. https://doi.org/10.1016/S1473-3099(20)30483-7.

Williams, Rebecca J., Tony Tse, Katelyn DiPiazza, and Deborah A. Zarin. 2015. "Terminated Trials in the ClinicalTrials.gov Results Database: Evaluation of Availability of Primary Outcome Data and Reasons for Termination." *PLOS ONE* 10 (5): e0127242. https://doi.org/10.1371/journal.pone.0127242.

Wong, Chi Heem, Kien Wei Siah, and Andrew W. Lo. 2019. "Estimation of Clinical Trial Success Rates and Related Parameters." *Biostatistics* 20 (2): 273–86. https://doi.org/10.1093/biostatistics/kxx069.

Woodcock, Janet, and Lisa M. LaVange. 2017. "Master Protocols to Study Multiple Therapies, Multiple Diseases, or Both." *New England Journal of Medicine* 377 (1): 62–70. https://doi.org/10.1056/NEJMra1510062.

World Health Organization. 2021. "Ethics and Governance of Artificial Intelligence for Health." https://iris.who.int/handle/10665/341996.

World Medical Association. 2013. "Declaration of Helsinki: Ethical Principles for Medical Research Involving Human Subjects." https://www.wma.net/policies-post/wma-declaration-of-helsinki/.

———. 2025. "World Medical Association Declaration of Helsinki: Ethical Principles for Medical Research Involving Human Participants." *JAMA* 333 (1): 71–74. https://doi.org/10.1001/jama.2024.21972.

Wu, Qingyun, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, et al. 2023. "AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation." *arXiv Preprint arXiv:2308.08155*. https://arxiv.org/abs/2308.08155.

Xu, Yang, Min Dong, and Xuemei Liu. 2020. "Characteristics and Trends of Clinical Studies Primarily Sponsored by China in WHO Primary Registries Between 2009 and 2018: A Cross-Sectional Survey." *BMJ Open* 10 (11): e037262. https://doi.org/10.1136/bmjopen-2020-037262.

Yao, Shunyu, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. "ReAct: Synergizing Reasoning and Acting in Language Models." *International Conference on Learning Representations*. https://arxiv.org/abs/2210.03629.

Yuan, Ying, Kenneth R. Hess, Susan G. Hilsenbeck, and Mark R. Gilbert. 2016. "Bayesian Optimal Interval Design: A Simple and Well-Performing Design for Phase I Oncology Trials." *Clinical Cancer Research* 22 (17): 4291–301. https://doi.org/10.1158/1078-0432.CCR-16-0592.

Zhou, Xian, Suyu Liu, Edward S. Kim, Roy S. Herbst, and J. Jack Lee. 2008. "Bayesian Adaptive Design for Targeted Therapy Development in Lung Cancer." *Clinical Trials* 5 (3): 181–93. https://doi.org/10.1177/1740774508091815.